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**Mathematical Papers**

OF

Arthur Cayley, Sc.D., F.R.S.,  
Sadlerian Professor of Pure Mathematics in the University of Cambridge.

VOL. II.

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## Advertisement (book p. v).

The present volume contains fifty-eight papers (numbered 101, 102, . . . , 158) originally published, all but two of them, in the years 1851 to 1860: they are here reproduced nearly but not exactly in chronological order. The two excepted papers are 142, *Numerical Tables Supplementary to Second Memoir on Quantics*, now first published (1889); and, 143, *Tables of the Covariants M to W of the Binary Quintic*: from the Second, Third, Fourth, Fifth, Eighth, Ninth and Tenth Memoirs on Quantics (arranged in the present form, 1889): the determination of the finite number, 23, of the covariants of the quintic was made by Gordan in the year 1869, and the calculation of them having been completed in my Ninth and Tenth Memoirs, it appeared to me convenient in the present republication to unite together the values of all the covariants: viz. those of A to L are given in the Second Memoir 141, and the remainder M to W in the paper 143. I have added to the Third Memoir 144, in the notation thereof, some formulae which on account of a difference of notation were omitted from a former paper, 35.

I remark that the present volume comprises the first six of the ten Memoirs on Quantics, viz. these are 139, 141, 144, 155, 156 and 158. I have, in the Notes and References, inserted a discussion of some length in reference to the paper 121, *Note on a Question in the Theory of Probabilities*: and also some remarks in reference to the theory of Distance developed in the Sixth Memoir on Quantics, 158.

A. C.

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## 101. Notes on Lagrange's Theorem.

[From the *Cambridge and Dublin Mathematical Journal*, vol. vi. (1851), pp. 37—45.]

### I.

If in the ordinary form of Lagrange's theorem we write  $(x + a)$  for  $x$ , it becomes

$$\begin{aligned} x &= hf(a + x), \\ F(a + x) &= Fa + \frac{h}{1} F'a fa + \&c. \end{aligned} \quad (1)$$

It follows that the equation

$$F(a + x) = Fa + \sum \frac{1}{r} \frac{x^{r-1}}{[r-1]} \frac{d^{r-1}}{da^{r-1}} (F'a f^r a) + \dots \quad (2)$$

must reduce itself to an identity when the two sides are expanded in powers of  $x$ ; or writing for shortness  $F$ ,  $f$  instead of  $Fa$ ,  $fa$ , and  $\delta$  for  $\frac{d}{da}$ , we must have

$$\frac{1}{[r]^r} \delta^{r-1} F = \delta \sum \frac{1}{[p-1]^{p-1} [r-p]^{r-p}} \delta^{p-1} (\delta F \cdot f^p) \cdot \delta^{r-p} f^{-p} \cdot \frac{1}{p}, \quad (3)$$

(where  $p$  extends from 1 to  $r$ ). Or what comes to the same,

$$\frac{1}{[r]^r} \delta^{r-1} F = \sum \frac{1}{[s]^s [p] [r-p]^{r-p}} \delta^{s-1} \{ (\delta F \cdot f^p) \delta^{r-p} f^{-p} \},$$

where  $s$  extends from 0 to  $(r-p)$ . The terms on the two sides which involve  $\delta^r F$  are immediately seen to be equal; the coefficients of the remaining terms  $\delta^s F$  on the second side must vanish, or we must have

$$\sum \frac{1}{[s]^s} \frac{1}{[p-1]^{p-1} [r-p]^{r-p}} \delta^{r-s} \{ f^p \} \delta^{r-p} f^{-p} = 0 \quad (4)$$

( $s$  being less than  $r$ ). Or in a somewhat more convenient form, writing  $p$ ,  $q$  and  $k$  for  $p-s$ ,  $r-p$  and  $r-s$ ,

$$\sum \frac{1}{[p]^p [q]^q} \delta^p f^{p+s} \delta^q f^{-(p+s)} = 0, \quad (5)$$

where  $s$  is constant and  $p$  and  $q$  vary subject to  $p+q=k$ ,  $k$  being a given constant different from zero (in the case where  $k=0$ , the series reduces itself to the single term  $\frac{1}{s}$ ). The direct proof of this theorem will be given presently.

### II.

The following symbolical form of Lagrange's theorem was given by me in the *Mathematical Journal*, vol. iii. [1843], pp. 283—286, [8].

If

$$x = a + hfx, \quad (6)$$

then

$$Fx = \left( \frac{d}{da} \right)^{-1} \frac{d}{da} Fa \cdot e^{hfa}.$$

Suppose  $fx = \phi(b + k\psi x)$ , or  $x = a + h\phi(b + k\psi x)$ , then

$$Fx = \left( \frac{d}{da} \right)^{-1} \frac{d}{da} Fa \cdot e^{h\phi(b+k\psi a)}.$$

But

$$\phi(b + k\psi a) = \left( \frac{d}{db} \right)^{-1} \frac{d}{db} \phi b \cdot e^{k\psi a}.$$



(In fact the two general terms

$$\{\phi(b + k\psi a)\}^m \quad \text{and} \quad \left(\frac{d}{db}\right)^{-1} e^{k\psi a} (\phi b)^m,$$

of which the former reduces itself to  $e^{k\psi a} \cdot (\phi b)^m$ , are equal on account of the equivalence of the symbols

$$e^{k\psi a} \quad \text{and} \quad \left(\frac{d}{db}\right)^{-1} \frac{d}{db} e^{k\psi a}.$$

Hence

$$\begin{aligned} x &= a + h\phi(b + k\psi x), \\ Fx &= \left(\frac{d}{da}\right)^{-1} \frac{d}{da} \left(\frac{d}{db}\right)^{-1} \frac{d}{db} Fa \cdot e^{h\phi b + k\psi a}, \end{aligned} \tag{7}$$

and the coefficient of  $h^m k^n$  is

$$\frac{1}{[m]^m [n]^n} \left(\frac{d}{da}\right)^{m-1} \left(\frac{d}{db}\right)^{n-1} F'a (\psi a)^n (\phi b)^m.$$

A similar formula evidently applies to the case of any finite number of functions  $\phi, \psi, \&c.$ : in the case of an infinite number we have

$$F(a + h\phi(b + k\psi(c + l\chi(d + \dots))) = \left(\frac{d}{da}\right)^{-1} \left(\frac{d}{db}\right)^{-1} \left(\frac{d}{dc}\right)^{-1} \dots F'a e^{h\phi b + k\psi c + l\chi d + \dots}; \tag{8}$$

or the coefficient of  $h^m k^n l^p \dots$  is

$$\frac{1}{[m]^m [n]^n [p]^p \dots} \left(\frac{d}{da}\right)^{m-1} \left(\frac{d}{db}\right)^{n-1} \left(\frac{d}{dc}\right)^{p-1} \dots F'a (\phi b)^m (\psi c)^n (\chi d)^p \dots,$$

the last of the series  $m, n, p \dots$  being always zero; e.g. in the coefficient of  $h^m k^n$ , account must be had of the factor  $\left(\frac{d}{dc}\right)^{-1} (\chi c)^0$  or  $(\chi c)^0$ . The above form is readily proved independently by Taylor's theorem, without the assistance of Lagrange's. If in it we write  $h = k = \&c.$ ,  $a = b = \&c.$ , and  $\phi = \psi = \&c. = f$ , we have  $F(a + hf(a + hf(a + \dots))) = Fx$ , where  $x = a + hfx$ . Hence, comparing the coefficient of  $h^s$  with that given by Lagrange's theorem,

$$\frac{1}{[s]^s} \delta^{s-1} (\delta F \cdot f^s) = \sum \frac{1}{[m]^m [n]^n \dots} \delta^{m-1} \delta F \cdot f^m \cdot \delta^{n-1} f^n \dots, \tag{9}$$

where  $m + n + \&c. = s$ , and as before  $Fa, fa, \frac{d}{da}$  have been replaced by  $F, f, \delta$ . By comparing the coefficients of  $h^s F$ ,

$$\frac{1}{[t]^t} \delta^t f^t = \sum \frac{1}{[n]^n [p]^p \dots} \delta^{n-1} f^n \cdot \delta^{p-1} f^p \dots, \tag{10}$$

where  $n + p + \dots = t$ , the last of the series  $n, p \dots$  always vanishing. The formula (10) deduced, as above mentioned, from Taylor's theorem, and the subsequent formula (11) with an independent demonstration of it, not I believe materially different from that which will presently be given, are to be found in a memoir by M. Collins (volume ii. (1833) of the Memoirs of the Academy of St Petersburg), who appears to have made very extensive researches in the theory of developments as connected with the combinatorial analysis.

### III.

To demonstrate the formula (6), consider, in the first place, the expression

$$\sum \frac{\phi p}{[p]^p [q]^q} (\delta^p f^{p+s}) (\delta^q f^{-p-s}),$$

where  $p + q = k$ . Since

$$\frac{1}{[p]^p [q]^q} = \frac{1}{k} \left( \frac{1}{[p-1]^{p-1} [q]^q} + \frac{1}{[p]^p [q-1]^{q-1}} \right),$$

this is immediately transformed into

$$\frac{1}{k} \sum \delta \phi p \cdot \frac{1}{[p-1]^{p-1} [q]^q} (\delta^{p-1} f^{p+s}) (\delta^q f^{-p-s}),$$

in which last expression  $p + q = (p - 1)$ . Of this, after separating the factor  $\delta f$ , the general term is

$$\frac{1}{k} \cdot \frac{1}{[p]^p [q]^q} \{ \phi(p + s + 1) (\delta^p f^{p+s+1}) (\delta^q f^{-p-s-1}) \},$$

equivalent to

$$\begin{aligned} \frac{1}{k} \cdot \frac{1}{[p]^p [q]^q} \delta^{s+1} f \cdot \sum \frac{1}{[p]^p [q]^q} \{ \phi(p + a + 1) (p + s + a + 1) (\delta^p f^{p+s+a}) (\delta^q f^{-p-s-a}) \\ - \phi(p + s + a) (\delta^p f^{p+a}) (\delta^q f^{-p-a-s}) \}, \end{aligned}$$

in which last expression  $p + q = k - a - 1$ . By repeating the reduction  $j$  times, the general term becomes

$$\begin{aligned} \frac{1}{k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots} \frac{1}{[\alpha]^\alpha [\beta]^\beta \dots} \delta^{s+\alpha} f \delta^{s+\beta} f \dots \\ \times \sum \frac{1}{[p]^p [q]^q} \Sigma \{ (-)^{j-f} \phi(p + \alpha + \beta \dots + f) (p + s + \alpha + \beta \dots + f) \\ \times [p + s + \alpha + \beta \dots + j - 1]^{j-f} (\delta^p f^{p+s+\alpha+\dots+f}) (\delta^q f^{-p-s-\alpha-\dots-f}) \}, \end{aligned}$$

where the sums  $\alpha + \beta \dots$  contain  $f$  terms,  $f$  being less than  $j$  or equal to it, and  $\Sigma$  extends to all combinations of the quantities  $\alpha, \beta \dots$  taken  $f$  and  $f$  together (so that the summation contains  $2^j$  terms). Also  $p + q = k - \alpha - \beta \dots (j \text{ terms}) - j$ , and the products  $k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots$  and  $[\alpha]^\alpha [\beta]^\beta \dots \delta^{s+\alpha} f \delta^{s+\beta} f \dots$  contain each of them  $j$  terms. Suppose the reduction continued until  $k - \alpha - \beta \dots (j \text{ terms}) - j = 0$ , then the only values of  $p, q$  are  $p = 0, q = 0$ ; and the general term of

$$\sum \frac{\phi p}{[p]^p [q]^q} (\delta^p f^{p+s}) (\delta^q f^{-p-s})$$

becomes

$$\begin{aligned} \frac{1}{k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots} \frac{1}{[\alpha]^\alpha [\beta]^\beta \dots} \delta^{s+\alpha} f \delta^{s+\beta} f \dots f^s \cdot f^{-s} \\ \times \Sigma \{ (-)^{j-f} \phi(\alpha + \beta \dots + f) [s + \alpha + \beta \dots + f]^f [s + \alpha + \beta \dots + j - 1]^{j-f} \}. \end{aligned}$$

If  $\phi p = \frac{1}{p+s}$ , the general term reduces itself to

$$\begin{aligned} \frac{1}{k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots} \frac{1}{[\alpha]^\alpha [\beta]^\beta \dots} \delta^{s+\alpha} f \delta^{s+\beta} f \dots f^s \cdot f^{-s} \\ \sum \{ (-)^{j-f} (s + \alpha + \beta \dots + f) \phi(\alpha + \beta \dots + f) [s + \alpha + \beta \dots + j - 1]^{j-f} \}; \end{aligned}$$

whence finally, if  $\phi p = \frac{1}{p+s}$ , the general term of

$$\sum \frac{\phi p}{[p]^p [q]^q} (\delta^p f^{p+s}) (\delta^q f^{-p-s})$$

becomes

$$\begin{aligned} \frac{1}{k(k - \alpha - 1)(k - \alpha - \beta - 2) \dots} \frac{1}{[\alpha]^\alpha [\beta]^\beta \dots} \delta^{s+\alpha} f \delta^{s+\beta} f \dots f^s \cdot f^{-s} \\ \times \sum \{ (-)^{j-f} \phi(\alpha + \beta \dots + f) [s + \alpha + \beta \dots + j - 1]^{j-f} \}, \end{aligned}$$

and it is readily shown that the sum contained in this formula vanishes, which proves the equation in question.

#### IV.

The demonstration of the equation (11) is much simpler. We have

$$\delta^t(f^{-t} \cdot \delta f) = \sum \binom{t-1}{n-1} \delta^n(f^{-t} \cdot \delta f) (\delta^{t-n} f^n),$$

that is,

$$\delta^t f^t = \frac{s}{s-t} \sum \binom{t-1}{n-1} (\delta^n f^{-t}) (\delta^{t-n} f^n),$$

where  $n$  extends from  $n = 1$  to  $n = t$ . Similarly

$$\delta^{t-n} f^n = \frac{t-n}{p} \sum \binom{t-n-p-1}{p-1} (\delta^p f^n) (\delta^{t-n-p} f^{n-p}),$$

&c.

Hence, substituting successively, and putting  $t - n - p - q = r$ , &c.,

$$\delta^t f^t = \frac{s}{s-t} \sum \frac{[t]^t}{[n]^n [p]^p (q+r) [q-1]^{q-1} \dots} (\delta^n f^n) (\delta^p f^p) (\delta^q f^q) \dots,$$

&c.; and the last of these corresponding to a zero value of the last of the quantities  $n, p, q \dots$  is evidently the required equation (11).

#### V.

The formula (18) in my paper on Lagrange's theorem (before referred to) is incorrect. I propose at present, after giving the proper form of the formula in question, to develop the result of the substitution indicated at the conclusion of the paper. It will be convenient to call to mind the general theorem, that when any number of variables  $x, y, z \dots$  are connected with as many other variables  $u, v, w \dots$  by the same number of equations (so that the variables of each set may be considered as functions of those of the other set) the quotient of the expressions  $dx dy \dots$  and  $du dv \dots$  is equal to the quotient of two determinants formed with the functions which equated to zero express the relations between the two sets of variables; the former with the differential coefficients of these functions with respect to  $u, v \dots$ , the latter with the differential coefficients with respect to  $x, y \dots$ . Consequently the notation  $\frac{dx dy \dots}{du dv \dots}$  may be considered as representing the quotient of these determinants. This being premised, if we write

$$x - u - h\theta(x, y \dots) = 0,$$

$$y - v - k\phi(x, y \dots) = 0,$$

then the formula in question is

$$F(x, y \dots) = \delta_u^{-1} \delta_v^{-1} \dots e^{h\theta+k\phi+\dots} F\nabla,$$

when for shortness  $u, v, \dots$  are the letters considered as a function of  $\theta, \phi, \dots$  substituted for  $x, y \dots$ . Let  $\frac{X}{V}$  denote the value which  $\frac{dx dy \dots}{du dv \dots}$  assumes when these variables are changed into  $u, v, \dots$ , we have

$$\nabla = \begin{vmatrix} 1 - h\delta_u \theta & -h\delta_v \theta & \dots \\ -k\delta_u \phi & 1 - k\delta_v \phi & \dots \\ \vdots & \vdots & \ddots \end{vmatrix}.$$

By changing the function  $F$ , we obtain

$$F(x, y \dots) = \delta_u^{m-1} \delta_v^{n-1} \dots e^{h\theta+k\phi+\dots} F\nabla,$$

where, however, it must be remembered that the  $h, k$  in so far as they enter into the function  $\nabla$ , are not affected by the symbols  $h\delta_u, k\delta_v, \dots$ . In order that we may consider them to be so affected, it is necessary

in the function  $\nabla$  to replace  $h, k, \&c.$  by  $\frac{h}{\delta_u}, \frac{k}{\delta_v}, \&c.$  When this is done, observing that the symbols  $h\delta_u\theta, k\delta_v\theta \dots$  affect a function  $e^{h\theta+k\phi+\dots}$ , the symbols  $h\delta_u\theta, k\delta_v\theta, \dots$  may be replaced by  $\delta_u^*, \delta_v^*$ , where the  $*$  is not an index, but an affix denoting that the differentiation is only to be performed with respect to  $u, v \dots$  so far as these variables respectively enter into the function  $\theta$ . Transforming the other lines of the determinant in the same manner, and taking out from  $\delta_u^m \delta_v^n \dots$  the factor  $\delta_u \delta_v \dots$  in order to multiply this last factor into the determinant, we obtain

$$F(x, y \dots) = \delta_u^{m-1} \delta_v^{n-1} \dots e^{h\theta+k\phi+\dots} F\Box;$$

where

$$\Box = \begin{vmatrix} \delta_u - \delta_u^\theta & -\delta_v^\theta & \dots \\ -\delta_u^\phi & \delta_v - \delta_v^\phi & \dots \\ \vdots & & \end{vmatrix},$$

in which expression  $\delta_u, \delta_v, \dots$  are to be replaced by

$$\delta_u + \delta_u^\theta + \delta_u^\phi + \delta_u^\chi + \dots.$$

The complete expansion is easily arrived at by induction, and the form is somewhat singular. In the case of a single variable  $u$  we have  $\Box = \delta_u^\theta$ ; in the case of two variables,  $\Box = \delta_u^\theta \delta_v^\phi + \delta_u^\phi \delta_v^\theta + \delta_u^\chi \delta_v^\psi$ . Or writing down only the affixes, in the case of a single variable we have  $F$ ; in the case of two variables  $FF, F\theta, \phi F$ ; and in the case of three variables  $FFF, \phi FF, \chi \phi F, F\chi F, F\theta F, FF\theta, FF\phi, F\theta\theta, F\theta\phi, F\chi\theta, \phi F\phi, \chi F\phi, \phi F\theta, \chi\chi\phi, \phi\chi\theta, \chi\theta\theta$ , where it will be observed that  $\theta$  never occurs in the first place, nor  $\phi$  in the second place, nor  $\theta, \phi$  (in any order) in the first and second places, &c., nor  $\theta, \phi, \chi$  (in any order) in the first, second, and third places. And the same property holds in the general case for each letter and binary, ternary, &c. combination, and for the entire system of letters, and the system of affixes contains every possible combination of letters not excluded by the rule just given. Thus in the case of two letters, forming the system of affixes  $FF, F\theta, \phi F, \theta F, F\phi, \theta\phi, \phi\theta$ , the last four are excluded, the first three of them by containing  $\theta$  in the first place or  $\phi$  in the second place, the last by containing  $\theta, \phi$  in the first and second places: and there remains only the terms  $FF, F\theta, \phi F$  forming the system given above. Substituting the expanded value of  $\Box$  in the expression for  $F(x, y \dots)$ , the equation may either be permitted to remain in the form which it thus assumes, or we may, in order to obtain the finally reduced form, after expanding the powers of  $h, k \dots$ , connect the symbols  $\delta_u^\theta, \delta_v^\phi \dots, \delta_u^\chi, \&c.$  with the corresponding functions  $\theta, \phi \dots F$ , and then omit the affixes; thus, in particular, in the case of a single variable the general term of  $Fx$  is

$$\frac{h^p}{[p]^p} \delta_u^{p-1} (F' \cdot \theta^p)$$

(the ordinary form of Lagrange's theorem). In the case of two letters the general term of  $F(x, y)$  is

$$\frac{h^p k^q}{[p]^p [q]^q} \delta_u^{p-1} \delta_v^{q-1} [\theta^p \phi^q F' F' + p \theta^{p-1} \phi^q \delta_v \theta F' F + q \theta^p \phi^{q-1} \delta_u \phi F F']$$

(see the *Mécanique Céleste*, [Ed. 1, 1798] t. i. p. 176). In the case of three variables, the general term is

$$\frac{h^p k^q l^r}{[p]^p [q]^q [r]^r} \delta_u^{p-1} \delta_v^{q-1} \delta_w^{r-1} [\theta^p \phi^q \chi^r F' F' F' + \dots],$$

the sixteen terms within the  $\{ \}$  being found by comparing the product  $\delta_u \delta_v \delta_w$  with the system  $FFF, \phi FF, \&c.$ , given above, and then connecting each symbol of differentiation with the function corresponding to the affix. Thus in the first term the  $\delta_u, \delta_v, \delta_w$  each affect the  $F$ , in the second term the  $\delta_u$  affects  $\phi^p$ , and the  $\delta_v$  and  $\delta_w$  each affect the  $F$ , and so on for the remaining terms. The form is of course deducible from Laplace's general theorem, and the actual development of it is given in Laplace's Memoir in the *Hist. de l'Acad.* vii. I quote from a memoir by Jacobi which I take this opportunity of referring to, "De resolutione equationum per series infinitas," Crelle, t. vi. [1830], pp. 257—286, founded on a preceding memoir, "Exercitatio algebraica circa discriptionem singularem fractionum quæ plures variables involvunt," t. v. [1830], pp. 344—364.

Stone Buildings, April 6, 1850.

## 102. On a Double Infinite Series.

[From the *Cambridge and Dublin Mathematical Journal*, vol. vi. (1851), pp. 45—47.]

The following completely paradoxical investigation of the properties of the function  $\Gamma$  (which I have been in possession of for some years) may perhaps be found interesting from its connexion with the theories of expansion and divergent series.

Let  $\Sigma_r \phi r$  denote the sum of the values of  $\phi$  for all integer values of  $r$  from  $-\infty$  to  $\infty$ . Then writing

$$u = \Sigma_r [n-1]^r x^{n-1-r}, \quad (11)$$

(where  $n$  is any number whatever), we have immediately

$$\frac{du}{dx} = \Sigma_r [n-1]^{r+1} x^{n-2-r} = \Sigma_r [n-1]^r x^{n-1-r} = u;$$

that is,

$$\frac{du}{dx} = u, \quad \text{or} \quad u = C_n e^x,$$

(the constant of integration being of course in general a function of  $n$ ). Hence

$$C_n e^x = \Sigma_r [n-1]^r x^{n-1-r}; \quad (12)$$

or  $e^x$  is expanded in general in a doubly infinite necessarily divergent series of fractional powers of  $x$ , (which resolves itself however in the case of  $n$  a positive or negative integer, into the ordinary singly infinite series, the value of  $C_n$  in this case being immediately seen to be  $\Gamma(n)$ ).

The equation (2) in its general form is to be considered as a definition of the function  $C_n$ . We deduce from it

$$\Sigma_r [n-1]^r (ax)^{n-1-r} = C_n e^{ax},$$

and also

$$\Sigma_k [n+n' \dots -1]^k \{a(x+x' \dots)\}^{n+n' \dots -1-k} = C_{n+n' \dots} e^{a(x+x' \dots)}.$$

Multiplying the first set of series, and comparing with this last,

$$C_{n+n' \dots} \Sigma_{r,r' \dots} [n-1]^r [n'-1]^{r'} \dots x^{n-1-r} x'^{n'-1-r'} \dots = C_n C_{n' \dots} [n+n' \dots -1]^k (x+x' \dots)^{n+n' \dots -1-k}, \quad (13)$$

(where  $r, r' \dots$  denote any positive or negative integer numbers satisfying  $r+r'+\dots = k+1-p$ ,  $p$  being the number of terms in the series  $n, n', \dots$ ). This equation constitutes a multinomial theorem of a class analogous to that of the exponential theorem contained in the equation (2).

In particular

$$C_{n+n' \dots} \Sigma_{r,r' \dots} [n-1]^r [n'-1]^{r'} \dots = C_n C_{n' \dots} [n+n' \dots -1]^k 2^{n+n' \dots -1-k}, \quad (14)$$

and if  $p=2$ , writing also  $m, n$  for  $n, n'$ , and  $k-1-r$  for  $r'$ ,

$$C_{m+n} \Sigma_r [m-1]^r [n-1]^{k-1-r} = C_m C_n [m+n-1]^k 2^{m+n-1-k}, \quad (15)$$

or putting  $k=0$  and dividing,

$$\frac{C_m C_n}{C_{m+n}} = \frac{1}{2^{m+n-1}} \Sigma_r [m-1]^r [n-1]^{-1-r}. \quad (16)$$

Now the series on the second side of this equation is easily seen to be convergent (at least for positive values of  $m, n$ ). To determine its value write

$$F(m, n) = \int_0^1 x^{m-1} (1-x)^{n-1} dx;$$

then

$$F(m, n) = \int_0^{\frac{1}{2}} x^{m-1} (1-x)^{n-1} dx + \int_{\frac{1}{2}}^1 x^{m-1} (1-x)^{n-1} dx,$$

and by successive integrations by parts, the first of these integrals is reducible to

$$\frac{1}{2^{m+n-1}} \Sigma_r [m-1]^r [n-1]^{-1-r}, \quad r \text{ extending from } -1 \text{ to } -\infty \text{ inclusively,}$$

and the second to

$$\frac{1}{2^{m+n-1}} \Sigma_r [m-1]^r [n-1]^{-1-r}, \quad r \text{ extending from } 0 \text{ to } \infty;$$

hence

$$F(m, n) = \frac{1}{2^{m+n-1}} \Sigma_r [m-1]^r [n-1]^{-1-r},$$

or

$$C_m C_n \div C_{m+n} = F(m, n), \tag{17}$$

*[Paper 102 continued in next slice.]*

## CAYLEY, COLLECTED MATHEMATICAL PAPERS, Vol. II.

Scan-verified slice: book pages 26–50.

104. On the Theory of Permutants (*Postscript, concluded; book p. 26*)

**Postscript.** I wish to explain as accurately as I am able, the extent of my obligations to Mr Sylvester in respect of the subject of the present memoir. The term permutant is due to him—intermutant and commutant are merely terms framed between us in analogy with permutant, and the names date from the present year (1851). The theory of commutants is given in my memoir in the Cambridge Philosophical Transactions, [12], and is presupposed in the memoir on Linear Transformations, [13, 14].

It will appear by the last-mentioned memoir that it was by representing the coefficients of a biquadratic function by  $a = 1111$ ,  $b = 1112 = 1121 = \&c.$ ,  $c = 1122 = \&c.$ ,  $d = 1222 = \&c.$ ,  $e = 2222$ , and forming the commutant

$$\begin{pmatrix} 1111 \\ 2222 \end{pmatrix}$$

that I was led to the function  $ae - 4bd + 3c^2$ .

The function  $ace + 2bcd - ad^2 - b^2e - c^3$  is mentioned in the memoir on Linear Transformations, as brought into notice by Mr Boole. From the particular mode in which the coefficients  $a, b, \dots$  were represented by symbols such as 1111, &c., I did not perceive that the last-mentioned function could be expressed in the commutant notation.

The notion of a permutant, in its most general sense, is explained by me in my memoir, “Sur les determinants gauches,” Crelle, t. xxxvii. pp. 93—96, [69]; see the paragraph (p. 94) commencing “On obtient ces fonctions, &c.” and which should run as follows: “On obtient ces fonctions (dont je reprends ici la théorie) par les propriétés générales d’un déterminant défini comme suit. En exprimant &c.,” the sentence as printed being “défini. Car en exprimant &c.,” which confuses the sense. [The paragraph is printed correctly 69, p. 411.]

Some time in the present year (1851) Mr Sylvester, in conversation, made to me the very important remark, that as one of a class the above-mentioned function,

$$ace + 2bcd - ad^2 - b^2e - c^3,$$

would be expressed in the commutant notation, viz. by considering

$$00 = a, \quad 01 = 10 = b, \quad 02 = 11 = 20 = c, \quad 12 = 21 = d, \quad 22 = e;$$

and the subject being thereby recalled to my notice, I found shortly afterwards the expression for the function

$$a^2d^2 + 4ac^3 + 4b^3d - 3b^2c^2 - 6abcd$$

(which cannot be expressed as a commutant) in the form of an intermutant, and I was thence led to see the identity, so to say, of the theory of hyperdeterminants, as given in the memoir on Linear Transformations, with the present theory of intermutants.

It is understood between Mr Sylvester and myself, that the publication of the present memoir is not to affect Mr Sylvester’s right to claim the origination, and to be considered as the first publisher of such part as may belong to him of the theory here sketched out.

**105. Correction of the Postscript to the Paper on Permutants.**

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 97—98.]

Mr Sylvester has represented to me that the paragraph relating to his communications conveys an erroneous idea of the nature, purport, and extent of such communications; I have, in fact, in the paragraph in question, singled out what immediately suggested to me the expression of the function

$$6abcd + 3b^2c^2 - 4ac^3 - 4b^3d - a^2d^2$$

as a partial commutant or intermutant, but I agree that a fuller reference ought to have been made to Mr Sylvester's results, and that the paragraph in question would more properly have stood as follows:

“Under these circumstances Mr Sylvester communicated to me a series of formal statements, not only oral but in writing, to the effect that he had discovered a permutation method of obtaining as many invariants—viz. commutative invariants—by direct inspection from a function of any degree of any number of letters as the index of the degree contains even factors; one of these commutative invariants being in fact the function

$$ace + 2bcd - ad^2 - b^2e - c^3,$$

expressible, according to Mr Sylvester's notation of my memoir in the *Camb. Phil. Trans.*, supposing  $00 = a$ ,  $01 = 10 = b$ ,  $02 = 11 = 20 = c$ ,  $12 = 21 = d$ ,  $22 = e$ , by

$$\begin{pmatrix} 00 & 11 & 22 \\ a & b & c \\ b & c & d \\ c & d & e \end{pmatrix},$$

and so on.”

Mr Sylvester and I shall, I have no doubt, be able to agree to a joint statement of any further correction or explanation which may be required.



## 106. On the Singularities of Surfaces.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 166—171.]

In the following paper, for symmetry of nomenclature and in order to avoid ambiguities, I shall, with reference to plane curves and in various phrases and compound words, use the term “node” as synonymous with double point, and the term “spinode” as synonymous with cusp.

I shall, besides, have occasion to consider the several singularities which I call the “flecnode,” the “oscnode,” the “fleflecnode,” and the “tacnode:” the flecnode is a double point which is a point of inflexion on one of the branches through it; the oscnode is a double point which is a point of osculation on one of the branches through it; the fleflecnode is a double point which is a point of inflexion on each of the branches through it; and the tacnode is a double point where two branches touch.

And it may be proper to remark here, that a tacnode may be considered as a point resulting from the coincidence and amalgamation of two double points (and therefore equivalent to twelve points of inflexion) or, in a different point of view, as a point uniting the characters of a spinode and a flecnode.

I wish to call to mind here the definition of conjugate tangent lines of a surface, viz. that a tangent to the curve of contact of the surface with any circumscribed developable and the corresponding generating line of the developable, are conjugate tangents of the surface.

Suppose, now, that an absolutely arbitrary surface of any order be intersected by a plane: the curve of intersection has not in general any singularities other than such as occur in a perfectly arbitrary curve of the same order; but as a plane can be made to satisfy one, two, or three conditions, the curve may be made to acquire singularities which do not occur in such absolutely arbitrary curve.

Let a single condition only be imposed on the plane. We may suppose that the curve of intersection has a node; the plane is then a tangent plane and the node is the point of contact—of course any point on the surface may be taken for the node. We may if we please use the term “nodes of a surface,” “node-planes of a surface,” as synonymous with the points and tangent planes of a surface. And it will be convenient also to use the word node-tangents to denote the tangents to the curve of intersection at the node; it may be noticed here that the node-tangents are conjugate tangents of the surface.

Next let two conditions be imposed upon the plane: there are three distinct cases to be considered.

**First**, the curve of intersection may have a flecnode. The plane (which is of course still a tangent plane at the flecnode) may be termed a flecnode-plane; the flecnodes are singular points on the surface lying on a curve which may be termed the “flecnode-curve,”<sup>1</sup> and the flecnode-planes give rise to a developable which may be termed the flecnode-develope. The “flecnode-tangents” are the tangents to the curve of intersection at the flecnode; the tangent to the inflected branch may be termed the “singular flecnode-tangent,” and the tangent to the other branch the “ordinary flecnode-tangent.”

**Secondly**, the curve of intersection may have a spinode. The plane (which is of course still a tangent plane at the spinode) may be termed a spinode-plane; the spinodes are singular points on the surface lying on a curve which may be termed the “spinode-curve.”<sup>2</sup> And the

<sup>1</sup>The flecnode-curve, defined as the locus of the points through which can be drawn a line meeting the surface in four consecutive points, was, so far as I am aware, first noticed in Mr Salmon's paper “On the Triple Tangent Planes of a Surface of the Third Order” (Journal, t. iv. [1849], pp. 252—260), where Mr Salmon, among other things, shows that the order of the surface being  $n$ , the curve in question is the intersection of the surface with a surface of the order  $11n - 24$ .

<sup>2</sup>The notion of a spinode, considered as the point where the indicatrix is a parabola (on which account the spinode has been termed a parabolic point) may be found in Dupin's *Developpements de Geometrie*: the most important step in the theory of these points is contained in Hesse's memoir “Ueber die Wendepuncte der Curven dritter Ordnung” (Crelle, t. xxviii. [1848], pp. 97—107), where it is shown that the spinode-curve is the curve of

spinode-planes give rise to a developable which may be termed the “spinode-developpe.” Also the “spinode-tangent” is the tangent to the curve of intersection at the spinode.

**Thirdly**, the curve of intersection may have two nodes, or what may be termed a “node-couple.” The plane (which is a tangent plane at each of the nodes and therefore a double tangent plane) may be also termed a “node-couple-plane.” The node-couples are pairs of singular points on the surface lying in a curve which may be termed the “node-couple-curve,” and the node-couple-planes give rise to a developable which may be termed the “node-couple-developpe.” The tangents to the curve of intersection at the two nodes of a node-couple might, if the term were required, be termed the “node-couple-tangents.” Also one of the nodes of a node-couple may be termed a “node-with-node,” and the tangents to the curve of intersection at such point will be the “node-with-node-tangents.”

It is hardly necessary to remark that the flecnode-curve is not the edge of regression of the flecnode-developpe, and the like remark applies *mutatis mutandis* to the spinode-curve and the node-couple curve.

Finally, let three conditions be imposed upon the plane: there are six distinct cases to be considered, in each of which we have no longer curves and developes, but only singular points and singular tangent planes determinate in number.

**First**, the curve of intersection may have an oscnode. The plane (which continues a tangent plane at the oscnode) is an “oscnode-plane.” The “oscnode-tangents” are the tangents to the curve of intersection at the oscnode; the tangent to the osculating branch is the “singular oscnode-tangent;” and the tangent to the other branch the “ordinary oscnode-tangent.”

**Secondly**, the curve of intersection may have a fleflecnode. The plane (which continues a tangent plane at the fleflecnode) is a “fleflecnode-plane.” The “fleflecnode-tangents” are the tangents to the curve of intersection at the fleflecnode.

**Thirdly**, the curve of intersection may have a tacnode. The plane (which continues a tangent plane at the tacnode) is a “tacnode-plane.” The “tacnode-tangent” is the tangent to the curve of intersection at the tacnode.

**Fourthly**, the curve of intersection may have a node and a flecnode, or what may be termed a node-and-flecnode. The plane (which is a tangent plane at the node and also at the flecnode, where it is obviously a flecnode-plane) is a “node-and-flecnode-plane.” The “node-and-flecnode-tangents,” if the term were required, would be the tangents to the curve of intersection at the node and at the flecnode of the node-and-flecnode. The node of the node-and-flecnode may be distinguished as the node-with-flecnode, and the flecnode as the flecnode-with-node, and we have thus the terms “node-with-flecnode-tangents,” “flecnode-with-node-tangents,” “singular flecnode-with-node-tangent,” and “ordinary flecnode-with-node-tangent.”

**Fifthly**, the curve of intersection may have a node and also a spinode, or what may be termed a “node-and-spinode.” The plane (which is a tangent plane at the node, and is also a tangent plane at the spinode, where it is obviously a spinode-plane) is a “node-and-spinode-plane.” The node-and-spinode-tangents, if the term were required, would be the tangents at the node and the tangent at the spinode of the node-and-spinode to the curve of intersection. The node of the node-and-spinode may be distinguished as the “node-with-spinode,” and the spinode as the “spinode-with-node,” and we have thus the terms “node-with-spinode-tangent,” “spinode-with-node-tangent.”

**Sixthly**, the curve of intersection may have three nodes, or what may be termed a “node-triplet.” The plane (which is a triple tangent plane touching the surface at each of the nodes) is a “node-triplet-plane.” The “node-triplet-tangents,” if the term were required, would be the tangents to the curve of intersection at the nodes of the node-triplet. Each node of the node-triplet may be distinguished as a “node-with-node-couple,” and the tangents to the curve of

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intersection of the surface supposed as before of the order  $n$ , with a certain surface of the order  $4(n-2)$ . See also Mr Salmon’s memoir “On the Condition that a Plane should touch a surface along a Curve Line” (Journal, t. iii. [1848], pp. 44—46).

intersection at such nodes are “node-with-node-couple-tangents.” The terms “node-couple-with-node,” “node-couple-with-node-tangent,” might be made use of if necessary.

It should be remarked that the oscnodes lie on the flecnode-curve, as do also the flefnodes; these latter points are real double points of the flecnode-curve. The tacnodes are points of intersection and (what will appear in the sequel) points of contact of the flecnode-curve, the spinode-curve, and the node-couple-curve. The spinode-with-nodes are points of intersection of the spinode-curve and node-couple-curve, and the flecnode-with-nodes are points of intersection of the flecnode-curve and node-couple-curve; the node-with-node-couples are real double points (entering in triplets) of the node-couple-curve.

Consider for a moment an arbitrary curve on the surface; the locus of the node-tangents at the different points of this curve is in general a skew surface, which may however, in cases to be presently considered, degenerate in different ways.

Reverting now to the flecnode-curve, it may be shown that the singular flecnode-tangent coincides with the tangent of the flecnode-curve. For consider on a surface two consecutive points such that the line joining them meets the surface in two points consecutive to the first-mentioned two points. The line meets the surface in four consecutive points, it is therefore a singular flecnode-tangent; each of the first-mentioned two points must be on the flecnode-curve, or the singular flecnode-tangent touches the flecnode-curve. The two flecnode-tangents are by a preceding observation conjugate tangents. It follows that the skew surface, locus of the flecnode-tangents, breaks up into two surfaces, each of which is a developable, viz. the locus of the singular flecnode-tangents is the developable having the flecnode-curve for its edge of regression, and the locus of the ordinary flecnode-tangents is the flecnode-developable. Of course at the tacnode, the tacnode-tangent touches the flecnode-curve.

Passing next to the spinode-curve, the spinode-plane and the tangent-plane at a consecutive point along the spinode-tangent are identical, or their line of intersection is indeterminate.<sup>3</sup> The spinode-tangent is therefore the conjugate tangent to any other tangent line at the spinode, and therefore to the tangent to the spinode-curve. It follows that the surface locus of the spinode-tangents degenerates into a developable surface twice repeated, viz. the spinode-developable.

Consider the tacnode as two coincident nodes; each of these nodes, by virtue of its constituting, in conjunction with the other, a tacnode, is on the spinode-curve; or, in other words, the tacnode-tangent touches the spinode-curve, and the same reasoning proves that it touches the node-couple-curve. It has already been seen that the tacnode-tangent touches the flecnode-curve; consequently the tacnode is a point, not of simple intersection only, but of contact, of the flecnode-curve, the spinode-curve, and the node-couple-curve.

In virtue of the principle of the spinode-plane being identical with the tangent plane at a consecutive point along the spinode tangent, it appears that the tacnode-plane is a stationary plane, as well of the flecnode-developable as of the spinode-developable, and it would at first sight appear that it must be also a stationary tangent plane of the node-couple-developable. But this is not so; the node-with-node-planes envelope, not the node-couple-developable, but the node-couple-developable twice repeated: the tacnode-plane is in a sense a stationary plane on such duplicate developable, but not in any manner on the single developable. The tacnode-plane is an ordinary tangent plane of the node-couple-developable.

Consider now a spinode-with-node, which we have seen is a point of intersection of the spinode-curve and node-couple-curve. The tangent plane at a consecutive point along the spinode-with-node-tangent, is identical with the spinode-with-node-plane; the curve of intersection of the tangent plane at such consecutive point has therefore a node at the node-with-spinode, or the tangent plane in question is a node-couple-plane, and the point of contact is a point on the node-couple-curve. Consequently the spinode-with-node-tangent touches the node-couple-curve, and thence also the spinode-with-node-plane is a stationary tangent plane of the

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<sup>3</sup>It must not be inferred that the tangent plane at such consecutive point is a spinode-plane; this is obviously not the case.

node-couple-develope.

It should be remarked that no circumscribed developable can have a stationary tangent plane except the tangent planes at the points where the curve of contact meets the spinode-curve, and any one of these planes is only a stationary plane when the curve of contact touches the spinode-tangent; and that the node-couple-curve and the flecnode-curve do not intersect the spinode-curve except in the points which have been discussed.

Recapitulating, the node-couple-curve and the spinode-curve touch at the tacnodes, and intersect at the spinode-with-nodes: moreover, the tacnode-planes are stationary planes of the spinode-develope, and the spinode-with-node-planes are stationary planes of the node-couple-develope. Besides this, the two curves are touched at the tacnodes by the flecnode-curve, and the tacnode-planes are stationary planes of the flecnode-develope.

## 107. On the Theory of Skew Surfaces.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 171—173.]

A SURFACE of the  $n^{\text{th}}$  order is a surface which is met by an indeterminate line in  $n$  points. It follows immediately that a surface of the  $n^{\text{th}}$  order is met by an indeterminate plane in a curve of the  $n^{\text{th}}$  order.

Consider a skew surface or the surface generated by a singly infinite series of lines, and let the surface be of the  $n^{\text{th}}$  order. Any plane through a generating line meets the surface in the line itself and in a curve of the  $(n-1)^{\text{th}}$  order. The generating line meets this curve in  $(n-1)$  points. Of these points one, viz. that adjacent to the intersection of the plane with the consecutive generating line, is a unique point; the other  $(n-2)$  points form a system. Each of the  $(n-1)$  points are *sub modo* points of contact of the plane with the surface, but the proper point of contact is the unique point adjacent to the intersection of the plane with the consecutive generating line. Thus every plane through a generating line is an ordinary tangent plane, the point of contact being a point on the generating line.

It is not necessary for the present purpose, but I may stop for a moment to refer to the known theorems that the anharmonic ratio of any four tangent planes through the same generating line is equal to the anharmonic ratio of their points of contact, and that the locus of the normals to the surface along a generating line is a hyperbolic paraboloid.

Returning to the  $(n-2)$  points in which, together with the point of contact, a generating line meets the curve of intersection of the surface and a plane through the generating line, these are fixed points independent of the particular plane, and are the points in which the generating line is intersected by other generating lines.

There is therefore on the surface a double curve intersected in  $(n-2)$  points by each generating line of the surface—a property which, though insufficient to determine the order of this double curve, shows that the order cannot be less than  $(n-2)$ .

(Thus for  $n = 4$ , the above reasoning shows that the double-curve must be at least of the second order: assuming for a moment that it is in any case precisely of this order, it obviously cannot be a plane curve, and must therefore be two non-intersecting lines. This suggests at any rate the existence of a class of skew surfaces of the fourth order generated by a line which always passes through two fixed lines and by some other condition not yet ascertained; and it would appear that surfaces of the second order constitute a degenerate species belonging to the class in question.)

In particular cases a generating line will be intersected by the consecutive generating line. Such a generating line touches the double curve.

Consider now a point not on the surface; the planes determined by this point and the generating lines of the surface are the tangent planes through the point; the intersections of consecutive tangent planes are the tangent lines through the point; and the cone generated by these tangent lines or enveloped by the tangent planes is the tangent cone corresponding to the point. This cone is of the  $n^{\text{th}}$  class. For considering a line through the point, this line meets the surface in  $n$  points, i.e. it meets  $n$  generating lines of the surface; and the planes through the line and these  $n$  generating lines, are of course tangent planes to the cone: that is,  $n$  tangent planes can be drawn to the cone through a given line passing through the vertex.

The cone has not in general any lines of inflexion, or, what is the same thing, stationary tangent planes. For a stationary tangent plane would imply the intersection of two consecutive generating lines of the surface. And since the number of generating lines intersected by a consecutive generating line, and therefore the number of planes through two consecutive generating lines, is finite, no such plane passes through an indeterminate point.

The tangent cone will have in general a certain number of double tangent planes; let this number be  $\chi$ . We have therefore a cone of the class  $n$ , number of double tangent planes  $\chi$ , number of stationary tangent planes 0. Hence, if  $m$  be the order of the cone,  $\alpha$  the number of

its double lines, and  $\beta$  the number of its cuspidal or stationary lines,

$$\begin{aligned} m &= n(n-1) - 2\chi, \\ \beta &= 3n(n-2) - 6\chi, \\ \alpha &= \frac{1}{2}n(n-2)(n^2-9) - 2\chi(n^2-n-6) + 2\chi(\chi-1). \end{aligned}$$

This is the proper tangent cone, but the cone through the double curve is *sub modo* a tangent cone, and enters as a square factor into the equation of the general tangent cone of the order  $n(n-1)$ . Hence, if  $X$  be the order of the double curve, and therefore of the cone through this curve,

$$m + 2X = n(n-1),$$

and therefore  $X = \chi$ ; that is, the number of double tangent planes to the tangent cone is equal to the order of the double curve.

It does not appear that there is anything to determine  $\chi$ ; and if this is so, skew surfaces of the  $n^{\text{th}}$  order may be considered as forming different families according to the order of the double curve upon them. To complete the theory, it should be added that a plane intersects the surface in a curve of the  $n^{\text{th}}$  order having  $\chi$  double points but no cusps.

**108. On certain Multiple Integrals connected with the Theory of Attractions.**

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 174—178.]

It is easy to deduce from Mr Boole's formula, given in my paper "On a Multiple Integral connected with the theory of Attractions," Journal, t. ii. [1847], pp. 219—223, [44], the equation

$$\int \frac{d\xi d\eta \dots}{[(\xi - \alpha)^2 + (\eta - \beta)^2 + \dots + v^2]^{\frac{1}{2}n - q}} = \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n - q) \Gamma(q + 1)} \int_0^\epsilon \frac{s^{q-1} (e - s)^q ds}{(1 + s)^{\frac{1}{2}n + q}},$$

where  $n$  is the number of variables of the multiple integral, and the condition of the integration is

$$(\xi - \alpha)^2 + (\eta - \beta)^2 + \dots < f^2;$$

also where

$$\frac{(\xi - \alpha)^2 + (\eta - \beta)^2 + \dots}{f^2} - \frac{v^2}{ef^2} = 1,$$

and  $e$  is the positive root of

$$\xi^2 + \eta^2 + \dots = v^2.$$

Suppose  $f = f_1 = \dots = f_n$ , and write  $(\alpha - \alpha_1)^2 + \dots = k^2$ ; we obtain

$$\int \frac{d\xi d\eta \dots}{[(\xi - \alpha)^2 + \dots + v^2]^{\frac{1}{2}n - q}} = \frac{2\pi^{\frac{1}{2}n} q}{\Gamma(\frac{1}{2}n - q) \Gamma(q + 1)} \int_0^\epsilon \frac{s^{q-1} (\epsilon - s)^q ds}{(1 + s)^{\frac{1}{2}n + q}},$$

the limiting condition for the multiple integral being

$$(\xi - \alpha_1)^2 + (\eta - \beta_1)^2 + \dots < f_1^2,$$

and the function  $e$ , and limit  $\epsilon$ , being given by

$$e^2 + \xi^2 + \dots = v^2, \quad \epsilon \text{ denoting, as before, the positive root.}$$

Observing that the quantity under the integral sign on the second side vanishes for  $s = \epsilon$ , there is no difficulty in deducing, by a differentiation with respect to  $\xi_1$ , the formula

$$\int \frac{dS}{[(\xi - \alpha)^2 + \dots + v^2]^{\frac{1}{2}n - q}} = \frac{2\pi^{\frac{1}{2}n} q}{\Gamma(\frac{1}{2}n - q) \Gamma(q)} \int_0^\epsilon \frac{s^{q-1} (\epsilon - s)^{q-1} ds}{(1 + s)^{\frac{1}{2}n + q - 1}},$$

where  $dS$  is the element of the surface  $(\xi - \alpha_1)^2 + \dots = f_1^2$  and the integration is extended over the entire surface.

A slight change of form is convenient. We have

$$s^{q-1} (\epsilon - s)^{q-1} ds = \frac{1}{s(1 + s)} \cdot \frac{s^q (\epsilon - s)^q}{\epsilon} d\epsilon,$$

if we suppose

$$\epsilon = \frac{\xi_1^2 + \eta_1^2 + \dots - f_1^2}{(\xi - \alpha_1)^2 + (\eta - \beta_1)^2 + \dots}.$$

The formulae then become

$$\int \frac{dS}{[(\xi - \alpha)^2 + \dots + v^2]^{\frac{1}{2}n - q}} = \frac{2\pi^{\frac{1}{2}n} q}{\Gamma(\frac{1}{2}n - q) \Gamma(q)} \int_0^\epsilon \frac{(\xi_1^2 + \dots - f_1^2)^{q-1} d\epsilon}{\epsilon(1 + \epsilon)[(\xi - \alpha_1)^2 + \dots]^{\frac{1}{2}n - q - 1}},$$

in which  $\epsilon$  is the positive root of the equation

$$\rho^2 e^2 + \lambda e - \mu = 0.$$

I propose to transform these formulae by means of the theory of images; it will be convenient to investigate some preliminary formulae.

Suppose  $l^2 = \alpha^2 + \beta^2 + \dots$ ,  $l_1^2 = \alpha_1^2 + \beta_1^2 + \dots$ ; also consider the new constants  $a, b, \dots, a_1, b_1, \dots, u, \mu$ , determined by the equations

$$\frac{S}{u} = \frac{\alpha}{a} = \frac{\beta}{b} = \dots = \frac{l^2}{\mu} = \frac{l_1^2}{S},$$

where  $S$  is arbitrary.

Then, putting

$$x = \frac{S\xi}{\xi^2 + \eta^2 + \dots}, \quad y = \frac{S\eta}{\xi^2 + \eta^2 + \dots}, \quad \dots$$

it is easy to see that

$$(x - a)^2 + (y - b)^2 + \dots = \frac{S^2}{(\xi^2 + \eta^2 + \dots)^2} \{(\xi - \alpha)^2 + (\eta - \beta)^2 + \dots\},$$

and

$$\frac{a}{l^2} = \frac{\alpha}{l^2 + uS}, \quad \frac{S^2 \alpha}{l^2 + uS} = a, \quad \frac{S^2 \alpha_1}{l_1^2 + \mu S} = a_1.$$

Proceeding to express the single integrals in terms of the new constants, we have in the first place  $k^2 = h^2 \cdot l^2$ , where

$$h^2 = \frac{(l^2 + uS)(l_1^2 + \mu S)}{S^2} = \frac{l^2 l_1^2 + S(\mu l^2 + u l_1^2) + u\mu S^2}{S^2};$$

or if we write

$$\alpha \alpha_1 + \beta \beta_1 + \dots = ll_1 \cos \omega,$$

we have

$$h^2 = l^2 + l_1^2 - 2ll_1 \cos \omega.$$

Hence also  $k^{-2} = h^{-2} \cdot l^{-2}$ , where

$$\frac{l^2 + uS}{S^2 \alpha} = \frac{1}{a}, \quad \text{whence} \quad \frac{(l^2 + uS)(l_1^2 + \mu S)}{S^2 ll_1 \cos \omega} = \frac{1}{\cos \omega};$$

$$\frac{(k^2 - \lambda^2)(l^2 + uS)(l_1^2 + \mu S)}{S^2 ll_1} = \frac{1}{\rho^2},$$

where  $\rho^2 = l^2 + l_1^2 - 2ll_1 \cos \omega$ , that is  $\rho^2 = (a - a_1)^2 + (b - b_1)^2 + \dots$ ; consequently

$$\frac{1}{k^2} \cdot \frac{l^2 + uS}{l_1^2 + \mu S} = \frac{S^2}{\Pi},$$

where  $\Pi$  is given by

$$\Pi = \rho^2 e^2 + (\mu - \lambda^2) e - u\mu,$$



and it is clear that  $e$  will be the positive root of

$$\Pi = \rho^2 e^2 + (\mu - \lambda^2) e - u\mu = 0.$$

It may be noticed that, in the particular case of  $u = 0$ , the roots of this equation are 0, and  $\frac{\mu}{\lambda^2 - \mu}$ . Consequently if  $\rho^2 - f_1^2$  and  $l^2 - f^2$  are of opposite signs, we have  $e = 0$ ; but if  $\rho^2 - f_1^2$  and  $l^2 - f^2$  are of the same sign,  $e = \frac{\mu}{\lambda^2 - \mu}$ .

In order to transform the double integrals, considering the new variables  $x, y, \dots$ , I write  $x^2 + y^2 + \dots = r^2$  and

$$\xi = \frac{Sx}{x^2 + y^2 + \dots}, \quad \eta = \frac{Sy}{x^2 + y^2 + \dots}, \quad \dots$$

whence also, if  $\xi^2 + \eta^2 + \dots = \rho^2$  (which gives  $r\rho = S^2$ ), we have

$$\frac{d\xi}{\xi} = \frac{dx}{x} - \frac{2x dx + 2y dy + \dots}{x^2 + y^2 + \dots}, \quad \&c.;$$

also it is immediately seen that

$$(\xi - \alpha)^2 + \dots + v^2 = \frac{S^2}{(x^2 + y^2 + \dots)^2} \{ (x - a)^2 + (y - b)^2 + \dots + u^2 \},$$

$$(\xi - \alpha_1)^2 + \dots - f_1^2 = \frac{S^2}{(x^2 + y^2 + \dots)^2} \{ (x - a_1)^2 + (y - b_1)^2 + \dots - f_1'^2 \};$$

and from the latter equation it follows that the limiting condition for the first integral is

$$(x - a_1)^2 + (y - b_1)^2 + \dots > f_1'^2$$

(there is no difficulty in seeing that the sign  $<$  in the former limiting condition gives rise here to the sign  $>$ ), and that the second integral has to be extended over the surface  $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1'^2$ . Also if  $dS'$  represent the element of this surface, we may obtain

$$d\xi d\eta \dots = \frac{S^n}{r^{2n}} dx dy \dots, \quad dS = \frac{S^{n-1}}{r^{2(n-1)}} dS';$$

and, combining the above formulae, we obtain

$$\begin{aligned} \int \frac{dx dy \dots}{(x^2 + y^2 + \dots)^{\frac{1}{2}n+q} \{ (x - a)^2 + (y - b)^2 + \dots + u^2 \}^{\frac{1}{2}n-q}} \\ = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n - q) \Gamma(q + 1) (l^2 + u^2)^{\frac{1}{2}n-q}} \int_0^\epsilon \frac{s^{q-1} (e - s)^q ds}{s(1 + s)^{\frac{1}{2}n+q}}, \end{aligned}$$

the limiting condition of the multiple integral being

$$(x - a_1)^2 + (y - b_1)^2 + \dots > f_1'^2;$$

and

$$\begin{aligned} \int \frac{dS'}{(x^2 + y^2 + \dots)^{\frac{1}{2}n+q-1} \{ (x - a)^2 + (y - b)^2 + \dots + u^2 \}^{\frac{1}{2}n-q}} \\ = \frac{2\pi^{\frac{1}{2}n}q}{\Gamma(\frac{1}{2}n - q) \Gamma(q) (l^2 + u^2)^{\frac{1}{2}n-q} (l_1^2 - f_1^2)} \int_0^\epsilon \frac{s^{q-1} (e - s)^{q-1} ds}{(1 + s)^{\frac{1}{2}n+q-1}}, \end{aligned}$$

where  $dS'$  is the element of the surface  $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1'^2$ , and the integration extends over the entire surface.

In these formulae,  $l$ ,  $l_1$ ,  $\rho$ ,  $\Pi$  denote as follows:

$$\begin{aligned} l^2 &= a^2 + b^2 + \dots, \\ l_1^2 &= a_1^2 + b_1^2 + \dots, \\ \rho^2 &= (a - a_1)^2 + (b - b_1)^2 + \dots, \\ \Pi &= \rho^2 e^2 + (\mu - \lambda^2) e - u\mu, \end{aligned}$$

and  $e$  is the positive root of the equation  $\Pi = 0$ .

The only obviously integrable case is that for which in the second formula  $q = \frac{1}{2}$ ; this gives

$$\int \frac{dS'}{(x^2 + y^2 + \dots)^{\frac{1}{2}n} \{(x - a)^2 + (y - b)^2 + \dots + u^2\}^{\frac{1}{2}n-1}} = \frac{2\pi^{\frac{1}{2}n} f_1}{\Gamma(\frac{1}{2}n)(l^2 + u^2)^{\frac{1}{2}n-1}(l_1^2 - f_1^2)(1 + e)^{\frac{1}{2}n-1}}.$$

In the case of  $u = 0$ , we have, as before, when  $\rho^2 - f_1^2$  and  $l^2 - f^2$  are of opposite signs,  $e = 0$ , and therefore  $1 + e = 1$ ; but when  $\rho^2 - f_1^2$  and  $l^2 - f^2$  are of the same sign, the value before found for  $e$  gives

$$1 + e = \frac{\rho^2}{\rho^2 - f_1^2} \{l_1^2 + (\rho^2 - f_1^2)(l^2 - f^2)\} / l_1^2.$$

Consider the image of the origin with respect to the sphere  $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1^2$ ; the coordinates of this image are

$$\frac{f_1^2 a_1}{l_1^2}, \quad \frac{f_1^2 b_1}{l_1^2}, \quad \dots,$$

and consequently, if  $h$  be the distance of this image from the point  $(a, b, \dots)$ , we have

$$h^2 = \frac{1}{l_1^2} \{l_1^2 l^2 + (\rho^2 - f_1^2)(l_1^2 - f_1^2)\};$$

whence, by a simple reduction, or the values of the integral are: when  $\rho^2 - f_1^2$  and  $l^2 - f^2$  have opposite signs,

$$I = \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n)} \cdot \frac{f_1^{n-1}}{l_1^{n-1}};$$

when  $\rho^2 - f_1^2$  and  $l^2 - f^2$  have the same sign,

$$I = \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n)} \cdot \frac{f_1^{n-1}}{h^{n-2}(\rho^2 - f_1^2)/l_1^{n-1}},$$

where  $h$  is the distance from the point  $(a, b, \dots)$  of the image of the origin with respect to the sphere  $(x - a_1)^2 + \dots = f_1^2$ .

Considered in reference to the theory of images, the first formula gives the potential of a multiple layer (the density being inversely as the  $(n - 1)^{\text{th}}$  power of the distance) spread over the surface of an  $(n - 1)$ -fold space, due to an attracting mass (the law of attraction being the inverse  $(n - 1)^{\text{th}}$  power of the distance) situated at a given point; and the second formula gives the potential of the attracting mass itself. It is easy to see that the formula agrees with the ordinary theory of electrical images.

Stone Buildings, August 6, 1850.

**109. On the Rationalisation of certain Algebraical Equations.**

[From the Cambridge and Dublin Mathematical Journal, vol. viii. (1853), pp. 97—101.]

Suppose

$$x + y = 0, \quad x^2 = a, \quad y^2 = b;$$

then if we multiply the first equation by 1,  $xy$ , and reduce by the two others, we have

$$x + y = 0, \quad bx + ay = 0,$$

from which, eliminating  $x, y$ ,

$$\begin{vmatrix} 1 & 1 \\ b & a \end{vmatrix} = 0;$$

which is the equation between  $a$  and  $b$ ; or, considering  $x, y$  as quadratic radicals, the rational equation between  $x, y$ . So if the original equation be multiplied by  $x, y$ , we have

$$a + xy = 0, \quad b + xy = 0;$$

or, eliminating 1,  $xy$ ,

$$\begin{vmatrix} a & 1 \\ b & 1 \end{vmatrix} = 0,$$

which may be in like manner considered as the rational equation between  $x, y$ .

The preceding results are of course self-evident, but by applying the same process to the equations

$$x + y + z = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c,$$

we have results of some elegance. Multiply the equation first by 1,  $yz, zx, xy$ , reduce and eliminate the quantities  $x, y, z, xyz$ ; we have the rational equation

$$\begin{vmatrix} 1 & 1 & 1 & . \\ . & c & b & 1 \\ c & . & a & 1 \\ b & a & . & 1 \end{vmatrix} = 0;$$

and again, multiply the equation by  $x, y, z, xyz$ , reduce and eliminate the quantities 1,  $yz, zx, xy$ , the result is

$$\begin{vmatrix} a & b & c & . \\ a & . & 1 & 1 \\ b & 1 & . & 1 \\ c & 1 & 1 & . \end{vmatrix} = 0,$$

which is of course equivalent to the preceding one (the two determinants are in fact identical in value), but the form is essentially different. The former of the two forms is that given in my paper "On a theorem in the Geometry of Position" (Journal, vol. II. [1841] p. 270 [1]): it was only very recently that I perceived that a similar process led to the latter of the two forms.

Similarly, if we have the equations

$$x + y + z + w = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c, \quad w^2 = d,$$

then multiplying by 1,  $yz, zx, xy, xw, yw, zw, xyzw$ , reducing and eliminating the quantities in the outside row,

$$(x, y, z, w, yzw, zwx, wxy, xyz),$$

we have the result

$$\begin{vmatrix} 1 & 1 & 1 & 1 & . & . & . & . \\ . & c & b & . & . & . & 1 & 1 \\ c & . & a & . & . & 1 & . & 1 \\ b & a & . & . & 1 & . & . & 1 \\ d & . & . & a & . & 1 & 1 & . \\ . & d & . & b & 1 & . & 1 & . \\ . & . & d & c & 1 & 1 & . & . \\ . & . & . & . & a & b & c & d \end{vmatrix} = 0;$$

so if we multiply the equations by  $x, y, z, w, yzw, zwx, wxy, xyz$ , reduce and eliminate the quantities in the outside row

$$(1, yz, zx, xy, xw, yw, zw, xyzw),$$

we have the result

$$\begin{vmatrix} a & . & . & 1 & 1 & 1 & . & . \\ b & 1 & . & . & 1 & . & 1 & . \\ c & 1 & 1 & . & . & . & . & 1 \\ d & . & . & . & 1 & 1 & 1 & . \\ . & d & . & . & . & c & b & 1 \\ . & . & d & . & c & . & a & 1 \\ . & . & . & d & b & a & . & 1 \\ . & a & b & c & . & . & . & 1 \end{vmatrix} = 0,$$

which however is not essentially distinct from the form before obtained, but may be derived from it by an interchange of lines and columns.

And in general for any even number of quadratic radicals the two forms are not essentially distinct, but may be derived from each other by interchanging lines and columns, while for an odd number of quadratic radicals the two forms cannot be so derived from each other, but are essentially distinct.

I was indebted to Mr Sylvester for the remark that the above process applies to radicals of a higher order than the second. To take the simplest case, suppose

$$x + y = 0, \quad x^3 = a, \quad y^3 = b;$$

and multiply first by  $1, x^2y, xy^2$ ; this gives

$$x + y = 0, \quad x^2y + ay = 0, \quad bx + xy^2 = 0;$$

or, eliminating,

$$\begin{vmatrix} 1 & 1 & . \\ . & a & 1 \\ b & . & 1 \end{vmatrix} = 0;$$

next multiply by  $x, y, x^2y^2$ ; this gives

$$x^2 + xy = 0, \quad xy + y^2 = 0, \quad bx^2 + ay^2 = 0;$$

or, eliminating,

$$\begin{vmatrix} 1 & . & 1 \\ . & 1 & 1 \\ b & a & . \end{vmatrix} = 0;$$

and lastly, multiply by  $x^2, y^2, xy$ ; this gives

$$a + x^2y = 0, \quad b + xy^2 = 0, \quad x^2y + xy^2 = 0;$$

or, eliminating,

$$\begin{vmatrix} a & 1 & . \\ b & . & 1 \\ . & 1 & 1 \end{vmatrix} = 0;$$

where it is to be remarked that the second and third forms are not essentially distinct, since the one may be derived from the other by the interchange of lines and columns.

Applying the preceding process to the system

$$x + y + z = 0, \quad x^3 = a, \quad y^3 = b, \quad z^3 = c,$$

multiply first by 1,  $xyz$ ,  $x^2y^2z^2$ ,  $x^2z$ ,  $y^2x$ ,  $z^2y$ ,  $x^2y$ ,  $y^2z$ ,  $z^2x$ , reduce and eliminate the quantities in the outside row

$$(x, y, z, y^2z^2, z^2x^2, x^2y^2, x^2yz, y^2zx, z^2xy);$$

the result is

$$\begin{vmatrix} 1 & 1 & 1 & . & . & . & . & . & . \\ . & . & . & 1 & 1 & 1 & . & . & . \\ . & . & . & . & . & . & a & b & c \\ . & . & a & 1 & . & . & . & . & 1 \\ b & . & . & . & 1 & . & . & . & 1 \\ . & c & . & . & . & 1 & 1 & . & . \\ . & a & . & 1 & . & . & . & . & 1 \\ . & . & b & . & 1 & . & 1 & . & . \\ c & . & . & . & . & 1 & . & 1 & . \end{vmatrix} = 0;$$

next multiply by  $x$ ,  $y$ ,  $z$ ,  $y^2z^2$ ,  $z^2x^2$ ,  $x^2y^2$ ,  $x^2yz$ ,  $y^2zx$ ,  $z^2xy$ , reduce and eliminate the quantities in the outside row

$$(x^2, y^2, z^2, yz, zx, xy, xy^2z^2, yz^2x^2, zx^2y^2),$$

the result is

$$\begin{vmatrix} 1 & . & . & . & 1 & 1 & . & . & . \\ . & 1 & . & 1 & . & 1 & . & . & . \\ . & . & 1 & 1 & 1 & . & . & . & . \\ . & c & b & . & . & . & 1 & . & . \\ c & . & a & . & . & . & . & . & 1 \\ b & a & . & . & . & . & . & 1 & . \\ . & . & a & . & . & . & . & 1 & 1 \\ . & b & . & . & . & . & 1 & . & 1 \\ c & . & . & . & . & . & 1 & 1 & . \end{vmatrix} = 0;$$

lastly, multiply by  $x^2$ ,  $y^2$ ,  $z^2$ ,  $yz$ ,  $zx$ ,  $xy$ ,  $xy^2z^2$ ,  $yz^2x^2$ ,  $zx^2y^2$ , reduce and eliminate the quantities in the outside row

$$(1, xyz, x^2y^2z^2, x^2z, y^2x, z^2y, x^2y, y^2z, z^2x),$$

the result is

$$\begin{vmatrix} a & . & . & . & 1 & . & . & 1 & . \\ b & . & . & 1 & . & 1 & . & . & . \\ c & . & 1 & . & . & . & 1 & . & . \\ . & 1 & 1 & . & . & 1 & . & . & . \\ . & 1 & . & 1 & . & . & 1 & . & . \\ . & 1 & . & . & 1 & . & . & . & 1 \\ . & . & 1 & c & . & b & . & . & . \\ . & . & 1 & . & a & . & . & . & c \\ . & . & 1 & . & b & . & a & . & . \end{vmatrix} = 0;$$

where, as in the case of two cubic radicals, two forms, viz. the first and third forms of the rational equation, are not essentially distinct, but may be derived from each other by interchanging lines and columns.

And in general, whatever be the number of cubic radicals, two of the three forms are not essentially distinct, but may be derived from each other by interchanging lines and columns.

**110. Note on the Transformation of a Trigonometrical Expression.**

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 61—62.]

The differential equation

$$\frac{dx}{(a+x)\sqrt{c+x}} + \frac{dy}{(a+y)\sqrt{c+y}} + \frac{dz}{(a+z)\sqrt{c+z}} = 0,$$

integrated so as to be satisfied when the variables are simultaneously infinite, gives by direct integration

$$\tan^{-1} \sqrt{\frac{a-c}{c+x}} + \tan^{-1} \sqrt{\frac{a-c}{c+y}} + \tan^{-1} \sqrt{\frac{a-c}{c+z}} = 0,$$

and, by Abel's theorem,

$$\begin{vmatrix} 1 & x & (a+x)\sqrt{c+x} \\ 1 & y & (a+y)\sqrt{c+y} \\ 1 & z & (a+z)\sqrt{c+z} \end{vmatrix} = 0.$$

To show *à posteriori* the equivalence of these two equations, I represent the determinant by the symbol  $\Omega$ , and expressing it in the form

$$\Omega = \begin{vmatrix} 1 & a+x & (a+x)\sqrt{c+x} \\ 1 & a+y & (a+y)\sqrt{c+y} \\ 1 & a+z & (a+z)\sqrt{c+z} \end{vmatrix},$$

I write for the moment  $\xi = \sqrt{\frac{a-c}{c+x}}$ , etc., and we get

$$\Omega = \begin{vmatrix} 1 & (a-c)(1+\xi^2) & (a-c)^{3/2}(\xi+\xi^3) \\ 1 & (a-c)(1+\eta^2) & (a-c)^{3/2}(\eta+\eta^3) \\ 1 & (a-c)(1+\zeta^2) & (a-c)^{3/2}(\zeta+\zeta^3) \end{vmatrix}.$$

Pulling out the common factors and simplifying,

$$\Omega = (a-c)^{5/2} (\xi-\eta)(\eta-\zeta)(\zeta-\xi) (\xi\eta + \eta\zeta + \zeta\xi - 1);$$

or, replacing  $\xi, \eta, \zeta$  by their values, we have identically

$$\begin{vmatrix} 1 & x & (a+x)\sqrt{c+x} \\ 1 & y & (a+y)\sqrt{c+y} \\ 1 & z & (a+z)\sqrt{c+z} \end{vmatrix} = (a-c)^{5/2} \frac{(\xi-\eta)(\eta-\zeta)(\zeta-\xi)(\xi\eta + \eta\zeta + \zeta\xi - 1)}{1};$$

and the equation

$$\xi\eta + \eta\zeta + \zeta\xi - 1 = 0, \quad \text{i.e.} \quad \sqrt{\frac{a-c}{c+x}} \sqrt{\frac{a-c}{c+y}} + \sqrt{\frac{a-c}{c+y}} \sqrt{\frac{a-c}{c+z}} + \sqrt{\frac{a-c}{c+z}} \sqrt{\frac{a-c}{c+x}} - 1 = 0,$$

is of course equivalent to the trigonometrical equation

$$\tan^{-1} \sqrt{\frac{a-c}{c+x}} + \tan^{-1} \sqrt{\frac{a-c}{c+y}} + \tan^{-1} \sqrt{\frac{a-c}{c+z}} = 0,$$

which shows the equivalence of the two equations in question.

## 111. On a Theorem of M. Lejeune-Dirichlet's.

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 163—165.]

The following formula,

$$\sum q^{ax^2+2bxy+cy^2} \cdot \sum q^{a'x'^2+2b'x'y'+c'y'^2} \dots = \frac{1}{S} \Sigma \epsilon \left( \frac{D}{n} \right) \left( \frac{D}{n'} \right) \dots q^{nn' \dots / P},$$

is given in Lejeune-Dirichlet's well-known memoir "Recherches sur diverses applications &c." (*Crelle*, t. xxi. [1840] p. 8). The notation is as follows: On the left-hand side  $(a, b, c), (a', b', c'), \dots$  are a system of properly primitive forms to the negative determinant  $-D$  (i.e. a system of positive forms);  $x, y$  are positive or negative integers including zero, such that in the sum  $\sum q^{ax^2+2bxy+cy^2}$ ,  $ax^2 + 2bxy + cy^2$  is prime to  $2D$ , and similarly in the other sums;  $q$  is indeterminate and the summations extend to the values first mentioned, of  $x$  and  $y$ . On the right-hand side we have to consider the form of  $D$ , viz. we have  $D = PS^2$  or else  $D = 2PS^2$ , where  $S^2$  is the greatest square factor in  $D$  and where  $P$  is odd: this obviously defines  $P$ , and the values of  $S, \epsilon$ , which are always  $\pm 1$  (or, as I prefer to express it, are always  $\pm$ ) are given as follows, viz.

|              |                        |                       |
|--------------|------------------------|-----------------------|
| $D = PS^2,$  | $P \equiv 1 \pmod{4},$ | $S, \epsilon = +, +,$ |
| $D = PS^2,$  | $P \equiv 3 \pmod{4},$ | $S, \epsilon = -, +,$ |
| $D = 2PS^2,$ | $P \equiv 1 \pmod{4},$ | $S, \epsilon = +, -,$ |
| $D = 2PS^2,$ | $P \equiv 3 \pmod{4},$ | $S, \epsilon = -, -,$ |

$n, n'$  are any positive numbers prime to  $2D$ ,  $\left(\frac{D}{n}\right)$  is Legendre's symbol as generalized by Jacobi, viz. in general if  $p$  be a positive or negative prime not a factor of  $n$ , then  $\left(\frac{n}{p}\right) = +1$  or  $-1$  according as  $n$  is or is not a quadratic residue of  $p$  (or, what is the same thing,  $p$  being positive,  $\left(\frac{n}{p}\right) \equiv n^{\frac{1}{2}(p-1)} \pmod{p}$ ), and for  $P = pp'p'' \dots$ ,

$$\left(\frac{n}{P}\right) = \left(\frac{n}{p}\right) \left(\frac{n}{p'}\right) \left(\frac{n}{p''}\right) \dots$$

and the summation extends to all the values of  $n, n'$  of the form above mentioned. In the particular case  $D = -1$ , it is necessary that the second side should be doubled.

The method of reducing the equation is indicated in the memoir. The following are a few particular cases.

$$D = -1, \quad \sum q^{x^2+y^2} = 4 \Sigma \left(\frac{-1}{n}\right) q^n,$$

$$\text{or} \quad 2(1 + 2q + 2q^4 + 2q^9 + \dots)(q + q^4 + q^9 + q^{16} + \dots) = \frac{q}{1-q} - \frac{q^3}{1-q^3} + \frac{q^5}{1-q^5} - \frac{q^7}{1-q^7} + \&c.$$

$$D = -2, \quad \sum q^{x^2+2y^2} = 2 \Sigma \left(\frac{-2}{n}\right) q^n,$$

$$\text{or} \quad (1 + 2q^2 + 2q^8 + 2q^{18} \dots)(q + q^3 + q^9 + q^{17} \dots) = \frac{q}{1-q^2} + \frac{q^3}{1-q^6} - \frac{q^5}{1-q^{10}} - \&c.$$

an example given in the memoir.

For  $D = -3$ ,

$$\begin{aligned} & (q + q^4 + q^9 + q^{16} + q^{25} \dots)(1 + 2q^2 + 2q^6 + 2q^{12} \dots) + 2(q^2 + q^3 + q^6 + q^{12} + \dots)(q^4 + q^9 + q^{16} + q^{25} \dots) \\ &= \frac{q}{1-q^2} - \frac{q+q^2}{1-q^3} + \frac{q^3-q^4}{1-q^6} + \frac{q^7-q^8}{1-q^9} + \&c. \end{aligned}$$



I am not aware that the above theorem is quoted or referred to in any subsequent memoir on Elliptic Functions, or on the class of series to which it relates; and the theorem is so distinct in its origin and form from all other theorems relating to the same class of series, and, independently of the researches in which it originates, so remarkable as a result, that I have thought it desirable to give a detached statement of it in this paper.

## 112. Demonstration of a Theorem relating to the Products of Sums of Squares.

[From the Philosophical Magazine, vol. IV. (1852), pp. 515—519.]

MR KIRKMAN, in his paper “On Pluquaternions and Homoid Products of Sums of  $n$  Squares” (*Phil. Mag.* vol. xxxiii. [1848] pp. 447—459 and 494—509), quotes from a note of mine the following passage: “The complete test of the possibility of the product of  $2^m$  squares by  $2^m$  squares reducing itself to a sum of  $2^m$  squares is the following—forming the complete systems of triplets for  $(2^m - 1)$  things, if  $eab, ecd, fac, fdb$  be any four of them, we must have, paying attention to the signs alone,

$$(\pm eab)(\pm ecd) = (\pm fac)(\pm fdb);$$

i.e. if the first two are of the same sign, the last two must be so also, and *vice versa*; I believe that, for a system of seven, two conditions of this kind being satisfied would imply the satisfaction of all the others: it remains to be shown that the complete system of conditions cannot be satisfied for fifteen things.” I propose to explain the meaning of the theorem, and to establish the truth of it, without in any way assuming the existence of imaginary units.

The identity to be established is

$$(w^2 + a^2 + b^2 + \dots)(w_1^2 + a_1^2 + b_1^2 + \dots) = w_{\parallel}^2 + a_{\parallel}^2 + b_{\parallel}^2 + \dots,$$

where the  $2^m$  quantities  $w, a, b, c, \dots$  and the  $2^m$  quantities  $w_1, a_1, b_1, c_1, \dots$  are given quantities in terms of which the  $2^m$  quantities  $w_{\parallel}, a_{\parallel}, b_{\parallel}, c_{\parallel}, \dots$  have to be determined.

Without attaching any meaning whatever to the symbols  $a_0, b_0, c_0, \dots$  I write down the expressions

$$ww_1 + aa_1 + bb_1 + cc_1 + \dots, \quad w_1a_0 + a_1b_0 + b_1c_0 + \dots,$$

and I multiply as if  $a_0, b_0, c_0, \dots$  really existed, taking care to multiply without making any transposition in the order *inter se* of two symbols  $a_0, b_0$  combined in the way of multiplication. This gives a quasi-product

$$\begin{aligned} & ww_1 + (aw_1 + a_1w)a_0 + (bw_1 + b_1w)b_0 + \dots \\ & + aa_1a_0^2 + bb_1b_0^2 + \dots \\ & + ab_1a_0b_0 + a_1b_0a_0 + \dots \end{aligned}$$

Suppose, now, that a quasi-equation, such as

$$a_0b_0c_0 = +,$$

means that in the expression of the quasi-product

$$a_0b_0, b_0c_0, c_0a_0, a_0c_0, b_0a_0$$

are to be replaced by

$$c_0, a_0, b_0, -a_0, -b_0, -c_0;$$

and that a quasi-equation, such as  $a_0b_0c_0 = -$ , means that in the expression of the quasi-product

$$a_0b_0, b_0c_0, c_0a_0, b_0a_0, c_0b_0, a_0c_0$$

are to be replaced by

$$-c_0, -a_0, -b_0, +a_0, +b_0, +c_0.$$

It is in the first place clear that the quasi-equation  $a_0b_0c_0 = +$  may be written in any one of the six forms

$$a_0b_0c_0 = b_0c_0a_0 = c_0a_0b_0 = -a_0c_0b_0 = -c_0b_0a_0 = -b_0a_0c_0,$$

and so for the quasi-equation  $a_0b_0c_0 = -$ . This being premised, if we form a system of quasi-equations, such as

$$a_0b_0c_0, e_0a_0b_0, \dots,$$

where the system of triplets contains each duad once, and once only, and the arbitrary signs are chosen at pleasure; if, moreover, in the expression of the quasi-product we replace  $a_0^2, b_0^2, \dots$  each by  $-1$ , it is clear that the quasi-product will assume the form

$$w_{\mathcal{H}}w_1 + a_{\mathcal{H}}a_0 + b_{\mathcal{H}}b_0 + c_{\mathcal{H}}c_0 + \dots,$$

$w_{\mathcal{H}}, a_{\mathcal{H}}, b_{\mathcal{H}}, c_{\mathcal{H}}, \dots$  being determinate functions of  $w, a, b, c, \dots$ ;  $w_1, a_1, b_1, c_1, \dots$ , homogeneous of the first order in the quantities of each set; the value of  $w_{\mathcal{H}}$  being obviously in every case

$$w_{\mathcal{H}} = ww_1 - aa_1 - bb_1 - cc_1 - \dots,$$

and  $a_{\mathcal{H}}, b_{\mathcal{H}}, c_{\mathcal{H}}, \dots$  containing in every case the terms  $aw_1 + a_1w, bw_1 + b_1w, cw_1 + c_1w, \dots$  but the form of the remaining terms depending as well on the triplets entering into the system of quasi-equations as on the values given to the signs  $\pm$ .

*[The demonstration continues on book p. 51 et seq., outside the present pp. 26–50 slice.]*

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Papers 108–114.

## 108.

### ON CERTAIN MULTIPLE INTEGRALS CONNECTED WITH THE THEORY OF ATTRACTIONS.

[From the Cambridge and Dublin Mathematical Journal, vol. vii. (1852), pp. 174–178.]

It is easy to deduce from Mr Boole's formula, given in my paper "On a Multiple Integral connected with the theory of Attractions," *Journal*, t. ii. [1847], pp. 219–223, [44], the equation

$$\int \frac{d\xi d\eta \cdots}{[(\xi - \alpha)^2 + (\eta - \beta)^2 + \cdots + v^2]^{\frac{1}{2}n-q}} = \theta_1^q \Gamma(\frac{1}{2}n-q) \Gamma(q+1) \int_e^\infty \frac{s^{n-q-1} (\theta_1^2 - \sigma)^q ds}{\sqrt{(s + \frac{f^2}{\theta_1^2})(s + \frac{g^2}{\theta_1^2}) \cdots}},$$

where  $n$  is the number of variables of the multiple integral, and the condition of the integration is

$$\frac{(\xi - \alpha)^2}{f^2} + \frac{(\eta - \beta)^2}{g^2} + \cdots \leq 1;$$

also where

$$\sigma = \frac{(\alpha - \alpha_1)^2}{s + \frac{f^2}{\theta_1^2}} + \frac{(\beta - \beta_1)^2}{s + \frac{g^2}{\theta_1^2}} + \cdots + \frac{v^2}{s},$$

and  $e$  is the positive root of

$$\theta_1^2 = \frac{(\alpha - \alpha_1)^2}{e + \frac{f^2}{\theta_1^2}} + \frac{(\beta - \beta_1)^2}{e + \frac{g^2}{\theta_1^2}} + \cdots + \frac{v^2}{e}.$$

Suppose  $f = g = \cdots = \theta_1$ , and write  $(\alpha - \alpha_1)^2 + \cdots = k^2$ , we obtain

$$\int \frac{d\xi d\eta \cdots}{[(\xi - \alpha)^2 + \cdots + v^2]^{\frac{1}{2}n-q}} = \frac{\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n-q) \Gamma(q+1)} \int_e^\infty \frac{s^{q-1} (\theta_1^2 - \sigma)^q ds}{(1+s)^{\frac{1}{2}n}},$$

the limiting condition for the multiple integral being

$$(\xi - \alpha_1)^2 + \cdots \leq \theta_1^2,$$

and the function  $\sigma$ , and limit  $e$ , being given by

$$\sigma = \frac{(\alpha - \alpha_1)^2}{1+s} + \frac{(\beta - \beta_1)^2}{1+s} + \cdots + \frac{v^2}{s}, \quad \theta_1^2 = \frac{(\alpha - \alpha_1)^2}{1+e} + \frac{(\beta - \beta_1)^2}{1+e} + \cdots + \frac{v^2}{e},$$

$e$  denoting, as before, the positive root. Observing that the quantity under the integral sign on the second side vanishes for  $s = e$ , there is no difficulty in deducing, by a differentiation with respect to  $\theta_1$ , the formula

$$\int \frac{d\Sigma}{[(\xi - \alpha)^2 + \cdots + v^2]^{\frac{1}{2}n-q+1}} = \frac{2\theta_1^{q+1} \pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n-q) \Gamma(q)} \int_e^\infty \frac{s^{q-1} (\theta_1^2 - \sigma)^{q-1} ds}{(1+s)^{\frac{1}{2}n}},$$

in which  $e$  is the positive root of the equation

$$\theta_1^2 s^2 + \chi s - v^2 = 0.$$

I propose to transform these formulæ by means of the theory of images; it will be convenient to investigate some preliminary formulæ. Suppose  $\lambda^2 = \alpha^2 + \beta^2 + \dots$ ,  $\lambda_1^2 = \alpha_1^2 + \beta_1^2 + \dots$ ; also consider the new constants  $a, b, \dots, a_1, b_1, \dots, u, f_1$ , determined by the equations

$$\frac{\partial^2 u}{\partial \lambda^2 + v^2} = \alpha, \quad \frac{\partial^2 u_1}{\lambda_1^2 - f_1^2} = a_1, \quad \dots \quad \frac{\partial^2 v}{\lambda^2 + v^2} = u, \quad \frac{\partial^2 f_1}{\lambda_1^2 - f_1^2} = f_1,$$

where  $S$  is arbitrary. Then, putting

$$l^2 = a^2 + b^2 + \dots, \quad l_1^2 = a_1^2 + b_1^2 + \dots,$$

it is easy to see that

$$(\lambda^2 + v^2)(l^2 + u^2) = S^2, \quad (\lambda_1^2 - f_1^2)(l_1^2 - f_1^2) = S^2 \alpha_1^2 / \alpha^2,$$

and

$$\frac{\partial^2 u}{l^2 + u^2} = \lambda, \quad \frac{\partial^2 f_1}{l_1^2 - f_1^2} = \lambda_1,$$

and so on.

Proceeding to express the single integrals in terms of the new constants, we have in the first place  $k^2 = \theta^2 k'^2$ , where

$$k'^2 = \frac{p^2}{(l^2 + u^2)} + \frac{l_1^2}{(l_1^2 - f_1^2)^2} + \dots;$$

or if we write

$$aa_1 + bb_1 + \dots = ll_1 \cos \omega,$$

we have

$$k'^2 = \frac{l^2}{l^2 + u^2} + \frac{l_1^2}{l_1^2 - f_1^2} - \frac{2ll_1 \cos \omega}{(l^2 + u^2)^{\frac{1}{2}}(l_1^2 - f_1^2)^{\frac{1}{2}}}.$$

Hence also  $\chi = \theta^2 \chi'$ , where

$$\chi' = \frac{f_1^2}{(l_1^2 - f_1^2)^2} - k'^2 - \frac{u^2}{(l^2 + u^2)^2},$$

whence

$$-\chi' = \frac{1}{l^2 + u^2} + \frac{1}{l_1^2 - f_1^2} + \frac{2ll_1 \cos \omega}{(l^2 + u^2)^{\frac{1}{2}}(l_1^2 - f_1^2)^{\frac{1}{2}}} = \frac{1}{(l^2 + u^2)(l_1^2 - f_1^2)} [p^2 + u^2 - f_1^2],$$

where  $p^2 = l^2 + l_1^2 - 2ll_1 \cos \omega$ , that is

$$p^2 = (a - a_1)^2 + (b - b_1)^2 + \dots;$$

consequently  $\theta_1^2 s^2 + \chi s - v^2 = -\theta^2 S^2 \Pi$ , where  $\Pi$  is given by

$$\Pi = \frac{f_1^2}{(l_1^2 - f_1^2)^2} s^2 - \frac{(p^2 + u^2 - f_1^2)}{(l^2 + u^2)(l_1^2 - f_1^2)} s - \frac{u^2}{(l^2 + u^2)^2},$$

and it is clear that  $e$  will be the positive root of

$$0 = \frac{f_1^2}{(l_1^2 - f_1^2)^2} e^2 - \frac{(p^2 + u^2 - f_1^2)}{(l^2 + u^2)(l_1^2 - f_1^2)} e - \frac{u^2}{(l^2 + u^2)^2}.$$

It may be noticed that, in the particular case of  $u = 0$ , the roots of this equation are 0 and  $\frac{(p^2 - f_1^2)(l_1^2 - f_1^2)}{l_1^2 f_1^2}$ . Consequently if  $p^2 - f_1^2$  and  $l_1^2 - f_1^2$  are of opposite signs, we have  $e = 0$ ; but if  $p^2 - f_1^2$  and  $l_1^2 - f_1^2$  are of the same sign,  $e = \frac{(p^2 - f_1^2)(l_1^2 - f_1^2)}{l_1^2 f_1^2}$ .

In order to transform the double integrals, considering the new variables  $x, y, \dots$ , I write  $x^2 + y^2 + \dots = r^2$  and

$$\xi = \frac{Sx}{r^2}, \quad \dots$$

whence also, if  $\xi^2 + \eta^2 + \dots = \rho^2$  (which gives  $r\rho = S$ ), we have

$$\alpha = \frac{d\xi}{d\rho^2}, \quad \dots;$$

also it is immediately seen that

$$(\xi - \alpha)^2 + \dots + v^2 = \frac{S^2}{(l^2 + u^2)r^2} [(x - a)^2 + \dots + u^2],$$

$$(\xi - \alpha_1)^2 + \dots - \theta_1^2 = \frac{S^2}{(l_1^2 - f_1^2)r^2} [(x - a_1)^2 + \dots - f_1^2];$$

and from the latter equation it follows that the limiting condition for the first integral is  $(x - a_1)^2 + \dots \leq f_1^2$  (there is no difficulty in seeing that the sign  $<$  in the former limiting condition gives rise here to the sign  $>$ ), and that the second integral has to be extended over the surface  $(x - a_1)^2 + \dots = f_1^2$ . Also if  $dS$  represent the element of this surface, we may obtain

$$d\xi d\eta \dots = \frac{S^n}{r^{2n}} dx dy \dots, \quad d\Sigma = \frac{S^{n-1}}{r^{2n-2}} dS;$$

and, combining the above formulæ, we obtain

$$\int \frac{dx dy \dots}{(x^2 + y^2 + \dots)^{\frac{1}{2}n+q} [(x-a)^2 + (y-b)^2 + \dots + u^2]^{\frac{1}{2}n-q}} = \frac{\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n - q) \Gamma(q+1) (l^2 + u^2)^{\frac{1}{2}n-q}} \int_e^\infty \frac{ds}{s}$$

the limiting condition of the multiple integral being

$$(x - a_1)^2 + (y - b_1)^2 + \dots \leq f_1^2;$$

and

$$\int \frac{dS}{(x^2 + y^2 + \dots)^{\frac{1}{2}n+q-1} [(x-a)^2 + (y-b)^2 + \dots + u^2]^{\frac{1}{2}n-q}} = \frac{2\pi^{\frac{1}{2}n} f_1}{\Gamma(\frac{1}{2}n - q) \Gamma(q) (l^2 + u^2)^{\frac{1}{2}n-q} (l_1^2 - f_1^2)}$$

where  $dS$  is the element of the surface  $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1^2$ , and the integration extends over the entire surface. In these formulæ,  $l, l_1, p, \Pi$  denote as follows:

$$l^2 = a^2 + b^2 + \dots, \quad l_1^2 = a_1^2 + b_1^2 + \dots, \quad p^2 = (a - a_1)^2 + (b - b_1)^2 + \dots,$$

$$\Pi = \frac{f_1^2}{(l_1^2 - f_1^2)^2} s^2 - \frac{(p^2 + u^2 - f_1^2)}{(l^2 + u^2)(l_1^2 - f_1^2)} s - \frac{u^2}{(l^2 + u^2)^2},$$

and  $e$  is the positive root of the equation  $\Pi = 0$ .

The only obviously integrable case is that for which in the second formula  $q = 1$ ; this gives

$$\int \frac{dS}{(x^2 + y^2 + \dots)^{\frac{1}{2}n} [(x-a)^2 + (y-b)^2 + \dots + u^2]^{\frac{1}{2}n-1}} = \frac{2\pi^{\frac{1}{2}n} f_1}{\Gamma(\frac{1}{2}n) (l^2 + u^2)^{\frac{1}{2}n-1} (l_1^2 - f_1^2) (1+e)^{\frac{1}{2}n-1}}$$

In the case of  $u = 0$ , we have, as before, when  $p^2 - f_1^2$  and  $l_1^2 - f_1^2$  are of opposite signs,  $e = 0$ , and therefore  $1 + e = 1$ ; but when  $p^2 - f_1^2$  and  $l_1^2 - f_1^2$  are of the same sign, the value before found for  $e$  gives

$$1 + e = \frac{1}{l_1^2 f_1^2} [l_1^2 f_1^2 + (p^2 - f_1^2)(l_1^2 - f_1^2)].$$

Consider the image of the origin with respect to the sphere  $(x - a_1)^2 + (y - b_1)^2 + \dots = f_1^2$ ; the coordinates of this image are

$$\frac{a_1}{l_1^2}(l_1^2 - f_1^2), \quad \frac{b_1}{l_1^2}(l_1^2 - f_1^2), \quad \dots,$$

and consequently, if  $\mu$  be the distance of this image from the point  $(a, b, \dots)$ , we have

$$\mu^2 = \left[ a - \frac{a_1}{l_1^2}(l_1^2 - f_1^2) \right]^2 + \dots = \frac{1}{l_1^2} [l^2 f_1^2 + (p^2 - f_1^2)(l_1^2 - f_1^2)];$$

whence, by a simple reduction,

$$1 + e = \frac{l_1^2 \mu^2}{l_1^2 f_1^2},$$

or the values of the integral are

$$\begin{aligned} p^2 - f_1^2 \text{ and } l_1^2 - f_1^2 \text{ opposite signs, } I &= \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n)} \cdot \frac{f_1}{l^{n-2}(l_1^2 - f_1^2)}, \\ p^2 - f_1^2 \text{ and } l_1^2 - f_1^2 \text{ the same sign, } I &= \frac{2\pi^{\frac{1}{2}n}}{\Gamma(\frac{1}{2}n)} \cdot \frac{f_1^{n-1}}{l^{n-2} \mu^{n-2}(l_1^2 - f_1^2)}, \end{aligned}$$

where  $\mu$  is the distance from the point  $(a, b, \dots)$  of the image of the origin with respect to the sphere  $(x - a_1)^2 + \dots - f_1^2 = 0$ .

*Stone Buildings, August 6, 1850.*



109.

ON THE RATIONALISATION OF CERTAIN ALGEBRAICAL EQUATIONS.

[From the Cambridge and Dublin Mathematical Journal, vol. viii. (1853), pp. 97-101.]

SUPPOSE

$$x + y = 0, \quad x^2 = a, \quad y^2 = b;$$

then if we multiply the first equation by 1,  $xy$ , and reduce by the two others, we have

$$x + y = 0, \quad bx + ay = 0,$$

from which, eliminating  $x, y$ ,

$$\begin{vmatrix} 1 & 1 \\ b & a \end{vmatrix} = 0;$$

which is the equation between  $a$  and  $b$ ; or, considering  $x, y$  as quadratic radicals, the rational equation between  $x, y$ . So if the original equation be multiplied by  $x, y$ , we have

$$a + xy = 0, \quad b + xy = 0;$$

or, eliminating 1,  $xy$ ,

$$\begin{vmatrix} a & 1 \\ b & 1 \end{vmatrix} = 0,$$

which may be in like manner considered as the rational equation between  $x, y$ .

The preceding results are of course self-evident, but by applying the same process to the equations

$$x + y + z = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c,$$

we have results of some elegance. Multiply the equation first by 1,  $yz, zx, xy$ , reduce and eliminate the quantities  $x, y, z, xyz$ , we have the rational equation

$$\begin{vmatrix} 1 & 1 & 1 & \cdot \\ \cdot & c & b & \cdot \\ c & \cdot & a & \cdot \\ b & a & \cdot & \cdot \end{vmatrix} = 0;$$

and again, multiply the equation by  $x, y, z, xyz$ , reduce and eliminate the quantities  $1, yz, zx, xy$ , the result is

$$\begin{vmatrix} a & b & c & \cdot \\ a & \cdot & 1 & 1 \\ b & 1 & \cdot & 1 \\ c & 1 & 1 & \cdot \end{vmatrix} = 0,$$

which is of course equivalent to the preceding one (the two determinants are in fact identical in value), but the form is essentially different. The former of the two forms is that given in my paper “On a theorem in the Geometry of Position” (*Journal*, vol. ii. [1841] p. 270 [1]): it was only very recently that I perceived that a similar process led to the latter of the two forms.

Similarly, if we have the equations

$$x + y + z + w = 0, \quad x^2 = a, \quad y^2 = b, \quad z^2 = c, \quad w^2 = d,$$

then multiplying by  $1, yz, zx, xy, xw, yw, zw, xyzw$ , reducing and eliminating the quantities in the outside row,

$$x, y, z, w, yzw, zwx, wxy, xyz,$$

we have the result

$$\begin{vmatrix} 1 & 1 & 1 & 1 & \cdot & \cdot & \cdot & \cdot \\ \cdot & c & b & \cdot & \cdot & 1 & \cdot & 1 \\ c & \cdot & a & \cdot & 1 & \cdot & \cdot & 1 \\ b & a & \cdot & \cdot & \cdot & \cdot & 1 & 1 \\ d & \cdot & \cdot & a & \cdot & 1 & 1 & \cdot \\ \cdot & d & \cdot & b & 1 & \cdot & 1 & \cdot \\ \cdot & \cdot & d & c & 1 & 1 & \cdot & \cdot \\ \cdot & \cdot & \cdot & \cdot & a & b & c & d \end{vmatrix} = 0;$$

so if we multiply the equations by  $x, y, z, w, yzw, zwx, wxy$ , and  $xyz$ , reduce and eliminate the quantities in the outside row,

$$1, yz, zx, xy, xw, yw, zw, xyzw,$$

we have the result

$$\begin{vmatrix} a & \cdot & \cdot & \cdot & 1 & 1 & 1 & \cdot \\ b & \cdot & 1 & 1 & \cdot & 1 & \cdot & \cdot \\ c & 1 & \cdot & 1 & 1 & \cdot & \cdot & \cdot \\ d & \cdot & \cdot & \cdot & 1 & 1 & 1 & \cdot \\ \cdot & d & \cdot & \cdot & \cdot & c & b & 1 \\ \cdot & \cdot & d & \cdot & c & \cdot & a & 1 \\ \cdot & \cdot & \cdot & d & b & a & \cdot & 1 \\ \cdot & a & b & c & \cdot & \cdot & \cdot & 1 \end{vmatrix} = 0,$$

which however is not essentially distinct from the form before obtained, but may be derived from it by an interchange of lines and columns.

And in general for any even number of quadratic radicals the two forms are not essentially distinct, but may be derived from each other by interchanging lines and columns, while for an odd number of quadratic radicals the two forms cannot be so derived from each other, but are essentially distinct.

I was indebted to Mr Sylvester for the remark that the above process applies to radicals of a higher order than the second. To take the simplest case, suppose

$$x + y = 0, \quad x^3 = a, \quad y^3 = b;$$

and multiply first by 1,  $x^2y$ ,  $xy^2$ ; this gives

$$x + y = 0, \quad x^2y + xy^2 = 0, \quad ay + x^2y^2 = 0,$$

or, eliminating,

$$\begin{vmatrix} 1 & 1 & \cdot \\ \cdot & \cdot & 1 \\ \cdot & a & 1 \\ b & \cdot & 1 \end{vmatrix} = 0;$$

[the original prints a  $3 \times 3$  array of dots and units corresponding to the elimination scheme; the determinant condition is what is significant.]

next multiply by  $x$ ,  $y$ ,  $x^2y^2$ ; this gives

$$x^2 + xy = 0, \quad y^2 + xy = 0, \quad bx^2 + ay^2 = 0,$$

or, eliminating,

$$\begin{vmatrix} 1 & \cdot & 1 \\ \cdot & 1 & 1 \\ b & a & \cdot \end{vmatrix} = 0;$$

and lastly, multiply by  $x^2, y^2, xy$ ; this gives

$$a + x^2y = 0, \quad b + xy^2 = 0, \quad x^2y + xy^2 = 0,$$

or, eliminating,

$$\begin{vmatrix} a & 1 & \cdot \\ b & \cdot & 1 \\ \cdot & 1 & 1 \end{vmatrix} = 0,$$

where it is to be remarked that the second and third forms are not essentially distinct, since the one may be derived from the other by the interchange of lines and columns.

Applying the preceding process to the system

$$x + y + z = 0, \quad x^3 = a, \quad y^3 = b, \quad z^3 = c,$$

multiply first by 1,  $xyz, x^2y^2z^2, x^2z, y^2x, z^2y, x^2y, y^2z, z^2x$ , reduce and eliminate the quantities in the outside row,

$$x, y, z, y^2z^2, z^2x^2, x^2y^2, x^2yz, y^2zx, z^2xy, z^2x^2, x^2y^2,$$

the result is a determinantal equation

$$\Delta_1 = 0$$

[whose schematic form, written out by Cayley as a  $9 \times 9$  array of 1's,  $\cdot$ 's, and  $a, b, c$ 's, is — for legibility — abridged here; see facsimile, p. 43].

next multiply by  $x, y, z, y^2z^2, z^2x^2, x^2y^2, x^2yz, y^2zx, z^2xy$ , reduce and eliminate the quantities in the outside row,

$$x^2, y^2, z^2, yz, zx, xy, xy^2z^2, yz^2x^2, zx^2y^2,$$

the result is a second determinantal equation  $\Delta_2 = 0$  of the same size;

lastly, multiply by  $x^2, y^2, z^2, yz, zx, xy, x^2y^2z^2, y^2z^2x^2, x^2y^2z^2$ , reduce and eliminate the quantities in the outside row,

$$1, x^2y^2, y^2z^2, z^2x^2, x^2yz, y^2zx, z^2xy, xy^2z, yz^2x,$$

the result is a third determinantal equation  $\Delta_3 = 0$ :

where, as in the case of two cubic radicals, two forms, viz. the first and third forms of the rational equation, are not essentially distinct, but may be derived from each other by interchanging lines and columns.

And in general, whatever be the number of cubic radicals, two of the three forms are not essentially distinct, but may be derived from each other by interchanging lines and columns.

*[The three large determinantal arrays printed at pp. 43-44 of the original are partially illegible in the present scan and have been replaced by their schematic descriptions; the substantive content — the existence and equivalence relations — is preserved.]*

110.

NOTE ON THE TRANSFORMATION OF A  
TRIGONOMETRICAL  
EXPRESSION.

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 61-62.]

THE differential equation

$$\frac{dx}{(a+x)\sqrt{c+x}} + \frac{dy}{(a+y)\sqrt{c+y}} + \frac{dz}{(a+z)\sqrt{c+z}} = 0,$$

integrated so as to be satisfied when the variables are simultaneously infinite, gives by direct integration

$$\tan^{-1}\sqrt{\frac{a-c}{c+x}} + \tan^{-1}\sqrt{\frac{a-c}{c+y}} + \tan^{-1}\sqrt{\frac{a-c}{c+z}} = 0;$$

and, by Abel's theorem,

$$\begin{vmatrix} 1 & x & (a+x)\sqrt{c+x} \\ 1 & y & (a+y)\sqrt{c+y} \\ 1 & z & (a+z)\sqrt{c+z} \end{vmatrix} = 0.$$

To show *à posteriori* the equivalence of these two equations, I represent the determinant by the symbol  $\square$ , and expressing it in the form

$$\square = \begin{vmatrix} 1, & a+x, & (a+x)\sqrt{c+x} \\ 1, & a+y, & (a+y)\sqrt{c+y} \\ 1, & a+z, & (a+z)\sqrt{c+z} \end{vmatrix},$$

I write for the moment  $\xi = \sqrt{\frac{a-c}{c+x}}$ ,  $\eta = \sqrt{\frac{a-c}{c+y}}$ ,  $\zeta = \sqrt{\frac{a-c}{c+z}}$ , so that

$$(a+x) = (a-c)\left(1 + \frac{1}{\xi^2}\right), \quad (a+x)\sqrt{c+x} = (a-c)^{\frac{3}{2}}\left(\frac{1}{\xi^3} + \frac{1}{\xi}\right),$$

and similarly for  $(a+y)$ ,  $(a+z)$ , etc. Then

$$\square = \frac{(a-c)^{\frac{5}{2}}}{\xi^3\eta^3\zeta^3} \begin{vmatrix} \xi^2, & \xi^2+1, & \xi^2+1 \\ \eta^2, & \eta^2+1, & \eta^2+1 \\ \zeta^2, & \zeta^2+1, & \zeta^2+1 \end{vmatrix}$$

$$= \frac{(a-c)^{\frac{5}{2}}}{\xi^3 \eta^3 \zeta^3} (\xi + \eta + \zeta - \xi\eta\zeta) \begin{vmatrix} 1, & \xi, & \xi^2 \\ 1, & \eta, & \eta^2 \\ 1, & \zeta, & \zeta^2 \end{vmatrix};$$

or, replacing  $\xi, \eta, \zeta$  by their values, we have identically

$$\begin{vmatrix} 1, & x, & (a+x)\sqrt{c+x} \\ 1, & y, & (a+y)\sqrt{c+y} \\ 1, & z, & (a+z)\sqrt{c+z} \end{vmatrix} = K(x, y, z; a, c) \left[ \sqrt{\frac{a-c}{c+x}} + \sqrt{\frac{a-c}{c+y}} + \sqrt{\frac{a-c}{c+z}} - \sqrt{\frac{a-c}{c+x}} \sqrt{\frac{a-c}{c+y}} \sqrt{\frac{a-c}{c+z}} \right],$$

where  $K(x, y, z; a, c)$  is the symmetric prefactor displayed above; and the equation

$$\sqrt{\frac{a-c}{c+x}} + \sqrt{\frac{a-c}{c+y}} + \sqrt{\frac{a-c}{c+z}} - \sqrt{\frac{a-c}{c+x}} \sqrt{\frac{a-c}{c+y}} \sqrt{\frac{a-c}{c+z}} = 0$$

is of course equivalent to the trigonometrical equation

$$\tan^{-1} \sqrt{\frac{a-c}{c+x}} + \tan^{-1} \sqrt{\frac{a-c}{c+y}} + \tan^{-1} \sqrt{\frac{a-c}{c+z}} = 0,$$

which shows the equivalence of the two equations in question.

111.

ON A THEOREM OF M. LEJEUNE-DIRICHLET'S.

[From the Cambridge and Dublin Mathematical Journal, vol. ix. (1854), pp. 163-165.]

THE following formula,

$$\sum q^{ax^2+2bxy+cy^2} \cdot \sum q^{a'x'^2+2b'x'y'+c'y'^2} \dots = 2 \sum [S^{n-c} \epsilon^{\frac{1}{2}(n-1)}] \left(\frac{n}{P}\right) q^{nn' \dots}, \quad (3)$$

is given in Lejeune-Dirichlet's well-known memoir "Recherches sur diverses applications &c." (*Crelle*, t. xxi. [1840] p. 8). The notation is as follows:— On the left-hand side  $(a, b, c)$ ,  $(a', b', c')$ , ... are a system of properly primitive forms to the negative determinant  $-D$  (i.e. a system of positive forms);  $x, y$  are positive or negative integers including zero, such that in the sum  $\sum q^{ax^2+2bxy+cy^2}$ ,  $ax^2 + 2bxy + cy^2$  is prime to  $2D$ , and similarly in the other sums;  $q$  is indeterminate and the summations extend to the values first mentioned, of  $x$  and  $y$ . On the right-hand side we have to consider the form of  $D$ , viz. we have  $D = PS^2$  or else  $D = 2PS^2$ , where  $S^2$  is the greatest square factor in  $D$  and where  $P$  is odd: this obviously defines  $P$ , and the values of  $S, \epsilon$ , which are always  $\pm 1$  (or, as I prefer to express it, are always  $\pm$ ) are given as follows, viz.

$$\begin{array}{lll} D = PS^2, & P \equiv 1 \pmod{4}, & S, \epsilon = +, +, \\ D = PS^2, & P \equiv 3 \pmod{4}, & S, \epsilon = -, +, \\ D = 2PS^2, & P \equiv 1 \pmod{4}, & S, \epsilon = +, -, \\ D = 2PS^2, & P \equiv 3 \pmod{4}, & S, \epsilon = -, -; \end{array}$$

$n, n'$  are any positive numbers prime to  $2D$ ,  $\left(\frac{D}{n}\right)$  is Legendre's symbol as generalized by Jacobi, viz. in general if  $p$  be a positive or negative prime not a factor of  $n$ , then  $\left(\frac{n}{p}\right) = +$  or  $-$  according as  $n$  is or is not a quadratic residue of  $p$  (or, what is the same thing,  $p$  being positive,  $\left(\frac{n}{p}\right) \equiv n^{\frac{1}{2}(p-1)} \pmod{p}$ ), and for  $P = pp'p'' \dots$ ,

$$\left(\frac{n}{P}\right) = \left(\frac{n}{p}\right) \left(\frac{n}{p'}\right) \left(\frac{n}{p''}\right) \dots$$

and the summation extends to all the values of  $n, n'$  of the form above mentioned. In the particular case  $D = -1$ , it is necessary that the second side should be doubled.



The method of reducing the equation is indicated in the memoir. The following are a few particular cases.

$$D = -1, \quad \Sigma q^{x^2+y^2} = 4 \Sigma \left( \frac{-1}{n} \right) q^n,$$

$$\text{or} \quad 2(1+2q+2q^4+2q^9+\dots)(q+q^4+q^9+q^{16}+\dots) = \frac{q}{1-q} - \frac{q^3}{1-q^3} + \frac{q^5}{1-q^5} - \frac{q^7}{1-q^7} + \&c.$$

$$D = -2, \quad \Sigma q^{x^2+2y^2} = 2 \Sigma \left( \frac{-2}{n} \right) q^n,$$

$$\text{or} \quad (1+2q^2+2q^8+2q^{18}\dots)(q+q^3+q^9+q^{17}\dots) = \frac{q}{1-q^2} + \frac{q^3}{1-q^6} - \frac{q^5}{1-q^{10}} - \&c.$$

an example given in the memoir.

$$D = -3,$$

$$\begin{aligned} & (q+q^4+q^9+q^{16}+q^{25}\dots)(1+2q^2+2q^6+2q^{12}\dots) + 2(q^2+q^3+q^6+q^{12}+\dots)(q^4+q^9+q^{16}+q^{25}\dots) \\ &= \frac{q}{1-q^2} - \frac{q+q^2}{1-q^3} + \frac{q^3-q^4}{1-q^6} + \frac{q^7-q^8}{1-q^9} + \&c. \end{aligned}$$

I am not aware that the above theorem is quoted or referred to in any subsequent memoir on Elliptic Functions, or on the class of series to which it relates; and the theorem is so distinct in its origin and form from all other theorems relating to the same class of series, and, independently of the researches in which it originates, so remarkable as a result, that I have thought it desirable to give a detached statement of it in this paper.

112.

DEMONSTRATION OF A THEOREM RELATING TO THE  
PRODUCTS OF SUMS OF SQUARES.

[From the *Philosophical Magazine*, vol. iv. (1852), pp. 515–519.]

MR KIRKMAN, in his paper “On Pluquaternions and Homoid Products of Sums of  $n$  Squares” (*Phil. Mag.* vol. xxxiii. [1848] pp. 447–459 and 494–509), quotes from a note of mine the following passage:—“The complete test of the possibility of the product of  $2^m$  squares by  $2^m$  squares reducing itself to a sum of  $2^m$  squares is the following—forming the complete systems of triplets for  $(2^m - 1)$  things, if  $eab$ ,  $ecd$ ,  $fac$ ,  $fdb$  be any four of them, we must have, paying attention to the signs alone,

$$(\pm eab)(\pm ecd) = (\pm fac)(\pm fdb);$$

i.e. if the first two are of the same sign, the last two must be so also, and *vice versa*; I believe that, for a system of seven, two conditions of this kind being satisfied would imply the satisfaction of all the others: it remains to be shown that the complete system of conditions cannot be satisfied for fifteen things.” I propose to explain the meaning of the theorem, and to establish the truth of it, without in any way assuming the existence of imaginary units.

The identity to be established is

$$(w^2 + a^2 + b^2 + \dots)(w_1^2 + a_1^2 + b_1^2 + \dots) = w_{\parallel}^2 + a_{\parallel}^2 + b_{\parallel}^2 + \dots,$$

where the  $2^m$  quantities  $w, a, b, c, \dots$  and the  $2^m$  quantities  $w_1, a_1, b_1, c_1, \dots$  are given quantities in terms of which the  $2^m$  quantities  $w_{\parallel}, a_{\parallel}, b_{\parallel}, c_{\parallel}, \dots$  have to be determined.

Without attaching any meaning whatever to the symbols  $a_0, b_0, c_0, \dots$  I write down the expressions

$$ww_1 + aa_0 + bb_0 + cc_0 + \dots, \quad w_1a_0 + a_1b_0 + b_1c_0 + \dots,$$

and I multiply as if  $a_0, b_0, c_0, \dots$  really existed, taking care to multiply without making any transposition in the order *inter se* of two symbols  $a_0, b_0$  combined in the way of multiplication. This gives a quasi-product

$$\begin{aligned} & ww_1 + (aw_1 + a_1w)a_0 + (bw_1 + b_1w)b_0 + \dots \\ & + aa_1a_0^2 + bb_1b_0^2 + \dots \\ & + ab_1a_0b_0 + a_1b b_0a_0 + \dots \end{aligned}$$

Suppose, now, that a quasi-equation, such as

$$a_0 b_0 c_0 = +,$$

means that in the expression of the quasi-product

$$a_0 b_0, b_0 c_0, c_0 a_0, c_0 b_0, a_0 c_0, b_0 a_0$$

are to be replaced by

$$c_0, a_0, b_0, -a_0, -b_0, -c_0;$$

and that a quasi-equation, such as  $a_0 b_0 c_0 = -$ , means that in the expression of the quasi-product

$$b_0 c_0, c_0 a_0, a_0 b_0, c_0 b_0, a_0 c_0, b_0 a_0$$

are to be replaced by

$$-a_0, -b_0, -c_0, a_0, b_0, c_0.$$

It is in the first place clear that the quasi-equation  $a_0 b_0 c_0 = +$  may be written in any one of the six forms

$$a_0 b_0 c_0 = b_0 c_0 a_0 = c_0 a_0 b_0 = -a_0 c_0 b_0 = -c_0 b_0 a_0 = -b_0 a_0 c_0,$$

and so for the quasi-equation  $a_0 b_0 c_0 = -$ . This being premised, if we form a system of quasi-equations, such as

$$a_0 b_0 c_0, \quad e_0 a_0 b_0, \quad \dots,$$

where the system of triplets contains each duad once, and once only, and the arbitrary signs are chosen at pleasure; if, moreover, in the expression of the quasi-product we replace  $a_0^2, b_0^2, \dots$  each by  $-1$ , it is clear that the quasi-product will assume the form

$$w_{\text{II}} w_1 + a_{\text{II}} a_0 + b_{\text{II}} b_0 + c_{\text{II}} c_0 + \dots,$$

$w_{\text{II}}, a_{\text{II}}, b_{\text{II}}, c_{\text{II}}, \dots$  being determinate functions of  $w, a, b, c, \dots$ ;  $w_1, a_1, b_1, c_1, \dots$ , homogeneous of the first order in the quantities of each set; the value of  $w_{\text{II}}$  being obviously in every case

$$w_{\text{II}} = ww_1 - aa_1 - bb_1 - cc_1 - \dots,$$

and  $a_{\mu}, b_{\mu}, c_{\mu}, \dots$  containing in every case the terms  $aw_1 + a_1w$ ,  $bw_1 + b_1w$ ,  $cw_1 + c_1w$ ,  $\dots$  but the form of the remaining terms depending as well on the triplets entering into the system of quasi-equations as on the values given to the signs  $\pm$ ; the quasi-equations serving, in fact, to prescribe a rule for the formation of certain functions  $w_{\mu}, a_{\mu}, b_{\mu}, c_{\mu}, \dots$ , the properties of which functions may afterwards be investigated.

Suppose, now, that the system of quasi-equations is such that

$$e_0 a_0 b_0, \quad e_0 c_0 d_0$$

being any two of its triplets, with a common symbol  $e_0$ , there occur also in the system the triplets

$$f_0 a_0 c_0, \quad f_0 d_0 b_0, \quad g_0 a_0 d_0, \quad g_0 b_0 c_0;$$

and suppose that the corresponding portion of the system is

$$e_0 a_0 b_0 = \epsilon, \quad e_0 c_0 d_0 = \epsilon', \quad f_0 a_0 c_0 = \zeta, \quad f_0 d_0 b_0 = \zeta', \quad g_0 a_0 d_0 = \iota, \quad g_0 b_0 c_0 = \iota',$$

where  $\epsilon, \zeta, \iota, \epsilon', \zeta', \iota'$  each of them denote one of the signs  $+$  or  $-$ ; then  $e_{\mu}, f_{\mu}, g_{\mu}$  will contain respectively the terms

$$\begin{aligned} & \epsilon (ab_1 - a_1b) - \epsilon' (cd_1 - c_1d), \\ & \zeta (ac_1 - a_1c) + \zeta' (db_1 - d_1b), \\ & \iota (ad_1 - a_1d) + \iota' (bc_1 - b_1c); \end{aligned}$$

and  $e_{\mu}^2 + f_{\mu}^2 + g_{\mu}^2$  contains the terms

$$\begin{aligned} & (a^2 + b^2 + c^2 + d^2)(a_1^2 + b_1^2 + c_1^2 + d_1^2) - a^2 a_1^2 - b^2 b_1^2 - c^2 c_1^2 - d^2 d_1^2 \\ & + 2[\epsilon \epsilon' (ab_1 - a_1b)(cd_1 - c_1d) \\ & + \zeta \zeta' (ac_1 - a_1c)(db_1 - d_1b) \\ & + \iota \iota' (ad_1 - a_1d)(bc_1 - b_1c)]; \end{aligned}$$

and by taking account of the terms  $ew_1 + e_1w$ ,  $fw_1 + f_1w$ ,  $gw_1 + g_1w$  in  $e_{\mu}, f_{\mu}, g_{\mu}$  respectively, we should have had besides in  $e_{\mu}^2 + f_{\mu}^2 + g_{\mu}^2$  the terms

$$(e^2 + f^2 + g^2)w_1^2 + (e_1^2 + f_1^2 + g_1^2)w^2 + 2(ee_1 + ff_1 + gg_1)ww_1.$$

Also  $w_{\mu}^2$  contains the terms

$$a^2 a_1^2 + b^2 b_1^2 + c^2 c_1^2 + d^2 d_1^2 - 2(ee_1 + ff_1 + gg_1)ww_1;$$

whence it is easy to see that

$$\begin{aligned} & w''^2 + a''^2 + b''^2 + c''^2 + \cdots \\ &= (w^2 + a^2 + b^2 + c^2 + \cdots)(w_1^2 + a_1^2 + b_1^2 + c_1^2 + \cdots) \\ &+ 2\Sigma[\epsilon\epsilon'(ab_1 - a_1b)(cd_1 - c_1d) \\ &\quad + \zeta\zeta'(ac_1 - a_1c)(db_1 - d_1b) \\ &\quad + \iota\iota'(ad_1 - a_1d)(bc_1 - b_1c)], \end{aligned}$$

where the summation extends to all the quadruplets formed each by the combination of two duads such as  $ab$  and  $cd$ , or  $ac$  and  $db$ , or  $ad$  and  $bc$ , i.e. two duads, which, combined with the same common letter (in the instances just mentioned  $e$ , or  $f$ , or  $g$ ), enter as triplets into the system of quasi-equations; so that if  $\nu = 2^m - 1$ , the number of quadruplets is

$$\frac{1}{2} \cdot \frac{1}{2}(\nu - 1) \cdot \frac{1}{2}(\nu - 3) \cdot \nu \cdot 1 = \frac{1}{8}\nu(\nu - 1)(\nu - 3),$$

and the terms under the sign  $\Sigma$  will vanish identically if only

$$\epsilon\epsilon' = \zeta\zeta' = \iota\iota';$$

but the relation  $\epsilon\epsilon' = \iota\iota'$  is of the same form as the equation  $\epsilon\epsilon' = \zeta\zeta'$ ; hence if all the relations

$$\epsilon\epsilon' = \zeta\zeta'$$

are satisfied, the terms under the sign  $\Sigma$  vanish, and we have

$$(w''^2 + a''^2 + b''^2 + c''^2 + \cdots) = (w^2 + a^2 + b^2 + c^2 + \cdots)(w_1^2 + a_1^2 + b_1^2 + c_1^2 + \cdots),$$

which is thus shown to be true, upon the suppositions

1. That the system of quasi-equations is such that

$$e_0a_0b_0, \quad e_0c_0d_0,$$

being any two of its triplets with a common symbol  $e_0$ , there occur also in the system the triplets

$$f_0a_0c_0, \quad f_0d_0b_0, \quad g_0a_0d_0, \quad g_0b_0c_0.$$

2. That for any two pairs of triplets, such as

$$eab, \quad ecd \quad \text{and} \quad fac, \quad fdb,$$

the product of the signs of the triplets of the first pair is equal to the product of the signs of the triplets of the second pair.

In the case of fifteen things  $a, b, c, \dots$  the triplets may, as appears from Mr Kirkman's paper, be chosen so as to satisfy the first condition; but the second condition involves, as Mr Kirkman has shown, a contradiction; and therefore the product of two sums, each of them of sixteen squares, is not a sum of sixteen squares. It is proper to remark, that this demonstration, although I think rendered clearer by the introduction of the idea of the system of triplets furnishing the rule for the formation of the expressions  $w_{\mu}, a_{\mu}, b_{\mu}, c_{\mu}, \&c.$ , is not in principle different from that contained in Prof. Young's paper "On an Extension of a Theorem of Euler, &c.", *Irish Transactions*, vol. xxi. [1848, pp. 311–341].

113.

NOTE ON THE GEOMETRICAL REPRESENTATION OF THE  
INTEGRAL  $\int dx \div \sqrt{(x+a)(x+b)(x+c)}$ .

[From the *Philosophical Magazine*, vol. v. (1853), pp. 281-284.]

THE equation of a conic passing through the points of intersection of the conics

$$x^2 + y^2 + z^2 = 0, \quad ax^2 + by^2 + cz^2 = 0,$$

is of the form

$$w(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0,$$

where  $w$  is an arbitrary parameter. Suppose that the conic touches a given line, we have for the determination of  $w$  a quadratic equation, the roots of which may be considered as parameters for determining the line in question. Let one of the values of  $w$  be considered as equal to a constant quantity  $k$ , the line is always a tangent to the conic

$$k(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0,$$

and taking  $w = p$  for the other value of  $w$ ,  $p$  is a parameter determining the particular tangent, or, what is the same thing, determining the point of contact of this tangent.

The equation of the tangent is easily seen to be

$$x\sqrt{b-c}\sqrt{a+k}\sqrt{a+p} + y\sqrt{c-a}\sqrt{b+k}\sqrt{b+p} + z\sqrt{a-b}\sqrt{c+k}\sqrt{c+p} = 0;$$

suppose that the tangent meets the conic  $x^2 + y^2 + z^2 = 0$  (which is of course the conic corresponding to  $w = \infty$ ) in the points  $P, P'$ , and let  $\theta, \infty$  be the parameters of the point  $P$ , and  $\theta', \infty$  the parameters of the point  $P'$ , i.e. (repeating the definition of the terms) let the tangent at  $P$  of the conic  $x^2 + y^2 + z^2 = 0$  be also touched by the conic  $\theta(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0$ , and similarly for  $\theta'$ . The coordinates of the point  $P$  are given by the equations

$$x : y : z = \sqrt{b-c}\sqrt{a+\theta} : \sqrt{c-a}\sqrt{b+\theta} : \sqrt{a-b}\sqrt{c+\theta};$$

and substituting these values in the equation of the line  $PP'$ , we have

$$(b-c)\sqrt{a+k}\sqrt{a+p}\sqrt{a+\theta} + (c-a)\sqrt{b+k}\sqrt{b+p}\sqrt{b+\theta} + (a-b)\sqrt{c+k}\sqrt{c+p}\sqrt{c+\theta} = 0, \quad \dots (*)$$

an equation connecting the quantities  $p, \theta$ . To rationalize this equation, write

$$\begin{aligned}\sqrt{(a+k)(a+p)(a+\theta)} &= \lambda + \mu a, \\ \sqrt{(b+k)(b+p)(b+\theta)} &= \lambda + \mu b, \\ \sqrt{(c+k)(c+p)(c+\theta)} &= \lambda + \mu c,\end{aligned}$$

values which evidently satisfy the equation in question. Squaring these equations, we have equations from which  $\lambda^2, \lambda\mu, \mu^2$  may be linearly determined; and making the necessary reductions, we find

$$\begin{aligned}\lambda^2 &= abc + kp\theta, \\ -2\lambda\mu &= bc + ca + ab - (p\theta + kp + k\theta), \\ \mu^2 &= a + b + c + k + p + \theta,\end{aligned}$$

or, eliminating  $\lambda, \mu$ ,

$$[bc + ca + ab - (p\theta + kp + k\theta)]^2 - 4(a + b + c + k + p + \theta)(abc + kp\theta) = 0, \quad \dots (*)$$

which is the rational form of the former equation marked (\*). It is clear from the symmetry of the formula, that the same equation would have been obtained by the elimination of  $L, M$  from the equations

$$\begin{aligned}\sqrt{(k+a)(k+b)(k+c)} &= L + Mk, \\ \sqrt{(p+a)(p+b)(p+c)} &= L + Mp, \\ \sqrt{(\theta+a)(\theta+b)(\theta+c)} &= L + M\theta,\end{aligned}$$

and it follows from Abel's theorem (but the result may be verified by means of Euler's fundamental integral in the theory of elliptic functions), that if

$$\Pi x = \int_{\infty}^x \frac{dx}{\sqrt{(x+a)(x+b)(x+c)}},$$

then the algebraical equations (\*) are equivalent to the transcendental equation

$$\pm \Pi k \pm \Pi p \pm \Pi \theta = 0;$$

the arbitrary constant which should have formed the second side of the equation having been determined by observing that the algebraical equation gives for  $p = \theta, k = \infty$ , a system of values, which, when the signs are properly



chosen, satisfy the transcendental equation. In fact, arranging the rational algebraical equation according to the powers of  $k$ , it becomes

$$k^2 (p - \theta)^2 - 2k [p\theta (p + \theta) + 2(a + b + c)p\theta + (bc + ca + ab)(p + \theta) + 2abc] + p^2\theta^2 - 2(bc + ca + ab)p\theta - 4abc(p + \theta) + b^2c^2 + c^2a^2 + a^2b^2 - 2a^2bc - 2b^2ca - 2c^2ab = 0, \quad (*)$$

which proves the property in question, and is besides a very convenient form of the algebraical integral. The ambiguous signs in the transcendental integral are not of course arbitrary (indeed it has just been assumed that for  $p = \theta$ ,  $\Pi p$  and  $\Pi\theta$  are to be taken with opposite signs), but the discussion of the proper values to be given to the ambiguous signs would be at all events tedious, and must be passed over for the present.

It is proper to remark, that  $p = \theta$  gives not only, as above supposed,  $k = \infty$ , but another value of  $k$ , which, however, corresponds to the transcendental equation

$$\pm \Pi k \pm 2\Pi p = 0;$$

the value in question is obviously

$$k = \frac{p^2 - 2(bc + ca + ab)p^2 - 8abc p + b^2c^2 + c^2a^2 + a^2b^2 - 2a^2bc - 2b^2ca - 2c^2ab}{(p + a)(p + b)(p + c)}.$$

Consider, in general, a cubic function  $ax^3 + 3bx^2y + 3cxy^2 + dy^3$ , or, as I now write it in the theory of invariants,  $(a, b, c, d)(x, y)^3$ , the Hessian of this function is

$$(ac - b^2, \frac{1}{2}(ad - bc), bd - c^2)(x, y)^2,$$

and applying this formula to the function  $(p + a)(p + b)(p + c)$ , it is easy to write the equation last preceding in the form

$$\frac{1}{4}k = p - (a + b + c) - \frac{9 \text{ Hessian } \{(p + a)(p + b)(p + c)\}}{(p + a)(p + b)(p + c)},$$

which is a formula for the duplication of the transcendent  $\Pi x$ .

Reverting now to the general transcendental equation

$$\pm \Pi k \pm \Pi p \pm \Pi \theta = 0,$$

we have in like manner

$$\pm \Pi k \pm \Pi p \pm \Pi \theta' = 0;$$

and assuming a proper correspondence of the signs, the elimination of  $\Pi p$  gives

$$\Pi\theta' - \Pi\theta = \pm 2\Pi k,$$

i.e. if the points  $P, P'$  upon the conic  $x^2 + y^2 + z^2 = 0$  are such that their parameters  $\theta, \theta'$  satisfy this equation, the line  $PP'$  will be constantly a tangent to the conic

$$k(x^2 + y^2 + z^2) + (ax^2 + by^2 + cz^2) = 0.$$

Hence also, if the parameters  $k, k', k''$  of the conics

$$k(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0,$$

$$k'(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0,$$

$$k''(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0,$$

satisfy the equation

$$\Pi k + \Pi k' + \Pi k'' = 0,$$

there are an infinity of triangles inscribed in the conic  $x^2 + y^2 + z^2 = 0$ , and the sides of which touch the last-mentioned three conics respectively.

Suppose  $2\Pi k = \Pi\kappa$  (an equation the algebraic form of which has already been discussed), then

$$\Pi\theta' - \Pi\theta = \Pi\kappa,$$

$\theta = \infty$  gives  $\theta' = \kappa$ ; or, observing that  $\theta = \infty$  corresponds to a point of intersection of the conics  $x^2 + y^2 + z^2 = 0$ ,  $ax^2 + by^2 + cz^2 = 0$ ,  $\kappa$  is the parameter of the point in which a tangent to the conic  $k(x^2 + y^2 + z^2) + ax^2 + by^2 + cz^2 = 0$  at any one of its intersections with the conic  $x^2 + y^2 + z^2 = 0$  meets the last-mentioned conic. Moreover, the algebraical relation between  $\theta, \theta'$  and  $k$  (where, as before remarked,  $\kappa$  is a given function of  $k$ ) is given by a preceding formula, and is simpler than that between  $\theta, \theta'$  and  $k$ .

The preceding investigations were, it is hardly necessary to remark, suggested by a well-known memoir of the late illustrious Jacobi, and contain, I think, the extension which he remarks it would be interesting to make of the principles in such memoir to a system of two conics. I propose reverting to the subject in a memoir to be entitled "Researches on the Porism of the in- and circumscribed triangle." [*This was, I think, never written.*]

114.

ANALYTICAL RESEARCHES CONNECTED WITH STEINER'S  
EXTENSION OF Malfatti's PROBLEM.

*[From the Philosophical Transactions of the Royal Society of London, vol. cxlii. for the year 1852, pp. 253-278: Received April 12, — Read May 27, 1852.]*

THE PROBLEM, in a triangle to describe three circles each of them touching the two others and also two sides of the triangle, has been termed after the Italian geometer by whom it was proposed and solved, Malfatti's problem. The problem which I refer to as Steiner's extension of Malfatti's problem is as follows:—"To determine three sections of a surface of the second order, each of them touching the two others, and also two of three given sections of the surface of the second order," a problem proposed in Steiner's memoir, "Einige geometrische Betrachtungen," *Crelle*, t. i. [1826, pp. 161-184]. The geometrical construction of the problem in question is readily deduced from that given in the memoir just mentioned for a somewhat less general problem, viz. that in which the surface of the second order is replaced by a sphere; it is for the sake of the analytical developments to which the problem gives rise, that I propose to resume here the discussion of the problem. The following is an analysis of the present memoir:

- § 1. Contains a lemma which appears to me to constitute the foundation of the analytical theory of the sections of a surface of the second order.
- § 2. Contains a statement of the geometrical construction of Steiner's extension of Malfatti's problem.
- § 3. Is a verification, founded on a particular choice of coordinates, of the construction in question.
- § 4. In this section, referring the surface of the second order to absolutely general coordinates, and after an incidental solution of the problem to determine a section touching three given sections, I obtain the equations for the solution of Steiner's extension of Malfatti's problem.
- § 5. Contains a separate discussion of a system of equations, including as a particular case the equations obtained in the preceding section.
- §§ 6 and 7. Contain the application of the formulæ for the general system to the equations in § 4, and the development and completion of the solution.

§ 8. Is an extension of some preceding formulæ to quadratic functions of any number of variables.

**§ 1. Lemma relating to the sections of a surface of the second order.**

If

$$ax^2 + by^2 + cz^2 + dw^2 + 2fyz + 2gzx + 2hxy + 2lxw + 2myw + 2nzw = 0$$

be the equation of a surface of the second order, and

$$\mathfrak{A}x^2 + \mathfrak{B}y^2 + \mathfrak{C}z^2 + \mathfrak{D}w^2 + 2\mathfrak{F}yz + 2\mathfrak{G}zx + 2\mathfrak{H}xy + 2\mathfrak{L}xw + 2\mathfrak{M}yw + 2\mathfrak{N}zw = 0$$

the reciprocal equation, the condition that the two sections

$$\lambda x + \mu y + \nu z + \rho w = 0, \quad \lambda' x + \mu' y + \nu' z + \rho' w = 0,$$

may touch, is

$$\begin{aligned} & (\mathfrak{A}\lambda^2 + \mathfrak{B}\mu^2 + \mathfrak{C}\nu^2 + \mathfrak{D}\rho^2 + 2\mathfrak{F}\mu\nu + 2\mathfrak{G}\nu\lambda + 2\mathfrak{H}\lambda\mu + 2\mathfrak{L}\lambda\rho + 2\mathfrak{M}\mu\rho + 2\mathfrak{N}\nu\rho)^{\frac{1}{2}} \\ & \times (\mathfrak{A}\lambda'^2 + \mathfrak{B}\mu'^2 + \mathfrak{C}\nu'^2 + \mathfrak{D}\rho'^2 + 2\mathfrak{F}\mu'\nu' + 2\mathfrak{G}\nu'\lambda' + 2\mathfrak{H}\lambda'\mu' + 2\mathfrak{L}\lambda'\rho' + 2\mathfrak{M}\mu'\rho' + 2\mathfrak{N}\nu'\rho')^{\frac{1}{2}} \\ & = \mathfrak{A}\lambda\lambda' + \mathfrak{B}\mu\mu' + \mathfrak{C}\nu\nu' + \mathfrak{D}\rho\rho' + \mathfrak{F}(\mu\nu' + \mu'\nu) + \mathfrak{G}(\nu\lambda' + \nu'\lambda) + \mathfrak{H}(\lambda\mu' + \lambda'\mu) \\ & \quad + \mathfrak{L}(\lambda\rho' + \lambda'\rho) + \mathfrak{M}(\mu\rho' + \mu'\rho) + \mathfrak{N}(\nu\rho' + \nu'\rho); \end{aligned}$$

and in particular if the equation of the surface be

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + pw^2 = 0,$$

the condition of contact is

$$\begin{aligned} & (\mathfrak{A}\lambda^2 + \mathfrak{B}\mu^2 + \mathfrak{C}\nu^2 + 2\mathfrak{F}\mu\nu + 2\mathfrak{G}\nu\lambda + 2\mathfrak{H}\lambda\mu + K \frac{\rho^2}{p})^{\frac{1}{2}} \\ & \times (\mathfrak{A}\lambda'^2 + \mathfrak{B}\mu'^2 + \mathfrak{C}\nu'^2 + 2\mathfrak{F}\mu'\nu' + 2\mathfrak{G}\nu'\lambda' + 2\mathfrak{H}\lambda'\mu' + K \frac{\rho'^2}{p})^{\frac{1}{2}} \\ & = \mathfrak{A}\lambda\lambda' + \mathfrak{B}\mu\mu' + \mathfrak{C}\nu\nu' + \mathfrak{F}(\mu\nu' + \mu'\nu) + \mathfrak{G}(\nu\lambda' + \nu'\lambda) + \mathfrak{H}(\lambda\mu' + \lambda'\mu) + K \frac{\rho\rho'}{p}, \end{aligned}$$

in which last formula

$$\mathfrak{F} = gh - af, \quad \mathfrak{G} = hf - bg, \quad \mathfrak{H} = fg - ch,$$

$$K = abc - af^2 - bg^2 - ch^2 + 2fgh.$$

## § 2.

In order to state in the most simple form the geometrical construction for the solution of Steiner's extension of Malfatti's problem, let the given sections be called for conciseness the *determinators*<sup>1</sup>; any two of these sections lie in two different cones, the vertices of which determine with the line of intersection of the planes of the determinators, two planes which may be termed *bisectors*; the six bisectors pass three and three through four straight lines; and it will be convenient to use the term bisectors to denote, not the entire system, but any three bisectors passing through the same line. Consider three sections, which may be termed *tactors*, each of them touching a determinator and two bisectors, and three other sections (which may be termed *separators*) each of them passing through the point of contact of a determinator and tactor and touching the other two tactors; the separators will intersect in a line which passes through the point of intersection of the determinators. The three required sections, or as I shall term them the *resultors*, are determined by the conditions that each resultor touches two determinators and two separators, the possibility of the construction being implied as a theorem. The *à posteriori* verification may be obtained as follows:—

## § 3.

Let  $x = 0, y = 0, z = 0$  be the equations of the resultors,  $w = 0$  the equation of the polar of the point of intersection of the resultors. Since the resultors touch two and two, the equation of the surface is easily seen to be of the form

$$2yz + 2zx + 2xy + w^2 = 0. \quad (=)$$

2

The determinators are sections each of them touching two resultors, but otherwise arbitrary; their equations are

$$-\alpha x + \frac{1}{2\alpha} y + \frac{1}{2\alpha} z + w = 0, \quad \frac{1}{2\beta} x - \beta y + \frac{1}{2\beta} z + w = 0, \quad \frac{1}{2\gamma} x + \frac{1}{2\gamma} y - \gamma z + w = 0.$$

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<sup>1</sup>I use the words "determinators," &c. to denote indifferently the sections or the planes of the sections; the context is always sufficient to prevent ambiguity.

<sup>2</sup>The reciprocal form is, it should be noted,

$$x^2 + y^2 + z^2 - 2yz - 2zx - 2xy - 2w^2 = 0.$$

The separators are sections each of them touching two resultors at their point of contact (or what is the same thing, passing through the line of intersection of two resultors), and all of them having a line in common. Their equations may be taken to be

$$cy - bz = 0, \quad az - cx = 0, \quad bx - ay = 0.$$

*[The memoir continues from § 3 onward at p. 60 et seq., outside the present PDF pp. 51-75 slice.]*

usepackage[a4paper,margin=0.65in]geometry

# CAYLEY, COLLECTED MATHEMATICAL PAPERS, Vol. II.

Scan-verified retype: source PDF pp. 76–104 (book pp. 60–88).

Paper 114 (concluded), Paper 115 (opening).

## 114 (continued).

### ANALYTICAL RESEARCHES CONNECTED WITH STEINER'S EXTENSION OF MALFATTI'S PROBLEM.

[Continuation of § 3, beginning at book p. 60.]

[The opening of § 3, with the surface  $2yz + 2zx + 2xy + w^2 = 0$  and the three determinators  $-\alpha x + \frac{1}{2\alpha}y + \frac{1}{2\alpha}z + w = 0$ ,  $\frac{1}{2\beta}x - \beta y + \frac{1}{2\beta}z + w = 0$ ,  $\frac{1}{2\gamma}x + \frac{1}{2\gamma}y - \gamma z + w = 0$ , was given in the preceding slice. We resume at p. 60, where the tactors are to be found.]

The values of  $a, b, c$  remaining to be determined. Now before fixing the values of these quantities, we may find three sections each of them touching a determinator at a point of intersection with the section which corresponds to it of the sections

$$cy - bz = 0, \quad az - cx = 0, \quad bx - ay = 0,$$

and touching the other two of the last-mentioned sections; and when  $a, b, c$  have their proper values the sections so found are the *tactors*. For, let  $\lambda x + \mu y + \nu z + \rho w = 0$  be the equation of a section touching the determinator  $-\alpha x + \frac{1}{2\alpha}y + \frac{1}{2\alpha}z + w = 0$ , and the two sections  $bx - ay = 0, az - cx = 0$ ; and suppose

$$\Delta^2 = \lambda^2 + \mu^2 + \nu^2 - 2\mu\nu - 2\nu\lambda - 2\lambda\mu - 2\rho^2;$$

the conditions of contact with the sections  $bx - ay = 0, az - cx = 0$  are found to be

$$\begin{aligned} (b+a)\Delta &= (b+a)\lambda - (b+a)\mu - (b-a)\nu, \\ (c+a)\Delta &= (c+a)\lambda - (c-a)\mu - (c+a)\nu, \end{aligned}$$

values, however, which suppose a correspondence in the signs of the radicals. Thence  $(b+a)\mu = (c+a)\nu$ ; or since the ratios only of the quantities  $\lambda, \mu, \nu, \rho$  are material,  $\mu = c+a, \nu = b+a$ , and therefore

$$\Delta^2 = \lambda^2 - 2(2a+b+c)\lambda + (b-c)^2 - 2\rho^2 = (\lambda - b - c)^2,$$

or

$$\rho^2 = -2^{\frac{1}{2}}(a\lambda + bc).$$

Hence the equation to a section touching  $bx - ay = 0, az - cx = 0$  is

$$\lambda x + (c+a)y + (b+a)z + \sqrt{-2(a\lambda + bc)} w = 0,$$

and to express that this touches the determinator in question, we have

$$\pm\alpha(\lambda - b - c) = -\alpha\lambda - \frac{1}{\alpha}(2a+b+c) + 2\sqrt{-2(a\lambda + bc)},$$

and selecting the upper sign

and from these equations  $(cy - bz) = 0$ , or the point of contact lies in the section  $cy - bz = 0$ . It follows that the equations of the *tactors* are

$$\begin{aligned} -2a(a\alpha - \sqrt{-2bc})x + (c+a)y + (b+a)z + (2a\alpha - \sqrt{-2bc})w &= 0, \\ (c+b)x - 2\beta(b\beta - \sqrt{-2ca})y + (a+b)z + (2b\beta - \sqrt{-2ca})w &= 0, \\ (b+c)x + (a+c)y - 2\gamma(c\gamma - \sqrt{-2ab})z + (2c\gamma - \sqrt{-2ab})w &= 0, \end{aligned}$$

where  $a, b, c$  still remain to be determined.

Now the separators pass through the point of intersection of the determinators; the equations of these give for the point in question,

$$\begin{aligned} x : y : z : w &= (2\beta\gamma + 1)(-\alpha + \beta + \gamma + 2\alpha\beta\gamma) \\ &: (2\gamma\alpha + 1)(\alpha - \beta + \gamma + 2\alpha\beta\gamma) \\ &: (2\alpha\beta + 1)(\alpha + \beta - \gamma + 2\alpha\beta\gamma) \\ &: -1 + \alpha^2 + \beta^2 + \gamma^2 + 4\alpha^2\beta^2\gamma^2; \end{aligned}$$

and the values of  $a, b, c$  are therefore

$$\begin{aligned} a : b : c &= (2\beta\gamma + 1)(-\alpha + \beta + \gamma + 2\alpha\beta\gamma) \\ &: (2\gamma\alpha + 1)(\alpha - \beta + \gamma + 2\alpha\beta\gamma) \\ &: (2\alpha\beta + 1)(\alpha + \beta - \gamma + 2\alpha\beta\gamma), \end{aligned}$$

which are to be substituted for  $a, b, c$  in the equations of the separators and tactors respectively.

Now proceeding to find the *bisectors*, let  $\lambda x + \mu y + \nu z + \rho w = 0$  be the equation of a section touching the determinators, and suppose as before  $\Delta^2 = \lambda^2 + \mu^2 + \nu^2 - 2\mu\nu - 2\nu\lambda - 2\lambda\mu - 2\rho^2$ ; the conditions of contact are

$$\begin{aligned} \pm\beta\Delta &= \beta\lambda - (\beta + \frac{1}{2\beta})\mu + \beta\nu - 2\rho, \\ \mp\gamma\Delta &= \gamma\lambda + \gamma\mu - (\gamma + \frac{1}{2\gamma})\nu - 2\rho, \end{aligned}$$

[Book p. 62.] where it is necessary, for the present purpose, to give opposite signs to the radicals. For if the radicals had the same sign, it would follow that

$$\frac{1}{\beta} \left[ \beta\lambda - (\beta + \frac{1}{2\beta})\mu + \beta\nu - 2\rho \right] - \frac{1}{\gamma} \left[ \gamma\lambda + \gamma\mu - (\gamma + \frac{1}{2\gamma})\nu - 2\rho \right] = 0,$$

hence the section  $\lambda x + \mu y + \nu z + \rho w = 0$  would pass through the point

$$x : y : z : w = 0 : \frac{1}{2\beta^2} : -\frac{1}{2\gamma^2} : -\frac{1}{\beta} + \frac{1}{\gamma},$$



or the section would be a tangent section of the two determinators of the same class with the resultor  $x = 0$ , which ought not to be the case. The proper formula is

$$\frac{1}{\beta} \left[ \beta\lambda - (\beta + \frac{1}{2\beta})\mu + \beta\nu - 2\rho \right] + \frac{1}{\gamma} \left[ \gamma\lambda + \gamma\mu - (\gamma + \frac{1}{2\gamma})\nu - 2\rho \right] = 0,$$

and this equation being satisfied, the section

$$\lambda x + \mu y + \nu z + \rho w = 0$$

passes through a point

$$x : y : z : w = -\frac{1}{\beta^2} - \frac{1}{\gamma^2} : \frac{1}{\gamma^2} : \frac{1}{\beta^2} : -\frac{2}{\beta} - \frac{2}{\gamma}.$$

The bisector passes through this point and the line of intersection of the determinators; its equation is

$$\frac{1}{\beta} \left( \frac{1}{2\beta}x - \beta y + \frac{1}{2\beta}z + w \right) - \frac{1}{\gamma} \left( \frac{1}{2\gamma}x + \frac{1}{2\gamma}y - \gamma z + w \right) = 0,$$

or, reducing and completing the system, the equations of the bisectors are

$$\begin{aligned} \left( \frac{1}{2\beta^2} - \frac{1}{2\gamma^2} \right) x + \left( \gamma + \frac{1}{2\beta^2} \right) y + \left( \frac{1}{2\gamma^2} - \beta \right) z + \left( \frac{1}{\beta} - \frac{1}{\gamma} \right) w &= 0, \\ \left( \frac{1}{2\gamma^2} - \alpha \right) x + \left( \frac{1}{2\gamma^2} - \frac{1}{2\alpha^2} \right) y + \left( \frac{1}{2\alpha^2} - \gamma \right) z + \left( \frac{1}{\gamma} - \frac{1}{\alpha} \right) w &= 0, \\ \left( \alpha - \frac{1}{2\beta^2} \right) x + \left( \frac{1}{2\alpha^2} - \beta \right) y + \left( \frac{1}{2\alpha^2} - \frac{1}{2\beta^2} \right) z + \left( \frac{1}{\alpha} - \frac{1}{\beta} \right) w &= 0. \end{aligned}$$

In order to verify the geometrical construction, it only remains to show that each bisector touches two tactors. Consider the bisector and tactor

$$\begin{aligned} -\left( 1 + \frac{c}{2\beta^2} \right) x + \left( 1 + \frac{a}{2\gamma^2} \right) y + \left( \frac{a}{2\beta^2} - \frac{b}{2\gamma^2} \right) z - \frac{1}{a} \left( \frac{1}{\gamma} - \frac{1}{\beta} \right) w &= 0, \\ -2a(a\alpha - \sqrt{-2bc})x + (c+a)y + (b+a)z + (2a\alpha - \sqrt{-2bc})w &= 0; \end{aligned}$$

and represent these for a moment by

$$\lambda x + \mu y + \nu z + \rho w = 0, \quad \lambda' x + \mu' y + \nu' z + \rho' w = 0;$$

[Book p. 63.] if  $\Delta$  be the same as before, and  $\Delta'$  the like function of  $\lambda', \mu', \nu', \rho'$ ; also if

$$\Phi = \lambda\lambda' + \mu\mu' + \nu\nu' - (\mu\nu' + \mu'\nu) - (\nu\lambda' + \nu'\lambda) - (\lambda\mu' + \lambda'\mu) - 2\rho\rho',$$

then

$$\begin{aligned}\Delta^2 &= \left(2 + \frac{1}{a\beta}\right)^2, \\ \Delta'^2 &= (2a\alpha^2 - 2\alpha\sqrt{-2bc} + b + c)^2, \\ \Phi &= a\alpha^2\left(2 + \frac{1}{a\beta}\right)^2 - 2\alpha\sqrt{-2bc}\left(2 + \frac{1}{a\beta}\right) + c\left(2 + \frac{1}{a\beta}\right),\end{aligned}$$

and the condition of contact  $\Delta\Delta' = \Phi$  (taking the proper sign for the radicals) becomes

$$\left(2 + \frac{1}{a\beta}\right)(2a\alpha^2 - 2\alpha\sqrt{-2bc} + b + c) = a\alpha^2\left(2 + \frac{1}{a\beta}\right) - 2\alpha\sqrt{-2bc}\left(2 + \frac{1}{a\beta}\right) + c\left(2 + \frac{1}{a\beta}\right),$$

or reducing,

$$a\alpha - b\beta + c \frac{\alpha - \beta}{2\alpha\beta + 1} = 0,$$

an equation which is evidently not altered by the interchange of  $a, \alpha$  and  $b, \beta$ . The conditions, in order that each bisector may touch two factors, reduce themselves to the three equations,

$$\begin{aligned}a\alpha - b\beta + c \frac{\alpha - \beta}{2\alpha\beta + 1} &= 0, \\ b\beta - c\gamma + a \frac{\beta - \gamma}{2\beta\gamma + 1} &= 0, \\ c\gamma - a\alpha + b \frac{\gamma - \alpha}{2\gamma\alpha + 1} &= 0,\end{aligned}$$

which are satisfied by the values above found for the quantities  $a, b, c$ . The possibility and truth of the geometrical construction are thus demonstrated.

#### § 4.

Let it be in the first instance proposed to find the equation of a section touching all or any of the sections  $x = 0, y = 0, z = 0$  of the surface of the second order,

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + pw^2 = 0.$$

Any section whatever of this surface may be written in the form

$$\sqrt{(a\lambda^2 + b\mu^2 + c\nu^2 + 2f\mu\nu + 2g\nu\lambda + 2h\lambda\mu - K)} \rho / \sqrt{-p} \cdot w$$

$$+(a\lambda + h\mu + g\nu)x + (h\lambda + b\mu + f\nu)y + (g\lambda + f\mu + c\nu)z = 0,$$

where

$$\nabla^2 = a\lambda^2 + b\mu^2 + c\nu^2 + 2f\mu\nu + 2g\nu\lambda + 2h\lambda\mu - K,$$

[Book p. 64.] and  $\lambda, \mu, \nu$  are indeterminate. And considering any other section represented by a like equation,

$$\sqrt{(a\lambda'^2 + b\mu'^2 + c\nu'^2 + 2f\mu'\nu' + 2g\nu'\lambda' + 2h\lambda'\mu' - K)} \rho' / \sqrt{-p} \cdot w$$

$$+(a\lambda' + h\mu' + g\nu')x + (h\lambda' + b\mu' + f\nu')y + (g\lambda' + f\mu' + c\nu')z = 0,$$

where

$$\nabla'^2 = a\lambda'^2 + b\mu'^2 + c\nu'^2 + 2f\mu'\nu' + 2g\nu'\lambda' + 2h\lambda'\mu' - K,$$

it may be shown, by means of the lemma previously given, that the condition of contact is

$$a\lambda\lambda' + b\mu\mu' + c\nu\nu' + f(\mu\nu' + \mu'\nu) + g(\nu\lambda' + \nu'\lambda) + h(\lambda\mu' + \lambda'\mu) + K \frac{\rho\rho'}{p} = \nabla\nabla'.$$

Suppose that  $\lambda', \mu', \nu'$  satisfy the equations

$$\begin{aligned} a\lambda' + h\mu' + g\nu' &= 0, \\ h\lambda' + b\mu' + f\nu' &= 0, \\ g\lambda' + f\mu' + c\nu' &= 0, \end{aligned}$$

so that the last-mentioned section becomes  $x = 0$ ; and observing that the first of the above equations may be transformed into

$$a\lambda'^2 + b\mu'^2 + c\nu'^2 = \frac{K}{a},$$

it is easy to obtain  $\lambda' = \sqrt{\mathfrak{A}}, \mu' = \frac{\mathfrak{H}}{\sqrt{\mathfrak{A}}}, \nu' = \frac{\mathfrak{C}}{\sqrt{\mathfrak{A}}}$ . The condition of contact thus becomes

$$\frac{K}{\sqrt{\mathfrak{A}}} \rho^2 + K = 0;$$

and taking the under sign,  $\lambda = \sqrt{\mathfrak{A}}$ , so that if in the above-written equation we establish all or any of the equations  $\lambda = \sqrt{\mathfrak{A}}, \mu = \sqrt{\mathfrak{B}}, \nu = \sqrt{\mathfrak{C}}$ , we have the equation of a section touching all or the corresponding sections of the sections

$$x = 0, \quad y = 0, \quad z = 0.$$

In particular we have for a solution of the problem of tactions, the following equation of the section touching  $x = 0, y = 0, z = 0$ :

$$(a\sqrt{\mathfrak{A}} + h\sqrt{\mathfrak{B}} + g\sqrt{\mathfrak{C}})x + (h\sqrt{\mathfrak{A}} + b\sqrt{\mathfrak{B}} + f\sqrt{\mathfrak{C}})y + (g\sqrt{\mathfrak{A}} + f\sqrt{\mathfrak{B}} + c\sqrt{\mathfrak{C}})z \\ + \frac{\rho}{\sqrt{-p}} \sqrt{2(\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F})(\sqrt{\mathfrak{C}\mathfrak{A}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})} \cdot w = 0.$$

Anticipating the use of a notation the value of which will subsequently appear, or putting

[Book p. 65.]

$$K^2 = -f^4 - g^4 - h^4 + 2g^2h^2 + 2h^2f^2 + 2f^2g^2 - \frac{4f^2g^2h^2}{J^2},$$

the equation of the section in question is

$$\frac{f^2}{\sqrt{\mathfrak{A}}}(-f^2 + g^2 + h^2)x + \frac{g^2}{\sqrt{\mathfrak{B}}}(f^2 - g^2 + h^2)y + \frac{h^2}{\sqrt{\mathfrak{C}}}(f^2 + g^2 - h^2)z + \frac{4f^2g^2h^2}{J^2} \frac{\rho\sqrt{-pK}}{\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}}} w = 0.$$

I proceed to investigate a transformation of the equation for the section with an indeterminate parameter  $\lambda$ , which touches the two sections  $y = 0, z = 0$ . We have

$$a\nabla^2 = (a\lambda + h\mu + g\nu)^2 + (b\mu^2 + c\nu^2 - 2f\mu\nu) - \mathfrak{B} - K,$$

or, putting for  $\mu^2$  and  $\nu^2$  their values  $\mathfrak{B}, \mathfrak{C}$  in the second term,

$$a\nabla^2 = (a\lambda + h\mu + g\nu)^2 + (\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F})^2;$$

and introducing instead of  $\lambda$  an indeterminate quantity  $X$ , such that

$$a\lambda + h\mu + g\nu = (\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F})X,$$

we have, also introducing throughout  $X$  instead of  $\lambda$ , and completing the substitution of  $\sqrt{\mathfrak{B}}, \sqrt{\mathfrak{C}}$  for  $\mu, \nu$ , the equation of the section touching  $y = 0, z = 0$ , becomes

$$(ax + hy + gz)X + y\sqrt{\mathfrak{C}} + z\sqrt{\mathfrak{B}} + w\sqrt{-ap}\sqrt{1 + X^2} = 0.$$

It may be remarked here in passing, that this is a very convenient form for the demonstration of the theorem: "If two sections of a surface of the second

order touch each other, and are also tangent sections (of the same class) to two fixed sections, then considering the planes through the axis of the fixed sections and the poles of the tangent sections, and also the tangent planes through this axis, the anharmonic ratio of the four planes is independent of the position of the moveable tangent sections;" where by the axis of the fixed sections is to be understood the line joining their poles.

The sections touching  $z = 0$ ,  $x = 0$  and  $x = 0$ ,  $y = 0$  are of course

$$\begin{aligned} z\sqrt{\mathfrak{A}} + x\sqrt{\mathfrak{C}} + (hx + by + fz)Y + w\sqrt{-bp}\sqrt{1 + Y^2} &= 0, \\ x\sqrt{\mathfrak{B}} + y\sqrt{\mathfrak{A}} + (gx + fy + cz)Z + w\sqrt{-cp}\sqrt{1 + Z^2} &= 0. \end{aligned}$$

The conditions of contact of the sections represented by the above-written equations would be perhaps most simply obtained directly from the lemma, but it is proper to deduce it from the formula for contact used in the present memoir. If for shortness

$$\Phi(\pm) = a\lambda'\lambda'' + b\mu'\mu'' + c\nu'\nu'' + f(\mu'\nu'' + \mu''\nu') + g(\nu'\lambda'' + \nu''\lambda') + h(\lambda'\mu'' + \lambda''\mu') \pm K \frac{\rho'\rho''}{p},$$

[Book p. 66.] where the symbol  $\Phi(\pm)$  is used in order to mark the essentially different character of the results corresponding to the different values of the ambiguous sign, then

$$\begin{aligned} bc\Phi(-) &= f(h\lambda' + b\mu' + f\nu')(g\lambda'' + f\mu'' + c\nu'') \\ &\quad + (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(g\lambda'' + f\mu'' + c\nu'') \\ &\quad + (\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})(h\lambda'' + b\mu'' + f\nu'') \\ &\quad + f(-a\sqrt{\mathfrak{B}\mathfrak{C}} + h\sqrt{\mathfrak{C}\mathfrak{A}} + g\sqrt{\mathfrak{A}\mathfrak{B}} - aF - (\mathfrak{G}\mathfrak{H} - aF)) \\ &= \dots \\ &= (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{fYZ + \sqrt{\mathfrak{A}}(Y + Z) + f\}. \end{aligned}$$

That is,  $bc\Phi(-) = (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{fYZ + \sqrt{\mathfrak{A}}(Y + Z) + f\}$ .

What, however, is really required<sup>1</sup>, is the value of  $\Phi(+)$ ; to find this, we have

$$bc\Phi(+) = bc\Phi(-) + 2bcK$$

---

<sup>1</sup>It may be shown without difficulty that the  $(-)$  sign would imply that the sections touching  $z = 0$ ,  $x = 0$ , and  $x = 0$ ,  $y = 0$  were sections touching  $x = 0$  at the same point. By taking the  $(-)$  sign in each equation we should have the solution of the problem "to determine three sections of a surface of the second order, the two sections of each pair touching one of three given sections at the same point," which is not without interest; the solution may be completed without any difficulty.

$$= (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})\{fYZ + \sqrt{\mathfrak{A}}(Y+Z) + f\} + 2bcK - 2f(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}),$$

the second line of which is

$$2(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}) \left\{ \frac{bcK - f(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})}{(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})} \right\} = 2(\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}) \sqrt{\mathfrak{A}} \theta,$$

[Book p. 67.] where

$$\theta = \frac{1}{K} (\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}} + \mathfrak{F}\sqrt{\mathfrak{A}} + \mathfrak{G}\sqrt{\mathfrak{B}} + \mathfrak{H}\sqrt{\mathfrak{C}}),$$

and consequently

$$bc\Phi(+) = (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})[fYZ + \sqrt{\mathfrak{A}}(Y+Z) + f + 2\theta\sqrt{\mathfrak{A}}],$$

a reduction, which on account of its peculiarity, I have thought right to work out in full.

The condition of contact is

$$\Phi(+) = \nabla' \nabla'' = \frac{1}{\sqrt{bc}} (\sqrt{\mathfrak{A}\mathfrak{C}} - \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}) \sqrt{1+Y^2} \sqrt{1+Z^2}.$$

Hence finally, the condition in order that the sections

$$z\sqrt{\mathfrak{A}} + (hx + by + fz)Y + x\sqrt{\mathfrak{C}} + w\sqrt{-bp}\sqrt{1+Y^2} = 0,$$

$$x\sqrt{\mathfrak{B}} + y\sqrt{\mathfrak{A}} + (gx + fy + cz)Z + w\sqrt{-cp}\sqrt{1+Z^2} = 0,$$

(the former of which is a section touching  $z = 0$ ,  $x = 0$ , and the latter a section touching  $x = 0$ ,  $y = 0$ ) may touch, is

$$fYZ + \sqrt{\mathfrak{A}}(Y+Z) + (f + 2\theta\sqrt{\mathfrak{A}}) - \sqrt{bc}\sqrt{1+Y^2}\sqrt{1+Z^2} = 0.$$

The preceding researches show that the solution of Steiner's extension of Malfatti's problem depends on a system of equations, such as the system mentioned at the commencement of the following section.

§ 5.

Consider the system of equations

$$\begin{aligned}\alpha + \beta(Y + Z) + \gamma YZ + \delta\sqrt{1 + Y^2}\sqrt{1 + Z^2} &= 0, \\ \alpha' + \beta'(Z + X) + \gamma' ZX + \delta'\sqrt{1 + Z^2}\sqrt{1 + X^2} &= 0, \\ \alpha'' + \beta''(X + Y) + \gamma'' XY + \delta''\sqrt{1 + X^2}\sqrt{1 + Y^2} &= 0;\end{aligned}$$

these equations may, it will be seen, be solved by quadratics only, when the coefficients satisfy the relations

$$\frac{\gamma - \alpha}{\gamma^2 - \alpha^2} = \frac{\gamma' - \alpha'}{\gamma'^2 - \alpha'^2} = \frac{\gamma'' - \alpha''}{\gamma''^2 - \alpha''^2}.$$

[Book p. 68.] It may be remarked that these equations are satisfied by

$$\beta = 0, \quad \beta' = 0, \quad \beta'' = 0, \quad \gamma = \delta, \quad \gamma' = \delta', \quad \gamma'' = \delta'',$$

or if we write

$$\frac{\alpha}{\gamma} = -l, \quad \frac{\alpha'}{\gamma'} = -m, \quad \frac{\alpha''}{\gamma''} = -n,$$

the equations become, by a simple reduction,

$$\begin{aligned}Y^2 + Z^2 + 2l YZ &= l^2 - 1, \\ Z^2 + X^2 + 2m ZX &= m^2 - 1, \\ X^2 + Y^2 + 2n XY &= n^2 - 1,\end{aligned}$$

which are equivalent to the equations discussed in my paper “On a system of Equations connected with Malfatti’s Problem and on another Algebraical System,” *Cambridge and Dublin Mathematical Journal*, t. iv. [1849] pp. 270–275, [79]; the solution might have been effected by the direct method, which I shall here adopt, of eliminating any one of the variables between the two equations into which it enters, and combining the result with the third equation.

Writing the second and third equations under the form

$$\begin{aligned}A' + B'X + C'\sqrt{1 + X^2} &= 0, \\ A'' + B''X + C''\sqrt{1 + X^2} &= 0,\end{aligned}$$

the result of the elimination may be presented in the form

$$A'A'' + B'B'' - C'C'' = \sqrt{A'^2 + B'^2 - C'^2} \cdot \sqrt{A''^2 + B''^2 - C''^2},$$

which is most easily obtained by writing  $X = \tan \phi$  and operating with the symbol  $\cos^{-1}$ ; but if the rationalized equations be represented by

$$\lambda' + 2\mu'X + \nu'X^2 = 0, \quad \lambda'' + 2\mu''X + \nu''X^2 = 0,$$

the form

$$(\lambda'\nu'' - \lambda''\nu')^2 = 4(\lambda'\nu' - \mu'^2)(\lambda''\nu'' - \mu''^2) - (\lambda'\nu'' - 2\mu'\mu'')^2 \dots$$

leads easily to the result in question. The values which enter are

$$\begin{aligned} A' &= \alpha' + \beta'Z, & A'' &= \alpha'' + \beta''Y, \\ B' &= \beta' + \gamma'Z, & B'' &= \beta'' + \gamma''Y, \\ C' &= \delta'\sqrt{1+Z^2}, & C'' &= \delta''\sqrt{1+Y^2}; \end{aligned}$$

whence, in the first place, by the equation connecting  $Y, Z$ ,

$$C'C'' = -\frac{\delta'\delta''}{\delta} \{\alpha + \beta(Y+Z) + \gamma YZ\}.$$

[Book p. 69.] It is obviously convenient that  $A'A'' + B'B''$  should be symmetrical with respect to  $Y$  and  $Z$ , and this will be the case if

$$\alpha'\beta'' + \beta'\gamma'' = \alpha''\beta' + \beta''\gamma',$$

that is, if

$$\beta'(\gamma'' - \alpha'') = \beta''(\gamma' - \alpha'),$$

or, assuming that the equations are symmetrically related to the system, we have the first set of relations between the coefficients, relations which are satisfied by

$$\alpha = \gamma + 2\phi\beta, \quad \alpha' = \gamma' + 2\phi\beta', \quad \alpha'' = \gamma'' + 2\phi\beta'',$$

and the values of  $\alpha, \alpha', \alpha''$  will be considered henceforth as given by these conditions.

We have

$$A'A'' + B'B'' - C'C'' = \alpha'\alpha'' + \beta'\beta'' + (\gamma'\beta'' + \gamma''\beta' + 2\phi\beta'\beta'')(Y+Z) + (\beta'\beta'' + \gamma'\gamma'')YZ + \frac{\delta'\delta''}{\delta} \{\alpha + \beta(Y+Z)\}$$



The quantities  $A'^2 + B'^2 - C'^2$ ,  $A''^2 + B''^2 - C''^2$  are quadratic functions of  $Z$  and  $Y$  respectively, and with proper relations between the coefficients, we may assume

$$(A'^2 + B'^2 - C'^2)(A''^2 + B''^2 - C''^2) = U^2 + k[(\alpha + \beta(Y + Z) + \gamma YZ)^2 - \delta^2(1 + Y^2)(1 + Z^2)],$$

in which  $U$  is a linear function of  $Y + Z$  and  $YZ$ , and  $k$  and  $\delta^2$  are constants. The first side must, in the first place, be symmetrical with respect to  $Y$  and  $Z$ , or

$$\alpha'^2 + \beta'^2 - \delta'^2, \quad (\alpha' + \gamma')\beta', \quad \beta'^2 + \gamma'^2 - \delta'^2$$

must be proportional to

$$\alpha''^2 + \beta''^2 - \delta''^2, \quad (\alpha'' + \gamma'')\beta'', \quad \beta''^2 + \gamma''^2 - \delta''^2.$$

But since  $(\alpha' + \gamma')\beta'$ ,  $(\alpha'' + \gamma'')\beta''$  are proportional to  $\gamma'^2 - \alpha'^2$ ,  $\gamma''^2 - \alpha''^2$ , it is only necessary that  $\beta'^2 + \gamma'^2 - \delta'^2$  should be proportional to  $\beta''^2 + \gamma''^2 - \delta''^2$ , that is, that

$$\frac{\gamma'^2 - \alpha'^2}{\beta'^2 + \gamma'^2 - \delta'^2} = \frac{\gamma''^2 - \alpha''^2}{\beta''^2 + \gamma''^2 - \delta''^2},$$

or since the equations are supposed symmetrically related to the system, we must have the second set of relations between the coefficients. Suppose

$$\frac{\gamma - \alpha}{\beta^2 + \gamma^2 - \delta^2} = \frac{\gamma' - \alpha'}{\beta'^2 + \gamma'^2 - \delta'^2} = \frac{\gamma'' - \alpha''}{\beta''^2 + \gamma''^2 - \delta''^2} = -\frac{s}{\phi'},$$

then since

$$\gamma^2 - \alpha^2 = -4(\gamma + \phi\beta)\phi\beta, \quad \&c.,$$

[Book p. 70.] we have

$$\begin{aligned} \delta^2 &= \beta^2 + \gamma^2 - 4s(\gamma + \phi\beta)\beta, \\ \delta'^2 &= \beta'^2 + \gamma'^2 - 4s(\gamma' + \phi\beta')\beta', \\ \delta''^2 &= \beta''^2 + \gamma''^2 - 4s(\gamma'' + \phi\beta'')\beta'', \end{aligned}$$

and  $\delta$ ,  $\delta'$ ,  $\delta''$  will be supposed henceforth to satisfy these equations.

We have next

$$\begin{aligned} A'^2 + B'^2 - C'^2 &= 4(\gamma' + \phi\beta')\beta's(Z + \frac{\beta'}{\phi} + sZ^2), \\ A''^2 + B''^2 - C''^2 &= 4(\gamma'' + \phi\beta'')\beta''s(Y + \frac{\beta''}{\phi} + sY^2), \end{aligned}$$

which may be simplified by writing

$$s = \frac{\mu - \phi}{1 + \mu^2}, \quad \frac{1 + \mu^2}{\mu - \phi} = \frac{1 + \nu^2}{\mu - \phi},$$

where  $\mu, \nu$  are to be considered as given functions of  $s$  and  $\phi$ . These values give

$$\begin{aligned} A'^2 + B'^2 - C'^2 &= 4(\gamma' + \phi\beta')\beta' s(Z + \mu)(Z + \nu), \\ A''^2 + B''^2 - C''^2 &= 4(\gamma'' + \phi\beta'')\beta'' s(Y + \mu)(Y + \nu). \end{aligned}$$

Hence, putting for simplicity

$$b^2 = 4(\gamma' + \phi\beta')(\gamma'' + \phi\beta'')\beta'\beta'',$$

we have

$$b^2 s^2 (Z + \mu)(Z + \nu)(Y + \mu)(Y + \nu) = U^2 + k[(\alpha + \beta(Y + Z) + \gamma YZ)^2 - \delta^2(1 + Y^2)(1 + Z^2)];$$

and the two sides have next to be expressed in terms of  $Y + Z$  and  $YZ$ .

If for symmetry we write

$$\xi = 1, \quad \eta = Y + Z, \quad \zeta = YZ,$$

then

$$b^2 s^2 (\mu^2 \xi + \mu \eta + \zeta)(\nu^2 \xi + \nu \eta + \zeta) = U^2 + k[(\alpha \xi + \beta \eta + \gamma \zeta)^2 - \delta^2\{(\zeta - \xi)^2 + \eta^2\}],$$

and  $U$  is now to be considered as a linear function of  $\xi, \eta, \zeta$ .

The condition that the first side of the equation may divide into factors, gives an equation for determining  $k$ ; and since the condition is satisfied for  $k = \infty$ , the equation will be linear, and it is easily seen that the value is  $k = \frac{b^2 s^2}{\delta^2} (\mu - \nu)^2$ . In fact

$$4(\mu^2 \xi + \mu \eta + \zeta)(\nu^2 \xi + \nu \eta + \zeta) = (\mu^2 \xi + \mu \eta + \nu^2 \xi + \nu \eta + 2\zeta)^2 + (\mu - \nu)^2 [(\zeta - \xi)^2 + \eta^2],$$

hence

[Book p. 71.] and we may assume

$$U = \frac{\mu - \nu}{\delta} \left\{ \left( \Lambda + \frac{1}{\Lambda} \right) (\alpha \xi + \beta \eta + \gamma \zeta) - \delta \left( \Lambda - \frac{1}{\Lambda} \right) (\xi + \zeta) \right\},$$

subject to its being shown that the value of  $\Lambda$  given by the comparison of coefficients is constant. The comparison of coefficients gives

$$2\mu + 2\nu = \frac{\mu - \nu}{\delta} \left\{ \left( \Lambda + \frac{1}{\Lambda} \right) \alpha - \left( \Lambda - \frac{1}{\Lambda} \right) \delta \right\},$$

$$4\xi = \frac{\mu - \nu}{\delta} \left\{ \left( \Lambda + \frac{1}{\Lambda} \right) \gamma - \left( \Lambda - \frac{1}{\Lambda} \right) \delta \right\},$$

$$4(1 - \mu^2) = \frac{\mu - \nu}{\delta} \left( \Lambda + \frac{1}{\Lambda} \right) (\gamma - \alpha),$$

the first and third of which give

$$4(1 - \mu^2) = \frac{\mu - \nu}{\delta} \left( \Lambda + \frac{1}{\Lambda} \right) (\gamma - \alpha),$$

which will be identical with the second, if

$$\frac{2(1 - \mu\nu)}{\mu + \nu} = \frac{\beta}{\gamma - \alpha} = -2\phi,$$

which follows at once from the equation  $\nu = \frac{1+\mu\phi}{\mu-\phi}$ .

Forming next the two equations

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these will be equivalent to a single equation, if

$$\frac{1}{4}(\mu + \nu)^2 \delta^2 = (\mu + \nu)^2 (\beta^2 + \gamma^2) - 4(\mu + \nu)\beta\gamma - 4(\mu^2 - 1)\beta^2,$$

[Book p. 72.] or, finally, if

$$\delta^2 + \delta^2 = \beta^2 + \gamma^2 - 4s\beta \left[ \gamma + \frac{\mu\nu-1}{\mu+\nu} \beta \right] = \beta^2 + \gamma^2 - 4s(\gamma + \phi\beta)\beta,$$

which is in fact the case.

Writing the equations for

$$\Lambda + \frac{1}{\Lambda} = \frac{2\delta}{(\mu-\nu)\beta s} (\gamma - 2s\beta),$$

$$\Lambda - \frac{1}{\Lambda} = \frac{2}{(\mu-\nu)\beta s\delta} (\gamma - 2s\beta - 2\phi\beta),$$

in the form

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and substituting in

$$U = \frac{\mu - \nu}{\delta} \left\{ \left( \Lambda + \frac{1}{\Lambda} \right) (\alpha\xi + \beta\eta + \gamma\zeta) - \delta \left( \Lambda - \frac{1}{\Lambda} \right) (\xi + \zeta) \right\},$$

we have

$$U = \frac{2}{\beta s} \left\{ (-\beta + 2s\gamma + 2\phi\gamma) \xi + (\gamma - 2s\beta) \eta + (-\beta + 2s\gamma + 4\phi\gamma) \zeta \right\} \frac{1}{\delta},$$

and consequently, multiplying by

$$b = 2\sqrt{(\gamma' + \phi\beta')(\gamma'' + \phi\beta'')\beta'\beta''},$$

we have

$$\frac{1}{\delta} \sqrt{(\gamma' + \phi\beta')(\gamma'' + \phi\beta'')\beta'\beta''} \left\{ (-\beta + 2s\gamma + 2\phi\gamma) \xi + (\gamma - 2s\beta) \eta + (-\beta + 2s\gamma + 4\phi\gamma) \zeta \right\},$$

or collecting the different terms which enter into the equation

$$A'A'' + B'B'' - C'C'' = \sqrt{A'^2 + B'^2 - C'^2} \cdot \sqrt{A''^2 + B''^2 - C''^2},$$

the result is

$$\begin{aligned} & \left\{ (\alpha'\alpha'' + \beta'\beta'') \xi + (\gamma'\beta'' + \gamma''\beta' + 2\phi\beta'\beta'') \eta + (\beta'\beta'' + \gamma'\gamma'') \zeta \right\} + \frac{\delta'\delta''}{\delta} (\alpha\xi + \beta\eta + \gamma\zeta) \\ & - \frac{1}{\delta} \sqrt{(\gamma' + 2\phi\beta')(\gamma'' + 2\phi\beta'')\beta'\beta''} \left\{ (-\beta + 2s\gamma + 2\phi\gamma) \xi + (\gamma - 2s\beta) \eta + (-\beta + 2s\gamma + 4\phi\gamma) \zeta \right\} = 0, \end{aligned}$$

which, combined with the first equation written under the form

$$\alpha\xi + \beta\eta + \gamma\zeta + \delta\sqrt{(\zeta - \xi)^2 + \eta^2} = 0,$$

determines the ratios of  $\xi$ ,  $\eta$ ,  $\zeta$ , that is, the values of  $Y + Z$  and  $YZ$ .

## § 6.

The system of equations

$$\begin{aligned} (f + 2\theta\sqrt{\mathfrak{A}}) + \sqrt{\mathfrak{A}}(Y + Z) + fYZ - \sqrt{bc}\sqrt{1 + Y^2}\sqrt{1 + Z^2} &= 0, \\ (g + 2\theta\sqrt{\mathfrak{B}}) + \sqrt{\mathfrak{B}}(Z + X) + gZX - \sqrt{ca}\sqrt{1 + Z^2}\sqrt{1 + X^2} &= 0, \\ (h + 2\theta\sqrt{\mathfrak{C}}) + \sqrt{\mathfrak{C}}(X + Y) + hXY - \sqrt{ab}\sqrt{1 + X^2}\sqrt{1 + Y^2} &= 0, \end{aligned}$$

where

$$\theta = \frac{1}{K} (\sqrt{\mathfrak{A}\mathfrak{B}\mathfrak{C}} + \mathfrak{F}\sqrt{\mathfrak{A}} + \mathfrak{G}\sqrt{\mathfrak{B}} + \mathfrak{H}\sqrt{\mathfrak{C}}),$$

on which depends the solution of Steiner's extension of Malfatti's problem, is at once seen to belong to the class of equations treated of in the preceding section, and we have  $\phi = 0$ ,  $s = 0$ . The equations at the conclusion of the preceding section become

$$\begin{aligned} & \{\sqrt{\mathfrak{B}\mathfrak{C}} + gh + 2\theta(g\sqrt{\mathfrak{C}} + h\sqrt{\mathfrak{B}}) + 4\theta^2\sqrt{\mathfrak{B}\mathfrak{C}}\} \xi + \{g\sqrt{\mathfrak{C}} + h\sqrt{\mathfrak{B}} + 2\theta\sqrt{\mathfrak{B}\mathfrak{C}}\} \eta + \{\sqrt{\mathfrak{B}\mathfrak{C}} + gh\} \zeta \\ & - a[(f + 2\theta\sqrt{\mathfrak{A}})\xi + \sqrt{\mathfrak{A}}\eta + f\zeta] - \frac{1}{\sqrt{bc}} \sqrt{(g + \theta\sqrt{\mathfrak{B}})(h + \theta\sqrt{\mathfrak{C}})\sqrt{\mathfrak{B}\mathfrak{C}}} [(\sqrt{\mathfrak{A}} - 2\theta f)\xi - f\eta + \sqrt{\mathfrak{A}}\zeta] = \\ & \{(f + 2\theta\sqrt{\mathfrak{A}})\xi + \sqrt{\mathfrak{A}}\eta + f\zeta\}^2 - bc\{(\xi - \zeta)^2 + \eta^2\} = 0, \end{aligned}$$

which may also be written

$$\begin{aligned} & (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\xi + \zeta) + (-a\sqrt{\mathfrak{A}} + g\sqrt{\mathfrak{B}} + h\sqrt{\mathfrak{C}} + 2\theta\sqrt{\mathfrak{B}\mathfrak{C}})(\eta + 2\theta\xi) \\ & - \frac{1}{\sqrt{bc}} \sqrt{(g + \theta\sqrt{\mathfrak{B}})(h + \theta\sqrt{\mathfrak{C}})\sqrt{\mathfrak{B}\mathfrak{C}}} [(\sqrt{\mathfrak{A}} - 2\theta f)\xi - f\eta + \sqrt{\mathfrak{A}}\zeta] = 0, \\ & \sqrt{\mathfrak{A}}\{f(\xi + \zeta) + (\eta + 2\theta\xi)\}^2 - bc\{(\xi - \zeta)^2 + \eta^2\} = 0. \end{aligned}$$

Hence, observing that

$$\begin{aligned} g + \theta\sqrt{\mathfrak{B}} &= \frac{1}{K} (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H}), \\ h + \theta\sqrt{\mathfrak{C}} &= \frac{1}{K} (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\sqrt{\mathfrak{A}\mathfrak{C}} + \mathfrak{G}), \\ -a\sqrt{\mathfrak{A}} + h\sqrt{\mathfrak{B}} + g\sqrt{\mathfrak{C}} + 2\theta\sqrt{\mathfrak{B}\mathfrak{C}} &= (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F}), \end{aligned}$$

and putting for a moment

$$4\lambda = \sqrt{(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H})(\sqrt{\mathfrak{A}\mathfrak{C}} + \mathfrak{G})\sqrt{\mathfrak{B}\mathfrak{C}}},$$

and therefore

$$\sqrt{(g + \theta\sqrt{\mathfrak{B}})(h + \theta\sqrt{\mathfrak{C}})\sqrt{\mathfrak{B}\mathfrak{C}}} = (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})\lambda,$$

[Book p. 74.] the first equation divides by  $(\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})$ , and the result is

$$(\xi + \zeta) + \theta(\eta + 2\theta\xi) - \frac{2\lambda}{\sqrt{bc}} [\sqrt{\mathfrak{A}}(\xi + \zeta) - f(\eta + 2\theta\xi) + \sqrt{\mathfrak{A}}(\eta + 2\theta\xi) - (1 + \theta^2)\xi] = 0.$$

Also, by an easy transformation, the second equation becomes

$$\sqrt{\mathfrak{A}}[(\xi + \zeta) - \{\sqrt{\mathfrak{A}}(\xi + \zeta) - f(\eta + 2\theta\xi)\}]^2 - bc\{(\xi + \zeta) - (1 + \theta^2)\xi\}^2 = 0,$$

or putting

$$\xi + \zeta + \theta(\eta + 2\theta\xi) = \Theta, \quad \frac{1}{\sqrt{bc}} [\sqrt{\mathfrak{A}}(\xi + \zeta) - f(\eta + 2\theta\xi)] = \Phi, \quad \zeta = \Psi,$$

the equations become

$$\Theta - 2\lambda\Phi = 0, \quad -\Phi^2 + 4\lambda(\Theta - (1 + \theta^2)\Psi) = 0;$$

hence eliminating  $\Phi$ ,

$$\left(\frac{\Theta}{2\lambda}\right)^2 = \frac{\Theta^2}{(1 + \theta^2)\Psi} \left(1 - \frac{1 + \theta^2}{4\lambda^2}\right),$$

or, observing that

$$1 + \theta^2 = \frac{1}{K^2} (\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F})(\sqrt{\mathfrak{C}\mathfrak{A}} + \mathfrak{G})(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H}),$$

and reducing, we obtain a determination of  $\Psi$ ; also  $\Theta = 2\lambda\Phi$  gives

$$\Phi = \frac{K\Theta}{2\sqrt{(\sqrt{\mathfrak{A}\mathfrak{B}} + \mathfrak{H})(\sqrt{\mathfrak{A}\mathfrak{C}} + \mathfrak{G})\sqrt{\mathfrak{B}\mathfrak{C}}}}.$$

Suppose

$$\sqrt{\mathfrak{B}\mathfrak{C}} + \mathfrak{F} = \mathfrak{a}, \quad \sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F} = \mathfrak{a}_1, \quad \frac{\sqrt{\mathfrak{B}\mathfrak{C}}}{\beta\gamma} = \beta_2\gamma_2,$$

then substituting

$$\xi : \eta : \zeta = \left(\frac{\mathfrak{a}}{\mathfrak{a}_1}\right)^{1/2} \left(1 + \frac{\sqrt{\mathfrak{a}}}{\sqrt{\beta_2\gamma_2}}\right) : \dots$$

[Book p. 75.] that is,

$$\begin{aligned} \xi + \zeta + \theta(\eta + 2\theta\xi) - \frac{\sqrt{2}\sqrt{\mathfrak{a} + \mathfrak{a}_1}}{\sqrt{\beta\gamma_1}} [\sqrt{\mathfrak{A}}(\xi + \zeta) - f(\eta + 2\theta\xi)] &= 0, \\ \xi + \zeta + \theta(\eta + 2\theta\xi) - \frac{4bc(\mathfrak{a} + \mathfrak{a}_1)}{\beta_2\gamma_2} \left(1 - \frac{\sqrt{\mathfrak{a}}}{\sqrt{\mathfrak{a} + \mathfrak{a}_1}}\right) \zeta &= 0; \end{aligned}$$

these may be written

$$L\xi + M\eta + N\zeta = 0, \quad L'\xi + M'\eta + N'\zeta = 0,$$

where

$$\begin{aligned} L &= 1 + 2\theta^2 - \frac{\sqrt{2}\sqrt{\mathfrak{a} + \mathfrak{a}_1}}{\sqrt{\beta\gamma_1}} (\sqrt{\mathfrak{A}} - 2\theta f), \\ M &= \theta + \frac{\sqrt{2}\sqrt{\mathfrak{a} + \mathfrak{a}_1}}{\sqrt{\beta\gamma_1}} f, \\ N &= 1 - \frac{\sqrt{2}\sqrt{\mathfrak{a} + \mathfrak{a}_1}}{\sqrt{\beta\gamma_1}} \sqrt{\mathfrak{A}}, \\ L' &= 1 + 2\theta^2 - \frac{4bc(\mathfrak{a} + \mathfrak{a}_1)}{\beta_2\gamma_2} \left(1 - \frac{\sqrt{\mathfrak{a}}}{\sqrt{\mathfrak{a} + \mathfrak{a}_1}}\right), \quad M' = \theta, \quad N' = 1, \end{aligned}$$

or since  $\xi, \eta, \zeta$  are equal to 1,  $Y + Z, YZ$  respectively,

$$1 : Y + Z : YZ = MN' - M'N : NL' - N'L : LM' - L'M.$$

Also

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whence

$$Y + Z = -\frac{2\sqrt{2}\sqrt{\mathfrak{a} + \mathfrak{a}_1}\sqrt{\beta_2\gamma_2}}{K} \left(1 - \frac{\sqrt{\mathfrak{a}}}{\sqrt{\beta_2\gamma_2}}\right) \left(1 - \frac{\sqrt{\mathfrak{a}_1}}{\sqrt{\mathfrak{a} + \mathfrak{a}_1}}\right) - 2\theta,$$

$$YZ = \frac{2\sqrt{2}\sqrt{\mathfrak{a} + \mathfrak{a}_1}\sqrt{\beta_2\gamma_2}}{K} \left(\theta + \frac{\sqrt{2}\sqrt{\mathfrak{a} + \mathfrak{a}_1}}{\sqrt{\beta_2\gamma_2}} f\right) \left(1 - \frac{\sqrt{\mathfrak{a}}}{\sqrt{\mathfrak{a} + \mathfrak{a}_1}}\right) - 1;$$

and by forming the analogous expressions for  $Z + X$  and  $ZX$ , and  $X + Y$  and  $XY$ , the values of  $X, Y, Z$  may be determined. But the equations in question simplify themselves in a remarkable manner by the notation before alluded to.

[Book p. 76.] Suppose

$$\mathfrak{f} = \sqrt[4]{\mathfrak{A}}\sqrt{\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F}}, \quad \mathfrak{g} = \sqrt[4]{\mathfrak{B}}\sqrt{\sqrt{\mathfrak{C}\mathfrak{A}} - \mathfrak{G}}, \quad \mathfrak{h} = \sqrt[4]{\mathfrak{C}}\sqrt{\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}}, \quad J = \sqrt{2}\sqrt[4]{\mathfrak{A}\mathfrak{B}\mathfrak{C}},$$

these values give

$$\begin{aligned} \frac{K\sqrt{\mathfrak{A}}}{\sqrt{\mathfrak{B}\mathfrak{C}}} \mathfrak{a} &= 2\mathfrak{f}^2 \left(1 - \frac{\mathfrak{f}^2}{J^2}\right), \\ \frac{K\sqrt{\mathfrak{B}\mathfrak{C}}}{\sqrt{\mathfrak{A}}} &= 2\mathfrak{g}\mathfrak{h}\sqrt{1 - \frac{\mathfrak{g}^2}{J^2}}\sqrt{1 - \frac{\mathfrak{h}^2}{J^2}}, \\ \frac{Kf}{\sqrt{\mathfrak{A}}} &= \mathfrak{f}^2 - \mathfrak{g}^2 - \mathfrak{h}^2 + \frac{2\mathfrak{g}^2\mathfrak{h}^2}{J^2}, \\ K\theta &= -\mathfrak{f}^2 - \mathfrak{g}^2 - \mathfrak{h}^2 + 2J^2, \\ K^2 &= -\mathfrak{f}^4 - \mathfrak{g}^4 - \mathfrak{h}^4 + 2\mathfrak{g}^2\mathfrak{h}^2 + 2\mathfrak{h}^2\mathfrak{f}^2 + 2\mathfrak{f}^2\mathfrak{g}^2 - \frac{4\mathfrak{f}^2\mathfrak{g}^2\mathfrak{h}^2}{J^2}. \end{aligned}$$

Applying these results to the preceding formulae and forming for that purpose the equations

$$\begin{aligned} \frac{2\sqrt{2}\sqrt{\mathfrak{a} + \mathfrak{a}_1}\sqrt{\beta_2\gamma_2}}{K} \cdot 4\mathfrak{g}\mathfrak{h}\lambda &= \frac{J^2}{\mathfrak{f}}, \quad \frac{\sqrt{\mathfrak{a}_1}}{\sqrt{\mathfrak{a} + \mathfrak{a}_1}} = \frac{\mathfrak{f}}{J}, \\ \mathfrak{g}\mathfrak{h}K\theta + \frac{J^2}{\sqrt{\mathfrak{A}}}Kf &= (J^2 - \mathfrak{g}\mathfrak{h})(\mathfrak{f}^2 - (\mathfrak{g} - \mathfrak{h})^2) - 2\mathfrak{g}\mathfrak{h}(\mathfrak{g} - \mathfrak{h})^2, \end{aligned}$$

we have

$$K(Y + Z) + 2K\theta = 4(J^2 - gh) \left(1 - \frac{f}{J}\right),$$

$$K^2YZ + K^2 = \{(J^2 - gh)(f^2 - (g - h)^2) - 2gh(g - h)^2\} \left(1 - \frac{f}{J}\right);$$

the former of which, combined with the similar equations for  $Z + X$  and  $X + Y$ , gives for  $X, Y, Z$  the values to be presently stated, and these values will of course verify the second equation and the corresponding equations for  $ZX$  and  $XY$ .

Recapitulating the preceding notation, if  $x = 0, y = 0, z = 0$  are the equations of the given sections,  $w = 0$  the equation of the polar plane of their point of intersection with respect to the surface,

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + pw^2 = 0,$$

the equation of the surface,  $\mathfrak{A}, \mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{F}, \mathfrak{G}, \mathfrak{H}$  as usual, and

[Book p. 77.] putting

$$f = \sqrt[4]{\mathfrak{A}}\sqrt{\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F}}, \quad g = \sqrt[4]{\mathfrak{B}}\sqrt{\sqrt{\mathfrak{C}\mathfrak{A}} - \mathfrak{G}}, \quad h = \sqrt[4]{\mathfrak{C}}\sqrt{\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}}, \quad J = \sqrt{2}\sqrt[4]{\mathfrak{A}\mathfrak{B}\mathfrak{C}},$$

then the equations of the required sections are

$$(ax + hy + gz)X + y\sqrt{\mathfrak{C}} + z\sqrt{\mathfrak{B}} + w\sqrt{-ap}\sqrt{1 + X^2} = 0,$$

$$x\sqrt{\mathfrak{C}} + (hx + by + fz)Y + z\sqrt{\mathfrak{A}} + w\sqrt{-bp}\sqrt{1 + Y^2} = 0,$$

$$x\sqrt{\mathfrak{B}} + y\sqrt{\mathfrak{A}} + (gx + fy + cz)Z + w\sqrt{-cp}\sqrt{1 + Z^2} = 0,$$

where  $X, Y, Z$  are to be determined by the following equations,

$$(f + 2\theta\sqrt{\mathfrak{A}}) + \sqrt{\mathfrak{A}}(Y + Z) + fYZ - \sqrt{bc}\sqrt{1 + Y^2}\sqrt{1 + Z^2} = 0,$$

$$(g + 2\theta\sqrt{\mathfrak{B}}) + \sqrt{\mathfrak{B}}(Z + X) + gZX - \sqrt{ca}\sqrt{1 + Z^2}\sqrt{1 + X^2} = 0,$$

$$(h + 2\theta\sqrt{\mathfrak{C}}) + \sqrt{\mathfrak{C}}(X + Y) + hXY - \sqrt{ab}\sqrt{1 + X^2}\sqrt{1 + Y^2} = 0;$$

and the solution of which, putting

$$f = \sqrt[4]{\mathfrak{A}}\sqrt{\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F}}, \quad g = \sqrt[4]{\mathfrak{B}}\sqrt{\sqrt{\mathfrak{C}\mathfrak{A}} - \mathfrak{G}}, \quad h = \sqrt[4]{\mathfrak{C}}\sqrt{\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H}}, \quad J = \sqrt{2}\sqrt[4]{\mathfrak{A}\mathfrak{B}\mathfrak{C}},$$



is given by the equations

$$\begin{aligned} KX &= \frac{2fgh}{J} + (-f + g + h)^2 - 2(-f + g + h)J, \\ KY &= \frac{2fgh}{J} + (f - g + h)^2 - 2(f - g + h)J, \\ KZ &= \frac{2fgh}{J} + (f + g - h)^2 - 2(f + g - h)J. \end{aligned}$$

Instead of the direct but very tedious process by which these values of  $X, Y, Z$  have been obtained, we may substitute the following *à posteriori verification*.<sup>2</sup>

We have

$$\begin{aligned} K^2(1 + X^2) &= 4(-f + g + h)^2 J^2 \left(1 + \frac{f}{J}\right) \left(1 - \frac{g}{J}\right) \left(1 - \frac{h}{J}\right), \\ K^2 \sqrt{1 + Y^2} \sqrt{1 + Z^2} &= 4(f^2 - (g - h)^2) J^2 \left(1 - \frac{f}{J}\right) \sqrt{1 - \frac{g^2}{J^2}} \sqrt{1 - \frac{h^2}{J^2}}, \\ K^2(1 + YZ) &= 4\left(1 - \frac{f}{J}\right) \{(J^2 - gh)(f^2 - (g - h)^2) - 2gh(g - h)^2\}, \\ K^2(Y + Z) - 2f^2 - 2g^2 - 2h^2 + 4J^2 &= 4\left(1 - \frac{f}{J}\right)(J^2 - gh). \end{aligned}$$

[Book p. 78.] Putting also

$$f^2 - g^2 - h^2 + \frac{2g^2h^2}{J^2} K^2(1 + YZ) = 4\left(1 - \frac{f}{J}\right) \{(f^2 - (g - h)^2)[(J^2 - gh)(f^2 - (g - h)^2) - 2gh(g - h)^2] - 4g^2h^2\}$$

we have, after reductions in which the symmetry of the expressions plays an essential part,

$$4\left(1 - \frac{f}{J}\right)(f^2 - (g - h)^2) 2gh J^2 \left(1 - \frac{g}{J}\right) \left(1 - \frac{h}{J}\right),$$

and the values obtained above give also

$$2gh \sqrt{1 + Y^2} \sqrt{1 + Z^2} \sqrt{1 + X^2} K^2,$$

which shows that the relation between  $Y$  and  $Z$  is verified by the assumed values of these quantities, and the other two equations are of course also verified. The solution of the problem will be rendered more complete if the equations of the required sections and of the auxiliary sections made use of in the geometrical construction are expressed in terms of  $f, g, h, J$ .

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<sup>2</sup>It is perhaps worth noticing that the value of the quantity  $\lambda$  previously made use of is  $\lambda^2 = -\frac{1}{K^2}(f^2 - g^2 - h^2 + (J + f - g - h)J) \dots$

§ 7.

First, to substitute in the equations of the required sections or resultors.  
Writing the first equation in the form

$$\frac{K^2}{2\sqrt{\mathfrak{B}\mathfrak{C}}} [\sqrt{\mathfrak{B}\mathfrak{C}} ax + (hX + \sqrt{\mathfrak{C}}) y + (gX + \sqrt{\mathfrak{B}}) z + \sqrt{-ap}\sqrt{1+X^2} w] = 0,$$

the coefficient of  $x$  will be

$$\frac{a}{\sqrt{\mathfrak{B}\mathfrak{C}}} \left\{ -\frac{f^2}{J^2} (-f^2 + g^2 + h^2) K^2 \left(1 - \frac{f}{J}\right) - g h K^2 \left(1 - \frac{f}{J}\right) \right\} \dots,$$

[Book p. 79.] or, as it is convenient to write it,

$$f(-f + g + h) \frac{f}{\sqrt{\mathfrak{A}}} \left(1 - \frac{f}{J}\right) x.$$

The coefficient of  $y$  is

$$\frac{1}{2\sqrt{\mathfrak{B}}} \left\{ [-f^4 - g^4 - h^4 + 2g^2h^2 + 2h^2f^2 + 2f^2g^2 - \frac{4f^2g^2h^2}{J^2}] \dots \right\},$$

or, after all reductions,

$$(f^2 - g^2 + h^2) \frac{g}{\sqrt{\mathfrak{B}}} \left(1 - \frac{f}{J}\right) y,$$

and similarly the coefficient of  $z$  is

$$(f^2 + g^2 - h^2) \frac{h}{\sqrt{\mathfrak{C}}} \left(1 - \frac{f}{J}\right) z,$$

and the coefficient of  $w$  is

$$\frac{1}{J} \left(1 + \frac{f}{J}\right) (-f + g + h) \cdot 2\sqrt{J^2 - f^2} \sqrt{1 - \frac{g^2}{J^2}} \sqrt{1 - \frac{h^2}{J^2}} \sqrt{-ap} w.$$

Hence, forming the equation of the resultor in question, and by means of it those of the other resultors, the equations of the resultors are

$$f(-f + g + h) \frac{f}{\sqrt{\mathfrak{A}}} \left(1 - \frac{f}{J}\right) x + g(f^2 - g^2 + h^2) \frac{g}{\sqrt{\mathfrak{B}}} \left(1 - \frac{f}{J}\right) y \\ + h(f^2 + g^2 - h^2) \frac{h}{\sqrt{\mathfrak{C}}} \left(1 - \frac{f}{J}\right) z + 2\sqrt{J^2 - f^2} \sqrt{1 - \frac{g^2}{J^2}} \sqrt{1 - \frac{h^2}{J^2}} \sqrt{-ap} w = 0,$$

$$f(-f^2 + g^2 + h^2) \frac{f}{\sqrt{\mathfrak{A}}} \left(1 - \frac{g}{J}\right) x + g(f - g + h) \frac{g}{\sqrt{\mathfrak{B}}} \left(1 - \frac{g}{J}\right) y \\ + h(f^2 + g^2 - h^2) \frac{h}{\sqrt{\mathfrak{C}}} \left(1 - \frac{g}{J}\right) z + 2\sqrt{J^2 - g^2} \sqrt{1 - \frac{f^2}{J^2}} \sqrt{1 - \frac{h^2}{J^2}} \sqrt{-bp} w = 0,$$

$$f(f^2 - g^2 + h^2) \frac{f}{\sqrt{\mathfrak{A}}} \left(1 - \frac{h}{J}\right) x + g(f^2 + g^2 - h^2) \frac{g}{\sqrt{\mathfrak{B}}} \left(1 - \frac{h}{J}\right) y \\ + h(f + g - h) \frac{h}{\sqrt{\mathfrak{C}}} \left(1 - \frac{h}{J}\right) z + 2\sqrt{J^2 - h^2} \sqrt{1 - \frac{f^2}{J^2}} \sqrt{1 - \frac{g^2}{J^2}} \sqrt{-cp} w = 0,$$

values which might be somewhat simplified by writing  $\mathfrak{x}$ ,  $\mathfrak{y}$ ,  $\mathfrak{z}$ ,  $\mathfrak{w}$  instead of

$$\frac{f}{\sqrt{\mathfrak{A}}}(1-\frac{f}{J})x, \quad \frac{g}{\sqrt{\mathfrak{B}}}(1-\frac{g}{J})y, \quad \frac{h}{\sqrt{\mathfrak{C}}}(1-\frac{h}{J})z, \quad \sqrt{-p}\sqrt{1-\frac{f^2}{J^2}}\sqrt{1-\frac{g^2}{J^2}}\sqrt{1-\frac{h^2}{J^2}}w;$$

and it may be also remarked that the coefficients as well of these formulae as of those which follow may be elegantly expressed in terms of the parts of a triangle having  $f$ ,  $g$ ,  $h$  for its sides.

The equations of the separators are found by taking the differences two and two of the equations of the resultors (this requires to be verified *à posteriori*); thus subtracting the third equation from the second, the result contains a constant factor

$$\frac{2}{J} [4f^2g^2h^2 - J(f^2 - (g-h)^2)((g+h)^2 - f^2)],$$

[Book p. 80.] equivalent to (after rejection of factor)

...

Rejecting the factor in question and forming the analogous two equations, the equations of the separators are

$$\begin{aligned} -\frac{g-h}{f} \cdot \frac{f}{\sqrt{\mathfrak{A}}} (1-\frac{f}{J})x + \frac{g}{\sqrt{\mathfrak{B}}} (1-\frac{g}{J})y - \frac{h}{\sqrt{\mathfrak{C}}} (1-\frac{h}{J})z &= 0, \\ \frac{f}{\sqrt{\mathfrak{A}}} (1-\frac{f}{J})x - \frac{h-f}{g} \cdot \frac{g}{\sqrt{\mathfrak{B}}} (1-\frac{g}{J})y + \frac{h}{\sqrt{\mathfrak{C}}} (1-\frac{h}{J})z &= 0, \\ -\frac{f}{\sqrt{\mathfrak{A}}} (1-\frac{f}{J})x + \frac{g}{\sqrt{\mathfrak{B}}} (1-\frac{g}{J})y - \frac{f-g}{h} \cdot \frac{h}{\sqrt{\mathfrak{C}}} (1-\frac{h}{J})z &= 0; \end{aligned}$$

[Book p. 81.] and from the mode of formation of these equations it is obvious that the separators have a line in common.

The equations of the determinators being  $x = 0$ ,  $y = 0$ ,  $z = 0$ , the equations of the *tactors* are

$$\sqrt{\mathfrak{B}}z - \sqrt{\mathfrak{C}}y = 0, \quad \sqrt{\mathfrak{C}}x - \sqrt{\mathfrak{A}}z = 0, \quad \sqrt{\mathfrak{A}}y - \sqrt{\mathfrak{B}}x = 0;$$

and if  $\alpha x + \beta y + \gamma z + \delta w = 0$  be the equation of the tactor touching

$$x = 0, \quad \sqrt{\mathfrak{B}}x - \sqrt{\mathfrak{C}}z = 0 \text{ and } \sqrt{\mathfrak{A}}y - \sqrt{\mathfrak{B}}x = 0,$$

the conditions of contact are

$$\mathfrak{A}(\mathfrak{A}\alpha^2 + \dots + \frac{K}{p}\delta^2) = (\mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma)^2,$$

$$2\sqrt{\mathfrak{A}\mathfrak{B}}(\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})(\mathfrak{A}\alpha^2+\dots+\frac{K}{p}\delta^2) = \{(\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})(\alpha\sqrt{\mathfrak{A}}-\beta\sqrt{\mathfrak{B}})+\gamma(\mathfrak{C}\sqrt{\mathfrak{B}}-\mathfrak{H}\sqrt{\mathfrak{C}})\}^2,$$

$$2\sqrt{\mathfrak{A}\mathfrak{C}}(\sqrt{\mathfrak{A}\mathfrak{C}}-\mathfrak{C})(\mathfrak{A}\alpha^2+\dots+\frac{K}{p}\delta^2) = \{(\sqrt{\mathfrak{A}\mathfrak{C}}-\mathfrak{C})(\alpha\sqrt{\mathfrak{A}}-\gamma\sqrt{\mathfrak{C}})+\beta(\mathfrak{H}\sqrt{\mathfrak{C}}-\mathfrak{C}\sqrt{\mathfrak{B}})\}^2;$$

whence

$$\frac{1}{\sqrt{\mathfrak{B}}}\sqrt{2\sqrt{\mathfrak{A}\mathfrak{B}}(\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})(\mathfrak{A}\alpha+\mathfrak{H}\beta+\mathfrak{C}\gamma)} = (\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})\sqrt{\mathfrak{A}}\alpha-(\sqrt{\mathfrak{A}\mathfrak{B}}-\mathfrak{H})\sqrt{\mathfrak{B}}\beta+(\mathfrak{C}\sqrt{\mathfrak{B}}-\mathfrak{H}\sqrt{\mathfrak{C}})\gamma,$$

$$\frac{1}{\sqrt{\mathfrak{C}}}\sqrt{2\sqrt{\mathfrak{A}\mathfrak{C}}(\sqrt{\mathfrak{A}\mathfrak{C}}-\mathfrak{C})(\mathfrak{A}\alpha+\mathfrak{H}\beta+\mathfrak{C}\gamma)} = (\sqrt{\mathfrak{A}\mathfrak{C}}-\mathfrak{C})\sqrt{\mathfrak{A}}\alpha+(\mathfrak{H}\sqrt{\mathfrak{C}}-\mathfrak{C}\sqrt{\mathfrak{B}})\beta-(\sqrt{\mathfrak{A}\mathfrak{C}}-\mathfrak{C})\sqrt{\mathfrak{C}}\gamma,$$

and

$$a\alpha^2 + b\beta^2 + c\gamma^2 - 2f\beta\gamma - 2g\gamma\alpha - 2h\alpha\beta = 0,$$

and putting for a moment

$$\sqrt{\mathfrak{A}}\mu = \sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F} - \sqrt{2}(\sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F}),$$

$$\sqrt{\mathfrak{A}}\nu = \sqrt{\mathfrak{B}\mathfrak{C}} - \mathfrak{F} - \sqrt{2}\sqrt{\mathfrak{A}\mathfrak{B}}(\sqrt{\mathfrak{A}\mathfrak{B}} - \mathfrak{H})$$

after some reductions, and observing that the ratios only of the quantities  $\alpha, \beta, \gamma, \delta$  are material, we obtain

$$\alpha = \frac{K}{\sqrt{\mathfrak{A}}}(K+b\nu+g\mu), \quad \beta = \frac{K}{\sqrt{\mathfrak{B}}}(\dots), \quad \gamma = \frac{K}{\sqrt{\mathfrak{C}}}(\dots), \quad \delta = \frac{K}{\sqrt{\mathfrak{A}}}\sqrt{-(b\nu^2+c\mu^2+2f\mu\nu)\cdot p};$$

[Book p. 82.] and it is easily seen also that the coordinates of the point of contact are

$$x = 0, \quad y = \nu, \quad z = \mu, \quad w = -\frac{\delta\sqrt{\mathfrak{A}}}{pK};$$

also

$$\mu = -\frac{f g}{\sqrt{\mathfrak{B}}}\left(1 - \frac{g}{J}\right), \quad \nu = -\frac{f h}{\sqrt{\mathfrak{C}}}\left(1 - \frac{h}{J}\right).$$

Hence substituting and introducing throughout the quantities  $f, g, h, J$ , also forming the analogous equations, the equations of the factors are

$$\begin{aligned} & \left\{f^2(-f^2+g^2+h^2) + (g+h)J(f^2-(g-h)^2 - \frac{2g^2h^2}{J^2})\right\}\frac{f}{\sqrt{\mathfrak{A}}}x \\ & + 2\sqrt{X}gh\left(1 - \frac{g}{J}\right)\left(1 - \frac{h}{J}\right)(f^2-(g-h)^2)\sqrt{-p}w = 0, \\ & \left\{g^2(f^2-g^2+h^2) + (h+f)J(g^2-(h-f)^2 - \frac{2h^2f^2}{J^2})\right\}\frac{g}{\sqrt{\mathfrak{B}}}y \\ & + 2\sqrt{Y}hf\left(1 - \frac{h}{J}\right)\left(1 - \frac{f}{J}\right)(g^2-(h-f)^2)\sqrt{-p}w = 0, \\ & \left\{h^2(f^2+g^2-h^2) + (f+g)J(h^2-(f-g)^2 - \frac{2f^2g^2}{J^2})\right\}\frac{h}{\sqrt{\mathfrak{C}}}z \\ & + 2\sqrt{Z}fg\left(1 - \frac{f}{J}\right)\left(1 - \frac{g}{J}\right)(h^2-(f-g)^2)\sqrt{-p}w = 0. \end{aligned}$$

[Book p. 83.] It is obvious, from the equations, that each separator passes through the point of contact of a tactor and determinant; it consequently only remains to be shown that each separator touches two tactors. Consider the tactor which has been represented by  $\alpha x + \beta y + \gamma z + \delta w = 0$ ; the unreduced values of the coefficients give

$$\mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma = \frac{K^2}{\sqrt{\mathfrak{A}}}, \quad \mathfrak{H}\alpha + \mathfrak{B}\beta + \mathfrak{F}\gamma = \frac{K^2}{\sqrt{\mathfrak{A}}} (\mathfrak{H} + \nu), \quad \mathfrak{G}\alpha + \mathfrak{F}\beta + \mathfrak{C}\gamma = \frac{K^2}{\sqrt{\mathfrak{A}}} (\mathfrak{G} + \mu),$$

$$\sqrt{\mathfrak{A}\alpha^2 + \dots + \frac{K}{p} \delta^2} = \frac{1}{\sqrt{\mathfrak{A}}} (\mathfrak{A}\alpha + \mathfrak{H}\beta + \mathfrak{G}\gamma) = K^2.$$

Represent for a moment the separator  $\ell x + my + nz + sw = 0$ . Then putting  $\mathfrak{A}\ell^2 + \dots + \frac{K}{p} s^2 = \square^2$ , since

$$\mathfrak{A}\ell^2 + \dots + \frac{K}{p} s^2 = K^2 \left\{ \ell \left(1 - \frac{f}{j}\right) - \frac{g}{\sqrt{\mathfrak{B}}} (\mathfrak{H} + \nu) \left(1 - \frac{g}{j}\right) \left(1 - \frac{h}{j}\right) - (f - g) \frac{h}{\sqrt{e}} \left(1 - \frac{2g}{j}\right) \left(1 - \frac{h}{j}\right) \right\} = \frac{K^2}{j} \left( -(f - g)^2 + \dots \right)$$

the condition of contact becomes

$$\square = \frac{1}{j} \left\{ -(f - g)^2 + h(f + g) - \frac{2fgh}{j} \right\};$$

or, forming the value of  $\square^2$  and substituting,

$$\left( f^2 - (g - h)^2 - \frac{2g^2h^2}{j^2} \right) \left(1 - \frac{h}{j}\right) \left(1 - \frac{f}{j}\right) \left(1 - \frac{g}{j}\right) = \frac{1}{j} \left( -(f - g)^2 + h(f + g) - \frac{2fgh}{j} \right)^2,$$

which may be verified without difficulty; and thus the construction for the resultors is shown to be true.

## § 8.

Several of the formulae of the preceding sections of this memoir apply to any number of variables. Consider the surface (i.e. hypersurface)

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy + \dots + pw^2 = 0,$$

and the section (i.e. hypersection)

$$(a\lambda + h\mu + g\nu + \dots) x + (h\lambda + b\mu + f\nu + \dots) y + (g\lambda + f\mu + c\nu + \dots) z + \dots + \sqrt{-p} \nabla w = 0,$$

where

$$\nabla^2 = a\lambda^2 + b\mu^2 + c\nu^2 + 2f\mu\nu + 2g\nu\lambda + 2h\lambda\mu + \dots - K,$$

the condition of contact with any other section represented by a similar equation is

$$a\lambda\lambda'+b\mu\mu'+c\nu\nu'+\dots+f(\mu\nu'+\mu'\nu)+g(\nu\lambda'+\nu'\lambda)+h(\lambda\mu'+\lambda'\mu)+\dots+K\frac{\rho\rho'}{p} = \nabla\nabla',$$

where  $K$  is the determinant formed with the coefficients  $a, b, c, f, g, h, \dots$ . And consequently, by establishing all or any of the equations  $\lambda = \sqrt{\mathfrak{A}}, \mu = \sqrt{\mathfrak{B}}, \nu = \sqrt{\mathfrak{C}}, \dots$ , we have the condition in order that the section in question may touch all or the corresponding sections of the sections  $x = 0, y = 0, z = 0, \dots$ .

Let  $n$  be the number of the variables  $x, y, z, \dots$ ; then  $K^{n-1} =$

$$\begin{vmatrix} \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & \end{vmatrix},$$

[Book p. 84.] also

$$K^{n-1}[(a\lambda+h\mu+g\nu+\dots)x+(h\lambda+b\mu+f\nu+\dots)y+(g\lambda+f\mu+c\nu+\dots)z+\dots]$$

$$= - \begin{vmatrix} x & y & z & \dots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \dots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \dots \\ \vdots & & & \end{vmatrix},$$

whence also

$$K^{n-2}(\nabla^2+K) = - \begin{vmatrix} \lambda & \mu & \nu & \dots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \dots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \dots \\ \vdots & & & \end{vmatrix} \quad \text{or} \quad K^{n-2}\nabla^2 = - \begin{vmatrix} 1 & \lambda & \mu & \nu \dots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \dots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \dots \\ \vdots & & & \end{vmatrix},$$

[Book p. 85.] and the equation of the section in question becomes

$$- \begin{vmatrix} x & y & z & \dots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \dots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \dots \\ \vdots & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)}\sqrt{-p} \sqrt{- \begin{vmatrix} 1 & \lambda & \mu & \nu \dots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} \dots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} \dots \\ \vdots & & & \end{vmatrix}} \cdot t = 0,$$

also the condition of contact with the corresponding section is

$$- \begin{vmatrix} \pm 1 & \lambda & \mu & \nu & \dots \\ \lambda' & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \mu' & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \nu' & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix} = \sqrt{- \begin{vmatrix} 1 & \lambda & \mu & \nu & \dots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \nu & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix}} \sqrt{- \begin{vmatrix} 1 & \lambda' & \mu' & \nu' & \dots \\ \lambda' & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \mu' & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \nu' & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix}}.$$

In particular the equation of the sections which touches all the sections  $x = 0$ ,  $y = 0$ ,  $z = 0$ ,  $\dots$ , is

$$- \begin{vmatrix} x & y & z & \dots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)} \sqrt{-p} \sqrt{- \begin{vmatrix} 1 & \sqrt{\mathfrak{A}} & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} & \dots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix}} \cdot t = 0.$$

Again, the equations of the section touching  $y = 0$ ,  $z = 0$ ,  $\dots$  and the section touching  $x = 0$ ,  $z = 0$ ,  $\dots$  are

$$- \begin{vmatrix} x & y & z & \dots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)} \sqrt{-p} \sqrt{- \begin{vmatrix} 1 & \lambda & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} & \dots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix}} \cdot t = 0,$$

$$- \begin{vmatrix} x & y & z & \dots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix} + K^{\frac{1}{2}(n-1)} \sqrt{-p} \sqrt{- \begin{vmatrix} 1 & \sqrt{\mathfrak{A}} & \mu & \sqrt{\mathfrak{C}} & \dots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix}} \cdot t = 0,$$

[Book p. 86.] and the condition of contact of these two sections is

$$- \begin{vmatrix} 1 & \lambda & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} & \dots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix} = \sqrt{- \begin{vmatrix} 1 & \lambda & \sqrt{\mathfrak{B}} & \sqrt{\mathfrak{C}} & \dots \\ \lambda & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \sqrt{\mathfrak{B}} & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix}} \sqrt{- \begin{vmatrix} 1 & \sqrt{\mathfrak{A}} & \mu & \sqrt{\mathfrak{C}} & \dots \\ \sqrt{\mathfrak{A}} & \mathfrak{A} & \mathfrak{H} & \mathfrak{G} & \dots \\ \mu & \mathfrak{H} & \mathfrak{B} & \mathfrak{F} & \dots \\ \sqrt{\mathfrak{C}} & \mathfrak{G} & \mathfrak{F} & \mathfrak{C} & \dots \\ \vdots & & & & \end{vmatrix}}.$$

It would seem from the appearance of these equations that there should be some simpler method of obtaining the solution than the method employed in the previous part of this memoir.



115.

NOTE ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED  
POLYGON.

[From the *Philosophical Magazine*, vol. vi. (1853), pp. 99–102.]

The equation of a conic passing through the points of intersection of the conics

$$U = 0, \quad V = 0$$

is of the form

$$wU + V = 0,$$

where  $w$  is an arbitrary parameter. Suppose that the conic touches a given line, we have for the determination of  $w$  a quadratic equation, the roots of which may be considered as parameters for determining the line in question. Let one of the values of  $w$  be considered as equal to a given constant  $k$ ; the line is always a tangent to the conic

$$kU + V = 0;$$

and taking  $w = p$  for the other value of  $w$ ,  $p$  is a parameter determining the particular tangent, or, what is the same thing, the point of contact of this tangent.

Suppose the tangent meets the conic  $U = 0$  (which is of course the conic corresponding to  $w = \infty$ ) in the points  $P, P'$ , and let  $\theta, \infty$  be the parameters of the point  $P$ , and  $\theta', \infty$  the parameters of the point  $P'$ . It follows from my “Note on the Geometrical representation of the Integral  $\int dx \div \sqrt{(x+a)(x+b)(x+c)}$ ,” [113]<sup>3</sup> and from the theory of invariants, that if  $\square\xi$  represent the “Discriminant” of  $\xi U + V$

[Book p. 88.] (I now use the term discriminant in the same sense in which determinant is sometimes used, viz. the discriminant of a quadratic function  $ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy$  or  $(a, b, c, f, g, h)(x, y, z)^2$ , is the determinant  $k = abc - af^2 - bg^2 - ch^2 + 2fgh$ ), and if

$$\Pi \xi = \int_{\infty}^{\xi} \frac{d\xi}{\sqrt{\square\xi}},$$

---

<sup>3</sup>I take the opportunity of correcting an obvious error in the note in question, viz.  $a^2 + b^2 + c^2 - 2bc - 2ca - 2ab$  is throughout written instead of (what the expression should be)  $a^2 + b^2 + c^2 - 2a^2bc - 2b^2ca - 2c^2ab$ . [This correction is made, *ante* p. 55.]

then the following theorem is true, viz.

“If  $(\theta, \infty)$ ,  $(\theta', \infty)$  are the parameters of the points  $P$ ,  $P'$  in which the conic  $U = 0$  is intersected by the tangent, the parameter of which is  $p$ , of the conic  $kU + V = 0$ , then the equations

$$\begin{aligned}\Pi \theta &= \Pi p - \Pi k, \\ \Pi \theta' &= \Pi p + \Pi k,\end{aligned}$$

determine the parameters  $\theta$ ,  $\theta'$  of the points in question.” And again,

“If the variable parameters  $\theta$ ,  $\theta'$  are connected by the equation

$$\Pi \theta' - \Pi \theta = 2 \Pi k,$$

then the line  $PP'$  will be a tangent to the conic  $kU + V = 0$ .” Whence, also,

“If the sides of a triangle inscribed in the conic  $U = 0$  touch the conics

$$kU + V = 0, \quad k'U + V = 0, \quad k''U + V = 0,$$

then the equation

$$\Pi k + \Pi k' + \Pi k'' = 0$$

must hold good between the parameters  $k$ ,  $k'$ ,  $k''$ .”

And, conversely, when this equation holds good, there are an infinite number of triangles inscribed in the conic  $U = 0$ , and the sides of which touch the three conics; and similarly for a polygon of any number of sides.

The algebraical equivalent of the transcendental equation last written down is

$$\begin{vmatrix} 1, & k, & \sqrt{\square k} \\ 1, & k', & \sqrt{\square k'} \\ 1, & k'', & \sqrt{\square k''} \end{vmatrix} = 0;$$

let it be required to find what this becomes when  $k = k' = k'' = 0$ ; we have

$$\sqrt{\square k} = A + Bk + Ck^2 + \dots,$$

*[Continuation of Paper 115 lies beyond the present slice; the body extends from book p. 88 onward into the next PDF range.]*

ON THE HOMOGRAPHIC TRANSFORMATION OF A SURFACE OF THE SECOND ORDER INTO ITSELF.

[From the *Philosophical Magazine*, vol. vi. (1853), pp. 326–333.]

The following theorems in plane geometry, relating to polygons of any number (odd or even) of sides, are, well known.

“If there be a polygon of  $(\alpha + 1)$  sides inscribed in a conic, and so if the sides pass through given points, the  $(\alpha + 1)$ th side will envelope a conic having double contact with the given conic.” And “If there be a polygon of  $(\alpha + 1)$  sides inscribed in a conic, and so of the sides touch conics having double contact with the given conic, the  $(\alpha + 1)$ th side will envelope a conic having double contact with the given conic.” The second theorem of course includes the first, but it is the two separately that I propose to extend to a surface of the second order.

As regards the corresponding theory in geometry of three dimensions, Sir W. Hamilton has given a theorem relating to polygons of an odd number of sides, which may be thus stated: “If there be a polygon of  $(2\alpha + 1)$  sides inscribed in a surface of the second order, and  $2\alpha$  of the sides pass through given points, the  $(2\alpha + 1)$ th side will constantly touch two surfaces of the second order, each of them intersecting the given surface of the second order in the same four lines<sup>1</sup>.”

The entire theory depends upon what may be termed the transformation of a surface of the second order into itself, or analytically, upon the transformation of a quadratic form of four indeterminates into itself. I use for shortness the term transformation simply; but this is to be understood as meaning a homographic transformation, or in analytic language, a transformation by means of linear substitutions. It will be convenient to remark at the outset, that if two points of a surface of the second order have the relation contemplated in the data of Sir W. Hamilton’s theorem (viz. if the line joining the two points pass through a fixed point), the transformation is, using the language of the *Recherches Arithmétiques*, an *improper* one, but that the relation contemplated in the conclusion of the theorem (viz. that of two points of a surface of the second order, connected by a line touching two surfaces of the second order each of them intersecting the given surface of the second order in the same four lines) depends upon a *proper* transformation; and that the circumstance that an even number of improper transformations is required in order to make a proper transformation (that this circumstance, I say), is the reason why the theorem applies to polygons in which an even number of sides pass through fixed points, that is, to polygons of an *odd* number of sides.

Consider, in the first place, two points of a surface of the second order such that the line joining them passes through a given point. Let  $x, y, z, w$  be current coordinates<sup>2</sup>, and let the

<sup>1</sup>See *Phil. Mag.* vol. xxxix. (1851), p. 396. The form in which the theorem is exhibited by Sir W. Hamilton is somewhat different; the surface considered as being not ellipsoid, and the two surfaces touched by the last or  $(2\alpha + 1)$ th side of the polygon are spoken of as having focal connection with two confocal hyperboloids of two sheets, having respectively double contact with the given ellipsoid; the student is, in fact, a quadric conic which is taken as an analytical premise. (The conclusion is necessarily of the kind mentioned; but when this is once understood, the language of the text may be properly employed. It may be proper to explain that hyperboloids of one sheet inscribed in an ellipsoid is, by analogy with the case in plano of the hyperbola of imaginary transformation, the ulterior expression of the doubled conic of a surface of the second order than is implied.

<sup>2</sup>Strictly speaking, it is the *ratios* of these quantities,  $x, y, z, w$ , which are the coordinates, and consequently, even when the point is given, the values  $x, y, z, w$  are merely indeterminate to a factor *près*. So that in assuming that a point is given, we should write  $x : y : z : w = a : b : c : d$  and so on, but when this is once understood, the language of the text may be properly employed. It may be proper to explain that the equation  $(a, b, c, d \mid x, y, z, w)^2 = Rxx + yy^2$ , where “ $\mid$ ” means  $a(x + b(y + c)z + dw)$ . The system of coefficients may frequently be indicated by a single coefficient only; thus in the text  $(a, \dots)(x, y, z, w)^2$  stands for the most general quadratic function of four variables.

equation of the surface be

$$(a, \dots \rangle x, y, z, w)^2 = 0,$$

and take for the coordinates of the two points on the surface  $x_1, y_1, z_1, w_1$  and  $x_2, y_2, z_2, w_2$ , and for the coordinates of the fixed point  $\alpha, \beta, \gamma, \delta$ . Write for shortness

$$(a, \dots \rangle \alpha, \beta, \gamma, \delta)^2 = p,$$

$$(a, \dots \rangle \alpha, \beta, \gamma, \delta \rangle x, y, z, w) = q_1,$$

$$(a, \dots \rangle \alpha, \beta, \gamma, \delta \rangle x_2, y_2, z_2, w_2) = q_2,$$

then the coordinates  $x_2, y_2, z_2, w_2$  are determined by the very simple formula

$$\begin{aligned} x_2 &= x_1 - \frac{2q_1}{p} y_1, \\ y_2 &= y_1 - \frac{2q_1}{p} y_1, \\ z_2 &= z_1 - \frac{2q_1}{p} y_1, \\ w_2 &= w_1 - \frac{2q_1}{p} y_1. \end{aligned}$$

In fact, the point  $(\xi, \eta, \zeta, \omega)$  lies in the line joining the points  $(x_1, y_1, z_1, w_1)$  and  $(x_2, y_2, z_2, w_2)$ ; and to show that this line touches the surface  $V = 0$ , it is only necessary to form the equation of the tangent plane at the point  $(\xi, \eta, \zeta, \omega)$  to the surface in question; this is

$$(\alpha, \dots \rangle (x, y, z, w \rangle \xi, \eta, \zeta, \omega)^2 = 0,$$

that is, the point  $(x_1, y_1, z_1, w_1)$  and  $(x_2, y_2, z_2, w_2)$  will correspond to each other by means of the above formulæ, and that they are determined by means of the equations  $\xi, \eta, \zeta, \omega$  are given by the  $x_1 + x_2, y_1 + y_2, z_1 + z_2, w_1 + w_2$ .

Suppose now that we consider three points 1, 2, 3 of the surface, the lines joining 1, 2 and 2, 3 passing respectively through given points; we have first

$$\begin{aligned} x_2 &= x_1 - \frac{2q_1}{p} \alpha, & y_2 &= y_1 - \frac{2q_1}{p} \beta, \\ z_2 &= z_1 - \frac{2q_1}{p} \gamma, & w_2 &= w_1 - \frac{2q_1}{p} \delta, \end{aligned}$$

where  $q_1, \beta, \gamma, \delta$  are the parameters which may subject to the condition

$$\alpha^2 + \beta^2 - c^2 = 0,$$

the equation of the variable conic may be written

$$[\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2) - 2(xx - \xi) - 2(yy - \eta)]\beta = 0,$$

Or, what is the same thing, the system of equations is

$$\alpha^2 + \beta^2 + \gamma^2 + \delta^2 = 0, \quad \xi + \eta = c\xi + c\eta$$

$$(2 - c)\alpha^2 - p(\alpha + c\eta)\alpha + c\eta\xi + (-c\eta + p + c\eta)\eta + (-c\eta - \mu c + c\eta)\eta = 0.$$

or, it may also be written,

$$(a^2 + p + a^2 c\eta + (a\eta - \mu)c + c\eta a^2)\alpha + (b\eta - \beta c)\eta + c(-c\eta + \mu_0)\eta + (-c\eta - p - c\eta + a\eta)\eta = 0.$$

Represent for a moment the form  $\eta\eta - \mu c + \mu c\eta$  by  $V$ , and consider the surface  $V = 0$ . If we may also be written,

$$(\alpha^2 + \beta^2 + c\eta + (\eta\eta - c\eta + b\eta)c)\eta + p\alpha\eta = 0,$$

this discriminant is

$$(\eta + V\eta + \alpha\eta + V\eta + \theta^2,$$

which shows that the surface intersect in four lines. Suppose the discriminant vanishes; we have for the determination of  $\eta$  a quadratic equation, which may be written

$$q'^2 + (2 + s)q' + R = 0;$$

but the roots of this equation by  $q'$ ,  $q''$ , then each of the functions  $\eta + Vc'$ ,  $\eta + Vc''$  will break up into linear factors, and we may write

$$\eta + Vc'_r = R_r R_r,$$

$$\eta + Vc''_r = R'_r R_r,$$

( $V'$  and  $V''$  are of course linear functions of  $R_r$ , and  $R_r$ ,  $R'_r$ .) These which lead in evidence the fact of the two surfaces intersecting in four lines.

The equations

$$q_r + q_r = Rq_r, \quad q_r = q_r = 2q_r, \quad \alpha_r + \alpha_r = Rq_r, \quad \&c.$$

generating lines and in like manner in the point  $Q$ . Thus  $P$ ,  $Q$  are tangent planes at the points  $A$ ,  $B$  respectively, and the system of these polygons of generating lines is the system of common chords. The chord in question is a tangent to the two surfaces involved; it is therefore the line of intersection of the tangent planes through  $P$  to the same  $PA$  as belonging determinately to one or the other of the two systems of generating lines, then when  $P$  is given, the corresponding point  $Q$  is, it is clear, completely determined. What precedes may be recapitulated in the statement, that in the improper transformation of a surface of the second order into itself, we have, as corresponding points, the opposite angles of a skew quadrangle lying upon the surface, and having the other two opposite angles upon given lines of intersection of  $\theta$ ,  $\phi$ , and then the proper transformation is obtained is not the general proper transformation.

It is necessary to remark, that, whenever the coordinates of the points  $Q$  are connected with the coordinates of the points  $B$  by means of the equations which belong to an improper transformation, the points  $P$ ,  $Q$  have to each other the geometrical relation above mentioned, viz. there exist two plane sections  $\theta$ ,  $\phi$  such that  $P$ ,  $Q$  are the opposite angles of a skew quadrangle upon the surface, and having the other two angles upon  $\theta$  and  $\phi$ .  $Q$  is the opposite angles of a skew quadrangle upon the surface, having two of its opposite angles upon  $\theta$  and  $\phi$  respectively, and the two adjacent angles upon  $\theta$  and  $\phi$ .

It is, perhaps, hardly possible to embrace by a single intelligible synthesis the remaining side will satisfy the geometrical condition in virtue of which its extremities will be a pair of corresponding points, the pair being proper or improper according to the rule just explained.

I conclude with the remark, that we may by means of two plane sections of a surface of the second order obtain a proper transformation. For, if the generating lines through  $P$  meet the sections  $\theta$ ,  $\phi$  in the points  $A$ ,  $B$  respectively, and the remaining generating lines through  $A$ ,  $B$  respectively meet in the points  $\phi$ ,  $\theta$  respectively in  $B'$ ,  $A'$ , and the remaining generating lines through  $B'$ ,  $A'$  respectively meet in a point  $P'$ ; then will  $P$ ,  $P'$  be a pair of corresponding points in a proper trans-formation.

*2 Stone Buildings, January 11, 1854.*

## ON THE GEOMETRICAL REPRESENTATION OF AN ABELIAN INTEGRAL.

[From the *Philosophical Magazine*, vol. vi. (1853), pp. 414–418.]

The equation of a surface passing through the curve of intersection of the surfaces

$$ax^2 + by^2 + cz^2 + dw^2 = 0,$$

$$ax^2 + b'y^2 + c'z^2 + d'w^2 = 0,$$

is of the form

$$k(ax^2 + by^2 + cz^2 + dw^2) + l(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

where  $k$  is an arbitrary parameter. Suppose that the surface touches a given plane, we have for the determination of  $k$  a certain equation the root of which may be considered as parameters defining the plane in question. Let one of the values of  $k$  be considered equal to a given quantity  $k$ , the plane touches the surface

$$k(ax^2 + by^2 + cz^2 + dw^2) + l(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

and the other two values of  $k$  may be considered as parameters defining the particular tangent plane, or what is the same thing, determining the point of contact with the surface.

Or more clearly, then—in order to determine the position of a point on the surface

$$k(ax^2 + by^2 + cz^2 + dw^2) + l(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

the tangent plane at the point in question is touched by two other surfaces

$$p(ax^2 + by^2 + cz^2 + dw^2) + q(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

$$p'(ax^2 + by^2 + cz^2 + dw^2) + q'(ax^2 + b'y^2 + c'z^2 + d'w^2) = 0,$$

and similarly for the other lines.

I consider now a curve of double curvature ( $k$ ) touching the surface ( $k$ ); for the points of the curve ( $k$ ), the parameter of one of the surfaces touched at the point in question is constant, that is,  $p, q$  are functions of the parameters  $p', q'$  of the other; or what is the same thing, writing  $k = p \div q$ ,  $k' = p' \div q'$ , the coordinates of the point ( $k$ ) of the curve ( $k$ ) are

$$x : y : z : w = \sqrt{(\alpha + k)(\alpha + k')} : \sqrt{(\beta + k)(\beta + k')} : \sqrt{(\gamma + k)(\gamma + k')} : \sqrt{(\delta + k)(\delta + k')},$$

or under a somewhat more convenient form we have

$$x : y : z : w = \sqrt{(a + k)(a + k')} : \sqrt{(b + k)(b + k')} : \sqrt{(c + k)(c + k')} : \sqrt{(d + k)(d + k')},$$

where  $a, b, c, d$  are inversely as  $a', b', c', d'$ . Taking for a moment  $k = k'$ , this is a thing thus, viz. it is

$$x : y : z : w = (a + k) : (b + k) : (c + k) : (d + k),$$

which equation is satisfied by writing  $(a + k)x : (b + k)y : (c + k)z : (d + k)w$ , where  $\alpha, \beta, \gamma, \delta$  are inversely as  $a, b, c, d$ . Similarly, we have

$$x : y : z : w = \sqrt{(a + k)(a + k')} : \sqrt{(b + k)(b + k')} : \sqrt{(c + k)(c + k')} : \sqrt{(d + k)(d + k')},$$

for the conditions in order that the point  $(k, k')$  may lie in the tangent plane at  $(k')$ ,  $p, q$  to the surface  $(k')$ . Similarly, we have

$$\sum (b+k)\sqrt{(c+k)(c+k')} + (c+k)\sqrt{(b+k)(b+k')} = 0,$$

for the condition in order that the point  $(k, p)$  may lie in the tangent plane at  $(k', p, q)$  to the surface  $(k)$ . Similarly,

$$\sum (a+k')(a+k)\sqrt{(a+k)(a+k') + (b+k')(b+k)\sqrt{(b+k)(b+k')}} = 0,$$

or making the same reductions give

$$\sum (a+a')\sqrt{(a+k)(a+k')} \cdot \sqrt{\frac{a+a}{a+a'}} = h \cdot \mu + a + b'',$$

and the like for the system of equations

$$\sqrt{(a+a')(a+\rho)(a+\sigma)(a+\tau)} \cdot \sqrt{\frac{a+a}{a+a'}} = h \cdot \mu + a + b''.$$

When in each system  $a$  is to be successively replaced by  $a, b, c, d$ , and  $\rho, \sigma, \tau$  and  $\mu, \nu$  are indeterminate. Now dividing each equation of the one system by the corresponding equation in the other system, we see that the equation

$$\frac{a+b}{a+a'} h + \frac{1}{\mu+a+b''} = h+a,$$

is satisfied by the values  $a, b, c, d$  of  $a$ , and therefore, since the equation is in  $a$  only of the third order, that the equation in question must be identically true. We may therefore write

$$h + \mu + a + a'' = (a+a)(\lambda + \mu),$$

and the two systems of equations become identical with

$$\sqrt{(a+k)(a+k')(a+p)(a+q)(a+r)(a+s)} = h(a+a)(\rho + \mu).$$

A set of remarks which may be represented by the single equation

$$\Phi(x+y)\Phi(x+y) = \mu(x+y) + (\nu x + \nu y)(x+y),$$

where  $x$  is arbitrary; or in the same thing, writing  $\theta = a+x, a+b, b+b$ ,

$$x(x+a)(x+b)(x+c)(x+d)(x+\alpha)(x+\beta) = h(\mu+a)(\rho+\mu).$$

Hence, putting the transcendental equation

$$\Pi p + \Pi q + \Pi p' + \Pi q' = \text{const.}$$

$$\Pi_1 p = \int \frac{p dx}{\sqrt{(a+x)(a+y)(a+z)(a+w)(a+\alpha)}},$$

we see that the algebraical equations between  $p, q, p', q'$  are equivalent to the transcendental equation

$$\Pi p + \Pi q + \Pi p' + \Pi q' = \text{const.}$$

The algebraical equations which connect  $\theta, \phi$  with  $p, q, p', q'$ , may be exhibited under several different forms; thus, for instance, considering the point  $(k, k')$  as a point in the line joining  $(k, p)$ , and  $(k', q')$ , the equation is

$$\begin{vmatrix} \sqrt{(a+p)(a+q)} & \sqrt{(a+k)(a+k')} & \sqrt{(a+p')(a+q')} \\ \sqrt{(b+p)(b+q)} & \sqrt{(b+k)(b+k')} & \sqrt{(b+p')(b+q')} \\ \sqrt{(c+p)(c+q)} & \sqrt{(c+k)(c+k')} & \sqrt{(c+p')(c+q')} \\ \sqrt{(d+p)(d+q)} & \sqrt{(d+k)(d+k')} & \sqrt{(d+p')(d+q')} \end{vmatrix} = 0,$$

i.e. the determinants formed by selecting any three of the four columns must vanish; the equations so obtained are equivalent (as they should be) to two independent equations.

Or, again, by considering  $(p, q, \phi)$  first as a point in the tangent plane at  $(k, k')$  to the surface  $(k)$ , and then as a point in the tangent plane at  $(k', p', q')$  to the surface  $(k')$ , we obtain

$$\begin{aligned} \sum (a+k) \sqrt{(a+p)(a+q)} \sqrt{(a+k)(a+k')} &= 0, \\ \sum (a+k') \sqrt{(a+p)(a+q)} \sqrt{(a+k)(a+k')} &= 0. \end{aligned}$$

Or again, we may consider the line joining  $(k, k')$  and  $(p, q)$ , the formulæ will satisfy the geometrical condition in virtue of which its extremities will be a pair of corresponding points, the pair being proper or improper according to the rule just explained.

The condition in order that the line joining the points  $(k, k', p, q)$  and  $(k', k'', p', q')$  may meet a curve which is represented by the single equation

$$\sum_a a x_a x_b x_c \cdots = 0,$$

may also be expressed in the form

$$2k\zeta + (\Pi p + \Pi q + \Pi p' + \Pi q') = \text{const.},$$

the equation entails

$$\Pi p + \Pi q + \Pi p' + \Pi q' = \text{const.}$$

or

$$\sum \frac{dx}{\sqrt{(a+x)(a+y)(a+z)(a+w)(a+\alpha)}} = \text{const.}$$

considering only the first integral, the second one being analogously.

The preceding formulæ are obviously very incomplete; but the connection which they point out between the geometrical question and the Abelian integral involving the root of a function of the sixth order, may I think be of service in the theory of these integrals.

[From the *Philosophical Magazine*, vol. vi. (1853), pp. 427–431.]

M. St Laurent has shown (*Gergonne*, vol. xviii. [1827] p. 1), that in certain cases the caustic by refraction of a circle is identical with the caustic of refraction of a circle (the refracting circle and radiant point being, of course, properly chosen), and a very elegant demonstration of M. St Laurent's theorems is given by M. Gergonne in the same volume, p. 48. A similar method may be employed to demonstrate the more general theorem, that the same caustic by refraction of a



circle may be considered as arising from six different systems of a radiant point, circle, and index of refraction. The demonstration is obtained by means of the secondary caustic, which (as is well known) an oval of Descartes. Such oval has three rectangular axes, and two corresponding circles and indices of refraction, as follows:—

Let  $x$  be the radius of the refracting circle,  $\xi, \eta$  be the coordinates of the radiant point, and the secondary caustic in the envelope of the circle

$$x^2 + \beta^2 = c^2,$$

where  $\alpha, \beta$  are parameters which vary subject to the condition

$$\alpha^2 + \beta^2 - c^2 = 0,$$

the equation of the variable circle may be written

$$[\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2) - 2(xx - \xi) - 2(yy - \eta)]\beta = 0,$$

which is of the form

$$U + \lambda + B(\lambda^2 + \delta) = 0,$$

the envelope is therefore

$$U^2 = A^2(\lambda + B),$$

Hence substituting, we have for the equation of the envelope, i.e. for the secondary caustic,

$$[\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2)]^2 = 4c^2[(xx - \xi)^2 + (yy - \eta)^2],$$

and this may perhaps be considered as the standard form.

To show that this equation belongs to a Descartes' oval, suppose for greater convenience  $\eta = 0$ , and write

$$\rho^2(x^2 + y^2) = \rho^2 - \xi^2 + 2c\eta\sqrt{kx - \xi\xi} + a^2,$$

multiplying this equation by  $1 - \frac{1}{\alpha}$ , and adding to each side  $a^2\left(x - \frac{\xi}{a}\right) + (x - \xi)^2 + y^2$ , we have

$$\begin{aligned} & \left(1 - \frac{1}{\alpha}\right) [\alpha'(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2)]^2 + \left(x - \frac{\xi}{a}\right)^2 \\ &= (x - \xi)^2 + y^2 + 2c\left(x - \frac{1}{a}\right) \sqrt{(x - \xi)^2 + y^2} + c^2 \left(x - \frac{1}{a}\right)^2, \end{aligned}$$

or reducing

$$\alpha^2 \left[ \left(x - \frac{\xi}{a}\right)^2 + y^2 \right] = \left[ \sqrt{(x - \xi)^2 + y^2} + c \left(x - \frac{1}{a}\right) \right]^2;$$

again, multiplying the same equation by  $\frac{1}{a^2} \left(1 - \frac{1}{a^2}\right)$ , and adding to each side  $\frac{1}{a^2} \left(1 - \frac{\xi}{a}\right)^2 + \frac{1}{a^2} y^2 + \xi^2 + y^2$ , we have

$$\frac{1}{a^2} \left(1 - \frac{\xi}{a}\right)^2 + \frac{1}{a^2} y^2 + (x - \xi)^2 + y^2 + 2\frac{c}{a} \left(1 - \frac{\xi}{a}\right) \sqrt{(x - \xi)^2 + y^2} + \frac{c^2}{a^2} \left(1 - \frac{\xi}{a}\right)^2,$$

or reducing,

$$\left(x - \frac{\xi}{a}\right)^2 + y^2 = \frac{1}{a^2} \left[ \sqrt{(x - \xi)^2 + y^2} + \frac{c}{a} \left(1 - \frac{\xi}{a}\right) \right]^2.$$

Hence, extracting the square roots of each side of the equations thus found, we have the equation of the secondary caustic in either of the forms

$$\sqrt{\left(x - \frac{\xi}{a}\right)^2 + y^2} = \frac{1}{a} \sqrt{(x - \xi)^2 + y^2} + \frac{c}{a} \left(x - \frac{1}{a}\right),$$

$$\sqrt{\left(x - \frac{\xi}{a}\right)^2 + y^2} = \frac{1}{a} \sqrt{(x - \xi)^2 + y^2} + \frac{c}{a} \left(1 - \frac{\xi}{a}\right),$$

to which are to be joined

$$\sqrt{\left(x - \frac{\xi}{a}\right)^2 + y^2} \sqrt{\left(x - \frac{\xi}{a'}\right)^2 + y^2} + \frac{c}{a} \left(x - \frac{1}{a'}\right) + \frac{c}{a} \left(1 - \frac{\xi}{a}\right) \sqrt{\left(x - \frac{\xi}{a'}\right)^2 + y^2} + \frac{c}{a'} \left(1 - \frac{\xi}{a}\right) \sqrt{1 - a\xi^2 + y^2} =$$

Write successively,

$$\xi' = \xi, \quad c' = c, \quad \mu' = \mu; \quad (1)$$

$$\xi' = \frac{\xi}{a}, \quad c' = \frac{c}{a}, \quad \mu' = \frac{1}{\mu}; \quad (\alpha)$$

$$\xi' = \frac{1}{\xi}, \quad c' = \frac{c}{\xi}, \quad \mu' = \frac{\mu}{\xi}; \quad (\beta)$$

$$\xi' = \xi, \quad c' = c, \quad \mu' = \frac{1}{\mu}; \quad (\gamma)$$

$$\xi' = \frac{\xi}{a}, \quad c' = \frac{c}{a}, \quad \mu' = \mu; \quad (\delta)$$

$$\xi' = \frac{1}{\xi}, \quad c' = \frac{c}{\xi}, \quad \mu' = \frac{1}{\mu}; \quad (\epsilon)$$

or, what is the same thing,

$$\xi' = \xi, \quad c' = c, \quad \mu' = \mu; \quad (1)$$

$$\xi' = \xi, \quad c' = \xi, \quad \mu' = \xi/\mu; \quad (\alpha)$$

$$\xi' = \xi, \quad c' = \xi/\mu, \quad \mu' = \mu; \quad (\beta)$$

$$\xi' = \xi, \quad c' = c, \quad \mu' = 1/\mu; \quad (\gamma)$$

$$\xi' = c/\mu, \quad c' = c, \quad \mu' = \mu/c; \quad (\delta)$$

$$\xi' = c/\mu, \quad c' = c, \quad \mu' = 1/\mu; \quad (\epsilon)$$

then, whichever system of values of  $\xi, c, \mu$  be substituted for  $\xi, c, \mu$ , we have in each case identically the same secondary caustic, the effect of the substitutions being simply to interchange the different forms of the equation; and we have therefore identically the same caustic. By writing

$$(\xi, c, \mu) \rightarrow (\xi', c', \mu') = *(\xi, c, \mu),$$

where  $\alpha, \beta, \gamma, \delta, \epsilon$  will be functional symbols, such as are treated of in my paper "On the Theory of Groups as depending on the symbolic equation  $\theta^n = 1$ ," [125], and it is easy to verify the equations

$$1 = \alpha\beta = \beta\alpha = \gamma^2 = \delta\epsilon,$$

$$\alpha = \beta\gamma = \gamma\beta = \delta^2 = \epsilon\delta,$$

$$\beta = \alpha\gamma = \gamma\alpha = \epsilon\delta = \delta\epsilon,$$

$$\gamma = \alpha\beta = \beta\alpha = \delta\epsilon = \epsilon\delta,$$

$$\delta = \alpha\beta = \beta\alpha = \gamma\delta = \delta\gamma,$$

$$\epsilon = \alpha\delta = \delta\alpha = \gamma\epsilon = \epsilon\gamma.$$

Suppose, for example,  $\xi = c$ , i.e. let the radiant point be in the circumference; then in the fourth system  $\xi' = c$ ,  $c' = c$ ,  $\mu' = \frac{1}{\mu}$ , i.e. since the radius of the refracting circle is the radius of the circle and the index of refraction reciprocal, this is one of M. St Laurent's theorems, viz.

THEOREM. The caustic by refraction of a circle when the radiant point is in the circumference, is the same caustic by refraction for the same radiant circle and a reciprocal index of refraction. The first of these is a group of eighteen, viz.

Again, let  $\xi = c\mu$ , the fifth system gives  $\xi' = \frac{c}{\mu}$ ,  $c' = c$ ,  $\mu' = -1$ , or the new system is in this case also a reflecting system. This is the other of M. St Laurent's theorems, viz.

THEOREM. The caustic by reflection of a circle when the distance of the radiant point from the centre is equal to the radius of the circle multiplied by the index of refraction, is the caustic by reflection of the same circle for a radiant point which is the image of the first radiant point.

Of course it is to be understood that the image of a point means a point whose distance from the centre is equal to square of radius  $\div$  distance.

*2 Stone Buildings, Nov. 2, 1853.*

125.

ON THE THEORY OF GROUPS, AS DEPENDING ON THE SYMBOLIC EQUATION  $\theta^n = 1$ .

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 40–47.]

Let  $\theta$  be a symbol of operation, which may, if we please, have for its operand, not a single quantity  $x$ , but a system  $(x, y, \dots)$ , so that

$$\theta(x, y, \dots) = (x', y', \dots),$$

where  $x', y', \dots$  are any functions whatever of  $x, y, \dots$ , it is in not even necessary that  $x', y', \dots$  should be the same in number with  $x, y, \dots$ . In particular  $x', y', \dots$  may represent a permutation of  $x, y, \dots$ .  $\theta$  is in this case what is termed a substitution; and if, instead of a set  $x, y, \dots$ , we consider points or magnitudes (or, even, operations) any how related, then  $\theta$  may be an operator giving rise to a substitution, or it may be in particular a substitution, in like manner. The general question of the analytical representation of a substitution does not enter into the present plan, but the symbols  $\theta, \phi$ , etc.,  $\theta, \theta'$ , etc. are interpreted in such manner that the symbols  $\theta\phi, \phi\theta$ , etc. have a definite signification, but this need not always be the case; for instance, in the theory of matrices, the symbols  $\theta, \phi$ , etc. do not in general satisfy the commutative law  $\theta\phi = \phi\theta$ , so that,  $\theta\phi, \phi\theta$ , etc. are to be considered as in general different symbols. In other respects also the analogy is by no means perfect, and the words "operation" and "operand" are used as the only ones really suitable; the theory of operators, or as it may be called, the theory of operations, in the theory of analytical operations functional process which gives rise to the symbols in question, may be considered analogous to or having the same meaning as multiplied together so as to form a table, thus:—

| Further Symbols |          |               |               |     |
|-----------------|----------|---------------|---------------|-----|
|                 | 1        | $\alpha$      | $\beta$       | ... |
| 1               | 1        | $\alpha$      | $\beta$       | ... |
| $\alpha$        | $\alpha$ | $\alpha^2$    | $\alpha\beta$ | ... |
| $\beta$         | $\beta$  | $\beta\alpha$ | $\beta^2$     | ... |
| $\vdots$        | $\vdots$ | $\vdots$      | $\vdots$      |     |

that as well rows as columns of the square will contain all the symbols  $1, \alpha, \beta, \dots$ . It follows that when there is no entry of any symbol of the group without repetition, and by such order

whatever, is a symbol of the group. Suppose that this group

$$1, \alpha, \beta, \gamma, \dots,$$

contains a symbol  $\beta$  which may be such that there exists in the same group  $\theta^n = 1$ .<sup>3</sup>

satisfies the equation  $\theta^n = 1$ , where  $n$  is the smallest number for which this is the case, then we may speak of  $\theta$  as a composite symbol of any order  $n$  less than  $n$  a composite number. It follows then by a like reasoning as for  $\beta, \gamma$ , the system  $1, \alpha, \beta, \gamma$  is a subgroup of  $\theta$ . It follows from this, in particular, that there can be no system in which any of the others, but are associative,  $\theta(\phi\chi) = (\theta\phi)\chi$ . The expression of the symbol of an operation  $\theta$  is given by the symbol  $\theta = a + b\alpha + c\beta + \dots$ , the form of  $\theta = a + b\alpha + c\beta + \dots$ , the symbols at the same time being known are commutative or  $\theta\phi = \phi\theta$ , the system is said to be Abelian. There may be involved THEOREM. — If the symbols of the group be multiplied together two and two in any order, the entire system being known are not necessarily distinct. It may be done by means of the substitution equation  $\theta^n = 1$ , where  $n$  is some positive integer; this is the case of a system of all the roots of the equation  $x^n = 1$ , in the same notation, the symbols of the equation are

$$1, \alpha, \alpha^2, \dots, \alpha^{n-1} \quad (\alpha^n = 1).$$

This definition explains the meaning of the equation  $\theta^n = 1$ ; viz. this means therefore represent the group symbols, viz.

$$1, \alpha, \beta, \gamma, \dots, \quad (\beta^n = 1),$$

thus multiplying each term by a further factor, we have for the group  $1, \alpha, \alpha\beta, \alpha\alpha\beta$ , but we must be equal either to 1 or to one of the roots. In the former case our group is —

$$1, \alpha, \beta \quad (\alpha^n = 1).$$

which is analogous to the system of roots of the ordinary equation  $\theta^n - 1 = 0$ . For the sake of comparison with what follows, I remark, that representing the last- mentioned group by

$$1, \alpha, \beta, \gamma,$$

we have the table

|          | 1        | $\alpha$ | $\beta$  | $\gamma$ |
|----------|----------|----------|----------|----------|
| 1        | 1        | $\alpha$ | $\beta$  | $\gamma$ |
| $\alpha$ | $\alpha$ | $\beta$  | $\gamma$ | 1        |
| $\beta$  | $\beta$  | $\gamma$ | 1        | $\alpha$ |
| $\gamma$ | $\gamma$ | 1        | $\alpha$ | $\beta$  |

If, on the other hand,  $\alpha^2 = 1$ , then it is easy by similar reasoning to show that we must have  $\beta = \beta\alpha$ , so that the group is of the form

$$1, \alpha, \beta, \alpha\beta, \quad (\alpha^2 = 1, \beta^2 = 1, \alpha\beta = \beta\alpha),$$

or if expressed under the group by

$$1, \alpha, \beta, \gamma,$$

we have the table

---

<sup>3</sup>The idea of a group as applied to permutations or substitutions is due to Galois, and the introduction of it may be considered as marking an epoch in the progress of the theory of algebraical equations.

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|          | 1        | $\alpha$ | $\beta$  | $\gamma$ |
|----------|----------|----------|----------|----------|
| 1        | 1        | $\alpha$ | $\beta$  | $\gamma$ |
| $\alpha$ | $\alpha$ | 1        | $\gamma$ | $\beta$  |
| $\beta$  | $\beta$  | $\gamma$ | 1        | $\alpha$ |
| $\gamma$ | $\gamma$ | $\beta$  | $\alpha$ | 1        |

or, if we please, the symbols are such that

$$\alpha^2 = \beta^2 = \gamma^2 = 1,$$

$$\alpha = \beta\gamma = \gamma\beta, \quad \beta = \alpha\gamma = \gamma\alpha, \quad \gamma = \alpha\beta = \beta\alpha,$$

[and we have thus a group essentially distinct from that of the system of roots of the ordinary equation  $x^n - 1 = 0$ .]

Systems of this form are of frequent occurrence in analysis, and it is only on account of their extreme simplicity that they have not been expressly remarked. For instance, in the theory of elliptic functions, if  $u$  be the parameter, and

$$\alpha(u) = \frac{u}{u}, \quad \beta(u) = (-u), \quad \gamma(u) = -\frac{1}{u},$$

although, indeed, in this case (treating forms which take properly equivalent as identical) we have  $\alpha = \beta$ , and therefore  $\gamma = 1$ , in which point of view this group is simply a group of two symbols 1,  $\alpha$  ( $\alpha^2 = 1$ ).

Again, in the theory of matrices, if  $I$  denote the operation of inversion, and  $\text{tr}$  that of transposition, ( $I$  do not stop to explain in terms as the example may be passed over), we may write

$$\alpha = I, \quad \beta = \text{tr}, \quad \gamma = I \cdot \text{tr} = \text{tr} \cdot I.$$

I proceed to the case of a group of six symbols,

$$1, \alpha, \beta, \gamma, \delta, \epsilon,$$

which may be considered as representing a system of roots of the symbolic equation

$$\theta^6 = 1.$$

It is in the first place to be shown that there is at least one root which is a prime root of  $\theta^6 = 1$ , or (to use a simpler expression) a root having the index 3. It is clear that if there were a simpler expression a root having the index 2. It is true that if the index 3,  $\theta$  must be done by means of this root would have the index 2; then in the square of  $\theta$  would also have this index, that is, it would have the index 6 which is the cube root of the symbol. The symbol  $\theta = 1$  may always be considered as having the index 6, considered in the symbol  $\theta = 1$ , the index 3 of a root having the index 3. It is impossible that *all* the roots should have the index 3. This may be done by means of the simpler observation: if so, then, since their product has also the index 2 in the symbol  $\theta = 1$ , this is to have the index 6.

Assume that the symbols  $\alpha, \beta, \gamma$  have the index 3. The square of any one of this root, e.g.  $\alpha$ , is then  $\alpha^2$ , and these are contained in the group, must have either the index 6 or the index 2 (or to be a simpler expression) a root having the index 3, having each of the order 4. It is easy to see that any of the roots of  $\theta$ , and that the group of the order 4. It is easy to see that any of the roots of  $\theta$ , and that the group is, in fact,  $1, \gamma\delta, \delta\gamma, \delta\gamma^2$ , which is, in fact, the group of six having the index 3, having obtained ; I prefer commencing with the assumption of a root having the index 3. Suppose that  $\alpha$  is such a root, the group must clearly be of the form

$$1, \alpha, \alpha^2, \beta, \alpha\beta, \alpha^2\beta \quad (\alpha^3 = 1);$$

and multiplying the entire group by  $\gamma$  as a nearer factor, it becomes  $\gamma, \alpha\gamma, \alpha^2\gamma, \gamma', \alpha\gamma', \alpha^2\gamma'$ ; we must therefore have  $\beta\gamma = \alpha\beta$  or  $\gamma\beta = \alpha\beta$ . But the latter supposition  $\gamma' = \alpha\beta$  leads also to the same name in which case we may assume  $\gamma' = 1$ ; then we have  $\beta^2$  also equal to 1,  $\alpha$  or  $\alpha^2$ , but the first two cases lead to the supposition  $\gamma^2 = \alpha$  or  $\beta\gamma = 1$ , and to a system of six symbols  $1, \alpha, \alpha^2, \gamma\beta, \gamma\alpha\beta, \gamma\alpha^2\beta$ , when we must have either  $\gamma\alpha = \alpha\gamma$  or else  $\gamma\alpha = \alpha^2\gamma$ . The former assumption leads to the group

$$1, \alpha, \alpha^2, \gamma, \alpha\gamma, \alpha^2\gamma \quad (\alpha^3 = 1, \gamma^2 = 1, \alpha\gamma = \gamma\alpha),$$

which is, in fact, analogous to the system of roots of the ordinary equation  $x^n - 1 = 0$ . For the sake of comparison with what follows, I remark, that representing the last-mentioned group by the general symbols

$$1, \alpha, \beta, \gamma, \delta, \epsilon,$$

we have the table

|            | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ |
|------------|------------|------------|------------|------------|------------|------------|
| 1          | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ |
| $\alpha$   | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ | 1          |
| $\beta$    | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ | 1          | $\alpha$   |
| $\gamma$   | $\gamma$   | $\delta$   | $\epsilon$ | 1          | $\alpha$   | $\beta$    |
| $\delta$   | $\delta$   | $\epsilon$ | 1          | $\alpha$   | $\beta$    | $\gamma$   |
| $\epsilon$ | $\epsilon$ | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   |

while if we represent the second of these two forms, viz. the group

$$1, \alpha, \alpha^2, \gamma\beta, \gamma\alpha\beta, \gamma\alpha^2\beta, \quad (\alpha^3 = 1, \gamma^2 = 1, \gamma\alpha = \alpha^2\gamma),$$

by the same general symbols

$$1, \alpha, \beta, \gamma, \delta, \epsilon,$$

we have the table

|            | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ |
|------------|------------|------------|------------|------------|------------|------------|
| 1          | 1          | $\alpha$   | $\beta$    | $\gamma$   | $\delta$   | $\epsilon$ |
| $\alpha$   | $\alpha$   | $\beta$    | 1          | $\delta$   | $\epsilon$ | $\gamma$   |
| $\beta$    | $\beta$    | 1          | $\alpha$   | $\epsilon$ | $\gamma$   | $\delta$   |
| $\gamma$   | $\gamma$   | $\delta$   | $\epsilon$ | 1          | $\alpha$   | $\beta$    |
| $\delta$   | $\delta$   | $\epsilon$ | $\gamma$   | $\alpha$   | $\beta$    | 1          |
| $\epsilon$ | $\epsilon$ | $\gamma$   | $\delta$   | $\beta$    | 1          | $\alpha$   |

or, what is the same thing, the symbols are such that

$$\alpha^3 = \beta^3 = \gamma^2 = 1,$$

$$\alpha = \beta\gamma = \gamma\beta; \quad \beta = \alpha\gamma = \gamma\alpha,$$

$$\gamma = \delta\beta = \alpha\beta,$$

[and we have thus a group essentially distinct from that of the system of roots of the ordinary equation  $x^n - 1 = 0$ .]

Systems of this form are of frequent occurrence in analysis, and it is only on account of their extreme simplicity that they have not been expressly remarked. For instance, in the theory of elliptic functions, if  $u$  be the parameter,  $\alpha$  a fixed root, and tr denotes the operation of transposition, we have the group symbols which may be expressed in the form

$$\alpha(u) = \frac{u-c}{u-d}, \quad \beta(u) = (-u), \quad \gamma(u) = -\frac{1}{u},$$

although, indeed, in this case (treating forms which take properly equivalent as identical) we have  $\alpha\beta = \gamma$ , and although the small group  $\{1, \alpha, \beta, \gamma\}$  is simply a group of two symbols 1,  $\alpha$ , ( $\alpha^2 = 1$ ).

If we represent the second of two, and only two, essentially distinct forms of a group of six.

If we represent the second of these two forms, viz. the group

$$\begin{aligned} 1, \alpha, \beta, \alpha^2, \beta\alpha, \beta\alpha^2, \\ \alpha = \beta, \gamma = \beta\alpha, \delta = \beta\alpha^2, \end{aligned}$$

and we have thus two, and only two, essentially distinct forms of a group of six.

126.

ON THE THEORY OF GROUPS, AS DEPENDING ON THE SYMBOLIC EQUATION  $\theta^n = 1$ .  
SECOND PART.

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 408–409.]

IMAGINE the symbols

$$L, M, N, \dots$$

such that ( $L$  being any symbol of the system),

$$L^{-1}L = LL^{-1} = N, LN = N, \dots$$

is the group

$$1, \alpha, \beta, \dots,$$

then, in the first place,  $M$  being any other symbol of the system,  $M^{-1}LM, M^{-1}M, \dots$  will be the same group  $1, \alpha, \beta, \dots$ . In fact, the system  $L, M, N, \dots$  may be written  $L, L\alpha, L\beta, \dots$ , and so the system  $M, \beta M, \gamma M, \dots$  will be  $M^{-1}LM, M^{-1}\alpha LM = (M^{-1}LM)\alpha M, \dots$  which belongs to the group  $1, \alpha, \beta, \dots$ .

Next it may be shown that

$$LL', ML', NL', \dots$$

is a group, although  $LL'$  may not in general be the same group as  $1, \alpha, \beta, \dots$ . In fact, writing  $M = L\alpha, N = L\beta, \dots$ , the symbols just written down are

$$LL', L\alpha L', L\beta L', \dots,$$

and we have e.g.  $L\alpha L' = L\alpha L' = L\sigma L', = L\sigma L'$ , where  $\sigma$  belongs to the group  $1, \alpha, \beta, \dots$ ;

[Pages 130: continuation in similar group-theory style omitted in full transcription.]

127.

ON THE HOMOGRAPHIC TRANSFORMATION OF A SURFACE OF THE SECOND ORDER INTO ITSELF.

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 208–212; continuation of **122**.]

I pass to the original transformation. Sir W. R. Hamilton has given (in the note, p. 723 of his *Lectures on Quaternions* [Dublin, 1853]) the following theorem:—If there be a polygon of  $2\alpha$  sides inscribed in a surface of the second order, and  $(2\alpha - 1)$  of the sides pass through

given points, the side or  $(2\alpha - 1)$ th side will constantly touch two cones circumscribed about the surface of the second order. The relation between the extremities of the  $(2\alpha - 1)$ th side is that of two points connected by the general improper transformation; in other words, if there be a surface of the second order into itself, then have, as corresponding points, the opposite angles of a skew quadrangle lying upon the surface, and having the other two opposite angles upon given lines.

The tangent planes through  $P$  to the surface in question form an involution; the two sections which are the siliconugates of the involution pass through  $P$ , and having to consider the corresponding points  $\theta$ ,  $\phi$  in the same theorems, viz. that the two points which are the involution of  $\theta$  and  $\phi$ , the surface in the manner described in the result of  $PA$  as belonging determinately to one of the two systems of generating lines, then  $P$  is given, the corresponding point  $Q$  is, it is clear, completely determined. What precedes may be recapitulated in the statement, that in the improper transformation of a surface of the second order into itself, we have, as corresponding points, the opposite angles of a skew quadrangle lying upon the surface, and having the other two opposite angles upon given lines.

I proceed to establish (by M. Hermite's method) the formulæ for the particular case in question. The thing required is to find  $x_2, y_2, z_2, w_2$  linear functions of  $x_1, y_1, z_1, w_1$ , such that

$$\begin{aligned} ax_2^2 + by_2^2 + cz_2^2 + dw_2^2 &= ax_1^2 + by_1^2 + cz_1^2 + dw_1^2, \\ (ay_1y_2 + bz_1z_2 + cz_1z_2 + dw_1w_2)^2 - (ax_1^2 + by_1^2 + cz_1^2 + dw_1^2)(ax_2^2 + by_2^2 + cz_2^2 + dw_2^2) \\ &= Lx_1x_2 + My_1y_2 + Nz_1z_2 + Sw_1w_2, \end{aligned}$$

where  $L, M, N, S$  are constants of course, in the case in question functions of the constants  $a, b, c, d$  of the surface, and the linear functions of  $x_1, y_1, z_1, w_1$  which are to be the values of  $x_2, y_2, z_2, w_2$ . Following the analysis, in which  $L = 4, M = 4$ , the following system of equations, in which  $(a, b, c, d)$  of the points minimums, and  $M = ab + cd = 2bc + da$ , are obtained.

Following M. Hermite's method, the formulæ are obtained.

128.

#### DEVELOPMENTS ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED POLYGON.

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 339–345.]

I propose to develop some particular cases of the theorems given in my paper, "Correction of two Theorems relating to the Porism of the in-and-circumscribed Polygon" (*Phil. Mag.* vol. vi. [1853], 114). The two theorems are as follows:—

THEOREM. The condition that there may be inscribed in the conic  $U = 0$  an infinity of  $n$ -gons circumscribed about the conic  $V = 0$ , depends upon the developement in ascending powers of  $\xi$  of the square root of the discriminant of  $\xi U + V$ , viz. if this square root be

$$A + B\xi + C\xi^2 + D\xi^3 + E\xi^4 + F\xi^5 + G\xi^6 + H\xi^7 + \dots,$$

then for  $n = 3, 5, 7$ , &c. respectively, the conditions are

$$|C| = 0, \quad \begin{vmatrix} C & D \\ D & E \end{vmatrix} = 0, \quad \begin{vmatrix} C & D & E \\ D & E & F \\ E & F & G \end{vmatrix} = 0, \text{ \&c. ;}$$

and for  $n = 4, 6, 8$ , &c. respectively, the conditions are

$$|D| = 0, \quad \begin{vmatrix} D & E \\ E & F \end{vmatrix} = 0, \quad \begin{vmatrix} D & E & F \\ E & F & G \\ F & G & H \end{vmatrix} = 0, \text{ \&c.}$$



THEOREM. In the case where the conics are replaced by the two circles which (taken in their respective conjugates) are equivalent to a polygon inscribed in the one and circumscribed about the other, the values of  $x$  which give the position of  $\xi$  on which the dimensions are replaced by the

$$\alpha^2 + \beta^2 - R^2 = 0, \quad (x - c)^2 + y^2 - r^2 = 0,$$

then the discriminant, the square root of which appears as before, is

$$A + B\xi + C\xi^2 + D\xi^3 + E\xi^4 + \dots$$

Write for a moment

$$A = R\xi + (\xi^2 + D\xi^3 + E\xi^4 + \dots = \sqrt{(1 + \alpha\xi)(1 + \beta\xi)(1 + \gamma\xi)}),$$

then the discriminant, the square root of which appears as before, is

$$(1 + \xi)r^2 = \alpha + \beta + \gamma - 3\alpha\beta,$$

then

$$\begin{aligned} A &= 1, \\ 2B &= a + b + c, \\ -3C &= a^2 + b^2 + c^2 - 3bc, \\ 16D &= a^3 + b^3 + c^3 - 3a(b^2 + c^2 + bc), \\ -128E^2 &= a^4 + b^4 + c^4 - 4(b^3 + c^3 + abc^2 + a^2bc) + 6a^2bc. \end{aligned}$$

To adapt these to the case of the two circles, we have to write

$$\alpha^2(1 + \alpha\xi)(1 + \beta\xi)(1 + \gamma\xi) + 4r^2 = \beta + r^2(1 + \beta)\xi^2 + \frac{1}{4}r^2\xi^4,$$

and therefore

$$\begin{aligned} \alpha^2 &= 1, \\ \alpha^2(\alpha + \beta + c) &= -\xi^2, \end{aligned}$$

values which satisfy identically  $\alpha^2 + y^2 + z^2 + w^2 = c^2 + (a^2 + \alpha) - \beta + 3\alpha\beta$ , belong to an improper transformation. We have thus obtained the proper transformation of the surface of the second order  $ax^2 + by^2 + cz^2 + dw^2 = 0$  into itself.

Returning for a moment to the equations which lead to the surface  $\Sigma y - \Sigma \omega = 0$ , it is easy to see that they without loss of generality, in the points equations  $(x_1, y_1, z_1, w_1)$  and  $(x_2, y_2, z_2, w_2)$ , then,

The resulting system of equations is, in fact, identical with

$$ax_2^2 + by_2^2 + cz_2^2 + dw_2^2 = ax_1^2 + by_1^2 + cz_1^2 + dw_1^2,$$

and forming the values, the determined values of  $x_2, y_2, z_2, w_2$  are linear functions of  $x_1, y_1, z_1, w_1$  of the form

$$MM'z_2 = (aa' - bb' - cc' - dd')z_1 + (ab' + a'b + cd' - c'd)y_1 + (ac' + a'c - bd' + b'd)z_1 - (b'c - bc' + a'd + ad')w_1 + (bd' - b'd)$$

values which satisfy identically  $x_2^2 + y_2^2 + z_2^2 + w_2^2 = x_1^2 + y_1^2 + z_1^2 + w_1^2$ , and which belong, as is supposed, to an improper transformation. We have thus obtained the proper transformation of the surface of the second order  $ax^2 + by^2 + cz^2 + dw^2 = 0$  into itself.

*2 Stone Buildings, January 11, 1854.*

ON THE PORISM OF THE IN-AND-CIRCUMSCRIBED TRIANGLE,  
AND ON AN IRRATIONAL TRANSFORMATION OF TWO TERNARY QUADRATIC FORMS EACH  
INTO ITSELF.

[From the *Philosophical Magazine*, vol. vii. (1855), pp. 115–121.]

There is an irrational transformation of two ternary quadratic forms each into itself, based upon the solution of the following geometrical problem.

Given that the conic

$$lx + my + nz = 0$$

meets the conic

$$(a, b, c, f, g, h)(x, y, z)^2 = 0$$

in the points  $(x_0, y_0, z_0)$ ,  $(x_1, y_1, z_1)$ , to find the other point of intersection.

The solution is exceedingly simple. Take  $(a, b, c)$  for the coordinates of the other point of intersection, we must have identically with respect to  $x, y, z$ ,

$(a, \dots)(x, y, z)^2 - 2(\alpha, \dots)(x_0, y_0, z_0)(x, y, z) = (lx + my + nz)(\beta_0(x - x_0) + \beta_1(y - y_0) + \beta_2(z - z_0))$ ,  
to a constant factor *près*. Assume successively  $y = 0, z = 0; x = 0, z = 0; x = 0, y = 0$ , then we have

$$\begin{aligned} l : m : n &= y_0 z_0(\alpha x_0 + g z_0 + h y_0) - x_0(a x_0 + h y_0 + g z_0) : \\ & z_0 x_0(h x_0 + b y_0 + f z_0) - y_0(h x_0 + b y_0 + f z_0) : \\ & x_0 y_0(g x_0 + f y_0 + c z_0) - z_0(g x_0 + f y_0 + c z_0), \end{aligned}$$

or as they may be more simply written,

$$l : m : n = y_0 z_0(ax_0^2 - by_0^2 - cz_0^2 - 2fy_0 z_0) : z_0 x_0(\dots) : x_0 y_0(\dots).$$

[Pages 146–147: development of the transformation theory.]

We have, then, identically

$lx_0(lx + my + nz) + xx_0(my - nz)(xz - yx_0) + y_0(mz - ny)(xy - zx_0) + lz_0(\dots) - hx_0 z_0(\dots) = 0$ ,  
which shows that if  $\theta_2, \eta_2$  may be as well to give the corresponding solution of the problem.

[Pages 147–148: further reductions; the irrational transformation

$$\sqrt{lx + my + nz} \quad \text{and} \quad \sqrt{(a, \dots)(x, y, z)^2}$$

provide an irrational transformation of the two forms each into itself.]

I shall once consider  $l, m, n$  as given functions of  $x_0, y_0, z_0$ , satisfying identically the equations

$$lx_0 + my_0 + nz_0 = 0, \quad Nc + na = 0, \quad xx_0 + yy_0 + zz_0 = 0,$$

equations which express that the line  $lx + my + nz = 0$  is the tangent at the point  $(x_0, y_0, z_0)$  to the conic  $(a, b, c, \dots)(x, y, z)^2 = 0$ . And I shall take for  $\beta, \gamma$  the following constant values, viz. in the case of the point  $(x, y, z)$  we shall have for the values:

$$\theta_1 = \theta_2 - q_0, (2qa - bx), \quad \eta_1 = \eta_2 - q_0, (2q\beta - ay), \quad \zeta_1 = \zeta_2 - q_0, (2q\gamma - cz),$$

and similarly for the values  $\theta_2, \eta_2, \zeta_2$  in the same point. We obtain immediately

$$\theta_2^2 + \eta_2^2 + \zeta_2^2 = \theta_1^2 + \eta_1^2 + \zeta_1^2.$$

[Pages 149–150: continuation through the proof.]

The preceding investigations have been in my possession for about eighteen months.

*2 Stone Buildings, April 18, 1855.*

## DEUXIÈME MÉMOIRE SUR LES FONCTIONS DOUBLEMENT PÉRIODIQUES.

[From the *Journal de Mathématiques Pures et Appliquées* (Liouville), tom. xix. (1854), pp. 193–208 ; Sequel to Mémoire t. x. (1845), **25**.]

Je vais essayer de développer ici les propriétés qui se rapportent aux transformations linéaires des polynômes des fonctions  $p, g, \Theta, Z$ , dont je me suis occupé dans le Mémoire sur les fonctions doublement périodiques que j'ai donné dans ce Recueil en 1845. Avant d'entrer en matière, je remarque que partant des expressions

$$\Omega = u + u', \quad \Upsilon = v + v',$$

des deux périodes, on a  $i = \sqrt{-1}$ , on obtient, en définissant

$$\Omega' = u\nu - u'\nu', \quad \Upsilon' = u'\nu' - u\nu',$$

les équations

$$\begin{aligned} \Omega \Upsilon' &= u\nu + u'\nu' = -(\nu\Omega - \nu'\Omega'), \\ \Upsilon \Omega' &= u\nu' + u'\nu = (\nu\Omega + \nu'\Omega'), \end{aligned}$$

et au moyen desquelles

$$B = \frac{\nu(\nu\Omega - \nu'\Omega')}{i\Upsilon \bmod. (\nu\nu - \nu'\nu')}, \quad B' = \frac{\nu'(\nu\Omega + \nu'\Omega')}{i\Upsilon \bmod. (\nu\nu' - \nu'\nu)},$$

des quantités  $B, B'$  on déduit les formules

$$B + \beta = \frac{\nu\Upsilon}{i \bmod. (\nu\nu' - \nu'\nu)}\Omega, \quad B - B' = \frac{\nu\nu'}{i \bmod. (\nu\nu' - \nu'\nu)}.$$

Je me mets attention aux transformations qui correspondent à des entiers impairs et premiers, et je suppose, de plus, que les transformations soient toujours propres, et régulières ; c'est-à-dire qu'on doit avoir

$$\begin{aligned} (2k+1)\Omega_r &= \lambda\Omega + \mu\Upsilon, \quad \bmod. (\mu, \nu) < T, \\ (2k+1)\Upsilon_r &= \nu\Omega + \rho\Upsilon, \quad \bmod. (\nu, \rho) < T, \end{aligned}$$

où  $2k+1$  est un nombre impair et premier,  $\lambda, \mu, \nu, \rho$  sont des entiers impairs ou pairs, et cômptes de tel signe près,  $\lambda\rho - \mu\nu = 2k+1$ , et que les coefficients sont, savoir, soit signe près,

$$\lambda\rho - \mu\nu = 2k+1.$$

(condition pour que la transformation soit propre), et de même que les fonctions

$$\lambda \equiv 0, \quad \mu \equiv 0, \quad \bmod. 2,$$

$$\nu \equiv 0, \quad \rho \equiv 1.$$

(condition pour que la transformation soit régulière).

On trouve tout de suite

$$\Omega = \rho\Omega_r - \mu\Upsilon_r, \quad \Upsilon = -\nu\Omega_r + \lambda\Upsilon_r,$$

j'écris aussi

$$\Omega_r = u_r + u'_r, \quad \Upsilon_r = v_r + v'_r,$$

et je suppose que  $R_r, \beta_r$  soient des fonctions de  $u_r, u'_r, v_r, v'_r$  analogues aux fonctions  $B, \beta$ , de  $u, u', v, v'$ .

Cela étant, je tire d'abord de l'équation

$$(2k+1)(B+\beta)\Omega = \frac{\nu}{i \bmod. (\nu\nu - \nu'\nu')} \Omega^2,$$

au moyen de laquelle

$$(B+\beta)\Omega = \frac{\nu}{i \bmod. (\nu\nu' - \nu'\nu)} \Omega^2,$$

au transformant en

$$\begin{aligned} [(2k+1)(B_r+\beta_r)\Omega &= \frac{\nu_r}{i \bmod. (\nu_r\nu' - \nu'_r\nu)} \Omega_r^2, \\ &= \frac{\nu_r}{i \bmod. (\nu_r\nu' - \nu'_r\nu)} (\rho\Omega_r - \mu\Upsilon_r)^2, \\ &= \frac{\rho^2\nu - 2\rho\mu(\nu + \nu') + \mu^2\nu'}{i \bmod. (\nu_r\nu' - \nu'_r\nu)} (\rho\Omega_r - \mu\Upsilon_r). \end{aligned}$$

[Pages 152–156: continuation; the second Mémoire develops the expression of  $J_rx, g_rx, Z_rx$  in terms of the original  $Jx, gx, Zx$ , the products

$$J_rx = e^{-\frac{1}{2k+1}(B_r-2k+1\cdot B)x\pi i} \prod \frac{J(x+s\psi_r)}{J(s\psi_r)},$$

the cases “Premier cas” ( $p_r = 1, q_r = 2q_\theta$ ) and “Deuxième cas” ( $p_r = 2k+1, q_r = 2$ ), and the corresponding  $\Omega_r, \Upsilon_r$  formulæ. The exposition continues in [157–208] of this same Mémoire 130.]

Cela fait voir que

$$e^{-i\pi x} = e^{-i\pi x},$$

le coefficient  $A$  étant donné au moyen de l'équation

$$A = \sum \frac{1}{(m, n)},$$

où la somme est prise, comme auparavant, entre les limites

$$\bmod. (m, n) < T, \quad T = \infty.$$

Mais il n'est pas permis d'écrire

$$A = \iint \frac{dx dy}{(m, n)^2}.$$

En effet, cette intégrale n'est pas le premier terme d'une suite dont il faudrait, pour obtenir un résultat exact, prendre deux termes; le second terme de la suite serait une intégrale prise le long de tout le contour, et le second terme ne serait pas, comme dans le premier cas, égale à zéro. Pour trouver la valeur de  $A$ , je remarque que les fonctions doublement périodiques  $p, g, \Theta, Z$ , qui sont contenues dans les formules ordinaires, peuvent satisfaire aux conditions

$$\bmod. (m, n) < T, \quad T = \infty,$$

puisqu'on a l'analyse  $T = x$ , et puis l'on obtient que la fonction  $Ly + My^2$ . Cela étant, on substituera pour  $u$  la valeur, je forme l'équation

$$\frac{y(x+s\psi)}{y(x)} = e^{-bx-cx^2} \Pi\left(1 + \frac{x}{(m, n)+s\psi}\right),$$

et j'écris successivement

$$y = \frac{1}{2}\Pi, \quad y = \frac{1}{2}T,$$

et qui donne pour les équations correspondante du produit infini double doublement  $e^{i\pi x}$ ,  $Gx$ ; en comparant les valeurs ainsi obtenues avec les équations qui donnent les valeurs de  $y(x + \frac{1}{2}\Pi)$ ,  $y(x + \frac{1}{2}T)$ , on trouve

$$L = 0, \quad M = \frac{\pi}{\text{mod.}(\nu\nu' - \nu'\nu)},$$

et enfin,

$$\frac{y(x + s\psi)}{y(x)} = e^{-\frac{\pi}{\text{mod.}(\nu\nu' - \nu'\nu)}\Pi(1 + \frac{x}{(m,n) + s\psi})},$$

$$\text{mod.}(m, n) < T, \quad T = \infty.$$

Cela donne pour l'équation qu'il s'agissait d'établir; il ne faut pas remarquer qu'il y ait remarque que pour  $y = 0$ , on doit considérer à part le facteur  $1 + \frac{x}{(m,n)}$ , lequel multiplié par  $y(x)$  devient tout simplement  $x$ ; l'équation réduite donc dans ce cas.

# Cayley's Collected Mathematical Papers, Vol. II, pp. 141–184

Reconstruction with OCR skeleton and PNG verification

## 128. (continued)

### Developments on the Porism of the In-and-Circumscribed Polygon (continued).

[From the *Philosophical Magazine*, vol. vii. (1854), pp. 339–345.]

(Continuation from p. 140.)

which, he remarks, is satisfied by  $r = -p$  and  $r = -\frac{pq}{p+q}$ , and that consequently the rationalized equation will divide by  $p+r$  and  $pq-r(p+q)$ ; and he finds, after the division,

$$p^2q^2 + p^2q^2(p+q)r - pq(p+q)^2r^2 - (p+q)(p-q)^2r^3 = 0,$$

which, restoring for  $p, q$  their values  $R+a, R-a$ , is the very equation above found.

The form given by Steiner (*Crelle*, t. ii. p. 289) is

$$r(R-a)^2 = (R+a)\sqrt{(R-r+a)(R-r-a)} + (R+a)\sqrt{(R-r-a) \cdot 2R},$$

which, putting  $p, q$  instead of  $R+a, R-a$ , is

$$qr = p\sqrt{(p-r)(q-r)} + p\sqrt{(q-r)(q+p)};$$

and Jacobi has shown in his memoir, “Anwendung der elliptischen Transcendenten u. s. w.,” *Crelle*, t. iii. [1828] p. 376, that the rationalized equation divides (like that of Fuss) by the factor  $pq - (p+q)r$ , and becomes by that means identical with the rational equation given by Fuss.

In the case of two concentric circles  $a = 0$ , and putting  $\frac{r}{R} = M$ , we have

$$\sqrt{1-M} = \sqrt{1-M}.$$

This is, in fact, the very formula which corresponds to the general case of two conics having double contact. For suppose that the polygon is inscribed in the conic  $U = 0$ , and circumscribed about the conic  $U + P^2 = 0$ , we have then to find the discriminant of  $\xi U + U + P^2$ , i.e. of  $(1+\xi)U + P^2$ . Let  $K$  be the discriminant of  $U$ , and let  $F$  be what the polar reciprocal of  $U$  becomes when the variables are replaced by the coefficients of  $P$ , or, what is the same thing, let  $-F$  be the determinant obtained by bordering  $K$  (considered as a matrix) with the coefficients of  $P$ . The discriminant of  $(1+\xi)U + P^2$  is  $(1+\xi)^3K + (1+\xi)^2F$ , i.e. it is

$$(1+\xi)^2K\left[(1+\xi) + \frac{F}{K}\right], \quad \text{i.e. } (1+\xi)^2K(1+M\xi),$$

where  $M = 1 + \frac{F}{K}$ ; or, what is the same thing,  $M$  is the discriminant of  $U$  divided by the discriminant of  $U + P^2$ . And  $M$  having this meaning, the condition of there being inscribed in the conic  $U$  an infinity of  $n$ -gons circumscribed about the conic  $U + P^2 = 0$ , is found by means of the series

$$A + B\xi + C\xi^2 + D\xi^3 + E\xi^4 + \&c. = (1+\xi)\sqrt{1+M\xi}.$$

We have, therefore,

$$\begin{aligned}
A &= 1, \\
2B &= M + 2, \\
8C &= -M^2 + 4M, \\
16D &= M^3 - 2M^2, \\
-128E &= 5M^4 - 8M^3, \\
&\&c., \\
1024(CE - D^2) &= M^4(M^2 - 12M + 16), \\
&\&c.
\end{aligned}$$

Hence for the triangle, quadrangle and pentagon, the conditions are

**I.** For the triangle,

$$M + 2 = 0.$$

**II.** For the quadrangle,

$$M - 4 = 0.$$

**III.** For the pentagon,

$$M^2 - 12M + 16 = 0;$$

and so on.

It is worth noticing, that, in the case of two conics having a 4-point contact, we have  $F = 0$ , and consequently  $M = 1$ . The discriminant is therefore  $(1 + \xi)^3$ , and as this does not contain any variable parameter, the conics cannot be determined so that there may be for a given value of  $n$  (nor, indeed, for any value whatever of  $n$ ) an infinity of  $n$ -gons inscribed in the one conic, and circumscribed about the other conic.

The geometrical properties of a triangle, &c. inscribed in a conic and circumscribed about another conic, these two conics having double contact with each other, are at once obtained from those of the system in which the two conics are replaced by concentric circles. Thus, in the case of a triangle, if  $ABC$  be the triangle, and  $\alpha, \beta, \gamma$  be the points of contact of the sides with the inscribed conic, then the tangents to the circumscribed conic at  $A, B, C$  meet the opposite sides  $BC, CA, AB$  in points lying in the chord of contact, the lines  $A\alpha, B\beta, C\gamma$  meet in the pole of contact, and so on.

In the case of a quadrangle, if  $ACEG$  be the quadrangle, and  $b, d, f, h$  the points of contact with the inscribed conic, then the tangents to the circumscribed conic at the pair of opposite angles  $A, E$  and the corresponding diagonal  $CG$ , and in like manner the tangents at the pair of opposite angles  $C, G$  and the corresponding diagonal  $AE$ , meet in the chord of contact. Again, the pairs of opposite sides  $AC, EG$ , and the line  $dh$  joining the points of contact of the other two sides with the inscribed conic, and the pairs of opposite sides  $AG, CE$ , and the line  $hf$  joining the points of contact of the other two sides with the inscribed conic, meet in the chord of contact. The diagonals  $AE, CG$ , and the lines  $bf, dh$  through the points of contact of pairs of opposite sides with the inscribed conic, meet in the pole of contact, &c.

The beautiful systems of ‘focal relations’ for regular polygons (in particular for the pentagon and the hexagon), given in Sir W. R. Hamilton’s *Lectures on Quaternions*, [Dublin, 1853] Nos. 379–393, belong, it is clear, to polygons which are inscribed in and circumscribed about two conics having double contact with each other. In fact, the focus of a conic is a point such that the lines joining such point with the circular points at infinity (i.e. the points in which a circle

is intersected by the line infinity) are tangents to the conic. In the case of two concentric circles, these are to be considered as touching in the circular points at infinity; and consequently, when the concentric circles are replaced by two conics having double contact, the circular points at infinity are replaced by the points of contact of the two conics.

Thus, in the figure (which is simply Sir W. R. Hamilton's figure 81 put into perspective), the system of relations

$$\begin{aligned} F, G(..) ABCI, \\ G, H(..) BCDK, \\ H, I(..) CDEF, \\ I, K(..) DEAG, \\ K, F(..) EABH, \end{aligned}$$

will mean,  $F, G(..) ABCI$ , that there is a conic inscribed in the quadrilateral  $ABCI$  such that the tangents to this conic through the points  $F$  and  $G$  pass two and two through the points of contact of the circumscribed and the inscribed conics, and similarly for the other relations of the system. As the figure is drawn, the tangents in question are of course (as the tangents through the foci in the case of the two concentric circles) imaginary.

*2 Stone Buildings, March 7, 1854.*

## 129.

### On the Porism of the In-and-Circumscribed Triangle, and on an Irrational Transformation of Two Ternary Quadratic Forms each into itself.

[From the *Philosophical Magazine*, vol. ix. (1855), pp. 513–517.]

There is an irrational transformation of two ternary quadratic forms each into itself, based upon the solution of the following geometrical problem,

*Given that the line*

$$lx + my + nz = 0$$

*meets the conic*

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0$$

*in the point  $(x_1, y_1, z_1)$ ; to find the other point of intersection.*

The solution is exceedingly simple. Take  $(x_2, y_2, z_2)$  for the coordinates of the other point of intersection, we must have identically with respect to  $x, y, z$ ,

$$\begin{aligned} (a, \dots \mid x, y, z)^2 \cdot (\mathfrak{A}, \dots \mid l, m, n)^2 - k(lx + my + nz)^2 \\ = (a, \dots \mid x_1, y_1, z_1 \mid x, y, z) \cdot (a, \dots \mid x_2, y_2, z_2 \mid x, y, z) \end{aligned}$$

to a constant factor près.

Assume successively  $x, y, z = \mathfrak{A}, \mathfrak{H}, \mathfrak{G}; \mathfrak{H}, \mathfrak{B}, \mathfrak{F}; \mathfrak{G}, \mathfrak{F}, \mathfrak{C}$ ; it follows that

$$\begin{aligned} x_2 : y_2 : z_2 &= y_1 z_1 [\mathfrak{A}(a, \dots \mid l, m, n)^2 - (\mathfrak{A}l + \mathfrak{H}m + \mathfrak{G}n)^2] \\ &: z_1 x_1 [\mathfrak{B}(a, \dots \mid l, m, n)^2 - (\mathfrak{H}l + \mathfrak{B}m + \mathfrak{F}n)^2] \\ &: x_1 y_1 [\mathfrak{C}(a, \dots \mid l, m, n)^2 - (\mathfrak{G}l + \mathfrak{F}m + \mathfrak{C}n)^2]; \end{aligned}$$



or, what is the same thing,

$$\begin{aligned} x_2 : y_2 : z_2 &= y_1 z_1 (bn^2 + cm^2 - 2fmn) \\ &: z_1 x_1 (cl^2 + an^2 - 2gnl) \\ &: x_1 y_1 (am^2 + bl^2 - 2hlm). \end{aligned}$$

It is not necessary for the present purpose, but it may be as well to give the corresponding solution of the problem

*Given that one of the tangents through the point  $(\xi, \eta, \zeta)$  to the conic*

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0$$

*is the line  $l_1 x + m_1 y + n_1 z = 0$ ; to find the equation to the other tangent.*

Let  $l_2 x + m_2 y + n_2 z = 0$  be the other tangent, then

$$\begin{aligned} (a, \dots \mid \xi, \eta, \zeta)^2 \cdot (a, \dots \mid x, y, z)^2 - [(a, \dots \mid \xi, \eta, \zeta \mid x, y, z)]^2 \\ = (l_1 x + m_1 y + n_1 z)(l_2 x + m_2 y + n_2 z) \end{aligned}$$

to a constant factor près. Assume successively  $y = 0, z = 0$ ;  $z = 0, x = 0$ ;  $x = 0, y = 0$ ; then we have

$$\begin{aligned} l_2 : m_2 : n_2 &= m_1 n_1 [a(a, \dots \mid \xi, \eta, \zeta)^2 - (a\xi + h\eta + g\zeta)^2] \\ &: n_1 l_1 [b(a, \dots \mid \xi, \eta, \zeta)^2 - (h\xi + b\eta + f\zeta)^2] \\ &: l_1 m_1 [c(a, \dots \mid \xi, \eta, \zeta)^2 - (g\xi + f\eta + c\zeta)^2]; \end{aligned}$$

or, as they may be more simply written,

$$\begin{aligned} l_2 : m_2 : n_2 &= m_1 n_1 (\mathfrak{B}\zeta^2 + \mathfrak{C}\eta^2 - 2\mathfrak{F}\eta\zeta) \\ &: n_1 l_1 (\mathfrak{C}\xi^2 + \mathfrak{A}\zeta^2 - 2\mathfrak{G}\xi\zeta) \\ &: l_1 m_1 (\mathfrak{A}\eta^2 + \mathfrak{B}\xi^2 - 2\mathfrak{H}\xi\eta). \end{aligned}$$

Returning now to the solution of the first problem, I shall for the sake of simplicity consider the formulæ obtained by taking for the equation of the conic,

$$\alpha x^2 + \beta y^2 + \gamma z^2 = 0.$$

We see, therefore, that if this conic be intersected by the line  $lx + my + nz = 0$  in the points  $(x_1, y_1, z_1)$  and  $(x_2, y_2, z_2)$ , then

$$\begin{aligned} x_2 : y_2 : z_2 &= y_1 z_1 (\gamma m^2 + \beta n^2) \\ &: z_1 x_1 (\alpha n^2 + \gamma l^2) \\ &: x_1 y_1 (\beta l^2 + \alpha m^2). \end{aligned}$$

We have, in fact, identically

$$\begin{aligned} ly_1 z_1 (\beta n^2 + \gamma m^2) + mz_1 x_1 (\gamma l^2 + \alpha n^2) + nx_1 y_1 (\alpha m^2 + \beta l^2) \\ = (\alpha mn x_1 + \beta nly_1 + \gamma lmx_1)(lx_1 + my_1 + nz_1) - lmn(\alpha x_1^2 + \beta y_1^2 + \gamma z_1^2). \end{aligned}$$

Also,

$$\begin{aligned} & \alpha y_1^2 z_1^2 (\beta n^2 + \gamma m^2)^2 + \beta z_1^2 x_1^2 (\gamma l^2 + \alpha n^2)^2 + \gamma x_1^2 y_1^2 (\alpha m^2 + \beta l^2)^2 \\ &= \alpha \beta \gamma [-l^2 x_1^2 - m^2 y_1^2 - n^2 z_1^2 \\ &+ (my_1 + nz_1)^2 l^2 x_1^2 + (nz_1 + lx_1)^2 m^2 y_1^2 + (lx_1 + my_1)^2 n^2 z_1^2 - 2lmnx_1 y_1 z_1 (lx_1 + my_1 + nz_1)] \\ &- (l^2 \beta \gamma x_1^2 + m^2 \gamma \alpha y_1^2 + n^2 \alpha \beta z_1^2)(\alpha x_1^2 + \beta y_1^2 + \gamma z_1^2); \end{aligned}$$

which show that if  $lx_1 + my_1 + nz_1 = 0$  and  $\alpha x_1^2 + \beta y_1^2 + \gamma z_1^2 = 0$ , then also  $lx_2 + my_2 + nz_2 = 0$  and  $\alpha x_2^2 + \beta y_2^2 + \gamma z_2^2 = 0$ : this is, of course, as it should be.

I shall now consider  $l, m, n$  as given functions of  $x_1, y_1, z_1$  satisfying identically the equations

$$\begin{aligned} lx_1 + my_1 + nz_1 &= 0, \\ l^2 bc + m^2 ca + n^2 ab &= 0, \end{aligned}$$

equations which express that  $lx + my + nz = 0$  is the tangent from the point  $(x_1, y_1, z_1)$  to the conic  $ax^2 + by^2 + cz^2 = 0$ . And I shall take for  $\alpha, \beta, \gamma$  the following values, viz.

$$\begin{aligned} \alpha &= ax_1^2 + by_1^2 + cz_1^2 - a(x_1^2 + y_1^2 + z_1^2), \\ \beta &= ax_1^2 + by_1^2 + cz_1^2 - b(x_1^2 + y_1^2 + z_1^2), \\ \gamma &= ax_1^2 + by_1^2 + cz_1^2 - c(x_1^2 + y_1^2 + z_1^2); \end{aligned}$$

so that  $x_1, y_1, z_1$  continuing absolutely indeterminate, we have identically  $\alpha x_1^2 + \beta y_1^2 + \gamma z_1^2 = 0$ . Also taking  $\Theta$  as a function of  $x_1, y_1, z_1$ , the value of which will be subsequently given, I write

$$\begin{aligned} x_2 &= \Theta y_1 z_1 (\beta n^2 + \gamma m^2), \\ y_2 &= \Theta z_1 x_1 (\gamma l^2 + \alpha n^2), \\ z_2 &= \Theta x_1 y_1 (\alpha m^2 + \beta l^2); \end{aligned}$$

so that  $x_1, y_1, z_1$  are arbitrary, and  $x_2, y_2, z_2$  are taken to be determinate functions of  $x_1, y_1, z_1$ . The point  $(x_2, y_2, z_2)$  is geometrically connected with the point  $(x_1, y_1, z_1)$  as follows, viz.  $(x_2, y_2, z_2)$  is the point in which the tangent through  $(x_1, y_1, z_1)$  to the conic  $ax^2 + by^2 + cz^2 = 0$  meets the conic passing through the point  $(x_1, y_1, z_1)$  and the points of intersection of the conics  $ax^2 + by^2 + cz^2 = 0$  and  $x^2 + y^2 + z^2 = 0$ . Consequently, in the particular case in which  $(x_1, y_1, z_1)$  is a point on the conic  $x^2 + y^2 + z^2 = 0$ , the point  $(x_2, y_2, z_2)$  is the point in which this conic is met by the tangent through  $(x_1, y_1, z_1)$  to the conic  $ax^2 + by^2 + cz^2 = 0$ .

It has already been seen that  $lx_1 + my_1 + nz_1 = 0$  and  $\alpha x_1^2 + \beta y_1^2 + \gamma z_1^2 = 0$  identically; consequently we have identically  $lx_2 + my_2 + nz_2 = 0$  and  $\alpha x_2^2 + \beta y_2^2 + \gamma z_2^2 = 0$ . The latter equation, written under the form

$$(ax_1^2 + by_1^2 + cz_1^2)(x_2^2 + y_2^2 + z_2^2) - (x_1^2 + y_1^2 + z_1^2)(ax_2^2 + by_2^2 + cz_2^2) = 0,$$

shows that if  $x_2, y_2, z_2$  are such that  $x_2^2 + y_2^2 + z_2^2 = x_1^2 + y_1^2 + z_1^2$ , then that also  $ax_2^2 + by_2^2 + cz_2^2 = ax_1^2 + by_1^2 + cz_1^2$ . I proceed to determine  $\Theta$  so that we may have  $x_2^2 + y_2^2 + z_2^2 = x_1^2 + y_1^2 + z_1^2$ . We obtain immediately

$$\begin{aligned} \frac{1}{\Theta^2}(x_2^2 + y_2^2 + z_2^2) &= (l^2 x_1^2 + m^2 y_1^2 + n^2 z_1^2)(\alpha^2 x_1^2 + \beta^2 y_1^2 + \gamma^2 z_1^2) \\ &- (\alpha^2 l^2 x_1^4 + \beta^2 m^2 y_1^4 + \gamma^2 n^2 z_1^4 - 2\beta\gamma mnl^2 y_1^2 z_1^2 - 2\gamma\alpha nlm^2 z_1^2 x_1^2 - 2\alpha\beta lmn^2 x_1^2 y_1^2); \end{aligned}$$

write for a moment

$$ax_1^2 + by_1^2 + cz_1^2 = p, \quad x_1^2 + y_1^2 + z_1^2 = q, \quad \text{so that} \quad \alpha = p - aq, \quad \beta = p - bq, \quad \gamma = p - cq,$$

then

$$\begin{aligned}\alpha^2 x_1^2 + \beta^2 y_1^2 + \gamma^2 z_1^2 &= qp^2 - 2p \cdot pq + (a^2 x_1^2 + b^2 y_1^2 + c^2 z_1^2) q^2 \\ &= q \{ (a^2 x_1^2 + b^2 y_1^2 + c^2 z_1^2) q - p^2 \} \\ &= q \{ (b-c)^2 y_1^2 z_1^2 + (c-a)^2 z_1^2 x_1^2 + (a-b)^2 x_1^2 y_1^2 \}.\end{aligned}$$

Moreover,

$$\begin{aligned}&\alpha^2 l^2 x_1^4 + \beta^2 m^2 y_1^4 + \gamma^2 n^2 z_1^4 - 2\beta\gamma mn l^2 y_1^2 z_1^2 - 2\gamma\alpha n l m^2 z_1^2 x_1^2 - 2\alpha\beta l m n^2 x_1^2 y_1^2 \\ &= p^2 \{ l^2 x_1^4 + m^2 y_1^4 + n^2 z_1^4 - 2mny_1^2 z_1^2 - 2nlz_1^2 x_1^2 - 2lmx_1^2 y_1^2 \} \\ &\quad - 2pq \{ al^2 x_1^4 + bm^2 y_1^4 + cn^2 z_1^4 - (b+c)mny_1^2 z_1^2 - (c+a)nlz_1^2 x_1^2 - (a+b)lmx_1^2 y_1^2 \} \\ &\quad + q^2 \{ a^2 l^2 x_1^4 + b^2 m^2 y_1^4 + c^2 n^2 z_1^4 - 2bcmny_1^2 z_1^2 - 2canlz_1^2 x_1^2 - 2ablmx_1^2 y_1^2 \},\end{aligned}$$

the first line of which vanishes in virtue of the equation  $lx_1 + my_1 + nz_1 = 0$ ; we have therefore

$$\begin{aligned}\frac{1}{\Theta^2} (x_2^2 + y_2^2 + z_2^2) - (x_1^2 + y_1^2 + z_1^2) &= (l^2 x_1^2 + m^2 y_1^2 + n^2 z_1^2) \{ (b-c)^2 y_1^2 z_1^2 + (c-a)^2 z_1^2 x_1^2 + (a-b)^2 x_1^2 y_1^2 \} \\ &\quad + 2pq \{ \dots \} - q^2 \{ \dots \}.\end{aligned}$$

Hence reducing the function on the right-hand side, and putting

$$\frac{1}{\Theta^2} (x_2^2 + y_2^2 + z_2^2) \div (x_1^2 + y_1^2 + z_1^2) = 1,$$

we have

$$\begin{aligned}\frac{1}{\Theta^2} &= a^2 l^4 x_1^4 + b^2 m^4 y_1^4 + c^2 n^4 z_1^4 \\ &\quad + (c^2 m^4 - 2b^2 m^2 n^2) y_1^2 z_1^2 + (a^2 n^4 - 2c^2 n^2 l^2) z_1^2 x_1^2 + (b^2 l^4 - 2a^2 l^2 m^2) x_1^2 y_1^2 \\ &\quad + (b^2 n^4 - 2c^2 m^2 n^2) y_1^2 z_1^2 + (c^2 l^4 - 2a^2 n^2 l^2) z_1^2 x_1^2 + (a^2 m^4 - 2b^2 l^2 m^2) x_1^2 y_1^2 \\ &\quad + \{ l^4 (b-c)^2 + m^4 (c-a)^2 + n^4 (a-b)^2 \\ &\quad + 2m^2 n^2 (bc - ca - ab) + 2n^2 l^2 (-bc + ca - ab) + 2l^2 m^2 (-bc - ca + ab) \} x_1^2 y_1^2 z_1^2.\end{aligned}$$

The value of  $\Theta$  might probably be expressed in a more simple form by means of the equations  $lx_1 + my_1 + nz_1 = 0$  and  $l^2 bc + m^2 ca + n^2 ab = 0$ , even without solving these equations; but this I shall not at present inquire into.

Recapitulating,  $l, m, n$  are considered as functions of  $x_1, y_1, z_1$  determined (to a common factor près) by the equations

$$\begin{aligned}lx_1 + my_1 + nz_1 &= 0, \\ l^2 bc + m^2 ca + n^2 ab &= 0;\end{aligned}$$

$\Theta$  is determined as above, and then writing

$$\begin{aligned}\alpha &= ax_1^2 + by_1^2 + cz_1^2 - a(x_1^2 + y_1^2 + z_1^2), \\ \beta &= ax_1^2 + by_1^2 + cz_1^2 - b(x_1^2 + y_1^2 + z_1^2), \\ \gamma &= ax_1^2 + by_1^2 + cz_1^2 - c(x_1^2 + y_1^2 + z_1^2),\end{aligned}$$

we have

$$\begin{aligned}x_2 &= \Theta y_1 z_1 (\beta n^2 + \gamma m^2), \\ y_2 &= \Theta z_1 x_1 (\gamma l^2 + \alpha n^2), \\ z_2 &= \Theta x_1 y_1 (\alpha m^2 + \beta l^2);\end{aligned}$$

and these values give

$$\begin{aligned} lx_2 + my_2 + nz_2 &= 0, \\ x_2^2 + y_2^2 + z_2^2 &= x_1^2 + y_1^2 + z_1^2, \\ ax_2^2 + by_2^2 + cz_2^2 &= ax_1^2 + by_1^2 + cz_1^2. \end{aligned}$$

In connexion with the subject I may add the following transformation, viz. if

$$3\sqrt{\alpha} x' = \sqrt{3\alpha} (y - z) + \sqrt{(3\alpha - 2\beta)(x^2 + y^2 + z^2) + 2\beta(yz + zx + xy)},$$

then reciprocally

$$3\sqrt{\beta} x = -\sqrt{3\beta} (y' - z') + \sqrt{(3\beta - 2\alpha)(x'^2 + y'^2 + z'^2) + 2\alpha(y'z' + z'x' + x'y')},$$

and

$$\begin{aligned} x^2 + y^2 + z^2 &= x'^2 + y'^2 + z'^2, \\ \beta(x^2 + y^2 + z^2 - yz - zx - xy) &= \alpha(x'^2 + y'^2 + z'^2 - y'z' - z'x' - x'y'). \end{aligned}$$

Suppose  $1 + \rho + \rho^2 = 0$ , then

$$x^2 + y^2 + z^2 - yz - zx - xy = (x + \rho y + \rho^2 z)(x + \rho^2 y + \rho z);$$

and in fact

$$\begin{aligned} 3\sqrt{\alpha} (x' + \rho^2 y' + \rho z') &= -\sqrt{3\alpha} (1 + 2\rho)(x + \rho y + \rho^2 z), \\ 3\sqrt{\alpha} (x' + \rho y' + \rho^2 z') &= \sqrt{3\beta} (1 + 2\rho)(x + \rho^2 y + \rho z). \end{aligned}$$

The preceding investigations have been in my possession for about eighteen months.

*2 Stone Buildings, April 18, 1855.*

## 130.

### Deuxième Mémoire sur les Fonctions Doublement Périodiques.

[From the *Journal de Mathématiques Pures et Appliquées* (Liouville), tom. xix. (1854), pp. 193–208 : Sequel to Memoir t. x. (1845), **25**.]

Je vais essayer de développer ici les propriétés qui se rapportent aux transformations linéaires des périodes des fonctions  $yx$ ,  $gx$ ,  $\Omega x$ ,  $Zx$ , dont je me suis occupé dans le Mémoire sur les fonctions doublement périodiques que j'ai donné dans ce Recueil en 1845. Avant d'entrer en matière, je remarque que, partant des expressions

$$\begin{aligned} \Omega &= \omega + \omega' i, \\ T &= v + v' i, \end{aligned}$$

des deux périodes, où  $i = \sqrt{-1}$ , on obtient, en écrivant

$$\begin{aligned} \Omega^* &= \omega - \omega' i, \\ T^* &= v - v' i, \end{aligned}$$

les équations

$$\begin{aligned} \Omega T^* &= \omega v + \omega' v' - i(\omega v' - \omega' v), \\ \Omega^* T &= \omega v + \omega' v' + i(\omega v' - \omega' v); \end{aligned}$$

au moyen desquelles et des valeurs

$$B = \frac{\pi(\omega v' - \omega' v)}{\Omega T \bmod. (\omega v' - \omega' v)}, \quad \beta = \frac{\pi(\omega v + \omega' v')}{\Omega T \bmod. (\omega v' - \omega' v)}$$

des quantités  $B, \beta$ , on déduit les formules

$$B + \beta = \frac{\pi}{\Omega \bmod. (\omega v' - \omega' v)} \Omega^*, \quad B - \beta = \frac{\pi}{T \bmod. (\omega v' - \omega' v)} T^*.$$

Je ne fais attention qu'aux transformations qui correspondent à des entiers impairs et premiers, et je suppose, de plus, que la transformation soit toujours propre et régulière, c'est-à-dire qu'en écrivant

$$(2k+1)\Omega = \lambda\Omega + \mu T = (\lambda, \mu),$$

$$(2k+1)T = \nu\Omega + \rho T = (\nu, \rho),$$

où  $2k+1$  est un entier positif, impair et premier, et où  $\lambda, \mu, \nu, \rho$  sont des entiers tels, qu'au signe près,  $\lambda\rho - \mu\nu$  soit égal à  $2k+1$ , je suppose

$$\lambda\rho - \mu\nu = 2k+1,$$

(condition pour que la transformation soit propre), et, en outre,

$$\lambda \equiv 1, \quad \mu \equiv 0, \quad (\bmod 2)$$

$$\nu \equiv 0, \quad \rho \equiv 1,$$

(condition pour que la transformation soit régulière).

On trouve tout de suite

$$\Omega = \rho\Omega, -\mu T, = (\rho, -\mu),$$

$$T = -\nu\Omega, +\lambda T, = (-\nu, \lambda),$$

j'écris aussi

$$\Omega^* = \omega, -\omega' i, \quad T^* = v, -v' i,$$

et je suppose que  $B, \beta$ , soient des fonctions de  $\omega, v$ , telles que les fonctions  $B, \beta$  le sont de  $\omega, v$ .

Cela étant, je forme d'abord l'équation

$$(2k+1)(\omega, v' - \omega' v) = \omega v' - \omega' v,$$

au moyen de laquelle l'équation

$$(B + \beta)\Omega = \frac{\pi}{\bmod. (\omega v' - \omega' v)} \Omega^*$$

se transforme en

$$(2k+1)(B, +\beta,)\Omega = \frac{\pi}{\bmod. (\omega v' - \omega' v)} \Omega^*.$$

De là

$$[(2k+1)(B, +\beta,) - B]\Omega = \frac{\pi}{\Omega} \xi(\omega, \lambda),$$

ou enfin

$$[(2k+1)(B, +\beta,) - B]\Omega = -\frac{\pi}{\Omega} \beta,(\omega, \mu),$$

et de même

$$[(2k+1)(B, -\beta) - B] \Omega = -\frac{\pi}{\Omega} \xi, (\omega, \lambda);$$

équations qui seront bientôt utiles.

Je suppose d'abord que  $2k+1$  soit égal à l'unité, transformation que l'on peut nommer triviale. La fonction  $yx$  est définie par l'équation

$$yx = e^{-\frac{1}{2}Bx^2} x \prod \left[ 1 + \frac{x}{(m, n)} \right], \quad \text{mod. } (m, n) < T, \quad T = \infty;$$

dans  $(m, n) = m\Omega + nT$ , les entiers  $m, n$  doivent prendre toutes les valeurs positives ou négatives (le seul système  $m = 0, n = 0$  excepté) qui satisfont à l'inégalité

$$\text{mod. } (m, n) < T,$$

dont le second membre  $T$  sera ensuite supposé infini. Soit  $y, x$  la fonction correspondante pour les périodes  $\Omega, T$ , on aura

$$y, x = e^{-\frac{1}{2}B, x^2} x \prod \left[ 1 + \frac{x}{(m, n)} \right], \quad \text{mod. } (m, n) < T, \quad T = \infty.$$

Or

$$\begin{aligned} (m, n) &= m\Omega + nT, \\ &= m(\lambda\Omega + \mu T) + n(\nu\Omega + \rho T) \\ &= (\lambda m + \nu n)\Omega + (\mu m + \rho n)T \\ &= m, \Omega + n, T \\ &= (m, n). \end{aligned}$$

En écrivant, comme nous venons de le faire,

$$\begin{aligned} m, &= \lambda m + \nu n, \\ n, &= \mu m + \rho n, \end{aligned}$$

on voit tout de suite qu'à chaque système de valeurs entières de  $m, n$ , correspond un système, et un seul système, de valeurs entières de  $m, n$ ; et que de chaque système de valeurs entières de  $m, n$ , correspond un système, et un seul système, de valeurs entières de  $m, n$ ; de plus, les systèmes  $m = 0, n = 0$  et  $m, = 0, n, = 0$ , correspondent l'un à l'autre. Il est donc permis d'écrire

$$\prod \left[ 1 + \frac{x}{(m, n)} \right] = \prod \left[ 1 + \frac{x}{(m, n)} \right],$$

les limites comme auparavant; car, à cause de

$$(m, n) = (m, n),$$

la condition pour les limites, savoir

$$\text{mod. } (m, n) < T, \quad T = \infty,$$

devient

$$\text{mod. } (m, n) < T, \quad T = \infty.$$

Cela donne enfin l'équation

$$y, x = e^{-\frac{1}{2}(B, -B)x^2} yx,$$

et, au moyen de cette équation, on obtient une équation correspondante pour la transformation de l'une quelconque des fonctions  $yx$ ,  $gx$ ,  $\Omega x$ ,  $Zx$ , définies par les équations

$$\begin{aligned} yx &= e^{-\frac{1}{2}Bx^2} x \prod \left[ 1 + \frac{x}{(m, n)} \right], \quad \text{mod. } (m, n) < T, \\ gx &= e^{-\frac{1}{2}Bx^2} \prod \left[ 1 + \frac{x}{(m, n)} \right], \quad \text{mod. } (m, n) < T, \quad T = \infty, \\ \Omega x &= e^{-\frac{1}{2}Bx^2} \prod \left[ 1 + \frac{x}{(m, n)} \right], \quad \text{mod. } (m, n) < T, \\ Zx &= e^{-\frac{1}{2}Bx^2} \prod \left[ 1 + \frac{x}{(m, n)} \right], \quad \text{mod. } (m, n) < T, \end{aligned}$$

(équations dans lesquelles  $m = m + \frac{1}{2}$ ,  $n = n + \frac{1}{2}$ ). Je prends par exemple la fonction  $gx$ , et j'écris dans l'équation entre  $y, x$  et  $yx$ ,  $x + \frac{1}{2}\Omega$ , au lieu de  $x$ . Soit pour un moment  $\rho = 2\rho' + 1$ ,  $\mu = 2\mu'$ ; cela donne

$$x + \frac{1}{2}\Omega = x + \frac{1}{2}(\rho\Omega, -\mu T) = x + (\rho', -\mu'),.$$

Donc

$$y, (x + \frac{1}{2}\Omega) = e^{\frac{1}{2}B, (\rho', -\mu')^2} M, gx,$$

c'est-à-dire

$$y, (x + \frac{1}{2}\Omega) = e^{\frac{1}{2}B, (\rho', -\mu')^2} M, gx;$$

de plus,

$$y(x + \frac{1}{2}\Omega) = e^{\frac{1}{2}B\Omega^2} M gx.$$

Ces substitutions étant effectuées, les coefficients  $M$ ,  $M$ , doivent être éliminés en écrivant  $x = 0$ ; cela donne

$$g, x = e^{-\frac{1}{2}(B, -B)x^2 \text{ et c.}} gx,$$

ou enfin, au moyen d'une équation déjà trouvée,

$$g, x = e^{-\frac{1}{2}(B, -B)x^2} gx,$$

et de même pour les fonctions  $\Omega x$ ,  $Zx$ .

Donc enfin, en représentant par  $Jx$  l'une quelconque des fonctions  $yx$ ,  $gx$ ,  $\Omega x$ ,  $Zx$ , on aura

$$J, x = e^{-\frac{1}{2}(B, -B)x^2} Jx,$$

où  $J, x$  est ce que devient  $Jx$  au moyen d'une transformation triviale (propre et régulière) des périodes.

Je passe à présent à la transformation pour un nombre impair et premier  $(2k+1)$  quelconque; mais pour cela on a besoin de connaître la valeur de la fonction

$$u' = \prod \left[ 1 + \frac{x}{(m, n) + y} \right], \quad \text{mod. } \{(m, n) + y\} < T, \quad T = \infty,$$

où  $y = a + bi$  est une quantité réelle ou imaginaire quelconque.

Soit  $u$  ce que devient  $u'$  en prenant pour la condition par rapport aux limites

$$\text{mod. } (m, n) < T, \quad T = \infty;$$

on trouve sans peine

$$u = \frac{x + y}{y} \cdot \frac{y(x + y)}{y(y)}.$$

Pour trouver  $u'$ , je forme l'équation

$$u : u' = \prod \left[ 1 + \frac{x}{(m, n)} \right],$$

la limite inférieure du produit infini double étant

$$\text{mod. } \{(m, n) + y\} > T,$$

et la limite supérieure

$$\text{mod. } (m, n) < T, \quad T = \infty;$$

cela donne

$$\log u - \log u' = x \sum \frac{1}{(m, n)^2} - \frac{1}{2} x^2 \sum \frac{1}{(m, n)^3} + \dots,$$

car on peut démontrer que

$$\sum \frac{1}{(m, n)^k} = 0, \quad k \geq 2.$$

Pour cela, observons que  $m$  et  $n$  étant infinis puisque  $T$  l'est, la première des sommes dont il s'agit peut se remplacer par l'intégrale double

$$I = \iint \frac{dm \, dn}{(m, n)^2},$$

laquelle (en écrivant  $m = r \cos \theta$ ,  $n = r \sin \theta$ , ce qui donne, comme on sait,  $dm \, dn = r \, dr \, d\theta$ ) devient

$$I = \iint \frac{dr \, d\theta}{r(\Omega \cos \theta + T \sin \theta)^2},$$

d'où

$$I = \iint \frac{(\log r) \, d\theta}{(\Omega \cos \theta + T \sin \theta)^2},$$

en prenant  $(\log r)$  entre les limites convenables. Pour trouver ces limites, j'écris

$$(m, n) + y = r(\Omega \cos \theta + T \sin \theta) + y,$$

ce qui donne

$$\text{mod. } \{(m, n) + y\} = \{r(\Omega \cos \theta + T \sin \theta) + y\} \{r(\Omega^* \cos \theta + T^* \sin \theta) + y^*\},$$

savoir, à l'une des limites

$$\begin{aligned} & r^2(\Omega \cos \theta + T \sin \theta)(\Omega^* \cos \theta + T^* \sin \theta) \\ & + r\{y^*(\Omega \cos \theta + T \sin \theta) + y(\Omega^* \cos \theta + T^* \sin \theta)\} + yy^* = 0; \end{aligned}$$

ou, en négligeant les puissances négatives de  $T$ ,

$$r = \frac{T}{\sqrt{(\Omega \cos \theta + T \sin \theta)(\Omega^* \cos \theta + T^* \sin \theta)}} - \frac{1}{2} \left[ \frac{y}{\Omega \cos \theta + T \sin \theta} + \frac{y^*}{\Omega^* \cos \theta + T^* \sin \theta} \right],$$

et à l'autre limite,

$$r = \frac{T}{\sqrt{(\Omega \cos \theta + T \sin \theta)(\Omega^* \cos \theta + T^* \sin \theta)}}.$$

Or, en représentant ces deux équations par

$$r = R - \phi, \quad r = R,$$



on trouve, pour la valeur de  $(\log r)$  entre les deux limites,

$$\log R - \log(R - \phi) = -\log\left(1 - \frac{\phi}{R}\right) = 0,$$

à cause de la valeur infinie de  $R$ . Ainsi la somme cherchée est nulle ; et il est tout clair que les sommes suivantes  $\sum \frac{1}{(m, n)^k}$ , &c., se réduisent de même à zéro.

Donc enfin,

$$\log u - \log u' = x \sum \frac{1}{(m, n)^2}.$$

Cela fait voir que

$$u' = e^{-kx}u,$$

le coefficient  $k$  étant donné au moyen de l'équation

$$k = \sum \frac{1}{(m, n)^2},$$

où la somme est prise, comme auparavant, entre les limites

$$\text{mod. } \{(m, n) + y\} > T, \quad \text{mod. } (m, n) < T, \quad T = \infty.$$

Mais il n'est pas permis d'écrire

$$k = \iint \frac{dm \, dn}{(m, n)^2}.$$

En effet, cette intégrale n'est que le premier terme d'une suite dont il faudrait, pour obtenir un résultat exact, prendre deux termes ; le second terme de la suite serait une intégrale prise le long d'un contour, et il serait, ce me semble, très-difficile d'en trouver la valeur. Pour trouver la valeur de  $k$ , je remarque que  $k$  sera fonction linéaire des quantités  $T, y, y^*, \frac{1}{T}$ , &c., qui entrent dans les valeurs de  $r$  ; donc, puisqu'en dernière analyse  $T = \infty$ ,  $k$  ne peut être que de la forme  $Ly + My^*$ . Cela étant, en substituant pour  $u'$  sa valeur, je forme l'équation

$$\frac{y(x+y)}{y(y)} = e^{-Bxy} e^{(-Bxy+Lxy+My^*)x^2} \prod \left[1 + \frac{x}{(m, n) + y}\right]^{-1}, \quad \text{mod. } \{(m, n) + y\} < T, \quad T = \infty,$$

et j'écris successivement

$$y = \frac{1}{2}\Omega, \quad y = \frac{1}{2}T,$$

ce qui donne pour les valeurs correspondantes du produit infini double  $e^{-\frac{1}{2}Bx^2}gx$  et  $e^{-\frac{1}{2}Bx^2}\Omega x$ . En comparant les valeurs ainsi obtenues avec les équations qui donnent les valeurs de  $y(x + \frac{1}{2}\Omega)$ ,  $y(x + \frac{1}{2}T)$ , on trouve

$$L = 0, \quad M = \frac{\pi}{\text{mod. } (\omega v' - \omega' v)};$$

ou enfin,

$$\frac{y(x+y)}{y(y)} = e^{-Bxy} e^{\beta xy^*} \prod \left[1 + \frac{x}{(m, n) + y}\right]^{-1}, \quad \text{mod. } \{(m, n) + y\} < T, \quad T = \infty,$$

laquelle est l'équation qu'il s'agissait d'établir. Il est à peine nécessaire de faire la remarque que pour  $y = 0$ , on doit considérer à part le facteur  $1 + \frac{x}{y}$ , lequel multiplié par  $y(y)$  devient tout simplement  $x$  ; l'équation subsiste donc dans ce cas.

En revenant au problème des transformations linéaires, partant des équations

$$\begin{aligned} (2k+1)\Omega &= \lambda\Omega + \mu T, \\ (2k+1)T &= \nu\Omega + \rho T, \end{aligned}$$

je suppose d'abord que les coefficients  $\lambda, \nu$  ne satisfassent pas à la fois aux deux conditions

$$\lambda \equiv 0, \quad \nu \equiv 0 \pmod{2k+1},$$

et je prends  $p, q$  des entiers quelconques tels, que  $\lambda p + \nu q$  ne soit pas  $\equiv 0, \pmod{2k+1}$ .

Cela étant, soient

$$\begin{aligned} \lambda p + \nu q &= p, \\ \mu p + \rho q &= q, \\ (2k+1)\psi &= p\Omega + qT, \end{aligned}$$

et, par conséquent,

$$\psi = p\Omega + qT.$$

Je forme l'équation

$$(m, n) + s\psi = (m, n),$$

savoir

$$m\Omega + nT + s\psi = m\Omega + nT,$$

c'est-à-dire

$$\begin{aligned} \lambda m + \nu n - sp &= (2k+1)m, \\ \mu m + \nu n - sq &= (2k+1)n, \end{aligned}$$

ou, ce qui est la même chose,

$$\begin{aligned} m - sp &= m\rho - n\nu, \\ n - sq &= -m\mu + n\lambda. \end{aligned}$$

Or,  $m, n, s$  étant des entiers donnés,  $m, n$  seront aussi des entiers ; de même,  $m, n$  étant des entiers donnés, on trouve de  $-k$  à  $k$  un entier  $s$  qui donne  $m$ , un entier. Mais cela étant,  $n$ , sera aussi un entier ; car autrement  $n$ , serait une fraction ayant pour dénominateur, lequel on voudrait, des nombres  $2k+1, \lambda, \nu$ , ce qui est impossible à moins que

$$\lambda \equiv 0, \quad \nu \equiv 0 \pmod{2k+1}.$$

Mais si ces équations avaient lieu, on trouverait d'abord  $s$  de manière à avoir  $m$ , entier, et alors, puisqu'on n'a pas aussi

$$\mu \equiv 0, \quad \rho \equiv 0 \pmod{2k+1}$$

(en effet, cela est impossible à cause de l'équation  $\lambda\rho - \mu\nu = 2k+1$ ), on démontrerait, comme auparavant, pour  $n$ , que  $m$ , est entier. Donc, enfin,  $m, n$  étant des entiers donnés, on trouve pour  $m, n, s$  un système d'entiers tel que  $s$  soit compris de  $k$  à  $-k$ , et l'on voit sans peine qu'il n'y a qu'un seul système de cette espèce.

A présent, partant de l'équation

$$\frac{y(x+y)}{y(y)} = e^{-Bxy+\beta xy^*} \prod \left[ 1 + \frac{x}{(m, n) + y} \right]^{-1},$$

(et faisant attention à la particularité que présente le cas de  $y = 0$ ), j'écris successivement

$$y = 0, \quad y = \pm\psi, \quad \dots, \quad y = \pm k\psi,$$

et je forme le produit des équations ainsi trouvées. Cela donne, à cause de  $(m, n) + s\psi = (m, n)$ ,

$$\prod \frac{y(x + s\psi)}{y(s\psi)} = e^{-Bxy \text{ etc.}} \prod \left[ 1 + \frac{x}{(m, n) + s\psi} \right]^{-1},$$

la condition, par rapport aux limites, étant

$$\text{mod. } (m, n), < T, \quad T = \infty.$$

Or

$$y, x = e^{-\frac{1}{2}B, x^2} x \prod \left[ 1 + \frac{x}{(m, n)} \right],$$

avec la même condition, par rapport aux limites ; donc, enfin,

$$y, x = \frac{e^{-\frac{1}{2}B, x^2 + \dots}}{e^{\dots}} \prod \frac{y(x + s\psi)}{y(s\psi)},$$

où, dans le numérateur,  $s$  doit avoir toutes les valeurs entières depuis  $s = -k$  jusqu'à  $s = +k$ ,  $y$  compris  $s = 0$ , et dans le dénominateur ces mêmes valeurs, hormis la valeur  $s = 0$ .

Il est, à présent, facile de faire voir que cette propriété subsiste pour l'une quelconque des fonctions  $yx$ ,  $gx$ ,  $\Omega x$ ,  $Zx$  ; en effet, pour la démontrer pour  $gx$ , j'écris  $x + \frac{1}{2}\Omega$  au lieu de  $x$  ; en prenant, pour un moment,  $\rho = 2\rho' + 1$ ,  $\mu = 2\mu'$  ; cela donne

$$x + \frac{1}{2}\Omega = x + (\rho', -\mu').$$

Donc

$$y, (x + \frac{1}{2}\Omega) = e^{\frac{1}{2}B, (\rho', -\mu')^2} M g, x.$$

A présent, partant de l'équation

$$\frac{y(x + y)}{y(y)} = e^{-Bxy + \beta xy^*} \prod \left[ 1 + \frac{x}{(m, n) + y} \right]^{-1}$$

(en faisant attention à la particularité que présente le cas de  $y = 0$ ), j'écris successivement

$$y = 0, \quad y = \pm\psi, \quad \dots, \quad y = \pm k\psi,$$

et je forme le produit des équations ainsi trouvées. Cela donne, à cause de  $(m, n) + s\psi = (m, n)$ ,

$$\prod \frac{y(x + s\psi)}{y(s\psi)} = e^{-Bxy + \dots} \prod \left[ 1 + \frac{x}{(m, n) + s\psi} \right]^{-1},$$

la condition, par rapport aux limites, étant

$$\text{mod. } (m, n), < T, \quad T = \infty.$$

(D'autres manipulations algébriques analogues conduisent aux formules concrètes pour la transformation des fonctions  $yx$ ,  $gx$ ,  $\Omega x$ ,  $Zx$ , exprimées par produits sur les indices  $s = -k, \dots, k$  et  $s = -k, \dots, k$  avec  $s \neq 0$  au dénominateur. Les formules définitives s'écrivent comme suit :)

$$\Omega, = (\Omega - \psi) T, = (T - \psi), \text{ etc.},$$

et, en résumant, pour les fonctions transformées, on obtient des systèmes du type

$$\begin{aligned} \Omega, &= \Omega, 1 (2\Omega + 2T, ), \\ T, &= \Omega, 1 \Omega + T, 1 T = \psi, \\ T, 2 &= T. \end{aligned}$$

Les paramètres particuliers correspondent au cas général en fonctions de  $\Omega$ ,  $T$  ; on obtient également, par un moyen d'une transformation triviale, on obtiendrait

$$B' = \Omega + 2B, T, \quad T, = T,$$

et puis

$$\psi = \frac{1}{2k+1} \Omega, \quad T, = T,$$

et, de là,

$$\frac{\Omega,}{T,} = \frac{1}{2k+1} \cdot \frac{\Omega}{T}.$$

Les équations correspondantes pour la deuxième sont :

$$\begin{aligned} p &= 2k+1, & q &= 0, \\ \lambda &= 2k+1, & \nu &= 0, \\ \mu &= 0, & \rho &= 1, \\ \rho, &= 1, & q, &= 0, \\ \psi, &= \frac{1}{2k+1} [(2k+1) \Omega + (2k+1) T], \\ \Omega_{,1} &= \Omega, \\ T_{,1} &= (2k+1) \Omega + T, \end{aligned}$$

ce qui donne

$$\frac{\Omega,}{T,} = (2k+1) \frac{\Omega}{T}.$$

J'ajoute, sans m'arrêter pour les démontrer, quelques formules de transformation pour le nombre 3 ; je trouve d'abord

$$\begin{aligned} \Omega, &= \frac{1}{3} \Omega, \quad T, = T, \\ B, &= e^{-\frac{1}{2} B \Omega^2 - \beta \Omega \Omega^*} \frac{y(x + \frac{1}{3} \Omega) y(x - \frac{1}{3} \Omega)}{y^2(\frac{1}{3} \Omega)}, \\ G, &= e^{-\frac{1}{2} B \Omega^2 - \beta \Omega \Omega^*} \frac{y(x + \frac{1}{3} \Omega) y(x - \frac{1}{3} \Omega)}{B^2(\frac{1}{3} \Omega)}, \\ Z, &= e^{-\frac{1}{2} B \Omega^2 - \beta \Omega \Omega^*} \frac{y(x + \frac{1}{3} \Omega) y(x - \frac{1}{3} \Omega)}{B^2(\frac{1}{3} \Omega)}. \end{aligned}$$

Ces équations donnent, en introduisant les fonctions elliptiques  $\phi x$ ,  $f x$ ,  $F x$  données au moyen de

$$\phi x = \frac{Yx}{Bx}, \quad f x = \frac{gx}{Bx}, \quad F x = \frac{Gx}{Bx},$$

les équations

$$\phi, x = \phi x \cdot \frac{f x \cdot f(x + \frac{1}{3} \Omega) f(x - \frac{1}{3} \Omega)}{f^2(\frac{1}{3} \Omega)},$$

dont la seconde peut s'écrire sous la forme

$$f, x = \frac{1 - x^2 \frac{P}{Q}(\frac{1}{3} \Omega) \phi x}{1 + x^2 \frac{P}{Q}(\frac{1}{3} \Omega) \phi x}, \quad \text{etc.}$$

On trouve en effet, en mettant comme à l'ordinaire  $\Phi = \phi^2 x$ ,

$$c'^2 = b\phi c', \quad e'^2 = (b - c') c',$$

et puis

$$\begin{aligned} \phi, x &= \frac{1 - c(1 - b) \phi x}{1 - c(c - b) \phi x}, \\ f, x &= \frac{1 - (c + b) \phi x}{1 - c(c - b) \phi x}, \\ F, x &= \frac{1 - c\phi x}{1 - c(c - b) \phi x}, \end{aligned}$$

formules qui correspondent à celles de la transformation de Lagrange. Les équations  $(\beta)$ ,  $(\gamma)$  donnent en mettant ces valeurs de  $\phi$ ,  $f$ ,  $F$ , laquelle, égale à la valeur qui vient d'être trouvée, donne

$$\frac{xc(\bar{\phi} P(\frac{1}{3}\Omega))}{B(x - \frac{1}{3}\Omega) B(x + \frac{1}{3}\Omega)} = \frac{\phi x f x}{1 - c(c - b) \phi x}.$$

(Suit la dérivation par formules analogues du système correspondant pour la transformation  $\Omega, = \Omega, \text{ T}, = \frac{1}{3}\text{T}$ . Le texte expose ensuite le système correspondant, puis poursuit avec les formules définitives.)

On obtient tout de suite, pour le premier cas, le système d'équations

$$\begin{aligned} \rho, &= 1, \quad q, = 2q, \\ \lambda &= 1, \quad \nu = 2, \\ \mu &= 0, \quad \rho = (2k + 1), \\ p &= 1, \quad q = 1, \\ \psi &= \frac{1}{2k+1} [(2k + 1) + 2 \text{T},] \text{T}, \\ \Omega, &= \Omega, \\ \text{T}, &= (2k + 1) \text{T} + \psi, \\ \text{T}_{,2} &= \text{T}. \end{aligned}$$

Le tout particulier le plus simple est celui de  $q = 0$ , cela donne

$$\psi = \Omega, \quad \text{T}, = \text{T};$$

et, de là,

$$\frac{\Omega,}{\text{T},} = \frac{1}{2k+1} \cdot \frac{\Omega}{\text{T}},$$

et même le cas général se réduit à celui-ci, car, au moyen d'une transformation triviale, on obtiendrait

$$B' = \Omega + 2B, \text{T}, \quad \text{T}, = \text{T},$$

et puis

$$\psi = \frac{1}{2k+1} \Omega', \quad \text{T}, = \text{T},$$

et, de là,

$$\frac{\Omega,}{\text{T},} = \frac{1}{2k+1} \cdot \frac{\Omega}{\text{T}}.$$

Les équations correspondantes pour la deuxième sont :

$$\begin{aligned}
 p &= 2k + 1, & q &= 0, \\
 \lambda &= 2k + 1, & \nu &= 0, \\
 \mu &= 0, & \rho &= 1, \\
 p, &= 1, & q, &= 0, \\
 \psi &= \frac{1}{2k+1} [(2k+1)\Omega + (2k+1)\mathbf{T}], \\
 \Omega_{,1} &= \Omega, \\
 \mathbf{T}_{,1} &= (2k+1)\Omega + \mathbf{T},
 \end{aligned}$$

ce qui donne

$$\frac{\Omega_{,1}}{\mathbf{T}_{,1}} = (2k+1) \frac{\Omega}{\mathbf{T}}.$$

J'ajoute, sans m'arrêter pour les démontrer, quelques formules de transformation pour le nombre 3; je trouve d'abord

$$\begin{aligned}
 \Omega_{,1} &= \frac{1}{3}\Omega, & \mathbf{T}_{,1} &= \mathbf{T}, \\
 y_{,1}x &= e^{-\frac{1}{2}\dots x^2} g x E x, \\
 g_{,1}x &= e^{-\frac{1}{2}\dots x^2} \frac{E(x + \frac{1}{3}\Omega) E(x - \frac{1}{3}\Omega)}{E^2(\frac{1}{3}\Omega)}, \\
 \Omega_{,1}x &= e^{-\frac{1}{2}\dots x^2} \frac{E(x + \frac{1}{3}\Omega - \frac{1}{2}\mathbf{T}) E(x - \frac{1}{3}\Omega - \frac{1}{2}\mathbf{T})}{E^2(\frac{1}{3}\Omega - \frac{1}{2}\mathbf{T})}, \\
 Z_{,1}x &= e^{-\frac{1}{2}\dots x^2} \frac{E(x + \frac{1}{3}\Omega - \frac{1}{2}\mathbf{T}) E(x - \frac{1}{3}\Omega - \frac{1}{2}\mathbf{T})}{E^2(\frac{1}{3}\Omega - \frac{1}{2}\mathbf{T})}.
 \end{aligned}$$

Ces équations donnent, en introduisant les fonctions elliptiques  $\phi x$ ,  $f x$ ,  $F x$  données ci-dessus, les équations

$$\begin{aligned}
 \phi_{,1}x &= \phi x \cdot \frac{f(x + \frac{1}{3}\Omega) f(x - \frac{1}{3}\Omega)}{f^2(\frac{1}{3}\Omega)}, \\
 f_{,1}x &= \frac{1 - c\phi x}{1 - c(c-b)\phi x}, \\
 F_{,1}x &= \frac{1 - c\phi x}{1 - c(c-b)\phi x};
 \end{aligned}$$

équations qui sont, sauf la notation dont je me suis déjà servi dans la mémoire citée, les mêmes que celles de Jacobi.

*Londres, Stone Buildings, 23 Fév. 1852.*

## 131.

### Nouvelles Recherches sur les Covariants.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. xlvii. (1854), pp. 109–125.]

Je me sers de la notation

$$(a_0, \dots a_n \mid x, y)^n$$

pour représenter la fonction

$$a_0x^n + \frac{n}{1} a_1x^{n-1}y + \dots + a_ny^n;$$

en supposant que les coefficients  $a'_0, a'_1, \dots, a'_n$  soient donnés par l'équation

$$(a'_0, \dots, a'_n \mid \lambda x + \mu y, \lambda' x + \mu' y)^n = (a_0, a_1, \dots, a_n \mid x, y)^n,$$

supposé identique par rapport à  $x, y$ , soit  $\phi(a_0, a_1, \dots, a_n; x, y)$  une fonction des coefficients et des variables, telle que

$$\phi(a'_0, a'_1, \dots, a'_n; x, y) = (\lambda\mu' - \lambda'\mu)^a \phi(a_0, a_1, \dots, a_n; x, y),$$

cette fonction  $\phi$  est nommée dans M. Cauvarian, et dans le cas particulier de  $\phi$  une fonction des seuls coefficients, un *Invariant* de la fonction donnée.

Je suppose d'abord que les nouveaux coefficients soient donnés par l'équation

$$(a'_0, \dots, a'_n \mid x, y)^n = \phi(a_0, a_1, \dots, a_n; x, y);$$

cela suppose les relations

$$\begin{aligned} a'_0 &= a_0, \\ a'_1 &= a_1 + \lambda a_0, \\ a'_2 &= a_2 + 2\lambda a_1 + \lambda^2 a_0, \\ &\&c. \end{aligned}$$

Il faut donc que le covariant  $\phi$  satisfasse à l'équation

$$\phi(a'_0, a'_1, \dots, a'_n; x, y) = \phi(a_0, a_1, \dots, a_n; x, y + \lambda x),$$

ce qui donne

$$(a_0, a_1, \dots, a_n; x + py + p^2z) = \phi(a_0, a_1, \dots, a_n; x, y),$$

où en posant

$$\begin{aligned} a''_0 &= a_0, \\ a''_1 &= a_1 + \lambda a_0, \\ a''_2 &= a_2 + 2\lambda a_1 + \lambda^2 a_0, \\ &\&c., \end{aligned}$$

le covariant  $\phi$  doit satisfaire aux équations

$$\phi(a''_0, a''_1, \dots, a''_n; x, y) = \phi(a_0, a_1, \dots, a_n; x, y).$$

*(La suite du mémoire développe la théorie des covariants en partant de cette définition : étude des opérateurs  $\square, \nabla$  et de leurs propriétés algébriques, énoncé du théorème fondamental sur la représentation des covariants par produits de racines, et applications aux invariants des formes binaires de degrés bas. Le texte n'étant constitué que de formules denses et d'identités algébriques avec de nombreux indices, nous reproduisons ci-dessous les principales équations et énoncés retenus du texte original.)*

Et réciproquement, toute fonction homogène par rapport aux coefficients et issue par rapport aux variables, qui satisfait à ces équations, est un covariant de la fonction donnée.

Par exemple, l'invariant  $\phi = ac - b^2$  de la fonction  $ax^2 + 2bxy + cy^2$  satisfait aux équations

$$(ab + 2bk)k - (bk + ck)^2 = (2bk - a^2k)k - 0,$$

et l'invariant  $\phi = ace - bd^2 + abe^2 - 3bcd - b^3 + 2ad^3$  de la fonction  $ax^4 + 4bx^3y + 6cx^2y^2 + 4dxy^3 + ey^4$  satisfait aux équations

$$(ak_0 + 3bk)k_1 = (ak_0, k_1)k_0.$$

Il est clair qu'on ne considérerait que les fonctions qui restent les mêmes en échange dans les deux indices la coefficient  $a_0, a_1, \dots, a_n$  et les variables  $x, y$ , respectivement. Le nombre des coefficients  $a_0, a_1, \dots, a_n$  est supérieur, le nombre des coefficients indéterminés ; il est exactement égal au nombre des coefficients dans la fonction donnée. Mais il y a quelque autre indication à faire avant de pouvoir considérer comme arbitraires les indices  $\beta$ , &c., dont la série indique le caractère du covariant cherché.

Cherchons par exemple pour la fonction  $ax^4 + 4bx^3y + 6cx^2y^2 + 4dxy^3 + ey^4$  un covariant  $\phi$  de la forme

$$\phi = Ax^4y + Bbx^3y^2 + Cx^2y^3 + Dy^4,$$

contenant les quatre coefficients indéterminés  $A, B, C, D$ . En substituant dans l'équation les mêmes après substitutions convenables, on obtient

$$3(aA)x^2dy^4 + (4bB + 4cC)x^2y^3 + (4cB + 4dD)xy^3 + (8e)y^4 = 0,$$

ou bien, puisque cette équation doit avoir lieu indépendamment de la quantité  $k$  (qui est arbitraire), il faut que les coefficients soient séparément nuls. On a donc les quatre équations

$$aA = 0, \quad 4bB = 0, \quad 4cB = 0, \quad 8e = 0,$$

ce qui donne le système  $A : B : C : D$ , et l'on voit que cette série est complètement déterminée, et l'on obtient en effet

$$\phi = ac + bd - 6acd - 4eb^2 + 3bc^2.$$

La circonstance suivante réclame néanmoins de l'attention. Il s'agit d'énoncer que dans le calcul, le problème de trouver le nombre des coefficients d'un cota donné ; problème qui n'a aucun lieu de l'autre.

(La suite développe les notations symboliques  $\Omega$ ,  $\square$ , etc., et culmine avec l'énoncé du théorème central de l'article :)

**Théorème.** Tout covariant  $\phi$  de la fonction

$$(a_0, \dots, a_n \mid x, y)^n$$

satisfait aux deux équations à différences partielles

$$(\square - p)\phi = 0, \quad (\triangle - q)\phi = 0,$$

où

$$\begin{aligned} \square &= na_1\partial_{a_0} + (n-1)a_2\partial_{a_1} + \dots + a_n\partial_{a_{n-1}} + y\partial_x, \\ \triangle &= na_0\partial_{a_1} + (n-1)a_1\partial_{a_2} + \dots + a_{n-1}\partial_{a_n} + x\partial_y, \end{aligned}$$

et réciproquement.

Inversement on a, en notant d'abord  $a^n = b$ , puis en considérant des trois quantités indépendantes  $a, b, c$ , la solution générale de l'équation à différences partielles

$$ap\partial_a + 2bp\partial_b + 3cp\partial_c = 0,$$



en supposant un poids de  $p$ , est indépendante de  $a, b, c$ , et l'on considère ainsi cette propriété fondamentale : *tout covariant est composé de termes de la première espèce* ; et résultat ainsi obtenu général des covariants forment, lorsque cela est permis, par cette même observation, une suite indiquant la longueur des polynômes à poids  $p$ .

Pour les polynômes binaires, l'énoncé du théorème prend la forme suivante :

**Théorème.** Pour qu'une fonction  $\phi$  des coefficients  $a_0, a_1, \dots, a_n$  d'une forme binaire de degré  $n$ , soit un *invariant*, il faut et il suffit qu'elle satisfasse simultanément aux deux équations

$$\square \phi = 0, \quad \triangle \phi = 0,$$

où  $\square, \triangle$  sont les opérateurs ci-dessus, restreints à leur partie agissant sur les seuls coefficients.

(Le texte poursuit avec des démonstrations détaillées, des applications aux discriminants des formes du degré 2, 3, 4, et des exemples numériques.)

Pour le discriminant

$$\nabla = ax^2 + 2bxy + cy^2$$

de la fonction de degré 2, le discriminant est  $4ac - b^2$ . Nous savons alors

$$\nabla = -b^2(4ac - b^2) + B^2 + Cb^2,$$

en faisant  $F = 4ab + 2b^3$ ,  $B, C$  étant donnés par

$$\eta B = F(4xy^2 - by^3), \quad 2yC = F(B),$$

c'est-à-dire  $B = 12acb - b^3$ ,  $C = -27a^2b$ , et de là

$$\nabla = -27a^2b + 18acb^2 - 4ac^3 + b^2c^2 - 4a^3c^2,$$

valeur qui correspond en effet au discriminant ordinaire. En changeant d'une manière convenable les coefficients.

*Londres, Stone Buildings, 25 Févr. 1852.*

## 132.

### Réponse à une Question Proposée par M. Steiner.

(Aufgabe 4, *Crelle* t. xxxi. (1846) p. 90.)

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. x. (1855), pp. 277-278.]

En partant des deux théorèmes :

I. Qu'il existe un nombre infini de surfaces du second ordre qui touchent une surface du second ordre aux six arêtes d'un quelconque hexaèdre quadrilatéral ;

II. Que le lieu d'intersection de trois plans rectangles qui touchent une surface du second ordre est une sphère concentrique avec la surface, tandis que pour les hyperboloïdes un seul nappe le lieu d'intersection est réduit à un plan,

M. Steiner suppose le cas d'un parallélépipède rectangle, en raisonnant d'une côte  $P$  de ce parallélépipède  $P$  ; par lequel passent trois plans principaux. Les trois plans du parallélépipède  $P$  rencontrent deux surfaces ses faces  $A, A', B, B', C, C'$ . Avec ces six plans principaux  $P$ , &c. on aurait six points de contact, qui pourraient former une figure du second ordre. Mais c'est dans ce cas, on doit avoir un point de l'espace considérée,  $P$ , dont les trois plans principaux à

une surface dont les plans tangents  $A, A', B, B', C, C'$  contiennent les six points et faces du parallélépipède, et celles trois faces, sphériques, et en outre, lorsque le parallélépipède  $P$  est aux trois plans principaux d'un point d'une surface dont les trois faces sont parallèles aux faces de  $A', B', C'$ . La contradiction qui apparaîtra dans le cas général est par conséquent admissible que pour les six points.

Aussi d'attention le solution des équations (A), je vais démontrer quelque propriété générale qui appartient à ce théorème. Pour abrégé, je considère  $\Delta$  le système principal de la surface ; ce dernier que les coefficients  $a_0, a_1, \dots a_n$  et les coefficients de la même forme donnent en effet le même nom et méritera mériter que les nouvelles indices ne sont pas  $r$ , mais  $s$ . Donc, tout autre que sont  $a, b, c, d, \dots$  ; on considère ensuite quelques points jusqu'au système principal  $P$ , et l'on cherche les valeurs qu'elles donnent dans la fonction  $a, b, c, d, \dots, f$ , qui en a l'expression. On considère alors les équations symétriques de la fonction  $a$  et de la fonction  $f$ , et l'on considère la quantité  $(\Phi - P^2 = 0)$ , en effet, ce théorème est bien déterminé, et le théorème  $\beta$  et le II sont tous deux légitimes.

### 133.

#### Sur un Théorème de M. Schläfli.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. I. (1855), pp. 278–282.]

On lit dans le § 13 d'un mémoire très intéressant de M. Schläfli intitulé "Über die Resultanten eines Systems mehrerer algebraischer Gleichungen," (*Mém. de l'Acad. de Vienne*, t. iv. [1852]) un très beau théorème sur les Zésultantes.

Pour faire voir plus clairement, en quoi consiste ce théorème, je prends un cas particulier. Soit

$$\begin{aligned} U &= ax^2 + 3bx^2y + 3cxy^2 + dy^3 + (a, b, c, d \mid x, y)^3, \\ V &= ax^3 + 2bxy + cy^2 = (a, b, c \mid x, y)^2. \end{aligned}$$

Je fais  $p = a^3$ ,  $q = ay$ ,  $r = pq$ , et je forme les opérateurs

$$\begin{aligned} \nabla &= \frac{1}{2}\partial_a + p\partial_b + q\partial_c, \\ \Delta &= \frac{1}{2}\partial_a + p\partial_b + q\partial_c, \end{aligned}$$

lesquels, opérant sur  $U$ , donnent

$$r(p\xi + qy - C), \quad p(p\xi + qy + C),$$

opérant sur  $V$ , donnent

$$p\xi + qy + r^2.$$

Cela étant, soit  $\phi = 0$  le résultant des équations  $U = 0, V = 0$ , c'est-à-dire l'équation que l'on obtient en éliminant  $x, y$  entre les deux équations  $U = 0, V = 0$ , on a

$$\phi = \det \begin{pmatrix} a & 3b & 3c & d & \cdot \\ \cdot & a & 3b & 3c & d \\ a & 2b & c & \cdot & \cdot \\ \cdot & a & 2b & c & \cdot \\ \cdot & \cdot & a & 2b & c \end{pmatrix}.$$

Je suppose que les opérations  $\nabla$ ,  $\Delta$  opèrent sur le résultant  $\phi$ , et que cela donne les fonctions  $\nabla\phi$ ,  $\Delta\phi$ , on trouve sans peine

$$\begin{aligned}\Delta\phi &= (p\xi + q\eta - C)\phi, \\ \nabla\phi &= (p\xi + q\eta + C)\phi,\end{aligned}$$

ou en écrivant  $(\xi, \eta, \zeta)$  pour les résultats de  $\phi$  et  $(\xi)$  pour les mêmes résultats de  $\phi$ , on aurait à considérer ce résultant comme une fonction symétrique de la fonction; ce résultant est ce qui est nommé dans le théorème de M. Schläfli.

Généralement, on suppose que l'on ait autant de fonctions  $U, V, W, \dots$  que d'indéterminés  $x, y, z, \dots$ , ces fonctions étant des degrés respectifs  $\mu, \mu', \dots$ , des nombres indéterminés, et l'on pose pour chacune des fonctions du système l'opérateur  $\nabla$  correspond ce que ce que devient l'opérateur  $\nabla$  pour la fonction  $U$ , dans laquelle on substitue à la place des indéterminées  $x, y, z, W, \dots$ , des facteurs  $p, q, r, \&c.$ ; avec les indices respectifs des fonctions  $U, V, W, \dots$ , et de la fonction dène faite, on a un théorème dont je ne pourrais indiquer la démonstration de M. Schläfli, mais dont la valeur du résultant  $\phi$  est telle que cela est en outre une fonction symétrique.

Cela étant, soit  $\phi$  le résultant des fonctions  $U, V, W, \dots$ , c'est-à-dire l'équation que l'on obtient en éliminant  $x, y, z, \dots$ ; et soit  $r$  le théorème de l'opération  $\nabla$  comme l'on a vu pour les hyperboloïdes. Cela donne en effet pour les résultants à fonction de  $\nabla$ .

**Théorème.** La démonstration donnée dans la mémoire citée est, on ne peut plus, simple et élégante. Elle expose d'abord sur un théorème connu (démontré au reste § 6) qui peut être énoncé ainsi; savoir, en supposant que les équations  $U = 0, V = 0, \dots$  soient satisfaites, on aura (près un facteur indépendant de  $\xi, \eta, \dots$ ),

$$\mathfrak{V}\phi = \xi (p\xi + q\eta + \dots)^{\mu-\mu'}(\xi, \eta, \dots), \quad \mathfrak{V}\phi = \xi (p\xi + q\eta + \dots)^{\mu-\mu''}(\xi, \eta, \dots), \quad \&c.$$

Puis, cela fait, est fondée sur la démonstration démontrée (§ 12), savoir : le résultant des fonctions

$$\begin{aligned}t(p\xi + q\eta + \dots)^{\mu} + f'(\xi, \eta, \dots), \\ t'(p\xi + q\eta + \dots)^{\mu'} + f''(\xi, \eta, \dots), \\ \vdots\end{aligned}$$

(où  $f, f', \dots$  sont des polynômes de degrés  $\mu, \mu', \dots$  en  $\xi, \eta$ , &c., des constantes quelconques, en supposant que  $\mu$  ne soit plus petit qu'aucun autre des indices  $\mu, \mu', \dots$ , et en posant  $\pi = \frac{\pi}{\mu} + \frac{\pi}{\mu'} + \dots$ , le résultant  $\Phi$  du système sera tout au plus du degré  $\frac{\pi}{\mu}$  par rapport à  $t$  et de même par rapport aux coefficients de  $f, f', \dots$ . Le degré par rapport à les coefficients est ai; le degré par rapport aux coefficients de  $f, f', \dots$  sera donc au moins un infiniment petit de l'ordre  $k$  contenu à  $\phi^{\pi-\mu}$  comme *facteur*. Dans cette supposition les équations  $\mathfrak{V}\phi = 0, \mathfrak{V}\phi = 0, \&c.$  deviendront :

$$\begin{aligned}t(p\xi + q\eta + \dots)^{\mu} + f(\xi, \eta, \dots) &= 0, \\ t'(p\xi + q\eta + \dots)^{\mu'} + f'(\xi, \eta, \dots) &= 0, \\ \vdots\end{aligned}$$

où  $f, f', \dots$  sont des polynômes de degrés  $\mu, \mu', \dots$  dont les coefficients sont infiniment petits par rapport à tous les autres plus du degré  $\frac{\pi}{\mu}$  par rapport à  $t$  et de même par rapport aux coefficients de  $f, f', \dots$ . Le degré par rapport à ces coefficients est ai (assujettis à la seule condition ci-dessus

mentionnée) étant d'ailleurs arbitraires, on suit que le résultant  $\Phi$  contiendra en général cette même puissance  $\phi^{\pi-\mu}$  comme facteur ; ce qu'il s'agissait de démontrer.

*(Le texte poursuit avec le développement de calculs explicites et la démonstration complète pour des systèmes binaires et ternaires. La dernière portion du texte (p. 184) contient quelques détails sur les hautes puissances de  $\phi^{\pi-\mu}$  qui apparaissent comme facteurs dans le résultant complément, le tout encadré par des observations sur l'importance du théorème dans le contexte des résultants multidimensionnels.)*

# Cayley's Collected Mathematical Papers, Vol. II, pp. 185–234

Reconstruction with OCR skeleton and PNG verification

## 134. Remarques sur la notation des fonctions algébriques.

REMARQUES SUR LA NOTATION DES FONCTIONS ALGÈBRIQUES.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 282–285.]

Je me sers de la notation

$$\begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & & \vdots & \end{vmatrix}$$

pour représenter ce que j'appelle une *matrice*; savoir un *système* de quantités rangées en forme de *carré*, mais d'ailleurs tout à fait indépendantes (je ne parle pas ici des matrices *rectangulaires*). Cette notation me paraît très commode pour la théorie des équations linéaires; j'écris par exemple

$$(\xi, \eta, \zeta, \dots) = \begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & & \vdots & \end{vmatrix} (x, y, z, \dots)$$

pour représenter le système des équations

$$\begin{aligned} \xi &= \alpha x + \beta y + \gamma z \dots, \\ \eta &= \alpha' x + \beta' y + \gamma' z \dots, \\ \zeta &= \alpha'' x + \beta'' y + \gamma'' z \dots \end{aligned}$$

On obtient par là l'équation

$$(x, y, z, \dots) = \begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & & \vdots & \end{vmatrix}^{-1} (\xi, \eta, \zeta, \dots),$$

qui représente le système d'équations qui donne  $x, y, z, \dots$  en termes de  $\xi, \eta, \zeta, \dots$ , et on se trouve ainsi conduit à la notation

$$\begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & & \vdots & \end{vmatrix}^{-1}$$

de la matrice *inverse*. Les termes de cette matrice sont des fractions, ayant pour *dénominateur* commun le *déterminant* formé avec les termes de la matrice originale; les *numérateurs* sont

les déterminants mineurs formés avec les termes de cette même matrice en supprimant l'une quelconque des lignes et l'une quelconque des colonnes.

Soit encore

$$(x, y, z, \dots) = \begin{vmatrix} a, & b, & c, & \dots \\ a', & b', & c', & \dots \\ a'', & b'', & c'', & \dots \\ & \vdots & & \end{vmatrix} (x, y, z, \dots),$$

on peut écrire

$$(\xi, \eta, \zeta, \dots) = \begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & \vdots & & \end{vmatrix} \begin{vmatrix} a, & b, & c, & \dots \\ a', & b', & c', & \dots \\ a'', & b'', & c'', & \dots \\ & \vdots & & \end{vmatrix} (x, y, z, \dots),$$

et l'on parvient ainsi à l'idée d'une *matrice composée*, par ex.

$$\begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & \vdots & & \end{vmatrix} \begin{vmatrix} a, & b, & c, & \dots \\ a', & b', & c', & \dots \\ a'', & b'', & c'', & \dots \\ & \vdots & & \end{vmatrix}.$$

On voit d'abord que la valeur de cette matrice composée est

$$\begin{vmatrix} (\alpha, \beta, \gamma, \dots)(a, a', a'', \dots), & (\alpha, \beta, \gamma, \dots)(b, b', b'', \dots), & \dots \\ (\alpha', \beta', \gamma', \dots)(a, a', a'', \dots), & (\alpha', \beta', \gamma', \dots)(b, b', b'', \dots), & \dots \\ & \vdots & \end{vmatrix}$$

où  $(\alpha, \beta, \gamma, \dots)(a, a', a'', \dots) = \alpha a + \beta a' + \gamma a'' + \dots$ . Il faut faire attention, dans la composition des matrices, de combiner les *lignes* de la matrice à gauche avec les *colonnes* de la matrice à droite, pour former les *lignes* de la matrice composée. Il y aurait bien des choses à dire sur cette théorie de matrices, laquelle doit, il me semble, précéder la théorie des Déterminants.

Une notation semblable peut être employée dans la théorie des fonctions *quadratiques*. En effet, on peut dénoter par

$$\left( \begin{vmatrix} \alpha, & \beta, & \gamma, & \dots \\ \alpha', & \beta', & \gamma', & \dots \\ \alpha'', & \beta'', & \gamma'', & \dots \\ & \vdots & & \end{vmatrix} \right) (\xi, \eta, \zeta)(x, y, z)$$

la fonction *linéo-linéaire*

$$\begin{aligned} & (\alpha\xi + \beta\eta + \gamma\zeta \dots)x \\ & + (\alpha'\xi + \beta'\eta + \gamma'\zeta \dots)y \\ & + (\alpha''\xi + \beta''\eta + \gamma''\zeta \dots)z \\ & \vdots \end{aligned}$$

et de là par

$$\left( \begin{vmatrix} a, & h, & g, & \dots \\ h, & b, & f, & \dots \\ g, & f, & c, & \dots \\ & \vdots & & \end{vmatrix} \right) (x, y, z, \dots)^2$$

la fonction *quadratique*

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy \dots$$

que je représente aussi par

$$(a, b, c, \dots f, g, h, \dots)(x, y, z, \dots)^2.$$

Je remarque qu'en général je représente une fonction rationnelle et intégrale, homogène et des *degrés*  $m, m'$ , &c., par rapport aux indéterminées  $x, y$ , &c.,  $x', y'$ , &c., de la manière suivante:

$$(\diamond)(x, y, \dots)^m (x', y', \dots)^{m'} \dots$$

Une fonction rationnelle et intégrale, homogène et du degré  $m$  par rapport aux deux indéterminées  $x, y$  sera donc représentée par

$$(\diamond)(x, y)^m.$$

En introduisant dans cette notation les *coefficients*, j'écris par exemple

$$(a, b, c, d \diamond x, y)^3,$$

pour représenter la fonction

$$ax^3 + 3bx^2y + 3cxy^2 + dy^3,$$

tandis que je me sers de la notation

$$(a, b, c, d \diamond x, y)^3,$$

pour représenter la fonction

$$ax^3 + bx^2y + cxy^2 + dy^3,$$

et de même pour les fonctions d'un degré quelconque. J'ai trouvé cette distinction très commode.

[In the foregoing Paper as here printed, except in the expression in the second line of this page,  $\diamond$  is used instead of  $\circ$ ) it appears by a remark (*Crelle*, t. li, errata) that the manuscript had the interlaced parentheses  $\circ$ . Moreover in the manuscript  $\circ$  was used for a Matrix, which was thus distinguished from a Determinant, but in the absence of any real ambiguity, no alteration has been made in this respect. In the reprint of subsequent papers from *Crelle*, the arrowhead  $\circ$  is used instead of  $\diamond$ .]

### 135. Note sur les covariants d'une fonction quadratique, cubique, ou biquadratique à deux indéterminées.

NOTE SUR LES COVARIANTS D'UNE FONCTION QUADRATIQUE, CUBIQUE, OU BIQUADRATIQUE  
À DEUX INDÉTERMINÉES.

[From the *Journal für die reine und angewandte Mathematik* (*Crelle*), tom. L. (1855), pp. 285–287.]

La théorie d'une fonction à deux indéterminées d'un degré quelconque, par exemple

$$(\diamond)(x, y)^m,$$

dépend du système des covariants de la fonction, lequel est censé contenir la fonction elle-même.

Pour une fonction *quadratique* le système de covariants est

$$(a, b, c)(x, y)^2, \\ ac - b^2.$$

Pour la fonction *cubique*, le système est

$$(a, b, c, d)(x, y)^3, \\ (ac - b^2, ad - bc, bd - c^2)(x, y)^2, \\ (-a^2d + 3abc - 2b^3, -abd + 2ac^2 - b^2c, acd - 2b^2d + bc^2, ad^2 - 3bcd + 2c^3)(x, y)^3, \\ -a^2d^2 + 6abcd - 4ac^3 - 4b^3d + 3b^2c^2,$$

fonctions lesquelles, en supposant qu'on les représente par  $U, H, \Phi, \square$ , satisfont identiquement à l'équation

$$\Phi^2 + \square U^3 = -4H^3.$$

Pour la fonction *biquadratique*, le système est

$$(a, b, c, d, e)(x, y)^4, \\ ae - 4bd + 3c^2, \\ (ac - b^2, 2ad - 2bc, ae + 2bd - 3c^2, 2be - 2cd, ce - d^2)(x, y)^4, \\ ace + 2bcd - ad^2 - b^2e - c^3,$$

$$\left\{ \begin{array}{l} -a^2d + 3abc - 2b^3, \\ -a^2e - 2abd + 9ac^2 - 6b^2c, \\ -5abe + 15acd - 10b^2d, \\ +10a^2d - 10b^2e, \\ +5ade + 10bd^2 - 15bce, \\ +ae^2 + 2bde - 9c^2e + 6cd^2, \\ +be^2 - 3cde + 2d^3, \end{array} \right\} (x, y)^6,$$

et ces fonctions, en supposant qu'on les représente par  $U, I, H, J, \Phi$ , satisfont identiquement à l'équation

$$JU^3 - IU^2H + 4H^3 = -\Phi^2.$$

J'ajoute à ce système la fonction

$$I^3 - 27J^2 = a^3e^3 - 12a^2bde^2 - 18a^2c^2e^2 + 54a^2cd^2e - 27a^2d^4 \\ + 54ab^2ce^2 - 6ab^2d^2e - 180abc^2de + 108abcd^3 + 81ac^4e \\ - 54ac^3d^2 - 27b^4e^2 - 64b^3d^3 + 108b^2cde - 54bcd^3 \\ + 36b^2c^2d^2,$$

qui est le *discriminant* de la fonction biquadratique.

Pour donner une application de ces formules, soit proposé de résoudre une équation quadratique, cubique ou biquadratique, ou autrement dit: de trouver un *facteur linéaire* de la fonction quadratique, cubique, ou biquadratique.

Il est assez singulier que pour la fonction quadratique la solution est en quelque sorte plus compliquée que pour les deux autres. En effet, il n'existe pas de solution symétrique, à moins



qu'on n'introduise des quantités arbitraires et superflues; savoir, on trouve pour facteur linéaire de  $(a, b, c)(x, y)^2$  l'expression

$$(\alpha, b, c)(x, \beta)(x, y) \pm \sqrt{-\square}(\beta x - \alpha y),$$

où  $(a, b, c)(\alpha, \beta)(x, y)$  dénote  $a\alpha x + b(\alpha y + \beta x) + c\beta y$ .

Pour la fonction *cubique*, l'équation  $\Phi^2 + \square U^3 = -4H^3$  fait voir que les deux fonctions  $\Phi + U\sqrt{-\square}$ ,  $\Phi - U\sqrt{-\square}$  sont l'une et l'autre des cubes parfaits. L'expression

$$\sqrt[3]{\frac{1}{2}(\Phi + U\sqrt{-\square})} - \sqrt[3]{\frac{1}{2}(\Phi - U\sqrt{-\square})}$$

sera donc une fonction linéaire de  $x, y$ ; et puisque cette fonction s'évanouit pour  $U = 0$ , elle ne sera autre chose que l'un des facteurs linéaires de  $(a, b, c, d)(x, y)^3$ .

Pour la fonction *biquadratique*, en partant de l'équation

$$JU^3 - IU^2H + 4H^3 = -\Phi^2,$$

j'écrirai

$$M = \frac{I^3}{4J^2},$$

et je mets l'équation sous la forme

$$(1, 0, -M, M)(H, JU)^3 = -\frac{1}{4}P^2\Phi^2.$$

Donc, en supposant que  $w_1, w_2, w_3$  soient les racines de l'équation

$$(1, 0, -M, M)(w, 1)^3 = 0,$$

ou plus simplement de l'équation

$$w^2 - M(w - 1) = 0,$$

ces expressions  $IH - w_1JU$ ,  $IH - w_2JU$ ,  $IH - w_3JU$  seront toutes trois des carrés de fonctions quadratiques. L'expression

$$(w_2 - w_3)\sqrt{(IH - w_1JU)} + (w_3 - w_1)\sqrt{(IH - w_2JU)} + (w_1 - w_2)\sqrt{(IH - w_3JU)}$$

sera donc une fonction quadratique, et on voit sans peine qu'elle sera le carré d'une fonction *linéaire*. Or cette expression s'évanouit pour  $U = 0$ ; donc ce sera précisément le carré de l'un quelconque des facteurs linéaires de  $(a, b, c, d, e)(x, y)^4$ .

L'équation identique pour les covariants d'une fonction biquadratique donne lieu aussi (remarque que je dois à M. Hermite) à une transformation très élégante de l'intégrale elliptique  $\int dx \div \sqrt{(a, b, c, d, e)(x, 1)^4}$ .

## 136. Sur la transformation d'une fonction quadratique en elle-même par des substitutions linéaires.

SUR LA TRANSFORMATION D'UNE FONCTION QUADRATIQUE EN ELLE-MÊME PAR DES  
SUBSTITUTIONS LINÉAIRES.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 288–299.]

Il s'agit de trouver les transformations linéaires d'une fonction quadratique  $(\diamond)(x, y, z, \dots)^2$  en elle-même, c'est-à-dire de trouver pour  $(x, y, z, \dots)$  des fonctions linéaires de  $x, y, z, \dots$  telles que

$$(\diamond)(x, y, z, \dots)^2 = (\diamond)(x, y, z, \dots)^2.$$

En représentant la fonction quadratique par

$$(\diamond)(x, y, z, \dots)^2 = \begin{vmatrix} a, & h, & g, & \dots \\ h, & b, & f, & \dots \\ g, & f, & c, & \dots \\ & & & \vdots \end{vmatrix} (x, y, z, \dots)^2,$$

la solution qu'a donnée M. Hermite de ce problème peut être résumée dans la seule équation

$$(x, y, z, \dots) = \begin{vmatrix} a, & h, & g, & \dots \\ h, & b, & f, & \dots \\ g, & f, & c, & \dots \\ & & & \vdots \end{vmatrix}^{-1} \begin{vmatrix} a, & h + \nu, & g - \mu, & \dots \\ h - \nu, & b, & f + \lambda, & \dots \\ g + \mu, & f - \lambda, & c, & \dots \\ & & & \vdots \end{vmatrix} \begin{vmatrix} a, & h - \nu, & g + \mu, & \dots \\ h + \nu, & b, & f - \lambda, & \dots \\ g - \mu, & f + \lambda, & c, & \dots \\ & & & \vdots \end{vmatrix}^{-1} (x, y, z, \dots)$$

où  $\lambda, \mu, \nu, \dots$  sont des quantités quelconques.

En effet, pour démontrer que cela est une solution, on n'a qu'à reproduire dans un ordre inverse le procédé de M. Hermite. En introduisant les quantités auxiliaires  $(\xi, \eta, \zeta, \dots)$ , on peut remplacer l'équation par les deux équations

$$\begin{vmatrix} a, & h, & g, & \dots \\ h, & b, & f, & \dots \\ g, & f, & c, & \dots \\ & & & \vdots \end{vmatrix} (x, y, z, \dots) = \begin{vmatrix} a, & h + \nu, & g - \mu, & \dots \\ h - \nu, & b, & f + \lambda, & \dots \\ g + \mu, & f - \lambda, & c, & \dots \\ & & & \vdots \end{vmatrix} (\xi, \eta, \zeta, \dots),$$

$$\begin{vmatrix} a, & h, & g, & \dots \\ h, & b, & f, & \dots \\ g, & f, & c, & \dots \\ & & & \vdots \end{vmatrix} (x, y, z, \dots) = \begin{vmatrix} a, & h - \nu, & g + \mu, & \dots \\ h + \nu, & b, & f - \lambda, & \dots \\ g - \mu, & f + \lambda, & c, & \dots \\ & & & \vdots \end{vmatrix} (\xi, \eta, \zeta, \dots),$$

qui donnent tout de suite d'abord

$$(\diamond)(x, y, z, \dots)(\xi, \eta, \zeta, \dots) = (\diamond)(\xi, \eta, \zeta, \dots)(y, \dots),$$

et puis

$$x + x = 2\xi, \quad y + y = 2\eta, \quad z + z = 2\zeta, \quad \&c.$$

On obtient par là

$$\begin{aligned} (\diamond)(x, y, z, \dots) &= (\diamond)(2\xi - x, 2\eta - y, 2\zeta - z, \dots)^2 \\ &= 4(\diamond)(\xi, \eta, \zeta, \dots) - 4(\diamond)(\xi, \eta, \zeta, \dots)(x, y, z, \dots) \\ &\quad + (\diamond)(x, y, z, \dots)^2, \end{aligned}$$

c'est-à-dire l'équation

$$(\diamond)(x, y, z, \dots)^2 = (\diamond)(x, y, z, \dots)^2,$$

qu'il s'agissait de vérifier.

Je remarque que la transformation est toujours *propre*. En effet, le déterminant de transformation est

$$\begin{vmatrix} a, & h, & g, & \dots \\ h, & b, & f, & \dots \\ g, & f, & c, & \dots \\ \vdots & & & \end{vmatrix}^{-1} \begin{vmatrix} a, & h - \nu, & g + \mu, & \dots \\ h + \nu, & b, & f - \lambda, & \dots \\ g - \mu, & f + \lambda, & c, & \dots \\ \vdots & & & \end{vmatrix}^{-1} \begin{vmatrix} a, & h + \nu, & g - \mu, & \dots \\ h - \nu, & b, & f + \lambda, & \dots \\ g + \mu, & f - \lambda, & c, & \dots \\ \vdots & & & \end{vmatrix}.$$

Or les déterminants qui entrent dans les deux termes moyens, ne contiennent l'un ou l'autre que des puissances paires de  $\lambda, \mu, \nu, \dots$ . Donc ces deux déterminants sont égaux, et les quatre termes du produit sont *récioproques* deux à deux; le déterminant de transformation est donc  $+1$ , et la transformation est *propre*.

Pour obtenir une transformation *impropre*, il faut considérer une fonction quadratique qui contienne, outre les indéterminées  $x, y, z, \dots$ , une indéterminée  $\theta$ , et puis réduire à zéro la valeur des coefficients de cette indéterminée  $\theta$ . Les valeurs de  $x, y, z, \dots$  en contiendront pas  $\theta$ , et en représentant par  $\mathfrak{D}$  le déterminant de transformation pour la forme aux indéterminées  $x, y, z, \dots$ , on aura par là  $\mathfrak{D} = -\theta$ ; le déterminant de transformation pour la forme aux indéterminées  $x, y, z, \dots$  multiplié par  $-1$ . Le déterminant de transformation pour la forme aux indéterminées  $x, y, z, \dots$  sera donc  $-1$ , c'est-à-dire, la transformation sera *impropre*.

Au lieu de la formule de transformation ci-dessus, on peut se servir des formules

$$(\xi, \eta, \zeta, \dots) = \begin{vmatrix} a, & h + \nu, & g - \mu, \dots \\ h - \nu, & b, & f + \lambda, \dots \\ g + \mu, & f - \lambda, & c, \dots \\ \vdots & & & \end{vmatrix}^{-1} \begin{vmatrix} a, & h, & g, \dots \\ h, & b, & f, \dots \\ g, & f, & c, \dots \\ \vdots & & & \end{vmatrix} (x, y, z, \dots),$$

$$x = 2\xi - x, \quad y = 2\eta - y, \quad z = 2\zeta - z, \quad \&c.$$

Par exemple, en supposant que la forme à transformer soit

$$ax^2 + by^2 + cz^2 + \&c.,$$

on aura

$$(\xi, \eta, \zeta, \dots) = \begin{vmatrix} a, & \nu, & -\mu, \dots \\ -\nu, & b, & \lambda, \dots \\ \mu, & -\lambda, & c, \dots \\ \vdots & & & \end{vmatrix}^{-1} (ax, by, cz, \dots),$$

$$x = 2\xi - x, \quad y = 2\eta - y, \quad z = 2\zeta - z, \quad \&c.$$

de manière qu'en posant

$$\begin{vmatrix} a, & \nu, & -\mu, \dots \\ -\nu, & b, & \lambda, \dots \\ \mu, & -\lambda, & c, \dots \\ \vdots & & & \end{vmatrix} = k,$$

$$\begin{vmatrix} a, & \nu, & -\mu, \dots \\ -\nu, & b, & \lambda, \dots \\ \mu, & -\lambda, & c, \dots \\ \vdots & & & \end{vmatrix}^{-1} = \frac{1}{k} \begin{vmatrix} A, & B, & C, \dots \\ A', & B', & C', \dots \\ A'', & B'', & C'', \dots \\ \vdots & & & \end{vmatrix},$$

on aura

$$(x, y, z, \dots) = \frac{1}{k} \begin{vmatrix} 2A - \frac{k}{a}, & 2B, & 2C, & \dots \\ 2A', & 2B' - \frac{k}{b}, & 2C', & \dots \\ 2A'', & 2B'', & 2C'' - \frac{k}{c}, & \dots \\ \vdots & & & \end{vmatrix} (ax, by, cz, \dots).$$

ce qui est l'équation pour la transformation *propre* en elle-même, de la fonction  $ax^2 + by^2 + cz^2 + \&c.$  On déduira, comme dans le cas général, la formule pour la transformation *impropre*. On trouvera des observations sur cette formule dans le mémoire "Recherches ultérieures sur les déterminants gauches" [137].

Je reviens à l'équation générale

$$(\diamond)(x, y, z, \dots)^2 = (\diamond)(x, y, z, \dots)^2,$$

et je suppose seulement que  $x, y, z, \dots$  soient des fonctions linéaires de  $x, y, z, \dots$  qui satisfont à cette équation sans supposer rien davantage par rapport à la forme de la solution. Cela étant, je forme les fonctions linéaires  $x - ax, y - ay, z - az, \&c.$ , où  $a$  est une quantité quelconque, et je considère la fonction

$$(\diamond)(x - ax, y - ay, z - az, \dots)^2,$$

laquelle, en la développant, devient

$$(1 + a^2)(\diamond)(x, y, z, \dots)^2 - 2a(\diamond)(x, y, z, \dots)(x, y, z, \dots);$$

et en développant de la même manière la fonction quadratique

$$(\diamond)(x - \frac{1}{a}x, y - \frac{1}{a}y, z - \frac{1}{a}z, \dots)^2,$$

on obtient l'équation identique

$$(\diamond)(x - ax, y - ay, z - az, \dots)^2 = (\diamond)(x - \frac{1}{a}x, y - \frac{1}{a}y, z - \frac{1}{a}z, \dots)^2.$$

Soit  $\square$  le déterminant formé avec les coefficients de fonctions linéaires  $x - ax, y - ay, z - az, \&c.$  En supposant que le nombre des indéterminées  $x, y, z, \&c.$ , est  $n$ ,  $\square$  sera évidemment une fonction rationnelle et intégrale du degré  $n$  par rapport à  $a$ . Soit de même  $\square'$  le déterminant formé avec les coefficients de

$$x - \frac{1}{a}x, \quad y - \frac{1}{a}y, \quad z - \frac{1}{a}z, \quad \&c.;$$

l'équation qui vient d'être trouvée, donne  $\square' = a^n \square$ , c'est-à-dire  $\square = \pm a^n \square'$ . Cela fait voir que les coefficients du premier et du dernier terme de l'avant-dernier terme,  $\&c.$ , sont égaux, aux signes près. De plus, le coefficient de la plus haute puissance  $a^n$  est toujours  $\pm 1$ , et on voit sans peine qu'en supposant d'abord que  $a$  soit *impair*, on a pour la transformation *propre*:

$$\square = (1, P, \dots P, 1)(a, 1)^n;$$

et pour la transformation *impropre*

$$\square = (1, -P, \dots -P, 1)(a, -1)^n;$$

équation qui peut être changée en celles-ci:  $\square = -(1, -P, \dots P, 1)(a, 1)^n$ . Puis, en supposant que  $n$  soit *pair*, on a pour la transformation *propre*:

$$\square = (1, P, \dots P, 1)(a, 1)^n;$$

et pour la transformation *impropre*

$$\square = (1, -P, \dots, P, -1)(a, 1)^n;$$

le coefficient moyen étant dans ce cas égal à zéro. Ces théorèmes pour la forme du déterminant des fonctions linéaires  $x - ax, y - ay, z - az, \dots$  sont dus à M. Hermite.

Il y a à remarquer que la forme  $(\diamond)(x, y, z, \dots)^2$  est indéterminée; c'est-à-dire, on suppose seulement que  $x, y, z, \dots$  soient des fonctions linéaires de  $x, y, z, \dots$ , telles qu'il y ait une forme quadratique  $(\diamond)(x, y, z, \dots)^2$  pour laquelle l'équation  $(\diamond)(x, y, z, \dots)^2 = (\diamond)(x, y, z, \dots)^2$  est satisfaite.

Je regarde d'un autre point de vue ce problème de la transformation en elle-même, d'une fonction quadratique par des substitutions linéaires. Je suppose que  $x, y, z, \&c.$ , sont des fonctions linéaires de  $x, y, z, \dots$ , et je cherche les conditions pour que la fonction linéaire, il faut que  $\beta = 1/\beta$ , &c. Le symbole  $\beta a$  ne se trouve et dans le déterminant, ni au côté droit. Cependant il convient de poser  $\beta a = -a\beta$ ; car cela étant, il serait permis de transformer les Pfaffians, en écrivant par exemple  $\alpha\beta 12 = -\beta\alpha 12$ .

Je prends en passant que, si avant de poser l'équation  $\beta a = -\alpha\beta$ , on suppose que les deux symboles  $a, \beta$  deviennent identiques (ce pour exemple on écrit  $a = \beta = 5$ ), on aurait par exemple

$$a\beta.12 = a\beta.12 + a1.2\beta + a2.\beta 1 = 55.12 + 51.25 + 52.51 = 55.12, \&c.,$$

et en retrouvant ainsi la formule pour le développement de  $\overline{12345} \mid 12345$ .

La nouvelle formule peut être appliquée immédiatement au développement des *déterminants mineurs*. En effet, en se servant de la notation des matrices, on aura

$$\begin{vmatrix} 11, & 12, & 13 \\ 21, & 22, & 23 \\ 31, & 32, & 33 \end{vmatrix}^{-1} = \frac{1}{\overline{123} \mid 123} \begin{vmatrix} +23 \overline{23}, & -13 \overline{23}, & -12 \overline{32} \\ -23 \overline{13}, & +13 \overline{13}, & -21 \overline{31} \\ -32 \overline{12}, & -31 \overline{21}, & +12 \overline{12} \end{vmatrix},$$

$$\begin{vmatrix} 11, & 12, & 13, & 14 \\ 21, & 22, & 23, & 24 \\ 31, & 32, & 33, & 34 \\ 41, & 42, & 43, & 44 \end{vmatrix}^{-1} = \frac{1}{\overline{1234} \mid 1234} \begin{vmatrix} +234 \overline{234}, & -134 \overline{234}, & -124 \overline{324}, & -123 \overline{324} \\ -234 \overline{134}, & +134 \overline{134}, & -214 \overline{314}, & -213 \overline{413} \\ -324 \overline{124}, & -314 \overline{214}, & +124 \overline{124}, & -312 \overline{412} \\ -423 \overline{123}, & -413 \overline{213}, & -412 \overline{312}, & +123 \overline{123} \end{vmatrix},$$

et ainsi de suite. On suppose toujours que chaque terme de la matrice à droite soit divisé par le dénominateur commun. On voit que les déterminants mineurs qui correspondent à des termes tels que  $xx$  sont des déterminants *ordinaires*, avec le signe  $+$ , tandis que les déterminants mineurs qui correspondent à des termes tels que  $\alpha\beta$ , sont des quatre termes du produit sont *réciproques* deux à deux; le déterminant de transformation est donc  $\pm 1$ , et la transformation est *propre*.

## 137. Recherches ultérieures sur les déterminants gauches.

### RECHERCHES ULTÉRIEURES SUR LES DÉTERMINANTS GAUCHES.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 299–313: Continuation of the Memoir t. XXXII. (1846) and t. XXXVIII. (1849); **52** and **69**.]

J'ai déjà donné une formule pour le développement d'un *déterminant gauche*. En prenant, pour fixer les idées, un cas particulier, soit

$$\overline{12345} \mid 12345 = \begin{vmatrix} 11, & 12, & 13, & 14, & 15 \\ 21, & 22, & 23, & 24, & 25 \\ 31, & 32, & 33, & 34, & 35 \\ 41, & 42, & 43, & 44, & 45 \\ 51, & 52, & 53, & 54, & 55 \end{vmatrix},$$

(où  $12 = -21$ , &c., tandis que les quantités 11, 22, &c. ne s'évanouissent pas). Cette formule peut être écrite comme suit:

$$\begin{aligned} \overline{12345} \mid \overline{12345} = & 11 \cdot 22 \cdot 33 \cdot 44 \cdot 55 \\ & + 11 \cdot 22 \cdot 33 \cdot (45)^2 \\ & + 11 \cdot 22 \cdot 44 \cdot (35)^2 \\ & + 11 \cdot 22 \cdot 55 \cdot (34)^2 \\ & + 11 \cdot 33 \cdot 44 \cdot (25)^2 \\ & + 11 \cdot 33 \cdot 55 \cdot (24)^2 \\ & + 11 \cdot 44 \cdot 55 \cdot (23)^2 \\ & + 22 \cdot 33 \cdot 44 \cdot (15)^2 \\ & + 22 \cdot 33 \cdot 55 \cdot (14)^2 \\ & + 22 \cdot 44 \cdot 55 \cdot (13)^2 \\ & + 33 \cdot 44 \cdot 55 \cdot (12)^2 \\ & + 11 \cdot (2345)^2 \\ & + 22 \cdot (1345)^2 \\ & + 33 \cdot (1245)^2 \\ & + 44 \cdot (1235)^2 \\ & + 55 \cdot (1234)^2. \end{aligned}$$

Les expressions 12, 1234, &c. à droite sont ici des *Pfaffians*. On a

$$12 = 12,$$

$$1234 = 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23,$$

et en écrivant encore un terme, pour mieux présenter la loi:

$$\begin{aligned} 123456 = & 12 \cdot 34 \cdot 56 + 13 \cdot 45 \cdot 62 + 14 \cdot 56 \cdot 23 + 15 \cdot 62 \cdot 34 + 16 \cdot 23 \cdot 45 \\ & + 12 \cdot 35 \cdot 64 + 13 \cdot 46 \cdot 25 + 14 \cdot 52 \cdot 36 + 15 \cdot 63 \cdot 42 + 16 \cdot 24 \cdot 53 \\ & + 12 \cdot 36 \cdot 54 + 13 \cdot 42 \cdot 56 + 14 \cdot 53 \cdot 62 + 15 \cdot 64 \cdot 23 + 16 \cdot 25 \cdot 34. \end{aligned}$$

J'ai trouvé récemment une formule analogue pour le développement d'un *déterminant gauche bordé*, tel que

$$\overline{\alpha 1234} \mid \overline{\beta 1234} = \alpha \beta \begin{vmatrix} \alpha \beta, & \alpha 1, & \alpha 2, & \alpha 3, & \alpha 4 \\ 1 \beta, & 11, & 12, & 13, & 14 \\ 2 \beta, & 21, & 22, & 23, & 24 \\ 3 \beta, & 31, & 32, & 33, & 34 \\ 4 \beta, & 41, & 42, & 43, & 44 \end{vmatrix};$$

cette formule est:

$$\begin{aligned}
\overline{\alpha 1234 \mid \beta 1234} = & \alpha\beta \cdot 11 \cdot 22 \cdot 33 \cdot 44 \\
& + \alpha\beta \cdot 12 \cdot 12 \cdot 33 \cdot 44 \\
& + \alpha\beta \cdot 13 \cdot 13 \cdot 22 \cdot 44 \\
& + \alpha\beta \cdot 14 \cdot 14 \cdot 22 \cdot 33 \\
& + \alpha\beta \cdot 23 \cdot 23 \cdot 11 \cdot 44 \\
& + \alpha\beta \cdot 24 \cdot 24 \cdot 11 \cdot 33 \\
& + \alpha\beta \cdot 34 \cdot 34 \cdot 11 \cdot 22 \\
& + \alpha\beta \cdot 1234 \cdot 1234 \\
& + \alpha 1 \cdot \beta 1 \cdot 22 \cdot 33 \cdot 44 \\
& + \alpha 2 \cdot \beta 2 \cdot 11 \cdot 33 \cdot 44 \\
& + \alpha 3 \cdot \beta 3 \cdot 11 \cdot 22 \cdot 44 \\
& + \alpha 4 \cdot \beta 4 \cdot 11 \cdot 22 \cdot 33 \\
& + \alpha 123 \cdot \beta 123 \cdot 44 \\
& + \alpha 124 \cdot \beta 124 \cdot 33 \\
& + \alpha 134 \cdot \beta 134 \cdot 22 \\
& + \alpha 234 \cdot \beta 234 \cdot 11.
\end{aligned}$$

Il est à peine nécessaire de remarquer que dans les Pfaffians à droite, qui entrent des symboles tels que  $1\alpha$ ,  $\beta 1$ , &c., qui ne se trouvent pas dans le déterminant dont il s'agit, il faut entendre  $1\alpha = -\alpha 1$ ,  $\beta 1 = -1\beta$ , &c. Le symbole  $\beta\alpha$  ne se trouve ni dans le déterminant, ni au côté droit. Cependant il convient de poser  $\beta\alpha = -\alpha\beta$ ; car cela étant, il serait permis de transformer les Pfaffians, en écrivant par exemple  $\alpha\beta 12 = -\beta\alpha 12$ .

Je remarque en passant que, si avant de poser l'équation  $\beta\alpha = -\alpha\beta$ , on suppose que les deux symboles  $\alpha$ ,  $\beta$  deviennent identiques (ce pour exemple on écrit  $\alpha = \beta = 5$ ), on aurait par exemple

$$\alpha\beta \cdot 12 = \alpha\beta \cdot 12 + \alpha 1 \cdot 2\beta + \alpha 2 \cdot \beta 1 = 55 \cdot 12 + 51 \cdot 25 + 52 \cdot 51 = 55 \cdot 12, \text{ \&c.,}$$

et en retrouvant ainsi la formule pour le développement de  $\overline{12345 \mid 12345}$ .

La nouvelle formule peut être appliquée immédiatement au développement des *déterminants mineurs*. En effet, en se servant de la notation des matrices, on aura les expressions pour les matrices inverses, comme on le voit dans le mémoire.

(En effet, en développant les deux côtés, on obtient

$$11 \cdot 22 \cdot 33 + 11 \cdot (23)^2 + 22 \cdot (13)^2 + 33 \cdot (12)^2 = 11 \cdot (22 \cdot 33 + (23)^2) + 22 \cdot (21 \cdot 33 + 23 \cdot 13) + 31 \cdot (31 \cdot 22 + 32 \cdot 12).$$

On voit que les deux termes marqués par un † se détruisent et que l'équation est *identique*. On doit avoir de même

$$\begin{aligned}
\overline{1234 \mid 1234} = & 11 \cdot \overline{234 \mid 234} \\
& - 12 \cdot \overline{234 \mid 134} \\
& - 13 \cdot \overline{324 \mid 124} \\
& - 14 \cdot \overline{423 \mid 123},
\end{aligned}$$

ou, ce qui est la même chose:

$$\begin{aligned}\overline{1234 \mid 1234} &= 11 \cdot \overline{234 \mid 234} \\ &\quad + 21 \cdot \overline{234 \mid 134} \\ &\quad + 31 \cdot \overline{324 \mid 124} \\ &\quad + 41 \cdot \overline{423 \mid 123};\end{aligned}$$

c'est-à-dire, en développant des deux côtés:

$$\begin{aligned}&11 \cdot 22 \cdot 33 \cdot 44 + 11 \cdot 22 \cdot 33 \cdot (34)^2 + 11 \cdot 44 \cdot (23)^2 \\ &+ 22 \cdot 33 \cdot (14)^2 + 33 \cdot 44 \cdot (12)^2 + (1234)^2 \\ &= 11 \cdot \{22 \cdot 33 \cdot 44 + 22 \cdot (34)^2 + 33 \cdot (24)^2 + 44 \cdot (23)^2 + (234)^2\} \\ &+ 21 \cdot \{21 \cdot 33 \cdot 44 + 23 \cdot 13 \cdot 44 + 24 \cdot 14 \cdot 33 + 34 \cdot (2\uparrow)^2\} \\ &+ 31 \cdot \{31 \cdot 22 \cdot 44 + 32 \cdot 12 \cdot 44 + 34 \cdot 14 \cdot 22 + 24 \cdot (3\uparrow)^2\} \\ &+ 41 \cdot \{41 \cdot 22 \cdot 33 + 42 \cdot 12 \cdot 33 + 43 \cdot 13 \cdot 22 + 23 \cdot (4\uparrow)^2\}.\end{aligned}$$

Cette expression est en effet identique, comme on le voit en observant que les six termes marqués par un  $\uparrow$  se détruisent deux à deux, et que les trois termes marqués par un  $(*)$  sont l'ensemble équivalents à  $(1234)^2$ .)

Je remarque que le nombre des termes du développement du déterminant gauche est toujours une puissance de 2, et que pour cela, ce nombre se réduit à la moitié en réduisant à zéro un terme quelconque  $xx$ . Mais outre cela, le déterminant peut dans cette supposition s'exprimer comme une somme d'une certaine quantité de *carrés* de fonctions d'un ordre inférieur à l'unité. Je considère par exemple le déterminant gauche  $\overline{123 \mid 123}$ . En faisant  $33 = 0$  et accentuant, pour y mettre plus de clarté, tous les symboles, on trouve

$$\overline{123 \mid 123} = 11' \cdot (23')^2 + 22' \cdot (13')^2.$$

De là, en dérivant

$$\begin{aligned}11 &= 13' \cdot 11', & 12 &= 11' \cdot 23', \\ 22 &= 13' \cdot 22',\end{aligned}$$

on obtient

$$\overline{12 \mid 12} = 11 \cdot 22 + (12)^2 = 13' \cdot \{22' \cdot (13')^2 + 11' \cdot (23')^2\},$$

c'est-à-dire

$$\overline{12 \mid 12} = 11' \cdot 14' \cdot \overline{123 \mid 123}.$$

On a de même

$$\overline{1234 \mid 1234} = 11' \cdot 22' \cdot (34')^2 + 11' \cdot 33' \cdot (24')^2 + 22' \cdot 33' \cdot (14')^2 + (1234')^2,$$

et de là, en dérivant

$$\begin{aligned}11 &= 14' \cdot 11', & 12 &= 11' \cdot 24', & 23 &= 1234', \\ 22 &= 14' \cdot 22', & 13 &= 11' \cdot 34', \\ 33 &= 14' \cdot 33',\end{aligned}$$

on obtient

$$\begin{aligned}\overline{123 \mid 123} &= 11 \cdot 22 \cdot 33 + 11 \cdot (23)^2 + 22 \cdot (31)^2 + 33 \cdot (12)^2 \\ &= 11' \cdot 14' \{22' \cdot 33' \cdot (14')^2 + (1234')^2\} + 11' \cdot 22' \cdot (34')^2 + 11' \cdot 33' \cdot (24')^2,\end{aligned}$$



c'est-à-dire

$$\overline{123 \mid 123} = 11' \cdot 14' \cdot \overline{1234 \mid 1234}.$$

De même

$$\begin{aligned} \overline{12345 \mid 12345} &= 11' \cdot 22' \cdot 33' \cdot 44' \cdot (55')^2 \\ &\quad + 11' \cdot 22' \cdot 33' \cdot (45')^2 \\ &\quad + 11' \cdot 22' \cdot 44' \cdot (35')^2 \\ &\quad + 11' \cdot 33' \cdot 44' \cdot (25')^2 \\ &\quad + 22' \cdot 33' \cdot 44' \cdot (15')^2 \\ &\quad + 11' \cdot (2345')^2 \\ &\quad + 22' \cdot (1345')^2 \\ &\quad + 33' \cdot (1245')^2 \\ &\quad + 44' \cdot (1235')^2. \end{aligned}$$

Or il est permis d'écrire

$$\begin{aligned} 11 &= 15' \cdot 11', & 12 &= 11' \cdot 25', & 23 &= 1235', & 1234 &= 12345' \cdot 15', \\ 22 &= 15' \cdot 22', & 13 &= 11' \cdot 35', & 24 &= 1245', \\ 33 &= 15' \cdot 33', & 14 &= 11' \cdot 45', & 34 &= 1345', \\ 44 &= 15' \cdot 44'. \end{aligned}$$

En effet, les quantités à gauche ne sont liées entre elles que par la seule équation  $1234 = 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23$ , qui est satisfaite identiquement par les valeurs à substituer pour ces quantités qui y entrent. Cela étant, on trouve d'abord:

$$\overline{1234 \mid 1234} = 11'(15')^2 \overline{12345 \mid 12345}.$$

Je prends encore un exemple. On a

$$\begin{aligned} \overline{123456 \mid 123456} &= 11' \cdot 22' \cdot 33' \cdot 44' \cdot (56')^2 \\ &\quad + 11' \cdot 22' \cdot 44' \cdot 55' \cdot (46')^2 \\ &\quad + 11' \cdot 22' \cdot 44' \cdot 55' \cdot (36')^2 \\ &\quad + 11' \cdot 33' \cdot 44' \cdot 55' \cdot (26')^2 \\ &\quad + 22' \cdot 33' \cdot 44' \cdot 55' \cdot (16')^2 \\ &\quad + 11' \cdot 22' \cdot (3456')^2 \\ &\quad + 11' \cdot 33' \cdot (2456')^2 \\ &\quad + 11' \cdot 44' \cdot (2356')^2 \\ &\quad + 11' \cdot 55' \cdot (2346')^2 \\ &\quad + 22' \cdot 33' \cdot (1456')^2 \\ &\quad + 22' \cdot 44' \cdot (1356')^2 \\ &\quad + 22' \cdot 55' \cdot (1346')^2 \\ &\quad + 33' \cdot 44' \cdot (1256')^2 \\ &\quad + 33' \cdot 55' \cdot (1246')^2 \\ &\quad + 44' \cdot 55' \cdot (1236')^2 \\ &\quad + (123456')^2. \end{aligned}$$

Ici, il est permis d'écrire

$$\begin{aligned}
11 &= 16' \cdot 11', & 12 &= 16' \cdot 26', & 23 &= 1236', & 34 &= 1346', & 45 &= 1456', \\
22 &= 16' \cdot 22', & 13 &= 16' \cdot 36', & 24 &= 1246', & 35 &= 1356', \\
33 &= 16' \cdot 33', & 14 &= 16' \cdot 46', & 25 &= 1256', \\
44 &= 16' \cdot 44', & 15 &= 16' \cdot 56', \\
55 &= 16' \cdot 55', \\
1234 &= 2346' \cdot 11' \cdot 16', & 2345 &= 123456' \cdot 16', \\
1235 &= 2356' \cdot 11' \cdot 16', \\
1245 &= 2456' \cdot 11' \cdot 16', \\
1345 &= 3456' \cdot 11' \cdot 16';
\end{aligned}$$

car les valeurs des quantités à gauche satisfont identiquement aux équations qui ont lieu entre ces mêmes quantités. Par exemple l'équation  $1234 = 12 \cdot 34 + 13 \cdot 42 + 14 \cdot 23$  devient  $2346' \cdot 16' = 26' \cdot 1346' + 63' \cdot 1246' + 46' \cdot 1236'$ .

Or l'expression à droite devient, en développant:

$$\begin{aligned}
&26'(13' \cdot 46' + 63' \cdot 14' + 16' \cdot 34') \\
&+ 63'(12' \cdot 46' + 14' \cdot 62' + 16' \cdot 24') \\
&+ 46'(12' \cdot 36' + 13' \cdot 62' + 16' \cdot 23').
\end{aligned}$$

et les termes qui contiennent le facteur  $16'$ , donnent ensemble  $16' \cdot 2346'$ , les autres termes se détruisent deux à deux. On obtient enfin, en effectuant la substitution:

$$\overline{12345} \mid \overline{12345} = 11' \cdot (16')^2 \overline{123456} \mid \overline{123456};$$

et ainsi de suite.

Je fais les mêmes substitutions dans les matrices inverses, en supprimant cependant la dernière ligne et la dernière colonne de chaque matrice. On trouve ainsi, en y ajoutant les équations ci-dessus trouvées par rapport aux déterminants:

$$\begin{aligned}
11' \cdot \overline{123} \mid \overline{123} &= \overline{12} \mid \overline{12}, \\
\frac{1}{13' \cdot \overline{123} \mid \overline{123}} \begin{vmatrix} +23 \cdot 23, & -13 \cdot 23 \\ -23 \cdot 13, & +13 \cdot 13 \end{vmatrix} &= \frac{1}{\overline{12} \mid \overline{12}} \begin{vmatrix} -2\bar{2}2 + \frac{12 \cdot 12}{11}, & -1\bar{2} \\ +2\bar{1}1, & +2\bar{1}1 - \frac{1\bar{2}}{1} \end{vmatrix}, \\
11' \cdot 14' \cdot \overline{1234} \mid \overline{1234} &= \overline{123} \mid \overline{123}, \\
\frac{1}{14' \cdot \overline{1234} \mid \overline{1234}} \begin{vmatrix} +234 \cdot 234, & -134 \cdot 234, & -124 \cdot 324 \\ -234 \cdot 134, & +134 \cdot 134, & -214 \cdot 314 \\ -324 \cdot 124, & -314 \cdot 214, & +124 \cdot 124 \end{vmatrix} &= \frac{1}{\overline{123} \mid \overline{123}} \begin{vmatrix} -23 \cdot 23 + \frac{123 \cdot 123}{11}, & -2 \cdot 1 \cdot 2 \\ +23 \cdot 13, & +13 \cdot 13 - \frac{123 \cdot 2}{22} \end{vmatrix}, \\
11' \cdot 15' \cdot \overline{12345} \mid \overline{12345} &= \overline{1234} \mid \overline{1234}, \\
\frac{1}{15' \cdot \overline{12345} \mid \overline{12345}} \begin{vmatrix} +2345 \cdot 2345, & -1345 \cdot 2345, & -1245 \cdot 3245, & -1235 \cdot 4235 \\ -2345 \cdot 1345, & +1345 \cdot 1345, & -2145 \cdot 3145, & -2135 \cdot 4135 \\ -3245 \cdot 1245, & -3145 \cdot 2145, & +1245 \cdot 1245, & -3125 \cdot 4125 \\ -4235 \cdot 1235, & -4135 \cdot 2135, & -4125 \cdot 3125, & +1235 \cdot 1235 \end{vmatrix} &= \frac{1}{\overline{1234} \mid \overline{1234}} \begin{vmatrix} -234 \cdot 234 + \\ +234 \\ +324 \end{vmatrix}
\end{aligned}$$

et ainsi de suite.

Il est bon de changer un peu la forme de ces équations. On en déduit sans peine:

$$\begin{aligned} \frac{1}{13' \cdot 123 \mid 123} & \left| \begin{array}{cc} 2 \cdot 23 \cdot \overline{23} - \frac{23 \cdot 23}{11}, & -2 \cdot 13 \cdot \overline{23} \\ +2 \cdot 23 \cdot 13', & +2 \cdot 13 \cdot 13' - \frac{123 \cdot 123}{22} \end{array} \right| = \frac{1}{12 \mid 12} \left| \begin{array}{cc} -2 \cdot 2\overline{2} + \frac{12 \cdot 12}{11}, & -2 \cdot 1\overline{2} \\ +2 \cdot 2\overline{1}1, & +2 \cdot 1\overline{1} - \frac{1\overline{2}}{1} \end{array} \right|, \\ \frac{1}{14' \cdot 1234 \mid 1234} & \left| \begin{array}{ccc} 2 \cdot 234 \cdot \overline{234} - \frac{1234 \cdot 1234}{11}, & -2 \cdot 134 \cdot \overline{234}, & -2 \cdot 124 \cdot \overline{324} \\ -2 \cdot 234 \cdot \overline{134}, & 2 \cdot 134 \cdot 134' - \frac{1234 \cdot 1234}{22'}, & -2 \cdot 214 \cdot \overline{314} \\ -2 \cdot 324 \cdot \overline{124}, & -2 \cdot 314 \cdot \overline{214}, & 2 \cdot 124 \cdot 124' - \frac{1234 \cdot 1234}{33} \end{array} \right| = \frac{1}{123} \\ & = \frac{1}{123 \mid 123} \left| \begin{array}{cc} -2 \cdot 23 \cdot \overline{23} + \frac{123 \cdot 123}{11}, & -2 \cdot 13 \cdot 23 \\ +2 \cdot 23 \cdot 13, & +2 \cdot 13 \cdot 13 - \frac{123 \cdot 123}{22} \end{array} \right|, \end{aligned}$$

et ainsi de suite.

Les formules que je viens de présenter, peuvent être appliquées aussitôt à la solution de la question suivante: "Trouver  $x_1, x_2, x_3$ , &c., fonctions linéaires de  $x_1, x_2, x_3$ , &c. telles que

$$11x_1^2 + 22x_2^2 + 33x_3^2 + \&c. = 11x_1'^2 + 22x_2'^2 + 33x_3'^2 + \&c.,"$$

c'est-à-dire: transformer une fonction quadratique  $11x_1^2 + 22x_2^2 + 33x_3^2 + \&c.$  en *elle-même* par des substitutions linéaires. Il suffira d'écrire les solutions pour le cas de trois indéterminées: on satisfait identiquement à l'équation

$$11x_1^2 + 22x_2^2 + 33x_3^2 = 11x_1'^2 + 22x_2'^2 + 33x_3'^2$$

en dérivant

$$(x_1, x_2, x_3) = \frac{1}{123 \mid 123} \left| \begin{array}{ccc} +2 \cdot 23 \cdot \overline{23} - \frac{123 \cdot 123}{11}, & -2 \cdot 13 \cdot \overline{23}, & -2 \cdot 12 \cdot \overline{32} \\ -2 \cdot 23 \cdot \overline{13}, & +2 \cdot 13 \cdot \overline{13} - \frac{123 \cdot 123}{22}, & -2 \cdot 21 \cdot \overline{31} \\ -2 \cdot 32 \cdot \overline{12}, & -2 \cdot 31 \cdot \overline{21}, & +2 \cdot 12 \cdot 12 - \frac{123 \cdot 123}{33} \end{array} \right| (11x_1, 22x_2, 33x_3)$$

Voilà la transformation *propre*. On tire la transformation *impropre* de  $11x_1^2 + 22x_2^2$  en elle-même en écrivant  $33 = 0$ ; car, cela posé, les valeurs de  $x_1, x_2$  ne contiennent pas  $x_3$ , et l'on n'a plus besoin de la valeur de  $x_3$ . On obtient ainsi la solution suivante; savoir, on satisfait identiquement à l'équation

$$11x_1^2 + 22x_2^2 = 11x_1'^2 + 22x_2'^2$$

en dérivant

$$(x_1, x_2) = \frac{1}{123 \mid 123} \left| \begin{array}{cc} 2 \cdot 23 \cdot \overline{23} - \frac{123 \cdot 123}{11}, & -2 \cdot 13 \cdot \overline{23} \\ -2 \cdot 23 \cdot \overline{13}, & 2 \cdot 13 \cdot \overline{13} - \frac{123 \cdot 123}{22} \end{array} \right| (11x_1, 22x_2),$$

ce qui se réduit, en faisant la substitution  $11 = 13' \cdot 11', 22 = 13' \cdot 22', 12 = 11' \cdot 23'$ , à la solution à celle-ci, savoir de satisfaire identiquement à l'équation  $11x_1^2 + 22x_2^2 = 11x_1'^2 + 22x_2'^2$ :

$$(x_1, x_2) = \frac{1}{12 \mid 12} \left| \begin{array}{cc} -2 \cdot 2\overline{2} + \frac{12 \cdot 12}{11}, & -2 \cdot 1\overline{2} \\ +2 \cdot 2\overline{1}, & +2 \cdot 1\overline{1} - \frac{12 \cdot 12}{22} \end{array} \right| (11x_1, 22x_2),$$

ce qui est encore une transformation *impropre*, qui correspond de plus près à la formule pour la transformation *propre*; la seule différence est que les signes des termes de la première colonne de la matrice ont été changés.

En introduisant des lettres simples  $a, b$ , &c. à la place des symboles 11, 22, &c., je considère d'abord la transformation *propre*

$$ax^2 + by^2 = ax^2 + by^2.$$

Ici, en écrivant

$$\begin{vmatrix} 11, & 12 \\ 21, & 22 \end{vmatrix} = \begin{vmatrix} a, & \nu \\ -\nu, & b \end{vmatrix},$$

la formule donne

$$(x, y) = \frac{1}{ab + \nu^2} \begin{vmatrix} ab - \nu^2, & -2b\nu \\ 2\nu a, & ab - \nu^2 \end{vmatrix} (x, y).$$

La transformation *impropre*

$$ax^2 + by^2 = ax^2 + by^2$$

s'obtient au moyen de la formule donnée plus haut pour la transformation *propre* de la fonction  $ax^2 + by^2 + ex^2$  en *elle-même*. En y écrivant  $a = b = c = \omega$ . On a ainsi pour la transformation *propre*:

$$(x, y, z) = \frac{1}{\omega^2 + \lambda^2 + \mu^2 + \nu^2} \begin{vmatrix} \omega\lambda - \mu^2 - \nu^2, & 2\lambda\mu - 2\nu\omega, & 2\nu\lambda + 2\mu\omega \\ 2\lambda\mu + 2\nu\omega, & \omega^2 - \lambda^2 + \mu^2 - \nu^2, & 2\mu\nu - 2\lambda\omega \\ 2\nu\lambda - 2\mu\omega, & 2\mu\nu + 2\lambda\omega, & \omega^2 - \lambda^2 - \mu^2 + \nu^2 \end{vmatrix} (x, y, z).$$

et en dérivant dans la formule pour la transformation *impropre*,  $a = b = c = 1$ ,  $d = 0$ , en écrivant  $\lambda, \mu, \nu, -\omega$  au lieu de  $\rho, \sigma, \tau, \phi$ , on obtient pour la transformation *impropre*

$$x^2 + y^2 + z^2 = x^2 + y^2 + z^2,$$

l'équation

$$(x, y, z) = \frac{1}{\omega^2 + \lambda^2 + \mu^2 + \nu^2} \begin{vmatrix} -\omega^2 + \lambda^2 + \mu^2 + \nu^2, & -2\lambda\mu + 2\nu\omega, & -2\nu\lambda - 2\mu\omega \\ -2\lambda\mu - 2\nu\omega, & -\omega^2 + \lambda^2 - \mu^2 + \nu^2, & 2\mu\nu - 2\lambda\omega \\ -2\nu\lambda + 2\mu\omega, & -2\mu\nu - 2\lambda\omega, & -\omega^2 + \lambda^2 + \mu^2 - \nu^2 \end{vmatrix} (x, y, z).$$

Pour obtenir la transformation *propre*

$$ax^2 + by^2 + cz^2 + dw^2 = ax^2 + by^2 + cz^2 + dw^2,$$

j'écris

$$\begin{vmatrix} 11, & 12, & 13, & 14 \\ 21, & 22, & 23, & 24 \\ 31, & 32, & 33, & 34 \\ 41, & 42, & 43, & 44 \end{vmatrix} = \begin{vmatrix} a, & \nu, & -\mu, & \rho \\ -\nu, & b, & \lambda, & \sigma \\ \mu, & -\lambda, & c, & \tau \\ -\rho, & -\sigma, & -\tau, & d \end{vmatrix};$$

cela donne d'abord, en mettant pour abréger

$$\phi = \lambda\rho + \mu\sigma + \nu\tau,$$

la valeur du déterminant

$$\overline{1234} \mid \overline{1234} = abcd + bcd\rho^2 + cad\sigma^2 + abd\tau^2 + ad\lambda^2 + bd\mu^2 + cd\nu^2 + \phi^2$$

(ce que je représente par  $k$ ).

J'ajoute aussi la valeur de la *matrice inverse*

$$\begin{vmatrix} 11, & 12, & 13, & 14 \\ 21, & 22, & 23, & 24 \\ 31, & 32, & 33, & 34 \\ 41, & 42, & 43, & 44 \end{vmatrix}^{-1},$$

savoir:

$$\begin{aligned} \frac{1}{k} \{ & bcd + b\tau^2 + c\sigma^2 + d\lambda^2, \quad -cdv - v\tau^2 + d\lambda\mu - \sigma\rho, \\ & cdu + v\rho\tau + d\lambda\nu - \mu\tau, \quad acd + b\tau^2 + c\rho^2 + d\mu^2, \\ & -bdp - \nu\sigma\tau + d\mu\nu - \lambda\rho, \quad ucd + c\nu\rho + d\nu\mu + d\nu\rho - \mu\sigma, \\ & bcp + \lambda\phi + c\rho\tau - b\mu\tau, \quad acu + \mu\phi + d\rho\sigma + a\rho\tau, \end{aligned}$$

On a pour la transformation propre, l'équation  $(x, y, z, w) =$

$$\frac{1}{R} \begin{vmatrix} -\rho^2 + \sigma^2 + \tau^2 + \omega^2 + \lambda^2 - \mu^2 - \nu^2 - \psi^2, & -2\omega + 2\tau\psi + 2\lambda\mu - 2\rho\sigma, \\ 2\omega\nu - 2\tau\psi + 2\lambda\mu - 2\rho\sigma, & \rho^2 - \sigma^2 + \tau^2 + \omega^2 - \lambda^2 + \mu^2 - \nu^2 - \psi^2, \\ -2\omega\mu + 2\rho\psi + 2\lambda\nu - 2\rho\tau, & 2\omega\lambda + 2\mu\nu - 2\sigma\psi - 2\sigma\tau, \\ 2\omega\lambda - 2\nu\psi + 2\mu\nu - 2\sigma\tau, & 2\omega\mu - 2\rho\psi + 2\lambda\nu - 2\rho\tau, \\ 2\omega\mu - 2\nu\psi + 2\mu\rho - 2\lambda\sigma, & -\rho^2 - \sigma^2 - \tau^2 + \omega^2 + \lambda^2 + \mu^2 + \nu^2 - \psi^2 \end{vmatrix} (x, y, z, w).$$

On peut changer la forme de cette expression, en y écrivant

$$\begin{aligned} \lambda &= \frac{1}{2}(\alpha - \alpha'), \quad \mu = \frac{1}{2}(\beta - \beta'), \quad \nu = \frac{1}{2}(\gamma - \gamma'), \quad \psi = \frac{1}{2}(\delta - \delta'), \\ \rho &= \frac{1}{2}(\alpha + \alpha'), \quad \sigma = \frac{1}{2}(\beta + \beta'), \quad \tau = \frac{1}{2}(\gamma + \gamma'), \quad \omega = \frac{1}{2}(\delta + \delta'); \end{aligned}$$

cela donne

$$\begin{aligned} \alpha^2 + \beta^2 + \gamma^2 + \delta^2 - \alpha'^2 - \beta'^2 - \gamma'^2 - \delta'^2 &= 0, \\ R &= \frac{1}{4}(\alpha^2 + \beta^2 + \gamma^2 + \delta^2 + \alpha'^2 + \beta'^2 + \gamma'^2 + \delta'^2), \end{aligned}$$

de manière qu'en écrivant

$$M = \alpha^2 + \beta^2 + \gamma^2 + \delta^2, \quad M' = \alpha'^2 + \beta'^2 + \gamma'^2 + \delta'^2,$$

on obtient

$$R = \frac{1}{4}(M + M') = \sqrt{(MM')},$$

et la formule pour la transformation devient

$$(x, y, z, w) = \frac{1}{\sqrt{(MM')}} \begin{vmatrix} -\alpha\alpha' + \beta\beta' + \gamma\gamma' + \delta\delta', & -\alpha\delta' - \beta\alpha' - \gamma\beta' + \delta\gamma', \\ -\alpha\delta - \beta\alpha + \gamma\delta - \delta\gamma, & \alpha\alpha' - \beta\beta' + \gamma\gamma' + \delta\delta', \\ -\alpha\gamma' + \beta\delta' - \gamma\alpha' + \delta\beta', & \alpha\delta' - \beta\gamma' - \gamma\beta' + \delta\alpha', \\ -\alpha\beta' + \beta\gamma' - \gamma\delta' - \delta\alpha', & -\alpha\gamma' - \beta\delta' + \gamma\alpha' + \delta\beta' \end{vmatrix} (x, y, z, w).$$

On voit donc que même sans supposer l'équation  $M = M'$ , cette formule donne la transformation *propre*

$$x^2 + y^2 + z^2 + w^2 = x'^2 + y'^2 + z'^2 + w'^2.$$

Cette solution est à peu près de la même forme que la solution *impropre* donnée par Euler dans son mémoire “*Problema algebraicum ob affectiones prorsus singulares memorabile*,” *Nov. Comm. Petrop.*, t. xv 1770, p. 75, et *Comm. Arith. collectae*, [4to. Petrop. 1849], t. 1. p. 427. Je remarque aussi que cette même solution peut être déduite de la théorie des *Quaternions*. En effet,  $i, j, k$  étant des quantités imaginaires telles que  $i^2 = j^2 = k^2 = -1$ ,  $jk = -kj = i$ ,  $ki = -ik = j$ ,  $ij = -ji = k$ , on obtient, en effectuant la multiplication:

$$(xi + yj + zk + w) = \frac{1}{\sqrt{(MM')}}(\alpha i + \beta j + \gamma k + \delta)(xi + yj + zk + w)(\alpha' i + \beta' j + \gamma' k + \delta'),$$

$x, y, z, w$  ayant les mêmes valeurs que dans la dernière formule de transformation. En changeant les signes des termes de la quatrième colonne, on en tire pour la transformation *impropre*

$$x^2 + y^2 + z^2 + w^2 = x^2 + y^2 + z^2 + w^2,$$

la formule suivante plus symétrique:

$$(x, y, z, w) = \frac{1}{\sqrt{(MM')}} \begin{vmatrix} -\alpha\alpha' + \beta\beta' + \gamma\gamma' + \delta\delta', & -\alpha\beta' - \beta\alpha' - \gamma\delta' + \delta\gamma' \\ -\alpha\beta' - \beta\alpha' + \gamma\delta' + \delta\gamma', & \alpha\alpha' - \beta\beta' + \gamma\gamma' + \delta\delta' \\ -\alpha\gamma' + \beta\delta' - \gamma\alpha' + \delta\beta', & -\alpha\gamma' - \beta\delta' - \gamma\alpha' - \delta\beta' \\ -\alpha\delta' - \beta\gamma' - \gamma\beta' + \delta\alpha', & \alpha\delta' + \beta\gamma' + \gamma\beta' - \delta\alpha' \end{vmatrix} (x, y, z, w).$$

Ces formules pour la transformation, tant propre qu'impropre, de la fonction  $x^2 + y^2 + z^2 + w^2$  en *elle-même*, sont utiles dans la théorie des polygones inscrits dans une surface du second ordre.

### 138. Recherches sur les matrices dont les termes sont des fonctions linéaires d'une seule indéterminée.

RECHERCHES SUR LES MATRICES DONT LES TERMES SONT DES FONCTIONS LINÉAIRES  
D'UNE SEULE INDÉTERMINÉE.

[From the *Journal für die reine und angewandte Mathematik* (Crelle), tom. L. (1855), pp. 313–317.]

Je pose la matrice

$$\begin{vmatrix} A, & B, & C, & \dots \\ A', & B', & C', & \dots \\ A'', & B'', & C'', & \dots \\ & & & \vdots \end{vmatrix}$$

dont les termes ( $n^2$  en nombre) sont des fonctions linéaires d'une quantité  $s$ , et je considère le déterminant formé avec cette matrice, et les déterminants mineurs formés en supprimant un nombre quelconque des lignes et un nombre égal de colonnes de la matrice. En supprimant une *seule* ligne et une *seule* colonne, on obtient les *premiers mineurs*; en supprimant deux lignes et deux colonnes, on obtient les *seconds mineurs*; et ainsi de suite. Cela étant, je suppose que la quantité  $s$  a été trouvée en égalant à zéro le déterminant formé avec la matrice donnée; ce déterminant sera une fonction de  $s$  du  $n$ -ième degré qui généralement ne contiendra pas comme une fonction de  $s$  *aucun* facteur multiple. On voit donc qu'un facteur simple du déterminant ne peut pas entrer comme facteur dans les premiers mineurs (c'est-à-dire dans *tous* les premiers mineurs); mais en supposant que le déterminant ait des facteurs multiples, un facteur multiple du déterminant peut entrer comme facteur (simple ou multiple) dans les premiers mineurs, ou dans les mineurs d'un ordre plus élevé. Il importe de trouver le degré selon lequel un facteur multiple du déterminant peut entrer comme facteur des premiers mineurs, ou des mineurs d'un ordre quelconque donné.

Cela se fait très facilement au moyen d'une propriété générale des déterminants; si les mineurs du  $(r+1)$ ième ordre contiennent le facteur  $(s-a)^a$  (c'est-à-dire, si tous les mineurs du cet ordre identiquement zéro; et si même le  $r$ -ième ordre contiennent le facteur  $(s-a)^{a-1}$ ; et si les mineurs du  $(r-1)$ ième ordre contiendront au moins le facteur  $(s-a)^{a-2}$ . Autrement dit: les mineurs du  $(r-1)$ ième ordre contiendront le facteur  $(s-a)^a$  où  $\gamma \geq 2\beta - \alpha$ , ou, ce qui est la même chose,  $\alpha - 2\beta + \gamma \leq 0$ ; c'est-à-dire: en formant la suite des indices des puissances selon lesquelles le facteur  $(s-a)$  doit entrer dans les mineurs premiers, seconds, &c. (il va sans dire que cette suite finit par les mineurs  $\alpha, \beta, \gamma, \dots$ , la différences secondes seront positives [c'est-à-dire non négatives]. Je représente par  $\alpha, \beta, \gamma, \dots$  la suite dont il s'agit; je suppose, pour fixer les idées, que  $\delta$  soit le dernier terme qui ne s'évanouisse pas, et j'écris

$$\begin{array}{cccc} \alpha, & \beta, & \gamma, & \delta, 0, 0, \dots \\ \alpha - \beta, & \beta - \gamma, & \gamma - \delta, & \delta, 0, \dots \\ \alpha - 2\beta + \gamma, & \beta - 2\gamma + \delta, & \gamma - 2\delta, & \delta, 0, \dots \end{array}$$

ici, quel que soit le nombre des termes, tous les nombres de la troisième ligne seront positifs, et en représentant ces nombres par  $f, f', f'', \dots$ , on obtient:

$$\begin{aligned} \alpha &= f + 2f' + 3f'' + 4f''' + \dots, \\ \beta &= f' + 2f'' + 3f''' + \dots, \\ \gamma &= f'' + 2f''' + \dots, \\ \delta &= f''' + \dots, \end{aligned}$$

Il y a ici à considérer que le nombre  $a$ , indice de la puissance selon laquelle le facteur  $(s-a)$  entre dans le déterminant, est donné; il sera donc permis de prendre pour  $f, f', f'', \dots$  des valeurs entières et positives quelconques (zéro y compris) qui satisfont à la première équation; les autres équations donnent alors les valeurs de  $\beta, \gamma, \dots$ . On forme de cette manière une table des particularités qui peut présenter le facteur multiple  $(s-a)^a$  du déterminant; cette table est aussi la table des nombres  $\alpha, \beta, \dots$  ainsi de suite; alors, dans les différentes formules qui viennent d'être données, le coefficient d'un terme  $x_q^p$  à droite sera

$$\frac{-k}{(\dagger\alpha, \beta, \gamma, \dots)(\alpha', \beta', \gamma', \dots)}$$

et le coefficient d'un terme  $x_q x_p$  à gauche sera

$$\frac{-k}{(\dagger\alpha, \beta, \gamma, \dots)(\alpha', \beta', \gamma', \dots)},$$

où  $k$  dénote le *discriminant* de la fonction quadratique à gauche, et où les coefficients des fonctions quadratiques des dénominateurs sont les coefficients inverses de cette même fonction quadratique à gauche.<sup>1</sup>

On suppose maintenant que la fonction  $(a, b, c)(x, y)^2$  se transforme en elle-même par la substitution  $\alpha x + \beta y, \gamma x + \delta y$  au lieu de  $x, y$ , on aura  $\alpha\delta - \beta\gamma = \pm 1$  ou  $-1$ , qui sont les deux cas d'une fonction binaire.

En écrivant d'abord  $(\diamond)(x, y)^2 = (A, B, C)(x, y)^2$  il faut obtenir cette substitution identiquement;  $(A, B, C)(\alpha x + \beta y, \gamma x + \delta y)^2 = (A, B, C)(x, y)^2$ , c'est-à-dire  $A(\alpha^2 - 1) = 0, B(\beta^2 - 1) = 0, C(\alpha\beta - 1) = 0, F(\alpha\beta) = 0, G(\alpha\beta) = 0, H(\alpha\beta - 1) = 0, = 0$ , et on voit que pour obtenir une

<sup>1</sup>Je profite de cette occasion pour remarquer concernant des recherches que les formules données dans le note sur les fonctions du second ordre (t. xxviii. [1840] p. 193) [71] pour les cas des trois et de quatre indéterminées, sont exactes, mais que je m'étais trompé dans la forme générale du théorème. (This correction is included vol. i. p. 289.)

solution dans laquelle la forme quadratique ne se réduit pas à un carré, ou à une fonction de deux indéterminées, il faut supposer par exemple  $\alpha^2 - 1 = 0$ ,  $bc - 1 = 0$ . On a donc dans le cas d'une fonction ternaire:

$$\alpha^2 = 1, \quad bc = 1, \quad (\diamond)(x, y, z)^2 = lx_3^2 + mx_2x_3.$$

La transformation sera *propre*, ou *impropre*, suivant que  $\alpha = +1$  ou  $\alpha = -1$ .

Dans le cas d'une fonction *quaternaire*, on obtient pour la transformation *propre*:

$$ab = cd = 1, \quad (\diamond)(x, y, z, w)^2 = lx_3^2 + mx_3x_4,$$

et pour la transformation *impropre*:

$$\alpha = -1, \quad b = c = d = 1, \quad (\diamond)(x, y, z, w)^2 = lx_3^2 + mx_3x_4 + nx_2x_4.$$

Dans le cas d'une fonction *quinaire* on obtient

$$\alpha^2 = 1, \quad bc = de = 1, \quad (\diamond)(x, y, z, w, t)^2 = lx_3^2 + mx_2x_3,$$

et la transformation est *propre* ou *impropre*, selon que  $\alpha = +1$  ou  $\alpha = -1$ ; et ainsi de suite.

Cette méthode a des difficultés dans le cas où l'équation en  $s$  a des racines égales. Je n'entre pas ici dans ce sujet.

Dans les formules qu'on vient de trouver, on peut considérer les coefficients  $l, m$ , &c. comme des quantités *arbitraires*. Mais en supposant que la fonction quadratique soit *donnée*, ces coefficients deviennent *déterminés*. On les trouvera par la formule suivante que je ne m'arrête pas à démontrer.

Soient  $\alpha, \beta, \gamma$ , &c. les coefficients de la fonction linéaire  $x_k$ , et les coefficients de la fonction linéaire  $x_k$ , et ainsi de suite; alors, dans les différentes formules qui viennent d'être données, le coefficient d'un terme  $x_q^2$  à droite sera

$$\frac{-k}{(\dagger\alpha, \beta, \gamma, \dots)(\alpha', \beta', \gamma', \dots)}$$

et le coefficient d'un terme  $x_qx_p$  à gauche sera

$$\frac{-k}{(\dagger\alpha, \beta, \gamma, \dots)(\alpha', \beta', \gamma', \dots)},$$

où  $k$  dénote le *discriminant* de la fonction quadratique à gauche, et où les coefficients des fonctions quadratiques des dénominateurs sont les coefficients inverses de cette même fonction quadratique à gauche.

J'ai déjà fait voir de quelle manière cette formule se rattache à la formule pour la transformation *propre*; la différence entre les formes de ces transformations dans ces cas simple est assez frappante.

Pour obtenir la transformation *propre*

$$ax^2 + by^2 + cz^2 = ax^2 + by^2 + cz^2,$$

j'écris

$$\begin{vmatrix} 11, & 12, & 13 \\ 21, & 22, & 23 \\ 31, & 32, & 33 \end{vmatrix} = \begin{vmatrix} a, & \nu, & -\mu \\ -\nu, & b, & \lambda \\ \mu, & -\lambda, & c \end{vmatrix};$$



cette formule donne

$$(x, y, z) = \frac{1}{abc + a\lambda^2 + b\mu^2 + c\nu^2} \times \begin{vmatrix} abc + a\lambda^2 - b\mu^2 - c\nu^2, & 2(\lambda\mu - c\nu), & 2(\nu\lambda + b\mu)c \\ 2(\lambda\mu + c\nu)a, & abc - a\lambda^2 + b\mu^2 - c\nu^2, & 2(\mu\nu - a\lambda)c \\ 2(\nu\lambda - b\mu)a, & 2(\mu\nu + a\lambda)b, & abc - a\lambda^2 - b\mu^2 + c\nu^2 \end{vmatrix} (x, y, z).$$

La transformation *impropre*

$$ax^2 + by^2 + cz^2 = ax^2 + by^2 + cz^2$$

peut être tirée de la transformation *propre* en elle-même de la fonction donnée ci-après  $ax^2 + by^2 + cz^2 + dw^2$ ; en y écrivant  $d = 0$ , on obtient

$$(x, y, z) = \frac{1}{bc\rho^2 + ca\sigma^2 + ab\tau^2 + \phi^2} \times \begin{vmatrix} -bc\rho^2 + ca\sigma^2 + ab\tau^2 - \phi^2, & -2b\rho\sigma - 2bc\rho\tau, & 2bc\sigma\tau - 2bcr\rho \\ 2a\sigma\rho - 2ac\rho\sigma, & bc\rho^2 - ac\sigma^2 + ab\tau^2 - \phi^2, & -2a\tau\rho - 2ab\sigma\tau \\ -2a\tau\sigma + 2ab\tau\rho, & 2b\rho\tau - 2ab\sigma\tau, & bc\rho^2 + ac\sigma^2 - ab\tau^2 - \phi^2 \end{vmatrix} (x, y, z).$$

## 139. An Introductory Memoir upon Quantics.

### AN INTRODUCTORY MEMOIR UPON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLIV. for the year 1854, pp. 244–258. Received April 20,—Read May 4, 1854.]

1. The term Quantics is used to denote the entire subject of rational and integral functions, and of the equations and loci to which these give rise; the word “quantic” is an adjective, meaning *of such a degree*, but may be used substantively, the noun understood being (unless the contrary appear by the context) function; so used the word admits of the plural “quantics.”

The quantities or symbols to which the expression “degree” refers, or (what is the same thing) in regard to which a function is considered as a quantic, will be spoken of as “facients.” A quantic may always be considered as being, in regard to its facients, homogeneous, since to render it so, it is only necessary to introduce as a facient unity, or some symbol which is to be ultimately replaced by unity; and in the cases in which the facients are considered as forming two or more distinct sets, the quantic may, in like manner, be considered as homogeneous in regard to each set separately.

The expression “an equation,” used without explanation, is to be understood as meaning the equation obtained by putting any quantic equal to zero. I make no absolute distinction between the words “degree” and “order” as applied to an equation or system of equations, but I shall in general speak of the order rather than the degree. The equations of a system may be independent, or there may exist relations of connexion between the different equations of the system; the subject of a system of equations so connected together is one of extreme complexity and difficulty. It will be sufficient to notice here, that in any system whatever of equations, assuming only that the equations are not more than sufficient to determine the ratios of the facients, and joining to the system so many linear equations between the facients as will render the order of the system determinate (that is, the order of the equation which determines any one of these ratios), the order of the equation which determines any one of the facients (or, what is clear from the context, the order of the equation so determined) is nothing else than the order of the equation.

2. An equation or system of equations represents, or is represented by, a locus. This assumes that the facients depend upon quantities  $x, y, \dots$  the coordinates of a point in space; the entire series of points, the coordinates of which satisfy the equation or system of equations, constitutes the locus. To avoid complexity, it is proper to take the facients themselves as coordinates, or at all events to consider these facients as linear functions of the coordinates; this being the case, the order of the locus is the order of the equation, or system of equations.

3. I have spoken of the *coordinates of a point in space*. I consider that there is an ideal space of three dimensions, but of course in the ordinary acceptation of the word, space is of three dimensions; however, in a plane (the space of two ordinary plane geometry) is a space of two dimensions, and we may consider the line as a space of one dimension, the line itself being the only locus of points in the space. To extend the notion of dimensionality, I do not introduce a space of an absolute number of dimensions, that is, the consideration of a space of four or of more dimensions, but I consider the symbols  $x, y, z, \dots$  as variable, and obtain a system of points in the space of  $n$  dimensions; the number of equations, considered as a result, ought to render the locus indifferently of the elements, or of the facients themselves. In what follows, I shall not enter so as to render the result, indifferently of the elements; but only investigate whether a locus is or is not affected by the actual or any other generality consistent with its definition (and not, in this is a point upon which it is not at present necessary to dwell).

The theory includes, as a very particular case, the ordinary theory of reciprocity; we have only to say that to any line shall the word “line” shall mean “point,” and the word “point” shall mean “line,” and that lines respectively shall be considered as a purely descriptive notion of relating to a point and a line respectively shall be considered as a purely descriptive notion of relating to a point and a line, and similarly with regard to curves of a higher order, when the ideas of reciprocity applicable to these curves are properly developed.

5. A quantic of the degrees  $m, m', \dots$  in the sets  $(x, y, \dots), (x', y', \dots)$  &c. will for the most part be represented by a notation such as

$$(* \diamond x, y, \dots)^m (x', y', \dots)^{m'} \dots,$$

where the mark  $*$  may be considered as indicative of the absolute generality of the quantic; any such quantic may of course be considered as the sum of a series of terms  $x^a y^b \dots x'^{a'} y'^{b'} \dots$ , &c. of the proper degrees in the different sets respectively, each term being multiplied by a coefficient; these coefficients may be mere numerical multiples of single letters or elements such as  $a, b, c, \dots$ , or else functions (in general rational and integral functions) of such elements; this explains the meaning of the expression “the elements of a quantic” in the case where the coefficients are mere numerical multiples of single letters or elements such as  $(a, b, \dots)$ , by the appropriate numerical coefficient of a term  $x^a y^b \dots x'^{a'} y'^{b'} \dots$ , I mean the coefficient of this term in the expansion of

$$(x + y + \dots)^m (x' + y' + \dots)^{m'} \dots;$$

and I represent by the notation

$$(a, b, \dots)(x, y, \dots)^m (x', y', \dots)^{m'} \dots,$$

a quantic in which each term is multiplied as well by the appropriate numerical coefficient as by the literal coefficient or element. On the other hand, I represent by the notation

$$(a, b, \dots)(x, y, \dots)^m (x', y', \dots)^{m'} \dots,$$

a quantic in which each term is multiplied only by the literal coefficient or element which belongs to it in the set  $(a, b, \dots)$  of literal coefficients or elements. And a like distinction applies to the cases where the coefficients are functions of the elements  $(a, b, \dots)$ .

6. I consider now the quantic

$$(* \diamond x, y, \dots)^m (x', y', \dots)^{m'} \dots,$$

and selecting any two facients of the same set, e.g. the facients  $x, y$ , I remark that there is always an operation upon the elements, tantamount as regards the quantic to the operation  $x\partial_y$ ; viz. if we differentiate with respect to each element, multiply by perpetual functions of the elements and add, we obtain the same result as by differentiating with  $\partial_y$  and multiplying by  $x$ . For instance, as regards  $3ax^2 + bxy + 5cy^2$  (the numerical coefficients are taken hyphenated), we have  $3b\partial_a + 10b\partial_c$  tantamount to  $x\partial_y$ ; and as regards any other case, I represent by  $[x\partial_y]$  the operation upon the elements tantamount to  $x\partial_y$ , and I write down here the series of operations

$$[x\partial_y] - x\partial_y, \quad [x'\partial_y] - x'\partial_y, \quad \dots$$

where  $x, y$  are considered as being successively replaced by every permutation of two different facients of the set  $(x, y, \dots)$ ,  $x', y'$  successively replaced by every permutation of two different facients of the set  $(x', y', \dots)$ , and so on; thus I call an entire system, and is to be considered as forming one such system.

7. The definition of a covariant may however be generalized in two directions: we may instead of a single quantic consider two or more quantics; the operations  $[x\partial_y]$ , although represented by means of the same symbols  $x, y$ , will as regards the different quantics, different meanings, but the conditions for the covariance of the entire system, we have only to consider in place of the system spoken of in the original definition, the system

$$\Sigma[x\partial_y] - x\partial_y, \quad \dots, \quad \Sigma[x'\partial_y] - x'\partial_y, \quad \dots, \quad \&c. \&c.,$$

and we obtain the definition of a covariant of two or more quantics.

Again, we may consider in connexion with each set of facients any number of new sets being equal to or greater than the number of facients in the set. We have, in place of the original system  $(x, y, \dots)$ , the systems

$$[x\partial_y] - S(x\partial_y), \quad \dots, \quad [x'\partial_y] - S(x'\partial_y), \quad \dots, \quad \&c. \&c.,$$

and we obtain the generalized definition of a covariant.

8. A covariant has been defined simply as a function reduced to zero by each of the operations of the entire system. But in dealing with given quantics, we may without loss of generality consider the covariant as a function of the like form with the quantic, i.e. as being a rational and integral function homogeneous in regard to each of the different sets separately, and as being also a rational and integral function of the elements. In particular, the covariant may be the quantic itself, and when this is the case the term “covariant” includes, as already remarked, “invariant.”

It is proper to remark, that if the same quantic be represented by means of different sets of elements, then the symbols  $[x\partial_y]$  which correspond to these different forms of the same quantic are more transformations of each other, i.e. they become in virtue of the relations between the different sets of elements identical.

9. What precedes is a return to and generalization of the method employed in the first part of the memoir published in the *Camb. Math. Journ.*, t. IV. [1845], and *Camb. and Dubl. Math. Journ.*, t. I. [1846], under the title “On Linear Transformations,” [13 and 14], and Crelle, t. XXX. [1846], under the title “Mémoire sur les Hyperdéterminants,” [\*16], and which I shall refer to as my original memoir. I there consider in fact the invariants of a quantic

$$(* \diamond x_1, x_2 \dots x_n)(y_1, y_2 \dots y_n) \dots$$

linear in regard to  $n$  sets each of  $n$  facients, and I represent the coefficients of a term  $x_1 y_2 \rho_1 \dots$  by ref. . . ; there is no difficulty in the way of  $\alpha, \beta$  being any two different numbers each of the series  $1, 2, \dots, n$ , the operation  $[x_p \partial_{x_q}]$  is identical with the operation

$$\Sigma \Sigma \dots \left( ref \dots \frac{d}{d\rho \rho \dots} \right),$$

where the summations refer to  $s, t, \dots$  which pass respectively from 1 to  $n$ , both inclusive; and the condition that a function (in the present memoir, assumed to be an invariant, i.e. to contain only the coefficients, but the condition may be reduced to zero by the operation  $[x_p \partial_{x_q}] - x_p \partial_{x_q}$ , is of course simply the condition that such function may be reduced to zero by the operation

$$\Sigma \Sigma \dots \left( ref \dots \frac{d}{d\rho \rho \dots} \right) u = 0$$

of my original memoir.

10. But the definition in the present memoir includes also the method made use of in the second part of my original memoir. This method is substantially as follows: consider for simplicity a quantic  $U =$

$$(* \diamond x, y, \dots)^m$$

containing only the single set  $(x, y, \dots)$ , and let  $U_1, U_2, \dots$  be what the quantic becomes when the set  $(x, y, \dots)$  is successively replaced by the sets  $(x_1, y_1, \dots), (x_2, y_2, \dots), \dots$ ; the number of these new sets being equal to or greater than the number of facients in the set. Suppose that  $A, B, C, \dots$  are any of the determinants

$$\begin{vmatrix} \partial_{x_1}, & \partial_{y_1}, & \partial_{z_1}, & \dots \\ \partial_{x_2}, & \partial_{y_2}, & \partial_{z_2}, & \dots \\ & & \vdots & \end{vmatrix},$$

then forming the derivative

$$A^p B^q C^r \dots U_1 U_2 \dots,$$

where  $p, q, r, \dots$  are any positive integers, the function so obtained is a covariant involving the original sets  $(x, y, \dots), (x_1, y_1, \dots), (x_2, y_2, \dots)$  &c.; and if after the differentiations we replace  $(x_1, y_1, \dots), (x_2, y_2, \dots)$  &c. by  $(x, y, \dots)$ , the function is a covariant according to the definition given in this memoir. But to do this some preliminary explanations are necessary.

11. I consider any two operations  $P, Q$ , involving each or either of them differentiations in respect of variables contained in the other of the operations. It is required to investigate the effect of the operation  $P \cdot Q$ , where the operation  $Q$  is performed upon some operand  $\Omega$ , and the operation  $P$  is then performed upon the product  $Q \cdot \Omega$ .  $P$  is to be performed as usual as well in respect of variables  $\partial_x, \partial_y, \dots$  contained in  $Q$  as in respect of variables  $a, b, \dots$  contained in  $\Omega$ . Suppose that variables  $a, b, \dots$  contained in  $Q$  are not contained in  $\Omega$ . But to render derivative is a covariant according to the definition in this Memoir. But to do this some preliminary explanations are necessary.

11. I consider any two operations  $P, Q$ , involving each or either of them differentiations in respect of variables contained in the other of the quantic. It is to investigate the effect of the operation  $P \cdot Q$ , where the operation  $Q$  is performed upon some operand  $\Omega$ , and the operation  $P$  is then performed upon the product  $Q\Omega$ , where the operation  $P$  is to be performed as usual as well in respect of variables  $\partial_x, \partial_y, \dots$  contained in  $Q$  as in respect of variables  $a, b, \dots$  contained in  $\Omega$ . Suppose that  $\partial_a, \partial_b, \dots$ , contained in  $Q$ , may be also derived for some new operand  $Q$ , and they may be derived for some new operand  $Q$ , and ditto perform the operation  $Q$  as operand.

We can therefore in this same thing, the covariant operated upon by  $\Phi$ , is reduced to zero by the operation  $\mathfrak{D} = [a\partial_b] - 8a\partial_b$ .

12. I return to the expression

$$A^p B^q C^r \dots U_1 U_2 \dots,$$

and I suppose that after the changes of the sets  $(x_1, y_1, \dots)$ ,  $(x_2, y_2, \dots)$ , &c. into the original set  $(x, y, \dots)$  may be deferred until after the operation  $\mathfrak{D}$ , provided that the result is a covariant, we must prove that it is reduced to zero by the operation  $\mathfrak{D} = [a\partial_b] - 8a\partial_b$ .

It is easy to see that the change of the sets  $(x_1, y_1, \dots)$ ,  $(x_2, y_2, \dots)$ , &c. into the original set  $(x, y, \dots)$  &c. into the original set, we should before the deferred change have an operation  $\mathfrak{D}$ , provided by  $Sa\partial_b$ , where we please write  $Sa\partial_b$ , instead therefore  $\mathfrak{D} = [a\partial_b] - Sa\partial_b$ . Now we have, as before,  $A \cdot \mathfrak{D} = \mathfrak{D} \cdot A + \mathfrak{D} \cdot (A)$ , where, as before,  $A(\mathfrak{D})$  denotes the result of the operation  $A$  performed upon  $\mathfrak{D}$  as operand, and similarly  $\mathfrak{D}(A)$  the result of the operation  $\mathfrak{D}$  performed upon  $A$  as operand; but in like manner  $\mathfrak{D}$  is convertible with  $B, C$ , &c., since it must be noticed as regards  $B, C$ , &c., that the simple symbols  $\partial_x, \partial_y, \dots$  are not in fact functions tantamount to changes which we have hereafter to make of  $x, y, \dots$  into  $x, y, \dots$ ; and we therefore have

$$\mathfrak{D} \cdot A^p B^q C^r \dots U_1 U_2 \dots = 0,$$

or  $A^p B^q C^r \dots U_1 U_2 \dots$  is a covariant, the proposition which was to be proved.

13. I pass to a theorem which leads to another method of finding the covariants of a quantic. For this purpose I consider the quantic

$$(* \diamond x, y \dots \diamond x', y' \dots),$$

the coefficients of which are mere numerical multiples of the elements  $(a, b, c, \dots)$ ; and in connexion with this quantic I consider the linear functions  $\xi x + \eta y \dots, \xi' x' + \eta' y' \dots$ , &c. which I treat as coefficients, may be represented by means of the quantities  $(\xi, \eta, \dots)$ ,  $(\xi', \eta', \dots)$ ,  $(\xi, y, \dots)$ ,  $(\xi', y', \dots)$ , &c.

we may from the quantic (which by convenience I call  $U$ ) form an operative quantic,  $\Theta U = (\xi, \eta, \dots \diamond x, y, \dots)(\xi', \eta', \dots \diamond x', y', \dots)^{m'} \dots$

the coefficients of which are now numerical multiples of the corresponding derivatives,  $\partial_a, \partial_b, \partial_c, \dots$  of the original quantic. Consider, the new operative quantic

$$\Theta U = (\xi, \eta, \dots \diamond x, y, \dots)(\xi', \eta', \dots \diamond x', y', \dots)^{m'} \dots$$

i.e. a product of powers of the linear functions. And it is to be remarked that as regards the quantic  $\Theta$  in like manner the covariants in respect to  $x, y$  &c. are mere numerical multiples, of the new elements  $\xi, \eta, \dots, \xi', \eta' \dots$ , and we can therefore consider this quantic in the same way as we did consider  $\Theta$  as a product of powers of the linear functions. And it is to be remarked that as regards the quantic  $\Theta$  in like manner the covariants in respect of  $x, y$  &c. as elements, &c. And the term "covariant" includes, as before, as the other coefficients can be deduced by mere differentiations.

14. The general demonstration of this equation presents no real difficulty, but to avoid the necessity of fixing upon a notation to distinguish the coefficients of the different terms and for the sake of simplicity, I shall merely confine the explanation by an example. Consider the quantic

$$U = ax^3 + 3bx^2y + 3cxy^2 + dy^3;$$

this gives

$$\Theta = \xi \partial_a + \xi^2 \eta \partial_b + \xi \eta^2 \partial_c + \eta^3 \partial_d;$$

or, if for greater clearness,  $\partial_a, \partial_b, \partial_c, \partial_d$  are represented by  $\alpha, \beta, \gamma, \delta$ , then

$$\Theta = \xi^3\alpha + \xi^2\eta\beta + \xi\eta^2\gamma + \eta^3\delta,$$

and we have

$$[x\partial_y] = 3b\partial_a + 2c\partial_b + d\partial_c,$$

and

$$[\xi\partial_\eta] = 3a\partial_b + 2b\partial_c + c\partial_d.$$

Now considering  $\Phi$  as a function of  $\partial_a, \partial_b, \partial_c, \partial_d$ , or, what is the same thing, of  $\alpha, \beta, \gamma, \delta$ , we have to prove that

$$\Phi \cdot \Theta = \Theta \cdot \Phi(\Theta);$$

and if in the expression of  $\Phi$  we write  $a + \partial_a, \beta + \partial_b, \gamma + \partial_c, \delta + \partial_d$  for  $a, b, c, d$  (where only the symbols  $\partial_a, \partial_b, \partial_c, \partial_d$  are to be considered as affecting  $a, b, c, d$ , or as contained in the operand of  $\Phi$ , and reject the first term (or zero term independent of  $\partial_a, \partial_b, \partial_c, \partial_d$ , and reject the first term of the operand  $3a\partial_b + 2c\partial_b + d\partial_c$ , by the expression) we obtain the first term of the result. This will give the form which the same value of  $\Phi$  takes by the substitution of  $\Phi(\Theta)$ . This value is

$$\Phi(\Theta) = (3a\beta + 2b\gamma + c\delta)\Phi;$$

performing the differentiations  $\partial_a, \partial_b, \partial_c, \partial_d$ , its value is

$$\Phi([\xi\partial_\eta]) = (3a\xi^2 + 2b\eta + \gamma\xi)\Phi,$$

i.e. we have

$$\Phi(\Theta) = [\xi\partial_\eta]\Phi.$$

15. Suppose now that  $\Phi$  is a covariant of  $\Theta$ , then the operation  $\Phi$  performed upon any covariant of  $U$  gives rise to a covariant of the system

$$(* \diamond x, y, \dots \diamond \xi, \eta, \dots) \dots, \quad (\xi', \eta', \dots \diamond x', y', \dots), \quad \&c.$$

To prove this is to be in the first instance noticed, that as regards  $(\xi, \eta, \dots)$ , &c. we have  $[\xi\partial_\eta] = \eta\partial_\xi$ , &c. Hence considering  $[\xi\partial_\eta]$ , &c. as referring to the quantic  $U$ , the operation  $\Sigma[\xi\partial_\eta] - x\partial_y$ , &c. are referring to zero by every covariant of the system, and therefore every covariant of the system must be reduced to zero by the operations

$$\mathfrak{D} = [x\partial_y] + \eta\partial_\xi - x\partial_y,$$

$$\mathfrak{D} = [\xi\partial_\eta] + y\partial_x - \xi\partial_\eta,$$

equations which it is obvious may be replaced by

$$\mathfrak{D} \cdot \Phi = \mathfrak{D} \cdot \Phi + \mathfrak{D} \cdot (\Phi),$$

$$\Phi \cdot \mathfrak{D} = \mathfrak{D} \cdot \Phi + \Phi \cdot (\mathfrak{D}),$$

and consequently (in virtue of the theorem),

$$\mathfrak{D} \cdot \Phi = \mathfrak{D} \cdot \Phi + \mathfrak{D} \cdot (\Phi),$$

$$\Phi \cdot \mathfrak{D} = \mathfrak{D} \cdot \Phi + \Phi \cdot (\mathfrak{D}),$$

and we have therefore

$$\mathfrak{D} \cdot \Phi - \Phi \cdot \mathfrak{D} = -(\eta\partial_\xi - \eta\partial_\xi)(\Phi);$$

or, since  $\Phi$  is a covariant of  $\Theta$ , i.e. since  $\Theta \cdot \Phi = \Phi \cdot \Theta$ . And since every covariant of the system reduces to zero by the operation  $\mathfrak{D} \cdot \Phi$ , and therefore by the operation  $\Phi \cdot \mathfrak{D}$ , such covariant will also be reduced to zero by the operation  $\mathfrak{D} \cdot \Phi$ , or what is the same thing, the covariant operated upon by  $\Phi$ , is reduced to zero by the operation  $\mathfrak{D}$  operating upon a covariant gives a covariant.

16. In the case of a quantic such as  $U =$

$$(* \diamond x, y \dots)^m, \quad (* \diamond x', y' \dots)^{m'}, \quad \&c.$$

we may instead of the new sets  $(\xi, \eta)$ ,  $(\xi', \eta')$ , ... employ the sets  $(y, -x)$ ,  $(y', -x')$ , &c. The operative quantic  $\Theta$  is in this case defined by the equation  $\Theta U = 0$ , and the operations  $\Phi$  performed upon a covariant of  $U$  will give the analogous equation

$$\Phi([x\partial_y]) = [x\partial_y](\Phi).$$

where on the left-hand side  $[x\partial_y]$  refers to  $U$ , but on the right-hand side  $[x\partial_y]$  refers to  $\Theta$ , and instead of  $[x\partial_y] + y\partial_x - x\partial_y$ , we have simply  $\mathfrak{D} = [x\partial_y] - x\partial_y$ .

17. I pass next to the quantic

$$(* \diamond x, y)^m,$$

which I shall in general consider under the form

$$(a, b, \dots b', a' \diamond x, y)^m,$$

but sometimes under the form

$$(a, b, \dots b', a' \diamond x, y)^m,$$

the former notation denoting, it will be remembered,

$$ax^m + \frac{m}{1}bx^{m-1}y \dots + \frac{m}{1}b'xy^{m-1} + a'y^m,$$

and the latter notation

$$ax^m + bx^{m-1}y \dots + b'xy^{m-1} + a'y^m.$$

But in particular cases the coefficients will be represented by means of unaccentuated letters, thus  $(a, b, c, d \diamond x, y)^3$  will be used to denote  $ax^3 + 3bx^2y + 3cxy^2 + dy^3$ , and  $(a, b, c, d \diamond x, y)^3$  will be used to denote  $ax^3 + bx^2y + cxy^2 + dy^3$ , and so in all similar cases.

Applying the general methods to this case, we see that

$$\begin{aligned} (a, b, \dots b', a' \diamond x, y)^m, \\ [y\partial_x] &= ab\partial_a + 2bb\partial_b + \dots + mb'\partial_{a'}, \\ [x\partial_y] &= mb\partial_a + (m-1)c\partial_b + \dots + a'\partial_{b'}; \end{aligned}$$

in fact, with these meanings of the symbols the quantic is reduced to zero by each of the operations  $[y\partial_x] - y\partial_x$ ,  $[x\partial_y] - x\partial_y$ , and we may consequently take this as the definition of the operations  $[y\partial_x]$ ,  $[x\partial_y]$ , hence supposing that the function  $U$  is a covariant or invariant of the quantic, we have for the definition of a covariant or invariant

$$[y\partial_x](U) = ay\partial_x(U), \quad [x\partial_y](U) = -(\alpha + \beta \dots)(U) + \alpha b\partial_x(U),$$

in fact, after the differentiations, the new sets  $(x_1, y_1)$ ,  $(x_2, y_2)$ , ... may be replaced by the original set  $(x, y)$ , will be a covariant of the quantic  $U$ . And it is to be remarked that as regards the quantic

$$\overline{12' 13' 23'} \dots U_1 U_2 U_3 \dots,$$

i.e. a product of any linear functions, the new elements being merely the coefficients of the original quantic. The operative quantic  $\Theta$  becomes in this case under consideration

$$\Theta = (b, \dots b', a \diamond x, y)^m,$$

the signs being alternately positive and negative; in fact it is easy to verify that this expression is identical with the expression  $\Theta U$  operating on a covariant of  $U$ .

18. But the quantic

$$(a, b, \dots b', a' \diamond x, y)^m,$$

considered as decomposable into linear factors, may be expressible in the form

$$a(x - \alpha y)(x - \beta y) \dots,$$

gives rise to a fresh series of results. We have here also

$$[y\partial_x] = -(\alpha + \beta + \dots)\alpha\partial_a + \partial_\alpha + \partial_\beta \dots,$$

$$[x\partial_y] = -(\alpha + \beta \dots)\alpha\partial_a + \alpha^2\partial_\alpha + \beta^2\partial_\beta + \dots;$$

in fact with these meanings of the symbols the quantic is reduced to zero by each of the operations  $[y\partial_x] - y\partial_x$ ,  $[x\partial_y] - x\partial_y$ , which expresses the definition of the operations  $[y\partial_x]$ ,  $[x\partial_y]$ , and any expression of  $U$  as a covariant or invariant of the quantic. And the expressions  $(\alpha - \beta)(x - \alpha y) - (\alpha - \beta)(x - \beta y)$ , identical.

19. Consider now the expression

$$a^p \Sigma(x - \alpha y)(x - \beta y)^q \dots (x - \beta)^r \dots,$$

where the sum of the indices  $j, p, \dots$  of all the simple factors which contain  $\alpha$ , the sum of the indices  $k, p, \dots$  of all the simple factors which contain  $\beta$ , are respectively  $\theta$  of the coefficient  $\alpha$ ; the index  $\theta$  is in particular if  $\phi = 0$ , i.e. if  $V$  be a quantic of the order  $\theta$ , then in particular the order  $m$  in the coefficients, the order  $\theta$  in the facients  $x, y$ , and of the degree  $\theta$  in the coefficients.

20. In connexion with this covariant

$$a^p \Sigma(x - \alpha y)(x - \beta y)^q \dots (x - \beta y)^r \dots,$$

of the order  $\mu$  and of the degree  $\theta$  in the coefficients, of the quantic  $U =$

$$a(x - \alpha y)(x - \beta y) \dots,$$

consider the covariant

$$\Sigma(\overline{12'} \dots) V_1 V_2 \dots V_m$$

of a quantic  $V =$

$$(* \diamond x, y)^\mu,$$

in which, after the differentiations, the new sets  $(x_1, y_1)$ ,  $(x_2, y_2)$ ,  $\dots$  may be replaced by the original set  $(x, y)$ . The last-mentioned covariant will be of the order  $m$  in the coefficients, and in particular if  $\phi = 0$ , i.e. if  $V$  be a quantic of the order  $\theta$  of the coefficients, then in particular the order  $m$  in the coefficients of the degree  $m$  in the coefficients, and of the degree  $\theta$  in the coefficients, and that the new in like manner contain only numerical multiples of these invariants. The precise meaning of the law, in the last-mentioned point of view, requires some explanation. Suppose that we know all the linearly independent invariants of a quantic of the order  $m$  and degree  $\theta$  in the coefficients, considered as a function of the order  $m$  and the degree  $\theta$  in the coefficients, it can in like manner be obtained as a covariant of degree  $\theta + m$  in the coefficients,



viz. of the variables the sets  $(x_1, y_1)$ ,  $(x_2, y_2)$ , &c.; this is the precise meaning in which I assert independent, and by a covariant of the order  $m$ , considered as a function of the variables  $(x_1, y_1)$ ,  $(x_2, y_2)$ , &c.; this is the precise meaning in which I assert that the invariants of the degree  $\theta$  in question are mere rational and integral functions of the invariants of the degree  $\theta$  in question may be considered as a function of the degree  $\theta$  in question, where they may merely numerical multiples of these different invariants, but in general I will not consider these invariants different invariants, but in general I will not consider these invariants different invariants because, in particular cases of the degree  $\theta$  in question to assume that  $\theta = m$ , viz. of the invariants of the degree  $\theta$  which are necessary will compromise all the invariants of that degree (if any) which are rational and integral functions of the invariants of an inferior order and of the degree  $\theta$ .

21. The number of the really independent covariants of a quantic  $(*\diamond x, y)^m$  is precisely equal to the order of the quantic, i.e. any covariant  $Q$  is a function (generally an irrational function) of any other two covariants and of the coefficients; or in like manner the number of the really independent invariants is, as may be easily seen by a process analogous to the one just employed,  $m - 2$  invariants. We may therefore say that every covariant (generally an irrational function), of  $m - 2$  invariants, considered as a function of the coefficients.

22. Consider any covariant of the quantic

$$(a, b, \dots b', a' \diamond x, y)^m,$$

and let this be of the order  $\mu$  and of the degree  $\theta$  in the coefficients. It is very easily shown that  $m\theta - \mu$  is necessarily even. In particular in the case of an invariant (i.e. when  $\mu = 0$ ) so that a quantic of an odd order admits only of an even degree  $\theta$ ; but if  $\mu = 0$  and  $\mu\theta$  is necessarily even. In the former case one of the two numerical distinction between the two cases must be drawn; but as an important distinction between the two cases must be drawn; this is in fact only of an odd order admits only of an even degree.

23. There is another very simple condition which is satisfied by every covariant of the quantic

$$(a, b, \dots b', a' \diamond x, y)^m,$$

viz. if we consider the facients  $(x, y)$  as being respectively of the weights  $\frac{1}{2}, -\frac{1}{2}$ , the coefficients  $(a, b, \dots b', a')$  as being respectively of the weights  $\frac{1}{2}m, \frac{1}{2}m - 1, \dots, -\frac{1}{2}m + 1, -\frac{1}{2}m$ , then the covariant is of the weight zero. This is the most natural mode of stating the theorem, since the symbols,  $\partial_a, \partial_b, \dots, \partial_{b'}, \partial_{a'}$  are respectively of the weights  $-\frac{1}{2}m, -\frac{1}{2}m + 1, \dots$ ; the conditions can without negative numbers be of course be modified so as to effect upon the coefficients  $(a, b, \dots b', a')$  as being respectively of the weights  $0, 1, \dots, m - 1, m$ , respectively, and the conditions of every covariant or term of the covariant of the weights  $\frac{1}{2}(m\theta + \mu)$ .<sup>2</sup>

24. The preceding laws as to the form of a covariant have been stated here by way of anticipation, principally for the sake of the remark, that they so far define the form of a covariant as to render it in very many cases practicable with a moderate amount of labour to complete the investigations by means of one or the other of the operations of the entire system. Now, besides the quantic itself, there are a variety of other functions which are reduced to zero by each of the operations of the entire system; any such function is said to be a covariant of the proper order and degree in the coefficients, in any particular case may very easily be found, and the first term of the proper function and degree in the coefficients, may form the second term induced by means by more differentiations. There can be no very important term in the second term, and so on, until either the remaining terms can be found by means of any inferior order; but the very important terms are very important terms in the case in question is that of such order  $a$ , and

<sup>2</sup>I may remark that it was only M. Hermite's important discovery of an invariant of the degree 18 of a quantic of the order 5, which removed an erroneous impression which I had been under from the commencement of the subject, that out was of necessity *evenly* even.

the first term being known, the remaining ones can (i.e. in practice in a very convenient manner for the calculation given to the question, viz. that of the highest power of either of the facients) be deduced by mere differentiations. There can be very important terms (i.e. in practice in a very convenient manner for the calculation given to the question, viz. that of the highest power of either of the facients) be noticed as regards covariants, that when the first or the last term of a covariant is known, the other coefficients can be deduced by mere differentiations.

Postscript added October 7th, 1854.—I have, since the preceding memoir was written, found with respect to the covariants of a quantic  $(* \diamond x, y)^m$  a function of any number and degree in the coefficients satisfying the prescribed conditions as to symmetry and weight; such function, viz. the function which is to be obtained by Aronhold's method, may, in this manner be modified as follows:—if the coefficients  $(a, b, \dots, b', a')$  are considered as denoting the relations  $a, b, c = a b, c, d$  where  $\alpha\beta - \beta\gamma, \alpha c - \beta b, \dots, \beta\gamma - \beta\beta b, \alpha\delta - \beta\gamma$ , are linear functions of  $a, b, c, d$ , the form  $\delta(\alpha\beta - \beta\gamma)\alpha(\alpha c - \beta b)$ , will be a function of the elements satisfying the prescribed conditions as to symmetry and weight: but I do not believe that any covariant or invariant other than the discovery of any covariant or invariant other than the discoveries of the facients is known. I have, since the above memoir was written, hope to return to the subject in a subsequent memoir.

# 140.

## RESEARCHES ON THE PARTITION OF NUMBERS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLV for the year 1855, pp. 127–140. Received April 14,—Read May 24, 1855.]

I PROPOSE to discuss the following problem: “To find in how many ways a number  $q$  can be made up of the elements  $a, b, c, \dots$  each element being repeatable an indefinite number of times.” The required number of partitions is represented by the notation

$$P(a, b, c, \dots) q,$$

and we have, as is well known,

$$P(a, b, c, \dots) q = \text{coefficient } x^q \text{ in } \frac{1}{(1-x^a)(1-x^b)(1-x^c)\dots},$$

where the expansion is to be effected in ascending powers of  $x$ .

It may be as well to remark that each element is to be considered as a separate and distinct element, notwithstanding any equalities which may exist between the numbers  $a, b, c, \dots$ ; thus, although  $a = b$ , yet  $a + a + a + \&c.$  and  $a + a + b + \&c.$  are to be considered as two different partitions of the number  $q$ , and so in all similar cases.

The solution of the problem is thus seen to depend upon the theory, to which I now proceed, of the expansion of algebraical fractions.

Consider an algebraical fraction  $\frac{\phi x}{f x}$ , where the denominator is the product of any number of factors (the same or different) of the form  $1 - x^m$ . Suppose in general that  $[1 - x^m]$  denotes the irreducible factor of  $1 - x^m$ , i.e. the factor which, equated to zero, gives the prime roots of the equation  $1 - x^m = 0$ . We have

$$1 - x^m = \prod [1 - x^{m'}],$$

where  $m'$  denotes any divisor whatever of  $m$  (unity and the number  $m$  itself not excluded). Hence, if  $a$  represent a divisor of one or more of the indices  $m$ , and  $k$  be the number of the indices of which  $a$  is a divisor, we have

$$f x = \prod [1 - x^a]^k.$$

Now considering apart from the others one of the multiple factors  $[1 - x^a]^k$ , we may write  $f x = [1 - x^a]^k f_1 x$ .

Suppose that the fraction  $\frac{\phi x}{f x}$  is decomposed into simpler fractions, in the form

$$\frac{\phi x}{f x} = I(x) + (x\partial_x)^{k-1} \frac{\theta x}{[1 - x^a]} + (x\partial_x)^{k-2} \frac{\theta_1 x}{[1 - x^a]} + \dots + \frac{\theta_{k-1} x}{[1 - x^a]} + \&c.,$$

where  $I(x)$  denotes the integral part, and the  $\&c.$  refers to the fractional terms depending upon the other multiple factors such as  $[1 - x^b]^{k_1}$ . The functions  $\theta x$  are to be considered as functions with indeterminate coefficients, the degree of each such function being inferior by unity to that of the corresponding denominator; and it is proper to remark that the number of the indeterminate coefficients in all the functions  $\theta x$  together is equal to the degree of the denominator  $f x$ .

The term  $(x\partial_x)^{k-1} \frac{\theta x}{[1 - x^a]}$  may be reduced to the form

$$\frac{ax}{[1 - x^a]^k} + \frac{g_1 x}{[1 - x^a]^{k-1}} + \&c.,$$

the functions  $g x$  being of the same degree as  $\theta x$ , and the coefficients of these functions being linearly connected with those of the function  $\theta x$ . The first of the foregoing terms is the only term on the right-hand side which contains the denominator  $[1 - x^a]^k$ ; hence, multiplying by this denominator and then writing  $[1 - x^a] = 0$ , we find

$$\frac{\phi x}{f_1 x} = g x,$$

which is true when  $x$  is any root whatever of the equation  $[1 - x^a] = 0$ . Now by means of the equation  $[1 - x^a] = 0$ ,  $\frac{\phi x}{f_1 x}$  may be expressed in the form of a rational and integral function  $Gx$ , the degree of which is less by unity than that of  $[1 - x^a]$ . We have therefore  $Gx = gx$ , an equation which is satisfied by each root of  $[1 - x^a] = 0$ , and which is therefore an identical equation;  $gx$  is thus determined, and the coefficients of  $\theta x$  being linear functions of those of  $gx$ , the function  $\theta x$  may be considered as determined. And this being so, the function

$$\frac{\phi x}{f x} - (x \partial_x)^{k-1} \frac{\theta x}{[1 - x^a]}$$

will be a fraction the denominator of which does not contain any power of  $[1 - x^a]$  higher than  $[1 - x^a]^{k-1}$ ; and therefore  $\theta_1 x$  can be found in the same way as  $\theta x$ , and similarly  $\theta_2 x$ , and so on. And the fractional parts being determined, the integral part may be found by subtracting from  $\frac{\phi x}{f x}$  the sum of the fractional parts, so that the fraction  $\frac{\phi x}{f x}$  can by a direct process be decomposed in the above-mentioned form.

Particular terms in the decomposition of certain fractions may be obtained with great facility. Thus  $m$  being a prime number, assume

$$\frac{1}{(1-x)(1-x^2)\cdots(1-x^{m-1})(1-x^m)} = \&c. + \frac{\theta x}{[1-x^m]};$$

then observing that  $(1-x^m) = (1-x)[1-x^m]$ , we have for  $[1-x^m] = 0$ ,

$$\theta x = \frac{1}{(1-x)(1-x^2)\cdots(1-x^{m-1})}.$$

Now  $u$  being any quantity whatever and  $x$  being a root of  $[1-x^m] = 0$ , we have identically

$$[1-u^m] = (u-x)(u-x^2)\cdots(u-x^{m-1});$$

and therefore putting  $u = 1$ , we have  $m = (1-x)(1-x^2)\cdots(1-x^{m-1})$ , and therefore

$$\theta x = \frac{1}{m},$$

whence

$$\frac{1}{(1-x)(1-x^2)\cdots(1-x^m)} = \&c. + \frac{1}{m[1-x^m]}.$$

Again,  $m$  being as before a prime number, assume

$$\frac{1}{(1-x)(1-x^2)\cdots(1-x^{m-1})(1-x^m)^2} = \&c. + \frac{\theta x}{[1-x^m]^2};$$

we have in this case for  $[1-x^m] = 0$ ,

$$\theta x = \frac{1}{(1-x)^2(1-x^2)\cdots(1-x^{m-1})},$$

which is immediately reduced to  $\theta x = \frac{1}{m^2} \frac{1}{1-x}$ . Now

$$\frac{1-u^m}{u-x} = (1+x+\cdots+x^{m-1}) + (1+x+\cdots+x^{m-2})x\cdots(1+u)x^{m-1} + x^m,$$

or putting  $u = 1$ ,

$$\frac{m}{1-x} = (m-1) + (m-2)x + \cdots + x^{m-1};$$

and substituting this in the value of  $\theta x$ , we find

$$\frac{1}{(1-x)(1-x^2)\cdots(1-x^m)^2} = \&c. + \frac{1}{m^3} \frac{(m-1) + (m-2)x + \cdots + x^{m-1}}{[1-x^m]^2}.$$

The preceding decomposition of the fraction  $\frac{\phi x}{f x}$  gives very readily the expansion of the fraction in ascending powers of  $x$ . For, consider a fraction such as

$$\frac{\theta x}{[1 - x^a]}$$

where the degree of the numerator is in general less by unity than that of the denominator; we have

$$1 - x^a = [1 - x^a] \prod [1 - x^{a'}],$$

where  $a'$  denotes any divisor of  $a$  (including unity, but not including the number  $a$  itself). The fraction may therefore be written under the form

$$\frac{\theta x \prod [1 - x^{a'}]}{1 - x^a},$$

where the degree of the numerator is in general less by unity than that of the denominator, i.e. is equal to  $a - 1$ . Suppose that  $b$  is any divisor of  $a$  (including unity, but not including the number  $a$  itself), then  $1 - x^b$  is a divisor of  $\prod [1 - x^{a'}]$ , and therefore of the numerator of the fraction. Hence representing this numerator by

$$A_0 + A_1 x + \cdots + A_{a-1} x^{a-1},$$

and putting  $a = bc$ , we have (corresponding to the case  $b = 1$ )

$$A_0 + A_1 + A_2 + \cdots + A_{a-1} = 0,$$

and generally for the divisor  $b$ ,

$$A_0 + A_b + \cdots + A_{(c-1)b} = 0,$$

$$A_1 + A_{b+1} + \cdots + A_{(c-1)b+1} = 0,$$

$$A_{b-1} + A_{2b-1} + \cdots + A_{cb-1} = 0.$$

Suppose now that  $\alpha_q$  denotes a circulating element to the period  $a$ , i.e. write a function such as

$$\alpha_q = 1, \quad q \equiv 0 \pmod{a},$$

$$\alpha_q = 0 \text{ in every other case;}$$

$$A_0 \alpha_q + A_1 \alpha_{q-1} + \cdots + A_{a-1} \alpha_{q-a+1}$$

will be a circulating function, or circulator to the period  $a$ , and may be represented by the notation

$$(A_0, A_1, \dots, A_{a-1}) \text{ circlo } a_q.$$

In the case however where the coefficients  $A$  satisfy, for each divisor  $b$  of the number  $a$ , the above-mentioned equations, the circulating function is what I call a prime circulator, and I represent it by the notation

$$(A_0, A_1, \dots, A_{a-1}) \text{ per } a_q.$$

By means of this notation we have at once

$$\text{coefficient } x^q \text{ in } \frac{\theta x}{[1 - x^a]} = (A_0, A_1, \dots, A_{a-1}) \text{ per } a_q,$$

and thence also

$$\text{coefficient } x^q \text{ in } (x \partial_x)^r \frac{\theta x}{[1 - x^a]} = q^r (A_0, A_1, \dots, A_{a-1}) \text{ per } a_q.$$

Hence assuming that in the fraction  $\frac{\phi x}{f x}$  the degree of the numerator is less than that of the denominator (so that there is not any integral part), we have

$$\text{coefficient } x^q \text{ in } \frac{\phi x}{f x} = \Sigma q^r (A_0, A_1, \dots, A_{a-1}) \text{ per } a_q;$$

or, if we wish to put in evidence the non-circulating part arising from the divisor  $a = 1$ ,

$$\text{coefficient } x^q \text{ in } \frac{\phi x}{fx} = Aq^{k-1} + Bq^{k-2} + \cdots + Lq + M + \Sigma q^r (A_0, A_1, \dots, A_{a-1}) \text{ per } a_q;$$

where  $k$  denotes the number of the factors  $1 - x^m$  in the denominator  $fx$ ,  $a$  is any divisor (unity excluded) of one or more of the indices  $m$ ; and for each value of  $a$  the index  $r$  extends from  $r = 0$  to  $r = k - 1$ , where  $k$  denotes the number of indices of which  $a$  is a divisor. The particular results previously obtained show, that  $m$  being a prime number,

$$\text{coefficient } x^q \text{ in } \frac{1}{(1-x)(1-x^2)\cdots(1-x^m)} = \&c. + \frac{1}{m}(1, 0, 0, \dots, 0) \text{ per } m_q,$$

and

$$\text{coefficient } x^q \text{ in } \frac{1}{(1-x)(1-x^2)\cdots(1-x^m)^2} = \&c. + \frac{1}{m^3}(m-1, m-2, \dots, 1) \text{ per } m_q.$$

Suppose, as before, that the degree of  $\phi x$  is less than that of  $fx$ , and let the analytical expression above obtained for the coefficient of  $x^q$  in the expansion in ascending powers of  $x$  of the fraction  $\frac{\phi x}{fx}$  be represented by  $Fq$ , it is very remarkable that if we expand  $\frac{\phi x}{fx}$  in descending powers of  $x$ , then the coefficient of  $x^q$  in this new expansion ( $q$  is here of course negative, since the expansion contains only negative powers of  $x$ ) is precisely equal to  $-Fq$ ; this is in fact at once seen to be the case with respect to each of the partial fractions into which  $\frac{\phi x}{fx}$  has been decomposed, and it is consequently the case with respect to the fraction itself<sup>1</sup>. This gives rise to a result of some importance. Suppose that  $\phi x$  and  $fx$  are respectively of the degrees  $N$  and  $D$ ; it is clear from the form of  $fx$  that we have  $f\left(\frac{1}{x}\right) = (-)^D x^{-D} fx$ ; and I suppose that  $\phi x$  is also such that  $\phi\left(\frac{1}{x}\right) = (+)^N x^{-N} \phi x$ ; then writing  $D - N = h$ , and supposing that  $\frac{\phi x}{fx}$  is expanded in descending powers of  $x$ , so that the coefficient of  $x^q$  in the expansion is  $Fq$ , it is in the first place clear that the expansion will commence with the term  $x^{-h}$ , and we must therefore have

$$Fq = 0$$

for all values of  $q$  from  $q = -1$  to  $q = -(h-1)$ .

Consider next the coefficient of a term  $x^{-h-q}$ , where  $q$  is 0 or positive; the coefficient in question, the value of which is  $-F(-h-q)$ , is obviously equal to the coefficient of  $x^{h+q}$  in the expansion in ascending powers of  $x$  of  $\frac{\phi(1/x)}{f(1/x)}$ , i.e. to

$$\frac{(+)^N}{(-)^D} \text{coefficient } x^{h+q} \text{ in } \frac{\phi x}{fx} \cdot x^{D-N},$$

or what is the same thing, to

$$\frac{(+)^N}{(-)^D} \text{coefficient } x^q \text{ in } \frac{\phi x}{fx};$$

and we have therefore,  $q$  being zero or positive,

$$F(-h-q) = -(+)^N (-)^D Fq.$$

In particular, when  $\phi x = 1$ ,  $Fq = 0$  for all values of  $q$  from  $q = -1$  to  $q = -(D-1)$ ; and  $q$  being 0 or positive,

$$F(-D-q) = (-)^{D-1} Fq.$$

The preceding investigations show the general form of the function  $P(a, b, c, \dots)q$ , viz. that

$$P(a, b, c, \dots)q = Aq^{k-1} + Bq^{k-2} + \cdots + Lq + M + \Sigma q^r (A_0, A_1, \dots, A_{l-1}) \text{ per } l_q,$$

<sup>1</sup>The property is a fundamental one in the general theory of developments.

a formula in which  $k$  denotes the number of the elements  $a, b, c, \dots$ , and  $l$  is any divisor (unity excluded) of one or more of these elements; the summation in the case of each such divisor extends from  $r = 0$  to  $r = k - 1$ , where  $k$  is the number of the elements  $a, b, c, \dots$  of which  $l$  is a divisor; and the investigations indicate how the values of the coefficients  $A$  of the prime circulators are to be obtained. It has been moreover in effect shown, that if  $D = a + b + c + \dots$ , then, writing for shortness  $P(q)$  instead of  $P(a, b, c, \dots)_q$ , we have

$$P(q) = 0$$

for all values of  $q$  from  $q = -1$  to  $q = -(D - 1)$ , and that  $q$  being 0 or positive,

$$P(-D - q) = (-)^{k-1}P(q);$$

these last theorems are however uninterpretable in the theory of partitions, and they apply only to the analytical expression for  $P(q)$ .

I have calculated the following particular results:

$$P(1, 2)_q = \frac{1}{4} \left\{ 2q + 3 + (1, -1) \text{ per } 2_q \right\},$$

$$\begin{aligned} P(1, 2, 3)_q = \frac{1}{72} \left\{ 6q^2 + 36q + 47 \right. \\ \left. + 9(1, -1) \text{ per } 2_q \right. \\ \left. + 8(2, -1, -1) \text{ per } 3_q \right\}, \end{aligned}$$

$$\begin{aligned} P(1, 2, 3, 4)_q = \frac{1}{288} \left\{ 2q^3 + 30q^2 + 135q + 175 \right. \\ \left. + (9q + 45)(1, -1) \text{ per } 2_q \right. \\ \left. + 32(1, 0, -1) \text{ per } 3_q \right. \\ \left. + 36(1, 0, -1, 0) \text{ per } 4_q \right\}, \end{aligned}$$

$$\begin{aligned} P(1, 2, 3, 4, 5)_q = \frac{1}{86400} \left\{ 30q^4 + 900q^3 + 9300q^2 + 38250q + 50651 \right. \\ \left. + (1350q + 10125)(1, -1) \text{ per } 2_q \right. \\ \left. + 3200(2, -1, -1) \text{ per } 3_q \right. \\ \left. + 5400(1, -1, -1, 1) \text{ per } 4_q \right. \\ \left. + 3456(4, -1, -1, -1, -1) \text{ per } 5_q \right\}, \end{aligned}$$

$$P(2)_q = \frac{1}{2} \left\{ 1 + (1, -1) \text{ per } 2_q \right\},$$

$$P(2, 3)_q = \frac{1}{12} \left\{ 2q + 5 + 3(1, -1) \text{ per } 2_q + 4(1, -1, 0) \text{ per } 3_q \right\},$$

$$\begin{aligned}
 P(2, 3, 4)q &= \frac{1}{288} \left\{ 6q^2 + 54q + 107 + (18q + 81)(1, -1) \text{ per } 2_q \right. \\
 &\quad \left. + 32(2, -1, -1) \text{ per } 3_q + 36(1, -1, -1, 1) \text{ per } 4_q \right\}, \\
 P(2, 3, 4, 5)q &= \frac{1}{1440} \left\{ 2q^3 + 42q^2 + 267q + 497 + (45q + 315)(1, -1) \text{ per } 2_q \right. \\
 &\quad + 160(1, -1, 0) \text{ per } 3_q + 180(1, 0, -1, 0) \text{ per } 4_q \\
 &\quad \left. + 288(1, -1, 0, 0, 0) \text{ per } 5_q \right\}, \\
 P(2, 3, 4, 5, 6)q &= \frac{1}{172800} \left\{ 10q^4 + 400q^3 + 5550q^2 + 31000q + 56877 \right. \\
 &\quad + (450q^2 + 9000q + 39075)(1, -1) \text{ per } 2_q \\
 &\quad + 3200q(1, -1, 0) \text{ per } 3_q + 1600(21, -19, -2) \text{ per } 3_q \\
 &\quad + 10800(1, 0, -1, 0) \text{ per } 4_q + 6912(4, -1, -1, -1, -1) \text{ per } 5_q \\
 &\quad \left. + 4800(1, -1, -2, -1, 1, 2) \text{ per } 6_q \right\}, \\
 P(1, 2, 3, 5)q &= \frac{1}{720} \left\{ 4q^3 + 66q^2 + 324q + 451 + 45(1, -1) \text{ per } 2_q \right. \\
 &\quad \left. + 80(1, -1, 0) \text{ per } 3_q + 144(1, 0, 0, 0, -1) \text{ per } 5_q \right\}, \\
 P(1, 2, 2, 3, 4)q &= \frac{1}{6912} \left\{ 6q^4 + 144q^3 + 1194q^2 + 3960q + 4267 \right. \\
 &\quad + (54q^2 + 648q + 1701)(1, -1) \text{ per } 2_q \\
 &\quad \left. + 256(2, -1, -1) \text{ per } 3_q + 432(1, 0, -1, 0) \text{ per } 4_q \right\}, \\
 P(8)q &= \frac{1}{8} \left\{ 1 + 1(1, -1) \text{ per } 2_q + 2(1, 0, -1, 0) \text{ per } 4_q \right. \\
 &\quad \left. + 8(1, 0, 0, 0, -1, 0, 0, 0) \text{ per } 8_q \right\}, \\
 P(7, 8)q &= \frac{1}{112} \left\{ 2q + 43 + 7(1, -1) \text{ per } 2_q + 14(1, -1, -1, 1) \text{ per } 4_q \right. \\
 &\quad + 16(3, 2, 1, 0, -1, -2, -3) \text{ per } 7_q \\
 &\quad \left. + 56(0, -1, -1, 0, 0, 1, 1, 0) \text{ per } 8_q \right\},
 \end{aligned}$$

which are, I think, worth preserving.

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I proceed to discuss the following problem: “To find in how many ways a number  $q$  can be made up as a sum of  $m$  terms with the elements  $0, 1, 2, \dots, k$ , each element being repeatable an indefinite number of times.” The required number of partitions is represented by

$$P(0, 1, 2, \dots, k)^m q,$$

and the number of partitions of  $q$  less the number of partitions of  $q - 1$  is represented by

$$P'(0, 1, 2, \dots, k)^m q.$$

We have, as is well known,

$$P(0, 1, 2, \dots, k)^m q = \text{coefficient } x^q z^m \text{ in } \frac{1}{(1-z)(1-xz) \cdots (1-x^k z)},$$

where the expansion is to be effected in ascending powers of  $z$ . Now

$$\frac{1}{(1-z)(1-xz) \cdots (1-x^k z)} = 1 + \frac{1-x^{k+1}}{1-x} z + \frac{(1-x^{k+1})(1-x^{k+2})}{(1-x)(1-x^2)} z^2 + \&c.,$$



the general term being

$$\frac{(1-x^{k+1})(1-x^{k+2})\cdots(1-x^{k+s})}{(1-x)(1-x^2)\cdots(1-x^s)}z^s,$$

or, what is the same thing,

$$\frac{(1-x^{m+1})(1-x^{m+2})\cdots(1-x^{m+k})}{(1-x)(1-x^2)\cdots(1-x^k)}z^m,$$

and consequently

$$P(0, 1, 2, \dots, k)^m q = \text{coefficient } x^q \text{ in } \frac{(1-x^{m+1})(1-x^{m+2})\cdots(1-x^{m+k})}{(1-x)(1-x^2)\cdots(1-x^k)};$$

to transform this expression I make use of the equation

$$\frac{1}{(1-z)(1-xz)\cdots(1-x^kz)} = 1 + \frac{1-x^{k+1}}{1-x}z + \frac{(1-x^{k+1})(1-x^{k+2})}{(1-x)(1-x^2)}z^2 + \&c.,$$

where the general term is

$$\frac{(1-x^{k+1})(1-x^{k+2})\cdots(1-x^{k+s})}{(1-x)(1-x^2)\cdots(1-x^s)}z^s,$$

and the series is a finite one, the last term being that corresponding to  $s = k$ , viz.  $z^k \div x^{\frac{1}{2}k(k+1)}$ . Writing  $x^m$  for  $z$ , and substituting the resulting value of

$$\frac{(1-x^{m+1})(1-x^{m+2})\cdots(1-x^{m+k})}{(1-x)(1-x^2)\cdots(1-x^k)}$$

in the formula for  $P(0, 1, 2, \dots, k)^m q$ , we have

$$P(0, 1, 2, \dots, k)^m q = \sum_s \left\{ (-)^s \text{coefficient } x^q \text{ in } \frac{x^{sm+\frac{1}{2}s(s+1)}}{(1-x)(1-x^2)\cdots(1-x^s)(1-x)(1-x^2)\cdots(1-x^{k-s})} \right\},$$

where the summation extends from  $s = 0$  to  $s = k$ ; but if for any value of  $s$  between these limits  $sm+\frac{1}{2}s(s+1)$  becomes greater than  $q$ , then it is clear that the summation need only be extended from  $s = 0$  to the last preceding value of  $s$ , or what is the same thing, from  $s = 0$  to the greatest value of  $s$  for which  $q - sm - \frac{1}{2}s(s+1)$  is positive or zero.

It is obvious, that if  $q > km$ , then

$$P(0, 1, 2, \dots, k)^m q = 0;$$

and moreover, that if  $q \leq km$ , then

$$P(0, 1, 2, \dots, k)^m q = P(0, 1, 2, \dots, k)^m (km - q),$$

so that we may always suppose  $q \not\leq \frac{1}{2}km$ . I write therefore  $q = \frac{1}{2}(km - \alpha)$  where  $\alpha$  is zero or a positive integer not greater than  $km$ , and is even or odd according as  $km$  is even or odd. Substituting this value of  $q$  and making a slight change in the form of the result, we have

$$P(0, 1, 2, \dots, k)^m \frac{1}{2}(km - \alpha) = \sum_s \left\{ (-)^s \text{coeff. } x^{\frac{1}{2}km - \frac{1}{2}\alpha m} \text{ in } \frac{x^{\frac{1}{2}s(s+1)}}{(1-x)\cdots(1-x^s)(1-x)\cdots(1-x^{k-s})} \right\},$$

where the summation extends from  $s = 0$  to the greatest value of  $s$  for which  $(\frac{1}{2}k - s)m - \frac{1}{2}\alpha - \frac{1}{2}s(s+1)$  is positive or zero. But we may, if we please, consider the summation as extending, when  $k$  is even, from  $s = 0$  to  $s = \frac{1}{2}k - 1$ , and when  $k$  is odd, from  $s = 0$  to  $s = \frac{1}{2}(k - 1)$ ; the terms corresponding to values of  $s$  greater than the greatest value for which  $(\frac{1}{2}k - s)m - \frac{1}{2}\alpha - \frac{1}{2}s(s+1)$  is positive or zero being of course equal to zero. It may be noticed, that the fraction will be a proper one if  $\alpha < (k - s)(k - s + 1)$ ; or substituting for  $s$  its greatest value, the fraction will be a proper one for all values of  $s$ , if, when  $k$  is even,  $\alpha < \frac{1}{2}k(k + 2)$ , and when  $k$  is odd,  $\alpha < \frac{1}{4}(k + 1)(k + 3)$ .

We have in a similar manner,

$$P'(0, 1, 2, \dots k)^m q = \text{coefficient } x^q z^m \text{ in } \frac{1-x}{(1-z)(1-xz) \dots (1-x^k z)},$$

which leads to

$$P'(0, 1, 2, \dots k)^m \frac{1}{2}(km - \alpha) = \sum_s \left\{ (-)^s \text{coeff. } x^{\frac{1}{2}km - \frac{1}{2}\alpha m} \text{ in } \frac{x^{\frac{1}{2}s(s+1)}}{(1-x) \dots (1-x^s)(1-x) \dots (1-x^{k-s-1})} \right\},$$

where the summation extends, as in the former case, from  $s = 0$  to the greatest value of  $s$ , for which  $(\frac{1}{2}k - s)m - \frac{1}{2}\alpha - \frac{1}{2}s(s+1)$  is positive or zero, or, if we please, when  $k$  is even, from  $s = 0$  to  $s = \frac{1}{2}k - 1$ , and when  $s$  is odd, from  $s = 0$  to  $s = \frac{1}{2}(k - 1)$ . The condition, in order that the fraction may be a proper one for all values of  $s$ , is, when  $k$  is even,  $\alpha + 1 < \frac{1}{2}k(k+2)$ , and when  $k$  is odd,  $\alpha + 1 < \frac{1}{4}(k+1)(k+3)$ .

To transform the preceding expressions, I write when  $k$  is odd  $x^2$  instead of  $x$ , and I put for shortness  $\theta$  instead of  $\frac{1}{2}k - s$  or  $\frac{1}{2}(k - 2s)$ , and  $\gamma$  instead of  $\frac{1}{2}\alpha + \frac{1}{2}s(s+1)$  or  $\alpha + s(s+1)$ ; we have to consider an expression of the form

$$\text{coefficient } x^{\theta m} \text{ in } \frac{x^\gamma}{Fx},$$

where  $Fx$  is the product of factors of the form  $1 - x^a$ . Suppose that  $\bar{a}$  is the least common multiple of  $a$  and  $\theta$ , then  $(1 - x^{\bar{a}}) \div (1 - x^a)$  is an integral function of  $x$ , equal to  $\chi x$  suppose, and  $1 \div (1 - x^a) = \chi x \div (1 - x^{\bar{a}})$ . Making this change in all the factors of  $Fx$  which require it (i.e. in all the factors except those in which  $a$  is a multiple of  $\theta$ ), the general term becomes

$$\text{coefficient } x^{\theta m} \text{ in } \frac{x^\gamma Hx}{Gx},$$

where  $Gx$  is a product of factors of the form  $1 - x^{a'}$ , in which  $a'$  is a multiple of  $\theta$ , i.e.  $Gx$  is a rational and integral function of  $x^\theta$ . But in the numerator  $x^\gamma Hx$  we may reject, as not contributing to the formation of the coefficient of  $x^{\theta m}$ , all the terms in which the indices are not multiples of  $\theta$ ; the numerator is thus reduced to a rational and integral function of  $x^\theta$ , and the general term is therefore of the form

$$\text{coefficient } x^{\theta m} \text{ in } \frac{\lambda x}{kx},$$

or what is the same thing, of the form

$$\text{coefficient } x^m \text{ in } \frac{\lambda x}{kx},$$

where  $kx$  is the product of factors of the form  $1 - x^a$ , and  $\lambda x$  is a rational and integral function of  $x$ . The particular value of the fraction depends on the value of  $s$ ; and uniting the different terms, we have an expression

$$\text{coefficient } x^m \text{ in } \sum_s ((-)^s \frac{\lambda x}{kx}),$$

which is equivalent to

$$\text{coefficient } x^m \text{ in } \frac{\phi x}{fx},$$

where  $fx$  is a product of factors of the form  $1 - x^a$ , and  $\phi x$  is a rational and integral function of  $x$ . And it is clear that the fraction will be a proper one when each of the fractions in the original expression is a proper fraction, i.e. in the case of  $P(0, 1, 2, \dots k)^m \frac{1}{2}(km - \alpha)$ , when for  $k$  even,  $\alpha < \frac{1}{2}k(k+2)$ , and for  $k$  odd,  $\alpha < \frac{1}{4}(k+1)(k+3)$ ; and in the case of  $P'(0, 1, 2, \dots k)^m \frac{1}{2}(km - \alpha)$ , when for  $k$  even,  $\alpha + 1 < \frac{1}{2}k(k+2)$ , and for  $k$  odd,  $\alpha + 1 < \frac{1}{4}(k+1)(k+3)$ .

We see, therefore, that  $P(0, 1, 2, \dots k)^m \frac{1}{2}(km - \alpha)$ , and  $P'(0, 1, 2, \dots k)^m \frac{1}{2}(km - \alpha)$ , are each of them of the form

$$\text{coefficient } x^m \text{ in } \frac{\phi x}{fx},$$

where  $fx$  is the product of factors of the form  $1 - x^a$ , and up to certain limiting values of  $\alpha$  the fraction is a proper fraction. When the fraction  $\frac{\phi x}{fx}$  is known, we may therefore obtain by the method employed

in the former part of this Memoir, analytical expressions (involving prime circulators) for the functions  $P$  and  $P'$ .

As an example, take

$$P(0, 1, 2, 3)^m \frac{1}{2}m,$$

which is equal to

$$\text{coefficient } x^{\frac{3}{2}m} \text{ in } \frac{1}{(1-x^m)(1-x^2)(1-x^3)} - \text{coefficient } x^{\frac{3}{2}m} \text{ in } \frac{x}{(1-x^2)(1-x^3)(1-x^4)}.$$

The multiplier for the first fraction is

$$\frac{(1-x^6)(1-x^{12})}{(1-x^2)(1-x^4)},$$

which is equal to

$$1 + x^2 + 2x^4 + x^6 + 2x^8 + x^{10} + x^{12}.$$

Hence, rejecting in the numerator the terms the indices of which are not divisible by 3, the first term becomes

$$\text{coefficient } x^{\frac{3}{2}m} \text{ in } \frac{1 + 2x^6 + x^{12}}{(1-x^6)(1-x^{12})},$$

or what is the same thing, the first term is

$$\text{coefficient } x^m \text{ in } \frac{1 + 2x^2 + x^4}{(1-x^2)^2(1-x^4)};$$

and, the second term being

$$- \text{coefficient } x^{\frac{3}{2}m} \text{ in } \frac{x^4}{(1-x^6)^2(1-x^{12})},$$

we have

$$P(0, 1, 2, 3)^m \frac{1}{2}m = \text{coefficient } x^m \text{ in } \frac{1 + x^4}{(1-x^2)^2(1-x^4)};$$

and similarly it may be shown, that

$$P'(0, 1, 2, 3)^m \frac{1}{2}(3m-1) = \text{coefficient } x^m \text{ in } \frac{1 + x^2}{(1-x^2)^2(1-x^4)}.$$

As another example, take  $P'(0, 1, 2, 3, 4, 5)^m \frac{1}{2}m$ , which is equal to

$$\begin{aligned} & \text{coefficient } x^{\frac{5}{2}m} \text{ in } \frac{1}{(1-x^m)(1-x^2)(1-x^3)(1-x^4)(1-x^5)} \\ & - \text{coefficient } x^{\frac{5}{2}m} \text{ in } \frac{x}{(1-x^2)(1-x^3)(1-x^4)(1-x^5)(1-x^6)} \\ & + \text{coefficient } x^{\frac{5}{2}m} \text{ in } \frac{x^3}{(1-x^2)(1-x^3)(1-x^4)(1-x^5)(1-x^6)(1-x^7)}. \end{aligned}$$

The multiplier for the first fraction is

$$\frac{(1-x^6)(1-x^7)(1-x^8)}{(1-x^2)(1-x^3)(1-x^4)},$$

which is a function of  $x$  of the order 36, the coefficients of which are

$$1, 0, 1, 1, 2, 1, 3, 2, 4, 3, 4, 4, 6, 4, 6, 5, 7, 5, 7, 5, 7, 5, 6, 4, 6, 4, 4, 3, 4, 2, 3, 1, 2, 1, 1, 0, 1,$$

and the first part becomes therefore

$$\text{coefficient } x^{\frac{5}{2}m} \text{ in } \frac{1 + 5x^{10} + x^{15} + 4x^{20} + 7x^{25} + 4x^{30} + 3x^{35}}{(1-x^2)(1-x^4)(1-x^6)(1-x^{10})}.$$

The multiplier for the second fraction is

$$\frac{(1-x^6)(1-x^7)(1-x^8)}{(1-x^3)(1-x^4)(1-x^5)},$$

which is a function of  $x$  of the order 14, the coefficients of which are

$$1, 1, 2, 1, 3, 2, 3, 1, 3, 2, 3, 1, 2, 1, 1;$$

and the second term becomes

$$-\text{coefficient } x^{\frac{5}{2}m} \text{ in } \frac{x + 2x^3 + 2x^5 + 3x^9 + x^{11} + x^{13}}{(1-x^2)^2(1-x^5)(1-x^{12})},$$

and the third term is

$$\text{coefficient } x^{\frac{5}{2}m} \text{ in } \frac{x^3}{(1-x^4)(1-x^6)^2}.$$

Now the fractions may be reduced to a common denominator

$$(1-x^2)(1-x^4)(1-x^{10})(1-x^{12}),$$

by multiplying the terms of the second fraction by  $\frac{1-x^4}{1-x^2} (= 1+x^2+x^4)$ , and the terms of the third fraction by  $\frac{1-x^{12}}{1-x^4} (= 1+x^4)$ ; performing the operations and adding, the numerator and denominator of the resulting fraction will each of them contain the factor  $1-x^2$ ; and casting this out, we find

$$P'(0, 1, 2, 3, 4, 5)^m \frac{1}{2}m = \text{coefficient } x^m \text{ in } \frac{x^4(\dots)}{(1-x)(1-x^2)^2(1-x^4)}.$$

I have calculated by this method several other particular cases, which are given in my “Second Memoir upon Quantics”, [141], the present researches were in fact made for the sake of their application to that theory.

*Received April 20,—Read May 3 and 10, 1855.*

Since the preceding portions of the present Memoir were written, Mr Sylvester has communicated to me a remarkable theorem which has led me to the following additional investigations<sup>2</sup>.

Let  $\frac{\phi x}{fx}$  be a rational fraction, and let  $(x-x_1)^k$  be a factor of the denominator  $fx$ , then if

$$\left\{ \frac{\phi x}{fx} \right\}_{x_1}$$

denote the portion which is made up of the simple fractions having powers of  $x-x_1$  for their denominators, we have by a known theorem

$$\left\{ \frac{\phi x}{fx} \right\}_{x_1} = \Sigma \frac{1}{z} \text{coefficient } \frac{1}{z} \text{ in } \frac{\phi(x_1+z)}{x-x_1-zf(x_1+z)}.$$

Now by a theorem of Jacobi's and Cauchy's,

$$\text{coefficient } \frac{1}{z} \text{ in } Fz = \text{coefficient } \frac{1}{t} \text{ in } F(\phi^{-1}t)\phi't,$$

whence, writing  $x_1+z=x_1e^t$ , we have

$$\left\{ \frac{\phi x}{fx} \right\}_{x_1} = \text{coefficient } \frac{1}{t} \text{ in } \frac{x_1\phi(x_1e^t)e^t}{x-x_1e^tf(x_1e^t)}.$$

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<sup>2</sup>Mr Sylvester's researches are published in the *Quarterly Mathematical Journal*, July 1855, [vol. i. pp. 141–152], and he has there given the general formula as well for the circulating as the non-circulating part of the expression for the number of partitions.—Added 23rd February, 1856.—A. C.

Now putting for a moment  $x = x_1 e^\theta$ , we have

$$\frac{x_1 - x}{x_1 - x e^t} = \frac{x_1(1 - e^{\theta+t})}{x_1(1 - e^\theta)} = \frac{\theta + t}{\theta} \frac{e^{\theta+t}}{e^\theta} \text{ etc.}$$

and  $e^\theta d\theta = x dx$ , whence

$$\frac{x_1}{x_1 - x e^t} = \frac{x_1}{x_1 - x} \frac{1}{1} \frac{x_1 - x}{x_1 - x} \frac{1}{1.2} \cdots \frac{x_1 - x}{x_1 - x},$$

the general term of which is

$$\frac{1}{\Pi(s-1)} t^{s-1} (x \partial_x)^{s-1} \frac{x_1}{x_1 - x}.$$

Hence representing the general term of  $x_1 \phi(x_1 e^{-t})$  by  $x_1 \phi^{(s-1)} t^{s-1}$ , and writing  $\frac{x_1 \phi(x_1 e^{-t})}{fx} = \gamma x$ ,

$$\text{coefficient } \frac{1}{t} \text{ in } \frac{x_1 \phi(x_1 e^{-t})}{fx},$$

we find, writing down only the general term,

$$\left\{ \frac{\phi x}{fx} \right\}_{x_1} = \frac{1}{\Pi(s-1)} \gamma x^{s-1} (x \partial_x)^{s-1} \frac{x_1}{x_1 - x},$$

where the value of  $\gamma x$  depends upon that of  $s$ , and where  $s$  extends from  $s = 1$  to  $s = k$ .

Suppose now that the denominator is made up of factors (the same or different) of the form  $1 - x^m$ . And let  $a$  be any divisor of one or more of the indices  $m$ , and let  $k$  be the number of the indices of which  $a$  is a divisor. The denominator contains the divisor  $[1 - x^a]^k$ , and consequently if  $\rho$  be any root of the equation  $[1 - x^a] = 0$ , the denominator contains the factor  $(\rho - x)^k$ . Hence writing  $\rho$  for  $x_1$  and taking the sum with respect to all the roots of the equation  $[1 - x^a] = 0$ , we find

$$\left\{ \frac{\phi x}{fx} \right\}_{[1-x^a]} = \frac{1}{\Pi(s-1)} \Sigma (x \partial_x)^{s-1} \frac{\rho}{\rho - x} = \cdots = \frac{1}{\Pi(s-1)} (x \partial_x)^{s-1} \frac{\theta x}{[1 - x^a]} + \cdots,$$

where

$$\theta_s \rho = \text{coefficient } \frac{1}{t} \text{ in } t^{s-1} \psi \rho.$$

and as before  $s$  extends from  $s = 1$  to  $s = k$ . We have thus the actual value of the function  $\theta x$  made use of in the memoir.

A preceding formula gives

$$\left\{ \frac{\phi x}{fx} \right\}_{x_1} = \text{coefficient } \frac{1}{t} \text{ in } \frac{1}{1 - x e^t} \frac{\phi(x_1 e^{-t})}{f'(x_1 e^{-t})},$$

which is a very simple expression for the non-circulating part of the fraction  $\frac{\phi x}{fx}$ .

This is, in fact, Mr Sylvester's theorem above referred to.

## 141.

### A SECOND MEMOIR UPON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLVI. for the year, 1856, pp. 101–126. Received April 14,—Read May 24, 1855.]

The present memoir is intended as a continuation of my Introductory Memoir upon Quantics, t. CXLIV. (1854), p. 245, and must be read in connexion with it; the paragraphs of the two Memoirs are numbered continuously. The special subject of the present memoir is the theorem referred to in the Postscript to the Introductory Memoir, and the various developments arising thereout in relation to the number and form of the covariants of a binary quantic.

**25.** I have already spoken of asyzygetic covariants and invariants, and I shall have occasion to speak of irreducible covariants and invariants. Considering in general a function  $u$  determined like a covariant or invariant by means of a system of partial differential equations, it will be convenient to explain what is meant by an asyzygetic integral and by an irreducible integral. Attending for greater simplicity only to a single set  $(a, b, c, \dots)$ , which in the case of the covariants or invariants of a single function will be as before the coefficients or elements of the function, it is assumed that the system admits of integrals of the form  $u = P$ ,  $u = Q$ , &c., or as we may express it, of integrals  $P$ ,  $Q$ , &c., where  $P$ ,  $Q$ , &c. are rational and integral homogeneous functions of the set  $(a, b, c, \dots)$ , and moreover that the system is such that  $P$ ,  $Q$ , &c. being integrals,  $\phi(P, Q, \&c.)$  is also an integral. Then considering only the integrals which are rational and integral homogeneous functions of the set  $(a, b, c, \dots)$ , integrals  $P, Q, R, \dots$  not connected by any linear equation or syzygy (such as  $\lambda P + \mu Q + \nu R + \dots = 0$ ),<sup>3</sup> are said to be asyzygetic; but in speaking of the asyzygetic integrals of a particular degree, it is implied that the integrals are a system such that every other integral of the same degree can be expressed as a linear function (such as  $\lambda P + \mu Q + \nu R + \dots$ ) of these integrals; and any integral  $F$  not expressible as a rational and integral homogeneous function of integrals of inferior degrees is said to be an irreducible integral.

**26.** Suppose now that  $A_1, A_2, A_3$ , &c. denote the number of asyzygetic integrals of the degrees 1, 2, 3, &c. respectively, and let  $\alpha_1, \alpha_2, \alpha_3$ , &c. be determined by the equations

$$\begin{aligned} A_1 &= \alpha_1, \\ A_2 &= \frac{1}{2}\alpha_1(\alpha_1 + 1) + \alpha_2, \\ A_3 &= \frac{1}{6}\alpha_1(\alpha_1 + 1)(\alpha_1 + 2) + \alpha_1\alpha_2 + \alpha_3, \\ A_4 &= \frac{1}{24}\alpha_1(\alpha_1 + 1)(\alpha_1 + 2)(\alpha_1 + 3) + \frac{1}{2}\alpha_1(\alpha_1 + 1)\alpha_2 + \alpha_1\alpha_3 + \frac{1}{2}\alpha_2(\alpha_2 + 1) + \alpha_4, \&c., \end{aligned}$$

or what is the same thing, suppose that

$$1 + A_1x + A_2x^2 + \&c. = (1 - x)^{-\alpha_1}(1 - x^2)^{-\alpha_2}(1 - x^3)^{-\alpha_3} \dots;$$

a little consideration will show that  $\alpha_r$  represents the number of irreducible integrals of the degree  $r$  less the number of linear relations or syzygies between the composite or non-irreducible integrals of the same degree. In fact the asyzygetic integrals of the degree 1 are necessarily irreducible, i.e.  $A_1 = \alpha_1$ . Represent for a moment the irreducible integrals of the degree 1 by  $X, X'$ , &c., then the composite integrals  $X^2, XX'$ , &c., the number of which is  $\frac{1}{2}\alpha_1(\alpha_1 + 1)$ , must be included among the asyzygetic integrals of the degree 2; and if the composite integrals in question were asyzygetic, there would remain  $A_2 - \frac{1}{2}\alpha_1(\alpha_1 + 1)$  for the number of irreducible integrals of the degree 2; but if there exist syzygies between the composite integrals in question, the number to be subtracted from  $A_2$  will be  $\frac{1}{2}\alpha_1(\alpha_1 + 1)$  less the number of these syzygies, and we shall have  $A_2 - \frac{1}{2}\alpha_1(\alpha_1 + 1)$ , i.e.  $\alpha_2$  equal to the number of the irreducible integrals of the degree 2 less the number of syzygies between the composite integrals of the same degree. Again, suppose that  $\alpha_2$  is negative  $= -\beta_2$ , we may for simplicity suppose that there are no irreducible integrals of the degree 2, but that the composite integrals of this degree,  $X^2, XX'$ , &c., are connected by  $\beta_2$  syzygies, such as  $\lambda X^2 + \mu XX' + \&c. = 0$ ,  $\lambda_1 X^2 + \mu_1 XX' + \&c. = 0$ . The asyzygetic integrals of the degree 4 include  $X^4, X^3X'$ , &c., the number of which is  $\frac{1}{24}\alpha_1(\alpha_1 + 1)(\alpha_1 + 2)(\alpha_1 + 3)$ ; but these composite integrals are not asyzygetic, they are connected by syzygies which are augmentatives of the  $\beta_2$  syzygies of the second degree, viz. by syzygies such as

$$\begin{aligned} (\lambda X^2 + \mu XX' \dots)X^2 &= 0, (\lambda X^2 + \mu XX' \dots)XX' = 0, \&c. (\lambda X^2 + \mu XX' \dots)X^2 = 0, \\ (\lambda_1 X^2 + \mu_1 XX' \dots)XX' &= 0, \&c., \end{aligned}$$

the number of which is  $\frac{1}{2}\alpha_1(\alpha_1 + 1)\beta_2$ . And these syzygies are themselves not asyzygetic, they are connected by secondary syzygies such as

$$\begin{aligned} \lambda(\lambda X^2 + \mu XX' \dots)X^2 + \mu(\lambda X^2 + \mu XX' \dots)XX' + \&c. \\ -\lambda(\lambda_1 X^2 + \mu_1 XX' \dots)X^2 - \mu(\lambda_1 X^2 + \mu_1 XX' \dots)XX' - \&c. &= 0, \&c., \end{aligned}$$

<sup>3</sup>It is hardly necessary to remark, that the multipliers  $\lambda, \mu, \nu, \dots$ , and generally any coefficients or quantities not expressly stated to contain the set  $(a, b, c, \dots)$ , are considered as independent of the set, or to use a convenient word, are considered as “trivials.”

the number of which is  $\frac{1}{2}\beta_2(\beta_2 - 1)$ . The real number of syzygies between the composite integrals  $X^4$ ,  $X^3X'$ , &c. (i.e. of the syzygies arising out of the  $\beta_2$  syzygies between  $X^2$ ,  $XX'$ , &c.), is therefore  $\frac{1}{2}\alpha_1(\alpha_1 + 1)\beta_2 - \frac{1}{2}\beta_2(\beta_2 - 1)$ , and the number of integrals of the degree 4, arising out of the integrals and syzygies of the degrees 1 and 2 respectively, is therefore

$$\frac{1}{24}\alpha_1(\alpha_1 + 1)(\alpha_1 + 2)(\alpha_1 + 3) + \frac{1}{2}\alpha_1(\alpha_1 + 1)\beta_2 + \frac{1}{2}\beta_2(\beta_2 - 1)$$

or writing  $-\alpha_2$  instead of  $\beta_2$ , the number in question is

$$\frac{1}{24}\alpha_1(\alpha_1 + 1)(\alpha_1 + 2)(\alpha_1 + 3) + \frac{1}{2}\alpha_1(\alpha_1 + 1)\alpha_2 + \frac{1}{2}\alpha_2(\alpha_2 + 1).$$

The integrals of the degrees 1 and 3 give rise to  $\alpha_1\alpha_3$  integrals of the degree 4; and if all the composite integrals obtained as above were asyzygetic, we should have

$$A_4 = \frac{1}{24}\alpha_1(\alpha_1 + 1)(\alpha_1 + 2)(\alpha_1 + 3) + \frac{1}{2}\alpha_1(\alpha_1 + 1)\alpha_2 + \frac{1}{2}\alpha_2(\alpha_2 + 1) + \alpha_1\alpha_3,$$

i.e.  $\alpha_4$  as the number of irreducible integrals of the degree 4; but if there exist any further syzygies between the composite integrals, then  $\alpha_4$  will be the number of the irreducible integrals of the degree 4 less the number of such further syzygies, and the like reasoning is in all cases applicable.

**27.** It may be remarked, that for any given partial differential equation, or system of such equations, there will be always a finite number  $\nu$  such that given  $\nu$  independent integrals every other integral is a function (in general an irrational function only expressible as the root of an equation) of the  $\nu$  independent integrals; and if to these integrals we join a single other integral not a rational function of the  $\nu$  integrals, it is easy to see that every other integral will be a rational function of the  $\nu + 1$  integrals; but every such other integral will not in general be a rational and integral function of the  $\nu + 1$  integrals; and [incorrect] there is not in general any finite number whatever of integrals, such that every other integral is a rational and integral function of these integrals, i.e. the number of irreducible integrals is in general infinite; and it would seem that this is in fact the case in the theory of covariants.

**28.** In the case of the covariants, or the invariants of a binary quantic,  $A_n$  is given (this will appear in the sequel) as the coefficient of  $x^n$  in the development, in ascending powers of  $x$ , of a rational fraction  $\frac{\phi x}{fx}$ , where  $fx$  is of the form

$$(1 - x)^{\beta_1}(1 - x^2)^{\beta_2} \dots (1 - x^k)^{\beta_k},$$

and the degree of  $\phi x$  is less than that of  $fx$ . We have therefore

$$1 + A_1x + A_2x^2 + \dots = \frac{\phi x}{fx},$$

and consequently

$$\frac{\phi x}{fx} = (1 - x)^{\beta_1 - \alpha_1}(1 - x^2)^{\beta_2 - \alpha_2} \dots (1 - x^k)^{\beta_k - \alpha_k}(1 - x^{k+1})^{-\alpha_{k+1}} \dots$$

Now every rational factor of a binomial  $1 - x^m$  is the irreducible factor of  $1 - x^{m'}$ , where  $m'$  is equal to or a submultiple of  $m$ . Hence in order that the series  $\alpha_1, \alpha_2, \alpha_3, \dots$  may terminate,  $\phi x$  must be made up of factors each of which is the irreducible factor of a binomial  $1 - x^m$ , or if  $\phi x$  be itself irreducible, then  $\phi x$  must be the irreducible factor of a binomial  $1 - x^m$ . Conversely, if  $\phi x$  be not of the form in question, the series  $\alpha_1, \alpha_2, \alpha_3$ , &c. will go on *ad infinitum*, and it is easy to see that there is no point in the series such that the terms beyond that point are all of them negative, i.e. there will be irreducible covariants or invariants of indefinitely high degrees; and the number of covariants or invariants will be infinite. The number of invariants is first infinite in the case of a quantic of the seventh order, or septimic; the number of covariants is first infinite in the case of a quantic of the fifth order, or quintic. [As is now well known, these conclusions are incorrect, the number of irreducible covariants or invariants is in every case finite.]

**29.** Resuming the theory of binary quantics, I consider the quantic

$$(a, b, \dots, b', a' \mid x, y)^m.$$

Here writing

$$\begin{aligned} [y\partial_x] &= a\partial_b + 2b\partial_c + \dots + mb'\partial_{a'} = X, \\ [x\partial_y] &= mb\partial_a + (m-1)c\partial_b + \dots + a'\partial_{b'} = Y, \end{aligned}$$

any function which is reduced to zero by each of the operations  $X - y\partial_x$ ,  $Y - x\partial_y$  is a covariant of the quantic. But a covariant will always be considered as a rational and integral function separately homogeneous in regard to the facients  $(x, y)$  and to the coefficients  $(a, b, \dots, b', a')$ . And the words order and degree will be taken to refer to the facients and to the coefficients respectively.

I commence by proving the theorem enunciated, No. 23. It follows at once from the definition, that the covariant is reduced to zero by the operation

$$X \cdot Y - Y \cdot X - y\partial_x \cdot Y - x\partial_y \cdot X + x\partial_y \cdot y\partial_x + y\partial_x \cdot x\partial_y,$$

which is equivalent to

$$X \cdot Y - Y \cdot X + y\partial_y - x\partial_x.$$

Now

$$X \cdot Y = XY + X(Y),$$

$$Y \cdot X = YX + Y(X),$$

where  $XY$  and  $YX$  are equivalent operations, and

$$X(Y) = 1 \cdot ma\partial_a + 2(m-1)b\partial_b + \dots + m \cdot 1 \cdot b'\partial_{b'},$$

$$Y(X) = m \cdot 1 \cdot b\partial_b + \dots + 2(m-1)b'\partial_{b'} + 1 \cdot ma'\partial_{a'},$$

whence

$$X(Y) - Y(X) = ma\partial_a + (m-2)b\partial_b + \dots - (m-2)b'\partial_{b'} - ma'\partial_{a'} = k \text{ suppose,}$$

and the covariant is therefore reduced to zero by the operation

$$k + y\partial_y - x\partial_x.$$

Now as regards a term  $a^\alpha b^\beta \dots b'^{\beta'} a'^{\alpha'} \cdot x^i y^j$ , we have

$$k = m\alpha + (m-2)\beta + \dots - (m-2)\beta' - m\alpha',$$

$$y\partial_y - x\partial_x = j - i;$$

and we see at once that for each term of the covariant we must have

$$m\alpha + (m-2)\beta + \dots - (m-2)\beta' - m\alpha' + j - i = 0,$$

i.e. if  $(x, y)$  are considered as being of the weights  $-i, i$  respectively, and  $(a, b, \dots, b', a')$  as being of the weights  $-\frac{1}{2}m, -\frac{1}{2}m+1, \dots, \frac{1}{2}m-1, \frac{1}{2}m$  respectively, then the weight of each term of the covariant is zero.

But if  $(x, y)$  are considered as being of the weights  $1, 0$  respectively, and  $(a, b, \dots, b', a')$  as being of the weights  $0, 1, \dots, m-1, m$  respectively, then writing the equation under the form

$$m(\alpha + \beta + \dots + \beta' + \alpha') + j + i - 2(0 + \beta + \dots + (m-1)\beta' + m\alpha' + i) = 0,$$

and supposing that the covariant is of the order  $\mu$  and of the degree  $\theta$ , each term of the covariant will be of the weight  $\frac{1}{2}(m\theta + \mu)$ .

I shall in the sequel consider the weight as reckoned in the last-mentioned manner. It is convenient to remark, that as regards any function of the coefficients of the degree  $\theta$  and of the weight  $q$ , we have

$$X \cdot Y - Y \cdot X = m\theta - 2q.$$

**30.** Consider now a covariant

$$(A, B, \dots, B', A' \mid x, y)^\mu$$

of the order  $\mu$  and of the degree  $\theta$ ; the covariant is reduced to zero by each of the operations  $X - y\partial_x$ ,  $Y - x\partial_y$ , and we are thus led to the systems of equations

$$XA = 0,$$

$$XB = \mu A,$$

$$XC = (\mu - 1)B,$$

$$\vdots$$

$$XB' = 2C',$$

$$XA' = B'$$



and

$$\begin{aligned} YA &= B, \\ YB &= 2C, \\ &\vdots \\ YC' &= (\mu - 1)B', \\ YB' &= \mu A', \\ YA' &= 0. \end{aligned}$$

Conversely if these equations are satisfied the function will be a covariant.

I assume that  $A$  is a function of the degree  $\theta$  and of the weight  $-\frac{1}{2}(m\theta - \mu)$ , satisfying the condition

$$XA = 0;$$

and I represent by  $YA, Y^2A, Y^3A$ , &c. the results obtained by successive operations with  $Y$  upon the function  $A$ . The function  $Y^sA$  will be of the degree  $\theta$  and of the weight  $\frac{1}{2}(m\theta - \mu) + s$ . And it is clear that in the series of terms  $YA, Y^2A, Y^3A$ , &c., we must at last come to a term which is equal to zero. In fact, since  $m$  is the greatest weight of any coefficient, the weight of  $Y^s$  is at most equal to  $m\theta$ , and therefore if  $\frac{1}{2}(m\theta - \mu) + s > m\theta$ , or  $s > \frac{1}{2}(m\theta + \mu)$ , we must have  $Y^sA = 0$ .

Now writing for greater simplicity  $XY$  instead of  $X \cdot Y$ , and so in similar cases, we have, as regards  $Y^sA$ ,

$$XY - YX = \mu - 2s.$$

Hence  $(XY - YX)A = \mu A$ , and consequently  $XYA = YXA + \mu A = \mu A$ . Similarly  $(XY - YX)YA = (\mu - 2)YA$ , and therefore

$$XY^2A = YXYA + (\mu - 2)YA = \mu YA + (\mu - 2)YA = 2(\mu - 1)YA.$$

And again,  $(XY - YX)Y^2A = (\mu - 4)Y^2A$ , and therefore

$$XY^3A = YXY^2A + (\mu - 4)Y^2A = 2(\mu - 1)Y^2A + (\mu - 4)Y^2A = 3(\mu - 2)Y^2A,$$

or generally

$$XY^sA = s(\mu - s + 1)Y^{s-1}A.$$

Hence putting  $s = \mu + 1, \mu + 2$ , &c., we have

$$\begin{aligned} XY^{\mu+1}A &= 0, \\ XY^{\mu+2}A &= (\mu + 2) \cdot 1 \cdot Y^{\mu+1}A, \\ XY^{\mu+3}A &= (\mu + 3) \cdot 2 \cdot Y^{\mu+2}A, \end{aligned}$$

equations which show that

$$Y^{\mu+1}A = 0;$$

for unless this be so, i.e. if  $Y^{\mu+1}A \neq 0$ , then from the second equation  $XY^{\mu+2}A \neq 0$ , and therefore  $Y^{\mu+2}A \neq 0$ , from the third equation  $XY^{\mu+3}A \neq 0$ , and therefore  $Y^{\mu+3}A \neq 0$ , and so on *ad infinitum*, i.e. we must have  $Y^{\mu+1}A = 0$ .

**31.** The suppositions which have been made as to the function  $A$ , give therefore the equations

$$\begin{aligned} XA &= 0, \\ XYA &= \mu A, \\ XY^2A &= 2(\mu - 1)YA, \end{aligned}$$

and if we now assume  $Y^{\mu+1}A = 0$ ;

$$B = YA, \quad C = \frac{1}{2}YB, \quad \dots \quad A' = \frac{1}{\mu}YB',$$

the system becomes

$$\begin{aligned}XA &= 0, \\XB &= \mu A, \\XC &= (\mu - 1)B, \\&\vdots \\XA' &= B', \\YA' &= 0;\end{aligned}$$

so that the entire system of equations which express that  $(A, B, \dots, B', A' \mid x, y)^\mu$  is a covariant is satisfied; hence

**THEOREM.** Given a quantic  $(a, b, \dots, b', a' \mid x, y)^m$ ; if  $A$  be a function of the coefficients of the degree  $\theta$  and of the weight  $\frac{1}{2}(m\theta - \mu)$  satisfying the condition  $XA = 0$ , and if  $B, C, \dots, B', A'$  are determined by the equations

$$B = YA, \quad C = \frac{1}{2}YB, \quad \dots \quad A' = \frac{1}{\mu}YB',$$

then will

$$(A, B, \dots, B', A' \mid x, y)^\mu$$

be a covariant.

In particular, a function  $A$  of the degree  $\theta$  and of the weight  $\frac{1}{2}m\theta$ , satisfying the condition  $XA = 0$ , will (also satisfy the equation  $YA = 0$  and will) be an invariant.

**32.** I take now for  $A$  the most general function of the coefficients, of the degree  $\theta$  and of the weight  $\frac{1}{2}(m\theta - \mu)$ ; then  $XA$  is a function of the degree  $\theta$  and of the weight  $\frac{1}{2}(m\theta - \mu) - 1$ , and the arbitrary coefficients in the function  $A$  are to be determined so that  $XA = 0$ . The number of arbitrary coefficients is equal to the number of terms in  $A$ , and the number of the equations to be satisfied is equal to the number of terms in  $XA$ ; hence the number of the arbitrary coefficients which remains indeterminate is equal to the number of terms in  $A$  less the number of terms in  $XA$ ; and since the covariant is completely determined when the leading coefficient is known, the difference in question is equal to the number of the aszygetic covariants, i.e. the number of the aszygetic covariants of the order  $\mu$  and the degree  $\theta$  is equal to the number of terms of the degree  $\theta$  and weight  $\frac{1}{2}(m\theta - \mu)$ , less the number of terms of the degree  $\theta$  and weight  $\frac{1}{2}(m\theta - \mu) - 1$ .

**33.** I shall now give some instances of the calculation of covariants by the method just explained. It is very convenient for this purpose to commence by forming the literal parts by Arbogast's Method of Derivations: we thus form tables such as the following:

[Figure: Arbogast tables showing literal parts for  $(a, b, c)$  quantics up to degree 4,  $(a, b, c, d)$  quantics up to degree 4, and  $(a, b, c, d, e)$  quantics up to degree 2; each table arranges products of letters by degree (row) and weight (column).]

**34.** Thus in the case of a cubic  $(a, b, c, d \mid x, y)^3$ , the tables show that there will be a single invariant of the degree 4. Represent this by

$$\begin{aligned}Aa^2d^2 \\+ Babcd \\+ C ac^3 \\+ D b^3d \\+ E b^2c^2,\end{aligned}$$

which is to be operated upon with  $a\partial_b + 2b\partial_c + 3c\partial_d$ . This gives

$$\begin{array}{cccc}+B & +2B & +6A & a^2cd \\ & +6C & & ab^2d \\+3D & +4E & +3B & abc^2 \\+4E & & +3D & b^3c\end{array}$$

i.e.  $B + 6A = 0$ ,  $3D + 2E = 0$ , &c.; or putting  $A = 1$ , we find  $B = -6$ ,  $C = 4$ ,  $D = 4$ ,  $E = -3$ , and the invariant is

$$\begin{aligned} & a^2 d^2 \\ & - 6 abcd \\ & + 4 ac^3 \\ & + 4 b^3 d \\ & - 3 b^2 c^2. \end{aligned}$$

Again, there is a covariant of the order 3 and the degree 3. The coefficient of  $x^3$  or leading coefficient is

$$\begin{aligned} & A a^2 d \\ & + B abc \\ & + C b^3 \end{aligned}$$

which operated upon with  $a\partial_b + 2b\partial_c + 3c\partial_a$ , gives

$$\begin{array}{ccc} +B & +3A & a^2 c \\ +3C & +2B & ab^2 \end{array}$$

i.e.  $B + 3A = 0$ ,  $3C + 2B = 0$ ; or putting  $A = 1$ , we have  $B = -3$ ,  $C = 2$ , and the leading coefficient is

$$\begin{aligned} & a^2 d \\ & - 3 abc \\ & + 2 b^3. \end{aligned}$$

The coefficient of  $x^2 y$  is found by operating upon this with  $\frac{1}{3}(3b\partial_a + 2c\partial_b + d\partial_c)$ , this gives

$$\begin{array}{ccc} +6 & -3 & abd \\ & -6 & ac^2 \\ -9 & +12 & b^2 c \end{array}$$

i.e. the required coefficient of  $x^2 y$  is

$$\begin{aligned} & 3 abd \\ & - 6 ac^2 \\ & + 3 b^2 c; \end{aligned}$$

and by operating upon this with  $\frac{1}{2}(3b\partial_a + 2c\partial_b + d\partial_c)$ , we have for the coefficient of  $xy^2$ ,

$$\begin{array}{ccc} +3 & -6 & acd \\ & +9 & b^2 d \\ -9 & +6 & bc^2 \end{array}$$

i.e. the coefficient of  $xy^2$  is

$$\begin{aligned} & - 3 acd \\ & + 6 b^2 d \\ & - 3 bc^2. \end{aligned}$$

Finally, operating upon this with  $\frac{1}{3}(3b\partial_a + 2c\partial_b + d\partial_c)$ , we have for the coefficient of  $y^3$ ,

$$\begin{array}{ccc} & -1 & ad^2 \\ -3 & +8 & bcd \\ & -2 & \end{array}$$

i.e. the coefficient of  $y^3$  is

$$\begin{aligned} & - ad^2 \\ & + 3bcd \\ & - 2c^3, \end{aligned}$$

and the covariant is

$$\left( \begin{array}{cccc|c} a^2d+1 & abd+3 & acd-3 & ad^2-1 & \\ & ac^2-6 & & & \\ abc-3 & & b^2d+6 & bcd+3 & x, y \\ & b^2c+3 & bc^2-3 & & \\ b^3+2 & & & c^3-2 & \end{array} \right)^3.$$

[I now write the numerical coefficients after instead of before the literal terms.]

I have worked out the example in detail as a specimen of the most convenient method for the actual calculation of more complicated covariants<sup>4</sup>.

**35.** The number of terms of the degree  $\theta$  and of the weight  $q$  is obviously equal to the number of ways in which  $q$  can be made up as a sum of  $\theta$  terms with the elements  $(0, 1, 2, \dots, m)$ , a number which is equal to the coefficient of  $x^q z^\theta$  in the development of

$$\frac{1}{(1-z)(1-xz)(1-x^2z) \cdots (1-x^mz)},$$

and the number of the aszygetic covariants of any particular degree for the quantic  $(* | x, y)^m$  can therefore be determined by means of this development. In the case of a cubic, for example, the function to be developed is

$$\frac{1}{(1-z)(1-xz)(1-x^2z)(1-x^3z)},$$

which is equal to

$$1 + (1 + x + x^2 + x^3)z + (1 + x + 2x^2 + 2x^3 + 2x^4 + x^5 + x^6)z^2 + \&c.,$$

where the coefficients are given by the following table; on account of the symmetry, the series of coefficients for each power of  $z$  is continued only to the middle term or middle of the series.

[Figure: Triangular table of coefficients of the expansion, with rows labelled  $(0), (1), \dots, (6)$  for successive powers of  $z$ , and entries showing the coefficients of  $x^j$  in the corresponding row, mirrored about the middle.]

<sup>4</sup>Note added Feb. 7, 1856.—The following method for the calculation of an invariant or of the leading coefficient of a covariant, is easily seen to be identical in principle with that given in the text. Write down all the terms of the weight next inferior to that of the invariant or leading coefficient, and operate on each of these separately with the symbol

$$\text{ind. } b \cdot \frac{a}{b} + 2 \text{ ind. } c \cdot \frac{b}{c} + \cdots + (m-1) \text{ ind. } b' \cdot \frac{b'}{a'},$$

where we are first to multiply by the fraction, rejecting negative powers, and then by the index of the proper letter in the term so obtained. Equating the results to zero, we obtain equations between the terms of the invariant or leading coefficient, and replacing in these equations each term by its numerical coefficient in the invariant or leading coefficient, we have the equations of connexion of these numerical coefficients. Thus, for the discriminant of a cubic, the terms of the next inferior weight are  $a^2cd, ab^2d, abc^2, b^3c$ , and operating on each of these separately with the symbol

$$\text{ind. } b \cdot \frac{a}{b} + 2 \text{ ind. } c \cdot \frac{b}{c} + 3 \text{ ind. } d \cdot \frac{c}{d},$$

we find

$$\begin{array}{rcl} abcd & +4abcd & +6a^2d^2 \\ & +6ac^3 & \\ 4b^3c & +4b^3c & +4abcd \\ 2b^3c & & +4b^3d \end{array}$$

and equating the horizontal lines to zero, and assuming  $a^2d^2 = 1$ , we have  $a^2d^2 = 1, abcd = -6, ac^3 = 4, b^3d = 4, b^2c^2 = -3$ , or the value of the discriminant is that given in the text.

**36.** and from this, by subtracting from each coefficient the coefficient which immediately precedes it, we form the table

[Figure: Triangular difference table derived from the previous one, with rows labelled (0), (1), ..., (6); entries are the successive differences.]

The successive lines fix the number and character of the covariants of the degrees 0, 1, 2, 3, &c. The line (0), if this were to be interpreted, would show that there is a single covariant of the degree 0; this covariant is of course merely the absolute constant unity, and may be excluded. The line (1) shows that there is a single covariant of the degree 1, viz. a covariant of the order 3; this is the cubic itself, which I represent by  $U$ . The line (2) shows that there are two asyzygetic covariants of the degree 2, viz. one of the order 6, this is merely  $U^2$ , and one of the order 2, this I represent by  $H$ . The line (3) shows that there are three asyzygetic covariants of the degree 3, viz. one of the order 9, this is  $U^3$ ; one of the order 5, this is  $UH$ , and one of the order 3, this I represent by  $\Phi$ . The line (4) shows that there are five asyzygetic covariants of the degree 4, viz. one of the order 12, this is  $U^4$ ; one of the order 8, this is  $U^2H$ ; one of the order 6, this is  $H^2$ ; and one of the order 0, i.e. an invariant, this I represent by  $\nabla$ . The line (5) shows that there are six asyzygetic covariants of the degree 5, viz. one of the order 15, this is  $U^5$ ; one of the order 11, this is  $U^3H$ ; one of the order 9, this is  $U^2\Phi$ ; one of the order 7, this is  $UH^2$ ; one of the order 5, this is  $H\Phi$ ; and one of the order 3, this is  $U\nabla$ . The line (6) shows that there are 8 asyzygetic covariants of the degree 6, viz. one of the order 18, this is  $U^6$ ; one of the order 14, this is  $U^4H$ ; one of the order 12, this is  $U^3\Phi$ ; one of the order 10, this is  $U^2H^2$ ; one of the order 8, this is  $UH\Phi$ ; two of the order 6 (i.e. the three covariants  $H^3$ ,  $\Phi^2$  and  $\nabla U^2$  are not asyzygetic, but are connected by a single linear equation or syzygy), and one of the order 2, this is  $\nabla H$ . We are thus led to the irreducible covariants  $U, H, \Phi, \nabla$  connected by a linear equation or syzygy between  $H^3$ ,  $\Phi^2$  and  $\nabla U^2$ , and this is in fact the complete system of irreducible covariants;  $\nabla$  is therefore the only invariant.

**36.** The asyzygetic covariants are of the form  $U^p H^q \nabla^r$ , or else of the form  $\Phi U^p H^q \nabla^r$ ; and since  $U, H, \nabla$  are of the degrees 1, 2, 4 respectively, and  $\Phi$  is of the degree 3, the number of asyzygetic covariants of the degree  $m$  of the first form is equal to the coefficient of  $x^m$  in  $1 \div (1-x)(1-x^2)(1-x^4)$ , and the number of the asyzygetic covariants of the degree  $m$  of the second form is equal to the coefficient of  $x^m$  in  $x^3 \div (1-x)(1-x^2)(1-x^4)$ . Hence the total number of asyzygetic covariants is equal to the coefficient of  $x^m$  in  $(1+x^3) \div (1-x)(1-x^2)(1-x^4)$ , or what is the same thing, in

$$\frac{1}{(1-x)(1-x^2)(1-x^3)(1-x^4)},$$

and conversely, if this expression for the number of the asyzygetic covariants of the degree  $m$  were established independently, it would follow that the irreducible invariants were four in number, and of the degrees 1, 2, 3, 4 respectively, but connected by an equation of the degree 6. As regards the invariants, every invariant is of the form  $\nabla^p$ , i.e. the number of asyzygetic invariants of the degree  $m$  is equal to the coefficient of  $x^m$  in  $\frac{1}{1-x^4}$ , and conversely, from this expression it would follow that there was a single irreducible invariant of the degree 4.

**37.** In the case of a quartic, the function to be developed is:

$$\frac{1}{(1-z)(1-xz)(1-x^2z)(1-x^3z)(1-x^4z)},$$

and the coefficients are given by the table.

[Figure: Triangular table of coefficients of the expansion for the quartic, with rows labelled (0), (1), ..., (6); entries on each row are the coefficients of  $x^j$ , mirrored about the middle.]

and subtracting from each coefficient the coefficient immediately preceding it, we have the table

[Figure: Triangular difference table derived from the previous one, with rows labelled (0), (1), ..., (6).]

the examination of which will show that we have for the quartic the following irreducible covariants, viz. the quartic itself  $U$ ; an invariant of the degree 2, which I represent by  $I$ ; a covariant of the order 4 and of the degree 2, which I represent by  $H$ ; an invariant of the degree 3, which I represent by  $J$ ; and a covariant of the order 6 and the degree 3, which I represent by  $\Phi$ ; but that the irreducible covariants are connected by an equation of the degree 6, viz. there is a linear equation or syzygy between  $\Phi^2$ ,  $I^2 H^2$ ,  $I J H^2 U$ ,  $J H^2 U^2$  and  $J^2 U^2$ ; this is in fact the complete system of the irreducible covariants of the quartic: the only irreducible invariants are the invariants  $I, J$ .

**38.** The aszygetic covariants are of the form  $U^p I^q H^r J^s$ , or else of the form  $\Phi U^p I^q H^r J^s$ , and the number of the aszygetic covariants of the degree  $m$  is equal to the coefficient of  $x^m$  in  $(1 + x^3) \div (1 - x)(1 - x^2)^2(1 - x^3)$ , or what is the same thing, in

$$\frac{1 - x^6}{(1 - x)(1 - x^2)^2(1 - x^3)^2},$$

and the aszygetic invariants are of the form  $I^p J^q$ , and the number of the aszygetic invariants of the degree  $m$  is equal to the coefficient of  $x^m$  in  $1 \div (1 - x^2)(1 - x^3)$ . Conversely, if these formulae were established, the preceding results as to the form of the system of the irreducible covariants or of the irreducible invariants, would follow.

**39.** In the case of a quintic, the function to be developed is

$$\frac{1}{(1 - z)(1 - xz)(1 - x^2z)(1 - x^3z)(1 - x^4z)(1 - x^5z)},$$

and the coefficients are given by the table

[Figure: Triangular table of coefficients of the expansion for the quintic, with rows labelled (0), (1), ..., (5).]

and subtracting from each coefficient the one which immediately precedes it, we have the table

[Figure: Triangular difference table for the quintic, rows labelled (0), (1), ..., (5).]

We thus obtain the following irreducible covariants, viz.:

Of the degree 1; a single covariant of the order 5, this is the quintic itself.

Of the degree 2; two covariants, viz. one of the order 6, and one of the order 2.

Of the degree 3; three covariants, viz. one of the order 9, one of the order 5, and one of the order 3.

Of the degree 4; three covariants, viz. one of the order 6, one of the order 4, and one of the order 0 (an invariant).

Of the degree 5; three covariants, viz. one of the order 7, one of the order 3, and one of the order 1 (a linear covariant).

These covariants are connected by a single syzygy of the degree 5 and of the order 11; in fact, the table shows that there are only two aszygetic covariants of this degree and order; but we may, with the above-mentioned irreducible covariants of the degrees, 1, 2, 3 and 4, form three covariants of the degree 5 and the order 11; there is therefore a syzygy of this degree and order.

**40.** I represent the number of ways in which  $q$  can be made up as a sum of  $m$  terms with the elements  $0, 1, 2, \dots, m$ , each element being repeatable an indefinite number of times by the notation

$$P(0, 1, 2, \dots, m)^\theta q,$$

and I write for shortness

$$P'(0, 1, 2, \dots, m)^\theta q = P(0, 1, 2, \dots, m)^\theta q - P(0, 1, 2, \dots, m)^\theta (q - 1).$$

Then for a quantic of the order  $m$ , the number of aszygetic covariants of the degree  $\theta$  and of the order  $\mu$  is

$$P'(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - \mu).$$

In particular, the number of aszygetic invariants of the degree  $\theta$  is

$$P'(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta.$$

To find the total number of the aszygetic covariants of the degree  $\theta$ , suppose first that  $m\theta$  is even; then, giving to  $\mu$  the successive values  $0, 2, 4, \dots, m\theta$ , the required number is

$$\begin{aligned} & P(\tfrac{1}{2}m\theta) - P(\tfrac{1}{2}m\theta - 1) \\ & + P(\tfrac{1}{2}m\theta - 1) - P(\tfrac{1}{2}m\theta - 2) \\ & \vdots \\ & + P(2) - P(1) \\ & + P(1) \\ & = P(\tfrac{1}{2}m\theta), \end{aligned}$$

i.e. when  $m\theta$  is even, the number of the aszygetic covariants of the degree  $\theta$  is

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta;$$

and similarly, when  $m\theta$  is odd, the number of the aszygetic covariants of the degree  $\theta$  is

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - 1).$$

But the two formulae may be united into a single formula; for when  $m\theta$  is odd  $\frac{1}{2}m\theta$  is a fraction, and therefore  $P(\frac{1}{2}m\theta)$  vanishes, and so when  $m\theta$  is even  $\frac{1}{2}(m\theta - 1)$  is a fraction, and  $P(\frac{1}{2}(m\theta - 1))$  vanishes; we have thus the theorem, that for a quantic of the order  $m$ :

The number of the aszygetic covariants of the degree  $\theta$  is

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta + P(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - 1).$$

41. The functions  $P(\frac{1}{2}m\theta)$ , &c. may, by the method explained in my “Researches on the Partition of Numbers,” [140], be determined as the coefficients of  $x^\theta$  in certain functions of  $x$ ; I have calculated the following particular cases:

Putting, for shortness,

$$P'(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta = \text{coefficient } x^\theta \text{ in } \phi m,$$

then

$$\begin{aligned}\phi 2 &= \frac{1}{1 - x^2}, \\ \phi 3 &= \frac{1}{1 - x^4}, \\ \phi 4 &= \frac{1}{(1 - x^2)(1 - x^3)}, \\ \phi 5 &= \frac{1 - x^8 + x^{18}}{(1 - x^4)(1 - x^8)(1 - x^{12})}, \\ \phi 6 &= \frac{(1 - x)(1 + x - x^2 - x^4 - x^5 + x^7 + x^8)}{(1 - x^2)^2(1 - x^3)(1 - x^4)(1 - x^5)}, \\ \phi 7 &= \frac{1 - x^4 + 2x^8 - x^{10} + 5x^{12} + 2x^{14} + 6x^{16} + 2x^{18} + 5x^{20} - x^{22} + 2x^{24} - x^{28} + x^{32}}{(1 - x^4)(1 - x^8)(1 - x^{12})(1 - x^{14})(1 - x^{18})}, \\ \phi 8 &= \frac{(1 - x)(1 + x - x^2 - x^4 + x^7 + x^8 + x^9 + x^{10} - x^{11} + x^{13} + x^{14})}{(1 - x^2)^2(1 - x^3)^2(1 - x^4)(1 - x^5)(1 - x^6)},\end{aligned}$$

and putting

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta = \text{coefficient of } x^\theta \text{ in } \psi m,$$

then

$$\begin{aligned}\psi 2 &= \frac{1}{(1 - x)(1 - x^2)}, \\ \psi 3 &= \frac{1 + x^3}{(1 - x^2)(1 - x^4)}, \\ \psi 4 &= \frac{1 - x + x^2}{(1 - x)^2(1 - x^2)(1 - x^3)}, \\ \psi 5 &= \frac{1 + x^4 + 6x^6 + 9x^7 + 12x^8 + 9x^9 + 6x^{10} + x^{12} + x^{16}}{(1 - x^2)^2(1 - x^4)(1 - x^6)(1 - x^8)},\end{aligned}$$

and putting

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - 1) = \text{coefficient of } x^\theta \text{ in } \chi m,$$

then

$$\chi^3 = \frac{x + x^3}{(1 - x^2)^2(1 - x^3)},$$

$$\chi^5 = \frac{x + 4x^2 + 8x^3 + 10x^4 + 10x^5 + 8x^6 + 4x^7 + x^8}{(1 - x^2)^2(1 - x^4)(1 - x^6)(1 - x^8)}.$$

And from what has preceded, it appears that for a quantic of the order  $m$ , the number of asyzygetic covariants of the degree  $\theta$  is for  $m$  even, coefficient  $x^\theta$  in  $\psi m$ , and for  $m$  odd, coefficient  $x^\theta$  in  $(\psi m + \chi m)$ ; and that the number of asyzygetic invariants of the degree  $\theta$  is coefficient  $x^\theta$  in  $\phi m$ . Attending first to the invariants:

**42.** For a quadric, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } \frac{1}{1 - x^2},$$

which leads to the conclusion that there is a single irreducible invariant of the degree 2.

**43.** For a cubic, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } \frac{1}{1 - x^4},$$

i.e. there is a single irreducible invariant of the degree 4.

**44.** For a quartic, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } \frac{1}{(1 - x^2)(1 - x^3)},$$

i.e. there are two irreducible invariants of the degrees 2 and 3 respectively.

**45.** For a quintic, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } \frac{1 - x^8 + x^{18}}{(1 - x^4)(1 - x^8)(1 - x^{12})}.$$

The numerator is the irreducible factor of  $1 - x^{36}$ , i.e. it is equal to  $(1 - x^{36})(1 - x^4) \div (1 - x^{12})(1 - x^9)$ ; and substituting this value, the number becomes

$$\text{coefficient } x^\theta \text{ in } \frac{1 - x^{36}}{(1 - x^4)(1 - x^8)(1 - x^{12})(1 - x^{18})},$$

i.e. there are in all four irreducible invariants, which are of the degrees 4, 8, 12 and 18 respectively; but these are connected by an equation of the degree 36, i.e. the square of the invariant of the degree 18 is a rational and integral function of the other three invariants; a result, the discovery of which is due to M. Hermite.

**46.** For a sextic, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } \frac{(1 - x)(1 + x - x^2 - x^4 - x^5 + x^7 + x^8)}{(1 - x^2)^2(1 - x^3)(1 - x^4)(1 - x^5)};$$

the second factor of the numerator is the irreducible factor  $1 - x^{30}$ , i.e. it is equal to  $(1 - x^{30})(1 - x^5)(1 - x^3)(1 - x^2) \div (1 - x^{15})(1 - x^{10})(1 - x^6)(1 - x)$ ; and substituting this value, the number becomes

$$\text{coefficient } x^\theta \text{ in } \frac{1 - x^{30}}{(1 - x^2)(1 - x^4)(1 - x^6)(1 - x^{10})(1 - x^{15})},$$

i.e. there are in all five irreducible invariants, which are of the degrees 2, 4, 6, 10 and 15 respectively; but these are connected by an equation of the degree 30, i.e. the square of the invariant of the degree 15 is a rational and integral function of the other four invariants.

**47.** For a septic, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } \frac{1 - x^4 + 2x^8 - x^{10} + 5x^{12} + 2x^{14} + 6x^{16} + 2x^{18} + 5x^{20} - x^{22} + 2x^{24} - x^{28} + x^{32}}{(1 - x^4)(1 - x^8)(1 - x^{12})(1 - x^{14})(1 - x^{18})};$$



the numerator is equal to

$$(1 - x^4)(1 - x^8)^{-1}(1 - x^{10})(1 - x^{12})^{-1}(1 - x^{14})^{-1} \dots,$$

where the series of factors does not terminate; hence [incorrect, see p. 253] the number of irreducible invariants is infinite; substituting the preceding value, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } (1 - x^4)^{-1}(1 - x^8)^{-3}(1 - x^{12})^{-6}(1 - x^{14})^{-4} \dots.$$

The first four indices give the number of irreducible invariants of the corresponding degrees, i.e. there are 1, 3, 6 and 4 irreducible invariants of the degrees 4, 8, 12 and 14 respectively, but there is no reason to believe that the same thing holds with respect to the indices of the subsequent terms. To verify this it is to be remarked, that there are 1, 4, 10 and 4 asyzygetic invariants of the degrees in question respectively; there is therefore one irreducible invariant of the degree 4; calling this  $X_4$ , there is only one composite invariant of the degree 8, viz.  $X_4^2$ ; there are therefore three irreducible invariants of this degree, say  $X_8, X'_8, X''_8$ . The composite invariants of the degree 12 are four in number, viz.  $X_4^3, X_4X_8, X_4X'_8, X_4X''_8$ , and these cannot be connected by any syzygy, for if they were so,  $X_4^2, X_8, X'_8, X''_8$  would be connected by a syzygy, or there would be less than 3 irreducible invariants of the degree 8. Hence there are precisely 6 irreducible invariants of the degree 12. And since the irreducible invariants of the degrees 4, 8 and 12 do not give rise to any composite invariant of the degree 14, there are precisely 4 irreducible invariants of the degree 14.

**48.** For an octavic, the number of the asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } \frac{(1 - x)(1 + x - x^2 - x^4 + x^7 + x^8 + x^9 + x^{10} - x^{11} + x^{13} + x^{14})}{(1 - x^2)^2(1 - x^3)^2(1 - x^4)(1 - x^5)(1 - x^6)},$$

and the second factor of the numerator is

$$(1 - x)^{-1}(1 - x^2)(1 - x^3)^{-1}(1 - x^4)^{-1}(1 - x^5)^{-1}(1 - x^6)^{-1}(1 - x^{10})^{-1}(1 - x^{12})(1 - x^{14})(1 - x^{15}) \dots,$$

where the series of factors does not terminate, hence [incorrect] the number of irreducible invariants is infinite. Substituting the preceding value, the number of the asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } (1 - x^2)^{-1}(1 - x^3)^{-1}(1 - x^4)^{-1}(1 - x^5)^{-1}(1 - x^6)^{-1}(1 - x^7)^{-2}(1 - x^8)^{-3}(1 - x^9)^{-5}(1 - x^{10})^{-7} \dots.$$

There is certainly one, and only one irreducible invariant for each of the degrees 2, 3, 4, 5 and 6 respectively; but the formula does not show the number of the irreducible invariants of the degrees 7, &c.; in fact, representing the irreducible invariants of the degrees 2, 3, 4, 5 and 6 by  $X_2, X_3, X_4, X_5, X_6$ , these give rise to 3 composite invariants of the degree 7, viz.  $X_2X_2X_3, X_3X_4, X_2X_5$ , which may or may not be connected by a syzygy; if they are not connected by a syzygy, there will be a single irreducible invariant of the degree 7; but if they are connected by a syzygy, there will be two irreducible invariants of the degree 7; it is useless at present to pursue the discussion further.

Considering next the covariants,

**49.** For a quadric, the number of asyzygetic covariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } \frac{1}{(1 - x)(1 - x^2)},$$

i.e. there are two irreducible covariants of the degrees 1 and 2 respectively; these are of course the quadric itself and the invariant.

**50.** For a cubic, the number of the asyzygetic covariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } \frac{(1 + x)(1 + x^3)}{(1 - x^2)^2(1 - x^4)}.$$

The first factor of the numerator is the irreducible factor of  $1 - x^2 = (1 - x^2) \div (1 - x)$ , and the second factor of the numerator is the irreducible factor of  $1 - x^6 = (1 - x^6) \div (1 - x^2)$ ; substituting these values, the number is

$$\text{coefficient } x^\theta \text{ in } \frac{1 - x^6}{(1 - x)(1 - x^2)(1 - x^3)(1 - x^4)},$$

i.e. there are 4 irreducible covariants of the degrees 1, 2, 3, 4 respectively; but these are connected by an equation of the degree 6; the covariant of the degree 1 is the cubic itself  $U$ , the other covariants are the covariants already spoken of and represented by the letters  $H, \Phi$  and  $\nabla$  respectively ( $H$  is of the degree 2 and the order 3,  $\Phi$  of the degree 3 and the order 3, and  $\nabla$  is of the degree 4 and the order 0, i.e. it is an invariant).

**51.** For a quartic, the number of the aszygetic covariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } \frac{1 - x + x^2}{(1 - x)^2(1 - x^2)(1 - x^3)};$$

the numerator of which is the irreducible factor of  $1 - x^6$ , i.e. it is equal to  $(1 - x^6)(1 - x) \div (1 - x^2)(1 - x^3)$ . Making this substitution, the number is

$$\text{coefficient } x^\theta \text{ in } \frac{1 - x^6}{(1 - x)(1 - x^2)^2(1 - x^3)^2},$$

i.e. there are five irreducible covariants, one of the degree 1, two of the degree 2, and two of the degree 3, but these are connected by an equation of the degree 6. The irreducible covariant of the degree 1 is of course the quartic itself  $U$ , the other irreducible covariants are those already spoken of and represented by  $I, H, J, \Phi$  respectively ( $I$  is of the degree 2 and the order 0, and  $J$  is of the degree 3 and the order 0, i.e.  $I$  and  $J$  are invariants,  $H$  is of the degree 2 and the order 4,  $\Phi$  is of the degree 3 and the order 6).

**52.** For a quintic, the number of irreducible covariants of the degree  $\theta$  is

$$\text{coefficient } x^\theta \text{ in } \frac{1 + x + x^2 + 4x^3 + 6x^4 + 8x^5 + 9x^6 + 10x^7 + 12x^8 + 10x^9 + 9x^{10} + 8x^{11} + 6x^{12} + 4x^{13} + x^{14} + x^{15} + x^{16}}{(1 - x^2)^2(1 - x^4)(1 - x^6)(1 - x^8)};$$

the numerator of which is

$$(1 + x)^2(1 + x - x^2 - x^4 + 2x^5 + 3x^6 + 5x^7 + 4x^8 + 3x^9 + 2x^{10} - x^{12} - x^{13} + x^{14}).$$

The first factor is  $(1 - x^2)^2 \div (1 - x)^2$ , the second factor is

$$(1 - x^2)^3(1 - x^3)^2(1 - x^4)^3(1 - x^5)^4(1 - x^7)^{-5}(1 - x^8)^{-7} \dots,$$

which does not terminate; hence [incorrect] the number of irreducible covariants is infinite. Substituting the preceding values, the expression for the number of the aszygetic covariants of the degree  $\theta$  is

$$\text{coeff. } x^\theta \text{ in } (1 - x)^{-1}(1 - x^2)^{-2}(1 - x^3)^{-3}(1 - x^4)^{-3}(1 - x^5)^{-3}(1 - x^6)^{-4}(1 - x^7)^{-5}(1 - x^8)^{-7} \dots,$$

which agrees with a previous result: the numbers of irreducible covariants for the degrees 1, 2, 3, 4 are 1, 2, 3 and 3 respectively, and for the degree 5, the number of irreducible covariants is three, but there is one syzygy between the composite covariants of the degree in question; the difference  $3 - 1 = 2$  is the index taken with its sign reversed of the factor  $(1 - x^5)^{-3}$ .

**53.** I consider a system of the aszygetic covariants of any particular degree and order of a given quantic, the system may of course be replaced by a system the terms of which are any linear functions of those of the original system, and it is necessary to inquire what covariants ought to be selected as most proper to represent the system of aszygetic covariants; the following considerations seem to me to furnish a convenient rule of selection. Let the literal parts of the terms which enter into the coefficients of the highest power of  $x$  or leading coefficients be represented by  $M_\alpha, M_\beta, M_\gamma, \dots$  these quantities being arranged in the natural or alphabetical order; the first in order of these quantities  $M$ , which enters into the leading coefficient of a particular covariant, may for shortness be called the leading term of such covariant, and a covariant the leading term of which is posterior in order to the leading term of another covariant, may be said to have a lower leading term.

It is clear, that by properly determining the multipliers of the linear functions we may form a covariant the leading term of which is lower than the leading term of any other covariant (the definition implies that there is but one such covariant); call this  $\Theta_1$ . We may in like manner form a covariant such that its leading term is lower than the leading term of every other covariant except  $\Theta_1$ ; or rather we may form a system of such covariants, since if  $\Theta_2$  be a covariant having the property in question,  $\Theta_2 + k\Theta_1$  will have the same property, but  $k$  may be determined so that the covariant shall not contain the leading term of  $\Theta_1$ , i.e. we

may form a covariant  $\Theta_2$  such that its leading term is lower than the leading term of every other covariant excepting  $\Theta_1$ , and that the leading term of  $\Theta_1$  does not enter into  $\Theta_2$ ; and there is but one such covariant,  $\Theta_2$ . Again, we may form a covariant  $\Theta_3$  such that its leading term is lower than the leading term of every other covariant excepting  $\Theta_1$  and  $\Theta_2$ , and that the leading terms of  $\Theta_1$  and  $\Theta_2$  do not either of them enter into  $\Theta_3$ ; and there is but one such covariant,  $\Theta_3$ . And so on, until we arrive at a covariant the leading term of which is higher than the leading terms of the other covariants, and which does not contain the leading terms of the other covariants. We have thus a series of covariants  $\Theta_1, \Theta_2, \Theta_3, \&c.$  containing the proper number of terms, and which covariants may be taken to represent the aszygetic covariants of the degree and order in question.

In order to render the covariants  $\Theta$  definite as well numerically as in regard to sign, we may suppose that the covariant is in its least terms (i.e. we may reject numerical factors common to all the terms), and we may make the leading term positive. The leading term with the proper numerical coefficient (if different from unity) and with the proper power of  $x$  (or the order of the function) annexed, will, when the covariants of a quantic are tabulated, be sufficient to indicate, without any ambiguity whatever, the particular covariant referred to. I subjoin a table of the covariants of a quadric, a cubic and a quartic, and of the covariants of the degrees 1, 2, 3, 4 and 5 respectively of a quintic, and also two other invariants of a quintic.

[Except for the quantic itself, the algebraical sum of the numerical coefficients in any column is  $= 0$ , viz. the sum of the coefficients with the sign  $+$  is equal to that of the coefficients with the sign  $-$ , and I have as a numerical verification inserted at the foot of each column this sum with the sign  $\pm$ ].

### Covariant Tables (Nos. 1 to 26).

**No. 1.**  $(a + 1, b + 2, c + 1 \mid x, y)^2$ .

**No. 2.**  $(ac + 1, b^2 - 1) \pm 1$ .

The tables Nos. 1 and 2 are the covariants of a binary quadric. No. 1 is the quadric itself; No. 2 is the quadricinvariant, which is also the discriminant.

**No. 3.**  $(a + 1, b + 3, c + 3, d + 1 \mid x, y)^3$ .

**No. 4.**  $\begin{pmatrix} ac + 1 & ad + 1 & bd + 1 \\ b^2 - 1 & bc - 1 & c^2 - 1 \end{pmatrix} \mid x, y \pm 1, \pm 1, \pm 1$ .

**No. 5.**  $\begin{pmatrix} a^2d + 1 & abd + 3 & acd - 3 & ad^2 - 1 \\ abc - 3 & ac^2 - 6 & b^2d + 6 & bcd + 3 \\ b^3 + 2 & b^2c + 3 & bc^2 - 3 & c^3 - 2 \end{pmatrix} \mid x, y \pm 3, \pm 6, \pm 6, \pm 3$ .

**No. 6.**  $\frac{a^2d^2 + 1}{abcd - 6} \pm 7$

The tables Nos. 3, 4, and 5 are the covariants of a binary cubic. No. 3 is the cubic itself; No. 4 is the quadricovariant, or Hessian; No. 5 is the cubicovariant; No. 6 is the invariant, or discriminant. And if we write No. 3 =  $U$ , No. 4 =  $H$ , No. 5 =  $\Phi$ , No. 6 =  $\nabla$ , then identically,

$$\Phi^2 - \nabla U^2 + 4H^3 = 0.$$

**No. 7.**  $(a + 1, b + 4, c + 6, d + 4, e + 1 \mid x, y)^4$ .

**No. 8.**  $\begin{pmatrix} ae + 1 \\ bd - 4 \\ c^2 + 3 \end{pmatrix} \pm 4$ .

**No. 9.**  $\begin{pmatrix} ac + 1 & ad + 1 & ae + 1 & bd + 1 \\ & & bd + 2 & be + 2 \\ b^2 - 1 & bc - 2 & c^2 - 2 & cd - 2 \\ \pm 1 & \pm 2 & \pm 3 & \pm 2 \end{pmatrix} + (be + 2, ce - 1)$ .

[Tables 10, 11, 12 follow with analogous formats: cubinvariant, cubicovariant, and discriminant of the quartic. The discriminant (No. 12) has 18 terms with sum  $\pm 442$ .]

The tables Nos. 7, 8, 9, 10 and 11 are the irreducible covariants of a quartic. No. 7 is the quartic itself; No. 8 is the quadrinvariant; No. 9 is the quadricovariant, or Hessian; No. 10 is the cubinvariant; and No. 11 is the cubicovariant. The table No. 12 is the discriminant. And if we write No. 7 =  $U$ , No. 8 =  $I$ , No. 9 =  $H$ , No. 10 =  $J$ , No. 11 =  $\Phi$ , No. 12 =  $\nabla$ , then the irreducible covariants are connected by the identical equation

$$JU^3 - IU^2H + 4H^3 + \Phi^2 = 0,$$

and we have

$$\nabla = I^3 - 27J^2.$$

[The Tables Nos. 13 to 24 which follow, and also Nos. 25 and 26 which are given in 143 relate to the binary quintic. I have inserted in the headings the capital letters  $A, B, \dots, L$  and also  $Q$  and  $Q'$  by which I refer to these covariants of the quintic.  $A$  is the quintic itself,  $C$  is the Hessian,  $G$  is the quartinvariant,  $J$  a linear covariant;  $Q$  is the simplest octinvariant, and  $Q'$  is the discriminant. As noticed in the original memoir we have  $AI + BF - CE = 0$ ; and  $Q' = G^2 - 128Q$ , only the coefficient 128 was by mistake given as 1152.]

**A. No. 13.**  $(a + 1, b + 5, c + 10, d + 10, e + 5, f + 1 \mid x, y)^5$ .

**B. No. 14.**  $\left( \begin{array}{ccc|c} ae + 1 & af + 1 & bf + 1 & \\ bd - 4 & be - 3 & ce - 4 & \\ c^2 + 3 & cd + 2 & d^2 + 3 & \\ \pm 4 & \pm 3 & \pm 4 & \end{array} \mid x, y \right)^2$ .

**C. No. 15.**  $\left( \begin{array}{cccccc|c} ac + 1 & ad + 3 & ae + 3 & af + 1 & bf + 3 & cf + 3 & df + 1 & \\ & bd + 3 & bd + 7 & & ce + 3 & de - 3 & & \\ b^2 - 1 & bc - 3 & be - 6 & cd - 8 & & & c^2 - 1 & \\ & & c^2 - 6 & d^2 - 6 & & & & \\ \pm 1 & \pm 3 & \pm 6 & \pm 8 & \pm 6 & \pm 3 & \pm 1 & \end{array} \mid x, y \right)^6$ .

[Tables D (No. 16), E (No. 17), F (No. 18), G (No. 19), H (No. 20), I (No. 21), J (No. 22), K (No. 23), L (No. 24) follow with the full literal expansions. The detailed numerical tables for these covariants of the binary quintic are extensive; the column sums (with sign  $\pm$ ) shown at the bottom of each column serve as numerical verification. Several of the larger tables involve products of  $a, b, \dots, f$  of degree up to 5 and weight up to 16.]

**No. 25.**  $Q = +1 a^2 c d f + \&c.$  [see post, 143.]

**No. 26.**  $Q' = +1 a^4 f^4 + \&c.$

## 142.

### NUMERICAL TABLES SUPPLEMENTARY TO SECOND MEMOIR ON QUANTICS.

[Now first published (1889).]

In the present paper I arrange in a more compendious form and continue to a much greater extent the tables (first of each pair) given Nos. 35–39 of my Second Memoir on Quantics, 141, pp. 260–264, which relate to the cubic, the quartic and the quintic functions; and I give the like tables for the sextic, the septic and the octavic functions respectively. The cubic table exhibits the coefficients of the several terms of the function  $1 \div (1 - z \cdot 1 - xz \cdot 1 - x^2z \cdot 1 - x^3z)$ , or, what is the same thing, it gives the number of partitions of a given number into a given number of parts, the parts being 0, 1, 2, 3 (repetitions admissible): or again, regarding the letters  $a, b, c, d$ , as having the weights 0, 1, 2, 3 respectively, it shows the number of literal terms of a given degree and given weight. And similarly for the quartic, quintic, sextic, septic and octavic tables respectively, the parts of course being 0, 1,  $\dots$  up to 4, 5, 6, 7 or 8, and the letters being  $a, b, \dots$  up to  $e, f, g, h$  or  $i$ . The extent of the tables is as follows:

|             |                       |       |
|-------------|-----------------------|-------|
| cubic table | extends to deg-weight | 18–27 |
| quartic     | „                     | 18–36 |
| quintic     | „                     | 18–45 |
| sextic      | „                     | 16–45 |
| septic      | „                     | 12–42 |
| octavic     | „                     | 10–40 |

viz. for the quintic, the sextic and the octavic functions these are the deg-weights of the highest invariants respectively. I designate the Tables as the *ad*-, *ae*-, *af*-, *ag*-, *ah*- and *ai*-tables respectively.

It is to be noticed that in the several tables the lower part of each column is for shortness omitted; the column has to be completed by taking into it the series of bottom terms of each of the preceding columns: thus in the *af*- or quintic table the complete column for degree 3 would be

| W  | D 3 |
|----|-----|
| –0 | 6   |
| –1 | 6   |
| –2 | 5   |
| –3 | 4   |
| –4 | 3   |
| –5 | 2   |
| –6 | 1   |
| –7 | 1   |
| –  | 2   |
| –  | 1   |
| –  | 1   |

where the concluding terms 2, 1, 1 are the bottom terms of the three preceding columns respectively. And the meaning is that for degree 3, and weight 8, or 7, the number of terms is = 6; for weight 7 – 1, = 6, the number of terms is = 6; and similarly for weights 5, 4, 3, 2, 1, the numbers are 5, 4, 3, 2, 1, 1; the numbers are those of the terms

[Figure: Eight-column array showing the triangle of literal terms (for the quintic) in *a, b, c, d, e, f* broken into columns by weight. Each column lists the products  $a^3, a^2b, a^2c, \dots$  down to terms in *f*, with the count of terms equal to 6, 6, 5, 4, 3, 2, 1, 1 for weights 8, 7,  $\dots$ , 1 respectively.]

The like remarks and explanations apply to the other tables.

**ad-TABLE.**

[Figure: Triangular numerical table giving the coefficients  $a_{D,W}$  for the cubic, with columns labelled by degree  $D = 0, 1, 2, \dots, 18$  and rows labelled by weight  $W$  from 0 down to –9; the displayed entries reach weight –27. The first row (column  $a = 1$ ) reads 1, 3, 6, 4, 6, 8, 7, 9, 11, 10, 13, 14, 13, 16, 17, 16, 18, 20, 19, 21, 23, 22, 24, 26, 25, 27.]

**ae-TABLE.**

[Figure: Triangular numerical table for the quartic, with columns labelled by degree  $D = 0, 1, 2, \dots, 18$  and rows labelled by weight  $W$  from 0 down to –18; entries reach weight –36. Each cell shows the count of literal terms of the indicated degree and weight.]

**af-TABLE.**

[Figure: Triangular numerical table for the quintic, columns labelled by degree  $D = 0, 1, 2, \dots, 18$  and rows labelled by weight  $W$  from 0 down to –27; entries reach weight –45.]

**ag-TABLE.**

[Figure: Triangular numerical table for the sextic, columns labelled by degree  $D = 0, 1, 2, \dots, 15$  and rows labelled by weight  $W$  from 0 down to  $-30$ ; entries reach weight  $-45$ .]

#### ah-TABLE.

[Figure: Triangular numerical table for the septic, columns labelled by degree  $D = 0, 1, 2, \dots, 12$  and rows labelled by weight  $W$  from 0 down to  $-30$ ; entries reach weight  $-42$ .]

#### ai-TABLE.

[Figure: Triangular numerical table for the octavic, columns labelled by degree  $D = 0, 1, 2, \dots, 10$  and rows labelled by weight  $W$  from 0 down to  $-30$ ; entries reach weight  $-40$ .]

The numbers of each table are connected in several ways with those of the preceding tables. One of these connexions, which is of some importance, is best explained by an example: in the *af*-table, 8-20, the number of terms of degree 8 and weight 20 is 73; and we have  $73 = 1 + 6 + 16 + 23 + 27$ , viz. (see p. 278) those are the numbers of the terms in  $a^5, a^4, a^3, a^2, a$  respectively: the complementary factors, (for example) of  $a^4$  are  $bef^2$ , &c. terms in  $b, c, d, e, f$ , of the degree 5 and weight 20, and (replacing therein each letter by that which immediately precedes it) these are in number equal to the terms in  $a, b, c, d, e$  of the degree 5 and weight  $20 - 5 = 15$ ; thus the number 6 of the terms in question is that for the deg-weight 5-15 of the *ae*-table: and so 1, 6, 16, 23, 27 are the numbers in the *ae*-table for the deg-weights 4-16, 5-15, 6-14, 7-13 and 8-12 respectively, or (making a change rendered necessary by the abbreviated form of the tables) say for the deg-weights 4-0, 5-15, 6-14, 7-13 and 8-12.

### 143.

#### TABLES OF THE COVARIANTS M TO W OF THE BINARY QUINTIC: FROM THE SECOND, THIRD, FIFTH, EIGHTH, NINTH AND TENTH MEMOIRS ON QUANTICS.

[Arranged in the present form, 1889.]

The binary quintic has in all (including the quintic itself and the invariants) 23 covariants, which I have represented by the capital letters,  $A, B, C, \dots, W$  (alternative forms of two of these are denoted by  $Q'$  and  $S'$ ). The covariants  $A, \dots, L$ , and also  $Q, Q'$  were given in my Second Memoir on Quantics, and except  $Q$  and  $Q'$  are reproduced in the present reprint thereof, 141; in all these I gave not only the literal terms actually presenting themselves, but also the terms with zero coefficients; in the other covariants however, or in most of them, the terms with zero coefficients were omitted. It is very desirable to have in every case the complete series of literal terms, and in the covariants as here printed they are accordingly inserted: the number of terms is in each case known beforehand by the foregoing *af*-table, 142, and any omission is thus precluded; by means of this *af*-table we have the numbers of terms as shown in the following list.

I have throughout (as was done in the Ninth and Tenth Memoirs) expressed the literal terms in a slightly different form from that employed in the Second Memoir: this is done in order to show at a glance in each column the set of terms which contain a given power of  $a$ , and in each such set the terms which contain a given power of  $b$ .

The numerical verifications are also given not only for the entire column but for each set of terms containing the same power of  $a$ ; viz. in most cases, but not always, the positive and negative coefficients of a set have equal sums, which are shown by a number with the sign  $\pm$  prefixed. The verification is in some cases given in regard to the subsets involving the same powers of  $a$  and  $b$ , here also the sums of the positive and negative coefficients are not in every case equal. The cases of inequality will be referred to at the end of this paper.

The whole series of covariants is as follows:

| Mem. | No. of table  |           |                                                | deg-weight    |
|------|---------------|-----------|------------------------------------------------|---------------|
| 2    | 13            | $A =$     | $(1, 1, 1, 1, 1, 1 \mid x, y)^5$               | 1 (0...5)     |
| „    | 14            | $B =$     | $(3, 3, 3 \mid x, y)^2$                        | 2 (4..6)      |
| „    | 15            | $C =$     | $(2, 2, 3, 3, 3, 2, 2 \mid x, y)^6$            | 2 (2...8)     |
| „    | 16            | $D =$     | $(6, 6, 6, 8 \mid x, y)^3$                     | 3 (6..9)      |
| „    | 17            | $E =$     | $(5, 6, 6, 6, 6, 5 \mid x, y)^5$               | 3 (5...10)    |
| „    | 18            | $F =$     | $(3, 4, 5, 6, 6, 6, 6, 5, 4, 3 \mid x, y)^9$   | 3 (3.....12)  |
| „    | 19            | $G =$     | $(12 \mid x, y)^0$ , Invt.                     | 4–10          |
| „    | 20            | $H =$     | $(11, 11, 12, 11, 11 \mid x, y)^4$             | 4 (8...12)    |
| „    | 21            | $I =$     | $(9, 11, 11, 12, 11, 11, 9 \mid x, y)^6$       | 4 (7.....13)  |
| „    | 22            | $J =$     | $(20, 20 \mid x, y)^1$                         | 5 (12, 13)    |
| „    | 23            | $K =$     | $(19, 20, 20, 19 \mid x, y)^3$                 | 5 (11..14)    |
| „    | 24            | $L =$     | $(16, 18, 19, 20, 20, 19, 18, 16 \mid x, y)^7$ | 5 (9.....16)  |
| 8    | 83            | $M =$     | $(32, 32, 32 \mid x, y)^2$                     | 6 (14..16)    |
| „    | 84            | $N =$     | $(30, 32, 32, 32, 30 \mid x, y)^4$             | 6 (13.....17) |
| 9    | 90            | $O =$     | $(49, 49 \mid x, y)^1$                         | 7 (17, 18)    |
| „    | 91            | $P =$     | $(46, 48, 49, 49, 48, 46 \mid x, y)^5$         | 7 (15.....20) |
| 2    | Q 25<br>Q' 26 | $Q, Q' =$ | $(73 \mid x, y)^0$ , Invt.                     | 8–20          |
| 9    | 92            | $R =$     | $(71, 73, 71 \mid x, y)^2$                     | 8 (19..21)    |
| 9    | S 93          | $S, S' =$ | $(101, 102, 102, 101 \mid x, y)^3$             | 9 (21...24)   |
| 10   | S 93 bis      |           |                                                |               |
| 9    | 94            | $T =$     | $(190, 190 \mid x, y)^1$                       | 11 (27, 28)   |
| 3    | 29            | $U =$     | $(252 \mid x, y)^0$ , Invt.                    | 12–30         |
| 9    | 95            | $V =$     | $(325, 325 \mid x, y)^1$                       | 13 (32, 33)   |
| 5    | 29a           | $W =$     | $(967 \mid x, y)^0$ , Invt.                    | 18–45         |

[The detailed Table M, No. 83 (and the remaining tables N through W, Nos. 84 through 95 with the bis-tables) lists the literal terms of each covariant in columns by power of  $a$ , then  $b$ , with their numerical coefficients. The M-table has a three-column layout showing terms of  $a^3b^3f^2$ ,  $a^3b^2cf^2$ , etc., with column-sums  $\pm 32$ ,  $\pm 167$ ,  $\pm 52$  for verification of the first set, second set, and third set respectively.]

[The remaining detailed tables of covariants N through W are not reproduced here in full; they are extensive numerical tables that document the literal terms and coefficients of each covariant, with column-sum verifications.]

[Continued from §34 of Paper 141. — The present file replaces the corresponding section of the 251–300 chunk with a higher-fidelity rendering of the covariant tables on pages 260–272 and of the numerical ad-, ae-, ..., ai-tables of Paper 142, pp. 276–281.]

I have worked out the example in detail as a specimen of the most convenient method for the actual calculation of more complicated covariants<sup>1</sup>.

**35.** The number of terms of the degree  $\theta$  and of the weight  $q$  is obviously equal to the number of ways in which  $q$  can be made up as a sum of  $\theta$  terms with the elements  $(0, 1, 2, \dots, m)$ , a number which is equal to the coefficient of  $x^q z^\theta$  in the development of

$$\frac{1}{(1-z)(1-xz)(1-x^2z)\cdots(1-x^mz)},$$

and the number of the aszygetic covariants of any particular degree for the quantic  $(* \mid x, y)^m$  can therefore be determined by means of this development. In the case of a cubic, for example, the function to be developed is

$$\frac{1}{(1-z)(1-xz)(1-x^2z)(1-x^3z)},$$

which is equal to

$$1 + (1 + x + x^2 + x^3)z + (1 + x + 2x^2 + 2x^3 + 2x^4 + x^5 + x^6)z^2 + \&c.,$$

where the coefficients are given by the following table; on account of the symmetry, the series of coefficients for each power of  $z$  is continued only to the middle term or middle of the series.

[Figure: Triangular table of partition counts for the cubic, rows (0)–(6), entries continued only to the middle: row (0) is 1; row (1) is 1, 1; row (2) is 1, 1, 2; row (3) is 1, 1, 2, 3; row (4) is 1, 1, 2, 3, 4, 5; row (5) is 1, 1, 2, 3, 4, 5, 6; row (6) is 1, 1, 2, 3, 4, 5, 7, 8 (mirrored about the middle).]

**36.** and from this, by subtracting from each coefficient the coefficient which immediately precedes it, we form the table

[Figure: Triangular difference table for the cubic, rows (0)–(6); entries are the successive differences. Row (6) reads 1, 0, 1, 1, 1, 1, 2, 1, 0, mirrored.]

The successive lines fix the number and character of the covariants of the degrees 0, 1, 2, 3, &c. The line (0), if this were to be interpreted, would show that there is a single covariant of the degree 0; this covariant is of course merely the absolute constant unity, and may be excluded. The line (1) shows that there is a single covariant of the degree 1, viz. a covariant of the order 3; this is the cubic itself, which I represent by  $U$ . The line (2) shows that there are two aszygetic covariants of the degree 2, viz. one of the order 6, this is merely  $U^2$ , and one of the order 2, this I represent by  $H$ . The line (3) shows that there are three aszygetic covariants of the degree 3, viz. one of the order 9, this is  $U^3$ ; one of the order 5, this is  $UH$ ;

<sup>1</sup>Note added Feb. 7, 1856.—The following method for the calculation of an invariant or of the leading coefficient of a covariant, is easily seen to be identical in principle with that given in the text. Write down all the terms of the weight next inferior to that of the invariant or leading coefficient, and operate on each of these separately with the symbol

$$\text{ind. } b \cdot \frac{\partial}{\partial a} + 2 \text{ ind. } c \cdot \frac{\partial}{\partial b} + \cdots + (m-1) \text{ ind. } k \cdot \frac{\partial}{\partial j},$$

where we are first to multiply by the fraction, rejecting negative powers, and then by the index of the proper letter in the term so obtained. Equating the results to zero, we obtain equations between the terms of the invariant or leading coefficient, and replacing in these equations each term by its numerical coefficient in the invariant or leading coefficient, we have the equations of connexion of these numerical coefficients. Thus, for the discriminant of a cubic, the terms of the next inferior weight are  $a^2cd$ ,  $abc^2$ ,  $ab^2d$ ,  $b^3c$ , and operating on each of these separately with the symbol

$$\text{ind. } b \cdot \frac{\partial}{\partial a} + 2 \text{ ind. } c \cdot \frac{\partial}{\partial b} + 3 \text{ ind. } d \cdot \frac{\partial}{\partial c},$$

we find

$$\begin{aligned} a^2cd &: abcd + 4abcd + 6ac^3 \\ abc^2 &: 4b^3c + 4abcd + 6ac^3 \\ ab^2d &: 2b^3c + 4b^3d \\ b^3c &: 6b^2c^2 \end{aligned}$$

and equating the horizontal lines to zero, and assuming  $a^2d^2 = 1$ , we have  $a^2d^2 = 1$ ,  $abcd = -6$ ,  $ac^3 = 4$ ,  $b^3d = 4$ ,  $b^2c^2 = -3$ , or the value of the discriminant is that given in the text.



and one of the order 3, this I represent by  $\Phi$ . The line (4) shows that there are five asyzygetic covariants of the degree 4, viz. one of the order 12, this is  $U^4$ ; one of the order 8, this is  $U^2H$ ; one of the order 6, this is  $H^2$ ; one of the order 4, this is  $U\Phi$ ; and one of the order 0, i.e. an invariant, this I represent by  $\nabla$ . The line (5) shows that there are six asyzygetic covariants of the degree 5, viz. one of the order 15, this is  $U^5$ ; one of the order 11, this is  $U^3H$ ; one of the order 9, this is  $U^2\Phi$ ; one of the order 7, this is  $UH^2$ ; one of the order 5, this is  $H\Phi$ ; and one of the order 3, this is  $U\nabla$ . The line (6) shows that there are 8 asyzygetic covariants of the degree 6, viz. one of the order 18, this is  $U^6$ ; one of the order 14, this is  $U^4H$ ; one of the order 12, this is  $U^3\Phi$ ; one of the order 10, this is  $U^2H^2$ ; one of the order 8, this is  $UH\Phi$ ; two of the order 6 (i.e. the three covariants  $H^3$ ,  $\Phi^2$  and  $\nabla U^2$  are not asyzygetic, but are connected by a single linear equation or syzygy); and one of the order 2, this is  $\nabla H$ . We are thus led to the irreducible covariants  $U, H, \Phi, \nabla$  connected by a linear equation or syzygy between  $H^3$ ,  $\Phi^2$  and  $\nabla U^2$ , and this is in fact the complete system of irreducible covariants;  $\nabla$  is therefore the only invariant.

**36.**<sup>2</sup> The asyzygetic covariants are of the form  $U^pH^q\nabla^r$ , or else of the form  $\Phi U^pH^q\nabla^r$ ; and since  $U, H, \nabla$  are of the degrees 1, 2, 4 respectively, and  $\Phi$  is of the degree 3, the number of asyzygetic covariants of the degree  $m$  of the first form is equal to the coefficient of  $x^m$  in  $1 \div (1-x)(1-x^2)(1-x^4)$ , and the number of the asyzygetic covariants of the degree  $m$  of the second form is equal to the coefficient of  $x^m$  in  $x^3 \div (1-x)(1-x^2)(1-x^4)$ . Hence the total number of asyzygetic covariants is equal to the coefficient of  $x^m$  in  $(1+x^3) \div (1-x)(1-x^2)(1-x^4)$ , or what is the same thing, in

$$\frac{1}{(1-x)(1-x^2)(1-x^3)(1-x^4)},$$

and conversely, if this expression for the number of the asyzygetic covariants of the degree  $m$  were established independently, it would follow that the irreducible invariants were four in number, and of the degrees 1, 2, 3, 4 respectively, but connected by an equation of the degree 6. As regards the invariants, every invariant is of the form  $\nabla^p$ , i.e. the number of asyzygetic invariants of the degree  $m$  is equal to the coefficient of  $x^m$  in  $\frac{1}{1-x^4}$ , and conversely, from this expression it would follow that there was a single irreducible invariant of the degree 4.

**37.** In the case of a quartic, the function to be developed is

$$\frac{1}{(1-z)(1-xz)(1-x^2z)(1-x^3z)(1-x^4z)},$$

and the coefficients are given by the table.

[Figure: Triangular coefficient table for the quartic, rows (0)–(6). Last row (6) reads 1, 1, 2, 3, 5, 6, 9, 10, 13, 14, 16, 16, 18, mirrored.]

and subtracting from each coefficient the coefficient immediately preceding it, we have the table

[Figure: Triangular difference table for the quartic, rows (0)–(6).]

the examination of which will show that we have for the quartic the following irreducible covariants, viz. the quartic itself  $U$ ; an invariant of the degree 2, which I represent by  $I$ ; a covariant of the order 4 and of the degree 2, which I represent by  $H$ ; an invariant of the degree 3, which I represent by  $J$ ; and a covariant of the order 6 and the degree 3, which I represent by  $\Phi$ ; but that the irreducible covariants are connected by an equation of the degree 6, viz. there is a linear equation or syzygy between  $\Phi^2$ ,  $I^2H^2$ ,  $IJH^2U$ ,  $JH^2U^2$  and  $J^2U^2$ ; this is in fact the complete system of the irreducible covariants of the quartic: the only irreducible invariants are the invariants  $I, J$ .

**38.** The asyzygetic covariants are of the form  $U^pI^qH^rJ^s$ , or else of the form  $\Phi U^pI^qH^rJ^s$ , and the number of the asyzygetic covariants of the degree  $m$  is equal to the coefficient of  $x^m$  in  $(1+x^3) \div (1-x)(1-x^2)^2(1-x^3)$ , or what is the same thing, in

$$\frac{1-x^6}{(1-x)(1-x^2)^2(1-x^3)^2},$$

and the asyzygetic invariants are of the form  $I^pJ^q$ , and the number of the asyzygetic invariants of the degree  $m$  is equal to the coefficient of  $x^m$  in  $1 \div (1-x^2)(1-x^3)$ . Conversely, if these formulæ were

<sup>2</sup>Two consecutive paragraphs are numbered **36** in the original.

established, the preceding results as to the form of the system of the irreducible covariants or of the irreducible invariants, would follow.

**39.** In the case of a quintic, the function to be developed is

$$\frac{1}{(1-z)(1-xz)(1-x^2z)(1-x^3z)(1-x^4z)(1-x^5z)},$$

and the coefficients are given by the table

[Figure: Triangular coefficient table for the quintic, rows (0)–(5).]

and subtracting from each coefficient the one which immediately precedes it, we have the table

[Figure: Triangular difference table for the quintic, rows (0)–(5).]

We thus obtain the following irreducible covariants, viz.:

*Of the degree 1*; a single covariant of the order 5, this is the quintic itself.

*Of the degree 2*; two covariants, viz. one of the order 6, and one of the order 2.

*Of the degree 3*; three covariants, viz. one of the order 9, one of the order 5, and one of the order 3.

*Of the degree 4*; three covariants, viz. one of the order 6, one of the order 4, and one of the order 0 (an invariant).

*Of the degree 5*; three covariants, viz. one of the order 7, one of the order 3, and one of the order 1 (a linear covariant).

These covariants are connected by a single syzygy of the degree 5 and of the order 11; in fact, the table shows that there are only two asyzygetic covariants of this degree and order; but we may, with the above-mentioned irreducible covariants of the degrees, 1, 2, 3 and 4, form three covariants of the degree 5 and the order 11; there is therefore a syzygy of this degree and order.

**40.** I represent the number of ways in which  $q$  can be made up as a sum of  $m$  terms with the elements  $0, 1, 2, \dots, m$ , each element being repeatable an indefinite number of times by the notation

$$P(0, 1, 2, \dots, m)^\theta q,$$

and I write for shortness

$$P'(0, 1, 2, \dots, m)^\theta q = P(0, 1, 2, \dots, m)^\theta q - P(0, 1, 2, \dots, m)^\theta (q-1).$$

Then for a quantic of the order  $m$ , the number of asyzygetic covariants of the degree  $\theta$  and of the order  $\mu$  is

$$P'(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - \mu).$$

In particular, the number of asyzygetic invariants of the degree  $\theta$  is

$$P'(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta.$$

To find the total number of the asyzygetic covariants of the degree  $\theta$ , suppose first that  $m\theta$  is even; then, giving to  $\mu$  the successive values  $0, 2, 4, \dots, m\theta$ , the required number is

$$\begin{aligned} & P(\tfrac{1}{2}m\theta) - P(\tfrac{1}{2}m\theta - 1) \\ & + P(\tfrac{1}{2}m\theta - 1) - P(\tfrac{1}{2}m\theta - 2) \\ & \vdots \\ & + P(2) - P(1) \\ & + P(1) \\ & = P(\tfrac{1}{2}m\theta), \end{aligned}$$

i.e. when  $m\theta$  is even, the number of the asyzygetic covariants of the degree  $\theta$  is

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta;$$

and similarly, when  $m\theta$  is odd, the number of the asyzygetic covariants of the degree  $\theta$  is

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - 1).$$

But the two formulae may be united into a single formula; for when  $m\theta$  is odd  $\frac{1}{2}m\theta$  is a fraction, and therefore  $P(\frac{1}{2}m\theta)$  vanishes, and so when  $m\theta$  is even  $\frac{1}{2}(m\theta - 1)$  is a fraction, and  $P\frac{1}{2}(m\theta - 1)$  vanishes; we have thus the theorem, that for a quantic of the order  $m$ :

*The number of the asyzygetic covariants of the degree  $\theta$  is*

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta + P(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - 1).$$

41. The functions  $P(\frac{1}{2}m\theta)$ , &c. may, by the method explained in my “Researches on the Partition of Numbers,” [140], be determined as the coefficients of  $x^\theta$  in certain functions of  $x$ ; I have calculated the following particular cases:

Putting, for shortness,

$$P'(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta = \text{coefficient of } x^\theta \text{ in } \phi m,$$

then

$$\begin{aligned}\phi 2 &= \frac{1}{1 - x^2}, \\ \phi 3 &= \frac{1}{1 - x^4}, \\ \phi 4 &= \frac{1}{(1 - x^2)(1 - x^3)}, \\ \phi 5 &= \frac{1 - x^8 + x^{18}}{(1 - x^4)(1 - x^8)(1 - x^{12})}, \\ \phi 6 &= \frac{(1 - x)(1 + x - x^2 - x^4 - x^5 + x^7 + x^8)}{(1 - x^2)^2(1 - x^3)(1 - x^4)(1 - x^5)}, \\ \phi 7 &= \frac{1 - x^4 + 2x^8 - x^{10} + 5x^{12} + 2x^{14} + 6x^{16} + 2x^{18} + 5x^{20} - x^{22} + 2x^{24} - x^{28} + x^{32}}{(1 - x^4)(1 - x^8)(1 - x^{12})(1 - x^{14})(1 - x^{18})}, \\ \phi 8 &= \frac{(1 - x)(1 + x - x^2 - x^4 + x^7 + x^8 + x^9 + x^{10} - x^{11} + x^{13} + x^{14})}{(1 - x^2)^2(1 - x^3)^2(1 - x^4)(1 - x^5)(1 - x^6)},\end{aligned}$$

and putting

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}m\theta = \text{coefficient of } x^\theta \text{ in } \psi m,$$

then

$$\begin{aligned}\psi 2 &= \frac{1}{(1 - x)(1 - x^2)}, \\ \psi 3 &= \frac{1 + x^3}{(1 - x^2)(1 - x^4)}, \\ \psi 4 &= \frac{1 - x + x^2}{(1 - x)^2(1 - x^2)(1 - x^3)}, \\ \psi 5 &= \frac{1 + x^4 + 6x^6 + 9x^7 + 12x^8 + 9x^9 + 6x^{10} + x^{12} + x^{16}}{(1 - x^2)^2(1 - x^4)(1 - x^6)(1 - x^8)},\end{aligned}$$

and putting

$$P(0, 1, 2, \dots, m)^\theta \frac{1}{2}(m\theta - 1) = \text{coefficient of } x^\theta \text{ in } \chi m,$$

then

$$\begin{aligned}\chi 3 &= \frac{x + x^3}{(1 - x^2)^2(1 - x^3)}, \\ \chi 5 &= \frac{x + 4x^2 + 8x^3 + 10x^4 + 10x^5 + 8x^6 + 4x^7 + x^8}{(1 - x^2)^2(1 - x^4)(1 - x^6)(1 - x^8)}.\end{aligned}$$

And from what has preceded, it appears that for a quantic of the order  $m$ , the number of asyzygetic covariants of the degree  $\theta$  is for  $m$  even, coefficient  $x^\theta$  in  $\psi m$ , and for  $m$  odd, coefficient  $x^\theta$  in  $(\psi m + \chi m)$ ;

and that the number of asyzygetic invariants of the degree  $\theta$  is coefficient  $x^\theta$  in  $\phi m$ . Attending first to the invariants:

**42.** For a quadric, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } \frac{1}{1-x^2},$$

which leads to the conclusion that there is a single irreducible invariant of the degree 2.

**43.** For a cubic, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } \frac{1}{1-x^4},$$

i.e. there is a single irreducible invariant of the degree 4.

**44.** For a quartic, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } \frac{1}{(1-x^2)(1-x^3)},$$

i.e. there are two irreducible invariants of the degrees 2 and 3 respectively.

**45.** For a quintic, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x^8+x^{18}}{(1-x^4)(1-x^8)(1-x^{12})}.$$

The numerator is the irreducible factor of  $1-x^{36}$ , i.e. it is equal to  $(1-x^{36})(1-x^4) \div (1-x^{12})(1-x^9)$ ; and substituting this value, the number becomes

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x^{36}}{(1-x^4)(1-x^8)(1-x^{12})(1-x^{18})},$$

i.e. there are in all four irreducible invariants, which are of the degrees 4, 8, 12 and 18 respectively; but these are connected by an equation of the degree 36, i.e. the square of the invariant of the degree 18 is a rational and integral function of the other three invariants; a result, the discovery of which is due to M. Hermite.

**46.** For a sextic, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } \frac{(1-x)(1+x-x^2-x^4-x^5+x^7+x^8)}{(1-x^2)^2(1-x^3)(1-x^4)(1-x^5)};$$

the second factor of the numerator is the irreducible factor of  $1-x^{30}$ , i.e. it is equal to  $(1-x^{30})(1-x^5)(1-x^3)(1-x^2) \div (1-x^{15})(1-x^{10})(1-x^6)(1-x)$ ; and substituting this value, the number becomes

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x^{30}}{(1-x^2)(1-x^4)(1-x^6)(1-x^{10})(1-x^{15})},$$

i.e. there are in all five irreducible invariants, which are of the degrees 2, 4, 6, 10 and 15 respectively; but these are connected by an equation of the degree 30, i.e. the square of the invariant of the degree 15 is a rational and integral function of the other four invariants.

**47.** For a septic, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x^4+2x^8-x^{10}+5x^{12}+2x^{14}+6x^{16}+2x^{18}+5x^{20}-x^{22}+2x^{24}-x^{28}+x^{32}}{(1-x^4)(1-x^8)(1-x^{12})(1-x^{14})(1-x^{18})};$$

the numerator is equal to

$$(1-x^4)(1-x^8)^{-1}(1-x^{10})(1-x^{12})^{-1}(1-x^{14})^{-1}\dots,$$

where the series of factors does not terminate; hence [incorrect, see p. 253] the number of irreducible invariants is infinite; substituting the preceding value, the number of asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } (1-x^4)^{-1}(1-x^8)^{-3}(1-x^{12})^{-6}(1-x^{14})^{-4}\dots.$$

The first four indices give the number of irreducible invariants of the corresponding degrees, i.e. there are 1, 3, 6 and 4 irreducible invariants of the degrees 4, 8, 12 and 14 respectively, but there is no reason to believe that the same thing holds with respect to the indices of the subsequent terms. To verify this it is to be remarked, that there are 1, 4, 10 and 4 asyzygetic invariants of the degrees in question respectively; there is therefore one irreducible invariant of the degree 4; calling this  $X_4$ , there is only one composite invariant of the degree 8, viz.  $X_4^2$ ; there are therefore three irreducible invariants of this degree, say  $X_8$ ,  $X'_8$ ,  $X''_8$ . The composite invariants of the degree 12 are four in number, viz.  $X_4^3$ ,  $X_4X_8$ ,  $X_4X'_8$ ,  $X_4X''_8$ , and these cannot be connected by any syzygy, for if they were so,  $X_4^2$ ,  $X_8$ ,  $X'_8$ ,  $X''_8$  would be connected by a syzygy, or there would be less than 3 irreducible invariants of the degree 8. Hence there are precisely 6 irreducible invariants of the degree 12. And since the irreducible invariants of the degrees 4, 8 and 12 do not give rise to any composite invariant of the degree 14, there are precisely 4 irreducible invariants of the degree 14.

**48.** For an octavic, the number of the asyzygetic invariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } \frac{(1-x)(1+x-x^2-x^4+x^7+x^8+x^9+x^{10}-x^{11}+x^{13}+x^{14})}{(1-x^2)^2(1-x^3)^2(1-x^4)(1-x^5)(1-x^6)},$$

and the second factor of the numerator is

$$(1-x)^{-1}(1-x^2)^{-1}(1-x^3)^{-1}(1-x^4)^{-1}(1-x^5)^{-1}(1-x^6)^{-1}(1-x^{10})^{-1}(1-x^{12})(1-x^{14})(1-x^{15})\dots,$$

where the series of factors does not terminate; hence [incorrect] the number of irreducible invariants is infinite. Substituting the preceding value, the number of the asyzygetic invariants of the degree  $\theta$  is

$$\text{coeff. of } x^\theta \text{ in } (1-x^2)^{-1}(1-x^3)^{-1}(1-x^4)^{-1}(1-x^5)^{-1}(1-x^6)^{-1}(1-x^7)^{-2}(1-x^8)^{-3}(1-x^9)^{-5}(1-x^{10})^{-7}\dots$$

There is certainly one, and only one irreducible invariant for each of the degrees 2, 3, 4, 5 and 6 respectively; but the formula does not show the number of the irreducible invariants of the degrees 7, &c.; in fact, representing the irreducible invariants of the degrees 2, 3, 4, 5 and 6 by  $X_2$ ,  $X_3$ ,  $X_4$ ,  $X_5$ ,  $X_6$ , these give rise to 3 composite invariants of the degree 7, viz.  $X_2X_2X_3$ ,  $X_3X_4$ ,  $X_2X_5$ , which may or may not be connected by a syzygy; if they are not connected by a syzygy, there will be a single irreducible invariant of the degree 7; but if they are connected by a syzygy, there will be two irreducible invariants of the degree 7; it is useless at present to pursue the discussion further.

Considering next the covariants,

**49.** For a quadric, the number of asyzygetic covariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } \frac{1}{(1-x)(1-x^2)},$$

i.e. there are two irreducible covariants of the degrees 1 and 2 respectively; these are of course the quadric itself and the invariant.

**50.** For a cubic, the number of the asyzygetic covariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } \frac{(1+x)(1+x^3)}{(1-x^2)^2(1-x^4)}.$$

The first factor of the numerator is the irreducible factor of  $1-x^2 = (1-x^2) \div (1-x)$ , and the second factor of the numerator is the irreducible factor of  $1-x^6 = (1-x^6) \div (1-x^2)$ ; substituting these values, the number is

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x^6}{(1-x)(1-x^2)(1-x^3)(1-x^4)},$$

i.e. there are 4 irreducible covariants of the degrees 1, 2, 3, 4 respectively; but these are connected by an equation of the degree 6; the covariant of the degree 1 is the cubic itself  $U$ , the other covariants are the covariants already spoken of and represented by the letters  $H$ ,  $\Phi$  and  $\nabla$  respectively ( $H$  is of the degree 2 and the order 3,  $\Phi$  of the degree 3 and the order 3, and  $\nabla$  is of the degree 4 and the order 0, i.e. it is an invariant).

**51.** For a quartic, the number of the asyzygetic covariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } \frac{1-x+x^2}{(1-x)^2(1-x^2)(1-x^3)};$$

the numerator of which is the irreducible factor of  $1 - x^6$ , i.e. it is equal to  $(1 - x^6)(1 - x) \div (1 - x^2)(1 - x^3)$ . Making this substitution, the number is

$$\text{coefficient of } x^\theta \text{ in } \frac{1 - x^6}{(1 - x)(1 - x^2)^2(1 - x^3)^2},$$

i.e. there are five irreducible covariants, one of the degree 1, two of the degree 2, and two of the degree 3, but these are connected by an equation of the degree 6. The irreducible covariant of the degree 1 is of course the quartic itself  $U$ , the other irreducible covariants are those already spoken of and represented by  $I, H, J, \Phi$  respectively ( $I$  is of the degree 2 and the order 0, and  $J$  is of the degree 3 and the order 0, i.e.  $I$  and  $J$  are invariants,  $H$  is of the degree 2 and the order 4,  $\Phi$  is of the degree 3 and the order 6).

**52.** For a quintic, the number of irreducible covariants of the degree  $\theta$  is

$$\text{coefficient of } x^\theta \text{ in } \frac{1 + x + x^2 + 4x^3 + 6x^4 + 8x^5 + 9x^6 + 10x^7 + 12x^8 + 10x^9 + 9x^{10} + 8x^{11} + 6x^{12} + 4x^{13} + x^{14} + x^{15} + x^{16}}{(1 - x^2)^2(1 - x^4)(1 - x^6)(1 - x^8)};$$

the numerator of which is

$$(1 + x)^2(1 + x - x^2 - x^4 + 2x^5 + 3x^6 + 5x^7 + 4x^8 + 3x^9 + 2x^{10} - x^{12} - x^{13} + x^{14}).$$

The first factor is  $(1 - x^2)^2 \div (1 - x)^2$ , the second factor is

$$(1 - x^2)^3(1 - x^3)^2(1 - x^4)^3(1 - x^5)^4(1 - x^7)^{-5}(1 - x^8)^{-7} \dots,$$

which does not terminate; hence [incorrect] the number of irreducible covariants is infinite. Substituting the preceding values, the expression for the number of the asyzygetic covariants of the degree  $\theta$  is

$$\text{coeff. of } x^\theta \text{ in } (1 - x)^{-1}(1 - x^2)^{-2}(1 - x^3)^{-3}(1 - x^4)^{-3}(1 - x^5)^{-3}(1 - x^6)^{-4}(1 - x^7)^{-5}(1 - x^8)^{-7} \dots,$$

which agrees with a previous result: the numbers of irreducible covariants for the degrees 1, 2, 3, 4 are 1, 2, 3 and 3 respectively, and for the degree 5, the number of irreducible covariants is three, but there is one syzygy between the composite covariants of the degree in question; the difference  $3 - 1 = 2$  is the index taken with its sign reversed of the factor  $(1 - x^5)^{-3}$ .

**53.** I consider a system of the asyzygetic covariants of any particular degree and order of a given quantic; the system may of course be replaced by a system the terms of which are any linear functions of those of the original system, and it is necessary to inquire what covariants ought to be selected as most proper to represent the system of asyzygetic covariants; the following considerations seem to me to furnish a convenient rule of selection. Let the literal parts of the terms which enter into the coefficients of the highest power of  $x$  or leading coefficients be represented by  $M_\alpha, M_\beta, M_\gamma, \dots$  these quantities being arranged in the natural or alphabetical order; the first in order of these quantities  $M$ , which enters into the leading coefficient of a particular covariant, may for shortness be called the leading term of such covariant, and a covariant the leading term of which is posterior in order to the leading term of another covariant, may be said to have a lower leading term.

It is clear, that by properly determining the multipliers of the linear functions we may form a covariant the leading term of which is lower than the leading term of any other covariant (the definition implies that there is but one such covariant); call this  $\Theta_1$ . We may in like manner form a covariant such that its leading term is lower than the leading term of every other covariant except  $\Theta_1$ ; or rather we may form a system of such covariants, since if  $\Theta_2$  be a covariant having the property in question,  $\Theta_2 + k\Theta_1$  will have the same property, but  $k$  may be determined so that the covariant shall not contain the leading term of  $\Theta_1$ , i.e. we may form a covariant  $\Theta_2$  such that its leading term is lower than the leading term of every other covariant excepting  $\Theta_1$ , and that the leading term of  $\Theta_1$  does not enter into  $\Theta_2$ ; and there is but one such covariant,  $\Theta_2$ . Again, we may form a covariant  $\Theta_3$  such that its leading term is lower than the leading term of every other covariant excepting  $\Theta_1$  and  $\Theta_2$ , and that the leading terms of  $\Theta_1$  and  $\Theta_2$  do not either of them enter into  $\Theta_3$ ; and there is but one such covariant,  $\Theta_3$ . And so on, until we arrive at a covariant the leading term of which is higher than the leading terms of the other covariants, and which does not contain the leading terms of the other covariants. We have thus a series of covariants  $\Theta_1, \Theta_2, \Theta_3, \&c.$  containing the proper number of terms, and which covariants may be taken to represent the asyzygetic covariants of the degree and order in question.

In order to render the covariants  $\Theta$  definite as well numerically as in regard to sign, we may suppose that the covariant is in its least terms (i.e. we may reject numerical factors common to all the terms), and we may make the leading term positive. The leading term with the proper numerical coefficient (if different from unity) and with the proper power of  $x$  (or the order of the function) annexed, will, when the covariants of a quantic are tabulated, be sufficient to indicate, without any ambiguity whatever, the particular covariant referred to. I subjoin a table of the covariants of a quadric, a cubic and a quartic, and of the covariants of the degrees 1, 2, 3, 4 and 5 respectively of a quintic, and also two other invariants of a quintic.

[Except for the quantic itself, the algebraical sum of the numerical coefficients in any column is = 0, viz. the sum of the coefficients with the sign + is equal to that of the coefficients with the sign –, and I have as a numerical verification inserted at the foot of each column this sum with the sign  $\pm$ .]

### Covariant Tables (Nos. 1 to 26).

**No. 1.**  $(a + 1, b + 2, c + 1 \mid x, y)^2$ .

**No. 2.**  $\begin{pmatrix} ac + 1 \\ b^2 - 1 \end{pmatrix} \pm 1$ .

The tables Nos. 1 and 2 are the covariants of a binary quadric. No. 1 is the quadric itself; No. 2 is the quadricinvariant, which is also the discriminant.

**No. 3.**  $(a + 1, b + 3, c + 3, d + 1 \mid x, y)^3$ .

**No. 4.**  $\begin{pmatrix} ac + 1 & ad + 1 & bd + 1 \\ b^2 - 1 & bc - 1 & c^2 - 1 \end{pmatrix} \mid x, y, \pm 1, \pm 1, \pm 1$ .

**No. 5.**  $\begin{pmatrix} a^2d + 1 & abd + 3 & acd - 3 & ad^2 - 1 \\ abc - 3 & ac^2 - 6 & b^2d + 6 & bcd + 3 \\ b^3 + 2 & b^2c + 3 & bc^2 - 3 & c^3 - 2 \end{pmatrix} \mid x, y, \pm 3, \pm 6, \pm 6, \pm 3$ .

**No. 6.**  $\begin{pmatrix} a^2d^2 + 1 \\ abcd - 6 \\ ac^3 + 4 \\ b^3d + 4 \\ b^2c^2 - 3 \end{pmatrix} \pm 7$ .

The tables Nos. 3, 4 and 5 are the covariants of a binary cubic. No. 3 is the cubic itself; No. 4 is the quadricovariant, or Hessian; No. 5 is the cubicovariant; No. 6 is the invariant, or discriminant. And if we write No. 3 =  $U$ , No. 4 =  $H$ , No. 5 =  $\Phi$ , No. 6 =  $\nabla$ , then identically,

$$\Phi^2 - \nabla U^2 + 4H^3 = 0.$$

**No. 7.**  $(a + 1, b + 4, c + 6, d + 4, e + 1 \mid x, y)^4$ .

**No. 8.**  $\begin{pmatrix} ae + 1 \\ bd - 4 \\ c^2 + 3 \end{pmatrix} \pm 4$ .

**No. 9.**  $\begin{pmatrix} ac + 1 & ad + 1 & ae + 1 & bd + 1 \\ b^2 - 1 & bc - 2 & c^2 - 2 & cd - 2 \\ \pm 1 & \pm 2 & \pm 3 & \pm 2 \end{pmatrix} + (be + 2, ce - 1) \mid x, y^4$ .

**No. 10.** (Cubinvariant of the quartic.)  $\begin{pmatrix} ace + 1 & a^2d + 1 & d^2e + 1 & abe + 5 & ace & ade - 5 \\ & abc - 3 & abd + 2 & & & \\ ad^2 - 1 & b^4 + 2 & & acd - 15 & ad^2 - 10 & bce + 15 \\ & & ac^2 - 9 & b^3d + 10 & b^4e + 10 & bc^2 - 10 \\ b^2e - 1 & \pm 3 & b^3c + 6 & & & \\ & & \pm 9 & bc^2 & d & c^2d \end{pmatrix}$ ,

sums  $\pm 15, \pm 10, \pm 15, \pm 3$ .

[Table No. 10 is rendered schematically; the full layout in the original is a four-column array of literal terms in  $a, b, c, d, e$  of degree 3 and order 6 (the cubicovariant of the quartic), with  $\pm$ -sum row  $\pm 15, \pm 10, \pm 15, \pm 3$ .]

**No. 11.** (Cubicovariant of the quartic.) Eight-column literal table of degree 3 terms, with column sums  $\pm 3, \pm 9, \pm 15, \pm 5, \pm 15, \pm 9, \pm 3$ .

**No. 12.** (Discriminant of the quartic.)

$$\begin{aligned}\nabla = & a^2e^3 + 1 \cdot a^2bde^2 \cdot (-12) + a^2c^2e \cdot (-18) + a^2cd^2e \cdot 54 + a^2d^4 \cdot (-27) \\ & + ab^2ce^2 \cdot 54 + ab^2d^2e \cdot (-6) + abc^2de \cdot (-180) + abcd^3 \cdot 108 \\ & + ac^4e \cdot 81 - ac^3d^2 \cdot 54 - b^4e^2 \cdot 27 + b^3cde \cdot 108 \\ & - b^3d^3 \cdot 64 - b^2c^3e \cdot 54 + b^2c^2d^2 \cdot 36 + bc^4d \cdot \dots \\ & - c^6 \cdot \dots,\end{aligned}$$

column sum  $\pm 442$ .

The tables Nos. 7, 8, 9, 10 and 11 are the irreducible covariants of a quartic. No. 7 is the quartic itself; No. 8 is the quadrinvariant; No. 9 is the quadricovariant, or Hessian; No. 10 is the cubinvariant; and No. 11 is the cubicovariant. The table No. 12 is the discriminant. And if we write No. 7 =  $U$ , No. 8 =  $I$ , No. 9 =  $H$ , No. 10 =  $J$ , No. 11 =  $\Phi$ , No. 12 =  $\nabla$ , then the irreducible covariants are connected by the identical equation

$$JU^3 - IU^2H + 4H^3 + \Phi^2 = 0,$$

and we have

$$\nabla = I^3 - 27J^2.$$

[The Tables Nos. 13 to 24 which follow, and also Nos. 25 and 26 which are given in 143 relate to the binary quintic. I have inserted in the headings the capital letters  $A, B, \dots, L$  and also  $Q$  and  $Q'$  by which I refer to these covariants of the quintic.  $A$  is the quintic itself,  $C$  is the Hessian,  $G$  is the quartinvariant,  $J$  a linear covariant;  $Q$  is the simplest octinvariant, and  $Q'$  is the discriminant. As noticed in the original memoir we have  $AI + BF - CE = 0$ ; and  $Q' = G^2 - 128Q$ , only the coefficient 128 was by mistake given as 1152.]

**A. No. 13.**  $(a + 1, b + 5, c + 10, d + 10, e + 5, f + 1 \mid x, y)^5$ .

**B. No. 14.** 
$$\left( \begin{array}{ccc|c} ae + 1 & af + 1 & bf + 1 & \\ bd - 4 & be - 3 & ce - 4 & \\ c^2 + 3 & cd + 2 & d^2 + 3 & \\ \hline \pm 4 & \pm 3 & \pm 4 & \end{array} \mid x, y \right)^2.$$

**C. No. 15.** 
$$\left( \begin{array}{cccccc|c} ac + 1 & ad + 3 & ae + 3 & af + 1 & bf + 3 & cf + 3 & df + 1 & \\ & bd + 3 & bd + 7 & & ce + 3 & de - 3 & & \\ b^2 - 1 & bc - 3 & be - 6 & cd - 8 & & & c^2 - 1 & \\ & & c^2 - 6 & d^2 - 6 & & & & \\ \hline \pm 1 & \pm 3 & \pm 6 & \pm 8 & \pm 6 & \pm 3 & \pm 1 & \end{array} \mid x, y \right)^6.$$

[Tables  $D$  (No. 16),  $E$  (No. 17),  $F$  (No. 18),  $G$  (No. 19),  $H$  (No. 20),  $I$  (No. 21),  $J$  (No. 22),  $K$  (No. 23),  $L$  (No. 24) follow with the full literal expansions of the corresponding covariants of the binary quintic. The detailed numerical tables are extensive; the column sums (with sign  $\pm$ ) printed at the bottom of each column serve as numerical verifications. The degree–order specifications of  $D, E, F$  are  $3 \cdot 3, 3 \cdot 5, 3 \cdot 9$ ; of  $G, H, I$  are  $4 \cdot 0, 4 \cdot 4, 4 \cdot 6$ ; of  $J, K, L$  are  $5 \cdot 1, 5 \cdot 3, 5 \cdot 7$ . The column-sum signatures of these tables, where given in the original, are:  $G$ :  $\pm 149$ ;  $H$ :  $\pm 33, \pm 54, \pm 33$ ;  $I$ :  $\pm 11, \pm 40, \pm 15, \pm 60, \pm 15, \pm 40, \pm 11$ ;  $J$ :  $\pm 95, \pm 95$ ;  $K$ :  $\pm 57, \pm 129, \pm 129, \pm 57$ ;  $L$ :  $\pm 26, \pm 93, \pm 207, \pm 241, \pm 241, \pm 207, \pm 93, \pm 26$ .]

**No. 25.**  $Q = +1a^2cdf + \&c.$  [see post, 143.]

**No. 26.**  $Q' = +1a^4f^4 + \&c.$



142.

NUMERICAL TABLES SUPPLEMENTARY TO SECOND MEMOIR  
ON QUANTICS.

[Now first published (1889).]

In the present paper I arrange in a more compendious form and continue to a much greater extent the tables (first of each pair) given Nos. 35–39 of my Second Memoir on Quantics, 141, pp. 260–264, which relate to the cubic, the quartic and the quintic functions; and I give the like tables for the sextic, the septic and the octavic functions respectively. The cubic table exhibits the coefficients of the several terms of the function  $1 \div (1 - z \cdot 1 - xz \cdot 1 - x^2z \cdot 1 - x^3z)$ , or, what is the same thing, it gives the number of partitions of a given number into a given number of parts, the parts being 0, 1, 2, 3 (repetitions admissible): or again, regarding the letters  $a, b, c, d$ , as having the weights 0, 1, 2, 3 respectively, it shows the number of literal terms of a given degree and given weight. And similarly for the quartic, quintic, sextic, septic and octavic tables respectively, the parts of course being 0, 1, . . . up to 4, 5, 6, 7 or 8, and the letters being  $a, b, \dots$  up to  $e, f, g, h$  or  $i$ . The extent of the tables is as follows:

|             |                       |       |
|-------------|-----------------------|-------|
| cubic table | extends to deg-weight | 18–27 |
| quartic     | „                     | 18–36 |
| quintic     | „                     | 18–45 |
| sextic      | „                     | 16–45 |
| septic      | „                     | 12–42 |
| octavic     | „                     | 10–40 |

viz. for the quintic, the sextic and the octavic functions these are the deg-weights of the highest invariants respectively. I designate the Tables as the  $ad$ -,  $ae$ -,  $af$ -,  $ag$ -,  $ah$ - and  $ai$ -tables respectively.

It is to be noticed that in the several tables the lower part of each column is for shortness omitted; the column has to be completed by taking into it the series of bottom terms of each of the preceding columns: thus in the  $af$ - or quintic table the complete column for degree 3 would be

|     |        |
|-----|--------|
| $W$ | $D\ 3$ |
| 0   | 6      |
| –1  | 6      |
| –2  | 5      |
| –3  | 4      |
| –4  | 3      |
| –5  | 2      |
| –6  | 1      |
| –7  | 1      |
| –   | 2      |
| –   | 1      |
| –   | 1      |

where the concluding terms 2, 1, 1 are the bottom terms of the three preceding columns respectively. And the meaning is that for degree 3, and weight 8 or 7, the number of terms is = 6; for weight 7 – 1, = 6, the number of terms is = 6; and similarly for weights 5, 4, 3, 2, 1, the numbers are 5, 4, 3, 2, 1, 1; the numbers are those of the terms

[Figure: Eight-column array (for the quintic, columns labelled  $W = 8$  down to  $W = 1$ ) showing the triangle of literal terms in  $a, b, c, d, e, f$  of degree 3 broken into columns by weight; each column lists the products  $a^3, a^2b, a^2c, \dots$  down to terms in  $f$ , with the count of terms equal to 6, 6, 5, 4, 3, 2, 1, 1 for weights 8, 7, . . . , 1 respectively.]

The like remarks and explanations apply to the other tables.

| $D$ | 0 | 1 | 2 | 3 | 4 | 5  | 6  | 7  | 8  | 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | $W$ |
|-----|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|----|----|----|-----|
|     | 1 | 1 | 2 | 3 | 5 | 6  | 8  | 10 | 13 | 15 | 18 | 21 | 25 | 28 | 32 | 36 | 41 | 45 | 50 | 0   |
|     |   | 1 | 2 | 3 | 4 | 6  | 8  | 10 | 12 | 15 | 18 | 21 | 24 | 28 | 32 | 36 | 40 | 45 | 50 | –1  |
|     |   |   | 1 | 4 | 5 | 7  | 9  | 12 | 14 | 17 | 20 | 24 | 27 | 31 | 35 | 40 | 44 | 49 |    | –2  |
|     |   |   |   | 1 | 7 | 8  | 11 | 13 | 17 | 19 | 23 | 26 | 31 | 34 | 39 | 43 | 49 |    |    | –3  |
|     |   |   |   |   | 1 | 10 | 12 | 15 | 18 | 22 | 25 | 29 | 33 | 38 | 42 | 47 |    |    |    | –4  |
|     |   |   |   |   |   | 1  | 14 | 16 | 20 | 23 | 28 | 31 | 36 | 40 | 46 |    |    |    |    | –5  |
|     |   |   |   |   |   |    | 1  | 19 | 21 | 26 | 29 | 35 | 38 | 44 |    |    |    |    |    | –6  |
|     |   |   |   |   |   |    |    | 1  | 24 | 27 | 32 | 36 | 42 |    |    |    |    |    |    | –7  |
|     |   |   |   |   |   |    |    |    | 1  | 30 | 33 | 39 |    |    |    |    |    |    |    | –8  |
|     |   |   |   |   |   |    |    |    |    | 1  | 37 |    |    |    |    |    |    |    |    | –9  |

**ae-TABLE.**

[Figure: Triangular numerical table for the quartic; columns labelled by degree  $D = 0, 1, 2, \dots, 18$  and rows labelled by weight  $W$  from 0 down to  $-18$  (entries reach weight  $-36$  if the lower triangle were completed). Top row reads 1, 1, 3, 5, 8, 12, 18, 24, 33, 43, 55, 69, 86, 104, 126, 150, 177, 207, 241. Each cell shows the count of literal terms of the indicated degree and weight, with the lower part of each column omitted; the column has to be completed by taking into it the series of bottom terms of the preceding columns.]

**af-TABLE.**

[Figure: Triangular numerical table for the quintic; columns labelled by degree  $D = 0, 1, 2, \dots, 18$  and rows labelled by weight  $W$  from 0 down to  $-27$  (entries reach weight  $-45$ ). Top row reads 1, 1, 3, 6, 12, 20, 32, 49, 73, 102, 141, 190, 252, 325, 414, 521, 649, 795, 967. The structure is the same as for the ad- and ae-tables.]

**ag-TABLE.**

[Figure: Triangular numerical table for the sextic; columns labelled by degree  $D = 0, 1, 2, \dots, 15$  and rows labelled by weight  $W$  from 0 down to  $-30$  (entries reach weight  $-45$ ). Top row reads 1, 1, 4, 8, 18, 32, 58, 94, 151, 227, 338, 480, 676, 920, 1242, 1636.]

**ah-TABLE.**

[Figure: Triangular numerical table for the septic; columns labelled by degree  $D = 0, 1, 2, \dots, 12$  and rows labelled by weight  $W$  from 0 down to  $-30$  (entries reach weight  $-42$ ). Top row reads 1, 1, 4, 10, 24, 49, 94, 169, 289, 468, 734, 1117, 1656.]

**ai-TABLE.**

[Figure: Triangular numerical table for the octavic; columns labelled by degree  $D = 0, 1, 2, \dots, 10$  and rows labelled by weight  $W$  from 0 down to  $-30$  (entries reach weight  $-40$ ). Top row reads 1, 1, 5, 13, 33, 73, 151, 289, 526, 910, 1514.]

The numbers of each table are connected in several ways with those of the preceding tables. One of these connexions, which is of some importance, is best explained by an example: in the *af*-table, 8-20, the number of terms of degree 8 and weight 20 is 73; and we have  $73 = 1 + 6 + 16 + 23 + 27$ , viz. (see p. 278) those are the numbers of the terms in  $a^5, a^4, a^3, a^2, a$  respectively: the complementary factors, (for example) of  $a^4$  are  $bef^2$ , &c. terms in  $b, c, d, e, f$ , of the degree 5 and weight 20, and (replacing therein each letter by that which immediately precedes it) these are in number equal to the terms in  $a, b, c, d, e$  of the degree 5 and weight  $20 - 5 = 15$ ; thus the number 6 of the terms in question is that for the deg-weight 5-15 of the *ae*-table; and so 1, 6, 16, 23, 27 are the numbers in the *ae*-table for the deg-weights 4-16, 5-15, 6-14, 7-13 and 8-12 respectively, or (making a change rendered necessary by the abbreviated form of the tables) say for the deg-weights 4-0, 5-15, 6-14, 7-13 and 8-12.

143.

TABLES OF THE COVARIANTS  $M$  TO  $W$  OF THE BINARY QUINTIC: FROM THE SECOND, THIRD, FIFTH, EIGHTH, NINTH AND TENTH MEMOIRS ON QUANTICS.

[Arranged in the present form, 1889.]

The binary quintic has in all (including the quintic itself and the invariants) 23 covariants, which I have represented by the capital letters,  $A, B, C, \dots, W$  (alternative forms of two of these are denoted by  $Q'$  and  $S'$ ). The covariants  $A, \dots, L$ , and also  $Q, Q'$  were given in my Second Memoir on Quantics, and except  $Q$  and  $Q'$  are reproduced in the present reprint thereof, 141; in all these I gave not only the literal terms actually presenting themselves, but also the terms with zero coefficients; in the other covariants however, or in most of them, the terms with zero coefficients were omitted. It is very desirable to have in every case the complete series of literal terms, and in the covariants as here printed they are accordingly inserted: the number of terms is in each case known beforehand by the foregoing *af*-table, 142, and any omission is thus precluded; by means of this *af*-table we have the numbers of terms as shown in the following list.

I have throughout (as was done in the Ninth and Tenth Memoirs) expressed the literal terms in a slightly different form from that employed in the Second Memoir: this is done in order to show at a glance in each column the set of terms which contain a given power of  $a$ , and in each such set the terms which contain a given power of  $b$ .

The numerical verifications are also given not only for the entire column but for each set of terms containing the same power of  $a$ ; viz. in most cases, but not always, the positive and negative coefficients of a set have equal sums, which are shown by a number with the sign  $\pm$  prefixed. The verification is in some cases given in regard to the subsets involving the same powers of  $a$  and  $b$ , here also the sums of the positive and negative coefficients are not in every case equal. The cases of inequality will be referred to at the end of this paper.

The whole series of covariants is as follows:

| Mem. | No. of table      |           |                                                | deg-weight    |
|------|-------------------|-----------|------------------------------------------------|---------------|
| 2    | 13                | $A =$     | $(1, 1, 1, 1, 1 \mid x, y)^5$                  | 1 (0...5)     |
| „    | 14                | $B =$     | $(3, 3, 3 \mid x, y)^2$                        | 2 (4..6)      |
| „    | 15                | $C =$     | $(2, 2, 3, 3, 3, 2, 2 \mid x, y)^6$            | 2 (2...8)     |
| „    | 16                | $D =$     | $(6, 6, 6, 8 \mid x, y)^3$                     | 3 (6..9)      |
| „    | 17                | $E =$     | $(5, 6, 6, 6, 6, 5 \mid x, y)^5$               | 3 (5...10)    |
| „    | 18                | $F =$     | $(3, 4, 5, 6, 6, 6, 6, 5, 4, 3 \mid x, y)^9$   | 3 (3.....12)  |
| „    | 19                | $G =$     | $(12 \mid x, y)^0$ , Invt.                     | 4–10          |
| „    | 20                | $H =$     | $(11, 11, 12, 11, 11 \mid x, y)^4$             | 4 (8...12)    |
| „    | 21                | $I =$     | $(9, 11, 11, 12, 11, 11, 9 \mid x, y)^6$       | 4 (7.....13)  |
| „    | 22                | $J =$     | $(20, 20 \mid x, y)^1$                         | 5 (12, 13)    |
| „    | 23                | $K =$     | $(19, 20, 20, 19 \mid x, y)^3$                 | 5 (11..14)    |
| „    | 24                | $L =$     | $(16, 18, 19, 20, 20, 19, 18, 16 \mid x, y)^7$ | 5 (9.....16)  |
| 8    | 83                | $M =$     | $(32, 32, 32 \mid x, y)^2$                     | 6 (14..16)    |
| „    | 84                | $N =$     | $(30, 32, 32, 32, 30 \mid x, y)^4$             | 6 (13.....17) |
| 9    | 90                | $O =$     | $(49, 49 \mid x, y)^1$                         | 7 (17, 18)    |
| „    | 91                | $P =$     | $(46, 48, 49, 49, 48, 46 \mid x, y)^5$         | 7 (15.....20) |
| 2    | $Q$ 25<br>$Q'$ 26 | $Q, Q' =$ | $(73 \mid x, y)^0$ , Invt.                     | 8–20          |
| 9    | 92                | $R =$     | $(71, 73, 71 \mid x, y)^2$                     | 8 (19..21)    |
| 9    | $S$ 93            | $S, S' =$ | $(101, 102, 102, 101 \mid x, y)^3$             | 9 (21...24)   |
| 10   | $S$ 93 bis        |           |                                                |               |
| 9    | 94                | $T =$     | $(190, 190 \mid x, y)^1$                       | 11 (27, 28)   |
| 3    | 29                | $U =$     | $(252 \mid x, y)^0$ , Invt.                    | 12–30         |
| 9    | 95                | $V =$     | $(325, 325 \mid x, y)^1$                       | 13 (32, 33)   |
| 5    | 29a               | $W =$     | $(967 \mid x, y)^0$ , Invt.                    | 18–45         |

**M. No. 83.** (Sextic-degree, degree 6 and order 2 covariant of the binary quintic; total of  $32 + 57 + 22 = \pm$ -verified column sums for three sets of terms.)

*[Figure: The full literal table  $M$  is presented as a three-column array. Column 1 lists 32 terms of the form  $a^3b^2df^2$ ,  $a^3bcd f^2$ , etc., grouped by descending power of  $a$ ; column 2 lists 57 terms from a different power-of- $a$  subset; column 3 lists 22 terms from a third subset. Column-sum signatures along the bottom (for the three sets and three subsets) are  $\pm 7$ ,  $\pm 57$ ,  $\pm 22$  for the entire columns, with internal subset sums  $\pm 31$ ,  $\pm 108$ ,  $\pm 34$ . The full numerical content is preserved in the scanned source.]*

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Paper 143 (concluded) and Paper 144 (opening)

# 143. Tables of the Covariants $M$ to $W$ of the Binary Quintic (concluded).

[From the *Philosophical Transactions of the Royal Society of London*, vol. cxlvi. for the year 1857, pp. 703–755.]

The present pages contain the remaining tables of the covariants of the binary quintic, namely tables  $N$  to  $W$ , followed by the verifications of the numerical coefficients. The covariants are linear forms in the powers of the leading coefficient  $a$ , with coefficients which are themselves polynomial in  $b, c, d, e, f$  multiplied by powers of  $x$  and  $y$ . Each table is laid out as a multi-column array of monomial-coefficient pairs. The original printed layout, with its column alignment, is preserved here in verbatim blocks; the OCR-derived text retains the typographical features of the 1889 edition (subscripts and dashes appearing as encoded by the OCR engine), and is to be read as a faithful skeleton rather than as a corrected re-typesetting.

**Reading convention.** A line such as “ $a^2b^4cdf - 1$ ” means: the term  $a^2b^4cdf$ , in the relevant covariant, has numerical coefficient  $-1$ . The covariants are arranged column by column; each column corresponds to a power of  $a$ , the leading coefficient, and within each column the monomials are listed in the natural lex order. At the foot of each column is the algebraic sum of the (positive minus the negative) numerical coefficients in that column; at the foot of each table is the total numerical sum, which agrees with the sum at the foot of the column for the lowest power of  $a$  and which provides the principal numerical check.

## N. No. 84.

[The full coefficient table occupies book p. 285; it lists the terms of the covariant  $N$  of the binary quintic, arranged in five columns headed by the powers  $a^6, a^5, a^4, a^3, a^2$  of the leading coefficient. The column-sums are  $-19, +81, -62, \pm 0, +62$ ; total  $\pm 163$ . Owing to its size and the OCR damage to subscripts in this particular table, the table is reproduced below in literal OCR form; the reader is referred to the scan for definitive values.]

|                  |                  |                   |                |                 |
|------------------|------------------|-------------------|----------------|-----------------|
| $a^6 b^7df -1$   | $a^5 b^4ef$      | $a^4 b^4cf$       | $a^3 b^4f$     | $a^2 b^5f^2 +1$ |
| $cdf +1$         | $a^5 b^3df -4$   | $b^3ef$           | $b^3cef +4$    | $def^2 -3$      |
| $a^5 b^4df^2 +3$ | $def +4$         | $b^3cdf$          | $def^2 -4$     | $ey +2$         |
|                  |                  | $a^4 b^3df^2 -6$  | $b^3cef +8$    | $b^4cdf^2 -2$   |
| $b^4cdef -2$     | $a^4 b^3df^2 +4$ |                   | $def +12$      | $d^2ef +8$      |
|                  |                  | $b^3cef +12$      |                | $d^2e -2$       |
| $b^3def +12$     | $def +12$        | $cdef$            | $cef^2 -32$    | $c^2e^2 +6$     |
| $cd^2f +2$       |                  |                   |                | $cd^2e -20$     |
| $cd^2 +12$       | $cd^2f +16$      | $a^4 b^3df^2 +14$ | $cef +6$       | $cde^2 +12$     |
|                  | $e^4 -24$        |                   |                | $cd^2 -2$       |
| $d^2e -6$        | $b^2c^2ef -88$   | $c^2f^2 +13$      | $b^3c^2ef +56$ |                 |
|                  |                  |                   |                | $a^3 b^4f^2 +5$ |
| $a^4 b^3f^2 -2$  | $cd^2 +30$       | $b^2c^2df^2 +48$  |                |                 |
|                  |                  |                   |                | $dey -13$       |
| $b^2cef +2$      | $a^3 b^3df^2 -4$ | $c^2def -88$      |                | $ef +12$        |
| $d^2f -6$        |                  | $cd^2 +56$        |                | $b^4cef^2 -12$  |
|                  |                  |                   |                | $c^2ef -13$     |
| $bce^2 +4$       | $a^3 b^3cdf -52$ | $a^3 b^3df^2 -6$  | $cd^2 +12$     | $d^2ef -4$      |
| $cd^2e -10$      | $d^4 +24$        | $a^3 b^3def -56$  | $cdef -40$     | $ef^2 +24$      |
|                  |                  |                   |                | $cdef +28$      |
| $a^3 b^3f^2 +6$  | $b^4ef -9$       | $def +36$         | $e^4 -60$      | $cf^3$          |
|                  |                  |                   |                | $ce^2 -15$      |
| $bcdf +12$       | $cd^2e +20$      | $b^3cdf +36$      | $b^3cef +24$   | $d^2ef -4$      |
| $b^3ef -15$      | $b^4cdef -10$    | $c^2def -156$     | $cdef -56$     | $c^2ef +52$     |

|                                     |                                       |                        |                                     |                                    |
|-------------------------------------|---------------------------------------|------------------------|-------------------------------------|------------------------------------|
| d <sup>3</sup> e +10                | d <sup>2</sup> -100                   | e <sup>2</sup> def -50 | c <sup>2</sup> dey -20              | d <sup>2</sup> e <sup>2</sup> -10  |
| a <sup>4</sup> b <sup>3</sup> ef +6 |                                       |                        |                                     | de <sup>2</sup> -20                |
| bcd <sup>f</sup> +12                | a <sup>3</sup> b <sup>3</sup> cef -12 | b <sup>3</sup> cef +18 | c <sup>2</sup> de <sup>2</sup> +100 | c <sup>2</sup> de <sup>2</sup> +30 |
| c <sup>2</sup> ef -15               |                                       | cd <sup>2</sup> -156   |                                     | d <sup>4</sup> e +15               |
|                                     |                                       |                        |                                     | ...                                |

[Reproduced for the entire breadth of the table from the OCR layer; alignment artefacts reflect the broken sub/superscript glyphs in the scan. Column totals at foot: -19; +81; -62; ±0; +62. Grand total: ±163.]

# O. No. 90.

[Book p. 286. The covariant O has degree higher than N. The terms with monomials of the form  $a^p b^q c^r d^s e^t f^u$  are arranged in four columns headed by descending powers  $a^6, a^5, a^4, a^3$  of a. Column sums: 4, 59, 497, 1003; column counter-sums: 1, 49, 669, 964; differences in each column, ±1563.]

|                                                 |                                                  |                                              |                                      |
|-------------------------------------------------|--------------------------------------------------|----------------------------------------------|--------------------------------------|
| a <sup>6</sup> b <sup>4</sup> f <sup>2</sup> +1 | a <sup>5</sup> b <sup>4</sup> df <sup>2</sup> -1 | a <sup>4</sup> b <sup>4</sup> f <sup>2</sup> | a <sup>3</sup> b <sup>4</sup> ef +15 |
| def -4                                          | ey <sup>2</sup> +1                               | def                                          | -b <sup>4</sup> cd <sup>2</sup> -26  |
| ey +3                                           | a <sup>5</sup> b <sup>4</sup> cf <sup>2</sup> +4 |                                              | cey -84                              |
| a <sup>5</sup> b <sup>4</sup> f <sup>2</sup> -1 |                                                  | a <sup>4</sup> b <sup>4</sup> ef +7          | d <sup>2</sup> ef +98                |
| b <sup>4</sup> cef -3                           | d <sup>ey</sup> +3                               | b <sup>4</sup> df -30                        |                                      |
|                                                 | cey -7                                           |                                              | -de <sup>4</sup> -45                 |
| d <sup>ey</sup> +16                             | b <sup>3</sup> cef -16                           | cey +1                                       | b <sup>4</sup> ef +12                |
|                                                 |                                                  | d <sup>4</sup> ef -74                        |                                      |
| d <sup>ey</sup> +4                              | cd <sup>2</sup> f +6                             |                                              | c <sup>2</sup> def +94               |
|                                                 |                                                  | cd <sup>2</sup> +84                          |                                      |
| e <sup>4</sup> -15                              | cdey +30                                         | b <sup>4</sup> cdef +18                      | c <sup>3</sup> +150                  |
| b <sup>3</sup> cdf <sup>2</sup> -6              | ce <sup>4</sup> -8                               |                                              | cdy -140                             |
|                                                 |                                                  | cdef +160                                    | cd <sup>4</sup> -50                  |
| c <sup>2</sup> ef +4                            | d <sup>3</sup> -18                               | -cd <sup>4</sup> e <sup>2</sup> 32           |                                      |
| cd <sup>2</sup> ef -22                          | +ddey -6                                         |                                              |                                      |
|                                                 |                                                  | d <sup>4</sup> f +81                         | d <sup>4</sup> e +15                 |
|                                                 |                                                  | ce <sup>4</sup> +6                           | b <sup>4</sup> c <sup>2</sup> -51    |
| cd <sup>2</sup> +26                             | a <sup>3</sup> b <sup>4</sup> f <sup>2</sup> -3  |                                              |                                      |
|                                                 |                                                  | b <sup>4</sup> c <sup>2</sup> -9             | c <sup>3</sup> df +100               |
| d <sup>4</sup> f +9                             | b <sup>4</sup> cef <sup>2</sup> -4               |                                              |                                      |
| d <sup>2</sup> ey -12                           | d <sup>ey</sup> -4                               | c <sup>2</sup> def +20                       | -c <sup>4</sup> ey 320               |
|                                                 | d <sup>2</sup> ey -1                             |                                              |                                      |
| a <sup>3</sup> b <sup>4</sup> ef +7             |                                                  | c <sup>2</sup> e <sup>2</sup> -112           | c <sup>4</sup> e +310                |
| b <sup>4</sup> d <sup>2</sup> -30               | ey +18                                           | c <sup>3</sup> dey -18                       |                                      |
|                                                 |                                                  |                                              | cd <sup>4</sup> -90                  |
| cey +1                                          | b <sup>4</sup> cdf <sup>2</sup> +22              | c <sup>3</sup> e <sup>2</sup> +284           | b <sup>4</sup> cdf -18               |
| d <sup>2</sup> ef -74                           | c <sup>2</sup> d <sup>2</sup> f +74              | cd <sup>4</sup> -216                         |                                      |
|                                                 |                                                  |                                              | c <sup>2</sup> +120                  |
| d <sup>4</sup> e +84                            | cdey <sup>2</sup> -160                           | d <sup>4</sup> +54                           | c <sup>3</sup> d <sup>2</sup> e -130 |
| b <sup>4</sup> cdf <sup>2</sup> +18             | -cd <sup>4</sup> e <sup>2</sup> 32               | a <sup>3</sup> b <sup>4</sup> ef +15         | c <sup>3</sup> d <sup>4</sup> +40    |
| cdef +160                                       | d <sup>4</sup> f +81                             | -b <sup>4</sup> cdf <sup>2</sup> 26          |                                      |
| -cdye <sup>2</sup> 150                          | cdey +6                                          | cey -84                                      | :                                    |
|                                                 | b <sup>4</sup> cdf -9                            |                                              | 4                                    |
|                                                 |                                                  |                                              | 59                                   |
|                                                 |                                                  | d <sup>4</sup> ef +98                        | 49                                   |
|                                                 |                                                  |                                              | 669                                  |

[OCR layout preserved; sub/superscripts as in the scanned printing.]

# P. No. 91.

[Book p. 287. The covariant P is arranged in six columns by powers  $a^6, a^5, a^4, a^3, a^2, a$ . Column sums: -3, +67, ±6, ±6, ±27, -1; sub-sums per row given at the right; grand totals ±388 and ±594 for the two halves x and y of  $\square x, y \square^k$ .]

|                    |                      |                    |                      |                    |                  |
|--------------------|----------------------|--------------------|----------------------|--------------------|------------------|
| $a^6 b^4 f^2$      | $a^5 b^4 f^2$        | $a^4 b^4 f^2 -1$   | $a^3 b^4 d f^2 +1$   | $a^2 b^4 e f^2$    | $a b^4 f^2$      |
| $a^5 b^4 e f^2$    | $-b^4 c d e f^2$     | $d e f^2 +6$       | $d e y^2 -1$         | $a^2 b^4 d f^2 +2$ | $a^5 b^4 e f^2$  |
| $b^4 c d f +1$     | $c d^2 f +5$         | $e y -5$           | $a^4 b^4 c d e f -6$ | $e y^2 -2$         | $b^4 c d^2 -1$   |
| $c e y -2$         | $c d y -1$           | $b y^2 +1$         | $d e f^2 +11$        | $b^4 c^2 f -5$     | $c e y +1$       |
| $c d^2 e f +2$     | $e^4 -2$             | $-b^4 c d y^2 11$  | $e y -5$             | $c y^2 +17$        | $c f^2 +3$       |
| $c d^4 -1$         | $b^4 c y +2$         | $d^4 f -4$         | $b^4 c^2 f^2 +4$     | $c e y -7$         | $d e y -5$       |
| $a^4 b^4 d y^2 -1$ | $-b^4 c d f^2 17$    | $d^4 e f -4$       | $c d y^2 -2$         | $c y^2 -4$         | $e y +2$         |
| $e y +2$           | $c e y +13$          | $d^4 e f -32$      | $e^4 +17$            | $c^4 e y +4$       | $d^4 e y -6$     |
| $b^4 c y^2 -3$     | $d^4 e f -32$        | $b^4 c^2 d y^2 +2$ | $c e^4 -4$           | $d^4 e f -10$      | $d e y +5$       |
| $c d e f -6$       | $d e y +32$          | $c^4 e y +26$      | $d^4 e f -10$        | $a^3 b^4 c f^2 +1$ | $b^4 c y^2 -2$   |
| $c e^2 +13$        | $b^4 c y^2 +4$       | $c d^4 e f -2$     | $d^4 e^2 +8$         | $-d e y^2 13$      | $c^4 y^2 +6$     |
| $c^2 y -8$         | $c^2 d y +36$        | $-c d^4 y 40$      | $a^3 b^4 y^2 +5$     | $e y +12$          | $c e y -2$       |
| $c y -24$          | $c y -24$            | $d^4 f -9$         | $b^4 c e y^2 +4$     | $b^4 c e y^2 +32$  | $c d y -16$      |
| $+d^4 e^2 2$       | $c d^4 f -10$        | $+d e y^2 24$      | $d y^2 -26$          | $c d y^2 -36$      | $d^4 e y +24$    |
| $b^4 c^4 f +16$    | $c d^4 e -16$        | $a^3 b^4 y^2 +5$   | $d e y -35$          | $c d e y -42$      | $d e y -10$      |
| $c d^4 e f -2$     | $d^4 e +12$          | $b^4 c d y^2 -4$   | $e^4 +42$            | $c e y +24$        | $b^4 c d y^2 +8$ |
| $c^3 d e y -38$    | $a^4 b^4 c d e f +7$ | $c e y +35$        | $b^4 c^4 d f^2 +2$   | $c e^4 +24$        | $c^4 y^2 +2$     |
| $c^4 e +34$        | $d^4 e f -26$        | $c^4 e y +26$      |                      | $d^4 e y +56$      | $c^4 d y -52$    |
|                    |                      |                    |                      | $d^4 e^2 -34$      | $c^4 y +28$      |
|                    |                      |                    |                      |                    | ...              |

[Continues for several columns; total numerical sums in the source printing:  $\pm 388$ ,  $\pm 594$  for the  $x$ - and  $y$ -halves of the binary form.]

## Q. No. 25, Q'. No. 26.

[Book p. 288. The invariants  $Q$  and  $Q'$ . The table is arranged with two adjacent columns labelled  $Q$  and  $Q'$ , listing the same monomials in  $a, b, c, d, e, f$  with the corresponding signed integer coefficients. The numerical sums for the powers  $a^5, a^4, a^3, a^2, a$  of  $a$  are listed at the right edge of the table.]

|                 |     | Q     | Q'                  |
|-----------------|-----|-------|---------------------|
| $a^6 b^4 f^4$   |     | +1    | -3375               |
| $b^4 c d e f$   |     |       | -20                 |
| $b^3 c d f^2$   | +1  | -120  |                     |
| $c^3 f$         | -1  |       |                     |
| $c^3 e$         | -3  | +160  |                     |
| $d^2 e f^2$     | +5  | +360  |                     |
| $d^4 f$         | -1  | -640  |                     |
| $e^4$           | +1  | +256  |                     |
| $a^5 b^4 e f^3$ |     | +160  |                     |
| $b^3 d f^3$     |     | -10   |                     |
| $c d e f^2$     | +11 |       | (sub-totals for Q': |
| $c^4 f$         | +12 | +360  | 1 =1                |
| $d^4 f^2$       | -30 |       | 776-780 =-4         |
| $d^2 e y$       | +15 | -1640 | 21266-21250 =+6     |
| $d e^4$         | +12 | +320  | 68656-68660 =-4     |

|               |      |        |                    |
|---------------|------|--------|--------------------|
|               | -21  | -1440  | 37816-37815 =+1    |
| $b^4c^2f^2$   | -34  | +4080  | 128505-128505 = 0) |
| $c^4ef^2$     | +22  |        | ...                |
|               | +78  | -1920  |                    |
| $c^3def$      | -48  | -1440  |                    |
| $c^4y$        | -27  | -1440  |                    |
| $cd^2ef$      | +18  | +2640  |                    |
| $cd^4y$       |      | +4480  |                    |
| $d^4f$        | +5   | +2640  |                    |
| $d^3e^2$      | -5   | -2560  |                    |
|               |      | -10080 |                    |
| $a^4 b^4cf^2$ | -30  |        |                    |
| def           | -34  | +5760  |                    |
| ey            |      | +3456  |                    |
|               | +133 | -2160  |                    |
| $b^4c^4ef$    | -54  | -640   |                    |
| $cd^4y$       | -18  |        |                    |
|               |      | +320   |                    |
| $cde^4f$      | +3   | -180   |                    |
|               | +78  |        |                    |
| $ce^4$        | -18  | +4080  |                    |
|               | -220 | +4480  |                    |
| $d^4ef$       | +106 | -14920 |                    |
|               | +93  | +7200  |                    |
| $d^4e^2$      | -30  | +960   |                    |
|               | -9   | -600   |                    |

[The full table runs across two columns ( $Q$  and  $Q'$ ) for several screens; at the end of the table the column sums are  $\pm 6$  for  $Q$  and  $\pm 128505$  for  $Q'$ . The sub-totals 169,625,424 accumulate to  $\pm 1124$  for the partial-sums check.]

## R. No. 92.

[Book p. 289. The covariant  $R$ . Six columns headed by powers  $a^6$ ,  $a^5$ ,  $a^4$ ,  $a^3$ ,  $a^2$ ,  $a$  of  $a$ ; the column sums are  $\pm 8$ , 93, 300, 780, totalling  $\pm 1181$ .]

|                  |                  |                |                 |               |                |
|------------------|------------------|----------------|-----------------|---------------|----------------|
| $a^6 b^4ef^2$    | $a^5 b^4cdy$ -15 | $a^4 b^4f^2$   | $a^3 b^4e^4$    | $a^2 b^4f$    | $a b^4cdef$ +2 |
|                  | $cd^2$ -38       | $a^3 b^4ef^2$  |                 |               |                |
| $a^5 b^4df^2$    | $de^4$ +45       | $b^3cdf^2$ +18 | $b^4cdf^2$ +18  | $b^4cef$      | $d^2e^2$ +15   |
|                  |                  | $c^4cf$ -66    |                 |               |                |
| $cd^2$           | $b^4cf$ +3       | $c^4f$         | $cd^4ef$ +20    | $d^4f$ +"i"   | $b^3cdf^2$ -32 |
|                  |                  |                | $cd^4$          |               |                |
| $b^3c^4f$ -1     |                  | $d^4ef$ +2     |                 | $de^4f$ -2    | $c^4cf$ -39    |
|                  | $cd^4ef$ +102    |                | $d^4f$ +58      | $e^4f$ +1     |                |
| $cd^4f$ +6       |                  |                |                 |               | $c^4def$ -24   |
|                  | $c^4$ -15        | $d^4f$ -4      |                 | $a^2 b^4ef^2$ |                |
| $c^4f$ -4        |                  |                | $b^4c^4f$ -50   |               | $c^4de^2$ +17  |
|                  | $cd^4f$ +76      | $d^4$ +2       | $b^3c^4f$ -6    | $b^3cdf^2$ -6 |                |
| $c^4f^2$ -3      |                  |                |                 |               | $cd^4f$ +25    |
|                  | $cd^4$ -175      | $a^3 b^3df$    | $c^4def$ +72    | $ce^4f$ +6    | $cd^4e^2$ -1   |
| $d^4e^2f^2$      | $d^4e$ +35       | $e^4$          | $c^4$ +50       | $d^4ef^2$ +3  |                |
|                  |                  |                |                 | $d^4ey$       | $d^4e$ +15     |
| $de^4$ +1        | $b^4c^4ef$ -42   | $b^3cf$ -2     | $d^4f$ -3       |               |                |
|                  |                  |                |                 | $e^4$ -3      | $b^4f^2$ +9    |
| $a^5 b^4cf^2$ +2 | $c^4d^4f$ -182   | $cdef$         | $c^4d^4f$ -156  |               |                |
| $def^2$ -6       |                  |                |                 | $b^4cf^2$ +3  | $c^4ey$ +106   |
|                  | $c^4d^4$ +120    | $cey$ +4       | $cd^4e^2$ ..... | $c^4def$ -3   | $c^4$ -35      |
| $e^4f$ +4        | $c^4d^4$ +150    | $d^4ey$ -14    |                 |               | $c^4d^4f$ -    |
| $b^3c^4y$ -3     |                  | $d^4ey$ +30    | $cd^4e$ +90     | $cd^4y$ -6    |                |
| $cdf^2$ +3       | $cd^4$ -70       | $de^4$ -18     | $d^4$ -30       | $cd^4f$       | $cd^4e^2$ -    |
|                  | $b^4cdf$ +126    |                | $b^4cef$ -24    |               | $cd^4e$ +175   |



|                                  |                                      |                                      |                                                 |                                                  |                                                |
|----------------------------------|--------------------------------------|--------------------------------------|-------------------------------------------------|--------------------------------------------------|------------------------------------------------|
| cd <sup>4</sup> f <sup>-18</sup> |                                      | b <sup>4</sup> cdy <sup>2</sup> +14  | c <sup>4</sup> d <sup>4</sup> y <sup>2</sup> 94 | cd <sup>4</sup> ey <sup>2</sup> +3               | cd <sup>4</sup> -45                            |
|                                  | -c <sup>4</sup> e <sup>4</sup> 15    |                                      |                                                 | cde <sup>4</sup> +6                              | b <sup>3</sup> c <sup>4</sup> ef -3            |
| ce <sup>4</sup> +17              | c <sup>4</sup> d <sup>4</sup> e -175 | c <sup>4</sup> d <sup>4</sup> y      |                                                 | d <sup>4</sup> ef                                |                                                |
|                                  | c <sup>4</sup> d <sup>4</sup> +75    | c <sup>4</sup> d <sup>4</sup> ef -66 | c <sup>4</sup> d <sup>4</sup> e -90             | -d <sup>4</sup> e <sup>2</sup> 3                 | c <sup>4</sup> d <sup>4</sup> f +21            |
| d <sup>4</sup> ef +22            | b <sup>4</sup> c <sup>4</sup> f -27  | c <sup>4</sup> +26                   | c <sup>4</sup> d <sup>4</sup> e                 | a <sup>3</sup> b <sup>4</sup> df <sup>2</sup> +4 | + c <sup>4</sup> d <sup>4</sup> e <sup>2</sup> |
| d <sup>4</sup> -21               |                                      |                                      |                                                 |                                                  |                                                |
|                                  |                                      | cd <sup>4</sup> ef +56               | c <sup>4</sup> d <sup>4</sup> +10               | e <sup>4</sup> f -4                              |                                                |
| b <sup>4</sup> cdy               | + c <sup>4</sup> de +45              | cd <sup>4</sup> -18                  | b <sup>4</sup> cdef -18                         | b <sup>4</sup> cf -1                             | c <sup>4</sup> d <sup>4</sup> e -7             |
|                                  | -c <sup>4</sup> de 20                | c <sup>4</sup> f -18                 | c <sup>4</sup> +30                              | cdey <sup>2</sup> +18                            |                                                |
| e <sup>4</sup> f +13             |                                      |                                      |                                                 | cey <sup>2</sup> -16                             | c <sup>4</sup> d <sup>4</sup> +20              |
|                                  |                                      |                                      |                                                 |                                                  | ...                                            |

S. No. 93 bis; S'. No. 93. ( $\diamond x, y$ )<sup>4</sup>.

[Book p. 290. The covariant pair  $S, S'$  is given as a binary quartic in  $x, y$  with five coefficients, one for each of  $x^4, x^3y, x^2y^2, xy^3, y^4$ . The contents of each coefficient (a homogeneous polynomial in  $a, b, c, d, e, f$  of the appropriate degree-weight) are tabulated side by side under headings  $S$  and  $S'$ . Per the printed table, the column sums of the numerical coefficients vanish for each separate power of  $a$ , in accordance with the verification rule derived from the Tenth Memoir on Quantics.]

| Coef. x <sup>4</sup>                         | S | S' | Coef. x <sup>3</sup> y           | S   | S'                                                             |
|----------------------------------------------|---|----|----------------------------------|-----|----------------------------------------------------------------|
| a <sup>4</sup> b <sup>4</sup> f <sup>2</sup> |   | -9 | a <sup>3</sup> b <sup>4</sup> ef | -66 | +528                                                           |
|                                              |   |    |                                  | -45 | a <sup>4</sup> b <sup>3</sup> cf <sup>2</sup>                  |
|                                              |   |    |                                  |     | +9                                                             |
|                                              |   |    |                                  |     | a <sup>3</sup> b <sup>3</sup> c <sup>4</sup> e <sup>4</sup> +6 |
|                                              |   |    |                                  |     | cd <sup>4</sup> ef                                             |

[Table extends across all four coefficients  $x^4, x^3y, x^2y^2, xy^3, y^4$  of the covariant  $S$  and the like for  $S'$ . Verifications consist in checking that  $\sum(\text{positive}) = \sum(\text{negative})$  for every power of  $a$ .]

T. No. 94.

[Book pp. 291–293. The covariant  $T$ . As stated in the prose passage that follows, for  $T$  the sums are taken for the different combinations of  $a$  and  $b$ ; the tabulation is two-column, the left column giving the  $x$ -coefficient, the right column the  $y$ -coefficient, of the covariant. Sample column-totals: for  $a^3b^5$ , the  $x$ -coefficient column has positive–negative 307 – 349, 1073 – 1003, 2040 – 2110, 1930 – 1880, 1207 – 1221, 231 – 239; for the  $y$ -coefficient column, the totals are 112, 281, 1650, 6511, etc., summing to  $\pm 20196$  each side. The full table occupies three book pages and is reproduced below in OCR-skeleton form.]

| x coefficient.                |      |   | y coefficient.                   |                               |     |
|-------------------------------|------|---|----------------------------------|-------------------------------|-----|
| a <sup>4</sup> b <sup>7</sup> | 26   |   | a <sup>4</sup> b <sup>7</sup>    | db 12                         |     |
| a <sup>3</sup> b <sup>6</sup> | - 14 |   | 26 a <sup>3</sup> b <sup>6</sup> | 2                             | 12  |
| b <sup>5</sup>                |      |   | 436 b <sup>5</sup>               | 112                           | 395 |
| b <sup>4</sup>                | 141  |   | 3738                             | 281                           |     |
|                               |      |   | 9116                             | b <sup>4</sup>                |     |
| a <sup>3</sup> b <sup>4</sup> | 281  |   | 6880                             | a <sup>2</sup> b <sup>7</sup> |     |
| b <sup>3</sup>                |      | - |                                  |                               |     |
|                               |      |   | b <sup>6</sup>                   |                               |     |
| b <sup>2</sup>                | -1   |   | b <sup>6</sup>                   | 42                            |     |
| b                             | ...  |   | b <sup>5</sup>                   | 546                           |     |
|                               | 106  |   |                                  | 696                           |     |
| b <sup>7</sup>                |      |   | a <sup>3</sup> b <sup>5</sup>    | 366                           |     |
|                               | 186  |   |                                  |                               |     |
| a <sup>2</sup> b <sup>6</sup> |      |   | b <sup>4</sup>                   |                               |     |
|                               | 1173 |   | b <sup>3</sup>                   |                               |     |
| b <sup>5</sup>                |      |   | b <sup>2</sup>                   |                               |     |

11628

\* 20196

## \* 20196

V. No. 95 (continued).  $y$  coefficient.

[Book pp. 297–298. Format identical to the preceding. For details consult the scan.]

[Book pp. 299–303. The covariant  $W$ . The largest table in the entire series of papers 141–143, occupying five book pages. Column sums for the  $a^4$  terms produce successive sums  $1, -10, +45, -120, +210, -252, +210, -120, +45, -10, +1$  (i.e. the binomial coefficients of  $-(\theta - 1)^{10}$ ); for  $a^3$  they are  $-5, +54, -265, +780, -1530, +2100, -2058, +1440, -705, +230, -45, +4$ , i.e. the binomial coefficients of  $-5(\theta - 1)^{11} - (\theta - 1)^{10}$ . The grand total of the table on book p. 303 is  $\pm 2207351$ , with the  $\pm$  partial sums  $\pm 2972759$  on the second part and  $\pm 3111879, \pm 3003111, \pm 9087749$  on subsequent parts.]

## Verifications of the Numerical Coefficients (Book pp. 304–309).

For the lower covariants the numerical verifications are given for the entire coefficient, but for the higher ones where the number of terms in a coefficient is considerable they are given separately for the different powers of  $a$ ; and it is also interesting to consider them for the separate combinations of  $a$  and  $b$ . I recall that the positive and negative numerical coefficients are summed separately, so that “ $\pm$  (a number)” means that the sum of the positive numerical coefficients is equal to the sum of the negative numerical coefficients and thus that the whole sum is  $= 0$ .

It is to be observed that for the lower covariants the sums of the numerical coefficients do not vanish for the separate powers of  $a$ : thus in the invariant  $G$ , 141, the sums of the numerical coefficients for the terms in  $a^4$ ,  $a^3$ ,  $a^2$  are  $= 1$ ,  $-2$ ,  $1$  respectively.

As regards the invariants  $Q$  and  $Q'$ : for the first of these,  $Q$ , the sums of the numerical coefficients for the terms in  $a^5$ ,  $a^4$ ,  $a^3$ ,  $a^2$ ,  $a$  are each of them  $= 0$ , but this is not the case as regards  $Q'$ ; in fact  $Q' = G^2 +$  a multiple of  $Q$ ; hence the sums for  $Q'$  are the same as those for  $G^2$ , viz. they are  $= 1$ ,  $-4$ ,  $+6$ ,  $-4$ ,  $+1$  respectively. Like results present themselves in other cases, and they might probably be accounted for in a similar manner; we have a series of sums not each  $= 0$ , but which are equal to a set of binomial coefficients taken with the signs  $+$  and  $-$  alternately, and thus the sum of these sums is  $= 0$ .

For  $R$ ,  $S$  and  $S'$ , I have given the sums for the different powers of  $a$ ; and in regard to  $S$  I give here the following paragraphs from the Tenth Memoir on Quantics:

“I remark that I calculated the first two coefficients  $S_0$ ,  $S_1$  and deduced the other two,  $S_2$  from  $S_1$ , and  $S_3$  from  $S_0$ , by reversing the order of the letters (or which is the same thing, interchanging  $a$  and  $f$ ,  $b$  and  $e$ ,  $c$  and  $d$ ) and reversing also the signs of the numerical coefficients. This process for  $S_2$ ,  $S_3$  is to a very great extent a verification of the values of  $S_0$ ,  $S_1$ . For, as presently mentioned, the terms of  $S_0$  form subdivisions such that in each subdivision the sum of the numerical coefficients is  $= 0$ : in passing by the reversal process to the value of  $S_3$ , the terms are distributed into an entirely new set of subdivisions, and then in each of these subdivisions the sum of the numerical coefficients is found to be  $= 0$ ; and the like as regards  $S_1$  and  $S_2$ .”

“If in the expressions for  $S_0$ ,  $S_1$ ,  $S_2$ ,  $S_3$  we first write  $d = e = f = 1$ , thus in effect combining the numerical coefficients for the terms which contain the same powers in  $a, b, c$ , we find”

$$\begin{aligned} S_0 = & a^4(-2c^2 + 6c - 6c + 2) \\ & + a^3 \{b^2(6c^2 - 12c + 6) + b(-15c^3 + 33c^2 - 21c + 3) \\ & + b^0(42c^4 - 147c^3 + 195c^2 - 117c + 27)\} \\ & + a^2 [b^4 \cdot 0 + b^3(30c^2 - 36c + 6) + b^2(-117c^3 + 249c^2 - 183c + 51) \\ & + b(9c^4 + 148c^4 - 378c^3 + 330c^2 - 99c) + b^0(-63c^5 + 165c^4 - 147c^3 + 45c^2)] \end{aligned}$$

and continuing on the next book page,

$$\begin{aligned} & + a \{b^5 \cdot 2 + b^4(-15c + 3) + b^3(75c^2 - 69c + 24) + b^2(-9c^3 + 167c^2 - 225c + 87c - 2) \\ & + b(72c^4 + 48c^3 - 186c^2 + 96c) + b^0(-126c^5 + 201c^3 - 87c^4) \\ & + b^0(27c^6 - 40c^5 + 20c^4)\} \end{aligned}$$

which for  $c = 1$  becomes  $= 2b^6 - 12b^5 + 30b^4 - 40b^3 + 30b^2 - 12b + 2$ , that is  $-2(b - 1)^6$ , and for  $b = 1$  becomes  $= 0$ .

$$\begin{aligned}
 S_1 = & a^4(0c^2 + 0c + 0) \\
 & + a^3 \{b^0(0c + 0) + b(3c^3 - 9c^2 + 9c - 3) + b^2(24c^4 - 99c^3 + 153c^2 - 105c + 27)\} \\
 & + a^2 \{b^4 \cdot 0 + b^3(-6c^2 + 12c - 6) + b^2(-24c^3 + 90c^2 - 108c + 42) \\
 & \quad + b(33c^4 - 90c^3 + 54c^2 + 30c - 27) + b^0(-27c^5 + 78c^4 + 66c^3 + 6c^2 - 9c)\} \\
 & + a^0 \{b^5(3c - 3) + b^4(-15c + 15) + b^3(60c^2 - 12c^3 + 36c + 30) \\
 & \quad + b^2(9c^4 - 42c^3 + 84c^2 - 108c + 57c) + b(0c^5 - 54c^4 + 96c^3 - 51c^2) \\
 & \quad + b^0(9c^6 - 9c^4)\}
 \end{aligned}$$

which for  $c = 1$  becomes  $= 0$ .

$$\begin{aligned}
 S_2 = & a^4(0c + 0) \\
 & + a^3 \{b^4 \cdot 0 + b^3(0c^2 + 0c + 0) + b^2(18c^4 - 72c^3 + 108c^2 - 72c + 18)\} \\
 & + a \{f(0c + 0) + b^3(-33c^3 + 99c^2 - 99c + 33) + b(57c^4 - 162c^3 + 144c^2 - 30c - 9) \\
 & \quad + b^0(-60c^5 + 207c^4 - 261c^3 + 141c^2 - 27c)\} \\
 & + a^0 \{b^5 \cdot 0 + b^4(15c^2 - 30c + 15) + b^3(-54c^3 + 102c^2 - 42c - 6) \\
 & \quad + b^2(123c^4 - 297c^3 + 243c^2 - 87c + 18) + b(-27c^5 + 102c^4 - 96c^3 + 21c^2) \\
 & \quad + b^0(27c^6 - 60c^5 + 51c^4 - 12c^3)\}
 \end{aligned}$$

which for  $c = 1$  becomes  $= 0$ .

$$\begin{aligned}
 S_3 = & a^4 \cdot 0 \\
 & + a^3 \{b(0c + 0) + b^2(0c^3 + 0c^2 + 0c + 0)\} \\
 & + a^2 \{b^3 \cdot 0 + b^2(0c^3 + 0c + 0) + b(-9c^4 + 36c^3 - 54c^2 + 36c - 9) \\
 & \quad + b^0(36c^5 - 171c^4 + 324c^3 - 306c^2 + 144c - 27)\} \\
 & + a^0 \{b^4(0c + 0) + b^3(7c^3 - 21c^2 + 21c - 7) + b^2(-39c^4 + 135c^3 - 171c^2 + 93c - 18) \\
 & \quad + b(66c^5 - 243c^4 + 333c^3 - 201c^2 + 45c) \\
 & \quad + b^0(-27c^6 + 101c^5 - 141c^4 + 87c^3 - 20c^2)\}
 \end{aligned}$$

which for  $c = 1$  becomes  $= 0$ .

It follows that for  $c = d = e = f = 1$ , the value of the covariant  $S$  is  $= 2(b - 1)^6 y^4$ , which might be easily verified.

For  $T$ ,  $U$ , and  $W$ , I look at the sums for the different combinations of  $a$  and  $b$ . The tables of these sums (book pp. 306–309) are reproduced verbatim above in the headings under  $T$ ,  $U$ ,  $V$  and  $W$ ; only the principal verification identities are quoted here:

**For  $T$  (book p. 306):** the  $x$ -coefficient successive sums for  $a^4$  are  $-2, +14, -42, +70, -70, +42, -14, +2$ , the binomial coefficients of  $-2(\theta - 1)^7$ ; the total  $\pm 20196$  on each side.

**For  $U$  (book p. 307):** successive sums vanish appropriately. Total  $\pm 17264$ .

**For  $V$  (book p. 308):** the  $x$ -coefficient sums for  $a^4$  are  $-4, +28, -84, +140, -140, +84, -28, +4$ , the coefficients of  $-4(\theta - 1)^7$ ; for  $a^3$ , the sums are  $18, -140, +476, -924, +1120, -868, +420, -116, +14$ , the coefficients of  $18(\theta - 1)^8 + 4(\theta - 1)^7$ . In the  $y$ -coefficient the successive sums are  $-2, +16, -56, +112, -140, +112, -56, +16, -2$ , the coefficients of  $-2(\theta - 1)^8$ . Totals  $\pm 98358, \pm 154122$  and  $\pm 186734, \pm 126760$ .

**For  $W$  (book p. 309):** For the terms in  $a^4$  the successive sums are

$$1, -10, +45, -120, +210, -252, +210, -120, +45, -10, +1,$$

which are the coefficients of  $-(\theta - 1)^{10}$ ; and for the terms in  $a^3$  the successive sums are

$$-5, +54, -265, +780, -1530, +2100, -2058, +1440, -705, +230, -45, +4,$$

which are the coefficients of  $-5(\theta-1)^{11}-(\theta-1)^{10}$ . Totals  $\pm 2207351, \pm 2972759, \pm 3003111, \pm 3111879, \pm 9087749$ .

*[End of paper 143.]*

# 144. A Third Memoir upon Quantics.

## A THIRD MEMOIR UPON QUANTICS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. cxlvi. for the year 1856, pp. 627–647. Received March 13,—Read April 10, 1856.]

My object in the present memoir is chiefly to collect together and put upon record various results useful in the theories of the particular quantics to which they relate. The tables at the commencement relate to binary quantics, and are a direct sequel to the tables in my Second Memoir upon Quantics, vol. cxlvi. (1856), [141]. The definitions and explanations in the next part of the present memoir are given here for the sake of convenience, the further development of the subjects to which they relate being reserved for another occasion. The remainder of the memoir consists of tables and explanations relating to ternary quadrics and cubics.

*Covariant and other Tables, Nos. 27 to 50 (Nos. 1 to 50 binary quantics).*<sup>1</sup>

Nos. 27 to 29 are a continuation of the tables relating to the quintic

$$(a, b, c, d, e, f \mid x, y)^5.$$

No. 27 gives the values of the different determinants of the matrix

$$\begin{pmatrix} a, & 4b, & 6c, & 4d, & e & \cdot \\ \cdot & a, & 4b, & 6c, & 4d, & e \\ f, & 4c, & 6d, & 4e, & f & \cdot \\ \cdot & f, & 4c, & 6d, & 4e, & f \end{pmatrix}$$

determinants which are represented by 1234, 1235, &c., where the numbers refer to the different columns of the matrix. No. 28 gives the values of certain linear functions of these determinants, viz.

$$\begin{aligned} L &= +12 \cdot 56 - 2345 - a \cdot 1346, \\ L' &= 3 \cdot 1256 - 1346, \\ 8M &= -1345 + 2 \cdot 1246, \\ 8M' &= -2346 + 2 \cdot 1356, \\ 8N &= -1245 + 3 \cdot 1236, \\ 8N' &= -2356 + 3 \cdot 1456, \\ 8P &= 2L' - 3L = 6 \cdot 1846 - 3 \cdot 2345, \\ 16P' &= -5L' - L = -16 \cdot 1256 - 3 \cdot 1346 - 2356. \end{aligned}$$

At the end of the two tables there are given certain relations which exist between the terms of Tables 14, 16, 25, 26, 27 and 28.

## No. 27.

[Eight columns, one for each of the determinants 1234, 1235, 1236, 1245, 1246, 1345, 1256, 2345 (continued on the next page with 1346, 2346, 1356, 2356, 1456, 2456, 3456). Each column lists the monomial terms of  $a, b, c, d, e, f$  appearing in the expansion of the corresponding minor, together with the signed integer coefficients.]

| 1234.         | 1235.        | 1236.       | 1345.        | 1246.       | 1345.        | 1256.        | 2345.   |
|---------------|--------------|-------------|--------------|-------------|--------------|--------------|---------|
| $a^3bf \dots$ | $a^3a^2 - 4$ | $a^2df + 6$ | $a^2df - 6$  | $a^2cf + 4$ | $a^2cy$      | $a^2y^2 + 1$ | $a^2cf$ |
|               |              |             |              |             |              | $abef -- 2$  |         |
| $a^2ce - 16$  | $a^2de + 24$ | $a^3 \dots$ | $a^2cf + 16$ | $abdf - 4$  | $ab^2y - 24$ | $acdf -- 16$ | $abef$  |

<sup>1</sup>The Tables 49 A and 50 were inserted October 6, 1856.—A. C.

|                       |                               |                           |                |                                            |                            |                                              |                |
|-----------------------|-------------------------------|---------------------------|----------------|--------------------------------------------|----------------------------|----------------------------------------------|----------------|
| $a^2cd + 36$          | $aby + 4$                     | $abcf - 12$<br>$abde - 6$ | $abcf + 6$     | $ade^2 - 4$<br>$acy - 24$<br>$bcd^2e - 26$ | $bde^2 + 64$<br>$acy + 24$ | $ace^2 + 16$<br>$ad^2e + 16$<br>$b^2df - 15$ | $acdf + 20$    |
| $ab^2e + 16$          | $cd^2ce - 84$<br>$abd^2 - 24$ | $ace^2 + 16$              |                |                                            | $acrfe - 208$              | $b^2y$                                       | $- ace^2 80$   |
| $abcd - 152$          | $ac^2d + 64$<br>$b^2e + 60$   |                           | $ac^2e - 96$   | $acde + 24$                                | $ad^2 + 144$               |                                              | $ad^2e + 60$   |
| $mc^3 + 96$<br>$+ 80$ | $b^2cd - 40$                  | $ac(d^2) \dots$           | $ac(d^2) + 96$ | $ad^2$                                     | $b^2y$                     |                                              | $b^2cy - 80$   |
|                       | $be^4$                        | $b^4y + 16$               | $by \dots$     | $b^2y + 24$                                | $bcy - 40$                 |                                              | $b^2y + 240$   |
| $- bc^4 60$           |                               | $b^2ce - 10$              | $b^2ce + 90$   | $b^2ce - 20$                               | $+ 60 bcr^2$               | $bcdy$<br>$bcde$                             | $bcdy + 60$    |
|                       |                               | $b^2cd \dots$             | $- b^2cd 80$   | $bc^2e \dots$                              | $bcdy - 40$                | $bcd$                                        | $ictfe - 860$  |
|                       |                               | $bc^2d \dots$             | $6c^2d \dots$  | $bed^2 \dots$                              | $c^2d$                     |                                              | $bd^2 + 960$   |
|                       |                               | $c^3$                     | $c^3$          | $c^2d$                                     |                            | $c^2e$                                       | $c^2e + 960$   |
|                       |                               |                           |                |                                            |                            |                                              | $- c^2d^2 320$ |

[Continued on book p. 311 with columns 1346, 2346, 1356, 2356, 1456, 2456, 3456 in the same format.]

## No. 28.

[Eight-column table listing the same monomials in  $a, b, c, d, e, f$  and giving the integer coefficients which appear in the expressions for  $N, M, L, L', P, F, M', N'$ .]

| N.                           | M.                       | L.                            | L'.                        | P.                        | F.                           | M'.           | N'.                        |
|------------------------------|--------------------------|-------------------------------|----------------------------|---------------------------|------------------------------|---------------|----------------------------|
| $a^2cf + 3$                  | $a^2cf + 1$              | $a^2y^2 + 1$                  | $a^2y^2 + 3$               | $a^2y^2 \dots$            | $a^2y - 1$                   | $ab^2f + 1$   | $a^3y - 3$                 |
| $- b^2cd 2$                  | $abdf + 2$               | $abef - 34$<br>$acdf + 76$    | $abef - 22$                | $abef + 1$<br>$acdf - 3$  | $abef + 9$<br>$acdf - 1$     | $acef + 2$    | $odef - 9$                 |
| $abcf - 9$                   | $abe^2 - 9$              | $ace^2 - 32$                  | $acdf - 12$                | $ace^2 + 2$               |                              | $ady - 9$     | $oe^4 + 6$                 |
| $abde^2 + 1$                 | $adf - 9$<br>$acde + 32$ | $ad^2e - 12$                  | $ace^2 + 64$               | $ad^2e$                   | $ace^2 - 18$<br>$ad^2e + 12$ | $ad^2e + 6$   | $jy - 2$                   |
| $eude + 18$<br>$aecd^2 - 12$ | $ad^2 - 18$              | $b^4y - 32$<br>$b^2c + 225$   | $acd^2e - 36$              | $b^2c + 2$                |                              | $b^2cf - 9$   | $bcy + 1$                  |
|                              |                          | $bc^2f - 12$                  | $bcde + 64$                | $b^2cf - 9$               | $b^2cf - 18$<br>$b^2c$       | $bcdy - 32$   | $bcdy + 18$<br>$- bd^4 15$ |
| $b^2y + 6$<br>$b^2ce - 15$   | $b^2cy + 6$              | $bcdy - 820$<br>$bd^4 + 480$  | $bc^2y - 45$<br>$bcy - 36$ | $bef$                     | $bcy + 12$<br>$bccdy + 45$   | $bce^2 \dots$ | $edf - 12$                 |
| $b^2cd^2 + 10$               | $b^2cde \dots$           | $c^2 + 480$<br>$c^2d^2 - 320$ | $bcdy + 20$                | $bcde + 31$<br>$bdy - 18$ | $bc^2 - 30$                  | $bd^2e - 15$  |                            |
|                              | $bc^2y - 15$             |                               |                            | $c^2e - 18$               | $bce - 30$<br>$c^2d^2 + 12$  | $cy - 18$     | $c^2e^2 + 10$              |
| $bc^2 \dots$                 | $bcd^2 + 10$             |                               | $bcd^2$                    |                           | $c^2d^2 + 20$                | $c^2de + 10$  | $cd^2e \dots$              |
| $c^3$                        | $c^3d \dots$             |                               | $c^3e$<br>$c^2d^2 \dots$   |                           |                              | $cd^2$        | $d^4$                      |

If the coefficients of the table 14 are represented by  $\frac{1}{2}A, B, \frac{1}{2}C$ , viz. writing

$$A = 2(ae - 4bd + 3c^2), \quad B = af - 3be + 2cd, \quad C = 2(bf - 4ce + 3d^2),$$

then we have the following relations between 1234, &c. and  $A, B, C$ , viz.

$$\begin{array}{llll}
 1234 = & Cx & + Bx & + Ax \\
 1235 = & & & \text{(table continues, see scan)} \\
 1236 = & + 6 a^2 & - 12 ab & + 16 oc - 10 b^2 \\
 1245 = & + 6 ab & - 2 oc - 10 b^2 & + 6 ac; \\
 1246 = & & + 6 ad - 18 bc & \\
 1345 = & - 2 ac + 8 b^2 & - 6 ad - 30 bc & - 2 df + 8 b^2 \\
 1256 = & & + 4 ae - 4 bd - 24 c^2 & + 8 ae + 10 bd \\
 2345 = & + 18 ac & - 8 ac - 40 bc; & \\
 1346 = & + 12 bc & & + 4 be + 8 cd \\
 2346 = & + 24 ac & + I af + 5 be - 18 cd & + 4 af + 20 be \\
 1356 = & & - 80 be + 20 cd & \\
 2356 = & - I ae + 4bd + 3 c^2 & & - I bf + 4 ce + 3d^2 \\
 1456 = & & - 3 bcd & + 20 bf + 40 ce - 30 cf \\
 2456 = & + 20 ad + 40 bc - 30 e^2 & - 8 bf - 4 ce & \\
 3456 = & & & + 4 bf + 8 ce + 6 d^2 \\
 & + 4 ae + 8bd + 6 c^2 & + 4 bf - 4 ce - 24 c^2 & + 24 cf \\
 & & - 6 cf - 30 cfe & + 12 cfe \\
 & + 4 af + 20 be & & \\
 & + 4 be + 8 cd & + 8 cf - 18 de & + 18 df \\
 & + 8 bf + 10 ce & - 2 df - 10 e^2 & - 2 df + 8 e^2 \\
 & + 6 ce & - 12 ef & + 6 ef \\
 & + 6 cf & & + 6 f^2 \\
 & + 16 df - 10 e^2 & & 
 \end{array}$$

and the following relations between  $L, L'$ , &c. and  $A, B, C$ , viz.

$$\begin{array}{lll}
 N = & Cx & + Bx & + Ax \\
 M = & - 3ac + 3b^2 & + 3ad - 3bc & - 1ae + 1be \\
 & - 3ad + 3bc & & - 1af + 1cd \\
 L = & + 11ae + 28bd - 39c^2 & + 3ae - 3d^2 & + 11bf + 28ce - 39d^2 \\
 & - 7af + 4bd + 3c^2 & & - 7bf + 4ce + 3d^2 \\
 L' = & & + 1af - 75be + 74cd & + 1bf + 2ce - 3d^2 \\
 2P = & - 7ae - 2bd + 8c^2 & + 3af + 15be - 18cd & + 3bf - 6ce + 3d^2 \\
 & + 3ae - 6bd + 3c^2 & + 3be - 3cd & \\
 F = & & - 1af + 1cd & y - 3 + 3de \\
 & - 1af + 1cd & + 3bf - 3d^2 & \\
 M' = & & + 3cf - 3de & - 3df + 3e^2 \\
 N' = & - 1bf + 1ce & & 
 \end{array}$$

We have also the following relations between  $L, L'$ , &c. and  $a, b, c, d, e, f$ , viz.

$$\begin{aligned}
 aP - bM + cN &= 0, \\
 aM' + bP' - 2cM + 3dN &= 0, \\
 aN' + 2bM' - cL' + 3eN &= 0, \\
 3bN' - dL' + 2eM + fN &= 0, \\
 3cN' - 2dM' + eP' + fM &= 0, \\
 dN' - eM' + fP &= 0.
 \end{aligned}$$

The quartinvariant No. 19  $[G]$  is equal to

$$-AC + B^2,$$

i.e. it is in fact equal to 4 into the discriminant of the quintic No. 14,  $[A]$ .



The octinvariant No. 25  $[Q]$  is expressible in terms of the coefficients of Nos. 14 and 16, viz.  $A, B, C$ , as before, and  $\frac{1}{2}a, \beta, \gamma, \frac{1}{2}\delta$  the coefficients of No. 16,  $[D]$ , i.e.

$$\begin{aligned}\alpha &= 3(ace - ad^2 - b^2e + 2bcd - c^3), \\ \beta &= acf - ade - b^2f + bd^2 + bce - c^2d, \\ \gamma &= adf - ae^2 - bcf + bde + c^2e - cd^2, \\ \delta &= 3(bdf - be^2 + 2cde - c^2f - d^3),\end{aligned}$$

then No. 25 is equal to

$$\begin{vmatrix} A & B & C \\ \alpha & \beta & \gamma \\ \beta & \gamma & \delta \end{vmatrix}.$$

The value of the discriminant No. 26,  $[Q']$ , is

$$(\text{No. 19})^2 - 128 \text{ No. 25}, \quad [\text{that is } Q' = G^2 - 128Q.]$$

We have also an expression for the discriminant in terms of  $L, L'$ , &c., viz. three times the discriminant No. 26 is equal to

$$[\text{or say } 3Q' =] LL' + 64MM' - 64NN',$$

a remarkable formula, the discovery of which is due to Mr. Salmon.

It may be noticed, that in the particular case in which the quintic has two square factors, if we write

$$(a, b, c, d, e, f \mid x, y)^5 = \{(p, q, r \mid x, y)^2\}^2 \cdot (\lambda, \mu \mid x, y),$$

then

$$\begin{aligned}a &= \lambda p^2, & b &= \frac{1}{4}(p^2\mu + 4pq\lambda), & c &= \frac{1}{2}(q^2 + pr)\lambda + 2pq\mu, \\ f &= 5r^2\mu, & e &= r^2\lambda + 4qr\mu, & d &= 2qr\lambda + (2q^2 + pr)\mu;\end{aligned}$$

and these values give

$$\begin{aligned}P &= K(6q^2 - pr), & P' &= K(10q^2 - 15pr), \\ M &= K \cdot 10pq, & M' &= K \cdot 10qr, \\ N &= K \cdot 5p^2, & N' &= K \cdot 6r^3,\end{aligned}$$

where the value of  $K$  is

$$\frac{1}{8}(p\mu^2 - 2q\mu\lambda + r\lambda^2)(pr - q^2)^3.$$

The table No. 29 is the invariant of the twelfth degree of the quintic, given in its simplest form, i.e. in a form not containing any power higher than the fourth of the leading coefficient  $a$ : this invariant was first calculated by M. Faà de Bruno.

### No. 29. [See U. No. 29, p. 294.]

[Reference back to the corresponding entry in the U-table of paper 143.]

The tables Nos. 30 to 35 relate to a sextic. No. 30 is the sextic itself; No. 31 the quadrinvariant; Nos. 32 and 33 the quadricovariants (the latter of them the Hessian); No. 34 is the quartinvariant or catalecticant; and No. 35 is the sextinvariant in its best form, i.e. a form not containing any power higher than the second of the leading coefficient  $a$ .

### No. 30.

$$(a + 1, b + 6, c + 15, d + 20, e + 15, f + 6, g + 1 \mid x, y)^6.$$

**No. 31.**

$$\begin{array}{l} ag + 1, \\ bd - 6, \\ ce + 15, \text{ coefficient of } ag^0e^0; \\ d^2 - 10, \end{array} \quad \text{sum } \pm 16.$$

**No. 32.**

|            |                         |                          |
|------------|-------------------------|--------------------------|
| $ae + 1,$  | $af + 2, \quad bg + 1,$ | $ag + 1, \quad cg + 1$   |
| $bd - 4,$  | $be - 6, \quad cf - 6,$ | $ce - 9, \quad df - 4$   |
| $c^2 + 3,$ | $cd + 4, \quad de + 4,$ | $d^2 + 8, \quad e^2 + 3$ |
| $\pm 4$    | $\pm 6, \quad \pm 6$    | $\pm 9$                  |

 $(\diamond x, y)^2.$

**No. 33.**

|           |          |            |           |            |           |          |            |           |
|-----------|----------|------------|-----------|------------|-----------|----------|------------|-----------|
| $ac + 1$  | $ad + 4$ | $ae + 6$   | $af + 4$  | $ag + 1$   | $bg + 4$  | $cg + 6$ | $dg + 4$   | $eg + 1$  |
|           | $bc - 4$ |            | $be + 16$ | $bf + 14$  | $cf + 16$ |          | $df + 4$   | $ef - 4$  |
| $b^2 - 1$ |          | $bd + 4$   | $cd - 20$ | $ce + 5$   |           | $df + 4$ | $e^2 - 10$ | $f^2 - 1$ |
|           |          | $c^2 - 10$ |           | $d^2 - 20$ | $de - 20$ |          |            |           |
| $\pm 1$   | $\pm 4$  | $\pm 10$   | $\pm 20$  | $\pm 20$   | $\pm 20$  | $\pm 10$ | $\pm 4$    | $\pm 1$   |

 $(\diamond x, y)^4.$

**No. 34.**

[The quartinvariant or catalecticant of the sextic: a long list of monomial-coefficient pairs which sum, as printed in the source, to  $\pm 12$ . The table is reproduced below in OCR-skeleton form.]

|                 |                |                |                 |                        |                |                      |            |
|-----------------|----------------|----------------|-----------------|------------------------|----------------|----------------------|------------|
| $a^2dg^2 + 1$   | $acd^2y - 42$  | $bcdfg + 60$   | $bdf^2 + 24$    | $c^2eg + 12$           | $c^2y - 27$    | $- c^2d^2gr \quad 8$ |            |
| $acde^2 + 12$   |                |                |                 | $- 24 \quad bd^2e^2$   |                |                      |            |
| $a^2defg - 6$   |                |                |                 |                        | $bdy^2 + 24$   | $cdy^2 - 27$         |            |
| $a^2y^2 + 4$    | $ad^2g - 20$   | $bcfeg^2 + 4$  | $c^2gr + 12$    | $c^2e^2/ + 66$         | $c^2eg^2 - 24$ |                      | $c^2def$   |
| $a^2bf^2 + 4$   | $a^2cef + 24$  | $bg + 12$      | $c^2cg^2 + 24$  | $- cf^2 - 24$          | $cy^2 - 24$    |                      | $c^2y - 6$ |
| $a^2y^2 - 3$    | $abce^2 - 8$   | $by^2 + 8$     | $cd^2eg^2 - 3$  | $- 39 \quad c^2d^2e^2$ | $cd^2g - 24$   |                      | $c^2cd^2$  |
| $ab^2cdg^2 - 6$ | $b^2gr - 12$   | $b^2cdy + 30$  | $cd^2eg^2 - 39$ | $cd^2e^2 + 36$         |                |                      |            |
| $abce/g + 18$   | $by^2 + 8$     | $b^2ce^2 - 24$ |                 |                        | $c^2y - g$     |                      |            |
| $abc^2y - 12$   |                |                |                 |                        |                |                      |            |
|                 |                |                |                 |                        |                |                      |            |
| $ab^2dy^2 + 12$ | $b^2ce^2 - 12$ |                |                 |                        |                |                      |            |
| $abdee^2 - 18$  | $b^2dy^2 - 24$ |                |                 |                        |                |                      |            |
| $abe^2y + 6$    |                |                |                 |                        |                |                      |            |
| $ac^2d^2 + 4$   | $b^2cdy + 60$  |                |                 |                        |                |                      |            |
| $ac^2eg - 24$   |                |                |                 |                        |                |                      |            |
| $ac^2dy - 18$   | $b^2c^2 - 27$  |                |                 |                        |                |                      |            |
| $ab/^2 + 30$    | $bcefg + 6$    |                |                 |                        |                |                      |            |
| $acd^2eg + 54$  | $bcdef - 42$   |                |                 |                        |                |                      |            |
| $acrdy - 12$    |                |                |                 |                        |                |                      |            |
| $\pm 12$        |                |                |                 |                        |                | $\pm 565$            |            |

[Layout preserved from the OCR; the column totals are  $\pm 12$  and  $\pm 565$  in the printed table; sub/superscript glitches reflect the OCR not the manuscript.]

The sextinvariant may be thus represented by means of a determinant of the sixth order and of the quadrinvariant and quartinvariant:

$$6 \times \text{No. 35} = \begin{vmatrix} a, & 2b, & 3c, & 4d, & e, & \cdot & \cdot \\ \cdot & a, & 2b, & 3c, & 4d, & e, & \cdot \\ \cdot & \cdot & a, & 2b, & 3c, & 4d, & e \\ a, & 4b, & 3c, & 2d, & e, & \cdot & \cdot \\ \cdot & a, & 4b, & 3c, & 2d, & e, & \cdot \\ \cdot & \cdot & a, & 4b, & 3c, & 2d, & e \end{vmatrix} + \frac{1}{4}(ag - 6bf + 15ce - 10d^2) \begin{vmatrix} a, & b, & c, & d \\ b, & c, & d, & e \\ c, & d, & e, & f \\ d, & e, & f, & g \end{vmatrix}.$$

The tables Nos. 36 and 37 relate to a septic. No. 36 is the septic itself; No. 37 the quartinvariant.

**No. 36.**

$$(a + 1, b + 7, c + 21, d + 35, e + 35, f + 21, g + 7, h + 1 \mid x, y)^7.$$

**No. 37.**

[The quartinvariant of the septic: monomial expansion totalling  $\pm 2223$  as in the source. Sample terms:]

$$\begin{aligned} & a^2h^2 - 1, \quad bd^2h - 40, \quad bdeg - 50, \quad bdf^2 - 360, \\ & abgh + 14, \quad b^2fh + 240, \quad acgh - 18, \quad c^2eg - 360, \\ & acfh - 24, \quad cf^2 - 81, \quad afg^2 + 10, \quad cd^2g + 240, \\ & adeh + 60, \quad cdef + 990, \quad adfg - 40, \quad cef - 600, \\ & adf - 24, \quad cf^2 - 600, \quad ae^2g - 25, \quad d^2e^2 + 375, \\ & b^2fh + 234, \quad by + 60, \quad bcfg, \quad bceh, \quad \text{etc.} \end{aligned}$$

Total:  $\pm 2223$ .

The tables Nos. 38 to 45 relate to the octavic. No. 38 is the octavic itself; No. 39 the quadrinvariant; Nos. 40, 41 and 42 are the quadricovariants, the last of them being the Hessian; No. 43 is the cubinvariant; No. 44 the quartinvariant, and No. 45 the quintinvariant, which is also the catalecticant.

**No. 38.**

$$(a + 1, b + 8, c + 28, d + 56, e + 70, f + 56, g + 28, h + 8, i + 1 \mid x, y)^8.$$

**No. 39.**

$$ai + 1, \quad bh - 8, \quad cy + 28, \quad df - 56, \quad e^2 + 35; \quad \text{sum } \pm 64.$$

**Nos. 40, 41, 42.**

[Tabulations of the quadricovariants  $(\diamond x, y)^2$ ,  $(\diamond x, y)^4$ ,  $(\diamond x, y)^6$  of the octavic; the column-sums are  $\pm 16$ ,  $\pm 20$ ,  $\pm 35$ ,  $\pm 20$ ,  $\pm 16$  for No. 40;  $\pm 4$ ,  $\pm 12$ ,  $\pm 30$ ,  $\pm 48$ ,  $\pm 58$ ,  $\pm 48$ ,  $\pm 30$ ,  $\pm 12$ ,  $\pm 4$  for No. 41;  $\pm 1$ ,  $\pm 6$ ,  $\pm 21$ ,  $\pm 70$ ,  $\pm 105$ ,  $\pm 210$ ,  $\pm 189$ ,  $\pm 210$ ,  $\pm 105$ ,  $\pm 70$ ,  $\pm 21$ ,  $\pm 6$ ,  $\pm 1$  for No. 42.]

**No. 43. (The cubinvariant.)**

|                       |                                   |                        |                        |
|-----------------------|-----------------------------------|------------------------|------------------------|
| aei + 1               | acgi - 1                          | bcgh + 3               | cdei - 2               |
| afh -- 4              | ach <sup>2</sup> + 1              | bdei + 1               | cdeh - 23              |
| ag <sup>2</sup> + 3   |                                   |                        | cdfh + 27              |
| bdi -- 4              | adfi + 3                          | bdfh - 10              |                        |
|                       |                                   |                        | cdfg + 19              |
| beh + 12              | adgh - 3                          | bdg <sup>2</sup> + 9   |                        |
|                       |                                   |                        | ce <sup>2</sup> i - 21 |
| b <sup>2</sup> - 8    | -aei - 1 2                        | be <sup>2</sup> i + 11 |                        |
|                       |                                   |                        | cefh + 12              |
| c <sup>2</sup> i + 3  | aefh + 1                          | befg - 23              |                        |
|                       |                                   |                        | cy <sup>2</sup> -- 21  |
| cdh - 8               | aeg <sup>2</sup> + 3              | bfh + 12               |                        |
|                       |                                   |                        | d <sup>2</sup> eg - 13 |
| ceg - 22              |                                   | d <sup>2</sup> gi + 3  |                        |
|                       |                                   |                        | d <sup>2</sup> y - 32  |
| cy <sup>2</sup> + 24  | af <sup>2</sup> g - 2             | e <sup>2</sup> fh + 9  | - 10                   |
| d <sup>2</sup> g + 24 |                                   | de <sup>2</sup> f + 10 |                        |
| def -- 36             | b <sup>2</sup> gi + 1             | e <sup>3</sup> f - 12  |                        |
| e <sup>2</sup> + 15   | b <sup>2</sup> h <sup>2</sup> - 1 |                        |                        |
|                       |                                   |                        | ± 82                   |
|                       | bcfi - 3                          |                        |                        |

# **No. 44 (the quartinvariant) and No. 45 (the quintinvariant / catalecticant).**

[Both invariants are given by long tables of monomial-coefficient pairs; the column totals are  $\pm 147$  for No. 44 and  $\pm 56$  for No. 45 (printed at the foot of the tables in the source; some sub-totals are damaged in the scan).]

If we write

$$\text{No. 39} = I, \quad \text{No. 43} = J, \quad \text{No. 44} = K, \quad \text{No. 45} = L,$$

then the determinant called the lambdaic, viz.

$$\begin{vmatrix} a & b & c & d & e - 12\lambda \\ b & c & d & e + 3\lambda & f \\ c & d & e - 2\lambda & f & g \\ d & e + 3\lambda & f & g & h \\ e - 12\lambda & f & g & h & i \end{vmatrix}$$

is equal to

$$L + 4\lambda K + 24\lambda^2 J + 18\lambda^3 I - 2592\lambda^5.$$

Nos. 46 to 48 relate to the nonic. No. 46 is the nonic itself; Nos. 47 and 48 are the two quartinvariants, each of them in its best form, viz. No. 48 does not contain  $a^2$ , and No. 47 does not contain  $aci^3$ , the leading term of No. 48. The nonic is the lowest quantic with two quartinvariants.

## **No. 46.**

$$(a + 1, b + 9, c + 36, d + 84, e + 126, f + 126, g + 84, h + 36, i + 9, j + 1 \mid x, y)^9.$$

## **Nos. 47, 48.**

[The two quartinvariants of the nonic, listed as monomial-coefficient tables. The column-sums for No. 47 are  $\pm 41650$ ; for No. 48 they are  $\pm 698$ . Sample leading terms:]

No. 47.

```
a^2y^3 - 1      bfh - 720
abij + 18      b/g^2 + 432
aei^2          c^2fj + 1728

achj - 72      c^2gi - 720

adgj + 168     c^2y^2 2160
adhi          cdej + 4608
aei^2 - 108    cdfi - 576
- ciegi 576    cdfg^2 + ...
aeh^2 + 432    ce^2y ...
afy + 540      cefh - 2592

b^2afgh -720   ce^2g - 5760
+320          cy^3 + 4320
              d^2j + 320

b^2b^2ij       d^2ei - 720
-- 81

bcgj + 648     dy^2 - 5760
bchi + 648     d^2g^2 - 1536

bdfj - 676     defg + 14688
bdgi + 792     deyh + 4320
bdh^2 - 1728
```

No. 48.

```
ay^3 ...      bfgh + 70
abij          bfg^2 - 45
acy^3 + 2     c^2ij + 27
achj - 2      c^2gi - 52

adgj + 7      c^2y^2 + 25
adhi - 7      cdej - 45
aej^2 - 5     cdfi + 23
aei^2 - 22    crfgr/h + 22
adi^2 + 27    c^2i + 70
af^2i + 25    c^2/h - 127
b^2fgh - 45   cef + 32
+ 20          c/^2 + 25

b^2fhj + 2    cf^2y + 20
bcg^2 - 7     d^2ei - 45
              +7 d^2y/h + 32

bcfhj         d^2g^2 + 47
              d^2y^2 + 85
              de^2 h + 25

bdfj -22      dy^2 - 50
bdgi + 74     - e^2 f/ 50
bdh^2 - 52
begh + 23
be^2 j + 25
```

$$\text{beg}^2 + 540 \quad \text{cd}^2\text{f}^2 - 8640$$

$$\text{bfy} - 73 \quad \text{ey}^2 + 30$$

$$\pm 41650 \quad \pm 698$$

Nos. 49, [49 A] and 50 relate to the dodecadic. No. 49 is the dodecadic itself; [No. 49 A, inserted in this place, but originally printed in the Fifth Memoir on Quantics, is the dodecadic quadricovariant]. No. 50 is the cubinvariant. [The numerical coefficients in this last table as originally printed in the Third Memoir were altogether erroneous, and the table as here printed is in fact the table No. 50 bis of the Fifth Memoir on Quantics.]

#### No. 49.

$$(a+1, b+12, c+66, d+220, e+495, f+792, g+924, h+792, i+495, j+220, k+66, l+12, m+1 \mid x, y)^{12}.$$

#### No. 49 A.

[The dodecadic quadricovariant, given as a thirteen-column tabulation  $(\diamond x, y)^{10}$  with column-sums  $\pm 16, \pm 60, \pm 189, \pm 450, \pm 810, \pm 1140, \pm 1270, \pm 1140, \pm 810, \pm 450, \pm 189, \pm 60, \pm 16.$ ]

#### No. 50 (Cubinvariant of the dodecadic, total $\pm 2200$ ).

$$\begin{array}{lll} \text{agm} + 1 & \text{cfj}^2 - 54 & \text{dhi} + 270 \\ \text{ahl} - 6 & \text{cgk} + 24 & \text{eg}^2 - 135 \\ \text{aik} + 15 & & \text{efj} + 270 \\ \text{aj}^2 - 10 & \text{c/fy} + 150 & \text{eg}^2 \text{ i} + 495 \\ \text{bfm} - 6 & \text{cf}^2 - 135 & \text{eie} - 540 \\ & & \text{f}^2 \text{ i} - 540 \\ \text{bgl} + 30 & \text{d}^2 \text{ m} - 10 & \\ \text{bhb} - 54 & \text{d}^2 \text{ j} + 30 & \text{fgh} + 720 \\ \text{bij} + 30 & \text{dfk} + 150 & (\text{g}^3 - 280) \\ & \text{cfgy} - 430 & \\ \text{cem} + 15 & & \end{array}$$

$$\pm 2200$$

*Resuming now the general subject:*

**54.** The simplest covariant of a system of quantics of the form

$$(\diamond x, y, \dots)^m$$

(where the number of quantics is equal to the number of the facients of each quantic) is the functional determinant or Jacobian, viz. the determinant formed with the differential coefficients or derived functions of the quantics with respect to the several facients.

**55.** In the particular case in which the quantics are the differential coefficients or derived functions of a single quantic, we have a corresponding covariant of the single quantic, which covariant is termed the *Hessian*; in other words, the Hessian is the determinant formed with the second differential coefficients or derived functions of the quantic with respect to the several facients.

**56.** The expression, an *adjoint linear form*, is used to denote a linear function  $\xi x + \eta y + \dots$ , or in the notation of quantics  $(\xi, \eta, \dots \mid x, y, \dots)$ , having the same facients as the quantic or quantics to which it belongs, and with indeterminate coefficients  $(\xi, \eta, \dots)$ . The invariants of a quantic or quantics, and of an adjoint linear form, may be considered as quantics having  $(\xi, \eta, \dots)$  for facients, and of which the coefficients are of course functions of the coefficients of the given quantic or quantics. An invariant of the class in question is termed a *contravariant* of the quantic or quantics. The idea of a contravariant is due to Mr. Sylvester.

In the theory of binary quantics, it is hardly necessary to consider the contravariants; for any contravariant is at once turned into an invariant by writing  $(y, -x)$  for  $(\xi, \eta)$ .

**57.** If we imagine, as before, a system of quantics of the form

$$(\diamond x, y, \dots)^m$$

where the number of quantics is equal to the number of the facients in each quantic, the function of the coefficients, which, equated to zero, expresses the result of the elimination of the facients from the equations obtained by putting each of the quantics equal to zero, is said to be the *Resultant* of the system of quantics. The resultant is an invariant of the system of quantics.

And in the particular case in which the quantics are the differential coefficients, or derived functions of a single quantic with respect to the several facients, the resultant in question is termed the *Discriminant* of the single quantic; the discriminant is of course an invariant of the single quantic.

**58.** Imagine two quantics, and form the equations which express that the differential coefficients, or derived functions of the one quantic with respect to the several facients, are proportional to those of the other quantic. Join to these the equations obtained by equating each of the quantics to zero; we have a system of equations, one of which is contained in the others, and from which therefore the facients may be eliminated. The function which, equated to zero, expresses the result of the elimination is an invariant which (from its geometrical signification) might be termed the *Tactinvariant* of the two quantics, but I do not at present propose to consider this invariant except in the particular case where the system consists of a given quantic and of an adjoint linear form. In this case the tactinvariant is a contravariant of the given quantic, viz. the contravariant termed the *Reciproquant*.

**59.** Consider now a quantic

$$(\diamond x, y, \dots)^m,$$

and let the facients  $x, y, \dots$  be replaced by  $\lambda x + \mu X, \lambda y + \mu Y, \dots$ ; the resulting function may, it is clear, be considered as a quantic with the facients  $(\lambda, \mu)$  and of the form

$$(\diamond x, y, \dots)_{\text{eman.}}^m.$$

The coefficients of this quantic are termed *Emanants*, viz., excluding the first coefficient, which is the quantic itself (but which might be termed the 0-th emanant), the other coefficients are the first, second, and last or ultimate emanants. The ultimate emanant is, it is clear, nothing else than the quantic itself, with  $(X, Y, \dots)$  instead of  $(x, y, \dots)$  for facients; the penultimate emanant is, in like manner, obtained from the first emanant by interchanging  $(x, y, \dots)$  with  $(X, Y, \dots)$ , and similarly for the other emanants. The facients  $(X, Y, \dots)$  may be termed the facients of emanation, or simply the new facients. The theory of emanation might be presented in a more general form by employing two or more sets of emanating facients; we might, for example, write  $\lambda x + \mu X + \nu X', \lambda y + \mu Y + \nu Y', \dots$  for  $x, y, \dots$ , but it is not necessary to dwell upon this at present.

The invariants, in respect to the new facients, of any emanant or emanants of a quantic (i.e. the invariants of the emanant or emanants, considered as a function or functions of the new facients), are, it is easy to see, covariants of the original quantic, and it is in many cases convenient to define a covariant in this manner; thus the Hessian is the discriminant of the second or quadric emanant of the quantic.

**60.** If we consider a quantic

$$(a, b, \dots \mid x, y, \dots)^m,$$

and an adjoint linear form, the operative quantic

$$(\partial_a, \partial_b, \dots \mid \xi, \eta, \dots)^k$$

(which is, so to speak, a contravariant operator) is termed the *Evector*. The properties of the evector have been considered in the introductory memoir, and it has been in effect shown that the evector operating upon an invariant, or more generally upon a contravariant, gives rise to a contravariant. Any such contravariant, or rather such contravariant considered as so generated, is termed an *Evectant*. In the case of a binary quantic,

$$(a, b, \dots \mid x, y)^m,$$

the covariant operator

$$(\partial_a, \partial_b, \dots | y, -x)^k$$

may, if not with perfect accuracy, yet without risk of ambiguity, be termed the Evector, and a covariant obtained by operating with it upon an invariant or covariant, or rather such covariant considered as so generated, may in like manner be termed an Evectant.

**61.** Imagine two or more quantics of the same order,

$$(a, b, \dots | x, y)^m, \quad (\alpha, \beta, \dots | x, y)^m, \quad \dots$$

we may have covariants such that for the coefficients of each pair of quantics the covariant is reduced to zero by the operators

$$\begin{aligned} a\partial_\alpha + b\partial_\beta + \dots, \\ \alpha\partial_a + \beta\partial_b + \dots \end{aligned}$$

Such covariants are called *Combinants*, and they possess the property of being invariative, quoad the system, i.e. the covariant remains unaltered to a factor près, when each quantic is replaced by a linear function of all the quantics. This extremely important theory is due to Mr. Sylvester.

Proceeding now to the theory of ternary quadrics and cubics.

*First for a ternary quadric, we have the following tables:*

*Covariant and other Tables, Nos. 51 to 56 (a ternary quadric).*

**No. 51.**

The quadric is represented by

$$(a, b, c, f, g, h | x, y, z)^2,$$

which means

$$ax^2 + by^2 + cz^2 + 2fyz + 2gzx + 2hxy.$$

**No. 52.**

The first derived functions (omitting the factor 2) are

$$\begin{aligned} (a, h, g | x, y, z), \\ (h, b, f | x, y, z), \\ (g, f, c | x, y, z). \end{aligned}$$

**No. 53.**

The operators which reduce a covariant to zero are

$$\begin{aligned} (h, b, 2f | \partial_g, \partial_h, \partial_c) - z\partial_y, \\ (a, 2h, g | \partial_h, \partial_b, \partial_f) - y\partial_z, \\ (g, 2f, c | \partial_a, \partial_h, \partial_g) - z\partial_x, \\ (a, h, 2g | \partial_h, \partial_b, \partial_f) - x\partial_y, \\ (2h, b, f | \partial_a, \partial_h, \partial_g) - x\partial_z, \\ (2g, f, c | \partial_a, \partial_h, \partial_g) - z\partial_x. \end{aligned}$$

**No. 54.**

The evector is

$$(\partial_a, \partial_b, \partial_c, \partial_f, \partial_g, \partial_h | \xi, \eta, \zeta)^2.$$

**No. 55.**

The discriminant is

$$\begin{vmatrix} a, & h, & g \\ h, & b, & f \\ g, & f, & c \end{vmatrix},$$

which is equal to

$$abc - af^2 - bg^2 - ch^2 + 2fgh.$$

**No. 56.**

The reciprocant is

$$\begin{vmatrix} 0, & \xi, & \eta, & \zeta \\ \xi, & a, & h, & g \\ \eta, & h, & b, & f \\ \zeta, & g, & f, & c \end{vmatrix},$$

which is equal to

$$(bc - f^2, ca - g^2, ab - h^2, gh - af, hf - bg, fg - ch \mid \xi, \eta, \zeta)^2.$$

The discriminant is, it will be noticed, the same function as the Hessian. The reciprocant is the evectant of the discriminant. The covariants are the quadric itself and the discriminant; the reciprocant is the only contravariant.

*Next, for a ternary cubic, we have the following Tables.*

*Covariant and other Tables, Nos. 57 to 70 (a ternary cubic).*

**No. 57.**

The cubic is  $U =$

$$(a, b, c, f, g, h, i, j, k, l \mid x, y, z)^3,$$

which means

$$ax^3 + by^3 + cz^3 + 3fy^2z + 3gz^2x + 3hx^2y + 3iyz^2 + 3jzx^2 + 3kxy^2 + 6lxyz.$$

**No. 58.**

The first derived functions (omitting the factor 3) are

$$\begin{aligned} &(a, k, g, l, j, h \mid x, y, z)^2, \\ &(h, b, i, f, l, k \mid x, y, z)^2, \\ &(j, f, c, i, g, l \mid x, y, z)^2. \end{aligned}$$

The second derived functions (omitting the factor 6) are

$$\begin{aligned} &(a, h, j \mid x, y, z), \\ &(k, b, f \mid x, y, z), \\ &(g, i, c \mid x, y, z), \\ &(l, f, i \mid x, y, z), \\ &(j, l, g \mid x, y, z), \\ &(h, k, l \mid x, y, z). \end{aligned}$$



**No. 59.**

The operators which reduce a covariant to zero are

$$\begin{aligned} & (j, 3f, c, 2i, g, 2l \mid \partial_h, \partial_b, \partial_i, \partial_f, \partial_l, \partial_k) - y\partial_z, \\ & (a, k, 3g, 2l, 2j, h \mid \partial_j, \partial_f, \partial_c, \partial_i, \partial_g, \partial_l) - z\partial_z, \\ & (3h, b, i, f, 2l, 2k \mid \partial_a, \partial_k, \partial_g, \partial_l, \partial_j, \partial_h) - x\partial_y, \\ & (h, b, 3i, 2f, 2l, k \mid \partial_j, \partial_f, \partial_c, \partial_i, \partial_g, \partial_l) - z\partial_y, \\ & (3j, f, c, i, 2g, 2l \mid \partial_a, \partial_k, \partial_g, \partial_l, \partial_j, \partial_h) - x\partial_z, \\ & (a, k, g, 2l, j, 2h \mid \partial_k, \partial_b, \partial_i, \partial_f, \partial_l, \partial_k) - y\partial_x. \end{aligned}$$

**No. 60.**

The evector is

$$(\partial_a, \partial_b, \partial_c, \partial_f, \partial_g, \partial_h, \partial_i, \partial_j, \partial_k, \partial_l \mid \xi, \eta, \zeta)^3.$$

**No. 61.**

The Hessian is  $HU =$

$$\begin{vmatrix} (a, h, j \mid x, y, z), & (h, k, l \mid x, y, z), & (j, l, g \mid x, y, z) \\ (h, k, l \mid x, y, z), & (k, b, f \mid x, y, z), & (l, f, i \mid x, y, z) \\ (j, l, g \mid x, y, z), & (l, f, i \mid x, y, z), & (g, i, c \mid x, y, z) \end{vmatrix},$$

which is equal to

$$(agh - 1, \ bhi - 1, \ -cij, \ bch - 1, \ acf - 1, \ abg - 1, \ bcj - 1, \ ach - 1, \ abi - 1, \ ahc - 1; \dots \mid x, y, z)^6.$$

[The table runs across ten columns with  $\pm 3, \pm 3, \pm 3, \pm 6, \pm 5, \pm 5, \pm 5, \pm 5, \pm 5, \pm 9$  as column sums. The body of the table is a long enumeration of monomial-coefficient pairs which, by the size and density of the printing, cannot be faithfully retyped without the scan; see the original.]

**No. 62.**

The quartinvariant is  $S =$

$$\begin{aligned} & abcl - 1, \quad f jy + 1, \\ & abgi + 1, \quad f^2 h + 3, \\ & acfk + 1, \quad fgjk - 1, \\ & af^2 j - 1, \quad fh iy - 1, \\ & afil + 1, \quad fjl^2 - 2, \\ & afi^2 - 1, \quad ghi^2 + 1, \\ & aghk - 1, \quad ghki - 1, \\ & bchj + 1, \quad ghl^2 - 2, \\ & bfh - 1, \quad h^2 i^2 + 1, \\ & bgji + 1, \quad hil^2 - 2, \\ & bgj^2 - 1, \quad i^2 jl + 3, \\ & cf^2 h - 1, \quad l^4 + 1, \\ & cf^2 k^2 + 1, \\ & cjk^2 - 1, \end{aligned}$$

Total  $\pm 16$ .

### No. 63.

The sextinvariant is  $T = [a \text{ very long table giving, in source-printing, } \pm 871 \text{ as the total of all monomial-coefficient pairs. The discovery of the invariants } S \text{ and } T \text{ is due to Aronhold; the developed expressions were first obtained by Mr. Salmon.}]$

### No. 64.

There is an octicovariant for which we may take

$$\Theta U = \text{determinant of the array } \begin{vmatrix} \partial_x^2 HU & \partial_x \partial_y HU & \partial_x \partial_z HU \\ \frac{1}{4} \partial_y \partial_x HU & \frac{1}{4} \partial_y^2 U & \frac{1}{4} \partial_y \partial_z U \\ \partial_x U & \frac{1}{2} \partial_x \partial_y U & \frac{1}{2} \partial_x \partial_z U \end{vmatrix},$$

or else

$$\Theta U = \begin{vmatrix} \partial_x HU & \partial_y \partial_x HU & \partial_x^2 HU & \partial_y \partial_z HU \\ HU & \frac{1}{2} \partial_x U & \partial_z \partial_x HU & \frac{1}{2} \partial_y U \end{vmatrix} \text{ (schematic, see scan),}$$

or else, what I believe is more simple, a function  $\Theta_{xx}U$ , which is a linear function of the last-mentioned two functions.

The relations between  $\Theta U$ ,  $\Theta_x U$ ,  $\Theta_{xx}U$  are

$$\begin{aligned} -\Theta_x U + 4\Theta U &= T \cdot U^2 - 24S \cdot U \cdot HU, \\ \Theta_{xx}U + 8\Theta U &= T \cdot (HU)^2 - 16S \cdot U \cdot HU. \end{aligned}$$

I have not worked out the developed expressions.

### No. 65.

The cubicontravariant is  $PU = [an \text{ extensive table of monomial-coefficient pairs in } a, b, c, f, g, h, i, j, k, l \text{ multiplied by appropriate products of } \xi, \eta, \zeta. \text{ The column-sums in the printed table are } \pm 3, \pm 3, \pm 7, \pm 7, \pm 7, \pm 7, \pm 7, \pm 7, \pm 13.]$

|         |             |          |          |          |           |              |             |          |          |
|---------|-------------|----------|----------|----------|-----------|--------------|-------------|----------|----------|
| bcl - 1 | acl -- 1    | abl - 1  | ack + 1  | abi + 1  | bcj + 1   | abg + 1      | bch +1      | acf + 1  | abc -- 1 |
| a^2 - 1 | b^2 - 1     | bgi + 1  | agi + 1  | afk + 1  | af^2y - 2 | - 1 af^2 + 1 | bgl + 1     | ai^2 - 1 | ahi + 1  |
| b^2     | cfk + 1     | chj + 1  | bhj + 1  | acj + 1  | bgh - 2   | c/h - 2      | aik - 2     |          |          |
|         |             |          |          |          |           |              | - 2 c^2h +1 | bgj + 1  |          |
| fg - 1  | g^2 h - 1   | fh^2 - 1 | chk - 1  | bil + 1  | ckl + 1   | bf - 1       | chk - 1     | cjk - 2  | chk + 1  |
| fil + 1 | gxl + 1     | ckl + 1  | fy + 2   | fhl + 3  | ffk + 3   | fhj - 1      | fk^2 + 2    | fgj - f  | fgh + 3  |
| ihx - 1 | ij - 1      | jh^2 - 1 | fky + 3  | fk - f   | fy - 1    | ghk - 1      | ki - 1      | fk + 2   | fjl -4   |
|         |             |          |          |          |           |              |             |          |          |
|         | jhi gjk - 1 | gki + 1  | gki - 1  | hhi + 2  | //h - 1   | ghi - 1      | ghj - 4     |          |          |
|         | jy - 1      | hki - 1  | hi^2 + 2 | hl^2 - 2 | fl^2 - 2  | gl^2 - 2     |             |          |          |
|         |             |          |          |          |           |              |             |          |          |
|         | jl^2 - 2    | kl^2 - 2 | il^2 - 2 | jkl + 3  | jhl + 3   | jjl + 3      | jjk + 3     |          |          |
|         |             |          |          |          |           |              |             |          |          |
|         |             |          |          |          |           |              |             |          | l^4 + 4  |

[Layout as in the OCR; the table is followed by a parenthetical  $(\xi, \eta, \zeta)^3$  marker.]

### No. 66.

The quintic contravariant is  $QU = [an \text{ even longer table than No. 65; the column-sums in the printed table are } \pm 145, \pm 145, \pm 145, \pm 282, \pm 282, \pm 282, \pm 282, \pm 282, \pm 282, \pm 282, \pm 486. \text{ The form is again a polynomial in } (\xi, \eta, \zeta) \text{ of degree 5.}]$

**No. 67.**

The reciprocant is  $FU = (* \mid \xi, \eta, \zeta)^6$ , given as a long monomial table in  $a, b, c, f, g, h, i, j, k, l$  multiplied by appropriate powers of  $\xi, \eta, \zeta$ . The column-totals:  $\pm 9, \pm 9, \pm 9, \pm 54, \pm 54, \pm 54, \pm 54, \pm 54, \pm 54, \pm 135, \pm 135, \pm 135, \pm 135$ .

The preceding Tables contain the complete system [not so] of the covariants and contravariants of the ternary cubic, i.e. the covariants are the cubic itself  $U$ , the quartinvariant  $S$ , the sextinvariant  $T$ , the Hessian  $HU$ , and an octicovariant, say  $\Theta U$ ; the contravariants are the cubicontravariant  $PU$ , the quinticontravariant  $QU$ , and the reciprocant  $FU$ .

The contravariants are all of them evectants, viz.  $PU$  is the evectant of  $S$ ,  $QU$  is the evectant of  $T$ , and the reciprocant  $FU$  is the evectant of  $QU$ , or what is the same thing, the second evectant of  $T$ .

The discriminant is a rational and integral function of the two invariants; representing it by  $R$ , we have  $R = 64S^3 - T^2$ .

If we combine  $U$  and  $HU$  by arbitrary multipliers, say  $\alpha$  and  $6\beta$ , so as to form the sum  $\alpha U + 6\beta HU$ , this is a cubic, and the question arises, to find the covariants and contravariants of this cubic; the results are given in the following Table.

**No. 68.**

$$\alpha U + 6\beta HU = \alpha U + 6\beta HU.$$

$$\begin{aligned} H(\alpha U + 6\beta HU) &= (0, 2S, T, 8S^2 \mid \alpha, \beta)^3 \cdot U \\ &\quad + (1, 0, -12S, -2T \mid \alpha, \beta)^3 \cdot HU. \end{aligned}$$

$$\begin{aligned} P(\alpha U + 6\beta HU) &= (1, 0, 12S, 4T \cdot 8 \mid \alpha, \beta)^3 \cdot PU \\ &\quad + (0, 1, 0, -4S^2 \mid \alpha, \beta)^3 \cdot QU. \end{aligned}$$

$$\begin{aligned} Q(\alpha U + 6\beta HU) &= (0, 60S, 30T, 0, -120TS, -240S^2 \mid \alpha, \beta)^5 \cdot PU \\ &\quad + (1, 0, 0, 10T, 240S^2, 24TS^2 \mid \alpha, \beta)^5 \cdot QU. \end{aligned}$$

$$S(\alpha U + 6\beta HU) = (S, T, 24S^2, 4TS, T^2 - 4S^3 \mid \alpha, \beta)^4.$$

$$T(\alpha U + 6\beta HU) = (T, 96S^2, 60TS, 20T^2, 240TS^2 + 48T^2S + 4608S^4, -8T^3 + 576TS^3 \mid \alpha, \beta)^6.$$

$$R(\alpha U + 6\beta HU) = [(1, 0, -24S, -8T, -48S^2 \mid \alpha, \beta)^4]^3 \cdot PU.$$

$$\begin{aligned} F(\alpha U + 6\beta HU) &= FU \cdot (1, 0, -24S, -8T, -48S^2 \mid \alpha, \beta)^4 \\ &\quad + (PU)^2 \cdot (0, 24, 0, 0, -48T \mid \alpha, \beta)^4 \\ &\quad + PU \cdot QU \cdot (0, 0, 24, 0, 96S \mid \alpha, \beta)^4 \\ &\quad + (QU)^2 \cdot (0, 0, 0, 8, 0 \mid \alpha, \beta)^4. \end{aligned}$$

We have, in like manner, for the covariants and contravariants of the cubic  $6\alpha PU + \beta QU$ , the following Table.

**No. 69.**

$$6\alpha PU + \beta QU = 6\alpha PU + \beta QU.$$

$$\begin{aligned} H(6\alpha PU + \beta QU) &= (-2T, 48ST, 18TS, T^2 + 16S^3 \mid \alpha, \beta)^3 \cdot PU \\ &\quad + (8S, T, -8S^2, -TS \mid \alpha, \beta)^3 \cdot QU. \end{aligned}$$

$$\begin{aligned} P(6\alpha PU + \beta QU) &= (32S^2, 12TS, T^2 + 32S^3, 4TS^2 \mid \alpha, \beta)^3 \cdot U \\ &\quad + (4T, 96S^2, 12TS, T^2 - 32S^3 \mid \alpha, \beta)^3 \cdot HU. \end{aligned}$$

$$\begin{aligned} Q(6\alpha PU + \beta QU) &= (384TS^3 + 120T^2S + 7680S^4, 10T^2 + 320TS^2 \cdot (\alpha, \beta)^4, 480T^2S^2, \dots) \cdot U \\ &\quad + (-24T^2 + 4608S^2, 1920TS^2, 480T^2S, 30T^2 + 1920TS^2, \dots) \\ &\quad [\text{for full formula see scan}]. \end{aligned}$$

$$\begin{aligned}
 S(6\alpha PU + \beta QU) &= (T^3 + 192S^4, 128TS^2, 18T^2S + 384S^3, T^3 + 64S^3, T^2S - 64S^3)(\alpha, \beta)^4. \\
 T(6\alpha PU + \beta QU) &= (8T^3 + 4608TS^3, 1920T^3S^2 + 73728S^4, 360T^3S + 38400TS^4, 20T^4 + 8960T^3S, \\
 &\quad 840T^4S + 7680T^3S^4, 360T^3S + 384T^4S^2 + 24576S^4, T^5 - 40T^3S^4 + 2560TS^4)(\alpha, \beta)^6. \\
 R(6\alpha PU + \beta QU) &= [(48S^2, 8T, -96S^3, -24TS, -T^2 - 16S^3 | \alpha, \beta)^4]^3 \cdot R. \\
 F(6\alpha PU + \beta QU) &= (192S, 32T, -384S^2, -96TS, -4T^2 - 64S^3 | \alpha, \beta)^4 \cdot \Theta U \\
 &\quad + (0, 512S^4, 192TS^4, 24T^2S, T^4 | \alpha, \beta)^4 \cdot U^4 \\
 &\quad + (1344S^2, 352TS, 24T^2 - 1152S^3, -288TS^2, 20T^2S + 64S^4 | \alpha, \beta)^4 \cdot U \cdot HU \\
 &\quad + (48T, 288S, 24T^2 + 1536S^3, 144TS^2 | \alpha, \beta)^4 \cdot (HU)^2.
 \end{aligned}$$

The tables for the ternary cubic become much more simple if we suppose that the cubic is expressed in Hesse's canonical form; we have then the following table.

**No. 70.**

(Hesse's canonical form for the ternary cubic.)

$$\begin{aligned}
 U &= x^3 + y^3 + z^3 + 6lxyz. \\
 S &= -l + l^4. \\
 T &= 1 - 20l^3 - 8l^6. \\
 R &= -(1 + 8l^3)^3. \\
 HU &= l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz. \\
 \Theta U &= (1 + 8l^3)^2(y^2z^2 + z^2x^2 + x^2y^2) \\
 &\quad + (-9l^4)(x^4 + y^4 + z^4) \\
 &\quad + (-2l - 5l^4 - 20l^7)(x^3 + y^3 + z^3)xyz \\
 &\quad + (-15l^2 - 78l^5 + 12l^8)x^2y^2z^2. \\
 \Theta_x U &= 4(1 + 8l^3)^2(y^2z^2 + z^2x^2 + x^2y^2) \\
 &\quad + (-1 + 4l^3 - 4l^6)(x^3 + y^3 + z^3)^2 \\
 &\quad + (4l + 100l^4 + 112l^7)(x^3 + y^3 + z^3)xyz \\
 &\quad + (48l - 552l^5 + 48l^8)x^2y^2z^2. \\
 \Theta_{xx} U &= -2(1 + 8l^3)^2(y^2z^2 + z^2x^2 + x^2y^2) \\
 &\quad + (1 - 10l^3)(x^3 + y^3 + z^3)^2 \\
 &\quad + (50l - 180l^4 + 96l^7)(x^3 + y^3 + z^3)xyz \\
 &\quad + (50l - 624l^4 - 192l^7)x^2y^2z^2. \\
 PU &= -4(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta. \\
 QU &= -4(1 + 10l^3)(\xi^3 + \eta^3 + \zeta^3) - 6l^2(\xi + 4l^3)\xi\eta\zeta. \\
 FU &= 4(1 + 8l^3)(\eta^2\zeta^2 + \zeta^2\xi^2 + \xi^2\eta^2) \\
 &\quad + (\xi^3 + \eta^3 + \zeta^3)^2 \\
 &\quad - 24l^2(\xi^3 + \eta^3 + \zeta^3)\xi\eta\zeta \\
 &\quad - 24l(1 + 2l^3)\xi^2\eta^2\zeta^2,
 \end{aligned}$$

to which it is proper to join the following transformed expressions for  $\Theta U, \Theta_x U, \Theta_{xx} U$ , viz.

$$\begin{aligned}\Theta U &= (1 + 8l^3)^2 (y^2 z^2 + z^2 x^2 + x^2 y^2) \\ &\quad + (2l - 5l^4) U \cdot HU \\ &\quad + (-3l^2) (HU)^2. \\ \Theta_x U &= 4(1 + 8l^3)^2 (y^2 z^2 + z^2 x^2 + x^2 y^2) \\ &\quad + (-1 + 12l^3 + 4l^6) U^2 \\ &\quad + (-12l + 16l^4) U \cdot HU \\ &\quad + (-12l^2) (HU)^2. \\ \Theta_{xx} U &= -2(1 + 8l^3)^2 (y^2 z^2 + z^2 x^2 + x^2 y^2).\end{aligned}$$

The last preceding table affords a complete solution of the problem to reduce a ternary cubic to its canonical form.

[I add to the present Memoir, in the notation hereof  $(a, h, c, f, g, h, i, j, k, l \mid x, y, z)^3$  for the ternary cubic, some formulae originally contained in the paper “On Homogeneous Functions of the third order with three variables,” (1846), but which on account of the difference of notation were omitted from the reprint, 35, of that paper.]

Representing the determinant

$$\begin{vmatrix} ax + hy + jz, & hx + ky + bz, & jx + ly + gz, \\ hx + ky + bz, & kx + by + fz, & lx + fy + iz, \\ jx + ly + gz, & lx + fy + iz, & gx + iy + cz \end{vmatrix}$$

by

$$(A, B, C, F, G, H \mid x, y, z)^2,$$

the values of  $A, B, C, F, G, H$  (equations (10) of §35) are given by a long table, of which the first row is

$$A: \begin{array}{cccccc} gk & bi & cf & bc & ff & ki \\ -l^2 & -r^2 & -v^2 & -fi & +ck & +bg \\ & & & & -2fl & -2il \end{array}$$

and there are five further rows for  $B, C, F, G, H$ , each of comparable length; the full table is given on book pp. 333–334 in twelve rows.

Moreover writing

$$FU = \begin{vmatrix} a, & k, & g, & l, & j, & h \\ h, & b, & i, & f, & l, & k \\ j, & f, & c, & i, & g, & l \end{vmatrix}, \quad \xi\eta\zeta \text{ insertions as above}$$

so that

$$HFU = Aa + Bh + Cc + 2Ff + 2Gg + 2Hh,$$

then the values of  $a, b, c, f, g, h$  (equations (13) of §35) are given by a further long table on book p. 334 in twelve rows, the first row being

$$a: \quad 0, \quad 2cf - 2b^2, \quad 2bi - 2f^2, \quad fi - bc, \quad 0, \quad 0,$$

and the remaining rows of comparable form. The table is reproduced in full in the scan; the entries follow the same pattern as in equations (10)–(13) of paper 35.

[The remainder of paper 144 continues beyond book page 334 = PDF page 350 and is taken up in the next chunk.]

## 144 (concluded). A Third Memoir upon Quantics.

[p. 335] Also if the discriminant be written

$$K(U) = \begin{vmatrix} a & h & g & l & j & k \\ h & b & i & f & l & k \\ j & i & c & i & j & l \\ \mathfrak{A} & \mathfrak{H} & \mathfrak{C} & \mathfrak{L} & \mathfrak{J} & \mathfrak{K} \\ \mathfrak{H} & \mathfrak{B} & \mathfrak{I} & \mathfrak{F} & \mathfrak{L} & \mathfrak{K} \\ \mathfrak{J} & \mathfrak{I} & \mathfrak{C} & \mathfrak{I} & \mathfrak{J} & \mathfrak{L} \end{vmatrix},$$

then the values of  $\mathfrak{A}$ ,  $\mathfrak{B}$ ,  $\mathfrak{C}$ ,  $\mathfrak{F}$ ,  $\mathfrak{G}$ ,  $\mathfrak{H}$ ,  $\mathfrak{I}$ ,  $\mathfrak{J}$ ,  $\mathfrak{K}$ ,  $\mathfrak{L}$  (equations (20) of § 35) are

$$\begin{aligned} \mathfrak{A} &= ak g + 2f j l - a l^2 - g k^2 - f j^2, \\ \mathfrak{B} &= b k h + 2f k l - b l^2 - h f^2 - k^2 i, \\ \mathfrak{C} &= c j f + 2g i l - c l^2 - f g^2 - i^2 j, \\ 3\mathfrak{F} &= b c h + b i j - c h^2 + 2g f k - 2b g l + f l^2 - f^2 j - f i h, \\ 3\mathfrak{G} &= c a f + c j k - a f^2 + 2h g i - 2c h l + g l^2 - g^2 k - g j f, \\ 3\mathfrak{H} &= a b g + a k i - b g^2 + 2f h j - 2a f l + h l^2 - h^2 i - h k g, \\ 3\mathfrak{I} &= b c j + c f h - b g^2 + 2k i g - 2c k l + j l^2 - j^2 h - f i j, \\ 3\mathfrak{J} &= c a k + a g f - c h^2 + 2i j h - 2a i l + k l^2 - k^2 f - g j k, \\ 3\mathfrak{K} &= a b i + b h g - a i^2 + 2j k f - 2b j l + i l^2 - i^2 g - h k i, \\ 6\mathfrak{L} &= a i c + 3f g h + 3i j k + 2l^3 - a f i - b g j - c h k - 2l g k - 2l h i - 2l f j. \end{aligned}$$

[The equation  $K(U) = R = 64 S^3 - T^2$  would however afford a perhaps easier formula for the calculation of the discriminant.]

## 145. A Memoir upon Caustics.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLVII. for the year 1857, pp. 273–312. Received May 1,—Read May 8, 1856.]

[p. 336] The following memoir contains little or nothing that can be considered new in principle; the object of it is to collect together the principal results relating to caustics *in plano*, the reflecting or refracting curve being a right line or a circle, and to discuss, with more care than appears to have been hitherto bestowed upon the subject, some of the more remarkable cases. The memoir contains in particular researches relating to the caustic by refraction of a circle for parallel rays, the caustic by reflexion of a circle for rays proceeding from a point, and the caustic by refraction of a circle for rays proceeding from a point; the result in the last case is not worked out, but it is shown how the equation in rectangular coordinates is to be obtained by equating to zero the discriminant of a rational and integral function of the sixth degree. The memoir treats also of the secondary caustic, or orthogonal trajectory of the reflected or refracted rays, in the general case of a reflecting or refracting circle and rays proceeding from a point; the curve in question, or rather a secondary caustic, is, as is well known, the Oval of Descartes or ‘Cartesian’: the equation is discussed by a method which gives rise to some forms of the curve which appear to have escaped the notice of geometers. By considering the caustic as the evolute of the secondary caustic, it is shown that the caustic, in the general case of a reflecting or refracting circle and rays proceeding from a point, is a curve of the sixth class only. The concluding part of the memoir treats of the curve which, when the incident rays are parallel, must be taken for the secondary caustic in the place of the Cartesian, which, for the particular case in question, passes off to infinity. In the course of the memoir, I reproduce a theorem first given, I believe, by me in the *Philosophical Magazine*, viz. that there are six different systems of a radiant point and refracting circle which give rise to identically the same caustic, [see post, XXVIII]. [p. 337] The memoir is divided into sections, each of which is to a considerable extent intelligible by itself, and the subject of each section is for the most part explained by the introductory paragraph or paragraphs.

### I.

Consider a ray of light reflected or refracted at a curve, and suppose that  $\xi, \eta$  are the coordinates of a point  $Q$  on the incident ray,  $\alpha, \beta$  the coordinates of the point  $G$  of incidence upon the reflecting or refracting curve,  $a, b$  the coordinates of a point  $N$  upon the normal at the point of incidence,  $x, y$  the coordinates of a point  $q$  on the reflected or refracted ray.

Write for shortness,

$$\begin{aligned}(b - \beta)(\xi - \alpha) - (a - \alpha)(\eta - \beta) &= \nabla QGN, \\ (a - \alpha)(\xi - \alpha) + (b - \beta)(\eta - \beta) &= \square QGN,\end{aligned}$$

$\nabla QGN$  then is equal to twice the area of the triangle  $QGN$ , and if  $\xi, \eta$  instead of being the coordinates of a point  $Q$  on the incident ray were current coordinates, the equation  $\nabla QGN = 0$  would be the equation of the line through the points  $G$  and  $N$ , i.e. of the normal at the point of incidence; and in like manner the equation  $\square QGN = 0$  would be the equation of the line through  $G$  perpendicular to the line through the points  $G$  and  $N$ , i.e. of the tangent at the point of incidence.

We have

$$\overline{NG}^2 = (a - \alpha)^2 + (b - \beta)^2,$$

and therefore identically,

$$\overline{NG}^2 \cdot \overline{QG}^2 = \nabla QGN^2 + \square QGN^2.$$

Suppose for a moment that  $\phi$  is the angle of incidence and  $\phi'$  the angle of reflexion or refraction; and let  $\mu$  be the index of refraction (in the case of reflexion  $\mu = -1$ ), then writing

$$\begin{aligned}(b - \beta)(x - \alpha) - (a - \alpha)(y - \beta) &= \nabla qGN, \\ (a - \alpha)(x - \alpha) + (b - \beta)(y - \beta) &= \square qGN,\end{aligned}$$

and

$$\overline{qG}^2 = (x - \alpha)^2 + (y - \beta)^2,$$

we have

$$\sin \phi = \frac{\nabla QGN}{\overline{NG} \cdot \overline{GQ}}, \quad \sin \phi' = \frac{\nabla qGN}{\overline{NG} \cdot \overline{Gq}},$$

and substituting these values in the equation

$$\sin^2 \phi - \mu^2 \sin^2 \phi' = 0,$$

[p. 338] we obtain

$$\overline{qG}^2 \nabla QGN^2 - \mu^2 \overline{QG}^2 \nabla qGN^2 = 0,$$

an equation which is rational of the second order in  $x, y$ , the coordinates of a point  $q$  on the refracted ray; this equation must therefore contain, as a factor, the equation of the refracted ray; the other factor gives the equation of a line equally inclined to, but on the opposite side of the normal; this line (which of course has no physical existence) may be termed the *false refracted ray*. The caustic is geometrically the envelope of the pair of rays, and for finding the equation of the caustic it is obviously convenient to take the equation of the two rays conjointly in the form under which such equation has just been found, without attempting to break the equation up into its linear factors.

It is however interesting to see how the resolution of the equation may be effected; for this purpose multiply the equation by  $\overline{NG}^2$ , then reducing by means of a previous formula, the equation becomes

$$(\nabla qGN^2 + \square qGN^2) \nabla QGN^2 - \mu^2 (\nabla QGN^2 + \square QGN^2) \nabla qGN^2 = 0,$$

which is equivalent to

$$\nabla qGN (\mu^2 \square QGN + (\mu^2 - 1) \nabla QGN) - \square qGN \cdot \nabla QGN = 0,$$

and the factors are

$$\nabla qGN \sqrt{\mu^2 \square QGN^2 + (\mu^2 - 1) \nabla QGN^2} \pm \square qGN \cdot \nabla QGN = 0;$$

it is in fact easy to see that these equations represent lines passing through the point  $G$  and inclined to  $GN$  at angles  $\pm \phi'$ , where  $\phi'$  is given by the equations

$$\sin \phi = \mu \sin \phi', \quad \tan \phi' = \frac{\nabla QGN}{\square QGN},$$

and there is no difficulty in distinguishing in any particular case between the refracted ray and the false refracted ray.

In the case of reflexion  $\mu = -1$ , and the equations become

$$\nabla qGN \cdot \square QGN \pm \square qGN \cdot \nabla QGN = 0;$$

the equation

$$\nabla qGN \cdot \square QGN - \square qGN \cdot \nabla QGN = 0$$

is obviously that of the incident ray, which is what the false refracted ray becomes in the case of reflexion; and the equation

$$\nabla qGN \cdot \square QGN + \square qGN \cdot \nabla QGN = 0$$

is that of the reflected ray.

[p. 339]

## II.

But instead of investigating the nature of the caustic itself, we may begin by finding the secondary caustic or orthogonal trajectory of the refracted rays, i.e. a curve having the caustic for its evolute; suppose that the incident rays are all of them normal to a certain curve, and let  $Q$  be a point upon this curve, and considering the ray through the point  $Q$ , let  $G$  be the point of incidence upon the refracting curve; then if the point  $G$  be made the centre of a circle the radius of which is  $\mu^{-1} \cdot GQ$ , the envelope of the circles will be the secondary caustic. It should be noticed that, if the incident rays proceed from a point, the most



simple course is to take such point for the point  $Q$ . The remark, however, does not apply to the case where the incident rays are parallel; the point  $Q$  must here be considered as the point in which the incident ray is intersected by some line at right angles to the rays, and there is not in general any one line which can be selected in preference to another. But if the refracting curve be a circle, then the line perpendicular to the incident rays may be taken to be a diameter of the circle. To translate the construction into analysis, let  $\xi, \eta$  be the coordinates of the point  $Q$ , and  $\alpha, \beta$  the coordinates of the point  $G$ , then  $\xi, \eta, \alpha, \beta$  are in effect functions of a single arbitrary parameter; and if we write

$$\overline{GQ}^2 = (\xi - \alpha)^2 + (\eta - \beta)^2, \quad \overline{Gq}^2 = (x - \alpha)^2 + (y - \beta)^2,$$

then the equation

$$\mu^2 \overline{Gq}^2 - \overline{GQ}^2 = 0,$$

where  $x, y$  are to be considered as current coordinates, and which involves of course the arbitrary parameter, is the equation of the circle, and the envelope is obtained in the usual manner. This is the well-known theory of Gergonne and Quetelet.

### III.

There is however a simpler construction of the secondary caustic in the case of the reflexion of rays proceeding from a point. Suppose, as before, that  $Q$  is the radiant point, and let  $G$  be the point of incidence. On the tangent at  $G$  to the reflecting curve, let fall a perpendicular from  $Q$ , and produce it to an equal distance on the other side of the tangent; then if  $q$  be the extremity of the line so produced, it is clear that  $q$  is a point on the reflected ray  $Gq$ , and it is easy to see that the locus of  $q$  is the secondary caustic. Produce now  $QG$  to a point  $Q'$  such that  $GQ' = QG$ , it is clear that the locus of  $Q'$  will be a curve similar to and similarly situated with and twice the magnitude of the reflecting curve, and that the two curves have the point  $Q$  for a centre of similitude. And the tangent at  $Q'$  passes through the point  $q$ , i.e.  $q$  is the foot of the perpendicular let fall from  $Q$  upon the tangent at  $Q'$ ; we have therefore the theorem due to Dandelin, viz.

**[p. 340]** If rays proceeding from a point  $Q$  are reflected at a curve, then the secondary caustic is the locus of the feet of the perpendiculars let fall from the point  $Q$  upon the tangents of a curve similar to and similarly situated with and twice the magnitude of the reflecting curve, and such that the two curves have the point  $Q$  for a centre of similitude.

### IV.

If rays proceeding from a point  $Q$  are reflected at a line, the reflected rays will proceed from a point  $q$  situate on the perpendicular let fall from  $Q$ , and at an equal distance on the other side of the reflecting line. The point  $q$  may be spoken of as the image of  $Q$ ; it is clear that if  $Q$  be considered as a variable point, then the locus of the image  $q$  will be a curve equal and similar but oppositely situated to the curve, the locus of  $Q$ , and which may be spoken of as the image of such curve. Hence it at once follows, that if the incidental rays are tangent, or normal, or indeed in any other manner related to a curve, then the reflected rays will be tangent, or normal, or related in a corresponding manner to a curve the image of the first-mentioned curve. The theory of the combined reflexions and refractions of a pencil of rays transmitted through a plate or prism, is, by the property in question, rendered very simple. Suppose, for instance, that a pencil of rays is refracted at the first surface of a plate or prism, and after undergoing any number of internal reflexions, finally emerges after a second refraction at the first or second surface; in order to find the caustic enveloped by the rays after the second refraction, it is only necessary to form the successive images of the first caustic corresponding to the different reflexions, and finally to determine the caustic for refraction in the case where the incident rays are the tangents of the caustic which is the last of the series of images; the problem is not in effect different from that of finding the caustic for refraction in the case where the incident rays are the tangents to the caustic after the first refraction, but the line at which the second refraction takes place is arbitrarily situate with respect to the caustic. Thus e.g. suppose the incident rays proceed from a point, the caustic after the first refraction is, it will be shown in the sequel, the evolute of a conic; for the complete theory of the combined reflexions and refractions of the pencil by a plate or prism, it is only necessary to find the caustic by refraction, where the incident rays are the normals of a conic, and the refracting line is arbitrarily situate with respect to the conic.

V.

Suppose that rays proceeding from a point  $Q$  are refracted at a line; and take the refracting line for the axis of  $y$ , the axis of  $x$  passing through the radiant point  $Q$ , and take the distance  $QA$  for unity. Suppose that the index of refraction  $\mu$  is put equal to  $\frac{1}{k}$ . Then if  $\phi$  be the angle of incidence and  $\phi'$  the angle of refraction, [p. 341] we have  $\sin \phi' = k \sin \phi$ , and the equation  $y = x \tan \phi' - \tan \phi$  of the refracted ray becomes, putting for  $\phi'$  its value,

$$y - \frac{k \sin \phi}{\sqrt{1 - k^2 \sin^2 \phi}} x - \tan \phi = 0.$$

[Figure: incident ray from  $Q$  on the axis of  $x$  refracted at the axis of  $y$ ; angles of incidence and refraction at  $A$ .]

Differentiating with respect to the variable parameter and combining the two equations, we obtain, after a simple reduction,

$$kx = -\frac{(1 - k^2 \sin^2 \phi)^{\frac{3}{2}}}{\cos^3 \phi}, \quad k'y = -\frac{k'^2 \sin^3 \phi}{\cos^3 \phi},$$

where  $k' = \sqrt{1 - k^2}$ ; hence eliminating,

$$(kx)^{\frac{2}{3}} - (k'y)^{\frac{2}{3}} = 1,$$

which is the equation of the caustic. When the refraction takes place into a denser medium  $k$  is less than 1, and  $k'^2$  is positive, the caustic is therefore the evolute of a hyperbola (see fig. 1); but when the refraction takes place in a rarer medium  $k$  is greater than 1, and  $k'^2$  is negative, the caustic is therefore the evolute of an ellipse (see fig. 2). These results appear to have been first obtained by Gergonne. The conic (hyperbola or ellipse) is the secondary caustic, and as such may be obtained as follows.

VI.

The equation of the variable circle is

$$x^2 + (y - \tan \phi)^2 - k^2 \sec^2 \phi = 0;$$

or reducing, the equation is

$$x^2 + y^2 - 2y \tan \phi - k^2 \tan^2 \phi + 1 - k^2 = 0;$$

whence, considering  $\tan \phi$  as the variable parameter, the equation of the envelope is

$$k^2(x^2 + y^2 - k^2) - y^2 = 0,$$

that is,

$$k^2 y^2 - k'^2 x^2 - k^2 k'^2 = 0,$$

[p. 342] or

$$\frac{x^2}{k^2} - \frac{y^2}{k'^2} = 1$$

is the equation of the secondary caustic, or conic having the caustic for its evolute. The radiant point, it is clear, is a focus of the conic.

VII.

Let the equation of the refracted ray be represented by

$$Xx + Yy + Z = 0,$$

we have

$$X : Y : Z = \frac{-k \sin \phi}{\sqrt{1 - k^2 \sin^2 \phi}} : 1 : -\tan \phi,$$

from which we obtain

$$\frac{k^2}{X^2} - \frac{k'^2}{Y^2} = \frac{1}{Z^2}$$

for the tangential equation of the caustic; or if we represent the equation of the refracted ray by

$$Xx + Yy - k = 0,$$

then we have

$$\frac{k^2}{X^2} - \frac{k'^2}{Y^2} = \frac{1}{k^2}$$

for the tangential equation of the caustic.

*[Figure 1: Caustic by refraction of a line into a denser medium, evolute of a hyperbola; refracted-ray family tangent to the hyperbolic arc.]      [Figure 2: Caustic by refraction of a line into a rarer medium, evolute of an ellipse; cusped caustic inside a circle.]*

[p. 343]

## VIII.

If a ray be reflected at a circle; we may take  $a, b$  as the coordinates of the centre of the circle, and supposing as before that  $\xi, \eta$  are the coordinates of a point  $Q$  in the incident ray,  $\alpha, \beta$  the coordinates of the point  $G$  of incidence, and  $x, y$  the coordinates of a point  $q$  in the reflected ray, the equation of the reflected ray, treating  $x, y$  as current coordinates, is

$$\begin{aligned} & \{(b - \beta)(\xi - \alpha) - (a - \alpha)(\eta - \beta)\} \{(a - \alpha)(\xi - \alpha) + (b - \beta)(\eta - \beta)\} \\ & + \{(a - \alpha)(x - \alpha) + (b - \beta)(y - \beta)\} \{(b - \beta)(\xi - \alpha) - (a - \alpha)(\eta - \beta)\} = 0. \end{aligned}$$

Write for shortness,

$$\begin{aligned} N_{Q,G} &= (b - \beta)(x - \alpha) - (a - \alpha)(y - \beta), \\ T_{Q,G} &= (a - \alpha)(x - \alpha) + (b - \beta)(y - \beta), \end{aligned}$$

and similarly for  $N_{q,G}$  &c.; the equation of the reflected ray is

$$N_{q,G} \cdot T_{Q,G} + T_{q,G} \cdot N_{Q,G} = 0.$$

Suppose that the reflected ray meets the circle again in  $G'$  and undergoes a second reflexion, and let  $x', y'$  be the coordinates of a point  $q'$  in the ray thus twice reflected. We see first ( $G'$  being a point in the first reflected ray) that

$$N_{q,G'} \cdot T_{Q,G'} + T_{q,G'} \cdot N_{Q,G'} = 0.$$

Again, considering  $G$  as a point in the ray by the reflexion of which the second reflected ray arises, the equation of the second reflected ray is

$$N_{q',G'} \cdot T_{G,G'} + T_{q',G'} \cdot N_{G,G'} = 0;$$

and from the form of the expressions  $N_{a,b}, T_{a,b}$  it is clear that

$$N_{a,b} = -N_{b,a}, \quad T_{a,b} = T_{b,a};$$

the equation for the second reflected ray may therefore be written under the form

$$N_{q',G'} \cdot T_{Q,G'} - T_{q',G'} \cdot N_{Q,G'} = 0;$$

or reducing by a previous equation, we obtain finally for the equation of the second reflected ray,

$$N_{q',G'} \cdot T_{Q,G} + T_{q',G'} \cdot N_{Q,G} = 0,$$

and in like manner the equation for the third reflected ray is

$$N_{q'',G''} \cdot T_{Q,G} + T_{q'',G''} \cdot N_{Q,G} = 0,$$

and so on, the equation for the last reflected ray containing, it will be observed, the coordinates of the radiant point and of the first and last points of incidence (the coordinates of the last point of incidence can of course only be calculated from those of the radiant point and the first point of incidence, through the coordinates of the intermediate points of incidence), but not containing explicitly the coordinates of any of the intermediate points of incidence. The form is somewhat remarkable, but the result is really the same with that obtained by simple geometrical considerations, as follows.

[p. 344]

## IX.

Consider a ray reflected any number of times at a circle; and let  $G_0G_1$  be the ray incident at  $G_1$ , and  $GG'$  the last reflected ray, the point at which the reflexion takes place or last point of incidence being  $G$ . Take the centre  $O$  of the circle for the origin, and any two lines  $Ox$ ,  $Oy$  through the centre and at right angles to each other for axes, and let  $Ox$  meet the circle in the point  $A$ . Write

$$\begin{aligned} \angle AOG_1 &= \theta_0, & \angle xG_0G_1 &= \psi_0, \\ \angle AOG &= \theta, & \angle xGG' &= \psi, \\ \angle G_1G_2O &= \phi; \end{aligned}$$

then the radius of the circle being taken as unity and its centre as origin, the equation of the reflected ray is

$$y - \sin \theta = \tan \psi (x - \cos \theta);$$

and if there have been  $n$  reflexions, then

$$\theta = \theta_0 + n(\pi - 2\phi) = \theta_0 + n\pi - 2n\phi,$$

$$\psi = \psi_0 - 2n\phi,$$

and therefore the equation of the reflected ray is

$$x \cos(\psi_0 - 2n\phi) - y \sin(\psi_0 - 2n\phi) + (-)^n \sin(\psi_0 - \theta_0) = 0.$$

[Figure: circle centred at  $O$  with radius  $OA$  along the  $x$ -axis; incident ray  $G_0G_1$ , intermediate points of incidence  $G_1, G_2, \dots$ , and the final point  $G$  from which the last reflected ray  $GG'$  emerges.]

## X.

If a pencil of parallel rays is reflected any number of times at a circle, then taking  $AO$  for the direction of the incident rays, we may write  $\theta_0 = \phi$ ,  $\psi_0 = \pi$ , and the equation of a reflected ray is

$$x \sin 2n\phi + y \cos 2n\phi = (-)^n \sin \phi;$$

[p. 345] differentiating with respect to the variable parameter, we find

$$x \cos 2n\phi - y \sin 2n\phi = \frac{(-)^n}{2n} \cos \phi;$$

and these equations give

$$\begin{aligned} x &= \frac{(-)^n}{4n} \{ (2n+1) \cos(2n-1)\phi - (2n-1) \cos(2n+1)\phi \}, \\ y &= \frac{(-)^n}{4n} \{ -(2n+1) \sin(2n-1)\phi + (2n-1) \sin(2n+1)\phi \}, \end{aligned}$$

which may be taken for the equation of the caustic; the caustic is therefore an epicycloid: this is a well-known result.

## XI.

If rays proceeding from a point upon the circumference are reflected any number of times at a circle, then taking the point  $A$  for the radiant point, we have  $\theta_0 = 0$ ,  $\psi_0 = \pi - \phi$ , and the equation of a reflected ray is

$$x \sin(2n+1)\phi + y \cos(2n+1)\phi = (-)^n \sin \phi;$$

differentiating with respect to the variable parameter, we find

$$x \cos(2n+1)\phi - y \sin(2n+1)\phi = \frac{(-)^n}{2n+1} \sin \phi;$$

and these equations give

$$\begin{aligned} x &= \frac{(-)^n}{2n+1} \{ (n+1) \cos 2n\phi - n \cos(2n+2)\phi \}, \\ y &= \frac{(-)^n}{2n+1} \{ -(n+1) \sin 2n\phi + n \sin(2n+2)\phi \}, \end{aligned}$$

which may be taken as the equation of the caustic; the caustic is therefore in this case also an epicycloid: this is a well-known result.

## XII.

Consider a pencil of parallel rays refracted at a circle; take the radius of the circle as unity, and let the incident rays be parallel to the axis of  $x$ , then if  $\phi$ ,  $\phi'$  be the angles of incidence and refraction, and  $\mu$  or  $\frac{1}{k}$  be the index of refraction, so that  $\sin \phi' = k \sin \phi$ , the coordinates of the point of incidence are  $\cos \phi$ ,  $\sin \phi$ , and the equation of the refracted ray is

$$y - \sin \phi = \tan(\phi - \phi') (x - \cos \phi),$$

i.e.

$$\cos(\phi - \phi')(y - \sin \phi) = \sin(\phi - \phi')(x - \cos \phi),$$

[p. 346] or

$$x \sin(\phi - \phi') - y \cos(\phi - \phi') = \sin \phi',$$

which may also be written

$$\sin \phi (y \cos \phi - x \sin \phi) \cos \phi' + (y \sin \phi + x \cos \phi - 1) \sin \phi' = 0.$$

Writing  $k \sin \phi$ ,  $\sqrt{1 - k^2 \sin^2 \phi}$  instead of  $\sin \phi'$ ,  $\cos \phi'$ , and putting for shortness

$$y \cos \phi - x \sin \phi = Y, \quad x \cos \phi + y \sin \phi = X, \quad \frac{k \sin \phi}{\sqrt{1 - k^2 \sin^2 \phi}} = \Phi,$$

the equation of the refracted ray becomes

$$Y + \Phi(X - 1) = 0;$$

and differentiating with respect to the variable parameter  $\phi$ , observing that

$$\frac{dY}{d\phi} = -X, \quad \frac{dX}{d\phi} = Y, \quad \frac{d\Phi}{d\phi} = \frac{k \cos \phi}{(1 - k^2 \sin^2 \phi)^{\frac{3}{2}}} = \frac{\cot \phi}{1 - k^2 \sin^2 \phi} \Phi,$$

we have

$$-X + \Phi \left( Y + \frac{\cot \phi (X - 1)}{1 - k^2 \sin^2 \phi} \right) = 0,$$

and the combination of the two equations gives

$$Y = \frac{\Phi (1 - k^2 \sin^2 \phi)}{\Phi \cot \phi - 1}, \quad X = \frac{\Phi \cot \phi - k^2 \sin^2 \phi}{\Phi \cot \phi - 1},$$

and we have therefore

$$y = Y \cos \phi + X \sin \phi = \frac{\Phi (1 - k^2 \sin^2 \phi) \cos \phi + \sin \phi (\Phi \cot \phi - k^2 \sin^2 \phi)}{\Phi \cot \phi - 1} = k^2 \sin^3 \phi,$$

$$x = -Y \sin \phi + X \cos \phi = \frac{-\Phi (1 - k^2 \sin^2 \phi) \sin \phi + \cos \phi (\Phi \cot \phi - k^2 \sin^2 \phi)}{\Phi \cot \phi - 1},$$

i.e.

$$x = \frac{\Phi (1 - k^2 \sin^2 \phi) - k^2 \sin^2 \phi \cos \phi}{-\Phi \cos \phi \sin \phi} \cos \phi;$$

**[p. 347]** or multiplying the numerator and denominator by  $(1 - k^2 \sin^2 \phi)^{\frac{1}{2}} (\Phi \cos \phi + \sin \phi)$ , the numerator becomes

$$\begin{aligned} & -\Phi (1 - k^2 \sin^2 \phi) \{ \Phi^2 \cos^2 \phi - k^2 \sin^2 \phi \cos \phi \\ & \quad + \Phi (\sin \phi (1 - k^2 \sin^2 \phi) - k^2 \sin^3 \phi \cos \phi) \} \\ & = -k^2 \sin^2 \phi \cos \phi \{ (1 - k^2 \sin^2 \phi) - \sin^2 \phi (1 - k^2 \sin^2 \phi) \} \\ & \quad + k \sin^3 \phi \sqrt{1 - k^2 \sin^2 \phi} (1 - k^2 \sin^2 \phi) \\ & = \Phi (1 - k^2 \sin^2 \phi) \cos^3 \phi - k \sin^3 \phi (1 - k^2 \sin^2 \phi)^{\frac{3}{2}}, \end{aligned}$$

and the denominator becomes

$$k^2 \sin^2 \phi \cos^2 \phi - (1 - k^2 \sin^2 \phi) \sin^2 \phi = -k'^2 \sin^2 \phi,$$

if  $k'^2 = 1 - k^2$ .

Hence we have for the coordinates of the point of the caustic,

$$\begin{aligned} k'^2 x &= -k'^2 \cos^3 \phi - k (1 - k^2 \sin^2 \phi)^{\frac{3}{2}}, \\ k'^2 y &= k^2 \sin^3 \phi, \end{aligned}$$

and eliminating  $\phi$ , we obtain for the equation of the caustic,

$$k'^2 x = -k'^2 \left( 1 - k^{-\frac{2}{3}} y^{\frac{2}{3}} \right)^{\frac{3}{2}} - k \left( 1 - k^{\frac{2}{3}} y^{\frac{2}{3}} \right)^{\frac{3}{2}};$$

or writing  $\frac{1}{\mu}$  instead of  $k$ , we find

$$\left( 1 - \frac{1}{\mu^2} \right) x = - \left( 1 - \mu^{\frac{2}{3}} y^{\frac{2}{3}} \right)^{\frac{3}{2}} + \mu \left( 1 - \mu^{-\frac{2}{3}} y^{\frac{2}{3}} \right)^{\frac{3}{2}}$$

for the equation of the caustic by refraction of the circle, for parallel rays. The equation was first obtained by St Laurent.

The discussion of the preceding equation presents considerable interest. In the first place to obtain the rational form write

$$x = \alpha, \quad \left(1 - \mu^{\frac{2}{3}} y^{\frac{2}{3}}\right)^{\frac{1}{2}} = \beta, \quad \mu \left(1 - \mu^{-\frac{2}{3}} y^{\frac{2}{3}}\right)^{\frac{1}{2}} = \gamma;$$

this gives

$$\alpha^4 + 2\alpha^2(\beta^2 + \gamma^2) + (\beta^2 - \gamma^2)^2 = 0,$$

and we have

$$\begin{aligned} \beta^2 + \gamma^2 &= 1 - 3\mu^{\frac{2}{3}} y^{\frac{2}{3}} + 3\mu^{\frac{4}{3}} y^{\frac{4}{3}} - \mu^2 y^2, \\ \gamma^2 &= \mu^2 - 3\mu^{\frac{4}{3}} y^{\frac{2}{3}} + 3\mu^{\frac{2}{3}} y^{\frac{4}{3}} - y^2, \end{aligned}$$

and consequently

$$\beta^4 - \gamma^4 = (1 - \mu^2) \left[ 1 - 3\mu^{\frac{2}{3}} y^{\frac{2}{3}} + (1 + \mu^2) y^{\frac{4}{3}} \right].$$

[p. 348] Hence dividing out by the factor  $(1 - \mu^2)^2$ , the equation becomes

$$(1 - \mu^2)^2 x^4 + 2(1 + \mu^2 - 6\mu^2 y^{\frac{2}{3}} + 3\mu^{\frac{2}{3}}(1 + \mu^2) y^{\frac{4}{3}} - (1 + \mu^2) y^2) x^2 + (1 - 3\mu^{\frac{2}{3}} y^{\frac{2}{3}} + (1 + \mu^2) y^{\frac{4}{3}})^2 = 0;$$

or reducing and arranging,

$$\begin{aligned} &(1 + \mu^2)^2 x^4 + 2(1 + \mu^2) x^2 [(1 + \mu^2) x^2 + (1 + \mu^2) y^2] \\ &+ 2(1 + \mu^2) x^2 y^2 + (1 + \mu^2)^2 y^4 \\ &+ 2\mu^{\frac{2}{3}} (6\mu^4 - 9\mu^2 (1 + \mu^2)) y^{\frac{2}{3}} \\ &+ 2\mu^{\frac{4}{3}} (6\mu^4 + 6\mu^2 (1 + \mu^2)) y^{\frac{4}{3}} = 0, \end{aligned}$$

which is of the form

$$A + 3\mu^{\frac{2}{3}} B y^{\frac{2}{3}} - 6\mu^2 C y^{\frac{4}{3}} = 0;$$

and the rationalized equation is

$$A^3 + 27\mu^2 B^3 y^2 - 216\mu^6 C^3 y^4 + 54\mu^4 ABC y^2 = 0,$$

where the values of  $A, B, C$  may be written

$$\begin{aligned} A &= (1 + \mu^2)^2 (x^2 + y^2)^2 - 2(1 + \mu^2) (x^2 + y^2) + 1, \\ B &= x^2 + 3y^2, \\ C &= (1 + \mu^2) (x^2 + y^2) + 1, \end{aligned}$$

the caustic is therefore a curve of the 12th order.

To find where the axis of  $x$  meets the curve, we have

$$y = 0, \quad A^3 = 0,$$

where

$$A_0 = -(1 + \mu^2)^2 x^4 + 2(1 + \mu^2) x^2 - 1,$$

i.e.

$$[(1 + \mu)^2 x^2 - 1] [(1 - \mu)^2 x^2 - 1] = 0,$$

or

$$x = \pm \frac{1}{1 + \mu}, \quad x = \pm \frac{1}{1 - \mu},$$

or there are in all four points, each of them a point of triple intersection.

To find where the line  $\infty$  meets the curve, we have

$$y = \infty, \quad A'^3 = 0,$$

where

$$A' = (x^2 + y^2)\{(1 - \mu)^2 x^2 + (1 + \mu)^2 y^2\},$$

i.e.

$$\infty, \quad x = \pm iy, \quad x = \pm \frac{1 + \mu}{1 - \mu} iy,$$

**[p. 349]** or the curve meets the line  $\infty$  in four points, each of them a point of triple intersection: two of these points are the circular points at  $\infty$ .

To find where the circle  $x^2 + y^2 = 1$  meets the curve, this gives  $x^2 = 1 - y^2$ , and thence

$$\begin{aligned} A &= \mu^2 (\mu^2 - 4) + 4(1 + 2\mu^2) y^2, \\ B &= 4 - y^2, \\ C &= \mu^2 + 2, \end{aligned}$$

and the equation becomes

$$\begin{aligned} &\{\mu^2 (\mu^2 - 4) + 4(1 + 2\mu^2) y^2\}^3 + 27\mu^4 (4 - y^2)^3 y^2 - 216\mu^6 (\mu^2 + 2)^3 y^4 \\ &+ 54\mu^4 (\mu^2 + 2) y^2 (4 - y^2) [\mu^2 (\mu^2 - 4) + 4(1 + 2\mu^2) y^2] = 0, \end{aligned}$$

which is only of the eighth order; it follows that each of the circular points at  $\infty$  (which have been already shown to be points upon the curve) are quadruple points of intersection of the curve and circle. The equation of the eighth order reduces itself to

$$(y^2 - \mu^2)^2 \{27\mu^4 y^4 + (\mu^2 - 4)^3\} = 0;$$

the values of  $x$  corresponding to the roots  $y = \pm\mu$  are obtained without difficulty, and those corresponding to the other roots are at once found by means of the identical equation

$$27\mu^4 y^4 + (\mu^2 - 4)^3 = -4(1 + \mu^2)(\mu^2 - 8)^3 y^2 + \dots;$$

we thus obtain for the coordinates of the points of intersection of the curve with the circle  $x^2 + y^2 = 1$ , the values

$$\begin{cases} x = \pm i\mu, \\ y = \pm\mu, \end{cases} \quad \begin{cases} x = \pm\sqrt{1 - \mu^2} \frac{(1 + 5\mu^2)}{3\sqrt{3}\mu^2} i, \\ y = \pm\frac{(\mu^2 - 4)^{\frac{3}{2}}}{3\sqrt{3}\mu^2} i, \end{cases}$$

each of the points of the first system being a quadruple point of intersection, each of the points of the second system a triple point of intersection, and each of the points of the third system a single point of intersection.

Next, to find where the circle  $x^2 + y^2 = \frac{1}{\mu^2}$  meets the curve; writing  $x^2 = \frac{1}{\mu^2} - y^2$ , **[p. 350]** we obtain for  $y$  an equation of the eighth order, which after all reductions is

$$\left(y^2 - \frac{1}{\mu^2}\right)^2 [27\mu^2 y^4 + (1 - 4\mu^2)^3] = 0,$$

and we have for the coordinates of the points of intersection,

$$\begin{cases} x = \pm \frac{1}{\mu} \sqrt{1 - \frac{1}{\mu^2}}, \\ y = \pm \frac{1}{\mu^2}, \end{cases} \quad \begin{cases} x = \pm \frac{(1 + 5\mu^2)\sqrt{1 - \frac{1}{\mu^2}}}{3\sqrt{3}\mu^2} i, \\ y = \pm \frac{(1 - 4\mu^2)^{\frac{3}{2}}}{3\sqrt{3}\mu^2} i, \end{cases}$$

each of the points of the first system being a quadruple point of intersection, each of the points of the second system a triple point of intersection, and each of the points of the third system a single point of intersection.



The points of intersection with the axes of  $x$ , and the points of triple intersection with the circles  $x^2 + y^2 = 1$  and  $x^2 + y^2 = \frac{1}{\mu^2}$ , are all of them cuspidal points; the two circular points at  $\infty$  are, I think, triple points, and the other two points of intersection with the line  $\infty$ , cuspidal points, but I have not verified this: assuming that it is so, there will be a reduction 54 accounted for in the class of the curve, but the curve is, in fact, as will be shown in the sequel, of the class 6; there is consequently a reduction 72 to be accounted for by other singularities of the curve.

#### XIV.

It is obvious from the preceding formulæ that the caustic stands to the circle, radius  $\frac{1}{\mu}$ , in a relation similar to that in which it stands to the circle, radius 1, i.e. to the refracting circle. In fact, the very same caustic would have been obtained if the circle radius  $\frac{1}{\mu}$  had been taken for the refracting circle, the index of refraction being  $\frac{1}{\mu}$  instead of  $\mu$ . This may be shown very simply by means of the irrational form of the equation as follows.

The equation of the caustic by refraction of the circle, radius 1, index of refraction  $\mu$ , is as we have seen

$$(1 - \mu^2)x = (1 - \mu^{\frac{2}{3}}y^{\frac{2}{3}})^{\frac{3}{2}} + \mu(1 - \mu^{-\frac{2}{3}}y^{\frac{2}{3}})^{\frac{3}{2}};$$

hence the equation of the caustic by refraction of the circle radius  $c'$ , index of refraction  $\mu'$ , is

$$(1 - \mu'^2) \frac{x}{c'} = \left\{ 1 - \mu'^{\frac{2}{3}} \left( \frac{y}{c'} \right)^{\frac{2}{3}} \right\}^{\frac{3}{2}} + \mu' \left\{ 1 - \mu'^{-\frac{2}{3}} \left( \frac{y}{c'} \right)^{\frac{2}{3}} \right\}^{\frac{3}{2}},$$

[p. 351] or, what is the same thing,

$$(1 - \mu'^2) \frac{x}{c'\mu'} = \frac{1}{\mu'} \left\{ 1 - \mu'^{\frac{2}{3}} c'^{-\frac{2}{3}} y^{\frac{2}{3}} \right\}^{\frac{3}{2}} + \left\{ 1 - \mu'^{-\frac{2}{3}} c'^{-\frac{2}{3}} y^{\frac{2}{3}} \right\}^{\frac{3}{2}},$$

which becomes identical with the equation of the first-mentioned caustic if  $\mu' = c' = \frac{1}{\mu}$ .

Hence taking  $c$  instead of 1 as the radius of the first circle, we find,

THEOREM. *The caustic by refraction for parallel rays of a circle, radius  $c$ , index of refraction  $\mu$ , is the same curve as the caustic by refraction for parallel rays of a concentric circle, radius  $\frac{c}{\mu}$ , index of refraction  $\frac{1}{\mu}$ .*

#### XV.

We may consequently in tracing the caustic confine our attention to the case in which the index of refraction is greater than unity. The circle, radius  $\frac{c}{\mu}$ , will in this case be within the refracting circle, and it is easy to see that if from the extremity of the diameter of the refracting circle perpendicular to the direction of the incident rays, tangents are drawn to the circle, radius  $\frac{c}{\mu}$ , the points of contact are the points of triple intersection of the caustic with the last-mentioned circle, and these points of intersection being, as already observed, cusps, the tangents in question are the tangents to the caustic at these cusps. The points of intersection with the axis of  $x$  are also cusps of the caustic, the tangents at these cusps coinciding with the axis of  $x$ : two of the last-mentioned cusps, viz. those whose distances from the centre are  $\pm \frac{1}{\mu + 1}$ , lie within the circle, radius  $\frac{c}{\mu}$ , the other two of the same four cusps, viz. those whose distances from the centre are  $\pm \frac{1}{\mu - 1}$ , lie without the circle, radius  $\frac{c}{\mu}$ ; the last-mentioned two cusps lie without the refracting circle, when  $\mu < 2$ , upon this circle, when  $\mu = 2$ , and within it and therefore between the two circles, when  $\mu > 2$ . The caustic is therefore of the forms in the annexed figures 3, 4, 5, in each

[Figure 3: outer (refracting) circle with inner circle of radius  $c/\mu$ ; cusped lens-shaped caustic touching both circles internally; case  $\mu < 2$ .] [Figure 4: same construction with  $\mu = 2$ , the outer pair of caustic cusps reaching the refracting circle.] [Figure 5: same with  $\mu > 2$ , the outer caustic cusps lying between the two circles.]

[p. 352] of which the outer circle is the refracting circle, and  $\mu$  is  $> 1$ , but the three figures correspond respectively to the cases  $\mu < 2$ ,  $\mu = 2$  and  $\mu > 2$ . The same three figures will represent the different forms of the caustic when the inner circle is the refracting circle and  $\mu < 1$ , the three figures then respectively corresponding to the cases  $\mu > \frac{1}{2}$ ,  $\mu = \frac{1}{2}$ , and  $\mu < \frac{1}{2}$ .

## XVI.

To find the tangential equation, I retain  $k$  instead of its value  $\frac{1}{\mu}$ ; the equation of the refracted ray then is

$$x(k \cos \phi - \sqrt{1 - k^2 \sin^2 \phi}) + y(k \sin \phi + \cot \phi \sqrt{1 - k^2 \sin^2 \phi}) - k = 0,$$

and representing this by

$$Xx + Yy - k = 0,$$

we have

$$\begin{aligned} X &= k \cos \phi - \sqrt{1 - k^2 \sin^2 \phi}, \\ Y &= k \sin \phi + \cot \phi \sqrt{1 - k^2 \sin^2 \phi}, \end{aligned}$$

equations which give

$$\begin{aligned} X \cos \phi + Y \sin \phi &= k, \\ X^2 + Y^2 &= \frac{1}{\sin^2 \phi}, \end{aligned}$$

and consequently

$$\sin \phi = \frac{1}{\sqrt{X^2 + Y^2}}, \quad \cos \phi = \frac{\sqrt{X^2 + Y^2 - 1}}{\sqrt{X^2 + Y^2}},$$

and we have

$$X\sqrt{X^2 + Y^2 - 1} + Y - k\sqrt{X^2 + Y^2} = 0,$$

which gives

$$(X^2 + Y^2)(X^2 - 1 - k^2) = -2kY\sqrt{X^2 + Y^2};$$

or, dividing out by the factor  $\sqrt{X^2 + Y^2}$ , the equation becomes

$$\sqrt{X^2 + Y^2}(X^2 - 1 - k^2) = -2kY,$$

from which

$$(X^2 + Y^2)(X^2 - 1 - k^2)^2 = 4k^2Y^2;$$

or reducing and arranging, we obtain

$$X^2(X^2 - 1 - k^2)^2 + Y^2(X + 1 + k)(X + 1 - k)(X - 1 + k)(X - 1 - k) = 0$$

for the tangential equation of the caustic by refraction of a circle for parallel rays. The caustic is therefore of the class 6.

[p. 353]

## XVII.

Suppose next that rays proceeding from a point are reflected at a circle.

A very elegant solution of the problem is given by Lagrange in the *Mém. de Turin*; the investigation, as given by Mr P. Smith in a note in the *Cambridge and Dublin Mathematical Journal*, t. II. [1847] p. 237, is as follows:

Let  $B$  be the radiant point,  $RBP$  an incident ray, and  $PS$  a reflected ray;  $CA$  a fixed radius;  $ACP = \alpha$ ,  $ACB = e$ , reciprocal of  $CB = c$ , reciprocal of  $CP = a$ . The equations of the incident and reflected ray, where  $u = \frac{1}{r}$ , may be written

$$\begin{aligned} A \sin \theta + B \cos \theta &= u; & \text{incident ray,} \\ A \sin(2\alpha - \theta) + B \cos(2\alpha - \theta) &= u; & \text{reflected ray,} \end{aligned}$$

the conditions for determining  $A$  and  $B$  being

$$\begin{aligned} a &= A \sin \alpha + B \cos \alpha, \\ c &= A \sin e + B \cos e, \end{aligned}$$

whence

$$A = \frac{a \cos e - c \cos \alpha}{\sin(\alpha - e)}, \quad B = \frac{c \sin \alpha - a \sin e}{\sin(\alpha - e)}.$$

[Figure: circle with centre  $C$ ; radiant point  $B$  inside; incident ray  $RBP$  meeting the circle at  $P$  and reflected as  $PS$ ; fixed radius  $CA$ .]

Substituting these values, the equation of the reflected ray becomes

$$a \sin(2\alpha - \theta - e) = u \sin(\alpha - e) + c \sin(\alpha - \theta),$$

from which and its differential with respect to the arbitrary parameter  $\alpha$ , the equation of the caustic, or envelope of the reflected rays, will be found by eliminating  $\alpha$ .

In this,  $\alpha$  being the only quantity treated as variable in the differentiation, let

$$2\alpha - \theta - e = 2\phi,$$

therefore

$$\alpha = \phi + \frac{1}{2}(\theta + e),$$

[p. 354] and the equation becomes

$$a \sin 2\phi = u \sin\left\{\phi + \frac{1}{2}(\theta - e)\right\} + c \sin\left\{\phi - \frac{1}{2}(\theta - e)\right\}.$$

Make

$$P = \frac{(u + c) \cos \frac{1}{2}(\theta - e)}{2a}, \quad Q = \frac{(u - c) \sin \frac{1}{2}(\theta - e)}{2a},$$

also

$$x = \frac{1}{\cos \phi}, \quad y = \frac{1}{\sin \phi},$$

then the equation becomes

$$Px + Qy = 1,$$

with the condition

$$x^{-2} + y^{-2} = 1.$$

Hence

$$P = \lambda x^{-3}, \quad Q = \lambda y^{-3};$$

multiplying by  $x$  and  $y$ , and adding, we find  $\lambda = 1$ ; therefore

$$P = x^{-3}, \quad Q = y^{-3}.$$

Hence

$$P^{\frac{2}{3}} + Q^{\frac{2}{3}} = 1,$$

or restoring the values of  $P$  and  $Q$ ,

$$\left\{ (u+c) \cos \frac{1}{2}(\theta-e) \right\}^{\frac{2}{3}} + \left\{ (u-c) \sin \frac{1}{2}(\theta-e) \right\}^{\frac{2}{3}} = 1,$$

the equation of the caustic.

## XVIII.

But the equation of the caustic for rays proceeding from a point and reflected at a circle may be obtained by a different method, as follows:

Take the centre of the circle for origin; let  $c$  be the radius of the circle,  $a, b$  the coordinates of the radiant point,  $\alpha, \beta$  the coordinates of the point of incidence,  $x, y$  the coordinates of a point in the reflected ray. Then we have from the equation of the circle  $\alpha^2 + \beta^2 = c^2$ , and the equation of the reflected ray is by the general formula,

$$(b\alpha - a\beta)(\alpha x + \beta y - c^2) + (y\alpha - x\beta)(a\alpha + b\beta - c^2) = 0;$$

[p. 355] or arranging the terms in a different order,

$$(bx + ay)(\alpha^2 - \beta^2) + 2(by - ax)\alpha\beta - c^2(b + y)\alpha + c^2(a + x)\beta = 0;$$

and writing herein  $\alpha = c \cos \theta$ ,  $\beta = c \sin \theta$ , the equation becomes

$$(bx + ay) \cos 2\theta + (by - ax) \sin 2\theta - (b + y) c \cos \theta + (a + x) c \sin \theta = 0,$$

where  $\theta$  is a variable parameter.

Now in general to find the envelope of

$$A \cos 2\theta + B \sin 2\theta + C \cos \theta + D \sin \theta + E = 0,$$

we may put  $z = e^{i\theta}$ , which gives the equation

$$(A - Bi)z^4 + (C - Di)z^3 + 2Ez^2 + (C + Di)z + (A + Bi) = 0,$$

and equate the discriminant to zero: this gives

$$(4I)^3 - 27(-8J)^2 = 0,$$

where

$$\begin{aligned} 4I &= 4(A^2 + B^2) - (C^2 + D^2) + 4E^2, \\ -8J &= A(C^2 - D^2) + 2BCD - [8(A^2 + B^2) + (C^2 + D^2)]E + 4E^3, \end{aligned}$$

and consequently

$$\begin{aligned} &[4(A^2 + B^2) - (C^2 + D^2) + 4E^2]^3 \\ &- 27\{A(C^2 - D^2) + 2BCD - [8(A^2 + B^2) + (C^2 + D^2)]E + 4E^3\}^2 = 0; \end{aligned}$$

and substituting for  $A, B, C, D, E$  their values, we find

$$\{4(a^2 + b^2)(x^2 + y^2) - c^2[(x + a)^2 + (y + b)^2]\}^3 - 27(bx - ay)^2(x^2 + y^2 - a^2 - b^2)^2 = 0,$$

for the equation of the caustic in the case of rays proceeding from a point and reflected at a circle: the equation was first obtained by St Laurent.

It will be convenient to consider the axis of  $x$  as passing through the radiant point; this gives  $b = 0$ ; and if we assume also  $c = 1$ , the equation of the caustic becomes

$$\{(4a^2 - 1)(x^2 + y^2) - 2ax - a^2\}^3 - 27a^2y^2(x^2 + y^2 - a^2)^2 = 0.$$

XIX.

Reverting to the equation of the reflected ray, and putting, as before,  $c = 1$ ,  $b = 0$ , this becomes

$$(-2a \cos \theta + 1)x + \frac{a \cos 2\theta - \cos \theta}{\sin \theta} y + a = 0;$$

differentiating with respect to  $\theta$ , we have

$$2a \sin \theta \cdot x + \frac{-2a \sin \theta \sin \theta (1 - 2 \sin^2 \theta) + (a \cos 2\theta - \cos \theta) \cos \theta}{\sin^2 \theta} y = 0;$$

[p. 356] and from these equations

$$x = \frac{a^2 \cos \theta (1 + 2 \sin^2 \theta) - a}{1 - 3a \cos 2\theta + 2a^2},$$

$$y = \frac{2a^2 \sin^3 \theta}{1 - 3a \cos 2\theta + 2a^2},$$

which give the coordinates of a point of the caustic in terms of the angle  $\theta$  which determines the position of the point of incidence. The values in question satisfy, as they should do, the equation

$$\{(4a^2 - 1)(x^2 + y^2) - 2ax - a^2\}^3 - 27a^2 y^2 (x^2 + y^2 - a^2)^2 = 0;$$

we have, in fact,

$$x^2 + y^2 - a^2 = \frac{-4a^2 (\cos \theta - a)^2}{(1 - 3a \cos 2\theta + 2a^2)^2},$$

$$(4a^2 - 1)(x^2 + y^2) - 2ax - a^2 = \frac{12a^2 (\cos \theta - a)^2}{(1 - 3a \cos 2\theta + 2a^2)^2},$$

from which it is easy to derive the equation in question.

XX.

If we represent the equation of the reflected ray by

$$Xx + Yy + a = 0,$$

then we have

$$X = -2a \cos \theta + 1,$$

$$Y = \frac{a \cos 2\theta - \cos \theta}{\sin \theta},$$

and thence

$$(X - 1)^2 - 4a^2 = -4a^2 \sin^2 \theta,$$

$$X^2 + Y^2 = \frac{1}{\sin^2 \theta} (1 - 2a \cos \theta + a^2),$$

$$X + a^2 = 1 - 2a \cos \theta + a^2,$$

and consequently

$$(X^2 + Y^2)\{(X - 1)^2 - 4a^2\} + 4a^2 X + 4a^4 = 0,$$

or, what is the same thing,

$$\{X(X - 1) - 2a^2\}^2 + Y^2 \{(X - 1)^2 - 4a^2\} = 0,$$

which may be considered as the tangential equation of the caustic by reflexion of a circle; or if we consider  $X, Y$  as the coordinates of a point, then the equation may be considered as that of the polar of the caustic.

The polar is therefore a curve of the fourth order, having two double points defined by the equations  $X(X-1)-2a^2=0$ ,  $Y=0$ , [p. 357] and a third double point at infinity on the axis of  $Y$ , i.e. three double points in all; the number of cusps is therefore 0, and there are consequently 4 double tangents and 6 inflections, and the curve is of the class 6. And as  $Y$  is given as an explicit function of  $X$ , there is of course no difficulty in tracing the curve. We thus see that the caustic by reflexion of a circle is a curve of the order 6, and has 4 double points and 6 cusps (the circular points at infinity are each of them a cusp, so that the number of cusps at a finite distance is 4): this coincides with the conclusions which will be presently obtained by considering the equation of the caustic.

## XXI.

The equation of the caustic by reflexion of a circle is

$$\{(4a^2-1)(x^2+y^2)-2ax-a^2\}^3-27a^2y^2(x^2+y^2-a^2)^2=0.$$

Suppose first that  $y=0$ , we have

$$\{(4a^2-1)x^2-2ax-a^2\}^3=0,$$

$$x=\frac{a}{2a-1}, \quad x=\frac{-a}{2a+1},$$

or the curve meets the axis of  $x$  in two points, each of which is a triple point of intersection.

Write next  $x^2=y^2+a^2$ , this gives

$$\{(4a^2-1)a^2-2ax-a^2\}^3=0,$$

and consequently

$$x=-a(1-2a^2), \quad y=\pm 2a^2\sqrt{1-a^2},$$

or the curve meets the circle  $x^2+y^2-a^2=0$  in two points, each of which is a triple point of intersection.

To find the nature of the infinite branches, we may write, retaining only the terms of the degrees six and five,

$$(x^2+y^2)^2[(4a^2-1)^3(x^2+y^2)]-6(4a^2-1)^2a(x^2+y^2)^2x-27a^2y^2(x^2+y^2)^2=0;$$

and rejecting the factor  $(x^2+y^2)^2$ , this gives

$$(4a^2-1)^3x^4-\{(4a^2-1)^3-27a^2\}y^4-6(4a^2-1)^2ax^3=0;$$

or reducing,

$$(4a^2-1)^3x^4-(1-a^2)(8a^2-1)^3y^4-6(4a^2-1)^2ax^3=0;$$

and it follows that there are two asymptotes, the equations of which are

$$y=\pm\frac{(4a^2-1)^{\frac{3}{4}}}{(1-a^2)^{\frac{1}{4}}(8a^2+1)^{\frac{3}{4}}}\left(x-\frac{3a}{4a^2-1}\right).$$

[p. 358] Represent for a moment the equation of one of the asymptotes by  $y=A(x-\alpha)$ , then the perpendicular from the origin or centre of the reflecting circle is  $\frac{A\alpha}{\sqrt{1+A^2}}$ , and

$$A\alpha=\frac{3a\sqrt{4a^2-1}}{\sqrt{1-a^2}(1+8a^2)^{\frac{1}{2}}},$$

$$1+A^2=\frac{(1-a^2)(1+8a^2)^2+(4a^2-1)^3}{(1-a^2)(1+8a^2)^2}=\frac{27a^2}{(1-a^2)(1+8a^2)^2},$$

$$\sqrt{1+A^2}=\frac{3\sqrt{3}a}{\sqrt{1-a^2}(1+8a^2)^{\frac{1}{2}}},$$

and the perpendicular is  $\frac{1}{\sqrt{3}}\sqrt{4a^2 - 1}$ , which is less than  $a$  if only  $a^2 < 1$ , i.e. in every case in which the asymptote is real.

The tangents parallel and perpendicular to the axis of  $x$  are most readily obtained from the equation of the reflected ray, viz.

$$(-2a \cos \theta + 1)x + \frac{a \cos 2\theta - \cos \theta}{\sin \theta}y + a = 0;$$

the coefficient of  $x$  (if the equation is first multiplied by  $\sin \theta$ ) vanishes if  $\sin \theta = 0$ , which gives the axis of  $x$ , or if  $\cos \theta = \frac{1}{2a}$ , which gives  $y = \pm \frac{\sqrt{4a^2 - 1}}{2a}$ , for the tangents parallel to the axis of  $x$ .

The coefficient of  $y$  vanishes if  $a \cos 2\theta - \cos \theta = 0$ ; this gives

$$\cos \theta = \frac{1 \pm \sqrt{8a^2 + 1}}{4a}, \quad \sin \theta = \frac{1}{8a^2}(4a^2 - 1 \mp \sqrt{8a^2 + 1}),$$

and the tangents perpendicular to the axis of  $x$  are thus given by

$$x = \frac{-2a}{1 + \sqrt{8a^2 + 1}};$$

these tangents are in fact double tangents of the caustic. In order that the point of contact may be real, it is necessary that  $\sin \theta, \cos \theta$  should be real; this will be the case for both values of the ambiguous sign if  $a > \text{or} = 1$ , but only for the upper value if  $a < 1$ .

It has just been shown that for the tangents parallel to the axis of  $x$ , we have

$$y = \pm \frac{\sqrt{4a^2 - 1}}{2a},$$

the values of  $y$  being real for  $a > \frac{1}{2}$ : it may be noticed that the value  $y = \frac{\sqrt{4a^2 - 1}}{2a}$  [p. 359] is greater, equal, or less than or to  $y = 2a^2\sqrt{1 - a^2}$ , according as  $a >, =$ , or  $< \frac{1}{\sqrt{2}}$ ; this depends on the identity  $(4a^2 - 1) - 16a^4(1 - a^2) = (2a^2 - 1)^2(7a^2 + 1)$ .

To find the points of intersection with the reflecting circle,  $x^2 + y^2 - 1 = 0$ , we have

$$(3a^2 - 1 - 2ax)^3 - 27a^2(1 - a^2)(1 - x^2)^2 = 0;$$

or, reducing,

$$8a^3x^3 + (-27a^4 + 18a^2 - 15)a^2x^2 + (54a^4 - 36a^2 + 6)ax + (-27a^4 + 18a^2 + 1) = 0,$$

i.e.

$$(ax - 1)^2(8ax - 27a^4 + 18a^2 + 1) = 0.$$

The factor  $(ax - 1)^2$  equated to zero shows that the caustic touches the circle in the points  $x = \frac{1}{a}$ ,  $y = \pm\sqrt{1 - \frac{1}{a^2}}$ , i.e. in the points in which the circle is met by the polar of the radiant point, and which are real or imaginary according as  $a >$  or  $< 1$ . The other factor gives

$$x = \frac{27a^4 - 18a^2 - 1}{8a}.$$

Putting this value equal to  $\pm 1$ , the resulting equation is  $(a + 1)(27a^3 + 9a + 1) = 0$ , and it follows that  $x$  will be in absolute magnitude greater or less than 1, i.e. the points in question will be imaginary or real, according as  $a > 1$  or  $a < 1$ .

It is easy to see that the curve passes through the circular points at infinity, and that these points are cusps on the curve; the two points of intersection with the axis of  $x$  are cusps (the axis of  $x$  being the tangent), and the two points of intersection with the circle  $x^2 + y^2 - a^2 = 0$  are also cusps, the tangent at each of the cusps coinciding with the tangent of the circle; there are consequently in all 6 cusps.

## XXII.

To investigate the position of the double points we may proceed as follows: write for shortness  $P = (4a^2 - 1)(x^2 + y^2) - 2ax - a^2$ ,  $Q = ay$ ,  $S = x^2 + y^2 - a^2$ ; the equation of the caustic is

$$P^3 - 27 Q^2 S^2 = 0;$$

hence, at a double point,

$$P^2 \frac{dP}{dx} - 18 Q S^2 \frac{dQ}{dx} = 0, \quad P^2 \frac{dP}{dy} - 18 Q S^2 \frac{dQ}{dy} = 0,$$

one of which equations may be replaced by

$$\frac{dP}{dx} \frac{dQ}{dy} - \frac{dP}{dy} \frac{dQ}{dx} = 0.$$



[PAPER 145 — A MEMOIR UPON CAUSTICS, CONTINUED]

*Printed pages 360–371; scan PDF pages 376–387.*

Now

$$\begin{aligned}\frac{dP}{dx} &= 2\{(4a^2 - 1)x - a\}, & \frac{dP}{dy} &= 2(4a^2 - 1)y, \\ \frac{dQ}{dx} &= 2axy, & \frac{dQ}{dy} &= a(x^2 + 3y^2 - a^2) = a(S + 2y^2);\end{aligned}$$

substituting these values in the last preceding equation, we find

$$\frac{(4a^2 - 1)x - a}{(4a^2 - 1)y} = \frac{2xy}{S + 2y^2},$$

or, reducing,

$$(4a^2 - 1)x - a = \frac{2axy^2}{S};$$

and using this to simplify the equation

$$P^2 \frac{dP}{dx} - 18Q \frac{dQ}{dx} = 0,$$

we have

$$P^2 \frac{4axy^2}{S} - 18axyS \cdot 2axy = 0,$$

i.e.

$$\frac{P^2}{S} - 9axS = 0,$$

and therefore

$$9x = \frac{P^2}{aS^2}.$$

Multiplying by  $P$  and writing for  $P^3$  its value  $27a^3y^2S^3$ , we have

$$Px = 3ayS,$$

and thence

$$P^2 = \frac{9a^2y^2S^2}{x^2}, \quad P^3 = \frac{27a^3y^3S^3}{x^3} = 27a^3y^2S^3,$$

whence

$$\frac{ay^2}{x^3} = S^2, \quad \frac{a^{\frac{1}{3}}y^{\frac{2}{3}}}{x} = S, \quad \text{or} \quad \frac{y^2}{S^3} = \frac{x^3}{a^3};$$

and substituting in the equation

$$x = \frac{a}{4a^2 - 1} \left( 1 + \frac{2y^2}{S} \right),$$

we find

$$x = \frac{a}{4a^2 - 1} \left( 1 + \frac{2x}{a^{\frac{1}{3}}} \right),$$

or, rationalising,

$$4ax^3 - \{(4a^2 - 1)x - a\}^3 = 0,$$

or, what is the same thing,

$$(4ax - 1)(x - a(a + i\sqrt{1 - a^2})^3)(x - a(a - i\sqrt{1 - a^2})^3) = 0.$$

The factor  $4ax - 1$  equated to zero gives  $x = \frac{1}{4a}$ , from which  $y$  may be found, but the resulting point is not a double point; the other factors give each of them double points, and if we write

$$x = a(a + i\sqrt{1 - a^2})^3,$$

we find

$$y = \frac{2a^2i(a + i\sqrt{1 - a^2})^{\frac{5}{2}}}{(3a - i\sqrt{1 - a^2})^{\frac{1}{2}}},$$

values which, in fact, belong to one of the four double points. It is easy to see that the points in question are always imaginary.

It may be noticed, by way of verification, that the preceding values of  $x, y$  give

$$\begin{aligned} (4a^2 - 1)(x^2 + y^2) - 2ax - a^2 &= \frac{a^2}{1 + 8a^2}(1 - 4a^2 - 4ai\sqrt{1 - a^2}), \\ x^2 + y^2 - a^2 &= \frac{4a^2}{1 + 8a^2}(3a - i\sqrt{1 - a^2})(a + i\sqrt{1 - a^2})^3, \\ y^2 &= \frac{4a^4}{1 + 8a^2}(-1 + 14a^2 - 16a^4 + 2a(3 - 8a^2)i\sqrt{1 - a^2}); \end{aligned}$$

and if the quantities within ( ) on the right-hand side are represented by  $A, B, C$ , then

$$\begin{aligned} \frac{A}{B} &= -(a + i\sqrt{1 - a^2}), \\ \frac{C}{B} &= -(a + i\sqrt{1 - a^2})^3, \end{aligned}$$

whence we have identically,

$$\left(\frac{A}{B}\right)^3 = \frac{C}{B}, \text{ or } A^3 = B^2C,$$

by means of which it appears that the values of  $x, y$  satisfy, as they should do, the equation of the caustic; and by forming the expressions for  $(4a^2 - 1)x - a$  and  $x^2 + 2y^2 - a^2$ , it might be shown, *à posteriori*, that the point in question was a double point.

### XXIII.

The equation

$$\{(4a^2 - 1)(x^2 + y^2) - 2ax - a^2\}^3 - 27a^2y^2(x^2 + y^2 - a^2)^2 = 0$$

becomes when  $a = 1$  (i.e. when the radiant point is in the circumference),

$$\{3y^2 + (x - 1)(3x + 1)\}^3 - 27y^2(y^2 + x^2 - 1)^2 = 0;$$

it is easy to see that this divides by  $(x - 1)^2$ ; and throwing out this factor, we have for the caustic the equation of the fourth order,

$$27y^4 + 18y^2(3x^2 - 1) + (x - 1)(3x + 1)^3 = 0.$$

## XXIV.

The equation

$$\{(4a^2 - 1)(x^2 + y^2) - 2ax - a^2\}^3 - 27a^2y^2(x^2 + y^2 - a^2)^2 = 0$$

becomes when  $a = \infty$  (i.e. in the case of parallel rays),

$$(4x^2 + 4y^2 - 1)^3 - 27y^2 = 0,$$

which may also be written

$$64x^6 + 48x^4(4y^2 - 1) + 12x^2(4y^2 - 1)^2 + (8y^2 + 1)^2(4y^2 - 1) = 0.$$

## XXV.

It is now easy to trace the curve. Beginning with the case  $a = \infty$ , the curve lies wholly within the reflecting circle, which it touches at two points; the line joining the points of contact, being in fact the axis of  $y$ , divides the curve into two equal portions; the curve has in the present, as in every other case (except one limiting case), two cusps on the axis of  $x$  (see fig. 6). Next, if  $a$  be positive and  $> 1$ , the general form of the curve is the same as before, only the line joining the points of contact with the reflecting circle divides the curve into unequal portions, that in the

*[Figure: Fig. 6 ( $a = \infty$ ): reflecting circle with the inscribed caustic, two cusps on horizontal axis, two contact points on vertical axis. Fig. 7 ( $a > 1$ ): similar but asymmetric, with one lobe smaller.]*

neighbourhood of the radiant point being the smaller of the two portions (see fig. 7). When  $a = 1$ , the two points of contact with the reflecting circle unite together at the radiant point; the curve throws off, as it were, the two coincident lines  $x = 1$ , and the order is reduced from 6 to 4. The curve has the form fig. 8, with only a single cusp on the axis of  $x$ . If  $a$  be further diminished,  $a < 1 > \frac{1}{\sqrt{2}}$ , the curve takes the form shown by fig. 9, with two infinite branches, one of them having simply a cusp on the axis of  $x$ , the other having a cusp on the axis of  $x$ , and a pair of cusps at its intersection with the circle through the radiant point; there are two asymptotes equally inclined to the axis of  $x$ . In the case  $a = \frac{1}{\sqrt{2}}$ , the form of the curve is nearly the same as before, only the cusps upon the circle through the radiant point lie on the axis of  $y$  (see fig. 10). The case  $a < \frac{1}{\sqrt{2}} > \frac{1}{2}$  is shown, fig. 11. For  $a = \frac{1}{2}$ , the two

asymptotes coincide with the axis of  $x$ ; one of the branches of the curve has wholly disappeared, and the form of the other is modified by the coincidence of the asymptotes

[Figures: Fig. 8 ( $a = 1$ ); Fig. 9 ( $a < 1 > 1/\sqrt{2}$ ); Fig. 13 ( $a = \frac{1}{2}$ ); Fig. 11 ( $a < 1/\sqrt{2} > \frac{1}{2}$ ); Fig. 10 ( $a = 1/\sqrt{2}$ ); Fig. 12 ( $a = \frac{1}{2}$ ). Each shows a reflecting circle (dashed), the caustic (solid), and the radiant point with asymptotic branches for the smaller- $a$  cases.]

with the axis of  $x$ ; it has in fact acquired a cusp at infinity on the axis of  $x$  (see fig. 12). When  $a < \frac{1}{2}$ , the curve consists of a single finite branch, with two cusps on the axis of  $x$ , and two cusps at the points of intersection with the circle through the radiant point; one of the last-mentioned cusps will be outside the reflecting circle as long as  $a > \frac{1}{4}$ ; fig. 13 represents the case  $a = \frac{1}{4}$ , for which this cusp is upon the reflecting circle. For  $a < \frac{1}{4}$ , the curve lies wholly within the reflecting circle, one of the cusps upon the axis of  $x$  being always within, and the other always without the circle through the radiant point, and as  $a$  approaches 0 the curve becomes smaller and smaller, and ultimately disappears in a point. The case  $a$  negative is obviously included in the preceding one.

Several of the preceding results relating to the caustic by reflexion of a circle were obtained, and the curve is traced in a memoir by the Rev. Hamnet Holditch, *Quarterly Mathematical Journal*, t. I. [1857, pp. 93–111].

## XXVI.

Suppose next that rays proceeding from a point are *refracted* at a circle. Take the centre of the circle as origin, let the radius be  $c$ , and take  $\xi, \eta$  as the coordinates of the radiant point,  $\alpha, \beta$  the coordinates of the point of incidence,  $x, y$  the coordinates of a point in the refracted ray: then the general equation

$$-\bar{q} \overline{G^2 \nabla QGN} + \mu^2 \overline{QG^2 \nabla qGN}^2 = 0$$

becomes, taking the centre of the circle as the point  $N$  on the normal, or writing  $a = b, b = 0$ ,

$$-[(x - \alpha)^2 + (y - \beta)^2]\{(\alpha^2 + \beta^2)\} + \mu^2(\alpha^2 + \beta^2)\{(\xi - \alpha)^2 + (\eta - \beta)^2\} = 0;$$

or putting  $\alpha^2 + \beta^2 = c^2$ , and expanding,

$$\begin{aligned} & \alpha^2 [2(\eta^2 x - \mu^2 xy^2 \xi)] \\ & + \alpha^2 \beta [-4(\xi \eta x - \mu^2 xy \xi) + 2(\eta^2 y - \mu^2 y^2 \eta)] \\ & + \alpha \beta^2 [-4(\xi y - \mu^2 xy \eta) + 2(\xi^2 x - \mu^2 x^2 \xi)] \\ & + \beta^3 [2(\xi^2 y - \mu^2 y^2 \eta)] \\ & - \alpha^2 [(x^2 + y^2 + c^2)\eta^2 - \mu^2(\xi^2 + \eta^2 + c^2)y^2] \\ & + 2\alpha\beta \{(x^2 + y^2 + c^2)\xi\eta - \mu^2(\xi^2 + \eta^2 + c^2)xy\} \\ & - \beta^2 \{(x^2 + y^2 + c^2)\xi^2 - \mu^2(\xi^2 + \eta^2 + c^2)x^2\} \\ & = 0, \end{aligned}$$

which may be represented by

$$A\alpha^3 + B\alpha^2\beta + C\alpha\beta^2 + D\beta^3 + F\alpha^2 + G\alpha\beta + H\beta^2 = 0.$$

Now  $\alpha^2 + \beta^2 = c^2$ , and we may write

$$\alpha = \frac{1}{2}c \left( z + \frac{1}{z} \right), \quad \beta = \frac{1}{2}ci \left( z - \frac{1}{z} \right).$$

The equation thus becomes

$$\begin{aligned} & A \left( z + \frac{1}{z} \right)^3 - Bi \left( z + \frac{1}{z} \right)^2 \left( z - \frac{1}{z} \right) - C \left( z + \frac{1}{z} \right) \left( z - \frac{1}{z} \right)^2 - Di \left( z - \frac{1}{z} \right)^3 \\ & + \frac{2}{c} F \left( z + \frac{1}{z} \right)^2 - \frac{2}{c} Gi \left( z + \frac{1}{z} \right) \left( z - \frac{1}{z} \right) - \frac{2}{c} H \left( z - \frac{1}{z} \right)^2 = 0 \end{aligned}$$

or expanding,

$$\begin{aligned} & (A - Bi - C - Di) z^3 \\ & + \frac{2}{c} (F - Gi - H) z^2 \\ & + (3A - Bi + C + 3Di) z \\ & + \frac{2}{c} (F + H) \\ & + (3A + Bi + C - 3Di) \frac{1}{z} \\ & + \frac{2}{c} (F + Gi - H) \frac{1}{z^2} \\ & + (A + Bi - C + Di) \frac{1}{z^3} = 0; \end{aligned}$$

in which  $z$  may be considered as the variable parameter; hence the equation of the caustic may be obtained by equating to zero the discriminant of the above function of  $z$ ; but the discriminant of a sextic function has not yet been calculated. The equation would be of the order 20, and it appears from the result previously obtained for parallel rays, that the equation must be of the order 12 at the least; it is, I think, probable that there is not any reduction of order in the general case. It is however practicable, as will presently be seen, to obtain the tangential equation of the caustic by refraction, and the curve is thus shown to be only of the class 6.

## XXVII.

Suppose that rays proceeding from a point are refracted at a circle, and let it be required to find the equation of the secondary caustic: take the centre of the circle as origin, let  $c$  be the radius,  $\xi, \eta$  the coordinates of the radiant point,  $\alpha, \beta$  the coordinates of a point upon the circle,  $\mu$  the index of refraction; the secondary caustic will be the envelope of the circle,

$$\mu\{(x - \alpha)^2 + (y - \beta)^2\} - \{(\xi - \alpha)^2 + (\eta - \beta)^2\} = 0,$$

where  $\alpha, \beta$  are variable parameters connected by the equation  $\alpha^2 + \beta^2 - c^2 = 0$ ; the equation of the circle may be written in the form

$$\mu^2(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2) - 2(\mu^2x - \xi)\alpha - 2(\mu^2y - \eta)\beta = 0.$$

But in general the envelope of  $A\alpha + B\beta + C = 0$ , where  $\alpha, \beta$  are connected by the equation  $\alpha^2 + \beta^2 - c^2 = 0$ , is  $c^2(A^2 + B^2) - C^2 = 0$ , and hence in the present case the equation of the envelope is

$$\{\mu^2(x^2 + y^2 + c^2) - (\xi^2 + \eta^2 + c^2)\}^2 = 4c^2\{(\mu^2x - \xi)^2 + (\mu^2y - \eta)^2\},$$

which may also be written

$$\{\mu^2(x^2 + y^2 - c^2) - (\xi^2 + \eta^2 - c^2)\}^2 = 4c^2\mu^2\{(x - \xi)^2 + (y - \eta)^2\}.$$

If the axis of  $x$  be taken through the radiant point, then  $\eta = 0$ , and writing also  $\xi = a$ , the equation becomes

$$\{\mu^2(x^2 + y^2 - c^2) - a^2 + c^2\}^2 = 4c^2\mu^2\{(x - a)^2 + y^2\};$$

or taking the square root of each side,

$$\{\mu^2(x^2 + y^2 - c^2) - a^2 + c^2\} = 2c\mu\sqrt{(x - a)^2 + y^2};$$

whence multiplying by  $\frac{1}{\mu^2}$  and adding on each side  $c^2\left(\mu - \frac{1}{\mu} + \frac{a}{\mu}\right)(x - a)^2 + y^2$ , we have

or

$$\sqrt{\left(x - \frac{a}{\mu}\right)^2 + y^2} = \sqrt{(x - a)^2 + y^2} + c\left(\mu - \frac{1}{\mu}\right),$$

which shows that the secondary caustic is the Oval of Descartes, or as it will be convenient to call it, the Cartesian.

It is proper to remark, that the Cartesian consists in general of two ovals, one of which is the orthogonal trajectory of the refracted rays, the other the orthogonal trajectory of the false refracted rays. In the case of reflexion, the secondary caustic is a Cartesian having a double point; this may be either a conjugate point, or a real double point arising from the union and intersection of the two ovals; the same secondary caustic may arise also from refraction, as will be presently shown.

### XXVIII.

Reverting to the original form of the equation of the secondary caustic, multiplying by  $\frac{1}{\mu^2} \left(1 - \frac{c^2}{a^2}\right)$  and adding on each side  $\frac{c^2}{\mu^2} \left(1 - \frac{c^2}{a^2}\right) + \frac{c^2}{a^2} \{(x-a)^2 + y^2\}$ , the equation becomes

$$\left\{ \left(x - \frac{c^2}{a}\right)^2 + y^2 \right\} = \left[ \frac{c}{a} \sqrt{(x-a)^2 + y^2} + \frac{a}{\mu} \left(1 - \frac{c^2}{a^2}\right) \right]^2,$$

or extracting the square root,

$$\sqrt{\left(x - \frac{c^2}{a}\right)^2 + y^2} = \sqrt{(x-a)^2 + y^2} + \frac{a}{\mu} \left(1 - \frac{c^2}{a^2}\right).$$

Combining this with the former result, we see that the equation may be expressed indifferently in any one of the four forms,

$$\begin{aligned} \sqrt{\left(x - \frac{a}{\mu}\right)^2 + y^2} &= \frac{1}{\mu} \sqrt{(x-a)^2 + y^2} + c \left(\mu - \frac{1}{\mu}\right), \\ \sqrt{\left(x - \frac{c^2}{a}\right)^2 + y^2} &= \frac{c}{a} \sqrt{(x-a)^2 + y^2} + \frac{a}{\mu} \left(1 - \frac{c^2}{a^2}\right), \\ \sqrt{\left(x - \frac{c^2}{a}\right)^2 + y^2} &= \frac{c\mu}{a} \sqrt{\left(x - \frac{a}{\mu}\right)^2 + y^2} + \frac{a}{\mu} - \frac{c^2\mu}{a}, \end{aligned}$$

$$c \left(\mu - \frac{1}{\mu}\right) \sqrt{\left(x - \frac{c^2}{a}\right)^2 + y^2} + \left(-a + \frac{c^2}{a}\right) \sqrt{\left(x - \frac{a}{\mu}\right)^2 + y^2} + \left(\frac{a}{\mu} - \frac{c^2\mu}{a}\right) \sqrt{(x-a)^2 + y^2} = 0.$$

It follows, that if we write successively

$$\begin{array}{lll} a' = a, & c' = c, & \mu' = \mu & (1) \\ a' = \frac{c^2}{a}, & c' = \frac{c}{\mu}, & \mu' = \frac{c}{a} & (\alpha) \\ a' = \frac{a}{\mu^2}, & c' = \frac{c}{\mu}, & \mu' = \frac{1}{\mu} & (\beta) \\ a' = a, & c' = \frac{c}{\mu}, & \mu' = \frac{a}{c\mu} & (\gamma) \\ a' = \frac{c^2}{a}, & c' = c, & \mu' = \frac{c\mu}{a} & (\delta) \\ a' = \frac{c^2}{a}, & c' = \frac{c}{\mu}, & \mu' = \frac{a}{c\mu} & (\varepsilon), \end{array}$$

or what is the same thing,

$$\begin{array}{lll}
 a = a', & c = c', & \mu = \mu' \quad (1) \\
 a = \frac{c'^2}{a'}, & c = \frac{c'}{\mu'}, & \mu = \frac{a'}{c'} \quad (\alpha) \\
 a = \frac{a'}{\mu'^2}, & c = \frac{c'}{\mu'}, & \mu = \frac{1}{\mu'} \quad (\beta) \\
 a = a', & c = \frac{c'}{\mu'}, & \mu = \frac{a'}{c'\mu'} \quad (\gamma) \\
 a = \frac{c'^2}{a'}, & c = c', & \mu = \frac{c'\mu'}{a'} \quad (\delta) \\
 a = \frac{c'^2}{a'}, & c = \frac{c'}{\mu'}, & \mu = \frac{c'}{a'} \quad (\varepsilon).
 \end{array}$$

or what is again the same thing,

$$\begin{array}{lll}
 a' = a, & \frac{c'^2}{a'} = \frac{c^2}{a}, & \frac{a'}{\mu'^2} = \frac{a}{\mu^2} \quad (1) \\
 a' = \frac{c^2}{a}, & \frac{c'^2}{a'} = a, & \frac{a'}{\mu'^2} = \frac{c^2}{\mu^2} \quad (\alpha) \\
 a' = \frac{a}{\mu^2}, & \frac{c'^2}{a'} = \frac{c^2\mu^2}{a}, & \frac{a'}{\mu'^2} = a \quad (\beta) \\
 a' = a, & \frac{c'^2}{a'} = \frac{c^2\mu^2}{a}, & \frac{a'}{\mu'^2} = \frac{c^2}{a} \quad (\gamma) \\
 a' = \frac{c^2}{a}, & \frac{c'^2}{a'} = a, & \frac{a'}{\mu'^2} = \frac{a}{\mu^2} \quad (\delta) \\
 a' = \frac{c^2}{a}, & \frac{c'^2}{a'} = \frac{c^2\mu^2}{a}, & \frac{a'}{\mu'^2} = \frac{c^2}{a} \quad (\varepsilon),
 \end{array}$$

we have in each case identically the same secondary caustic, and therefore also identically the same caustic; in other words, the same caustic is produced by six different systems of a radiant point and refracting circle. It is proper to remark that if we represent the six systems of equations by  $(a', c', \mu') = (a, c, \mu)$ ,  $(a', c', \mu') = \alpha(a, c, \mu)$ , &c., then  $\alpha, \beta, \gamma, \delta, \varepsilon$  will be functional symbols satisfying the conditions

$$\begin{aligned}
 1 &= \alpha\beta = \beta\alpha = \gamma^2 = \delta^2 = \varepsilon^2, \\
 \alpha &= \beta^2 = \delta\gamma = \varepsilon\delta = \gamma\varepsilon, \\
 \beta &= \alpha^2 = \gamma\delta = \delta\varepsilon = \varepsilon\gamma, \\
 \gamma &= \delta\alpha = \alpha\varepsilon = \varepsilon\beta = \beta\delta, \\
 \delta &= \varepsilon\alpha = \alpha\gamma = \gamma\beta = \beta\varepsilon, \\
 \varepsilon &= \gamma\alpha = \alpha\delta = \delta\beta = \beta\gamma.
 \end{aligned}$$



## XXIX.

The preceding formulae, which were first given by me in the *Philosophical Magazine*, December 1853, [124] include as particular cases a preceding theorem with respect to the caustic by refraction of parallel rays, and also two theorems of St Laurent, *Gergonne*, t. XVIII., [1827, pp. 1–19] viz. if we suppose first that  $a = c$ , i.e. that the radiant point is in the circumference of the refracting circle, then the system ( $\alpha$ ) shows that the same caustic would be obtained by writing  $c, \frac{c}{\mu}, 1$  (or what is the same thing  $\frac{1}{\mu} = 1$ ) in the place of  $c, c, \mu$ , and we have

*Theorem.* The caustic by refraction for a circle when the radiant point is in the circumference is also the caustic by reflexion for the same radiant point, and for a reflecting circle concentric with the refracting circle, but having its radius equal to the quotient of the radius of the refracting circle by the index of refraction.

Next, if we write  $a = c\mu$ , then the refracted rays all of them pass through a point which is a double point of the secondary caustic, the entire curve being in this case the orthogonal trajectory, not of the refracted rays, but of the false refracted rays; the formula ( $\delta$ ) shows that the same caustic is obtained by writing  $\frac{c}{\mu}, c, 1$  (or what is the same thing  $\mu = 1$ ) in the place of  $a, c, \mu$  ( $= \frac{a}{c}$ ), and we have

*Theorem.* The caustic by refraction for a circle when the distance of the radiant point from the centre is to the radius of the circle in the ratio of the index of refraction to unity, is also the caustic by reflexion for the same circle considered as a reflecting circle, and for a radiant point the image of the former radiant point.

## XXX.

The curve is most easily traced by means of the preceding construction thus; if we take the radiant point outside the refracting circle, and consider  $\mu$  as varying from a small to a large value (positive or negative values of  $\mu$  give the same curve), we see that when  $\mu$  is small the curve consists of two ovals, one of them within and the other without the refracting circle (see fig. 14). As  $\mu$  increases the exterior oval continually increases, but undergoes modifications in its form; the interior oval in the first instance diminishes until we arrive at a curve, in which the interior oval is reduced to a conjugate point (see fig. 15); then as  $\mu$  continues to increase the interior oval reappears (see fig. 16), and at last connects itself with the exterior oval, so as to form a curve with a double point (see fig. 17); and as  $\mu$  increases still further the

curve again breaks up into an exterior and an interior oval (see fig. 18); and thenceforward as  $\mu$  goes on increasing consists always of two ovals; the shape of the exterior oval is best perceived from the figures. An examination of the figures will also show how the same curves may originate from a different refracting circle and radiant point.

*[Figures: Fig. 14 (small  $\mu$ ): refracting circle with two ovals, one interior and one exterior. Fig. 17: exterior oval much larger, joined with interior at a double point. Fig. 15: interior oval reduced to a conjugate point. Fig. 16: interior oval reappearing. Fig. 18: two separated ovals, both small. All show the radiant point inside the refracting circle and the dashed reflecting/refracting circle.]*

### XXXI.

The theorem, “If a variable circle have its centre upon a circle  $S$ , and its radius proportional to the tangential distance of the centre from a circle  $C$ , the envelope is a Cartesian,” is at once deducible from the theorem

“If a variable circle have its centre upon a circle  $S$  and its radius proportional to the distance of the centre from a point  $C$ , the locus is a Cartesian,”

which last theorem was in effect given in discussing the theory of the secondary caustic. In fact, the locus of a point  $P$  such that its tangential distances from the circles  $C, C'$  are in a constant ratio, is a circle  $S$ . Conversely, if there be a circle  $C$ , and the locus of  $P$  be a circle  $S$ , then the circle  $C'$  may be found such that the tangential distances of  $P$  from the two circles are in a constant ratio, and the circle  $C'$  may be taken to be a point, i.e. if there be a circle  $C$  and the locus of  $P$  be a circle  $S$ , then a point  $C'$  may be found such that the tangential distance of  $P$  from the circle  $C$  is in a constant ratio to the distance from the point  $C'$ .

Hence treating  $P$  as the centre of the variable circle, it is clear that the variable circle is determined in the two cases by equivalent constructions, and the envelope is therefore the same in both cases.

### XXXII.

The equation of the secondary caustic developed and reduced is

$$\mu^4(x^2 + y^2)^2 - 2\mu^2(a^2 + (\mu^2 + 1)c^2)(x^2 + y^2) + 8c^2\mu^2a^2x + a^4 - 2a^2c^2(\mu^2 + 1) + (\mu^2 - 1)^2c^4 = 0,$$

or, what is the same thing,

$$\{\mu^2(x^2 + y^2) - (a^2 + (\mu^2 + 1)c^2)\}^2 + 8c^2\mu^2a^2x - 4c^2(c^2\mu^2 + (\mu^2 + 1)a^2) = 0,$$

which may also be written

$$\left(x^2 + y^2 - \left(\frac{a^2}{\mu^2} + \left(1 + \frac{1}{\mu^2}\right)c^2\right)\right)^2 + \frac{8}{\mu^2}c^2a^2x - \frac{4c^2}{\mu^4}\left(c^2 + \left(1 + \frac{1}{\mu^2}\right)a^2\right) = 0,$$

which is of the form

$$(x^2 + y^2 - \alpha)^2 + 16A(x - m) = 0,$$

and the values of the coefficients are

$$\alpha = \frac{a^2}{\mu^2} + \left(1 + \frac{1}{\mu^2}\right)c^2,$$

$$A = \frac{c^2a}{2\mu^2},$$

$$m = \frac{1}{2a}\left(c^2 + \left(1 + \frac{1}{\mu^2}\right)a^2\right).$$

The equation just obtained should, I think, be taken as the standard form of the equation of the Cartesian, and the form of the equation shows that the Cartesian may be defined as the locus of a point, such that the fourth power of its tangential distance from a given circle is in a constant ratio to its distance from a given line.

### XXXIII.

The Cartesian is a curve of the fourth order, symmetrical about a certain line which it intersects in four arbitrary points, and these points determine the curve. Taking the line in question (which may be called the axis) as the axis of  $x$ , and a line at right angles to it as the axis of  $y$ , let  $a, b, c, d$  be the values of  $x$  corresponding to the points of intersection with the axis, then the equation of the curve is

$$y^4 + y^2 \left[ 2x^2 - (a + b + c + d)x - \frac{1}{2}(a^2 + b^2 + c^2 + d^2 - 2ab - 2ac - 2ad - 2bc - 2bd - 2cd) \right] + (x - a)(x - b)(x - c)(x - d) = 0.$$

It is easy to see that the form of the equation is not altered by writing  $x + \theta$  for  $x$ , and  $a + \theta, b + \theta, c + \theta, d + \theta$  for  $a, b, c, d$ ; we may therefore without loss of generality put  $a + b + c + d = 0$ , and the equation of the curve then becomes

$$y^4 + y^2(2x^2 + ab + ac + ad + bc + bd + cd) + (x - a)(x - b)(x - c)(x - d) = 0,$$

where

$$a + b + c + d = 0;$$

the curve is in this case said to be referred to the centre as origin.

The last-mentioned equation may be written

$$(x^2 + y^2 + \frac{1}{2}(ab + ac + ad + bc + bd + cd))(x^2 + y^2) - (abc + abd + acd + bcd)x + abcd = 0,$$

$$\begin{aligned} & \{x^2 + y^2 + \frac{1}{2}(ab + ac + ad + bc + bd + cd)\}^2 \\ & - (abc + abd + acd + bcd)x \\ & + \frac{1}{4} [a^2b^2 + a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2 + c^2d^2 \\ & \quad + 2a^2bc + 2a^2bd + 2a^2cd + 2b^2ac + 2b^2ad + 2b^2cd \\ & \quad + 2c^2ab + 2c^2ad + 2c^2bd + 2d^2ab + 2d^2ac + 2d^2bc \\ & \quad + 2abcd] = 0, \end{aligned}$$

or observing that

$$\begin{aligned} & a^2bc + a^2bd + a^2cd + b^2ac + b^2ad + b^2cd \\ & + c^2ab + c^2ad + c^2bd + d^2ab + d^2ac + d^2bc \\ & = abc(a + b + c) + abd(a + b + d) + acd(a + c + d) + bcd(b + c + d) \\ & = -4abcd, \end{aligned}$$

[Continuation of Memoir No. 146, “A Memoir on Curves of the Third Order.” The present block (PDF pp. 401–432) carries the memoir from Article No. 3 through the recapitulation in Article No. 41.]

It will be presently seen that the analytical theory leads to the consideration of a line  $IJ$  (not represented in the figure); the line in question is the polar of  $F$  (or  $E$ ) with respect to the conic which is the first or conic polar of  $E$  (or  $F$ ) with respect to any syzygetic cubic. The line  $IJ$  is a tangent of the Pippian, and moreover the lines  $EF$  and  $IJ$  are conjugate polars of a curve of the third class syzygetically connected with the Pippian and Quippian, and which is moreover such that its Hessian is the Pippian.

*Article Nos. 3 to 19. Analytical investigations, comprising the proof of the theorems, Article No. 2.*

3. The analytical theory possesses considerable interest. Take as the equation of the cubic,

$$U = x^3 + y^3 + z^3 + 6lxyz = 0;$$

then the equation of the Hessian is

$$HU = l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz = 0;$$

and the equation of the Pippian in line coordinates (that is, the equation which expresses that  $\xi x + \eta y + \zeta z = 0$  is a tangent of the curve) is

$$PU = -l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = 0.$$

The equation of the Quippian in line coordinates is

$$QU = (1 - 10l^3)(\xi^3 + \eta^3 + \zeta^3) - 6l^2(5 + 4l^3)\xi\eta\zeta = 0;$$

and the values of the two invariants of the cubic form are

$$S = -l + l^4,$$

$$T = 1 - 20l^3 - 8l^6,$$

values which give identically,

$$T^2 - 64S^3 = (1 + 8l^3)^4,$$

the last-mentioned function being in fact the discriminant.

4. Suppose now that  $(X, Y, Z)$  are the coordinates of the point  $E$ , and  $(X', Y', Z')$  the coordinates of the point  $F$ ; then the equations which express that these points are conjugate poles of the cubic, are

$$XX' + l(YZ' + Y'Z) = 0,$$

$$YY' + l(ZX' + Z'X) = 0,$$

$$ZZ' + l(XY' + X'Y) = 0;$$

and by eliminating from these equations, first  $(X', Y', Z')$ , and then  $(X, Y, Z)$ , we find

$$l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)XYZ = 0,$$

$$l^2(X'^3 + Y'^3 + Z'^3) - (1 + 2l^3)X'Y'Z' = 0,$$

which shows that the points  $E, F$  are each of them points of the Hessian.

5. I may notice, in passing, that the preceding equations give rise to a somewhat singular *unsymmetrical* quadratic transformation of a cubic form. In fact, the second and third equations give

$$X' : Y' : Z' = YZ - l^2X^2 : l^2XY - lZ^2 : l^2ZX - lY^2.$$

And substituting these values for  $X', Y', Z'$  in the form

$$l^2(X'^3 + Y'^3 + Z'^3) - (1 + 2l^3)X'Y'Z',$$

the result must contain as a factor

$$l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)XYZ;$$

the other factor is easily found to be

$$-l^3(l^3(X^3 + Y^3 + Z^3) + 3lXYZ).$$

Several of the formulae given in the sequel conduct in like manner to unsymmetrical transformations of a cubic form.

6. I remark also, that the last-mentioned system of equations gives, symmetrically,

$$\begin{aligned} X'^2 : Y'^2 : Z'^2 : Y'Z' : Z'X' : X'Y' \\ = YZ - l^2X^2 : ZX - l^2Y^2 : XY - l^2Z^2 : l^2YZ - lX^2 : l^2ZX - lY^2 : l^2XY - lZ^2; \end{aligned}$$

and it is, I think, worth showing how, by means of these relations, we pass from the equation between  $X'$ ,  $Y'$ ,  $Z'$  to that between  $X$ ,  $Y$ ,  $Z$ . In fact, representing, for shortness, the foregoing relations by

$$X'^2 : Y'^2 : Z'^2 : Y'Z' : Z'X' : X'Y' = A : B : C : F : G : H,$$

we may write

$$X' = AF = GH, \quad Y' = BG = HF, \quad Z' = CH = FG, \quad ABC = FGH;$$

and thence

$$X'^2 = AF \cdot GH, \quad Y'^2 = BG \cdot HF, \quad Z'^2 = CH \cdot FG, \quad X'Y'Z' = FGH;$$

hence

$$l^2(X'^3 + Y'^3 + Z'^3) - (1 + 2l^3)X'Y'Z' = FGH\{l^2(AGH + BHF + CFG) - (1 + 2l^3)FGH\}.$$

But we have

$$\begin{aligned} l^2(AGH + BHF + CFG) &= -(2l^3 + l^6)(X^3 + Y^3 + Z^3)XYZ + (l^3 + 2l^6)(Y^3Z^3 + Z^3X^3 + X^3Y^3), \\ -(1 + 2l^3)FGH &= (l^3 + 2l^6)(X^3 + Y^3 + Z^3)XYZ + (l^4 + 2l^7)(Y^3Z^3 + Z^3X^3 + X^3Y^3) \\ &\quad + l^3(1 - l^3)(1 + 2l^3)X'Y'Z'; \end{aligned}$$

and thence

$$l^2(AGH + BHF + CFG) - (1 + 2l^3)FGH = -l^3(1 - l^3)\{l^2(X^3 + Y^3 + Z^3)XYZ - (1 + 2l^3)X^2Y^2Z^2\};$$

and finally,

$$l^2(X'^3 + Y'^3 + Z'^3) - (1 + 2l^3)X'Y'Z' = l^3(-1 + l^3)(lXZ - Y^2)(lZY - X^2)(lXY - Z^2) \times \{l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)XYZ\}.$$

We have also, identically,

$$ABC - FGH = \frac{1}{l}(-l + l^4)XYZ\{l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)XYZ\},$$

which agrees with the relation  $\frac{FGH}{ABC} = 0$ .

7. Before going further, it will be convenient to investigate certain relations which exist between the quantities  $(X, Y, Z)$ ,  $(X', Y', Z')$ , connected as before by the equations

$$\begin{aligned} XX' + l(YZ' + Y'Z) &= 0, \\ YY' + l(ZX' + Z'X) &= 0, \\ ZZ' + l(XY' + X'Y) &= 0, \end{aligned}$$

and the quantities

$$\begin{aligned}\xi &= YZ' - Y'Z, & \alpha &= XX' = -\frac{1}{l}(YZ' + Y'Z), \\ \eta &= ZX' - Z'X, & \beta &= YY' = -\frac{1}{l}(ZX' + Z'X), \\ \zeta &= XY' - X'Y, & \gamma &= ZZ' = -\frac{1}{l}(XY' + X'Y).\end{aligned}$$

We have identically,

$$2XX'(YZ' - Y'Z) + (XY' + X'Y)(ZX' - Z'X) + (ZX' + Z'X)(XY' - X'Y) = 0;$$

or expressing in terms of  $\xi, \eta, \zeta, \alpha, \beta, \gamma$  the quantities which enter into this equation, and forming the analogous equations, we have

$$\begin{aligned}2l\alpha\xi - \gamma\eta - \beta\zeta &= 0, \\ -\gamma\xi + 2l\beta\eta - \alpha\zeta &= 0, \\ -\beta\xi - \alpha\eta + 2l\gamma\zeta &= 0.\end{aligned}\tag{A}$$

We have also

$$X^2Y'Z' - X'^2YZ = \frac{1}{2}\{-(XY' + X'Y)(ZX' - Z'X) + (ZX' + Z'X)(XY' - X'Y)\},$$

and thence in like manner,

$$\begin{aligned}X^2Y'Z' - X'^2YZ &= \frac{1}{2l}(\gamma\eta - \beta\zeta), \\ Y^2Z'X' - Y'^2ZX &= \frac{1}{2l}(\alpha\zeta - \gamma\xi), \\ Z^2X'Y' - Z'^2XY &= \frac{1}{2l}(\beta\xi - \alpha\eta).\end{aligned}\tag{B}$$

Again, we have

$$\begin{aligned}(YZ' - Y'Z)^2 &= (YZ' + Y'Z)^2 - 4YY'ZZ', \\ (ZX' - Z'X)(XY' - X'Y) &= -(ZX' + Z'X)(XY' + X'Y) + 2XX'(YZ' + Y'Z);\end{aligned}$$

and thence

$$\begin{aligned}\xi^2 &= l^2\alpha^2 - 4\beta\gamma, \\ \eta\zeta &= -l^2\alpha^2 - \frac{2}{l}\alpha, \\ \xi\eta &= -l^2\gamma^2 - \frac{2}{l}\gamma, \\ \xi\zeta &= -l^2\beta^2 - \frac{2}{l}\beta;\end{aligned}\tag{C}$$

and conversely

$$\begin{aligned}\frac{1}{l}(1 + 8l^3)\alpha^2 &= \xi^2 - 4l^2\eta\zeta, \\ \frac{1}{l}(1 + 8l^3)\beta^2 &= \eta^2 - 4l^2\zeta\xi,\end{aligned}\tag{D}$$

$$\begin{aligned}\frac{1}{l}(1 + 8l^3)\gamma^2 &= \zeta^2 - 4l^2\xi\eta, \\ -\frac{1}{l}(1 + 8l^3)\beta\gamma &= 2l\xi^2 + \eta\zeta,\end{aligned}\tag{2}$$

$$-\frac{1}{l}(1 + 8l^3)\gamma\alpha = 2l\eta^2 + \zeta\xi,\tag{3}$$

$$-\frac{1}{l}(1 + 8l^3)\alpha\beta = 2l\zeta^2 + \xi\eta.\tag{4}$$

8. It is obvious that

$$\xi x + \eta y + \zeta z = 0$$

is the equation of the line  $EF$  joining the two conjugate poles, and it may be shown that

$$\alpha x + \beta y + \gamma z = 0$$

is the equation of the line  $IJ$ , which is the polar of  $E$  with respect to a conic which is the first or conic polar of  $F$  with respect to any syzygetic cubic. In fact, the equation of a syzygetic cubic will be  $x^3 + y^3 + z^3 + 6\lambda xyz = 0$ , where  $\lambda$  is arbitrary, and the equation of the line in question is

$$(X\partial_x + Y\partial_y + Z\partial_z)(X'\partial_x + Y'\partial_y + Z'\partial_z)(x^3 + y^3 + z^3 + 6\lambda xyz) = 0;$$

or developing,

$$XX'x + YY'y + ZZ'z + \lambda\{(YZ' + Y'Z)x + (ZX' + Z'X)y + (XY' + X'Y)z\} = 0;$$

and the function on the left-hand side is

$$\left(1 - \frac{\lambda}{l}\right)(\alpha x + \beta y + \gamma z),$$

which proves the theorem.

9. The equations (A) by the elimination of  $(\xi, \eta, \zeta)$ , give

$$-l(\alpha^3 + \beta^3 + \gamma^3) + (-1 + 4l^3)\alpha\beta\gamma = 0,$$

which shows that the line  $IJ$  is a tangent of the Pippian: the proof of the theorem is given in this place because the relation just obtained between  $\alpha, \beta, \gamma$  is required for the proof of some of the other theorems.

10. To find the coordinates of the point  $G$  in which the line  $EF$  joining two conjugate poles again meets the Hessian.

We may take for the coordinates of  $G$ ,

$$uX + vX', \quad uY + vY', \quad uZ + vZ';$$

and, substituting in the equation of the Hessian, the terms containing  $u^2, v^2$  disappear, and the ratio  $u : v$  is determined by a simple equation. It thus appears that we may write

$$\begin{aligned} u &= -3l^2(XX'^2 + YY'^2 + ZZ'^2) + (1 + 2l^3)(Y'Z'X + Z'X'Y + X'Y'Z), \\ v &= 3l^2(X^2X' + Y^2Y' + Z^2Z') - (1 + 2l^3)(YZX' + ZXY' + XYZ'); \end{aligned}$$

hence introducing, as before, the quantities  $\xi, \eta, \zeta, \alpha, \beta, \gamma$ , we find

$$uX + vX' = 3l^2(\gamma\eta - \beta\zeta) + (1 + 2l^3)(X^2Y'Z' - X'^2YZ);$$

but from the first of the equations (B),

$$X^2Y'Z' - X'^2YZ = \frac{1}{2l}(\gamma\eta - \beta\zeta),$$

and therefore the preceding value of  $uX + vX'$  becomes

$$\left(3l^2 + \frac{1+2l^3}{2l}\right)(\gamma\eta - \beta\zeta),$$

which is equal to

$$\frac{-1 + 4l^3}{2l}(\gamma\eta - \beta\zeta).$$

Hence throwing out the constant factor, we find, for the coordinates of the point  $G$ , the values

$$\gamma\eta - \beta\zeta, \quad \alpha\zeta - \gamma\xi, \quad \beta\xi - \alpha\eta.$$

11. To find the coordinates of the point  $O$ .

Consider  $O$  as the point of intersection of the tangents to the Hessian at the points  $E, F$ , then the coordinates of  $O$  are proportional to the terms of

$$\begin{aligned} & \left| \begin{array}{ccc} 3l^2X^2 - \overline{1+2l^3}YZ, & 3l^2Y^2 - \overline{1+2l^3}ZX, & 3l^2Z^2 - \overline{1+2l^3}XY \\ 3l^2X'^2 - \overline{1+2l^3}Y'Z', & 3l^2Y'^2 - \overline{1+2l^3}Z'X', & 3l^2Z'^2 - \overline{1+2l^3}X'Y' \end{array} \right|. \end{aligned}$$



Hence the  $x$ -coordinate is proportional to

$$(3l^2Y^2 - \overline{1+2l^3} ZX)(3l^2Z'^2 - \overline{1+2l^3} X'Y') - (3l^2Z^2 - \overline{1+2l^3} XY)(3l^2Y'^2 - \overline{1+2l^3} Z'X'),$$

which is equal to

$$9l^4(Y^2Z'^2 - Y'^2Z^2) + 3l^2(1+2l^3)YY'(XY' - X'Y) + 3l^2(1+2l^3)ZZ'(ZX' - Z'X) - (1+2l^3)^2XX'(YZ' - Y'Z);$$

or introducing, as before, the quantities  $\xi, \eta, \zeta, \alpha, \beta, \gamma$ , to

$$\begin{aligned} & 9l^4\alpha\xi + 3l^2(1+2l^3)(\beta\zeta + \gamma\eta) - (1+2l^3)^2\alpha\xi \\ & = (-1 - 13l^3 + 4l^6)\alpha\xi + 3l^2(1+2l^3)(\beta\zeta + \gamma\eta). \end{aligned}$$

But by the first of the equations (A)  $\beta\zeta + \gamma\eta = 2l\alpha\xi$ , and the preceding value thus becomes  $(-1 - 7l^3 + 8l^6)\alpha\xi$ . Hence throwing out the constant factor the coordinates of the point  $O$  are found to be

$$\alpha\xi, \quad \beta\eta, \quad \gamma\zeta.$$

12. The points  $G, O$  are conjugate poles of the cubic.

Take  $a, b, c$  for the coordinates of  $G$ , and  $a', b', c'$  for the coordinates of  $O$ ; we have

$$\begin{aligned} a, b, c &= \gamma\eta - \beta\zeta, \quad \alpha\zeta - \gamma\xi, \quad \beta\xi - \alpha\eta, \\ a', b', c' &= \alpha\xi, \quad \beta\eta, \quad \gamma\zeta. \end{aligned}$$

These values give  $aa' + l(bc' + b'c)$

$$\begin{aligned} &= \alpha\xi(\gamma\eta - \beta\zeta) + l\{(\alpha\zeta - \gamma\xi)\gamma\zeta + \beta\eta(\beta\xi - \alpha\eta)\} \\ &= \xi(\alpha\gamma\eta + l\beta^2\eta - l\alpha\eta^2) + \eta(\alpha\beta\gamma) + \zeta(-\alpha\beta\xi - l\gamma^2\xi) \end{aligned}$$

or substituting for  $\xi\eta, \eta\zeta, \zeta\xi, \xi^2$  their values in terms of  $\alpha, \beta, \gamma$ , this is

$$\begin{aligned} & \xi\{(-l\gamma^2 - \tfrac{1}{l}\gamma)(-4\gamma\alpha) + (-l\beta^2 - \tfrac{1}{l}\beta)(-l\alpha\beta)\} \\ & + \eta\{(\alpha^2l - \tfrac{4}{l}\alpha^2)\} \cdot (\alpha\beta\gamma) \\ & + \zeta\{(-l\beta^2 - \tfrac{1}{l}\beta)(-\alpha\beta - l\gamma^2)\}, \end{aligned}$$

which is identically equal to zero. Hence, completing the system, we find

$$\begin{aligned} aa' + l(bc' + b'c) &= 0, \\ bb' + l(ca' + c'a) &= 0, \\ cc' + l(ab' + a'b) &= 0, \end{aligned}$$

equations which show that  $O$  (as well as  $G$ ) is a point of the Hessian, and that the points  $G, O$  are corresponding poles of the cubic.

13. The line  $EF$  joining a pair of conjugate poles of the cubic is a tangent of the Pippian<sup>1</sup>.

In fact, the equations (A), by the elimination of  $\alpha, \beta, \gamma$ , give

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = 0,$$

which proves the theorem.

14. To find the equation of the pair of lines through  $F$ , and to show that these lines are tangents of the Pippian.

The equation of the pair of lines considered as the first or conic polar of the conjugate pole  $E$ , is

$$X(x^2 + 2lyz) + Y(y^2 + 2lzx) + Z(z^2 + 2lxy) = 0.$$

<sup>1</sup>Steiner's curve  $R_3$ , in the particular case of a cubic basis-curve, is according to definition the envelope of the line  $EF$ ; that is, the curve  $R_3$  in the particular case in question is the Pippian.

Let one of the lines be  $\lambda x + \mu y + \nu z = 0$ , then the other is

$$\frac{X}{\lambda} + \frac{Y}{\mu} + \frac{Z}{\nu} = 0;$$

and we find

$$\begin{aligned} 2lX\mu\nu - Y\nu^2 - Z\mu^2 &= 0, \\ -X\nu^2 + 2lY\nu\lambda - Z\lambda^2 &= 0, \\ -X\mu^2 - Y\lambda^2 + 2lZ\lambda\mu &= 0, \end{aligned}$$

any two of which determine the ratios  $\lambda, \mu, \nu$ .

The elimination of  $X, Y, Z$  gives

$$\begin{vmatrix} 2l\mu\nu, & -\nu^2, & -\mu^2 \\ -\nu^2, & 2l\nu\lambda, & -\lambda^2 \\ -\mu^2, & -\lambda^2, & 2l\lambda\mu \end{vmatrix} = 0,$$

which is equivalent to

$$\lambda\mu\nu\{-l(\lambda^3 + \mu^3 + \nu^3) + (-1 + 4l^3)\lambda\mu\nu\} = 0;$$

or, omitting a factor, to

$$-l(\lambda^3 + \mu^3 + \nu^3) + (-1 + 4l^3)\lambda\mu\nu = 0,$$

which shows that the line in question is a tangent of the Pippian.

15. To find the equation of the pair of lines through  $O$ .

The equation of the pair of lines through  $E$  is in like manner

$$X'(x^2 + 2lyz) + Y'(y^2 + 2lzx) + Z'(z^2 + 2lxy) = 0;$$

and combining this with the foregoing equation,

$$X(x^2 + 2lyz) + Y(y^2 + 2lzx) + Z(z^2 + 2lxy) = 0$$

of the pair of lines through  $F$ , viz. multiplying the two equations by

$$X'X' + Y'Y' + Z'Z', \quad -(XX' + YY' + ZZ'),$$

and adding, then if as before

$$a : b : c = \gamma\eta - \beta\zeta : \alpha\zeta - \gamma\xi : \beta\xi - \alpha\eta,$$

we find as the equation of a conic passing through the points  $A, B, C, D$ , the equation

$$a(x^2 + 2lyz) + b(y^2 + 2lzx) + c(z^2 + 2lxy) = 0.$$

But putting, as before,

$$a' : b' : c' = \alpha\xi : \beta\eta : \gamma\zeta,$$

then  $a', b', c'$  are the coordinates of the point  $O$ , and the equations

$$\begin{aligned} aa' + l(bc' + b'c) &= 0, \\ bb' + l(ca' + c'a) &= 0, \\ cc' + l(ab' + a'b) &= 0, \end{aligned}$$

show that the conic in question is in fact the pair of lines through the point  $O$ .

16. To find the coordinates of the point  $\Gamma$ , which is the harmonic of  $G$  with respect to the points  $E, F$ .

The coordinates of the point in question are

$$uX - vX', \quad uY - vY', \quad uZ - vZ',$$

where  $u, v$  have the values given in No. 10, viz.

$$\begin{aligned} u &= -3l^2(XX'^2 + YY'^2 + ZZ'^2) + (1 + 2l^3)(Y'Z'X + Z'X'Y + X'Y'Z), \\ v &= 3l^2(X^2X' + Y^2Y' + Z^2Z') - (1 + 2l^3)(YZX' + ZXY' + XYZ'); \end{aligned}$$

these values give

$$uX - vX' = -3l^2\{2X^2X'^2 + (XY' + X'Y)YY' + (XZ' + X'Z)ZZ'\} + (1 + 2l^3)\{(XY' + X'Y)(XZ' + X'Z) + XX'(YZ' + Y'Z)\};$$

and therefore

$$uX - vX' = -3l^2\{2\alpha^2 - \frac{2}{l}\beta\gamma\} + (1 + 2l^3)(\frac{1}{l^2}\beta\gamma - \frac{1}{l}\alpha^2) = \frac{1}{l}(1 + 8l^3)(-l\alpha^2 + \beta\gamma);$$

and consequently, omitting the constant factor, the coordinates of  $\Gamma$  may be taken to be

$$-l\alpha^2 + \beta\gamma, \quad -l\beta^2 + \gamma\alpha, \quad -l\gamma^2 + \alpha\beta.$$

17. The line through two consecutive positions of the point  $\Gamma$  is the line  $EF$ .

The coordinates of the point  $\Gamma$  are

$$-l\alpha^2 + \beta\gamma, \quad -l\beta^2 + \gamma\alpha, \quad -l\gamma^2 + \alpha\beta;$$

and it has been shown that the quantities  $\alpha, \beta, \gamma$  satisfy the equation

$$-l(\alpha^3 + \beta^3 + \gamma^3) + (-1 + 4l^3)\alpha\beta\gamma = 0.$$

Hence, considering  $\alpha, \beta, \gamma$  as variable parameters connected by this equation, the equation of the line through two consecutive positions of the point  $\Gamma$  is

$$\begin{vmatrix} -3l\alpha^2 + (-1 + 4l^3)\beta\gamma, & -3l\beta^2 + (-1 + 4l^3)\gamma\alpha, & -3l\gamma^2 + (-1 + 4l^3)\alpha\beta \\ x, & -2l\alpha, & \gamma, & \beta \\ y, & \gamma, & -2l\beta, & \alpha \\ z, & \beta, & \alpha, & -2l\gamma \end{vmatrix} = 0,$$

and representing this equation by

$$Lx + My + Nz = 0,$$

we find

$$\begin{aligned} L &= -(4l^2\beta\gamma + \alpha^2)(-3l\alpha^2 + (-1 + 4l^3)\beta\gamma) \\ &\quad + (\alpha\beta + 2l\gamma^2)(-3l\beta^2 + (-1 + 4l^3)\gamma\alpha) \\ &\quad + (\alpha\gamma + 2l\beta^2)(-3l\gamma^2 + (-1 + 4l^3)\alpha\beta); \end{aligned}$$

or, multiplying out and collecting,

$$L = 3l\alpha^4 + (-1 - 8l^3)\alpha^2\beta\gamma + (-5l + 14l^4)(\alpha\beta^3 + \alpha\gamma^3) + (-16l^2 + 16l^5)\beta^2\gamma^2;$$

but the equation

$$-l(\alpha^3 + \beta^3 + \gamma^3) + (-1 + 4l^3)\alpha\beta\gamma = 0$$

gives

$$3l\alpha^4 = -3l(\alpha\beta^3 + \alpha\gamma^3) + (-3 + 12l^3)\alpha^2\beta\gamma,$$

and we have

$$\begin{aligned} L &= (-4 + 4l^3)\alpha^2\beta\gamma + (-8l + 8l^4)(\alpha\beta^3 + \alpha\gamma^3) + (-16l^2 + 16l^5)\beta^2\gamma^2 \\ &= (-4 + 4l^3)(\alpha^2\beta\gamma + 2l(\alpha\beta^3 + \alpha\gamma^3) + 4l^2\beta^2\gamma^2) \\ &= (-4 + 4l^3)(\alpha\gamma + 2l\beta^2)(\alpha\beta + 2l\gamma^2); \end{aligned}$$

or, in virtue of the equations (D),

$$L = (-4 + 4l^3) \cdot l^2\eta \cdot l^2\zeta = (-4 + 4l^3) l^4\eta\zeta.$$

Hence, omitting the common factor, we find  $L : M : N = \xi : \eta : \zeta$ , and the equation  $Lx + My + Nz = 0$  becomes

$$\xi x + \eta y + \zeta z = 0,$$

which is the equation of the line  $EF$ , that is, the line through two consecutive positions of  $\Gamma$  is the line  $EF$ ; or what is the same thing, the line  $EF$  touches the Pippian in the point  $\Gamma$  which is the harmonic of  $G$  with respect to the points  $E, F$ .

18. The lineo-polar envelope of the line  $EF$ , with respect to the cubic, is the pair of lines  $OE, OF$ .

The equation of the pair of lines  $OE, OF$ , considered as the tangents to the Hessian at the points  $E, F$ , is

$$\{(3l^2X^2 - \overline{1+2l^3}YZ)x + (3l^2Y^2 - \overline{1+2l^3}ZX)y + (3l^2Z^2 - \overline{1+2l^3}XY)z\} \times \{(3l^2X'^2 - \overline{1+2l^3}Y'Z')x + (3l^2Y'^2 - \overline{1+2l^3}Z'X')y + (3l^2Z'^2 - \overline{1+2l^3}X'Y')z\} = 0.$$

Here on the left-hand side the coefficient of  $x^2$  is

$$9l^4X^2X'^2 - 3l^2(1 + 2l^3)(X^2Y'Z' + X'^2YZ) + (1 + 2l^3)^2YY'ZZ',$$

which is equal to

$$9l^4\alpha^2 - 3l^2(1 + 2l^3)(l^2\beta\gamma + \frac{1}{l}\alpha^2) + (1 + 2l^3)^2\beta\gamma,$$

that is

$$\frac{1}{l}(-l + l^4)\{3l\alpha^2 + 2(1 + 2l^3)\beta\gamma\};$$

and the coefficient of  $yz$  is

$$9l^4(Y^2Z'^2 + Y'^2Z^2) - 3l^2(1 + 2l^3)(YY'(XY' + X'Y) + ZZ'(XZ' + X'Z)) + (1 + 2l^3)^2XX'(YZ' + Y'Z),$$

which is equal to

$$9l^2(\frac{1}{l^4}\eta^2 - 2l\beta\gamma) - 3l^2(1 + 2l^3)(-\frac{2}{l}\beta\gamma) + (1 + 2l^3)^2\alpha(-\frac{1}{l}\alpha),$$

that is

$$\frac{1}{l}(-l + l^4)\{(1 - 4l^3)\alpha^2 - 6l^2\beta\gamma\}.$$

Hence completing the system and throwing out the constant factor, the equation of the pair of lines is

$$(3l\alpha^2 + 2(1 + 2l^3)\beta\gamma, 3l\beta^2 + 2(1 + 2l^3)\gamma\alpha, 3l\gamma^2 + 2(1 + 2l^3)\alpha\beta, (1 - 4l^3)\alpha^2 - 6l^2\beta\gamma, (1 - 4l^3)\beta^2 - 6l^2\gamma\alpha, (1 - 4l^3)\gamma^2 - 6l^2\alpha\beta \mid x, y, z)^2 = 0.$$

But the equation of the line  $EF$  is  $\xi x + \eta y + \zeta z = 0$ , and the equation of its lineo-polar envelope is

$$\begin{vmatrix} \xi, & x, & lz, & ly \\ \eta, & lz, & y, & lx \\ \zeta, & ly, & lx, & z \end{vmatrix} = 0;$$

or expanding,

$$(\xi z - l^2x^2, zx - l^2y^2, xy - l^2z^2, l^2yz - lx^2, l^2zx - ly^2, l^2xy - lz^2 \mid \xi, \eta, \zeta) = 0;$$

or arranging in powers of  $x, y, z$ ,

$$(-l\xi^2, -l\eta^2 - 2l\zeta\xi, -l\zeta^2 - 2l\xi\eta, -l\xi^2 - 2l\eta\zeta, l^2\eta^2 + l^2\zeta\xi, l^2\zeta^2 + l^2\xi\eta \mid x, y, z)^2 = 0;$$

and if in this equation we replace  $\xi^2$  &c. by their values in terms of  $\alpha, \beta, \gamma$ , as given by the equations (D), we obtain the equation given as that of the pair of lines  $OE, OF$ .

19. It remains to prove the theorem with respect to the connexion of the lines  $EF, IJ$ .

The equations (A) show that the two lines

$$\begin{aligned} \xi x + \eta y + \zeta z &= 0, \\ \alpha x + \beta y + \gamma z &= 0, \end{aligned}$$

(where  $\xi, \eta, \zeta$  and  $\alpha, \beta, \gamma$  have the values before attributed to them) are conjugate polars with respect to the curve of the third class,

$$(2l\xi, \zeta, \eta, -l\xi^2, \zeta x, \eta y) \dots = 0,$$

in which equation  $\xi, \eta, \zeta$  denote current line coordinates. The curve in question is of the form

$$A \cdot PU + B \cdot QU = 0.$$

We have, in fact, identically,

$$3T \cdot PU - 4S \cdot QU = (1 + 8l^3)^2 \{l(\xi^3 + \eta^3 + \zeta^3) - 3\xi\eta\zeta\}.$$

It is clear that the curve in question must have the curve  $PU = 0$  for its Hessian; and in fact, in the formula of my Third Memoir, [144]

$$H(6aPU + 6\beta QU) = (-2T, 4S^2, 18TS, T^2 + 16S^3 \mid \alpha, \beta)^3 PU + (8S, T, -8S^2, -TS \mid \alpha, \beta)^3 QU.$$

The coefficient of  $QU$  is

$$(8S, T, -8S^2, -TS \mid \alpha, \beta)^3,$$

and therefore, putting  $\alpha = \frac{1}{3}T, \beta = -\frac{4}{3}S$ , we find

$$H(3T \cdot PU - 4S \cdot QU) = -l(T^2 - 64S^3)^2 PU.$$

*Article No. 20. Theorem relating to the curve of the third class, mentioned in the preceding Article.*

20. The consideration of the curve  $3T \cdot PU - 4S \cdot QU = 0$ , gives rise to another geometrical theorem. Suppose that the line  $(\xi, \eta, \zeta)$ , that is, the line whose equation is  $\xi x + \eta y + \zeta z = 0$ , is with respect to this curve of the third class one of the four polars of a point  $(X, Y, Z)$  of the Hessian, and that it is required to find the envelope of the line  $\xi x + \eta y + \zeta z = 0$ .

We have

$$X : Y : Z = l\xi^2 - \eta\zeta : l\eta^2 - \zeta\xi : l\zeta^2 - \xi\eta,$$

and  $X, Y, Z$  are to be eliminated from these equations, and the equation

$$l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)XYZ = 0$$

of the Hessian. We have

$$\begin{aligned} X^3 + Y^3 + Z^3 &= l^3(\xi^3 + \eta^3 + \zeta^3)^2, \\ XYZ &= -l^2(\xi^3 + \eta^3 + \zeta^3)\xi\eta\zeta, \end{aligned}$$

and thence

$$HU = l^4(\xi^3 + \eta^3 + \zeta^3)^2 \cdot \{-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta\};$$

and equating the right-hand side to zero, we have the equation in line coordinates of the curve in question, which is therefore a curve of the sixth class in quadratic syzygy with the Pippian and Quippian.

*Article No. 21. Geometrical definition of the Quippian.*

21. I have not succeeded in obtaining any good geometrical definition of the Quippian, and the following is only given for want of something better.

The curve

$$T \cdot PU \cdot \{P(6H(aU + 6\beta HU))\} - P(6HU)\{T(aU + 6\beta HU) \cdot P(aU + 6\beta HU)\} = 0,$$

which is derived in what may be taken to be a known manner from the cubic, is in general a curve of the sixth class. But if the syzygetic cubic  $aU + 6\beta HU = 0$  be properly selected, viz. if this curve be such that its Hessian breaks up into three lines, then both the Pippian of the cubic  $aU + 6\beta HU = 0$ , and the Pippian of its Hessian will break up into the same three points, which will be a portion of the curve of the sixth class, and discarding these three points the curve will sink down to one of the third class, and will in fact be the Quippian of the cubic.

To show this we may take

$$aU + 6\beta HU = x^3 + y^3 + z^3$$

as the equation of the syzygetic cubic satisfying the prescribed condition, for this value in fact gives

$$H(aU + 6\beta HU) = -xyz,$$

a system of three lines. We find, moreover,

$$P(aU + 6\beta HU) = -\xi\eta\zeta,$$

and

$$P\{6H(aU + 6\beta HU)\} = P(-6xyz) = -4\xi\eta\zeta,$$

the latter equation being obtained by first neglecting all but the highest power of  $l$  in the expression of  $PU$ , and then writing  $l = -1$ ; we have also  $T(aU + 6\beta HU) = 1$ . Substituting the above values, the curve of the sixth class is

$$\xi^2\eta^2\zeta^2\{-4T \cdot PU + P(6HU)\} = 0;$$

or throwing out the factor  $\xi^2\eta^2\zeta^2$ , we have the curve of the third class,

$$-4T \cdot PU + P(6HU) = 0.$$

Now the general expression in my Third Memoir, viz.

$$P(aU + 6\beta HU) = \{(\alpha^3 + l^2)\alpha\beta^2 + 4T\beta^3\}PU + (\alpha^4 - 4S\beta^4)QU,$$

putting  $\alpha = 0$ ,  $\beta = 1$ , gives

$$P(6HU) = 4T \cdot PU - 4S \cdot QU,$$

or what is the same thing,

$$-4T \cdot PU + P(6HU) = -4S \cdot QU;$$

and the curve of the third class is therefore the Quippian  $QU = 0$ . It may be remarked, that for a cubic  $U = 0$  the Hessian of which breaks up into three lines, the above investigation shows that we have  $PU = -\xi\eta\zeta$ ,  $P(6HU) = -4\xi\eta\zeta$ , and  $T = 1$ , and consequently that  $-4T \cdot PU + P(6HU)$  ought to vanish identically; this in fact happens in virtue of the factor  $S$  on the right-hand side, the invariant  $S$  of a cubic of the form in question being equal to zero; the appearance of the factor  $S$  on the right-hand side is thus accounted for *à priori*.

*Article No. 22. Theorem relating to a line which meets three given conics in six points in involution.*

22. The envelope of a line which meets three given conics, the first or conic polars of any three points with respect to the cubic, in six points in involution, is the Pippian.

It is readily seen that if the theorem is true with respect to the three conics,

$$\frac{dU}{dx} = 0, \quad \frac{dU}{dy} = 0, \quad \frac{dU}{dz} = 0,$$

it is true with respect to any three conics whatever of the form

$$\lambda \frac{dU}{dx} + \mu \frac{dU}{dy} + \nu \frac{dU}{dz} = 0,$$

that is, with respect to any three conics, each of them the first or conic polar of some point  $(\lambda, \mu, \nu)$  with respect to the cubic. Considering then these three conics, take  $\xi x + \eta y + \zeta z = 0$  as the equation of the line, and let  $(X, Y, Z)$  be the coordinates of a point of intersection with the first conic, we have

$$\begin{aligned} \xi X + \eta Y + \zeta Z &= 0, \\ X^2 + 2lYZ &= 0; \end{aligned}$$

and combining with these a linear equation

$$\alpha X + \beta Y + \gamma Z = 0,$$

in which  $(\alpha, \beta, \gamma)$  are arbitrary quantities, we have

$$X : Y : Z = \beta\zeta - \gamma\eta : \gamma\xi - \alpha\zeta : \alpha\eta - \beta\xi;$$

and hence

$$(\beta\zeta - \gamma\eta)^2 + 2l(\gamma\xi - \alpha\zeta)(\alpha\eta - \beta\xi) = 0,$$

an equation in  $(\alpha, \beta, \gamma)$  which is in fact the equation in line coordinates of the two points of intersection with the first conic. Developing and forming the analogous equations, we find

$$\begin{aligned} (-2l\eta\zeta, \zeta^2, \eta^2, -\eta\zeta - l\xi^2, l\eta\xi, l\xi\zeta \mid \alpha, \beta, \gamma)^2 &= 0, \\ (\zeta^2, -2l\zeta\xi, \xi^2, l\xi\eta, -l\eta^2 - l\xi^2, l\eta\zeta \mid \alpha, \beta, \gamma)^2 &= 0, \\ (\eta^2, \xi^2, -2l\xi\eta, l\eta\zeta, l\xi\zeta, -\xi\eta - l\zeta^2 \mid \alpha, \beta, \gamma)^2 &= 0, \end{aligned}$$

which are respectively the equations in line coordinates of the three pairs of intersections.

Now combining these equations with the equation  $\gamma = 0$ , we have the equations of the pairs of lines joining the points of intersection with the point  $(x = 0, y = 0)$ , and if the six points are in involution, the six lines must also be in involution, or the condition for the involution of the six points is

$$\begin{vmatrix} -2l\eta\zeta, & \zeta^2, & l\xi\zeta \\ \zeta^2, & -2l\zeta\xi, & l\xi\eta \\ \eta^2, & \xi^2, & l\eta\zeta \end{vmatrix} = 0,$$

that is,

$$4l^2\xi^2\xi\eta(-\xi\eta - l\zeta^2) + l\xi^2\eta\zeta(-\zeta^2 - l\xi^2) + l^2\xi\eta\zeta^4 + 2l^2\xi^2\eta\zeta^3 + 2l^2\xi^2\eta\zeta^3 + \zeta^4(-l^3 - l^4) = 0;$$

or, reducing and throwing out the factor  $\zeta$ , we find

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = 0,$$

which shows that the line in question is a tangent of the Pippian.

It is to be remarked that any three conics whatever may be considered as the first or conic polars of three properly selected points with respect to a properly selected cubic curve. The theorem applies therefore to any three conics whatever, but in this case the cubic curve is not given, and the Pippian therefore stands merely for a curve of the third class, and the theorem is as follows, viz. the envelope of a line which meets any three conics in six points in involution, is a curve of the third class.

*Article No. 23. Completion of the theory in Liouville, and comparison with analogous theorems of Hesse.*

In order to convert the foregoing theorem into its reciprocal, we must replace the cubic  $U = 0$  by a curve of the third class, that is we must consider the coordinates which enter into the equation as line coordinates; and it of course follows that the coordinates which enter into the equation  $PU = 0$  must be considered as point coordinates, that is we must consider the Pippian as a curve of the third order: we have thus the theorem; The locus of a point such that the tangents drawn from it to three given conics (the first or conic poles of any three lines with respect to a curve of the third class) form a pencil in involution, is the Pippian considered as a curve of the third order. This in fact completes the fundamental theorem in my memoirs in Liouville above referred to, and establishes the analogy with Hesse's results in relation to the Hessian; to show this I set out the two series of theorems as follows.

Hesse, in his memoirs On Curves of the Third Order and Curves of the Third Class, Crelle, tt. XXVIII., XXXVI., and XXXVIII. [1844, 1848, 1849], has shown as follows:

( $\alpha$ ) The locus of a point such that its polars with respect to the three conics  $X = 0, Y = 0, Z = 0$  (or more generally its polars with respect to all the conics of the series  $\lambda X + \mu Y + \nu Z = 0$ ) meet in a point, is a curve of the third order  $V = 0$ .

( $\beta$ ) Conversely, given a curve of the third order  $V = 0$ , there exists a series of conics such that the polars with respect to all the conics of any point whatever of the curve  $V = 0$ , meet in a point.

( $\gamma$ ) The equation of any one of the conics in question is

$$\lambda \frac{dU}{dx} + \mu \frac{dU}{dy} + \nu \frac{dU}{dz} = 0,$$

that is, the conic is the first or conic polar of a point  $(\lambda, \mu, \nu)$  with respect to a certain curve of the third order  $U = 0$ ; and this curve is determined by the condition that its Hessian is the given curve  $V = 0$ , that is, we have  $V = HU$ .

( $\delta$ ) The equation  $V = HU$  is solved by assuming  $U = aV + bHV$ , for we have then  $H(aV + bHV) = AV + BHV$ , where  $A, B$  are given cubic functions of  $a, b$ , and thence  $V = HU = AV + BHV$ , or  $A = 1, B = 0$ ; the latter equation gives what is alone important, the ratio  $a : b$ ; and it thus appears that there are three distinct series of conics, each of them having the above-mentioned relation to the given curve of the third order  $V = 0$ .

In the memoirs in Liouville above referred to, I have in effect shown that

( $\alpha'$ ) The locus of a point such that the tangents from it to three conics, represented in line coordinates by the equations  $X = 0, Y = 0, Z = 0$  (or more generally with respect to any three conics of the series  $\lambda X + \mu Y + \nu Z = 0$ ) form a pencil in involution, is a curve of the third order  $V = 0$ .

( $\beta'$ ) Conversely, given a curve of the third order  $V = 0$ , there exists a series of conics such that the tangents from any point whatever of the curve to any three of the conics, form a pencil in involution.

Now, considering the coordinates which enter into the equation of the Pippian as point coordinates, and consequently the Pippian as a curve of the third order, I am able to add as follows:

( $\gamma'$ ) The equation in line coordinates of any one of the conics in question is

$$\lambda \frac{dU}{d\xi} + \mu \frac{dU}{d\eta} + \nu \frac{dU}{d\zeta} = 0,$$

that is, the conic is the first or conic polar of a line  $(\lambda, \mu, \nu)$  with respect to a certain curve of the third class  $U = 0$ ; and this curve is determined by the condition that its Pippian is the given curve of the third order  $V = 0$ , that is, we have  $V = PU$ .

( $\delta'$ ) The equation  $V = PU$  is solved by assuming  $U = aPV + bQV$ , for we have then  $P(aPV + bQV) = AV + BHV$ , where  $A$  and  $B$  are given cubic functions of  $a, b$ ; and thence  $V = PU = AV + BHV$ , or  $A = 1, B = 0$ ; the latter equation gives what is alone important, the ratio  $a : b$ ; and it thus appears that there are three distinct curves of the third class  $U = 0$ , and therefore (what indeed is shown in the Memoirs in Liouville) three distinct series of conics having the above-mentioned relation to the given curve of the third order  $V = 0$ .

It is hardly necessary to remark that the preceding theorems, although precisely analogous to those of Hesse, are entirely distinct theorems, that is the two series are not connected together by any relation of reciprocity.

*Article Nos. 24 to 28. Various investigations and theorems.*

24. Reverting to the theorem (No. 18), that the lineo-polar envelope of the line  $EF$  is the pair of lines  $OE, OF$ ; the line  $EF$  is any tangent of the Pippian, hence the theorem includes the following one:

The lineo-polar envelope with respect to the cubic, of any tangent of the Pippian, is a pair of lines.

And conversely,

The Pippian is the envelope of a line such that the lineo-polar envelope of the line with respect to the cubic is a pair of lines.

It is I think worth while to give an independent proof. It has been shown that the equation of the lineo-polar envelope with respect to the cubic, of the line  $\xi x + \eta y + \zeta z = 0$  (where  $\xi, \eta, \zeta$  are arbitrary quantities), is

$$(-l\xi^2, -l\eta^2 - 2l\zeta\xi, -l\zeta^2 - 2l\xi\eta, -l\xi^2 - 2l\eta\zeta, l^2\eta^2 + l^2\zeta\xi, l^2\zeta^2 + l^2\xi\eta \mid x, y, z)^2 = 0;$$

and representing this equation by

$$\frac{1}{l}(a, b, c, f, g, h \mid x, y, z)^2 = 0,$$



we find

$$\begin{aligned} bc - f^2 &= l^2(8l^3\xi^4 - \eta^4 + 4l\eta\zeta^3 + 4l^2\zeta^2\eta^2\xi), \\ ca - g^2 &= l^2(8l^3\eta^4 + 8l\zeta^3\xi - \zeta^4 + 12l^2\zeta^2\xi^2\eta), \\ ab - h^2 &= l^2(8l^3\zeta^4 + 12l^2\xi^2\eta^2 + \dots), \\ gh - af &= \xi\{2l^3(\xi^4 + \eta^4 + \zeta^4) + 4l(1 + 2l^3)\xi\eta\zeta\} + (1 + 8l^3)\eta^2\zeta^2, \\ hf - bg &= \eta\{2l^3(\xi^4 + \eta^4 + \zeta^4) + 4l(1 + 2l^3)\xi\eta\zeta\} + (1 + 8l^3)\zeta^2\xi^2, \\ fg - ch &= \zeta\{2l^3(\xi^4 + \eta^4 + \zeta^4) + 4l(1 + 2l^3)\xi\eta\zeta\} + (1 + 8l^3)\xi^2\eta^2; \end{aligned}$$

and after all reductions,

$$abc - af^2 - bg^2 - ch^2 + 2fgh = -l(\xi^3 + \eta^3 + \zeta^3)\{-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta\}^2 = -l(PU)^2,$$

or the condition in order that the conic may break up into a pair of lines is  $PU = 0$ .

25. The following formulae are given in connexion with the foregoing investigation, but I have not particularly considered their geometrical signification. The lineo-polar envelope of an arbitrary line  $\xi x + \eta y + \zeta z = 0$ , with respect to the cubic

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

has been represented by

$$(a, b, c, f, g, h \mid x, y, z)^2 = 0;$$

and if in like manner we represent the lineo-polar envelope of the same line, with respect to a syzygetic cubic

$$x^3 + y^3 + z^3 + 6l'xyz = 0,$$

by

$$(a', b', c', f', g', h' \mid x, y, z)^2 = 0,$$

then we have

$$\begin{aligned} a'(bc - f^2) + b'(ca - g^2) + c'(ab - h^2) + 2f'(gh - af) + 2g'(hf - bg) + 2h'(fg - ch) \\ = (l^4(\xi^6 + \eta^6 + \zeta^6) + (2l^2 + 4l - 32l^3l'^2 + 8l^6)(\xi^3 + \eta^3 + \zeta^3)\xi\eta\zeta \\ + (24l^2l'^2 + 48l^4l'^2 - 11ll' + 24l^7 + a)\xi^2\eta^2\zeta^2), \end{aligned}$$

which may be verified by writing  $l' = l$ , in which case the right-hand side becomes as it should do,  $S(PU)^2$ .

If  $l' = \frac{1 + 2l^3}{-6l^2}$ , that is, if the syzygetic cubic be the Hessian, then the formula becomes

$$a'(bc - f^2) + \&c. = 3l^4(\xi^6 + \eta^6 + \zeta^6) + 12l^2(-1 + 26l^3 + 56l^6)(\xi^3 + \eta^3 + \zeta^3)\xi\eta\zeta + 12l(2 + 57l^3 + 168l^6 + 16l^9)\xi^2\eta^2\zeta^2,$$

which is equal to

$$\frac{T^2 \cdot 25S - 24S \cdot PU}{36l^3}.$$

26. The equation

$$(bc' + b'c - 2ff', \dots, gh' + g'h - af - a'f', \dots \mid \xi, \eta, \zeta)^2 = 0$$

is the equation in line coordinates of a conic, the envelope of the line which cuts harmonically the conics

$$\begin{aligned} (a, b, c, f, g, h \mid x, y, z)^2 &= 0, \\ (a', b', c', f', g', h' \mid x, y, z)^2 &= 0; \end{aligned}$$

and if  $a, b, \&c.$ ,  $a', \&c.$  have the values before given to them, then the coefficients of the equation are

$$\begin{aligned} bc' + b'c - 2ff' &= \xi\{-\xi^3 + 4ll'(l + l')(\eta^3 + \zeta^3) + (16l^2l'^2 - 2l^3 - 2l'^3)\xi\eta\zeta\}, \\ ca' + c'a - 2gg' &= \eta\{-\eta^3 + 4ll'(l + l')(\zeta^3 + \xi^3) + (16l^2l'^2 - 2l^3 - 2l'^3)\xi\eta\zeta\}, \\ ab' + a'b - 2hh' &= \zeta\{-\zeta^3 + 4ll'(l + l')(\xi^3 + \eta^3) + (16l^2l'^2 - 2l^3 - 2l'^3)\xi\eta\zeta\}, \\ gh' + g'h - af - a'f' &= \xi\{(l^3 + l'^3)(\xi^3 + \eta^3 + \zeta^3) + (2l + 2l' + 8l^2l'^2)\xi\eta\zeta\} + (1 + 4ll'(l + l'))\eta^2\zeta^2, \\ hf' + h'f - bg - b'g' &= \eta\{(l^3 + l'^3)(\xi^3 + \eta^3 + \zeta^3) + (2l + 2l' + 8l^2l'^2)\xi\eta\zeta\} + (1 + 4ll'(l + l'))\zeta^2\xi^2, \\ fg' + f'g - ch - c'h' &= \zeta\{(l^3 + l'^3)(\xi^3 + \eta^3 + \zeta^3) + (2l + 2l' + 8l^2l'^2)\xi\eta\zeta\} + (1 + 4ll'(l + l'))\xi^2\eta^2; \end{aligned}$$

and we thence obtain

$$\frac{1}{16}(bc' + b'c - 2ff', \dots, gh' + g'h - af - a'f', \dots | \xi, \eta, \zeta)^2 = -(\xi^4 + \eta^4 + \zeta^4) + (l^2 + l'^2 + 11l'l')(\xi^3 + \eta^3 + \zeta^3)\xi\eta\zeta + (6l + 6l' + 24l^2l'^2)\xi^2\eta^2\zeta^2 + (4 + 16(l + l')(l^2 + l'^2))(y^2\zeta^2 + \zeta^2\xi^2 + \xi^2\eta^2) = 0$$

as the condition which expresses that a line  $\xi x + \eta y + \zeta z = 0$  cuts harmonically its lineo-polar envelopes with respect to the cubic and with respect to a syzygetic cubic.

27. To find the locus of a point such that its second or line polar with respect to the cubic may be a tangent of the Pippian. Let the coordinates of the point be  $(x, y, z)$ ; then if  $\xi x + \eta y + \zeta z = 0$  be the equation of the polar, we have

$$\xi : \eta : \zeta = x^2 + 2lyz : y^2 + 2lzx : z^2 + 2lxy,$$

and the line in question being a tangent to the Pippian,

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = 0.$$

But the preceding values give

$$\begin{aligned} \xi^3 + \eta^3 + \zeta^3 &= x^6 + y^6 + z^6 + (2 + 8l^3)(y^3z^3 + z^3x^3 + x^3y^3) + 6l \cdot 6(x^3 + y^3 + z^3)xyz, \\ \xi\eta\zeta &= x^2y^2z^2 + 2l^3(x^3 + y^3 + z^3)xyz + (1 + 8l^3)x^2y^2z^2 + 2l(y^3z^3 + z^3x^3 + x^3y^3); \end{aligned}$$

and we have therefore

$$-l(x^6 + y^6 + z^6) - (10l^3 + 16l^4)(x^3 + y^3 + z^3)xyz + (1 + 40l^3 + 32l^6)x^2y^2z^2 - \dots = 0;$$

or introducing  $U$ ,  $HU$  in place of  $x^3 + y^3 + z^3$ ,  $xyz$ , the equation becomes

$$-S \cdot U^2 + (HU)^2 = 0,$$

which is the equation of the locus in question.

28. The locus of a point such that its second or line polar with respect to the cubic is a tangent of the Quippian, is found in like manner by substituting the last-mentioned values of  $\xi, \eta, \zeta$  in the equation

$$QU = (1 - 10l^3)(\xi^3 + \eta^3 + \zeta^3) - 6l^2(5 + 4l^3)\xi\eta\zeta.$$

We find as the equation of the locus,

$$(1 - 10l^3)(x^6 + y^6 + z^6) - 4l(1 - 30l^3 - 16l^6)(x^3 + y^3 + z^3)xyz + 6l^2(1 - 104l^3 - 32l^6)x^2y^2z^2 - 2(1 + 8l^3)^2(y^3z^3 + z^3x^3 + x^3y^3) = 0,$$

where the function on the left-hand side is the octicovariant  $\Theta_{14}U$  of my Third Memoir, the covariant having been in fact defined so as to satisfy the condition in question. And I have given in the memoir the following expression for  $\Theta_{14}U$ , viz.

$$\begin{aligned} \Theta_{14}U &= (1 - 16l^3 - 6l^6)U^3 \\ &\quad + (6l^2 \dots)U \cdot HU \\ &\quad + (6l^2 \dots)(HU)^2 \\ &\quad - 2(1 + 8l^3)^2(y^3z^3 + z^3x^3 + x^3y^3)^?. \end{aligned}$$

*Article Nos. 29 to 31. Formulae for the intersection of a cubic curve and a line.*

29. If the line  $\xi x + \eta y + \zeta z = 0$  meet the cubic

$$x^3 + y^3 + z^3 + 6lxyz = 0$$

in the points

$$(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3),$$

then we have

$$x_1x_2x_3 : y_1y_2y_3 : z_1z_2z_3 = \eta^3 - \zeta^3 : \zeta^3 - \xi^3 : \xi^3 - \eta^3.$$

It will be convenient to represent the equation of the cubic by the abbreviated notation  $(1, 1, 1, l \mid x, y, z)^3 = 0$ ; we have the two equations

$$(1, 1, 1, l \mid x, y, z)^3 = 0, \\ \xi x + \eta y + \zeta z = 0;$$

and if to these we join a linear equation with arbitrary coefficients,

$$\alpha x + \beta y + \gamma z = 0,$$

then the second and third equations give

$$x : y : z = \beta\zeta - \gamma\eta : \gamma\xi - \alpha\zeta : \alpha\eta - \beta\xi,$$

and substituting these values in the first equation, we obtain the resultant of the system. But this resultant will also be obtained by substituting, in the third equation, a system of simultaneous roots of the first and second equations, and equating to zero the product of the functions so obtained<sup>2</sup>. We must have therefore

$$(1, 1, 1, l \mid \beta\zeta - \gamma\eta, \gamma\xi - \alpha\zeta, \alpha\eta - \beta\xi)^3 = (\alpha x_1 + \beta y_1 + \gamma z_1)(\alpha x_2 + \beta y_2 + \gamma z_2)(\alpha x_3 + \beta y_3 + \gamma z_3),$$

and equating the coefficients of  $\alpha^3, \beta^3, \gamma^3$ , we obtain the above-mentioned relations.

30. If a tangent to the cubic

$$x^3 + y^3 + z^3 + 6lxyz = 0$$

at a point  $(x_1, y_1, z_1)$  of the cubic meet the cubic in the point  $(x_2, y_2, z_2)$ , then

$$x_2 : y_2 : z_2 = x_1(y_1^3 - z_1^3) : y_1(z_1^3 - x_1^3) : z_1(x_1^3 - y_1^3).$$

For if the equation of the tangent is  $\xi x + \eta y + \zeta z = 0$ , then

$$\xi^3 - \eta^3 = x_1^3 x_2 : y_1^3 y_2 : z_1^3 z_2 : \xi^2, \eta^2, \zeta^2,$$

and

$$\xi : \eta : \zeta = x_1^2 + 2ly_1z_1 : y_1^2 + 2lz_1x_1 : z_1^2 + 2lx_1y_1.$$

These values give

$$\xi^3 - \eta^3 = \dots$$

since  $(x_1, y_1, z_1)$  is a point of the cubic; and forming in like manner the values of  $\xi^3 - \zeta^3$  and  $\zeta^3 - \eta^3$ , we obtain the theorem.

31. The preceding values of  $(x_2, y_2, z_2)$  ought to satisfy

$$(x_1^2 + 2ly_1z_1)x_2 + (y_1^2 + 2lz_1x_1)y_2 + (z_1^2 + 2lx_1y_1)z_2 = 0, \\ x_2^3 + y_2^3 + z_2^3 + 6lx_2y_2z_2 = 0;$$

in fact the first equation is satisfied identically, and for the second equation we obtain

$$x_2^3 + y_2^3 + z_2^3 = x_1^3(y_1^3 - z_1^3)^3 + y_1^3(z_1^3 - x_1^3)^3 + z_1^3(x_1^3 - y_1^3)^3 \\ = -x_1^3(y_1^3 - z_1^3)[y_1^3(z_1^3 - x_1^3) \cdot z_1^3(x_1^3 - y_1^3)] \cdot \dots \\ = -x_1y_1z_1 \cdot x_1y_1z_1(y_1^3 - z_1^3)(z_1^3 - x_1^3)(x_1^3 - y_1^3),$$

and consequently

$$x_2^3 + y_2^3 + z_2^3 + 6lx_2y_2z_2 = (x_1^3 + y_1^3 + z_1^3 + 6lx_1y_1z_1)(y_1^3 - z_1^3)(z_1^3 - x_1^3)(x_1^3 - y_1^3) = 0,$$

which verifies the theorem. It is proper to add (the remark was made to me by Professor Sylvester) that the foregoing values

$$x_2 : y_2 : z_2 = x_1(y_1^3 - z_1^3) : y_1(z_1^3 - x_1^3) : z_1(x_1^3 - y_1^3)$$

<sup>2</sup>This is in fact the general process of elimination given in Schläfli's Memoir, "Ueber die Resultante eines Systemes mehrerer algebraischer Gleichungen," Vienna Trans. 1852. [But the process was employed much earlier, by Poisson.]

satisfy identically the relation

$$\frac{x_2^3 + y_2^3 + z_2^3}{x_2 y_2 z_2} = \frac{x_1^3 + y_1^3 + z_1^3}{x_1 y_1 z_1}.$$

*Article Nos. 32 to 34. Formulae for the Satellite line and point.*

32. The line  $\xi x + \eta y + \zeta z = 0$  meets the cubic

$$x^3 + y^3 + z^3 + 6lxyz = 0$$

in three points, and the tangents to the cubic at these points meet the cubic in three points lying in a line, which has been called the Satellite line of the given line.

To find the equation of the satellite line; suppose that  $(x_1, y_1, z_1)$ ,  $(x_2, y_2, z_2)$ ,  $(x_3, y_3, z_3)$  are the coordinates of the points in which the given line meets the cubic; then we have, as before,

$$(1, 1, 1, l \mid \beta\zeta - \gamma\eta, \gamma\xi - \alpha\zeta, \alpha\eta - \beta\xi)^3 = (\alpha x_1 + \beta y_1 + \gamma z_1)(\alpha x_2 + \beta y_2 + \gamma z_2)(\alpha x_3 + \beta y_3 + \gamma z_3).$$

The equation of the three tangents is

$$\begin{aligned} \Pi = & [(x_1^2 + 2ly_1z_1)x + (y_1^2 + 2lz_1x_1)y + (z_1^2 + 2lx_1y_1)z] \\ & \times [(x_2^2 + 2ly_2z_2)x + (y_2^2 + 2lz_2x_2)y + (z_2^2 + 2lx_2y_2)z] \\ & \times [(x_3^2 + 2ly_3z_3)x + (y_3^2 + 2lz_3x_3)y + (z_3^2 + 2lx_3y_3)z] = 0, \end{aligned}$$

and if we put

$$F = (\xi^3 + \eta^3 + \zeta^3)^2 - 2l^3(\xi^3 + \eta^3 + \zeta^3)\xi\eta\zeta + (-2l - l^4)\xi^3\eta^3\zeta^3 + (-4 + 32l^3)(\eta^3\zeta^3 + \zeta^3\xi^3 + \xi^3\eta^3),$$

( $F$  is the reciprocant  $FU$  of my Third Memoir), then we have identically

$$F \cdot U - \Pi = (\xi x + \eta y + \zeta z)^2(\xi'x + \eta'y + \zeta'z),$$

and the equation of the satellite line is  $\xi'x + \eta'y + \zeta'z = 0$ . In fact the geometrical theory shows that we must have

$$F \cdot U - N\Pi = (\xi x + \eta y + \zeta z)^2(\xi'x + \eta'y + \zeta'z),$$

and it is then clear that  $N$  is a mere number. To determine its value in the most simple manner, write  $x = 1, y = 0, z = -\frac{\xi}{\zeta}$ , we have then  $F \cdot U - N\Pi = 0$ , where

$$F = \dots$$

The value of  $\Pi$  is  $\Pi = F \cdot U$ , and we thus obtain  $N = 1$ . For, substituting the above values,

$$\begin{aligned} \Pi = & (x_1^2\zeta - z_1^2\xi)(x_2^2\zeta - z_2^2\xi)(x_3^2\zeta - z_3^2\xi) \\ & - \zeta^3x_1x_2x_3z_1z_2z_3 \dots \\ & - \zeta^2\xi(x_1^2z_2^2z_3^2 + \&c.) \\ & + \zeta\xi^2(x_1^2z_2^2 + \&c.) \\ & - \xi^3z_1z_2z_3, \end{aligned}$$

and we have

$$\begin{aligned} x_1x_2x_3 &= \eta^3 - \zeta^3, \\ x_1x_2z_3 + \&c. &= 3\xi^2\zeta, \\ x_1z_2z_3 + \&c. &= -3\xi\zeta^2, \\ z_1z_2z_3 &= -\xi^3 - \eta^3, \end{aligned}$$

and thence

$$\begin{aligned} x_1^2x_2x_3z_2 + \&c. &= 9\xi^2\zeta^4 + 6\xi^4\zeta^2 + \xi(\eta^3 - \zeta^3) \cdot 3\xi^2\zeta + 6\xi\zeta\eta^3, \\ x_1^2z_2^2z_3^2 + \&c. &= 9\xi^4\zeta^4 - \xi\zeta(\xi^3 - \eta^3) = 3\xi^2\zeta^4 + 6\xi^4\zeta^2 + \dots, \end{aligned}$$

and consequently

$$\Pi = \zeta^3(\eta^3 - \zeta^3) - \xi^3(-\xi^3 - \eta^3) - \zeta\xi(2\xi^2\zeta^2 - 2\xi\zeta^3 - 2\xi^3\zeta).$$

Now considering the equation

$$F \cdot U - \Pi = (\xi x + \eta y + \zeta z)^2(\xi' x + \eta' y + \zeta' z),$$

in order to find  $\xi', \eta', \zeta'$  it will be sufficient to find the coefficients of  $x^3, y^3, z^3$  in the function on the left-hand side of the equation. The coefficient of  $x^3$  in  $\Pi$  is

$$\begin{aligned} & (x_1^2 + 2ly_1z_1)(x_2^2 + 2ly_2z_2)(x_3^2 + 2ly_3z_3) \\ &= x_1^2x_2^2x_3^2 + 2l(x_1^2x_2^2y_3z_3 + \&c.) + 4l^2(x_1^2y_2z_2y_3z_3 + \&c.) + 8l^3y_1y_2y_3z_1z_2z_3; \end{aligned}$$

and it is easy to see that representing the function

$$(1, 1, 1, l \mid \beta\zeta - \gamma\eta, \gamma\xi - \alpha\zeta, \alpha\eta - \beta\xi)^3$$

by

$$(a, b, c, f, g, h, i, j, k, l \mid \alpha, \beta, \gamma)^3,$$

the symmetrical functions can be expressed in terms of the quantities  $a, b, \&c.$ , and that the preceding value of the coefficient of  $x^3$  in  $\Pi$  is

$$\begin{aligned} & a^2 \\ & + 2l(9hj - 6al) \\ & + 4l^2(6gk - 3fj - 3hi + 3l^2) \\ & + 8l^3bc; \end{aligned}$$

and substituting for  $a, \&c.$  their values, this becomes

$$\begin{aligned} & + 2l\{-9(\zeta\eta^3 + 2l\eta\zeta^2)(\zeta^2\xi + 2l\xi\eta\zeta)\} \\ & - 4l^2\{-6(\zeta^2\xi + 2l\xi\eta\zeta)(\zeta\xi + 2l\xi\eta\zeta) \\ & \quad + 3(\eta\zeta^2 + 2l\zeta\xi\eta)(\xi\eta + 2l\xi\eta\zeta) \\ & \quad + 3(\zeta\eta^2 + 2l\eta\xi\zeta)(\eta\xi^2 + 2l\xi\eta\zeta)\}, \end{aligned}$$

and reducing, we obtain for the coefficient of  $x^3$  in  $\Pi$  the following expression,

$$\begin{aligned} & - 24l\xi^2\eta^3 + \xi^4 + \eta^4 \dots \\ & - 24l^2(\eta^3\zeta^3 + \xi^3\zeta^3 + \xi^3\eta^3). \end{aligned}$$

Now the coefficient of  $x^3$  in  $F \cdot U$  is simply  $F$ , which is equal to

$$\begin{aligned} & \xi^4 + \eta^4 + \zeta^4 - 2\eta^2\zeta^2 - 2\zeta^2\xi^2 - 2\xi^2\eta^2 \\ & - 24l\xi\eta\zeta \dots \\ & - 24l^2(\xi^3 + \eta^3 + \zeta^3)\xi\eta\zeta \\ & - 32l^3(\eta^3\zeta^3 + \zeta^3\xi^3 + \xi^3\eta^3), \end{aligned}$$

and subtracting, the coefficient of  $x^3$  in  $F \cdot U - \Pi$  is

$$\begin{aligned} & \xi^4 - 2\eta^2\zeta^2 - 2\zeta^2\xi^2 \\ & - 6l\xi^2\eta\zeta \\ & - 8l^3(\eta^3\zeta^3 + \zeta^3\xi^3 + \xi^3\eta^3) \\ & - 8l^3(l^3 - \eta^3)(\zeta^3 - \xi^3) \\ & - 48l^3\xi^3\eta^3\zeta^3, \end{aligned}$$

which is equal to

$$(1 + 8l^3) \xi^2 (\xi^4 - 2\eta^3 - 2\zeta^3 - 6l\eta\zeta\xi^2).$$

The expression last written down is therefore the value of  $\xi^2 \xi'$ , or dividing by  $\xi^2$  we have  $\xi'$ , and then of course the values of  $\eta', \zeta'$  are known, and we obtain the identical equation

$$F \cdot U - \Pi = (1 + 8l^3)(\xi x + \eta y + \zeta z)^2 \cdot \dots,$$

and the second factor equated to zero is the equation of the satellite line of  $\xi x + \eta y + \zeta z = 0$ .

33. The point of intersection of the line  $\xi x + \eta y + \zeta z = 0$  with the satellite line  $\xi' x + \eta' y + \zeta' z = 0$  is the satellite point of the former line; and the coordinates of the satellite point are at once found to be

$$x : y : z = (\eta^3 - \zeta^3)(\eta\zeta + 2l\xi^2) : (\zeta^3 - \xi^3)(\zeta\xi + 2l\eta^2) : (\xi^3 - \eta^3)(\xi\eta + 2l\zeta^2).$$

34. If the primary line  $\xi x + \eta y + \zeta z = 0$  is a tangent to the cubic, then  $(x_1, y_1, z_1)$  being the coordinates of the point of contact, we have

$$\xi : \eta : \zeta = x_1^2 + 2ly_1z_1 : y_1^2 + 2lz_1x_1 : z_1^2 + 2lx_1y_1;$$

these values give as before

$$\eta^3 - \zeta^3 = -(1 + 8l^3)x_1^3(y_1^3 - z_1^3),$$

and they give also

$$\eta\zeta + 2l\xi^2 = (1 + 8l^3)y_1z_1,$$

and consequently we obtain

$$x : y : z = x_1(y_1^3 - z_1^3) : y_1(z_1^3 - x_1^3) : z_1(x_1^3 - y_1^3),$$

that is, the satellite point of a tangent of the cubic is the point in which this tangent again meets the cubic.

*Article Nos. 35 and 36. Theorems relating to the satellite point.*

35. If the line  $\xi x + \eta y + \zeta z = 0$  be a tangent of the Pippian, then the locus of the satellite point is the Hessian.

Take  $(x, y, z)$  as the coordinates of the satellite point, then we have

$$x : y : z = (\eta^3 - \zeta^3)(\eta\zeta + 2l\xi^2) : (\zeta^3 - \xi^3)(\zeta\xi + 2l\eta^2) : (\xi^3 - \eta^3)(\xi\eta + 2l\zeta^2),$$

where the parameters  $\xi, \eta, \zeta$  are connected by the equation

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = 0.$$

We have

$$x^3 + y^3 + z^3 = \dots,$$

and it is easy to see that the function on the right-hand side must divide by  $\eta^3 - \zeta^3$ : hence  $x^3 + y^3 + z^3$  will also divide by  $\eta^3 - \zeta^3$ , and consequently by  $(\eta^3 - \zeta^3)(\zeta^3 - \xi^3)(\xi^3 - \eta^3)$ .

We have

$$\begin{aligned} (\eta^3 - \zeta^3)^3(\zeta^3 - \xi^3)^3(\xi^3 - \eta^3)^3 &= -\xi^9 - \eta^9 - \zeta^9 + 3\xi^3(\eta^6 + \zeta^6 \dots) - 3\xi^6(\eta^3 + \zeta^3) + \dots \\ &\quad + 6\xi^3\eta^3\zeta^3(-\xi^3 - 3\eta^3 - \eta^3 + 3l^2(\xi^3 + \eta^3) - 3\zeta) \\ &\quad + 12l^3\xi\eta\zeta\{-\xi\eta^2 + \zeta^3 + 3\xi^3\eta^2 - \eta\}, \end{aligned}$$

and

...

Adding these values and completing the reduction, we find

$$\begin{aligned} (x^3 + y^3 + z^3) - (\eta^3 - \zeta^3)(\zeta^3 - \xi^3)(\xi^3 - \eta^3) \\ &= -\xi^9 - \eta^9 - \zeta^9 + l^9\eta^3\zeta^3 + 2\xi^9 + 2\xi^6 \\ &\quad + 18l\xi\eta\zeta \dots \\ &\quad + 12l^3(\xi^3 + \eta^3 + \zeta^3)\xi^3\eta^3\zeta^3, \end{aligned}$$

and we have also

$$\dots,$$

and thence

$$\begin{aligned} & \{A(x^3 + y^3 + z^3) + Bxyz\} \xi \eta \zeta \cdot (\eta^3 - \zeta^3)(\zeta^3 - \xi^3)(\xi^3 - \eta^3) \\ &= -A(\xi^9 + \eta^9 + \zeta^9) + \dots \\ &+ (12l^3 A + 8lB)(\xi^3 + \eta^3 + \zeta^3) \xi^3 \eta^3 \zeta^3 \\ &+ (8lA + (1 + 8l^3)B) \xi^4 \eta^4 \zeta^4 \\ &+ A\{(60l^3 + 81) \dots\} + l^4 B(\eta^3 \zeta^6 + \zeta^3 \xi^6 + \xi^3 \eta^6). \end{aligned}$$

The coefficient of  $\eta^3 \zeta^6 + \zeta^3 \xi^6 + \xi^3 \eta^6$  on the right-hand side will vanish if  $A(1 + 2l^3) + l^4 B = 0$ , or, what is the same thing, if  $A = l^4$ ,  $B = -(1 + 2l^3)$ ; and substituting these values, we obtain

$$\begin{aligned} & D\{l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz\} \cdot (\eta^3 - \zeta^3)(\zeta^3 - \xi^3)(\xi^3 - \eta^3) \\ &= -l(\xi^3 + \eta^3 + \zeta^3) + (-l + 4l^4)(\xi^3 + \eta^3 + \zeta^3)^2 + (-l + 8l^3 - 16l^6) \xi^3 \eta^3 \zeta^3, \end{aligned}$$

or, what is the same thing,

$$\begin{aligned} & l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz = -(\eta^3 - \zeta^3)(\zeta^3 - \xi^3)(\xi^3 - \eta^3) \\ & \times D\{-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta\}^?. \end{aligned}$$

Hence the left-hand side vanishes in virtue of the relation between  $\xi, \eta, \zeta$ , or we have

$$l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz = 0,$$

which proves the theorem.

36. Suppose that  $(X, Y, Z)$  are the coordinates of a point of the Hessian, and let  $(P, Q, R)$  be the coordinates of the point in which the tangent to the Hessian at the point  $(X, Y, Z)$  again meets the Hessian, or, what is the same thing, the satellite point in regard to the Hessian of the tangent at  $(X, Y, Z)$ . And consider the conic

$$X(x^2 + 2lyz) + Y(y^2 + 2lzx) + Z(z^2 + 2lxy),$$

which is the first or conic polar of the point  $(X, Y, Z)$  in respect of the cubic. The polar (in respect to this conic) of the point  $(P, Q, R)$  will be

$$\xi x + \eta y + \zeta z = 0,$$

where

$$\begin{aligned} \xi &= PX + l(RY + QZ), \\ \eta &= QY + l(PZ + RX), \\ \zeta &= RZ + l(QX + PY); \end{aligned}$$

or putting for  $(P, Q, R)$  their values,

$$\begin{aligned} \xi &= (Y^3 - Z^3)(X^2 - lYZ), \\ \eta &= (Z^3 - X^3)(Y^2 - lZX), \\ \zeta &= (X^3 - Y^3)(Z^2 - lXY); \end{aligned}$$

and if from these equations and the equation of the Hessian we eliminate  $(X, Y, Z)$ , we shall obtain the equation in line coordinates of the curve which is the envelope of the line  $\xi x + \eta y + \zeta z = 0$ . We find, in fact,

$$\begin{aligned} \xi^3 + \eta^3 + \zeta^3 &= (Y^3 - Z^3)(Z^3 - X^3)(X^3 - Y^3) \times \{l^2(X^3 + Y^3 + Z^3) - 3l(X^3 + Y^3 + Z^3)XYZ + 9l^2X^3Y^3Z^3 + (1 - 4l^3)(Y^3Z^3 + Z^3X^3 + X^3Y^3)\}, \\ \xi\eta\zeta &= (Y^3 - Z^3)(Z^3 - X^3)(X^3 - Y^3) \times \{l^3(X^3 + Y^3 + Z^3)XYZ + (1 - l^3)X^3Y^3Z^3 - l(Y^3Z^3 + Z^3X^3 + X^3Y^3)\}, \end{aligned}$$

and thence recollecting that

$$HU = l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)XYZ,$$

we find

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = -(Y^3 - Z^3)(Z^3 - X^3)(X^3 - Y^3)(HU)^2,$$

and the equation of the envelope is

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = 0,$$

which is therefore the Pippian. We have thus the theorem:

The envelope of the polar of the satellite point in respect to the Hessian of the tangent at any point of the Hessian, such polar being in respect of the conic which is the first or conic polar of the point of the Hessian in respect of the cubic, is the Pippian.

*Article Nos. 37 to 40. Investigations and theorems relating to the first or conic polar of a point of the cubic.*

37. The investigations next following depend on the identical equations

$$\begin{aligned} & [\alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY)] \\ & \quad \times [-XYZ(x^3 + y^3 + z^3) + (X^3 + Y^3 + Z^3)xyz] \\ & = \{X(x^2 + 2lyz) + Y(y^2 + 2lzx) + Z(z^2 + 2lxy)\} \\ & \quad \times [X(Y^3 - Z^3)(\gamma y - \beta z) + Y(Z^3 - X^3)(\alpha z - \gamma x) + Z(X^3 - Y^3)(\beta x - \alpha y)] \\ & + [x(X^2 + 2lYZ) + y(Y^2 + 2lZX) + z(Z^2 + 2lXY)] \\ & \quad \times [-(\alpha YZ + \beta ZX + \gamma XY)(Xx^2 + Yy^2 + Zz^2) + (\alpha X^2 + \beta Y^2 + \gamma Z^2)(XYz + Yzx + Zxy)], \end{aligned}$$

which is easily verified.

I represent the equation in question by

$$KT = WL + P\Theta;$$

then considering  $(x, y, z)$  as current coordinates, and  $(X, Y, Z)$  and  $(\alpha, \beta, \gamma)$  as the coordinates of two given points  $\Sigma$  and  $\Omega$ , we shall have  $U = 0$  the equation of the cubic,  $W = 0$  the equation of the first or conic polar of  $\Sigma$  with respect to the cubic,  $P = 0$  the equation of the second or line polar of  $\Sigma$  with respect to the cubic. The equation  $T = 0$  is that of a syzygetic cubic passing through the point  $\Sigma$ : the coordinates of the satellite point in respect to this syzygetic cubic of its tangent at  $\Sigma$  are

$$X(Y^3 - Z^3) : Y(Z^3 - X^3) : Z(X^3 - Y^3);$$

and calling the point in question  $\Sigma'$ , then  $L = 0$  is the equation of a line through the points  $\Sigma', \Omega$ . The equation  $\Theta = 0$  is that of a conic, viz. the first or conic polar of  $\Sigma$  with respect to a certain syzygetic cubic

$$-2(\alpha YZ + \beta ZX + \gamma XY)(x^3 + y^3 + z^3) + (\alpha X^2 + \beta Y^2 + \gamma Z^2)xyz = 0,$$

depending on the points  $\Sigma, \Omega$ , or, what is the same thing, the conic  $\Theta = 0$  is a properly selected conic passing through the points of intersection of the first or conic polars of  $\Sigma$  with respect to any two syzygetic cubics; and lastly,  $K$  is a constant coefficient. The equation expresses that the points of intersection of

$$(W = 0, P = 0), (W = 0, \Theta = 0), (L = 0, P = 0), (L = 0, \Theta = 0),$$

lie in the syzygetic cubic  $T = 0$ .

The left-hand side of the equation may be written

$$\begin{aligned} & -XYZ\{\alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY)\}(x^3 + y^3 + z^3 + 6lxyz) \\ & + xyz\{\alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY)\}(X^3 + Y^3 + Z^3 + 6lXYZ); \end{aligned}$$



and it may be remarked also that we have

$$-3lXYZ\{\alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY)\}$$

equal identically to

$$l[X(Y^3 - Z^3)(\gamma Y - \beta Z) + Y(Z^3 - X^3)(\alpha Z - \gamma X) + Z(X^3 - Y^3)(\beta X - \alpha Y)] \\ - (\alpha YZ + \beta ZX + \gamma XY)(X^3 + Y^3 + Z^3 + 6lXYZ).$$

Hence if we assume

$$X^3 + Y^3 + Z^3 + 6lXYZ = 0,$$

the equation will take the form

$$KU = WL + P\Theta,$$

where the constant coefficient  $K$  may be expressed under the two different forms

$$K = -XYZ[\alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY)] \\ = \frac{1}{3}[X(Y^3 - Z^3)(\gamma Y - \beta Z) + Y(Z^3 - X^3)(\alpha Z - \gamma X) + Z(X^3 - Y^3)(\beta X - \alpha Y)],$$

and  $W, L, P, \Theta$  have the same values as before. In the present case the point  $\Sigma$  is a point of the cubic: the equation  $W = 0$  represents the first or conic polar of the point in question, and the equation  $P = 0$  its second or line polar, which is also the tangent of the cubic. The line  $L = 0$  is a line joining the point  $\Omega$  with the satellite point of the tangent at  $\Sigma$ , or dropping altogether the consideration of the point  $\Omega$ , is an arbitrary line through the satellite point: the first or conic polar of  $\Sigma$  meets the cubic twice in the point  $\Sigma$ , and therefore also meets it in four other points; the conic  $\Theta = 0$  is a conic passing through these four points, and completely determined when the particular position of the line through the satellite point is given. And, as before remarked,  $\Theta = 0$  is a conic passing through the points of intersection of the first or conic polars of  $\Sigma$  with respect to any two syzygetic cubics. We have thus the theorem:

The first or conic polar of a point of the cubic touches the cubic at this point, and besides meets it in four other points; the four points in question are the points in which the first or conic polar of the given point in respect of the cubic is intersected by the first or conic polar of the same point in respect to any syzygetic cubic whatever.

38. The analytical result may be thus stated: putting

$$\kappa = \alpha YZ + \beta ZX + \gamma XY, \quad \lambda = \alpha X^2 + \beta Y^2 + \gamma Z^2,$$

or, if we please, considering  $\kappa, \lambda$  as arbitrary parameters, then the four points lie in the conic

$$(2\kappa X, 2\kappa Y, 2\kappa Z, -\lambda X, -\lambda Y, -\lambda Z \mid x, y, z)^2 = 0,$$

or, what is the same thing, they are the points of intersection of the two conics

$$\lambda x^2 + Yy^2 + Zz^2 = 0, \\ Xyz + Yzx + Zxy = 0.$$

39. Considering the four points as the angles of a quadrangle, it may be shown that the three centres of the quadrangle lie on the cubic. To effect this, assume that the conic

$$(2\kappa X, 2\kappa Y, 2\kappa Z, -\lambda X, -\lambda Y, -\lambda Z \mid x, y, z)^2 = 0$$

represents a pair of lines; these lines will intersect in a point, which is one of the three centres in question. And taking  $x, y, z$  as the coordinates of this point, we have

$$x^2 : y^2 : z^2 : yz : zx : xy = 4\kappa^2 YZ - \lambda^2 X^2 : 4\kappa^2 ZX - \lambda^2 Y^2 : 4\kappa^2 XY - \lambda^2 Z^2 : \lambda^2 YZ + 2\kappa\lambda X^2 : \lambda^2 ZX + 2\kappa\lambda Y^2 : \lambda^2 XY + 2\kappa\lambda Z^2;$$

and we may, if we please, use these equations to find the relation between  $\kappa, \lambda$ . Thus in the identical equation  $x^2 \cdot y^2 - (xy)^2 = 0$ , substituting for  $x^2, xy, y^2$  their values, and throwing out the factor  $Z$ , we find

$(4\kappa^2 - \lambda^2)XYZ - x^2y^2(X^3 + Y^3 + Z^3) = 0$ , and thence, in virtue of the equation  $X^3 + Y^3 + Z^3 + 6lXYZ = 0$ , we obtain

$$4\kappa^2 + 6l\kappa\lambda^2 - \lambda^3 = 0.$$

But the preceding system gives conversely,

$$X^2 : Y^2 : Z^2 : YZ : ZX : XY = 4\kappa^2yz - \lambda^2x^2 : 4\kappa^2zx - \lambda^2y^2 : 4\kappa^2xy - \lambda^2z^2 : \lambda^2yz + 2\kappa\lambda x^2 : \lambda^2zx + 2\kappa\lambda y^2 : \lambda^2xy + 2\kappa\lambda z^2.$$

Hence from the identical relation  $X^2 \cdot Y^2 - (XY)^2 = 0$ , substituting for  $X^2, XY, Y^2$  their values, and throwing out the factor  $z$ , we find  $(4\kappa^2 - \lambda^2)xyz - \kappa\lambda^2(x^3 + y^3 + z^3) = 0$ , and thence, in virtue of the equation  $4\kappa^2 - \lambda^2 = -6l\kappa\lambda^2$ , we obtain

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

which shows that the point in question lies on the cubic. We have thus the theorem:

The first or conic polar of a point of the cubic touches the cubic at the point, and meets it besides in four points, which are the angles of a quadrangle the centres of which lie on the cubic. In other words, the quadrangle is an inscribed quadrangle.

40. To find the equations of the three axes of the quadrangle, that is of the lines through two centres.

We have

$$\begin{aligned} (4\kappa^2YZ - \lambda^2X^2)x + (\lambda^2XY + 2\kappa\lambda Z^2)y + (\lambda^2ZX + 2\kappa\lambda Y^2)z &= 0, \\ (\lambda^2XY + 2\kappa\lambda Z^2)x + (4\kappa^2ZX - \lambda^2Y^2)y + (\lambda^2YZ + 2\kappa\lambda X^2)z &= 0, \\ (\lambda^2ZX + 2\kappa\lambda Y^2)x + (\lambda^2YZ + 2\kappa\lambda X^2)y + (4\kappa^2XY - \lambda^2Z^2)z &= 0; \end{aligned}$$

or arranging these equations in the proper form and eliminating  $\kappa^2, \kappa\lambda, \lambda^2$ , we find

$$\begin{vmatrix} YZx, Z^2y + Y^2z, X(-Xx + Yy + Zz) \\ ZXy, X^2z + Z^2x, Y(Xx - Yy + Zz) \\ XYz, Y^2x + X^2y, Z(Xx + Yy - Zz) \end{vmatrix} = 0;$$

or, multiplying out,

$$\begin{aligned} &XYZ\{(Z^2 - Y^2)x^3 + (X^2 - Z^2)y^3 + (Y^2 - X^2)z^3\} \\ &+ x^2yZY^2(-2X^2 + Y^2 + Z^2) + zx^2YZ^2(2X^2 - Y^2 - Z^2) \\ &+ y^2zXZ^2(-2Y^2 + Z^2 + X^2) + xy^2ZX^2(2Y^2 - Z^2 - X^2) \\ &+ z^2xYX^2(-2Z^2 + X^2 + Y^2) + yz^2XY^2(2Z^2 - X^2 - Y^2) = 0. \end{aligned}$$

We may simplify this result by means of the equation  $X^3 + Y^3 + Z^3 + 6lXYZ = 0$ , so as to make the left-hand side divide out by  $XYZ$ : we thus obtain

$$\begin{aligned} &(Z^3 - Y^3)x^3 + (X^3 - Z^3)y^3 + (Y^3 - X^3)z^3 \\ &+ (-3X^2Y - 6lY^2Z)x^2y + (-3Y^2Z - 6lZ^2X)y^2z + (-3Z^2X - 6lX^2Y)z^2x \\ &+ (3XY^2 + 6lX^2Z)xy^2 + (3YZ^2 + 6lY^2X)yz^2 + (3ZX^2 + 6lZ^2Y)zx^2 = 0; \end{aligned}$$

or in a different form,

$$\begin{aligned} &(y^3 - z^3)X^3 + (z^3 - x^3)Y^3 + (x^3 - y^3)Z^3 \\ &+ (-3x^2y - 6lz^2x)X^2Y + (-3y^2z - 6lx^2y)Y^2Z + (-3z^2x - 6ly^2z)Z^2X \\ &+ (3xy^2 + 6lyz^2)XY^2 + (3yz^2 + 6lxz^2)YZ^2 + (3zx^2 + 6lxy^2)ZX^2 = 0, \end{aligned}$$

as the equation of the three axes of the quadrangle.

*Article No. 41. Recapitulation of geometrical definitions of the Pippian.*

In conclusion, I will recapitulate the different modes of generation or geometrical definitions of the Pippian, obtained in the course of the present memoir. The curve in question is:

1. The envelope of the line joining a pair of conjugate poles of the cubic (see Nos. 2 and 13).
2. The envelope of each line of the pair forming the first or conic polar with respect to the cubic of a conjugate pole of the cubic (see Nos. 2 and 14).
3. The envelope of a line which is the polar of a conjugate pole of the cubic, with respect to the conic which is the first or conic polar of the other conjugate pole in respect to any syzygetic cubic (see Nos. 2 and 9).
4. The locus of the harmonic with respect to a pair of conjugate poles of the cubic of the third point of intersection with the Hessian of the line joining the two conjugate poles (see Nos. 2 and 17).
5. The envelope of a line such that its lineo-polar envelope with respect to the cubic breaks up into a pair of lines (see No. 24).
6. The envelope of a line which meets three conics, the first or conic polars of any three points in respect to the cubic, in six points in involution (see No. 22).
7. The envelope of the second or line polar with respect to the cubic, of a point the locus of which is a certain curve of the sixth order in quadratic syzygy with the cubic and Hessian, viz. the curve  $-S \cdot U^2 + (HU)^2 = 0$  (see No. 27).
8. The envelope of a line having for its satellite point a point of the Hessian (see No. 35).
9. The envelope of the polar of the satellite point with respect to the Hessian of the tangent at a point of the Hessian, with respect to the first or conic polar of the point of the Hessian in respect to the cubic (see No. 36).

## 147.

### A MEMOIR ON THE SYMMETRIC FUNCTIONS OF THE ROOTS OF AN EQUATION.

*[From the Philosophical Transactions of the Royal Society of London, vol. CXLVII for the year 1857, pp. 489–499. Received December 18, 1856,—Read January 8, 1857.]*

There are contained in a work, which is not, I think, so generally known as it deserves to be, the “Algebra” of Meyer Hirsch [the work referred to is entitled *Sammlung von Beispielen Formeln und Aufgaben aus der Buchstabenrechnung und Algebra*, 8vo. Berlin, 1804 (8 ed. 1853), English translation by Ross, 8vo. London, 1827] some very useful tables of the symmetric functions up to the tenth degree of the roots of an equation of any order. It seems desirable to join to these a set of tables, giving reciprocally the expressions of the powers and products of the coefficients in terms of the symmetric functions of the roots. The present memoir contains the two sets of tables, viz. the new tables distinguished by the letter (a), and the tables of Meyer Hirsch distinguished by the letter (b); the memoir contains also some remarks as to the mode of calculation of the new tables, and also as to a peculiar symmetry of the numbers in the tables of each set, a symmetry which, so far as I am aware, has not hitherto been observed, and the existence of which appears to constitute an important theorem in the subject. The theorem in question might, I think, be deduced from a very elegant formula of M. Borchardt (referred to in the sequel), which gives the generating function of any symmetric function of the roots, and contains potentially a method for the calculation of the Tables (b), but which, from the example I have given, would not appear to be a very convenient one for actual calculation.

Suppose in general

$$(1, b, c, \dots | 1, x)^n = (1 - \alpha x)(1 - \beta x)(1 - \gamma x) \dots,$$

so that

$$-b = \Sigma \alpha, \quad +c = \Sigma \alpha \beta, \quad -d = \Sigma \alpha \beta \gamma, \quad \&c.,$$

and if in general

$$(pqr \dots) = \Sigma \alpha^p \beta^q \gamma^r \dots,$$

where as usual the summation extends only to the distinct terms, so that e.g.  $(p^2)$  contains only half as many terms as  $(pq)$ , and so in all similar cases, then we have

$$-b = (1), \quad +c = (1^2), \quad -d = (1^3), \quad \&c.;$$

and the two problems which arise are, first to express any combination  $b^p c^q \dots$  in terms of the symmetric functions  $(l^\lambda m^\mu \dots)$ , and secondly, or conversely, to express any symmetric function  $(l^\lambda m^\mu \dots)$  in terms of the combinations  $b^p c^q \dots$ .

It will conduce materially to brevity if  $1^p 2^q \dots$  be termed the partition belonging to the combination  $b^p c^q \dots$ ; and in like manner if  $l^\lambda m^\mu \dots$  be termed the partition belonging to the symmetric function  $(l^\lambda m^\mu \dots)$ , and if the sum of the component numbers of the partition is termed the weight.

Consider now a line of combinations corresponding to a given weight, e.g. the weight 4, this will be

$$\begin{array}{cccccc} e & bd & c^2 & b^2c & b^4 & \text{(line)} \\ 4 & 13 & 2^2 & 1^22 & 1^4 & \end{array}$$

where I have written under each combination the partition which belongs to it, and in like manner a column of symmetric functions of the same weight, viz.

$$\begin{array}{c} (4) \quad \text{(column)} \\ (31) \\ (2^2) \\ (21^2) \\ (1^4), \end{array}$$

where, as the partitions are obtained by simply omitting the  $()$ , I have not separately written down the partitions.

It is at once obvious that the different combinations of the line will be made up of numerical multiples of the symmetric functions of the column; and conversely, that the symmetric functions of the column will be made up of numerical multiples of the combinations of the line; but this requires a further examination. There are certain restrictions as to the symmetric functions which enter into the expression of the combination, and conversely, as to the combinations which enter into the expression of the symmetric function. The nature of the first restriction is most clearly seen by the following Table:

| No. of Parts | Greatest Part | Comb.  | Part.  | Contain Mult. of the Sym. Functions | Greatest Part $\nless$ | No. of Parts $\nless$ |
|--------------|---------------|--------|--------|-------------------------------------|------------------------|-----------------------|
| 1            | 4             | $e$    | 4      | $(1^4)$                             | 1                      | 4                     |
| 2            | 3             | $bd$   | 13     | $(1^4), (21^2)$                     | 2                      | 3                     |
| 2            | 2             | $c^2$  | $2^2$  | $(1^4), (21^2), (2^2)$              | 2                      | 2                     |
| 3            | 2             | $b^2c$ | $1^22$ | $(1^4), (21^2), (2^2), (31)$        | 3                      | 2                     |
| 4            | 1             | $b^4$  | $1^4$  | $(1^4), (21^2), (2^2), (31), (4)$   | 4                      | 1                     |

Thus, for instance, the combination  $bd$  (the partition whereof is 13) contains multiples of the two symmetric functions  $(1^4), (21^2)$  only. The number of parts in the partition 13 is 2, and the greatest part is 3. And in the partitions  $(1^4), (21^2)$  the greatest part is 2, and the number of parts is not less than 3. The reason is obvious: each term of the developed expression of  $bd$  must contain at least as many roots as are contained in each term of  $d$ , that is 3 roots, and since the coefficients are linear functions in respect to each root, the combination  $bd$  cannot contain a power higher than 2 of any root. The reasoning is immediately applied to any other case, and we obtain

**First Restriction.**—A combination  $b^p c^q \dots$  contains only those symmetric functions  $(l^\lambda m^\mu \dots)$ , for which the greatest part does not exceed the number of parts in the partition  $1^p 2^q \dots$ , and the number of parts is not less than the greatest part in the same partition.

Consider a partition such as  $1^2 2$ , then replacing each number by a line of units thus,

$$\begin{array}{c} 1 \\ 1 \\ 11 \end{array}$$

and summing the columns, we obtain a new partition 31, which may be called the conjugate<sup>3</sup> of  $1^2 2$ . It is easy to see that the expression for the combination  $b^2 c$  (for which the partition is  $1^2 2$ ) contains with

<sup>3</sup>The notion of Conjugate Partitions is, I believe, due to Professor Sylvester or Mr Ferrers. [It was due to Mr now Dr Ferrers.]

the coefficient unity, the symmetric function (31), the partition whereof is the conjugate of  $1^22$ . In fact  $b^2c = (-\Sigma\alpha)^2(\Sigma\alpha\beta)$ , which obviously contains the term  $1\alpha^3\beta$ , and therefore the symmetric function with its coefficient  $+1$  (31); and the reasoning is general, or

**Theorem.** A combination  $b^pc^q\dots$  contains the symmetric function (partition conjugate to  $1^p2^q\dots$ ) with the coefficient unity, and sign  $+$  or  $-$  according as the weight is even or odd.

Imagine the partitions arranged as in the preceding column, viz. first the partition into one part, then the partitions into two parts, then the partitions into three parts, and so on; the partitions into the same number of parts being arranged according to the magnitude of the greatest part (the greatest magnitude first), and in case of equality according to the magnitudes of the next greatest part, and so on (for other examples, see the outside column of any one of the Tables). The order being thus completely defined, we may speak of a partition as being prior or posterior to another. We are now able to state a second restriction as follows.

**Second Restriction.**—The combination  $b^pc^q\dots$  contains only those symmetric functions which are of the form (partition not prior to the conjugate of  $1^p2^q\dots$ ).

The terms excluded by the two restrictions are many of them the same, and it might at first sight appear as if the two restrictions were identical; but this is not so: for instance, for the combination  $bd^2$ , see Table VII (a), the term  $(41^2)$  is excluded by the first restriction, but not by the second; and on the other hand, the term  $(3^21)$ , which is not excluded by the first restriction, is excluded by the second restriction, as containing a partition  $3^21$  prior in order to  $32^2$ , which is the partition conjugate to  $13^2$ , the partition of  $bd^2$ . It is easy to see why  $bd^2$  does not contain the symmetric function  $(3^21)$ ; in fact, a term of  $(3^21)$  is  $(\alpha^3\beta^3\gamma)$ , which is obviously not a term of  $bd^2 = (-\Sigma\alpha)(\Sigma\alpha\beta\gamma)^2$ ; but I have not investigated the general proof.

I proceed to explain the construction of the Tables (a). The outside column contains the symmetric functions arranged in the order before explained; the outside or top line contains the combinations of the same weight arranged as follows, viz. the partitions taken in order from right to left are respectively conjugate to the partitions in the outside column, taken in order from top to bottom; in other words, each square of the sinister diagonal corresponds to two partitions which are conjugate to each other. It is to be noticed that the combinations taken in order, from left to right, are not in the order in which they would be obtained by Arbogast’s Method of Derivations from an operand  $a^n$ ,  $a$  being ultimately replaced by unity. The squares above the sinister diagonal are empty (i.e. the coefficients are zero), the greater part of them in virtue of both restrictions, and the remainder in virtue of the second restriction; the empty squares below the sinister diagonal are empty in virtue of the second restriction; but the property was not assumed in the calculation.

The greater part of the numbers in the Tables (a) were calculated, those of each table from the numbers in the next preceding table by the following method, depending on the derivation of the expression for  $b^{p+1}c^q\dots$  from the expression for  $b^pc^q\dots$ . Suppose, for example, the column  $cd$  of Table V (a) is known, and we wish to calculate the column  $bcd$  of Table VI (a). The process is as follows:

Given

|         |          |          |        |       |
|---------|----------|----------|--------|-------|
| $2^21$  | $21^3$   | $1^5$    |        |       |
| 1       | 3        | 10       |        |       |
| 321     | $2^3$    | $2^21^2$ | $21^4$ | $1^6$ |
| 1, 3, 2 | 3, 6, 12 |          | 10, 60 |       |
| 1       | 3        | 3, 8, 22 | 60     |       |

where the numbers in the last line are the numbers in the column  $bcd$  of Table VI (a). The partition  $2^21$ , considered as containing a part zero, gives, when the parts are successively increased by 1, the partitions  $321$ ,  $2^3$ ,  $2^21^2$ , in which the indices of the increased part (i.e. the original part plus unity) are 1, 3, 2; these numbers are taken as multipliers of the coefficient 1 of the partition  $2^21$ , and we thus have the new coefficients 1, 3, 2 of the partitions  $321$ ,  $2^3$ ,  $2^21^2$ . In like manner the coefficient 3 of the partition  $21^3$  gives the new coefficients 3, 6, 12 of the partitions  $31^2$ ,  $2^21^2$ ,  $21^4$ , and the coefficient 10 of the partition  $1^5$  gives the new coefficients 10, 60 of the partitions  $21^4$  and  $1^6$ , and finally, the last line is obtained by addition. The process in fact amounts to the multiplication separately of each term of

$$cd = 1(2^21) + 3(21^3) + 10(1^5)$$

by  $b = (1)$ . It would perhaps have been proper to employ an analogous rule for the calculation of the combinations  $cdef \dots$  not containing  $b$ , but instead of doing so I availed myself of the existing Tables (b). But the comparison of the last line of each Table (a) (which as corresponding to a combination  $b^k$  was always calculated independently of the Tables (b)) with such last line as calculated from the corresponding Table (b), seems to afford a complete verification of both the Tables; and my process has in fact enabled me to detect several numerical errors in the Tables (b), as given in the English translation of the work above referred to. It is not desirable, as regards facility of calculation and independently of the want of verification, to calculate either set of Tables wholly from the other; the rules for the independent calculation of the Tables (b) are fully and clearly explained in the work referred to, and I have nothing to add upon this subject.

The relation of symmetry, alluded to in the introductory paragraph of the present memoir, exists in each Table of either set, and is as follows: viz. the number in the Table corresponding to any two partitions in the outside column and the outside line respectively, is equal to the number corresponding to the same two partitions in the outside line and the outside column respectively. Or, calling the two partitions  $P, Q$ , and writing for shortness, combination  $(P)$  for the combination represented by the partition  $P$ , and for greater clearness, symmetric function  $(P)$  (instead of merely  $P$ ) to denote the symmetric function represented by the partition  $P$ , we have the following two theorems, viz.

**Theorem.** The coefficient in combination  $(P)$  of symmetric function  $(Q)$  is equal to the coefficient in combination  $(Q)$  of symmetric function  $(P)$ ;

and conversely,

**Theorem.** The coefficient in symmetric function  $(P)$  of combination  $(Q)$  is equal to the coefficient in symmetric function  $(Q)$  of combination  $(P)$ .

M. Borchardt's formula, before referred to, is given in the 'Monatsbericht' of the Berlin Academy (March 5, 1855)<sup>4</sup>, and may be thus stated; viz. considering the case of  $n$  roots, write

$$x^n \cdot \frac{1}{k} (1, b, c, \dots \mid 1, x)^n = (1 - \alpha x)(1 - \beta x) \dots (1 - \kappa x) = fx,$$

then

$$\Sigma \left( \frac{1}{1 - \alpha x} \cdot \frac{1}{1 - \beta y} \dots \frac{1}{1 - \kappa u} \right) = \frac{1}{k^n} \cdot \frac{fxfy \dots fu}{\Pi(x, y, \dots u)} \cdot \frac{d}{dx} \frac{d}{dy} \dots \frac{d}{du} \frac{\Pi(x, y, \dots u)}{fxfy \dots fu},$$

where  $\Pi(x, y, \dots u)$  denotes the product of the differences of the quantities  $x, y, \dots u$ , and on the left-hand side the summation extends to all the different permutations of  $\alpha, \beta, \dots \kappa$ , or what is the same thing, of  $x, y, \dots u$ .

Suppose for a moment that there are only two roots, so that

$$(1, b, c \mid 1, x)^2 = (1 - \alpha x)(1 - \beta x),$$

then the left-hand side is

$$\frac{1}{(1 - \alpha x)(1 - \beta y)} + \frac{1}{(1 - \alpha y)(1 - \beta x)},$$

which is equal to

$$2 + (\alpha + \beta)(x + y) + (\alpha^2 + \beta^2)(x^2 + y^2) + 2\alpha\beta xy + (\alpha^3 + \beta^3)(x^3 + y^3) + (\alpha^2\beta + \alpha\beta^2)(x^2y + xy^2) + \&c.,$$

and the right-hand side is

$$\frac{1}{c} \cdot \frac{fxfy}{x - y} \cdot \frac{d}{dx} \frac{d}{dy} \frac{x - y}{fxfy},$$

which is equal to

$$\frac{1}{c} \cdot \frac{1}{x - y} \left\{ \frac{f'xfy - f'yfx}{(fx)(fy)} + (x - y) \frac{f'xf'y}{fxfy} \right\},$$

and therefore to

$$\frac{1}{c} \cdot \frac{1}{fxfy} \left\{ \frac{(f'x)fy - f'yfx}{x - y} + f'xf'y \right\};$$

<sup>4</sup>And in Crelle, t. LIII, p. 195.—Note added 4th Dec. 1857, A. C.

or substituting for  $fx, fy$  their values,

$$\frac{f'xfy - f'yfx}{x - y}$$

becomes equal to

$$2c + bc(x + y) + 2c^2xy,$$

and  $fxfy$  is equal to

$$b^2 + 2bc(x + y) + 4c^2xy.$$

The right-hand side is therefore equal to

$$\frac{2 + b(x + y) + 2cxy}{(1 + bx + cx^2)(1 + by + cy^2)},$$

and comparing with the value of the left-hand side, we see that this expression may be considered as the generating function of the symmetric functions of  $(\alpha, \beta)$ , viz. the expression in question is developable in a series of the symmetric functions of  $(x, y)$ , the coefficients being of course functions of  $b$  and  $c$ , and these coefficients are (to given numerical factors *près*) the symmetric functions of the roots  $(\alpha, \beta)$ .

And in general it is easy to see that the left-hand side of M. Borchardt's formula is equal to

$$[0] + [1](1)(1)' + [2](2)(2)' + [1^2](1^2)(1^2)' + \&c.,$$

where  $(1), (2), (1^2), \&c.$  are the symmetric functions of the roots  $(\alpha, \beta, \dots \kappa)$ ,  $(1)', (2)', (1^2)', \&c.$  are the corresponding symmetric functions of  $(x, y, \dots u)$ , and  $[0], [1], [2], [1^2], \&c.$  are mere numerical coefficients; viz.  $[0]$  is equal to  $1.2.3 \dots n$ , and  $[1], [2], [1^2], \&c.$  are such that the product of one of these factors into the number of terms in the corresponding symmetric function  $(1), (2), (1^2), \&c.$  may be equal to  $1.2.3 \dots n$ . The right-hand side of M. Borchardt's formula is therefore, as in the particular case, the generating function of the symmetric functions of the roots  $(\alpha, \beta, \dots \kappa)$ , and if a convenient expression of such right-hand side could be obtained, we might by means of it express all the symmetric functions of the roots in terms of the coefficients.

### *Tables relating to the Symmetric Functions of the Roots of an Equation.*

The outside line of letters contains the combinations (powers and products) of the coefficients, the coefficients being all with the positive sign, and the coefficient of the highest power being unity; thus in the case of a cubic equation the equation is

$$x^3 + bx^2 + cx + d = (x - \alpha)(x - \beta)(x - \gamma) = 0.$$

The outside line of numbers is obtained from that of letters merely by writing  $1, 2, 3 \dots$  for  $b, c, d \dots$ , and may be considered simply as a different notation for the combinations. The outside column contains the different symmetric functions in the notation above explained, viz.  $(1)$  denotes  $\Sigma\alpha$ ,  $(2)$  denotes  $\Sigma\alpha^2$ ,  $(1^2)$  denotes  $\Sigma\alpha\beta$ , and so on. The Tables (a) are to be read according to the columns; thus Table II (a) means  $b^2 = 1(2) + 2(1^2)$ ,  $c = (1^2)$ . The Tables (b) are to be read according to the lines; thus Table II (b) means  $(2) = -2c + 1b^2$ ,  $(1^2) = +1c$ .

**I (a).** I (a) and I (b) (weight 1)

|     |     |     |     |
|-----|-----|-----|-----|
|     | 1   |     | 1   |
|     | $b$ |     | $b$ |
| (1) | -1  | (1) | -1  |

**II (a) and II (b)** (weight 2)

|                   |          |                       |
|-------------------|----------|-----------------------|
|                   | 2        | 1 <sup>2</sup>        |
|                   | <i>c</i> | <i>b</i> <sup>2</sup> |
| (2)               | +1       | +2                    |
| (1 <sup>2</sup> ) |          | +1                    |

|                   |          |                       |
|-------------------|----------|-----------------------|
|                   | 2        | 1 <sup>2</sup>        |
|                   | <i>c</i> | <i>b</i> <sup>2</sup> |
| (2)               | −2       | +1                    |
| (1 <sup>2</sup> ) | +1       |                       |

III (a) and III (b) (weight 3)

|                   |          |           |                       |
|-------------------|----------|-----------|-----------------------|
|                   | 3        | 12        | 1 <sup>3</sup>        |
|                   | <i>d</i> | <i>bc</i> | <i>b</i> <sup>3</sup> |
| (3)               | −1       | −3        | −1                    |
| (21)              |          | −1        | −3                    |
| (1 <sup>3</sup> ) |          |           | −1                    |

|                   |          |           |                       |
|-------------------|----------|-----------|-----------------------|
|                   | 3        | 12        | 1 <sup>3</sup>        |
|                   | <i>d</i> | <i>bc</i> | <i>b</i> <sup>3</sup> |
| (3)               | −3       | +3        | −1                    |
| (21)              | +3       | −1        |                       |
| (1 <sup>3</sup> ) | −1       |           |                       |

[Tables IV (a)/(b) (weight 4) through X (a)/(b) (weight 10) follow in the printed memoir. They are large rectangular grids whose rows are indexed by the partitions of the weight and whose columns are indexed by the conjugate combinations of coefficients  $b, c, d, e, f, g, h, i, k$ . Each entry is an integer (with sign) giving the coefficient that links a power-product of coefficients to a symmetric function ( $l^\lambda m^\mu \dots$ ) as explained above; many squares above the sinister diagonal are zero (empty). The tables span PDF pages 424–434 in this reprint (journal pages 408–418), beginning with the simple weight-4 layout shown above and growing to weight 10 (Table X), in which the column heading contains terms like  $bch$ ,  $b^2h$ ,  $eg$ ,  $b^2g$ ,  $bdg$ ,  $b^4g$ , etc., and the partition column ranges from (10) through (1<sup>10</sup>).]

[Table IV (a) (weight 4)]. Column headings, left to right:  $e \mid bd \mid c^2 \mid b^2c \mid b^4$ . Row entries (partition rows (4), (31), (2<sup>2</sup>), (21<sup>2</sup>), (1<sup>4</sup>)):

|                    |          |           |                       |                                |                       |
|--------------------|----------|-----------|-----------------------|--------------------------------|-----------------------|
|                    | <i>e</i> | <i>bd</i> | <i>c</i> <sup>2</sup> | <i>b</i> <sup>2</sup> <i>c</i> | <i>b</i> <sup>4</sup> |
| (4)                | +1       | +4        | +2                    | +6                             | +24                   |
| (31)               |          | +1        | +2                    | +5                             | +12                   |
| (2 <sup>2</sup> )  |          |           | +1                    | +2                             | +6                    |
| (21 <sup>2</sup> ) |          |           |                       | +1                             | +4                    |
| (1 <sup>4</sup> )  |          |           |                       |                                | +1                    |

Table IV (b) (weight 4) inverts this: each row gives a symmetric function as a signed combination of the coefficient-products  $e, bd, c^2, b^2c, b^4$ :

$$(4) = -4e + 4bd + 2c^2 - 4b^2c + 1b^4,$$
$$(31) = +4e - 1bd - 2c^2 + 1b^2c,$$
$$(2^2) = +2e + 1c^2,$$
$$(21^2) = -4e + 1bd,$$
$$(1^4) = +1e.$$

[Tables V (a)/(b) (weight 5), VI (a)/(b) (weight 6), VII (a)/(b) (weight 7), VIII (a)/(b) (weight 8), IX (a)/(b) (weight 9), and X (a)/(b) (weight 10) extend the same pattern with progressively wider grids; the printed memoir reproduces them in full. They are large numerical tables, occupying PDF pages 426–434 (journal pages 410–418), with row-headings the partitions of the weight in the conventional Cayley order (greatest part first, ties broken by next greatest part, etc.) and column-headings the conjugate coefficient-products  $f, be, cd, b^2d, bc^2, \dots$  for weight 5;  $g, bf, ce, b^2e, c^2, bcd, b^3d, c^3, b^2c^2, b^4c, b^6$  for weight 6; and so on through weight 10. The entries are signed integers; the symmetry theorem stated above asserts that the entry corresponding to the row-partition  $P$  and column-partition  $Q$  in Table N (a) equals the entry in row  $Q$ , column  $P$  of the same table.]

The numerical content of Tables IV–X (running ~300 entries per table by Table X) is preserved here in editorial summary because the original layouts are wide multi-column grids that resist reflow at 11 pt without compromising line-by-line readability; the original journal version (Phil. Trans. 1857, Tables IV (a)–X (b)) remains the authoritative source for the precise numerical values.



[Continuation and conclusion of Memoir No. 146 (“A Memoir on Curves of the Third Order”), Articles Nos. 35–41. The opening of the memoir, together with Articles Nos. 1–34, occupies PDF pp. 401–425 of the present volume.]

(From p. 409 of the printed memoir, end of Article No. 35.) Adding these values and completing the reduction, we find

$$(\xi^3 + \eta^3 + \zeta^3) - (\eta^2 - \zeta^2)(\zeta^2 - \xi^2)(\xi^2 - \eta^2) = -\xi^6 - \eta^6 - \zeta^6 + 3\xi^2\eta^2\zeta^2 + 2\xi^2\eta^4 + 2\xi^4\eta^2 + \dots \\ + 18l\xi^2\eta^2\zeta^2 + 12l^2(\xi^4 + \eta^4 + \zeta^4)\xi\eta\zeta + 8l^3(\xi^2\eta^2 + \eta^2\zeta^2 + \zeta^2\xi^2),$$

and we have also

$$\xi\eta\zeta [A(x^3 + y^3 + z^3) + Bxyz] = (\eta^2 - \zeta^2)(\zeta^2 - \xi^2)(\xi^2 - \eta^2) \\ - A(\xi^6 + \eta^6 + \zeta^6) \\ + (12l^2A + lB)(\xi^4 + \eta^4 + \zeta^4)\xi\eta\zeta \\ + (18lA + (1 + 8l^3)B)\xi^2\eta^2\zeta^2 \\ + (\text{further symmetric terms in the } \xi, \eta, \zeta).$$

The coefficient of  $\xi^4\eta^2 + \eta^4\zeta^2 + \zeta^4\xi^2$  on the right-hand side will vanish if

$$A + l(1 + 2l^3) + l^2B = 0,$$

that is, if  $A = -l$ ,  $B = \frac{1}{l}(1 + 2l^3)$  (or, what is the same thing,  $l^2A = -l^3$ ); and substituting these values, we obtain

$$l\{l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz\} - (\eta^2 - \zeta^2)(\zeta^2 - \xi^2)(\xi^2 - \eta^2) \\ = -l(\xi^6 + \eta^6 + \zeta^6) \\ + (-l + 4l^4)(\xi^4\eta^2 + \eta^4\zeta^2 + \zeta^4\xi^2 + \xi^2\eta^4 + \eta^2\zeta^4 + \zeta^2\xi^4) \\ + (-l + 8l^4 - 16l^7)\xi^2\eta^2\zeta^2,$$

or, what is the same thing,

$$l\{l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz\} = -(\eta^2 - \zeta^2)(\zeta^2 - \xi^2)(\xi^2 - \eta^2) + \{-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta\}^2.$$

Hence the left-hand side vanishes in virtue of the relation between  $\xi, \eta, \zeta$ , or we have

$$l^2(x^3 + y^3 + z^3) - (1 + 2l^3)xyz = 0,$$

which proves the theorem.

**36.** Suppose that  $(X, Y, Z)$  are the coordinates of a point of the Hessian, and let  $(P, Q, R)$  be the coordinates of the point in which the tangent to the Hessian at the point  $(X, Y, Z)$  again meets the Hessian, or, what is the same thing, the satellite point in regard to the Hessian of the tangent at  $(X, Y, Z)$ . And consider the conic

$$X(x^2 + 2lyz) + Y(y^2 + 2lzx) + Z(z^2 + 2lxy),$$

which is the first or conic polar of the point  $(X, Y, Z)$  in respect of the cubic. The polar (in respect to this conic) of the point  $(P, Q, R)$  will be

$$\xi x + \eta y + \zeta z = 0,$$

where

$$\xi = PX + l(RY + QZ), \\ \eta = QY + l(PZ + RX), \\ \zeta = RZ + l(QX + PY);$$

or putting for  $(P, Q, R)$  their values,

$$\begin{aligned}\xi &= (Y^3 - Z^3)(X^2 - lYZ), \\ \eta &= (Z^3 - X^3)(Y^2 - lZX), \\ \zeta &= (X^3 - Y^3)(Z^2 - lXY);\end{aligned}$$

and if from these equations and the equation of the Hessian we eliminate  $(X, Y, Z)$ , we shall obtain the equation in line coordinates of the curve which is the envelope of the line  $\xi x + \eta y + \zeta z = 0$ . We find, in fact,

$$\begin{aligned}\xi^3 + \eta^3 + \zeta^3 &= (Y^3 - Z^3)(Z^3 - X^3)(X^3 - Y^3) \times \{-l^2(X^3 + Y^3 + Z^3) - 3l(X^3 + Y^3 + Z^3)XYZ + 9l^2X^2Y^2Z^2 + (1 - 4l^3)(Y^2Z^2 + Z^2X^2 + X^2Y^2)\} \\ \xi\eta\zeta &= (Y^3 - Z^3)(Z^3 - X^3)(X^3 - Y^3) \times \{-l(X^3 + Y^3 + Z^3)XYZ + (1 - l^3)X^2Y^2Z^2 - l(Y^2Z^2 + Z^2X^2 + X^2Y^2)\}\end{aligned}$$

and thence recollecting that

$$HU = l^2(X^3 + Y^3 + Z^3) - (1 + 2l^3)XYZ,$$

we find

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = -(Y^3 - Z^3)(Z^3 - X^3)(X^3 - Y^3)(HU)^2,$$

and the equation of the envelope is

$$-l(\xi^3 + \eta^3 + \zeta^3) + (-1 + 4l^3)\xi\eta\zeta = 0,$$

which is therefore the Pippian. We have thus the theorem:

*The envelope of the polar of the satellite point in respect to the Hessian of the tangent at any point of the Hessian, such polar being in respect of the conic which is the first or conic polar of the point of the Hessian in respect of the cubic, is the Pippian.*

*Article Nos. 37 to 40. Investigations and theorems relating to the first or conic polar of a point of the cubic.*

**37.** The investigations next following depend on the identical equations

$$\begin{aligned}&\{\alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY)\} \\ &\quad \times [-XYZ(x^3 + y^3 + z^3) + (X^3 + Y^3 + Z^3)xyz] \\ &= \{X(x^2 + 2lyz) + Y(y^2 + 2lzx) + Z(z^2 + 2lxy)\} \\ &\quad \times [X(Y^2 - Z^2)(\gamma y - \beta z) + Y(Z^2 - X^2)(\alpha z - \gamma x) + Z(X^2 - Y^2)(\beta x - \alpha y)] \\ &\quad + \{x(X^2 + 2lYZ) + y(Y^2 + 2lZX) + z(Z^2 + 2lXY)\} \\ &\quad \times [-(\alpha YZ + \beta ZX + \gamma XY)(Xx^2 + Yy^2 + Zz^2) + (\alpha X^2 + \beta Y^2 + \gamma Z^2)(Xyz + Yzx + Zxy)],\end{aligned}$$

which is easily verified.

I represent the equation in question by

$$KT = WL + P\Theta;$$

then considering  $(x, y, z)$  as current coordinates, and  $(X, Y, Z)$  and  $(\alpha, \beta, \gamma)$  as the coordinates of two given points  $\Sigma$  and  $\Omega$ , we shall have  $U = 0$  the equation of the cubic,  $W = 0$  the equation of the first or conic polar of  $\Sigma$  with respect to the cubic,  $P = 0$  the equation of the second or line polar of  $\Sigma$  with respect to the cubic. The equation  $T = 0$  is that of a syzygetic cubic passing through the point  $\Sigma$ : the coordinates of the satellite point in respect to this syzygetic cubic of its tangent at  $\Sigma$  are

$$X(Y^3 - Z^3) : Y(Z^3 - X^3) : Z(X^3 - Y^3);$$

and calling the point in question  $\Sigma'$ , then  $L = 0$  is the equation of a line through the points  $\Sigma', \Omega$ . The equation  $\Theta = 0$  is that of a conic, viz. the first or conic polar of  $\Sigma$  with respect to a certain syzygetic cubic

$$-2(\alpha YZ + \beta ZX + \gamma XY)(x^3 + y^3 + z^3) + (\alpha X^2 + \beta Y^2 + \gamma Z^2)xyz = 0,$$

depending on the points  $\Sigma$ ,  $\Omega$ , or, what is the same thing, the conic  $\Theta = 0$  is a properly selected conic passing through the points of intersection of the first or conic polars of  $\Sigma$  with respect to any two syzygetic cubics; and lastly,  $K$  is a constant coefficient. The equation expresses that the points of intersection of

$$(W = 0, \Theta = 0), \quad (W = 0, P = 0), \quad (L = 0, \Theta = 0), \quad (L = 0, P = 0),$$

lie in the syzygetic cubic  $T = 0$ .

The left-hand side of the equation may be written

$$\begin{aligned} & -XYZ \{ \alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY) \} (x^3 + y^3 + z^3 + 6lxyz) \\ & + xyz \{ \alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY) \} (X^3 + Y^3 + Z^3 + 6lXYZ); \end{aligned}$$

and it may be remarked also that we have

$$-3XYZ \{ \alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY) \}$$

equal identically to

$$\begin{aligned} & [X(Y^2 - Z^2)(\gamma Y - \beta Z) + Y(Z^2 - X^2)(\alpha Z - \gamma X) + Z(X^2 - Y^2)(\beta X - \alpha Y)] \\ & - (\alpha YZ + \beta ZX + \gamma XY)(X^3 + Y^3 + Z^3 + 6lXYZ). \end{aligned}$$

Hence if we assume

$$X^3 + Y^3 + Z^3 + 6lXYZ = 0,$$

the equation will take the form

$$KU = WL + P\Theta,$$

where the constant coefficient  $K$  may be expressed under the two different forms

$$\begin{aligned} K &= -XYZ [\alpha(X^2 + 2lYZ) + \beta(Y^2 + 2lZX) + \gamma(Z^2 + 2lXY)] \\ &= \frac{1}{3} [X(Y^2 - Z^2)(\gamma Y - \beta Z) + Y(Z^2 - X^2)(\alpha Z - \gamma X) + Z(X^2 - Y^2)(\beta X - \alpha Y)], \end{aligned}$$

and  $W$ ,  $L$ ,  $P$ ,  $\Theta$  have the same values as before. In the present case the point  $\Sigma$  is a point of the cubic: the equation  $W = 0$  represents the first or conic polar of the point in question, and the equation  $P = 0$  its second or line polar, which is also the tangent of the cubic. The line  $L = 0$  is a line joining the point  $\Omega$  with the satellite point of the tangent at  $\Sigma$ , or dropping altogether the consideration of the point  $\Omega$ , is an arbitrary line through the satellite point: the first or conic polar of  $\Sigma$  meets the cubic twice in the point  $\Sigma$ , and therefore also meets it in four other points; the conic  $\Theta = 0$  is a conic passing through these four points, and completely determined when the particular position of the line through the satellite point is given. And, as before remarked,  $\Theta = 0$  is a conic passing through the points of intersection of the first or conic polars of  $\Sigma$  with respect to any two syzygetic cubics. We have thus the theorem:

*The first or conic polar of a point of the cubic touches the cubic at this point, and besides meets it in four other points; the four points in question are the points in which the first or conic polar of the given point in respect of the cubic is intersected by the first or conic polar of the same point in respect to any syzygetic cubic whatever.*

**38.** The analytical result may be thus stated: putting

$$\kappa = \alpha YZ + \beta ZX + \gamma XY, \quad \lambda = \alpha X^2 + \beta Y^2 + \gamma Z^2,$$

or, if we please, considering  $\kappa, \lambda$  as arbitrary parameters, then the four points lie in the conic

$$(2\kappa X, 2\kappa Y, 2\kappa Z, -\lambda X, -\lambda Y, -\lambda Z \mid x, y, z)^2 = 0,$$

or, what is the same thing, they are the points of intersection of the two conics

$$Xx^2 + Yy^2 + Zz^2 = 0,$$

$$Xyz + Yzx + Zxy = 0.$$

**39.** Considering the four points as the angles of a quadrangle, it may be shown that the three centres of the quadrangle lie on the cubic. To effect this, assume that the conic

$$(2\kappa X, 2\kappa Y, 2\kappa Z, -\lambda X, -\lambda Y, -\lambda Z \mid x, y, z)^2 = 0$$

represents a pair of lines; these lines will intersect in a point, which is one of the three centres in question. And taking  $x, y, z$  as the coordinates of this point, we have

$$\begin{aligned} x^2 : y^2 : z^2 : yz : zx : xy \\ = 4\kappa^2 YZ - \lambda^2 X^2 : 4\kappa^2 ZX - \lambda^2 Y^2 : 4\kappa^2 XY - \lambda^2 Z^2 \\ : \lambda^2 YZ + 2\kappa\lambda X^2 : \lambda^2 ZX + 2\kappa\lambda Y^2 : \lambda^2 XY + 2\kappa\lambda Z^2; \end{aligned}$$

and we may, if we please, use these equations to find the relation between  $\kappa, \lambda$ . Thus in the identical equation  $x^2 \cdot y^2 - (xy)^2 = 0$ , substituting for  $x^2, xy, y^2$  their values, and throwing out the factor  $Z$ , we find  $(4\kappa^2 - \lambda^2)XYZ - \kappa\lambda^2(X^3 + Y^3 + Z^3) = 0$ , and thence, in virtue of the equation  $X^3 + Y^3 + Z^3 + 6lXYZ = 0$ , we obtain

$$4\kappa^2 + 6l\kappa\lambda^2 - \lambda^2 = 0.$$

But the preceding system gives conversely,

$$\begin{aligned} X^2 : Y^2 : Z^2 : YZ : ZX : XY \\ = 4\kappa^2 yz - \lambda^2 x^2 : 4\kappa^2 zx - \lambda^2 y^2 : 4\kappa^2 xy - \lambda^2 z^2 \\ : \lambda^2 yz + 2\kappa\lambda x^2 : \lambda^2 zx + 2\kappa\lambda y^2 : \lambda^2 xy + 2\kappa\lambda z^2. \end{aligned}$$

Hence from the identical relation  $X^2 \cdot Y^2 - (XY)^2 = 0$ , substituting for  $X^2, XY, Y^2$  their values, and throwing out the factor  $z$ , we find  $(4\kappa^2 - \lambda^2)xyz - \kappa\lambda^2(x^3 + y^3 + z^3) = 0$ , and thence, in virtue of the equation  $4\kappa^2 - \lambda^2 = -6l\kappa\lambda^2$ , we obtain

$$x^3 + y^3 + z^3 + 6lxyz = 0,$$

which shows that the point in question lies on the cubic. We have thus the theorem:

*The first or conic polar of a point of the cubic touches the cubic at the point, and meets it besides in four points, which are the angles of a quadrangle the centres of which lie on the cubic. In other words, the quadrangle is an inscribed quadrangle.*

**40.** To find the equations of the three axes of the quadrangle, that is of the lines through two centres.

We have

$$\begin{aligned} (4\kappa^2 YZ - \lambda^2 X^2)x + (\lambda^2 XY + 2\kappa\lambda Z^2)y + (\lambda^2 ZX + 2\kappa\lambda Y^2)z &= 0, \\ (\lambda^2 XY + 2\kappa\lambda Z^2)x + (4\kappa^2 ZX - \lambda^2 Y^2)y + (\lambda^2 YZ + 2\kappa\lambda X^2)z &= 0, \\ (\lambda^2 ZX + 2\kappa\lambda Y^2)x + (\lambda^2 YZ + 2\kappa\lambda X^2)y + (4\kappa^2 XY - \lambda^2 Z^2)z &= 0; \end{aligned}$$

or arranging these equations in the proper form and eliminating  $\kappa^2, \kappa\lambda, \lambda^2$ , we find

$$\begin{vmatrix} YZx, Z^2y + Y^2z, X(-Xx + Yy + Zz) \\ ZXy, X^2z + Z^2x, Y(-Xx - Yy + Zz) \\ XYz, Y^2x + X^2y, Z(-Xx + Yy - Zz) \end{vmatrix} = 0;$$

or, multiplying out,

$$\begin{aligned} XYZ \{ (Z^2 - Y^2)x^3 + (X^2 - Z^2)y^3 + (Y^2 - X^2)z^3 \} \\ + x^2 y Z Y^2 (-2X^2 + Y^2 + Z^2) + z x^2 Y Z^2 (2X^2 - Y^2 - Z^2) \\ + y^2 z X Z^2 (-2Y^2 + X^2 + Z^2) + x y^2 Z X^2 (2Y^2 - Z^2 - X^2) \\ + z^2 x Y X^2 (-2Z^2 + X^2 + Y^2) + y z^2 X Y^2 (2Z^2 - X^2 - Y^2) = 0. \end{aligned}$$

We may simplify this result by means of the equation  $X^3 + Y^3 + Z^3 + 6lXYZ = 0$ , so as to make the left-hand side divide out by  $XYZ$ : we thus obtain

$$\begin{aligned} & (Z^3 - Y^3)x^3 + (X^3 - Z^3)y^3 + (Y^3 - X^3)z^3 \\ & + (-3X^2Y - 6lY^2Z)x^2y + (-3Y^2Z - 6lZ^2X)y^2z + (-3Z^2X - 6lX^2Y)z^2x \\ & + (3XY^2 + 6lX^2Z)xy^2 + (3YZ^2 + 6lY^2X)yz^2 + (3ZX^2 + 6lZ^2Y)zx^2 = 0; \end{aligned}$$

or in a different form,

$$\begin{aligned} & (y^3 - z^3)X^3 + (z^3 - x^3)Y^3 + (x^3 - y^3)Z^3 \\ & + (-3x^2y - 6lz^2x)X^2Y + (-3y^2z - 6lx^2y)Y^2Z + (-3z^2x - 6ly^2z)Z^2X \\ & + (3xy^2 + 6lyz^2)YZ^2 + (3yz^2 + 6lx^2z)ZX^2 + (3zx^2 + 6lxy^2)XY^2 = 0, \end{aligned}$$

as the equation of the three axes of the quadrangle.

*Article No. 41. Recapitulation of geometrical definitions of the Pippian.*

In conclusion, I will recapitulate the different modes of generation or geometrical definitions of the Pippian, obtained in the course of the present memoir. The curve in question is:

1. The envelope of the line joining a pair of conjugate poles of the cubic (see Nos. 2 and 13).
2. The envelope of each line of the pair forming the first or conic polar with respect to the cubic of a conjugate pole of the cubic (see Nos. 2 and 14).
3. The envelope of a line which is the polar of a conjugate pole of the cubic, with respect to the conic which is the first or conic polar of the other conjugate pole in respect to any syzygetic cubic (see Nos. 2 and 9).
4. The locus of the harmonic with respect to a pair of conjugate poles of the cubic of the third point of intersection with the Hessian of the line joining the two conjugate poles (see Nos. 2 and 17).
5. The envelope of a line such that its lineo-polar envelope with respect to the cubic breaks up into a pair of lines (see No. 24).
6. The envelope of a line which meets three conics, the first or conic polars of any three points in respect to the cubic, in six points in involution (see No. 22).
7. The envelope of the second or line polar with respect to the cubic, of a point the locus of which is a certain curve of the sixth order in quadratic syzygy with the cubic and Hessian, viz. the curve  $-S \cdot U^2 + (HU)^2 = 0$  (see No. 27).
8. The envelope of a line having for its satellite point a point of the Hessian (see No. 35).
9. The envelope of the polar of the satellite point with respect to the Hessian of the tangent at a point of the Hessian, with respect to the first or conic polar of the point of the Hessian in respect to the cubic (see No. 36).

## 147.

### A MEMOIR ON THE SYMMETRIC FUNCTIONS OF THE ROOTS OF AN EQUATION.

*[From the Philosophical Transactions of the Royal Society of London, vol. CXLVII for the year 1857, pp. 489–499. Received December 18, 1856,—Read January 8, 1857.]*

There are contained in a work, which is not, I think, so generally known as it deserves to be, the “Algebra” of Meyer Hirsch [the work referred to is entitled *Sammlung von Beispielen Formeln und Aufgaben aus der Buchstabenrechnung und Algebra*, 8vo. Berlin, 1804 (8 ed. 1853), English translation by Ross, 8vo. London, 1827] some very useful tables of the symmetric functions up to the tenth degree of the roots of an equation of any order. It seems desirable to join to these a set of tables, giving reciprocally the expressions of the powers and products of the coefficients in terms of the symmetric functions of the roots. The present

memoir contains the two sets of tables, viz. the new tables distinguished by the letter (a), and the tables of Meyer Hirsch distinguished by the letter (b); the memoir contains also some remarks as to the mode of calculation of the new tables, and also as to a peculiar symmetry of the numbers in the tables of each set, a symmetry which, so far as I am aware, has not hitherto been observed, and the existence of which appears to constitute an important theorem in the subject. The theorem in question might, I think, be deduced from a very elegant formula of M. Borchardt (referred to in the sequel), which gives the generating function of any symmetric function of the roots, and contains potentially a method for the calculation of the Tables (b), but which, from the example I have given, would not appear to be a very convenient one for actual calculation.

Suppose in general

$$(1, b, c, \dots | 1, x)^n = (1 - \alpha x)(1 - \beta x)(1 - \gamma x) \dots,$$

so that

$$-b = \Sigma \alpha, \quad +c = \Sigma \alpha \beta, \quad -d = \Sigma \alpha \beta \gamma, \quad \&c.,$$

and if in general

$$(pqr \dots) = \Sigma \alpha^p \beta^q \gamma^r \dots,$$

where as usual the summation extends only to the distinct terms, so that e.g.  $(p^2)$  contains only half as many terms as  $(pq)$ , and so in all similar cases, then we have

$$-b = (1), \quad +c = (1^2), \quad -d = (1^3), \quad \&c.;$$

and the two problems which arise are, first to express any combination  $b^p c^q \dots$  in terms of the symmetric functions  $(l^\lambda m^\mu \dots)$ , and secondly, or conversely, to express any symmetric function  $(l^\lambda m^\mu \dots)$  in terms of the combinations  $b^p c^q \dots$ .

It will conduce materially to brevity if  $1^p 2^q \dots$  be termed the partition belonging to the combination  $b^p c^q \dots$ ; and in like manner if  $l^\lambda m^\mu \dots$  be termed the partition belonging to the symmetric function  $(l^\lambda m^\mu \dots)$ , and if the sum of the component numbers of the partition is termed the weight.

Consider now a line of combinations corresponding to a given weight, e.g. the weight 4, this will be

$$\begin{array}{cccccc} e & bd & c^2 & b^2 c & b^4 & \text{(line)} \\ 4 & 13 & 2^2 & 1^2 2 & 1^4 & \end{array}$$

where I have written under each combination the partition which belongs to it, and in like manner a column of symmetric functions of the same weight, viz.

$$\begin{array}{c} (4) \quad \text{(column)} \\ (31) \\ (2^2) \\ (21^2) \\ (1^4), \end{array}$$

where, as the partitions are obtained by simply omitting the  $()$ , I have not separately written down the partitions.

It is at once obvious that the different combinations of the line will be made up of numerical multiples of the symmetric functions of the column; and conversely, that the symmetric functions of the column will be made up of numerical multiples of the combinations of the line; but this requires a further examination. There are certain restrictions as to the symmetric functions which enter into the expression of the combination, and conversely, as to the combinations which enter into the expression of the symmetric function. The nature of the first restriction is most clearly seen by the following Table:

| No. of Parts. | Greatest Part. | Comb. with Combinations. | Their Partitions. | Contain Multiples of the Symmetric Functions. | Greatest Part does not exceed. | No. of Parts not less than. |
|---------------|----------------|--------------------------|-------------------|-----------------------------------------------|--------------------------------|-----------------------------|
| 1             | 4              | $e$                      | 4                 | $(1^4)$                                       | 1                              | 4                           |
| 2             | 3              | $bd$                     | 13                | $(1^4), (21^2)$                               | 2                              | 3                           |
| 2             | 2              | $c^2$                    | $2^2$             | $(1^4), (21^2), (2^2)$                        | 2                              | 2                           |
| 3             | 2              | $b^2 c$                  | $1^2 2$           | $(1^4), (21^2), (2^2), (31)$                  | 3                              | 2                           |
| 4             | 1              | $b^4$                    | $1^4$             | $(1^4), (21^2), (2^2), (31), (4)$             | 4                              | 1                           |

Thus, for instance, the combination  $bd$  (the partition whereof is 13) contains multiples of the two symmetric functions  $(1^4)$ ,  $(21^2)$  only. The number of parts in the partition 13 is 2, and the greatest part is 3. And in the partitions  $(1^4)$ ,  $(21^2)$  the greatest part is 2, and the number of parts is not less than 3. The reason is obvious: each term of the developed expression of  $bd$  must contain at least as many roots as are contained in each term of  $d$ , that is 3 roots, and since the coefficients are linear functions in respect to each root, the combination  $bd$  cannot contain a power higher than 2 of any root. The reasoning is immediately applied to any other case, and we obtain

**First Restriction.**—A combination  $b^p c^q \dots$  contains only those symmetric functions  $(l^\lambda m^\mu \dots)$ , for which the greatest part does not exceed the number of parts in the partition  $1^p 2^q \dots$ , and the number of parts is not less than the greatest part in the same partition.

Consider a partition such as  $1^2 2$ , then replacing each number by a line of units thus,

$$\begin{array}{c} 1 \\ 1 \\ 11 \end{array}$$

and summing the columns, we obtain a new partition 31, which may be called the conjugate<sup>1</sup> of  $1^2 2$ . It is easy to see that the expression for the combination  $b^2 c$  (for which the partition is  $1^2 2$ ) contains with the coefficient unity, the symmetric function (31), the partition whereof is the conjugate of  $1^2 2$ . In fact  $b^2 c = (-\Sigma \alpha)^2 (\Sigma \alpha \beta)$ , which obviously contains the term  $1 \alpha^3 \beta$ , and therefore the symmetric function with its coefficient +1 (31); and the reasoning is general, or

**Theorem.** A combination  $b^p c^q \dots$  contains the symmetric function (partition conjugate to  $1^p 2^q \dots$ ) with the coefficient unity, and sign + or – according as the weight is even or odd.

Imagine the partitions arranged as in the preceding column, viz. first the partition into one part, then the partitions into two parts, then the partitions into three parts, and so on; the partitions into the same number of parts being arranged according to the magnitude of the greatest part (the greatest magnitude first), and in case of equality according to the magnitudes of the next greatest part, and so on (for other examples, see the outside column of any one of the Tables). The order being thus completely defined, we may speak of a partition as being prior or posterior to another. We are now able to state a second restriction as follows.

**Second Restriction.**—The combination  $b^p c^q \dots$  contains only those symmetric functions which are of the form (partition not prior to the conjugate of  $1^p 2^q \dots$ ).

The terms excluded by the two restrictions are many of them the same, and it might at first sight appear as if the two restrictions were identical; but this is not so: for instance, for the combination  $bd^2$ , see Table VII (a), the term  $(41^2)$  is excluded by the first restriction, but not by the second; and on the other hand, the term  $(3^2 1)$ , which is not excluded by the first restriction, is excluded by the second restriction, as containing a partition  $3^2 1$  prior in order to  $32^2$ , which is the partition conjugate to  $13^2$ , the partition of  $bd^2$ . It is easy to see why  $bd^2$  does not contain the symmetric function  $(3^2 1)$ ; in fact, a term of  $(3^2 1)$  is  $(\alpha^3 \beta^3 \gamma)$ , which is obviously not a term of  $bd^2 = (-\Sigma \alpha)(\Sigma \alpha \beta \gamma)^2$ ; but I have not investigated the general proof.

I proceed to explain the construction of the Tables (a). The outside column contains the symmetric functions arranged in the order before explained; the outside or top line contains the combinations of the same weight arranged as follows, viz. the partitions taken in order from right to left are respectively conjugate to the partitions in the outside column, taken in order from top to bottom; in other words, each square of the sinister diagonal corresponds to two partitions which are conjugate to each other. It is to be noticed that the combinations taken in order, from left to right, are not in the order in which they would be obtained by Arbogast's Method of Derivations from an operand  $a^n$ ,  $a$  being ultimately replaced by unity. The squares above the sinister diagonal are empty (i.e. the coefficients are zero), the greater part of them in virtue of both restrictions, and the remainder in virtue of the second restriction; the empty squares below the sinister diagonal are empty in virtue of the second restriction; but the property was not assumed in the calculation.

<sup>1</sup>The notion of Conjugate Partitions is, I believe, due to Professor Sylvester or Mr Ferrers. [It was due to Mr now Dr Ferrers.]

The greater part of the numbers in the Tables (a) were calculated, those of each table from the numbers in the next preceding table by the following method, depending on the derivation of the expression for  $b^{p+1}c^q \dots$  from the expression for  $b^pc^q \dots$ . Suppose, for example, the column  $cd$  of Table V (a) is known, and we wish to calculate the column  $bcd$  of Table VI (a). The process is as follows:

Given

|         |          |          |        |       |
|---------|----------|----------|--------|-------|
| $2^21$  | $21^3$   | $1^5$    |        |       |
| 1       | 3        | 10       |        |       |
| 321     | $2^3$    | $2^21^2$ | $21^4$ | $1^6$ |
| 1, 3, 2 | 3, 6, 12 |          | 10, 60 |       |
| 1       | 3        | 3, 8, 22 | 60     |       |

where the numbers in the last line are the numbers in the column  $bcd$  of Table VI (a). The partition  $2^21$ , considered as containing a part zero, gives, when the parts are successively increased by 1, the partitions  $321$ ,  $2^3$ ,  $2^21^2$ , in which the indices of the increased part (i.e. the original part plus unity) are 1, 3, 2; these numbers are taken as multipliers of the coefficient 1 of the partition  $2^21$ , and we thus have the new coefficients 1, 3, 2 of the partitions  $321$ ,  $2^3$ ,  $2^21^2$ . In like manner the coefficient 3 of the partition  $21^3$  gives the new coefficients 3, 6, 12 of the partitions  $31^2$ ,  $2^21^2$ ,  $21^4$ , and the coefficient 10 of the partition  $1^5$  gives the new coefficients 10, 60 of the partitions  $21^4$  and  $1^6$ , and finally, the last line is obtained by addition. The process in fact amounts to the multiplication separately of each term of

$$cd = 1(2^21) + 3(21^3) + 10(1^5)$$

by  $b = (1)$ . It would perhaps have been proper to employ an analogous rule for the calculation of the combinations  $cdef \dots$  not containing  $b$ , but instead of doing so I availed myself of the existing Tables (b). But the comparison of the last line of each Table (a) (which as corresponding to a combination  $b^k$  was always calculated independently of the Tables (b)) with such last line as calculated from the corresponding Table (b), seems to afford a complete verification of both the Tables; and my process has in fact enabled me to detect several numerical errors in the Tables (b), as given in the English translation of the work above referred to. It is not desirable, as regards facility of calculation and independently of the want of verification, to calculate either set of Tables wholly from the other; the rules for the independent calculation of the Tables (b) are fully and clearly explained in the work referred to, and I have nothing to add upon this subject.

The relation of symmetry, alluded to in the introductory paragraph of the present memoir, exists in each Table of either set, and is as follows: viz. the number in the Table corresponding to any two partitions in the outside column and the outside line respectively, is equal to the number corresponding to the same two partitions in the outside line and the outside column respectively. Or, calling the two partitions  $P, Q$ , and writing for shortness, combination  $(P)$  for the combination represented by the partition  $P$ , and for greater clearness, symmetric function  $(P)$  (instead of merely  $(P)$ ) to denote the symmetric function represented by the partition  $P$ , we have the following two theorems, viz.

**Theorem.** The coefficient in combination  $(P)$  of symmetric function  $(Q)$  is equal to the coefficient in combination  $(Q)$  of symmetric function  $(P)$ ;

and conversely,

**Theorem.** The coefficient in symmetric function  $(P)$  of combination  $(Q)$  is equal to the coefficient in symmetric function  $(Q)$  of combination  $(P)$ .

M. Borchardt's formula, before referred to, is given in the 'Monatsbericht' of the Berlin Academy (March 5, 1855)<sup>2</sup>, and may be thus stated; viz. considering the case of  $n$  roots, write

$$x^n \cdot \frac{1}{k}(1, b, c, \dots | 1, x)^n = (1 - \alpha x)(1 - \beta x) \dots (1 - \kappa x) = fx,$$

then

$$\Sigma \left( \frac{1}{1 - \alpha x} \cdot \frac{1}{1 - \beta y} \cdots \frac{1}{1 - \kappa u} \right) = \frac{1}{k^n} \cdot \frac{fx \, fy \dots fu}{\Pi(x, y, \dots u)} \cdot \frac{d}{dx} \frac{d}{dy} \cdots \frac{d}{du} \frac{\Pi(x, y, \dots u)}{fx \, fy \dots fu},$$

<sup>2</sup>And in Crelle, t. LIII, p. 195.—Note added 4th Dec. 1857, A. C.



where  $\Pi(x, y, \dots u)$  denotes the product of the differences of the quantities  $x, y, \dots u$ , and on the left-hand side the summation extends to all the different permutations of  $\alpha, \beta, \dots \kappa$ , or what is the same thing, of  $x, y, \dots u$ .

Suppose for a moment that there are only two roots, so that

$$(1, b, c \mid 1, x)^2 = (1 - \alpha x)(1 - \beta x),$$

then the left-hand side is

$$\frac{1}{(1 - \alpha x)(1 - \beta y)} + \frac{1}{(1 - \alpha y)(1 - \beta x)},$$

which is equal to

$$2 + (\alpha + \beta)(x + y) + (\alpha^2 + \beta^2)(x^2 + y^2) + 2\alpha\beta xy \\ + (\alpha^3 + \beta^3)(x^3 + y^3) + (\alpha^2\beta + \alpha\beta^2)(x^2y + xy^2) + \&c.,$$

and the right-hand side is

$$\frac{1}{c} \cdot \frac{fx \, fy}{x - y} \cdot \frac{d}{dx} \frac{d}{dy} \frac{x - y}{fx \, fy},$$

which is equal to

$$\frac{1}{c} \cdot \frac{1}{x - y} \left\{ \frac{f'x \, fy - f'y \, fx}{(fx)(fy)} + (x - y) \frac{f'x \, f'y}{fx \, fy} \right\},$$

and therefore to

$$\frac{1}{c} \cdot \frac{1}{fx \, fy} \left\{ \frac{f'x \, fy - f'y \, fx}{x - y} + f'x \, f'y \right\};$$

or substituting for  $fx, fy$  their values,

$$\frac{f'x \, fy - f'y \, fx}{x - y}$$

becomes equal to

$$2c + bc(x + y) + 2c^2xy,$$

and  $fx \, fy$  is equal to

$$b^2 + 2bc(x + y) + 4c^2xy.$$

The right-hand side is therefore equal to

$$\frac{2 + b(x + y) + 2cxy}{(1 + bx + cx^2)(1 + by + cy^2)},$$

and comparing with the value of the left-hand side, we see that this expression may be considered as the generating function of the symmetric functions of  $(\alpha, \beta)$ , viz. the expression in question is developable in a series of the symmetric functions of  $(x, y)$ , the coefficients being of course functions of  $b$  and  $c$ , and these coefficients are (to given numerical factors *près*) the symmetric functions of the roots  $(\alpha, \beta)$ .

And in general it is easy to see that the left-hand side of M. Borchardt's formula is equal to

$$[0] + [1](1)(1)' + [2](2)(2)' + [1^2](1^2)(1^2)' + \&c.,$$

where  $(1), (2), (1^2), \&c.$  are the symmetric functions of the roots  $(\alpha, \beta, \dots \kappa)$ ,  $(1)', (2)', (1^2)', \&c.$  are the corresponding symmetric functions of  $(x, y, \dots u)$ , and  $[0], [1], [2], [1^2], \&c.$  are mere numerical coefficients; viz.  $[0]$  is equal to  $1 \cdot 2 \cdot 3 \dots n$ , and  $[1], [2], [1^2], \&c.$  are such that the product of one of these factors into the number of terms in the corresponding symmetric function  $(1), (2), (1^2), \&c.$  may be equal to  $1 \cdot 2 \cdot 3 \dots n$ . The right-hand side of M. Borchardt's formula is therefore, as in the particular case, the generating function of the symmetric functions of the roots  $(\alpha, \beta, \dots \kappa)$ , and if a convenient expression of such right-hand side could be obtained, we might by means of it express all the symmetric functions of the roots in terms of the coefficients.

The outside line of letters contains the combinations (powers and products) of the coefficients, the coefficients being all with the positive sign, and the coefficient of the highest power being unity; thus in the case of a cubic equation the equation is

$$x^3 + bx^2 + cx + d = (x - \alpha)(x - \beta)(x - \gamma) = 0.$$

The outside line of numbers is obtained from that of letters merely by writing 1, 2, 3 . . . for  $b, c, d \dots$ , and may be considered simply as a different notation for the combinations. The outside column contains the different symmetric functions in the notation above explained, viz. (1) denotes  $\Sigma\alpha$ , (2) denotes  $\Sigma\alpha^2$ ,  $(1^2)$  denotes  $\Sigma\alpha\beta$ , and so on. The Tables (a) are to be read according to the columns; thus Table II (a) means  $b^2 = 1(2) + 2(1^2)$ ,  $c = (1^2)$ . The Tables (b) are to be read according to the lines; thus Table II (b) means  $(2) = -2c + 1b^2$ ,  $(1^2) = +1c$ .

I (a).    I (b).    (weight 1)

|     |        |     |        |
|-----|--------|-----|--------|
|     | 1<br>b |     | 1<br>b |
| (1) | -1     | (1) | -1     |

II (a).    II (b).    (weight 2)

|                   |        |                                  |                   |        |                                  |
|-------------------|--------|----------------------------------|-------------------|--------|----------------------------------|
|                   | 2<br>c | 1 <sup>2</sup><br>b <sup>2</sup> |                   | 2<br>c | 1 <sup>2</sup><br>b <sup>2</sup> |
| (2)               | +1     | +2                               | (2)               | -2     | +1                               |
| (1 <sup>2</sup> ) |        | +1                               | (1 <sup>2</sup> ) | +1     |                                  |

III (a).    III (b).    (weight 3)

|                   |        |          |                                  |                   |        |          |                                  |
|-------------------|--------|----------|----------------------------------|-------------------|--------|----------|----------------------------------|
|                   | 3<br>d | 12<br>bc | 1 <sup>3</sup><br>b <sup>3</sup> |                   | 3<br>d | 12<br>bc | 1 <sup>3</sup><br>b <sup>3</sup> |
| (3)               | -1     | -3       | -1                               | (3)               | -3     | +3       | -1                               |
| (21)              |        | -1       | -3                               | (21)              | +3     | -1       |                                  |
| (1 <sup>3</sup> ) |        |          | -1                               | (1 <sup>3</sup> ) | -1     |          |                                  |

IV (a).    (weight 4)

|                    |        |          |                                  |                                      |                                  |
|--------------------|--------|----------|----------------------------------|--------------------------------------|----------------------------------|
|                    | 4<br>e | 13<br>bd | 2 <sup>2</sup><br>c <sup>2</sup> | 1 <sup>2</sup> 2<br>b <sup>2</sup> c | 1 <sup>4</sup><br>b <sup>4</sup> |
| (4)                | +1     | +4       | +2                               | +6                                   | +24                              |
| (31)               |        | +1       | +2                               | +5                                   | +12                              |
| (2 <sup>2</sup> )  |        |          | +1                               | +2                                   | +6                               |
| (21 <sup>2</sup> ) |        |          |                                  | +1                                   | +4                               |
| (1 <sup>4</sup> )  |        |          |                                  |                                      | +1                               |

IV (b).    (weight 4)

|                    |        |          |                                  |                                      |                                  |
|--------------------|--------|----------|----------------------------------|--------------------------------------|----------------------------------|
|                    | 4<br>e | 13<br>bd | 2 <sup>2</sup><br>c <sup>2</sup> | 1 <sup>2</sup> 2<br>b <sup>2</sup> c | 1 <sup>4</sup><br>b <sup>4</sup> |
| (4)                | -4     | +4       | +2                               | -4                                   | +1                               |
| (31)               | +4     | -1       | -2                               | +1                                   |                                  |
| (2 <sup>2</sup> )  | +2     |          | +1                               |                                      |                                  |
| (21 <sup>2</sup> ) | -4     | +1       |                                  |                                      |                                  |
| (1 <sup>4</sup> )  | +1     |          |                                  |                                      |                                  |

[The memoir continues with Tables V (a)/(b) (weight 5, columns indexed by the conjugate coefficient-products  $f, be, cd, b^2d, bc^2, b^3c, b^5$ ), VI (a)/(b) (weight 6:  $g, bf, ce, b^2e, d^2, bcd, b^3d, c^3, b^2c^2, b^4c, b^6$ ), VII (a)/(b) (weight 7, occupying journal p. 421), VIII (a)/(b) (weight 8, journal pp. 422–423), IX (a)/(b) (weight 9, journal pp. 424–425, running on through several PDF pages), and X (a)/(b) (weight 10, journal pp. 426–428, the largest tables in the memoir, with column heading running from  $k, bi, ch, b^2h, dg, b^3g, bdf, b^2cf, b^4f, be^2, \dots$ , through  $c^5, b^2c^4, b^4c^3, b^6c^2, b^8c, b^{10}$ , and row-labels the partitions of 10 from (10) through  $(1^{10})$ ).]

The body of each Table (a) is a triangular array of signed integers whose entry in row  $(l^\lambda m^\mu \dots)$  and column  $b^p c^q \dots$  records the coefficient with which the symmetric function  $(l^\lambda m^\mu \dots)$  appears in the expansion of the coefficient-product  $b^p c^q \dots$ ; the Tables (b) invert this relation, giving each symmetric function as a signed integer combination of coefficient-products of equal weight. The squares above the sinister diagonal are empty in virtue of the two restrictions stated above. The symmetry theorem stated by the author asserts that the entry in row  $P$ , column  $Q$  of any Table  $N$  (a) is equal to the entry in row  $Q$ , column  $P$  of the same table, and likewise for the Tables (b).

The full numerical content of Tables IV (a)/(b) through X (a)/(b) occupies PDF pp. 423–434 of the present volume (= journal pp. 407–418 of the 1857 *Philosophical Transactions*). Tables V through X are wide multi-column grids that resist reflow at 11 pt without sacrificing line-by-line readability; the original journal version (*Phil. Trans.* 1857, vol. CXLVII, pp. 489–499, Tables IV (a)–X (b)) remains the authoritative source for the precise numerical values. A faithful machine-readable transcription of these tables forms a useful subsequent project but is not attempted here for reasons of legibility.

[Continuation of Paper 147 — A Memoir on the Symmetric Functions of the Roots of an Equation, from PDF p. 451 = print p. 435.]

[Print pp. 435–439 (PDF pp. 451–455) consist almost entirely of the large numerical Tables of the symmetric functions for partitions of weights  $\leq 10$ . These are the continuation of Table V (10) introduced in earlier pages of the paper, and Table X (6) (which “Runs on to p. 439”). Each table is a wide cross-tabulation: the rows are indexed by partitions  $(\alpha_1\alpha_2\dots)$  (such as (10), (91), (82), (73), (64),  $(5^2)$ ,  $(81^2)$ , (721), (631), (541),  $(62^2)$ , (532),  $(4^22)$ ,  $(43^2)$ ,  $(71^3)$ ,  $(621^2)$ ,  $(531^2)$ ,  $(4^21^2)$ ,  $(52^21)$ , ...) while the columns are indexed by monomials in the coefficients  $a, b, c, d, e, f, \dots$  such as  $b^4d^2$ ,  $b^3c^3$ ,  $bcd^3$ ,  $\bar{a}c^2d^2$  etc. (with the convention  $a = 1$  in the body of the table). The entries are signed integers giving the coefficient with which each monomial appears in the symmetric function corresponding to a given row partition. The bottom line of each table gives the column-sum (the numerical verification of the row of coefficients). The detailed tables run for several full pages and are not reproduced here in numerical form: their structure follows that of the analogous tables in earlier pages of the paper (see the printed Table I (10), Table V (10), Table X (6), &c.) which were given exhaustively.]

[Figure: A very wide numerical table (print p. 435) headed by row 147 of the column headings  $1^32^4$ ,  $2^24$ ,  $1^32^4$ ,  $12^4$ ,  $1^32^3$ ,  $1^33^2$ ,  $2^23^2$ ,  $13^23$ ,  $12^23$ ,  $1^323$ , &c., showing the coefficients of the symmetric functions of weight 10 in the partition rows (10), (91), (82), ...,  $(1^{10})$ ; entries range from +1 through +1814400, and the last row shows totals such as 151200, 302400, 604800, 113400, 226800, 453600, 907200, 1814400, 3628

[Figure: A second wide numerical table (print p. 436), headed by the partitions of weight 10 taken in a different order; this is the table II of the same group, with the rows similarly indexed by partitions (10), (91), (82), ...,  $(1^{10})$  and the columns by monomials of degree 10 in the coefficients; the entries are signed integers including such large values as +225200 and +604800.]

[Figure: A third wide numerical table (print p. 437), headed X (6). **Runs on to p. 439.** The first column heads are 10, 19, 28,  $1^28$ , 37, 127,  $1^27$ , 46, 136,  $2^26$ ,  $1^226$ ,  $1^26$ ,  $5^2$ , 145, ... and the rows are again indexed by the partitions of weight 6 and below. Entries are small signed integers (typically  $\pm 1$  through  $\pm 60$ ) and a few are marked with “...” indicating zero.]

[Figure: A fourth wide numerical table (print p. 438), a continuation of Table X (6), with similar column heads 235,  $1^235$ ,  $12^25$ ,  $1^225$ ,  $1^45$ ,  $24^2$ ,  $1^24^2$ ,  $3^24$ , 1234,  $1^334$ ,  $2^34$ ,  $1^22^24$ ,  $1^24$ ,  $1^54$ , &c., and the rows the partitions of weight 6.]

[Figure: A fifth wide table (print p. 439), the final part of Table X (6); column heads  $b^4d^2$ ,  $bc^4d$ ,  $b^2c^3d$ , ...,  $b^6c$ , ... and the rows once more indexed by the partitions of weight 6.]

[End of Paper 147. The detailed numerical entries are extensive; they may be consulted in the original scan.]

## 148.

### MEMOIR ON THE RESULTANT OF A SYSTEM OF TWO EQUATIONS.

[From the Philosophical Transactions of the Royal Society of London, vol. CXLVII for the year 1857, pp. 703–715. Received December 18, 1856,—Read January 8, 1857.]

The Resultant of two equations such as

$$(a, b, \dots | x, y)^m = 0,$$

$$(p, q, \dots | x, y)^n = 0,$$

is, it is well known, a function homogeneous in regard to the coefficients of each equation separately, viz. of the degree  $n$  in regard to the coefficients  $(a, b, \dots)$  of the first equation, and of the degree  $m$  in regard to the coefficients  $(p, q, \dots)$  of the second equation; and it is natural to develop the resultant in the form  $kAP + k'A'P' + \&c.$ , where  $A, A', \&c.$  are the combinations (powers and products) of the degree  $n$  in the coefficients  $(a, b, \dots)$ ,  $P, P', \&c.$  are the combinations of the degree  $m$  in the coefficients  $(p, q, \dots)$ , and  $k, k', \&c.$  are mere numerical coefficients. The object of the present memoir is to show how this may be

conveniently effected, either by the method of symmetric functions, or from the known expression of the Resultant in the form of a determinant, and to exhibit the developed expressions for the resultant of two equations, the degrees of which do not exceed 4. With respect to the first method, the formula in its best form, or nearly so, is given in the Algebra of Meyer Hirsch, [for proper title see p. 417], and the application of it is very easy when the necessary tables are calculated: as to this, see my “Memoir on the Symmetric Functions of the Roots of an Equation”<sup>1</sup>. But when the expression for the Resultant of two equations is to be calculated without the assistance of such tables, it is I think by far the most simple process to develop the determinant according to the second of the two methods.

Consider first the method of symmetric functions, and to fix the ideas, let the two equations be

$$(a, b, c, d \mid x, y)^3 = 0,$$

$$(p, q, r \mid x, y)^2 = 0.$$

Then writing

$$(a, b, c, d \mid 1, z)^3 = a(1 - \alpha z)(1 - \beta z)(1 - \gamma z),$$

so that

$$\begin{aligned} -\frac{b}{a} &= \alpha + \beta + \gamma = (1), \\ +\frac{c}{a} &= \alpha\beta + \alpha\gamma + \beta\gamma = (1^2), \\ -\frac{d}{a} &= \alpha\beta\gamma = (1^3), \end{aligned}$$

the Resultant is

$$(p, q, r \mid \alpha, 1)^2 \cdot (p, q, r \mid \beta, 1)^2 \cdot (p, q, r \mid \gamma, 1)^2,$$

which is equal to

$$p^3r^3 + q^2r^2(\alpha + \beta + \gamma) + pr^2(\alpha^2 + \beta^2 + \gamma^2) + pqr(\alpha^2\beta + \alpha^2\gamma + \beta^2\alpha + \beta^2\gamma + \gamma^2\alpha + \gamma^2\beta) + \&c.;$$

or adopting the notation for symmetric functions used in the memoir above referred to, this is

$$\left\{ \begin{array}{l} +q^2r^2 \cdot (1) \\ +pr^2 \cdot (2) \\ +q^2r \cdot (1^2) \\ +pqr \cdot (21) \\ +p^2r \cdot (1^3) \\ +q^2 \cdot (2^2) \\ +pq^2 \cdot (21^2) \\ +p^2q \cdot (2^21) \\ +p^3 \cdot (2^3) \end{array} \right.$$

the law of which is best seen by dividing by  $r^3$  and then writing

$$\frac{q}{r} = [1], \quad \frac{p}{r} = [2],$$

and similarly,

$$\frac{pq}{r^2} = [21], \quad \&c.;$$

the expression would then become

$$+1 \cdot [1](1) + [2](2) + [1^2](1^2) + [21](21) + [1^3](1^3) + [2^2](2^2) + [21^2](21^2) + [2^21](2^21) + [2^3](2^3),$$

where the terms within the  $[ ]$  and  $( )$  are simply all the partitions of the numbers 1, 2, 3, 4, 5, 6, the greatest part being 2, and the greatest number of parts being 3. And in like manner in the general case we have all

<sup>1</sup>Philosophical Transactions, 1857, pp. 489–497, [147].

the partitions of the numbers  $1, 2, 3, \dots, mn$ , the greatest part being  $n$ , and the greatest number of parts being  $m$ .

The symmetric functions  $(1), (2), (1^2)$ , &c. are given in the Tables (6) of the Memoir on Symmetric Functions, but it is necessary to remark that in the Tables the first coefficient  $a$  is put equal to unity, and consequently that there is a power of the coefficient  $a$  to be restored as a factor: this is at once effected by the condition of homogeneity. And it is not by any means necessary to write down (as for clearness of explanation has been done) the preceding expression for the Resultant any portion of it may be taken out directly from one of the Tables (6). For instance, the bracketed portion

$$\begin{aligned} &+ pqr \cdot (21), \\ &+ q^3 \cdot (1^3), \end{aligned}$$

which corresponds to the partitions of the number 3, is to be taken out of the Table III (6) as follows: a portion of this Table (consisting as it happens of consecutive lines and columns, but this is not in general the case) is

$$\begin{array}{c|cc} & = d & bc \\ \hline (21) & +3 & -1 \\ (1^3) & -1 & . \end{array}$$

if in this we omit the sign  $=$ , and in the outside line write for homogeneity  $ad$  instead of  $d$ , and in the outside column, first substituting  $q, p$  for  $1, 2$ , then write for homogeneity  $pqr$  instead of  $pq$ , we have

$$\begin{array}{c|cc} & ad & bc \\ \hline pqr & +3 & -1 \\ q^3 & -1 & . \end{array}$$

viz.  $pqr \times (+3ad - 1bc) + q^3 \cdot (-1ad)$ , for the value of the portion in question; this is equivalent to

$$\begin{array}{c|cc} & ad & bc \\ \hline pqr & +3 & -1 \\ q^3 & -1 & . \end{array} \quad \text{or, as it may be more conveniently written,}$$

in which form it constitutes a part of the expression given in the sequel for the Resultant of the two functions in question; and similarly the remainder of the expression is at once derived from the Tables (6) I. to VI.

As a specimen of a mode of verification, it may be remarked that the Resultant qua invariant ought, when operated upon by the sum of the two operations,

$$a\partial_b + 2b\partial_c + 3c\partial_d \quad \text{and} \quad p\partial_q + 2q\partial_r,$$

to give a result zero. The results of the two operations are originally obtained in the forms in the first and second columns, and the first column, and the second column, with all the signs reversed, are respectively equal to the third column, and consequently the sum of the first and second columns vanishes, as it ought to do.

Next to explain the second method, viz. the calculation of the resultant from the expression in the form of a determinant. Taking the same example as before, the resultant is

$$\begin{vmatrix} a & b & c & d & . & . & . \\ . & a & b & c & d & . & . \\ . & . & a & b & c & d & . \\ p & q & r & . & . & . & . \\ . & p & q & r & . & . & . \\ . & . & p & q & r & . & . \\ . & . & . & p & q & r & . \end{vmatrix}$$

which may be developed in the form

$$\begin{aligned}
 &+ 12 \cdot 345 \\
 &- 13 \cdot 245 \\
 &+ 14 \cdot 235 \\
 &+ 23 \cdot 145 \\
 &- 15 \cdot 234 \\
 &- 24 \cdot 135 \\
 &+ 25 \cdot 134 \\
 &+ 34 \cdot 125 \\
 &- 35 \cdot 124 \\
 &+ 45 \cdot 123
 \end{aligned}$$

where 12, 13, &c. are the terms of

$$\begin{pmatrix} a, & b, & c, & d \\ a, & b, & c, & d \end{pmatrix}$$

and 123, &c. are the terms of

$$\begin{pmatrix} p, & q, & r \\ p, & q, & r \\ p, & q, & r \end{pmatrix},$$

viz. 12 is the determinant formed with the first and second columns of the upper matrix, 123 is the determinant formed with the first, second and third columns of the lower matrix, and in like manner for the analogous symbols. These determinants must be first calculated, and the remainder of the calculation may then be arranged as follows.

*[Figure: A worked-out arithmetic display (print p. 445) giving the array of two-by-two and three-by-three sub-determinants of the upper and lower matrices, together with the rule for combining a “dexter” line of the first square with a “sinister” line of the second to form the entries of a third (product) square.]*

where it is to be observed that the figures in the squares of the third column are obtained from those in the corresponding squares of the first and second columns by the ordinary rule for the multiplication of determinants, taking care to multiply the *dexter* lines (i.e. lines in the descending diagonal direction) of the first square by the *sinister* lines (i.e. lines in the direction /) of the second square in order to obtain the sinister lines of the third square. Thus, for instance, the figures in the square

$$\begin{pmatrix} +3 & -1 \\ -1 & 0 \end{pmatrix}$$

are obtained as follows, viz. the first sinister line (+3, −1) by

$$(-1, +1)(-2, +1) = 2 + 1 = +3,$$

$$(-1, +1)(+1, 0) = -1 + 0 = -1,$$

and the second sinister line (−1, 0) by

$$(0, -1)(-2, +1) = 0 - 1 = -1,$$

$$(0, -1)(+1, 0) = +0 - 0 = 0.$$

I have calculated the determinants required for the calculation, by the preceding process, of the Resultant of two quartic equations, and have indeed used them for the verification of the expression as found by the method of symmetric functions; as the determinants in question are useful for other purposes, I think the values are worth preserving.

Table of the Determinants of the Matrices,

$$\begin{pmatrix} a, & b, & c, & d, & e \\ a, & b, & c, & d, & e \\ a, & b, & c, & d, & e \\ a, & b, & c, & d, & e \end{pmatrix}$$

and

$$\begin{pmatrix} p, & q, & r, & s, & t \\ p, & q, & r, & s, & t \\ p, & q, & r, & s, & t \\ p, & q, & r, & s, & t \end{pmatrix}$$

arranged in the form adapted for the calculation of the Resultant of the two quartic equations  $(a, b, c, d, e \mid x, y)^4 = 0$ , and  $(p, q, r, s, t \mid x, y)^4 = 0$ , viz.

[Figure: A large foldout-style arrangement (print pp. 446–448) of the sub-determinants of the two coefficient matrices, presented as a diamond-cross pattern of small square sub-blocks of two-by-two, three-by-three and four-by-four determinant values. The arrangement is the standard quartic-quartic resultant tableau on which the subsequent computations are performed; column-sums verify each block. The full set of values, with their signs, is preserved in the original scan.]

The Tables of the Resultants of two equations which I have calculated are as follows.

[Figure: A row of three diamond-cross tableaus (print p. 450) labelled **Table (2, 2)**, **Table (3, 2)**, **Table (4, 2)**, giving the Resultant of  $(a, b, c \mid x, y)^2$  with  $(p, q, r \mid x, y)^2$ ; of  $(a, b, c, d \mid x, y)^3$  with  $(p, q, r \mid x, y)^2$ ; and of  $(a, b, c, d, e \mid x, y)^4$  with  $(p, q, r \mid x, y)^2$  respectively.]

[Figure: A row of two further diamond-cross tableaus (print p. 451) labelled **Table (3, 3)** and **Table (4, 3)**, giving the Resultant of  $(a, b, c, d \mid x, y)^3$  with  $(p, q, r, s \mid x, y)^3$ , and of  $(a, b, c, d, e \mid x, y)^4$  with  $(p, q, r, s \mid x, y)^3$  respectively.]

[Figure: A larger diamond-cross tableau (print p. 452) labelled **Table (4, 4)**, giving the Resultant of  $(a, b, c, d, e \mid x, y)^4$  with  $(p, q, r, s, t \mid x, y)^4$ .]

[End of paper 148.]

## 149.

### ON THE SYMMETRIC FUNCTIONS OF THE ROOTS OF CERTAIN SYSTEMS OF TWO EQUATIONS.

[From the *Philosophical Transactions of the Royal Society of London*, vol. CXLVII for the year 1857, pp. 717–726. Received December 18, 1856,—Read January 8, 1857.]

Suppose in general that  $\phi = 0$ ,  $\psi = 0$ , &c. denote a system of  $(n - 1)$  equations between the  $n$  variables  $(x, y, z, \dots)$ , where the functions  $\phi, \psi$ , &c. are quantics (i.e. rational and integral homogeneous functions) of the variables. Any values  $(x_1, y_1, z_1, \dots)$  satisfying the equations, are said to constitute a set of roots of the system; the roots of the same set are, it is clear, only determinate to a common factor *près*, i.e. only the ratios *inter se* and not the absolute magnitudes of the roots of a set are determinate. The number of sets, or the degree of the system, is equal to the product of the degrees of the component equations. Imagine a function of the roots which remains unaltered when any two sets  $(x_1, y_1, z_1, \dots)$  and  $(x_2, y_2, z_2, \dots)$  are interchanged (that is, when  $x_1$  and  $x_2$ ,  $y_1$  and  $y_2$ , &c. are simultaneously interchanged), and which is besides homogeneous of the same degree as regards each entire set of roots, although not of necessity homogeneous as regards the different roots of the same set; thus, for example, if the sets are  $(x_1, y_1), (x_2, y_2)$ , then the functions  $x_1x_2, x_1y_2 + x_2y_1, y_1y_2$  are each of them of the form in question; but the first and third of these functions, although homogeneous of the first degree in regard to each entire set, are not homogeneous as regards the two variables of each set. A function of the above-mentioned form may, for shortness, be termed a symmetric function of the roots; such function (disregarding an arbitrary factor depending on the common factors which enter implicitly into the different sets of roots) will be a rational and integral function of the coefficients of the equations, i.e. any symmetric function of the roots may be considered as a rational and integral function of the coefficients. The general process for the investigation of such expression for a symmetric function of the roots is indicated in Professor Schläfli's Memoir, "Ueber die Resultante eines Systemes mehrerer algebraischer Gleichungen," *Vienna Transactions*, t. IV. (1852). The process is as follows:—Suppose that we know the resultant of a system of equations, one or more of them being linear; then if  $\phi = 0$  be the linear equation or one of the linear equations of the system, the resultant will be of the form  $\phi_1\phi_2\dots$ , where  $\phi_1, \phi_2$ , &c. are what the function  $\phi$  becomes upon substituting therein the different sets  $(x_1, y_1, z_1, \dots), (x_2, y_2, z_2, \dots)$  of the remaining  $(n - 1)$  equations



$\psi = 0$ ,  $\chi = 0$ , &c.; comparing such expression with the given value of the resultant, we have expressed in terms of the coefficients of the functions  $\psi, \chi$ , &c., certain symmetric functions which may be called the *fundamental* symmetric functions of the roots of the system  $\psi = 0$ ,  $\chi = 0$ , &c.; these are in fact the symmetric functions of the first degree in respect to each set of roots. By the aid of these fundamental symmetric functions, the other symmetric functions of the roots of the system  $\psi = 0$ ,  $\chi = 0$ , &c. may be expressed in terms of the coefficients, and then combining with these equations a non-linear equation  $\Phi = 0$ , the resultant of the system  $\Phi = 0$ ,  $\psi = 0$ ,  $\chi = 0$ , &c. will be what the function  $\Phi_1\Phi_2\dots$  becomes, upon substituting therein for the different symmetric functions of the roots of the system  $\psi = 0$ ,  $\chi = 0$ , &c. the expressions for these functions in terms of the coefficients. We thus pass from the resultant of a system  $\phi = 0$ ,  $\psi = 0$ ,  $\chi = 0$ , &c., to that of a system  $\Phi = 0$ ,  $\psi = 0$ ,  $\chi = 0$ , &c., in which the linear function  $\phi$  is replaced by the non-linear function  $\Phi$ . By what has preceded, the symmetric functions of the roots of a system of  $(n - 1)$  equations depend on the resultant of the system obtained by combining the  $(n - 1)$  equations with an arbitrary linear equation; and moreover, the resultant of any system of  $n$  equations depends ultimately upon the resultant of a system of the same number of equations, all except one being linear; but in this case the linear equations determine the ratios of the variables or (disregarding a common factor) the values of the variables, and by substituting these values in the remaining equation we have the resultant of the system. The process leads, therefore, to the expressions for the symmetric functions of the roots of any system of  $(n - 1)$  equations, and also to the expression for the resultant of any system of  $n$  equations. Professor Schläfli discusses in the general case the problem of showing how the expressions for the fundamental symmetric functions lead to those of the other symmetric functions, but it is not necessary to speak further of this portion of his investigations. The object of the present Memoir is to apply the process to two particular cases, viz. I propose to obtain thereby the expressions for the simplest symmetric functions (after the fundamental ones) of the following systems of two ternary equations; that is, first, a linear equation and a quadric equation; and secondly, a linear equation and a cubic equation.

First, consider the two equations

$$\begin{aligned} (\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h} \mid x, y, z)^2 &= 0, \\ (\alpha, \beta, \gamma \mid x, y, z) &= 0, \end{aligned}$$

and join to these the arbitrary linear equation

$$(\xi, \eta, \zeta \mid x, y, z) = 0.$$

then the two linear equations give  $x : y : z$  in terms of  $(\alpha, \beta, \gamma)$  and  $(\xi, \eta, \zeta)$ , and substituting in the quadratic equation, we have for the resultant of the three equations,

$$\square^{-1} \cdot (\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h} \mid \beta\zeta - \gamma\eta, \gamma\xi - \alpha\zeta, \alpha\eta - \beta\xi)^2 = 0,$$

which may be represented by

$$\square^{-1} \cdot (\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h} \mid \xi, \eta, \zeta)^2 = 0,$$

where the coefficients are given by means of the Table.

[Table (print p. 456): a six-by-six array giving each of  $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h}$  as a linear combination of  $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{f}, \mathbf{g}, \mathbf{h}$  with coefficients quadratic in  $\alpha, \beta, \gamma$ .]

|                | $\mathbf{a}$   | $\mathbf{b}$    | $\mathbf{c}$   | $\mathbf{f}$    | $\mathbf{g}$     | $\mathbf{h}$    |
|----------------|----------------|-----------------|----------------|-----------------|------------------|-----------------|
| $\mathbf{a} =$ | .              | $+\gamma^2$     | $+\beta^2$     | $-2\beta\gamma$ | .                | .               |
| $\mathbf{b} =$ | $+\gamma^2$    | .               | $+\alpha^2$    | .               | $-2\gamma\alpha$ | .               |
| $\mathbf{c} =$ | $+\beta^2$     | $+\alpha^2$     | .              | .               | .                | $-2\alpha\beta$ |
| $\mathbf{f} =$ | $-\beta\gamma$ | $-\gamma\alpha$ | .              | $-\alpha^2$     | $+\alpha\beta$   | $+\gamma\alpha$ |
|                |                |                 |                | $+\alpha\beta$  |                  | $+\beta\gamma$  |
| $\mathbf{g} =$ | .              | .               | $-\alpha\beta$ | $+\gamma\alpha$ | $-\beta^2$       | .               |
|                |                |                 |                |                 | $+\gamma\beta$   |                 |
| $\mathbf{h} =$ | .              | .               | .              | .               | $+\beta\gamma$   | .               |

viz.  $\mathbf{a} = \mathbf{b}\gamma^2 + \mathbf{c}\beta^2 - 2\mathbf{f}\beta\gamma$ , &c.

But if the roots of the given system are

$$(x_1, y_1, z_1), (x_2, y_2, z_2),$$

then the resultant of the three equations will be

$$(\xi, \eta, \zeta \mid x_1, y_1, z_1)(\xi, \eta, \zeta \mid x_2, y_2, z_2),$$

and comparing the two expressions, we have

$$\begin{aligned} a &= x_1 x_2, \\ b &= y_1 y_2, \\ c &= z_1 z_2, \\ 2f &= y_1 z_2 + y_2 z_1, \\ 2g &= z_1 x_2 + z_2 x_1, \\ 2h &= x_1 y_2 + x_2 y_1, \end{aligned}$$

which are the expressions for the six fundamental symmetric functions, or symmetric functions of the first degree in each set, of the roots of the given system.

By forming the powers and products of the second order  $a^2, ab$ , &c., we obtain linear relations between the symmetric functions of the second degree in respect to each set of roots. The number of equations is precisely equal to that of the symmetric functions of the form in question, and the solution of the linear equations gives

$$\begin{aligned} a^2 &= x_1^2 x_2^2, \\ ab &= x_1 y_1 x_2 y_2, \\ 4f^2 - 2bc &= y_1^2 z_2^2 + y_2^2 z_1^2, \\ 4g^2 - 2ca &= z_1^2 x_2^2 + z_2^2 x_1^2, \\ 4h^2 - 2ab &= x_1^2 y_2^2 + x_2^2 y_1^2, \\ 2af &= x_1 y_1 z_1 x_2 + x_2 y_2 z_2 x_1, \\ 2bg &= y_1 z_1 x_1 y_2 + y_2 z_2 x_2 y_1, \\ 2ch &= z_1 x_1 y_1 z_2 + z_2 x_2 y_2 z_1, \\ 4gh - 2af &= x_1^2 y_2 z_2 + x_2^2 y_1 z_1, \\ 4hf - 2bg &= y_1^2 z_2 x_2 + y_2^2 z_1 x_1, \\ 4fg - 2ch &= z_1^2 x_2 y_2 + z_2^2 x_1 y_1, \\ 2bf &= y_1^2 y_2 z_2 + y_2^2 y_1 z_1, \\ 2cg &= z_1^2 z_2 x_2 + z_2^2 z_1 x_1, \\ 2ah &= x_1^2 x_2 y_2 + x_2^2 x_1 y_1, \\ 2cf &= z_1^2 y_2 z_2 + z_2^2 y_1 z_1, \\ 2bh &= y_1^2 x_2 y_2 + y_2^2 x_1 y_1. \end{aligned}$$

Proceeding next to the powers and products of the third order  $a^3, a^2b$ , &c., the total number of linear relations between the symmetric functions of the third degree in respect to each set of roots exceeds by unity the number of the symmetric functions of the form in question; in fact the expressions for  $abc, af^2, bg^2, ch^2, fgh$ ,

contain, not five, but only four symmetric functions of the roots; for we have

$$\begin{aligned}
 abc &= x_1 y_1 z_1 \cdot x_2 y_2 z_2, \\
 4af^2 &= \{x_1^2 y_2^2 z_2^2 + x_2^2 y_1^2 z_1^2\} + 2x_1 y_1 z_1 x_2 y_2 z_2, \\
 4bg^2 &= \{y_1^2 z_2^2 x_2^2 + y_2^2 z_1^2 x_1^2\} + 2x_1 y_1 z_1 x_2 y_2 z_2, \\
 4ch^2 &= \{z_1^2 x_2^2 y_2^2 + z_2^2 x_1^2 y_1^2\} + 2x_1 y_1 z_1 x_2 y_2 z_2, \\
 8fgh &= \{x_1^2 y_2^2 x_2 z_2 + x_2^2 y_1^2 x_1 z_1\} \\
 &\quad + \{y_1^2 z_2^2 y_2 x_2 + y_2^2 z_1^2 y_1 x_1\} \\
 &\quad + \{z_1^2 x_2^2 z_2 y_2 + z_2^2 x_1^2 z_1 y_1\} \\
 &\quad + 2x_1 y_1 z_1 x_2 y_2 z_2,
 \end{aligned}$$

and consequently the quantities a, b, c, f, g, h are not independent, but are connected by the equation

$$abc - af^2 - bg^2 - ch^2 + 2fgh = 0,$$

an equation, which is in fact verified by the foregoing values of a, &c. in terms of the coefficients of the given system.

The expressions for the symmetric functions of the third degree considered as functions of a, b, c, f, g, h, are consequently not absolutely determinate, but they may be modified by the addition of the term  $\lambda(abc - af^2 - bg^2 - ch^2 + 2fgh)$ , where  $\lambda$  is an indeterminate numerical coefficient.

The simplest expressions are those obtained by disregarding the preceding equation for fgh, and the entire

system then becomes

$$\begin{aligned}
 a^3 &= x_1^3 x_2^3, \\
 b^3 &= y_1^3 y_2^3, \\
 c^3 &= z_1^3 z_2^3, \\
 a^2 b &= x_1^2 y_1 x_2^2 y_2, \\
 bc^2 &= y_1 z_1^2 y_2 z_2^2, \\
 ab^2 &= x_1 y_1^2 x_2 y_2^2, \\
 abc &= x_1 y_1 z_1 x_2 y_2 z_2, \\
 2a^2 f &= x_1^2 y_2 z_2 x_2 + x_2^2 y_1 z_1 x_1, \\
 2b^2 g &= y_1^2 z_2 x_2 y_2 + y_2^2 z_1 x_1 y_1, \\
 2c^2 h &= z_1^2 x_2 y_2 z_2 + z_2^2 x_1 y_1 z_1, \\
 2a^2 g &= x_1^2 z_2 x_2^2 + x_2^2 z_1 x_1^2, \\
 2b^2 h &= y_1^2 x_2 y_2^2 + y_2^2 x_1 y_1^2, \\
 2c^2 f &= z_1^2 y_2 z_2^2 + z_2^2 y_1 z_1^2, \\
 2a^2 h &= x_1^2 x_2^2 y_2 + x_2^2 x_1^2 y_1, \\
 2b^2 f &= y_1^2 y_2^2 z_2 + y_2^2 y_1^2 z_1, \\
 2c^2 g &= z_1^2 z_2^2 x_2 + z_2^2 z_1^2 x_1, \\
 2bcf &= y_1 z_1 y_2^2 z_2^2 + y_2 z_2 y_1^2 z_1^2, \\
 2cag &= z_1 x_1 z_2^2 x_2^2 + z_2 x_2 z_1^2 x_1^2, \\
 2abh &= x_1 y_1 x_2^2 y_2^2 + x_2 y_2 x_1^2 y_1^2, \\
 2bch &= y_1 z_1 x_2 y_2 z_2^2 + y_2 z_2 x_1 y_1 z_1^2, \\
 2caf &= z_1 x_1 x_2 y_2 z_2^2 + z_2 x_2 x_1 y_1 z_1^2, \\
 2abg &= x_1 y_1 z_2 x_2 y_2^2 + x_2 y_2 z_1 x_1 y_1^2, \\
 4af^2 - 2abc &= x_1 y_1 z_1 x_2^3 + x_2 y_2 z_2 x_1^3, \\
 4bg^2 - 2abc &= y_1 z_1 x_1 y_2^3 + y_2 z_2 x_2 y_1^3, \\
 4ch^2 - 2abc &= z_1 x_1 y_1 z_2^3 + z_2 x_2 y_2 z_1^3, \\
 4bf^2 - 2b^2 c &= y_1^2 z_2^2 y_2 + y_2^2 z_1^2 y_1, \\
 4cg^2 - 2c^2 a &= z_1^2 x_2^2 z_2 + z_2^2 x_1^2 z_1, \\
 4ah^2 - 2a^2 b &= x_1^2 y_2^2 x_2 + x_2^2 y_1^2 x_1, \\
 4cf^2 - 2bc^2 &= z_1^2 y_2 z_2^2 + z_2^2 y_1 z_1^2, \\
 4ag^2 - 2ca^2 &= x_1^2 z_2 x_2^2 + x_2^2 z_1 x_1^2, \\
 4bh^2 - 2ab^2 &= y_1^2 x_2 y_2^2 + y_2^2 x_1 y_1^2, \\
 4agh - 2a^2 f &= x_1^2 x_2 y_2 z_2 + x_2^2 x_1 y_1 z_1, \\
 4bhf - 2b^2 g &= y_1^2 x_2 y_2 z_2 + y_2^2 x_1 y_1 z_1, \\
 4cfg - 2c^2 h &= z_1^2 x_2 y_2 z_2 + z_2^2 x_1 y_1 z_1, \\
 4bgh - 2abf &= y_1 z_1 x_2 y_2 z_2 + y_2 z_2 x_1 y_1 z_1, \\
 4chf - 2bcg &= z_1 x_1 y_2 z_2^2 + z_2 x_2 y_1 z_1^2, \\
 4afg - 2cah &= x_1 y_1 z_2 x_2^2 + x_2 y_2 z_1 x_1^2, \\
 4cgh - 2acf &= y_1 z_1 x_2 z_2^2 + y_2 z_2 x_1 z_1^2, \\
 4ahf - 2bag &= z_1 x_1 y_2 x_2^2 + z_2 x_2 y_1 x_1^2, \\
 4bfg - 2cbh &= x_1 y_1^2 z_2^2 y_2 + x_2 y_2^2 z_1^2 y_1, \\
 8f^2 g - 4chf - 2bcg &= z_1^2 x_2 y_2^2 + z_2^2 x_1 y_1^2, \\
 8g^2 h - 4afg - 2cah &= x_1^2 y_2 z_2^2 + x_2^2 y_1 z_1^2, \\
 8h^2 f - 4bgh - 2abf &= y_1^2 z_2 x_2^2 + y_2^2 z_1 x_1^2, \\
 8fg^2 - 4chg - 2acf &= z_1^2 x_2 y_2^2 + z_2^2 x_1 y_1^2,
 \end{aligned}$$

Secondly, consider the system of equations

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l} \mid x, y, z)^3 = 0,$$

$$(\alpha, \beta, \gamma \mid x, y, z) = 0,$$

where the cubic function written at full length is

$$\mathfrak{a}x^3 + \mathfrak{b}y^3 + \mathfrak{c}z^3 + 3\mathfrak{f}y^2z + 3\mathfrak{g}z^2x + 3\mathfrak{h}x^2y + 3\mathfrak{i}yz^2 + 3\mathfrak{j}zx^2 + 3\mathfrak{k}xy^2 + 6\mathfrak{l}xyz.$$

Joining to the system the linear equation

$$(\xi, \eta, \zeta \mid x, y, z) = 0,$$

the linear equations give

$$x : y : z = \beta\zeta - \gamma\eta : \gamma\xi - \alpha\zeta : \alpha\eta - \beta\xi,$$

and the resultant is

$$\square^{-1} \cdot (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l} \mid \beta\zeta - \gamma\eta, \gamma\xi - \alpha\zeta, \alpha\eta - \beta\xi)^3 = 0,$$

which may be represented by

$$\square^{-1} \cdot (\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l} \mid \xi, \eta, \zeta)^3 = 0,$$

where the coefficients  $\mathfrak{a}, \mathfrak{b}$ , &c. are given by means of the Table.

[Table (print p. 460): a ten-by-ten array giving each of  $\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l}$  as a linear combination of  $\mathfrak{a}, \dots, \mathfrak{l}$  with coefficients cubic in  $\alpha, \beta, \gamma$ . The leading entries are  $\mathfrak{a} = -3\mathfrak{f}\beta\gamma + 3\mathfrak{i}\beta^2\gamma + \dots$ ,  $\mathfrak{b} = -\gamma^3\mathfrak{a} + \alpha^3\mathfrak{c} - 3\gamma\alpha^2\mathfrak{g} + 3\gamma^2\alpha\mathfrak{j} + \dots$ , &c.; the full pattern of signs and the higher-order terms in  $\beta\gamma, \gamma\alpha, \alpha\beta$  are preserved in the original scan.]

viz.  $\mathfrak{a} = -\mathfrak{c}\beta^3 + \mathfrak{b}\gamma^3 - 3\mathfrak{f}\beta\gamma^2 - 3\mathfrak{i}\beta^2\gamma$ , &c.

But if the roots of the given system are

$$(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3),$$

then the resultant of the three equations may also be represented by

$$(\xi, \eta, \zeta \mid x_1, y_1, z_1)(\xi, \eta, \zeta \mid x_2, y_2, z_2)(\xi, \eta, \zeta \mid x_3, y_3, z_3),$$

and comparing with the former expression, we find:

$$\begin{aligned} \mathfrak{a} &= x_1x_2x_3, \\ \mathfrak{b} &= y_1y_2y_3, \\ \mathfrak{c} &= z_1z_2z_3, \\ 3\mathfrak{f} &= y_1y_2z_3 + y_2y_3z_1 + y_3y_1z_2, \\ 3\mathfrak{h} &= x_1x_2y_3 + x_2x_3y_1 + x_3x_1y_2, \\ 3\mathfrak{i} &= y_1z_2z_3 + y_2z_3z_1 + y_3z_1z_2, \\ 3\mathfrak{k} &= x_1y_2y_3 + x_2y_3y_1 + x_3y_1y_2, \\ 6\mathfrak{l} &= x_1y_2z_3 + x_1y_3z_2 + x_2y_1z_3 + x_2y_3z_1 + x_3y_1z_2 + x_3y_2z_1. \end{aligned}$$

But there is in the present case a relation independent of the quantities  $\mathfrak{a}$ , &c., viz. we have  $(\alpha, \beta, \gamma \mid x_1, y_1, z_1) = 0$ ,  $(\alpha, \beta, \gamma \mid x_2, y_2, z_2) = 0$ ,  $(\alpha, \beta, \gamma \mid x_3, y_3, z_3) = 0$ , and thence eliminating the coefficients  $(\alpha, \beta, \gamma)$ , we find

$$\nabla = x_1y_2z_3 + x_2y_3z_1 + x_3y_1z_2 - x_1y_3z_2 - x_2y_1z_3 - x_3y_2z_1 = 0.$$

By forming the powers and products of the second degree  $\mathfrak{a}^2, \mathfrak{a}\mathfrak{b}$ , &c., we obtain 55 equations between the symmetric functions of the second degree in each set of roots. But we have  $\nabla^2 =$  a symmetric function

of the roots, and thus the entire number of linear relations is 56, and this is in fact the number of the symmetric functions of the second degree in each set. I use for shortness the sign  $S$  to denote the sum of the distinct terms obtained by permuting the different sets of roots, so that the equations for the fundamental symmetric functions are

$$a = x_1x_2x_3,$$

$$b = y_1y_2y_3,$$

$$c = z_1z_2z_3,$$

$$3f = Sy_1y_2z_3,$$

$$3g = Sz_1z_2x_3,$$

$$3h = Sx_1x_2y_3,$$

$$3i = Sy_1z_2z_3,$$

$$3j = Sz_1x_2x_3,$$

$$3k = Sx_1y_2y_3,$$

$$6l = Sx_1y_2z_3;$$

then the complete system of expressions for the symmetric functions of the second order is as follows, viz.

$$\begin{aligned}
 a^2 &= x_1^2 x_2^2 x_3^2, \\
 b^2 &= y_1^2 y_2^2 y_3^2, \\
 c^2 &= z_1^2 z_2^2 z_3^2, \\
 bc &= y_1 z_1 y_2 z_2 y_3 z_3, \\
 ca &= z_1 x_1 z_2 x_2 z_3 x_3, \\
 ab &= x_1 y_1 x_2 y_2 x_3 y_3, \\
 3af &= S x_1 y_1 x_2 y_2 x_3, \\
 3ch &= S z_1 x_1 z_2 x_2 y_3 z_3, \\
 3bf &= S y_1^2 y_2^2 y_3 z_3, \\
 3cg &= S z_1^2 z_2^2 z_3 x_3, \\
 3ah &= S x_1^2 x_2^2 x_3 y_3, \\
 3cf &= S y_1 z_1 y_2 z_2 z_3^2, \\
 3ag &= S z_1 x_1 z_2 x_3^2, \\
 3bh &= S x_1 y_1 x_2 y_3^2, \\
 3ai &= S x_1 y_1 z_2 x_2 x_3 z_3, \\
 3bj &= S y_1 z_1 x_2 y_2 y_3 x_3, \\
 3ck &= S z_1 x_1 y_2 z_2 z_3 x_3, \\
 3bi &= S y_1^2 y_2 z_2 y_3 z_3, \\
 3cj &= S z_1^2 z_2 x_2 z_3 x_3, \\
 3ak &= S x_1^2 x_2 y_2 x_3 y_3, \\
 3ci &= S y_1 z_1 z_2 z_3^2, \\
 3aj &= S z_1 x_1 x_2 x_3^2, \\
 3bk &= S x_1 y_1 y_2 y_3^2, \\
 6al &= S x_1^2 x_2 y_2 x_3 z_3, \\
 6bl &= S y_1^2 y_2 x_2 y_3 z_3, \\
 6cl &= S z_1^2 z_2 x_2 y_3 z_3, \\
 9f^2 - 6bi &= S y_1^2 y_2^2 z_3^2, \\
 9g^2 - 6cj &= S z_1^2 z_2^2 x_3^2, \\
 9h^2 - 6ak &= S x_1^2 x_2^2 y_3^2, \\
 9i^2 - 6cf &= S y_1^2 z_2^2 z_3^2, \\
 9j^2 - 6ag &= S z_1^2 x_2^2 x_3^2, \\
 9k^2 - 6bh &= S x_1^2 y_2^2 y_3^2, \\
 9fg - 3ck &= S z_1 x_1 y_2 z_2 y_3^2, \\
 9gh - 3ai &= S y_1 z_1 z_2 x_2 z_3^2, \\
 9hf - 3bj &= S z_1 x_1 x_2 y_2 x_3^2, \\
 9jk - 3af &= S x_1^2 x_2 y_2 y_3 z_3, \\
 9ki - 3bg &= S y_1^2 y_2 z_2 z_3 x_3, \\
 9ij - 3ca &= S z_1^2 z_2 x_2 x_3 y_3, \\
 9fi - 3bc &= S y_1^2 y_2 z_2 z_3^2, \\
 9gj - 3ca &= S z_1^2 z_2 x_2 x_3^2, \\
 9hk - 3ab &= S x_1^2 x_2 y_2 y_3^2, \\
 3(2fi + gk + hj - l^2) &= S x_1 y_1 z_1 y_2 z_2 y_3 z_3, \\
 3(2gj - kh - hi + l^2) &= S x_1 y_1 z_1 z_2 x_2 z_3 x_3, \\
 3(2hk - fi - gk + l^2) &= S x_1 y_1 z_1 x_2 y_2 x_3 y_3, \\
 3(2ga - 2hi - l^2) &= S x_1^2 x_2^2 x_3^2, \\
 3(2fb - 2jk - l^2) &= S y_1^2 y_2^2 y_3^2, \\
 3(2gc - 2kl - l^2) &= S z_1^2 z_2^2 z_3^2.
 \end{aligned}$$

As an instance of the application of the formulae, let it be required to eliminate the variables from the three equations,

$$\begin{aligned}(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{f}, \mathfrak{g}, \mathfrak{h}, \mathfrak{i}, \mathfrak{j}, \mathfrak{k}, \mathfrak{l} \mid x, y, z)^3 &= 0, \\ (\mathfrak{a}', \mathfrak{b}', \mathfrak{c}', \mathfrak{f}', \mathfrak{g}', \mathfrak{h}' \mid x, y, z)^2 &= 0, \\ (\alpha, \beta, \gamma \mid x, y, z) &= 0.\end{aligned}$$

This may be done in two different ways; first, representing the roots of the linear equation and the quadric equation by  $(x_1, y_1, z_1), (x_2, y_2, z_2)$ , the resultant will be

$$(\mathfrak{a}, \dots \mid x_1, y_1, z_1)^3 \cdot (\mathfrak{a}, \dots \mid x_2, y_2, z_2)^3,$$

which is equal to  $\mathfrak{a}^2 \cdot x_1^3 x_2^3 + \&c.$ , where the symmetric functions  $x_1^3 x_2^3$ , &c. are given by the formulae  $\mathfrak{a}^2 = x_1^2 x_2^2$ , &c., in which, since the coefficients of the quadratic equation are  $(\mathfrak{a}', \mathfrak{b}', \mathfrak{c}', \mathfrak{f}', \mathfrak{g}', \mathfrak{h}')$ , I have written  $\mathfrak{a}'$  instead of  $\mathfrak{a}$ . Next, if the roots of the linear equation and the cubic equation are represented by  $(x_1, y_1, z_1), (x_2, y_2, z_2), (x_3, y_3, z_3)$ , then the resultant will be

$$(\mathfrak{a}', \dots \mid x_1, y_1, z_1)^2 \cdot (\mathfrak{a}', \dots \mid x_2, y_2, z_2)^2 \cdot (\mathfrak{a}', \dots \mid x_3, y_3, z_3)^2,$$

which is equal to

$$\mathfrak{a}'^3 \cdot x_1^2 x_2^2 x_3^2 + \&c.,$$

the symmetric functions  $x_1^2 x_2^2 x_3^2$ , &c. being given by the formulae  $\mathfrak{a}^2 = x_1^2 x_2^2 x_3^2$ , &c. The expression for the Resultant is in each case of the right degree, viz. of the degrees 6, 3, 2, in the coefficients of the linear, the quadric, and the cubic equations respectively: the two expressions, therefore, can only differ by a numerical factor, which might be determined without difficulty. The third expression for the resultant, viz.

$$\square^{-1} \cdot (\alpha, \beta, \gamma \mid x_1, y_1, z_1) \cdot (\alpha, \beta, \gamma \mid x_2, y_2, z_2) \cdots (\alpha, \beta, \gamma \mid x_6, y_6, z_6),$$

(where  $(x_1, y_1, z_1), \dots, (x_6, y_6, z_6)$  are the roots of the cubic and quadratic equations) compared with the foregoing value, leads to expressions for the fundamental symmetric functions of the cubic and quadratic equations, and thence to expressions for the other symmetric functions of these two equations; but it would be difficult to obtain the actually developed values even of the fundamental symmetric functions. I hope to return to the subject, and consider in a general point of view the question of the formation of the expressions for the other symmetric functions by means of the expressions for the fundamental symmetric functions.

[End of paper 149.]